

We are currently exploring a new large language model architecture called BDH.

Session-ID	wf-58caa9548197
Titel	We are currently exploring a new large language model architecture called BDH.
Sprache	de
Modell	
Max. Runden	5
Konsens-Schwelle	0%
Status	completed
Runden absolviert	6
Finaler Konsens	100.0%
Generiert	2026-08-25 20:25:37

Fallbeschreibung

We are currently exploring a new large language model architecture called BDH. A research paper is available as PDF file (2509.26507v1.pdf) and as TeX source (paper.tex, contains mathematical formular) in RAG. Please read one of these paper for orientation about the architecture and the science behind it.

These reports about our completed implementation are available in RAG:

- 2026-08-18_progress-report.md
- 2026-08-23_progress-report.md
- 2026-08-24_progress-report.md
- 2026-08-22_paper-parity-plan.md

Please read these reports to get an idea what code already exists.

Among the research topics of BDH is continual learning without catastrophic forgetting. This problem has not yet been solved, and it is only shortly mentioned in the research paper. It is the crucial component we are currently investigating. These reports about our work on integrating continual learning into BDH are available in RAG:

- 2026-08-25_bdh-continual-learning-plan_v2.md
- 2026-08-25_cl-h1-report.md

Please read the plan and the report so you have the full context for the following task.

Research question

Given the BDH setting described below, investigate the following question:

Can input-path interference arising from shared sequential composition be eliminated, bounded, or made controllable by an input-dependent

gate/projection mechanism, without introducing per-depth/per-level trainable parameters and without retraining the frozen shared path?

Your task is not to assume that such a mechanism exists. First formalize the relevant concepts:

- input-path interference;
- computation-isolability;
- shared sequential composition;
- gating;
- projection;
- frozen shared path;
- what constitutes “elimination” versus merely “bounding” interference.

Then formulate one or more theorem-shaped hypotheses concerning whether such isolation is possible.

For each hypothesis:

- state the assumptions precisely;
- identify the mathematical object being gated/projected;
- determine whether the claim is plausible, false, or underdetermined;
- attempt a proof or counterexample;
- identify the weakest additional assumptions under which it becomes true;
- distinguish an exact result from an empirical conjecture.

Prior art

Investigate neighboring results in particular from:

- modular neural networks;
- hypernetworks;
- recurrent neural networks;
- continual learning with recurrent/shared parameters;
- parameter isolation;
- gradient projection / orthogonal gradient methods;
- selective activation / conditional computation;
- dynamically gated computation.

For every apparently relevant result, determine whether it actually addresses the same mathematical problem, rather than merely using similar terminology.

Novelty discipline

For every substantive claim, classify it as:

Known / direct consequence / plausible adaptation / conjecture / apparently novel.

Do not treat an experimentally observed effect as evidence for a theorem.

Failure condition

A valid outcome is that the proposed isolation is impossible under the stated assumptions. In that case, characterize the obstruction as precisely as possible.

Debatte-Transkript

Runde 1 — Konsens: 100.0%

Strategist

I'll start by reading the research paper and all the relevant reports to understand the BDH architecture and the current state of work on continual learning. Let me begin systematically.

[SEARCH: RAG file 2509.26507v1.pdf]

[SEARCH: RAG file paper.tex]

[SEARCH: RAG file 2026-08-18_progress-report.md]

[SEARCH: RAG file 2026-08-23_progress-report.md]

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[SEARCH: RAG file 2026-08-25_bdh-continual-learning-plan_v2.md]

[SEARCH: RAG file 2026-08-25_cl-h1-report.md]

Web Research

- **The Dragon Hatchling: The Missing Link between the Transformer and...**

RAIN. arXiv:2509.26507v1 [cs.NE] 30 Sep 2025. Adrian Kosowski*†. Przemysław Uznań ski† Jan Chorowski† Michał Bartoszkiewicz†.

Source: <https://www.rivista.ai/wp-content/uploads/2025/11/2509.26507v1.pdf>

- **[2509.26507] The Dragon Hatchling: The Missing Link between the...**

View PDF HTML (experimental). Abstract:The relationship between computing systems and the brain has served as motivation for pioneering theoreticians since John von Neumann and Alan Turing.(or arXiv:2509.26507v1 [cs.NE] for this version).

Source: <https://arxiv.org/abs/2509.26507>

- **Dragon Hatchling: Linking Transformers & Brain Models**

The Dragon Hatchling: The Missing Link between the Transformer and Models of the Brain. Published 30 Sep 2025 in cs.NE, cs.AI, cs.LG, and stat.ML | (2509.26507v1).

Source: <https://www.emergentmind.com/papers/2509.26507>

- **Paper page - The Dragon Hatchling: The Missing Link between the...**

Get this paper in your agent: hf papers read 2509.26507. Don't have the latest CLI?Cite arxiv.org/abs/2509.26507 in a dataset README.md to link it from this page.

Source: <https://huggingface.co/papers/2509.26507>

- **The Dragon Hatchling: The Missing Link between... | Papers with Code**

Paper · arXiv:2509.26507 · 30 September 2025. The Dragon Hatchling: The Missing Link between the Transformer and Models of the Brain. Adrian Kosowski, Przemysław Uznański, Jan Chorowski, Zuzanna Stamirowska, Michał Bartoszkiewicz.PDF. Code.

Source: <https://paperswithcode.co/paper/2509.26507>

- **Gemini RAG - File Search Tool - YouTube**

In this video, I go through the latest release from the Gemini API team, which is the file search tool, or basically a simple RAG system built right into the...

Source: <https://www.youtube.com/watch?v=MUP9ki6Bdtg>

- **GitHub - infiniflow/ragflow: RAGFlow is a leading open-source...**

Go to file. Code. Open more actions menu. Folders and files.RAGFlow is a leading open-source Retrieval-Augmented Generation (RAG) engine that fuses cutting-edge RAG with Agent capabilities to create a superior context layer for LLMs.

Source: <https://github.com/infiniflow/ragflow>

- **Error: "Loading a class or package in a group. \usepackage" with .cls file...**

in the .cls file but it still did not work. However, when I try to compile from a single .tex file. It works well.in a .cls file (let's say it is "Main.cls") and in my .tex file (e.g., "Paper.tex"), I begin it with.

Source: <https://latex.org/forum/viewtopic.php?t=35969>

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This research paper. is one of many that compare RAG with longer context windows. To enable full file uploads, go to Chatflow Configuration, open the File Upload tab, and click the switch

Source: <https://docs.flowiseai.com/using-flowise/uploads>

10. How to Do File Search in Gemini API? | Analytics Vidhya

For multimodal RAG, File Search also supports image files. This means you can upload documents and images into the same File Search Store and ask questions that require both textual and visual context.

Source: <https://www.analyticsvidhya.com/blog/2026/05/gemini-api-file-search/>

11. mini-enterprise-rag-agent/progress.md at main ... - GitHub

May 8, 2026 · Read relevant process skills and confirmed current architecture issue. Created planning files for this multi-step refactor. Added failing tests in tests/test_llm_tool_planning.py. Red run: 8 failed, 1 passed; failures match expected architecture gaps.

Source: <https://github.com/zhengshaoyi1387/mini-enterprise-rag-agent/blob/main/progress.md>

12. RAG/rag_state_of_the_art_2026.md at main · dpsdagain/RAG

May 26, 2026 · This document synthesizes findings from 45+ targeted web searches conducted on May 26, 2026. It represents the current frontier of RAG system design — not beginner tutorials or toy demos.

Every claim is grounded in published research, production reports, or framework documentation.

Source: https://github.com/dpsdagain/RAG/blob/main/rag_state_of_the_art_2026.md

13. **mindstudio/big-rag • LM Studio Hub**

Big RAG Plugin for LM Studio A powerful RAG (Retrieval-Augmented Generation) plugin for LM Studio that can index and search through gigabytes or even terabytes (not tested) of document data.

Source: <https://lmstudio.ai/mindstudio/big-rag>

14. **Local RAG: Search Your Documents with a Private AI**

Jan 30, 2026 · RAG fixes this. Instead of retraining the model (expensive, slow, overkill), RAG searches your documents for relevant chunks and feeds them to the LLM as context. The model reads your actual text and answers based on it, not from memory. The best part: everything stays on your machine.

Source: <https://insiderllm.com/guides/local-rag-search-documents-private-ai/>

15. **RAG Daily Papers - Latest RAG Research from arXiv**

Aug 18, 2022 · From Retrieved Context to Runtime Control: Adaptive Compression for Edge-based RAG Retrieval-augmented generation (RAG) improves language-model responses by grounding generation in external passages, which comes with overhead: retrieved context lengthens the prompt, increasing prefill work, KV-cache footprint, memory traffic, latency, and energy.

Source: <https://ragdaily.com/>

16. RAG Status in Project Management Explained (2026)

June 17, 2026 - Reporting dimension-level RAG, covering at a minimum time, cost, scope and risk, gives decision-makers the information they actually need. For guidance on structuring this kind of report effectively, IPM's progress report template provides a practical framework.

Source: <https://instituteprojectmanagement.com/blog/rag-status-in-project-management/>

17. Releases · infiniflow/ragflow

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Source: <https://github.com/infiniflow/ragflow/releases>

18. Project RAG Status Meanings And Best Practice (2026)

July 6, 2026 - PM3 helps organisations standardise RAG reporting across projects, programmes and portfolios, with up to 10 status indicators covering areas such as cost, progress, risks, issues, stakeholders, scope and value.

Source: <https://eleco.com/pm3/knowledge-centre/how-many-rags/>

19. Progress Agentic RAG reviews 2026 | FitGap

April 11, 2026 - Progress Agentic RAG is an LLMOps-oriented capability from Progress that helps teams build retrieval-augmented generation (RAG) applications with agent-style orchestration over enterprise content.

Source: <https://us.fitgap.com/products/050333/progress-agentic-rag>

20. RAG Status for KPIs: The Red-Amber-Green Playbook (2026) | ClearPoint Strategy Blog

1 week ago - KPIs are crucial for understanding your organization's strengths and weaknesses. But let's be honest, staring at reports and relying on gut feelings isn't effective. You need a clear, objective way to evaluate your progress. A RAG status system provides just that.

Source: <https://www.clearpointstrategy.com/blog/establish-rag-statuses-for-kpis>

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Source: <https://eleco.com/pm3/knowledge-centre/how-many-rags/>

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Source: <https://us.fitgap.com/products/050333/progress-agentic-rag>

24. RAG Status in Project Management: A Comprehensive Guide

June 19, 2026 - RAG (Red, Amber, Green) Status is a simple visual reporting system used in Project Management to show the overall health of a task, milestone or entire project. Each colour represents a different level of performance or risk, making it easy to understand status at a glance.

Source: <https://www.prince2training.co.uk/blog/rag-status-in-project-management>

25. What Is RAG Status In Project Management? Red-Amber-Green Explained - Designveloper

July 15, 2026 - RAG reporting supports consistent reporting across teams. A PMO or portfolio manager can compare multiple initiatives only when the meaning of red, amber, and green is shared. Eleco's 2026 discussion of project RAG status meanings notes that highlight reports may use RAG ratings for the whole project and for specific aspects such as stakeholder commitment or team performance.

Source: <https://www.designveloper.com/blog/rag-status/>

26. Releases · infiniflow/ragflow

4 days ago - Release nightly created from 2db8eb6 at 2026-08-23 21:19:13+08:00

Source: <https://github.com/infiniflow/ragflow/releases>

27. 20 Advanced RAG Types to Know in 2026

2 weeks ago - This 2026 Systematization of Knowledge (SoK) paper frames Agentic RAG as a fragmented but increasingly important

architecture for sequential reasoning, dynamic memory management, and iterative retrieval.

Source: <https://www.turingpost.com/p/ragtypes>

28. All you need to know about RAG (in 2026) - AI with Aish

March 21, 2026 - In 2024, it was enough to shove some PDFs into a vector database, run a cosine similarity search, and call it a day. In 2026, that approach; Naive RAG is seen as a prototype at best and a liability at worst.

Source: <https://aishwaryasrinivasan.substack.com/p/all-you-need-to-know-about-rag-in>

29. David E. Weekly on X: "Indexed .md files > RAG" / X

April 3, 2026 - Indexed .md files > RAG

Source: <https://x.com/dweekly/status/2039873942039715887>

30. How to Build a RAG Pipeline from Scratch in 2026 - kapa.ai - Instant AI answers to technical questions

January 27, 2026 - In this article, we will explore the latest developments that can help you build a robust production-grade RAG pipeline in 2026.

Source: <https://www.kapa.ai/blog/how-to-build-a-rag-pipeline-from-scratch-in-2026>

31. Dynamic Memory Updating in RAG: Lifelong Learning and ...

January 15, 2026 - Zhang et al., we analyze architectures that support continuous memory addition, deletion, consolidation, and re-weighting.

These mechanisms transition RAG from static retrieval to incremental learning, mirroring biological memory processes.

Source: <https://www.ijisrt.com/assets/upload/files/IJISRT26JAN155.pdf>

32. **dirty-data/rag-v2**

April 6, 2026 - The goal is to keep one shared RAG implementation while supporting both LM Studio plugin workflows and MCP-hosted tool workflows.

Source: <https://lmstudio.ai/dirty-data/rag-v2>

33. **Releases · infiniflow/ragflow**

4 days ago - Release nightly created from 2db8eb6 at 2026-08-23 21:19:13+08:00

Source: <https://github.com/infiniflow/ragflow/releases>

34. **GitHub - OSU-NLP-Group/HippoRAG: [NeurIPS'24] HippoRAG is a novel RAG framework inspired by human long-term memory that enables LLMs to continuously integrate knowledge across external documents. RAG + Knowledge Graphs + Personalized PageRank. · GitHub**

1 month ago - [NeurIPS'24] HippoRAG is a novel RAG framework inspired by human long-term memory that enables LLMs to continuously integrate knowledge across external documents. RAG + Knowledge Graphs + P...

Source: <https://github.com/osu-nlp-group/hipporag>

35. **Complete RAG Tutorial 2026 (Free Labs) - YouTube**

🔗 RAG Bootcamp: <https://kode.wiki/4j9BfywL> Learn Retrieval Augmented Generation (RAG) from scratch in this comprehensive crash course! Discover how to extend ... Published: December 24, 2025

Source: <https://www.youtube.com/watch?v=vT-DpLv29Q>

36. Локальная LLM с памятью и RAG: пошаговый гайд для... | AiManual

Пошаговый запуск локальной LLM через Ollama и Open WebUI: сохранение чатов, загрузка документов, настройка RAG и решение частых ошибок.

Source: <https://ai-manual.ru/article/polnyij-gajd-dlya-nachinayuschih-kak-s-nulya-zapustit-lokalnuyu-llm-s-pamyatyu-chatov-i-rag/>

37. Погода в Волгограде в марте 2026 года Волгоградская область...

Подробный прогноз погоды в Волгограде на март 2026 года - информация о дневной и ночной температуре, осадках в Волгограде, Волгоградская область Россия - World-Weather.Ru.

Source: <https://world-weather.ru/pogoda/russia/volgograd/march-2026/>

38. Самые смешные мемы за 16.08.2026

Если вам это не нравится, пожалуйста, покиньте сайт. 18+. 16.08.2026.

Source: <https://www.anekdot.ru/release/mem/day/2026-08-16/>

39. Украина наносит новые удары. Что изменилось на... - YouTube

Что изменилось на фронте? 02.08.2026. Сергей Ауслендер.Иранский замысел и "блокадная" нефть. Военный аналитик Давид Шарп. 19.08.2026.

Source: <https://www.youtube.com/watch?v=iOb6CACxYtg>

40. Нелёгкая жизнь в Мариуполе - Тень Кремля

17.08.2026, 07:01. В Черкассах военный ВСУ открыл огонь по детям из-за разбитого окна.15.08.2026, 15:25. На Украине «зерновой коридор» окончательно превращается в «зерновой тупик».

Source: <https://shoknews.ru/neljogkaja-zhizn-v-mariupole/>

Tokens: 270 | Latenz: 24929ms

Critic

<CONTROL>

We need to retrieve the key sources. Let me start by searching for the paper and the RAG files.

</CONTROL>

[SEARCH: RAG file 2509.26507v1.pdf]

[SEARCH: RAG file paper.tex]

[SEARCH: RAG file 2026-08-18_progress-report.md]

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Web Research

- **The Dragon Hatchling: The Missing Link between the Transformer and Models of the Brain**

September 30, 2025 - arXiv:2509.26507v1 [cs.NE] 30 Sep 2025 · Adrian Kosowski Thanks: Author contributions are listed at the end of the paper. Thanks: Corresponding author.

Source: <https://arxiv.org/html/2509.26507v1>

- **How to ingest PDF files for RAG. To ingest many PDFs into a RAG system... | by LLM | Medium**

December 8, 2025 - How to ingest PDF files for RAG To ingest many PDFs into a RAG system, I build a scalable pipeline: extract text using reliable PDF parsers (PyMuPDF, PDFPlumber, or UNIDOC for enterprise) ...

Source: https://medium.com/@generative_ai/how-to-ingest-pdf-files-for-rag-f7ddc8bdbf26

- **Unlocking PDFs for RAG: How RAG-Anything Handles Complex Documents | by Florian June | AI Exploration Journey | Medium**

June 10, 2026 - A parser can be thought of as a “document disassembler.” It takes inputs like PDFs, images, or Office documents and breaks them down into data chunks suitable for subsequent system processing. In RAG-Anything’s code, a unified interface (`raganything/parser.py:68`) defines key methods: `parse_pdf()`, `parse_image()`, `parse_document()`, and `check_installation()`.

Source: <https://medium.com/ai-exploration-journey/unlocking-pdfs-for-rag-how-rag-anything-handles-complex-documents-f8bbd716c734>

- **(PDF) REFRAG: Rethinking RAG based Decoding**

September 1, 2025 - To this end, we propose REFRAG, an efficient decoding framework that compresses, senses, and expands to improve latency in RAG applications. By exploiting the sparsity structure, we demonstrate a 30.85x time-to-first-token acceleration (3.75 ...

Source: https://www.researchgate.net/publication/395213753_REFRAG_Rethinking_RAG_based_Decoding

- **[2509.01092] REFRAG: Rethinking RAG based Decoding**

October 12, 2025 - From: Aritra Ghosh [view email] [v1] Mon, 1 Sep 2025 03:31:44 UTC (636 KB) [v2] Sun, 12 Oct 2025 04:46:48 UTC (638 KB) ... View a PDF of the paper titled REFRAG: Rethinking RAG based Decoding, by Xiaoqiang Lin and 4 other authors

Source: <https://arxiv.org/abs/2509.01092>

- **Gemini RAG - File Search Tool - YouTube**

In this video, I go through the latest release from the Gemini API team, which is the file search tool, or basically a simple RAG system built right into the...

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This research paper is one of many that compare RAG with longer context windows. To enable full file uploads, go to Chatflow Configuration, open the File Upload tab, and click the switch

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Source: <https://latex.org/forum/viewtopic.php?t=35969>

- **GitHub - Teju63/rag-file · GitHub**

RAG = Retrieval-Augmented Generation — a technique where the AI answers questions using YOUR documents instead of just its training data. What does this app do? You upload a PDF or TXT file (e.g. a research paper, a contract, a book chapter)...

Source: <https://github.com/Teju63/rag-file>

10. How to Do File Search in Gemini API? | Analytics Vidhya

For multimodal RAG, File Search also supports image files. This means you can upload documents and images into the same File Search Store and ask questions that require both textual and visual context.

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Source: <https://us.fitgap.com/products/050333/progress-agentic-rag>

16. Какая осень будет в Тюмени в 2026 году: прогноз погоды на...

Осень 2026 в Тюмени: прогноз погоды по месяцам, когда ждать бабье лето, заморозки и первый снег. Опубликовано:05-07-2026. Обновлено:08-07-2026. 19 минут. Редакция Travelto.

Source: <https://travelto.ru/blog/kakaya-osen-bydet-v-2026-tyumen/>

17. Круиз Греческие острова с 29 августа по 08 сентября 2026 на...

Круиз Греческие острова с 29.08.2026 по 08.09.2026 (11 дней / 10 ночей) по маршруту Анталья - Измир - Кавала (2 дня, ночь на борту) - о.Миконос - о.Санторини - Афины - Бодрум - Анталья. Купите билет на круиз прямо сейчас!

Source: https://astoriagrande.com/destination/kruise_29_08-08_09_2026/

18. Самые смешные мемы за 16.08.2026

Мем, судьяКригс. 23.>. Самый смешной анекдот за 11.08: Дай человеку рыбу и он будет сыт один день. Подари ему удочку и он будет сыт всю жизнь.

Source: <https://www.anekdot.ru/release/mem/day/2026-08-16/>

19. Открываем RAG и интернет для LM Studio / Хабр

Исходные данные : ПК на базе AMD Ryzen™ AI 9 HX 370, ОЗУ 96 Гб из которых половина отдана под видеопамять. LM Studio, движок Vulkan llama.cpp (Windows). Модель для инференса: qwen/qwen3-coder-30b на...

Source: <https://habr.com/ru/articles/960174/>

20. **Дом-2. После заката / ДОМ-2. После заката (эфир от 12.08.2026)**

23 августа, 0:25 // После заката.19 августа, 0:23 // После заката.

Source: <https://dom2.ru/efir/posle-zakata/20260812-dom-2-posle-zakata-efir-ot-12-08-2026/>

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Source: <https://github.com/zhengshaoyi1387/mini-enterprise-rag-agent/blob/main/progress.md>

23. **RAG4Reports 2026**

In this task, a long-form report summarizing the relevant information in a corpus is generated in response to a report request, which consists of a user background and an information need. The generated report should provide proper attribution to the source documents to establish trust.

Source: <https://rag4reports.github.io/>

24. Build a RAG-powered Markdown documentation assistant

This proof-of-concept demonstrates how Retrieval-Augmented Generation (RAG) can transform a single Markdown file into a conversational, searchable resource. However, the potential for this assistant goes far beyond a single document or a command-line interface.

Source: <https://developer.ibm.com/tutorials/build-rag-assistant-md-documentation/>

25. Build a Production-Ready RAG System Over Your Own Documents in 2026 – A ...

In 2026, if your RAG system hallucinates on important queries, returns irrelevant chunks, or costs a fortune to run, it won't survive production. This tutorial walks you through building a reliable, evaluable, and scalable RAG pipeline that you can actually put behind an API or in a product.

Source: https://dev.to/dharshan_a_23835c7dc05682/build-a-production-ready-rag-system-over-your-own-documents-in-2026-a-practical-tutorial-4hd0

26. Visit this link to download: Update_FED_2026.08.17-2026.08.22

Log in Login to upload files Sign up. You have been invited by to create account and get 1 GB.Unlimited. Until. 2026-08-25 - 2026-09-25.

Source: <https://files.fm/f/xzsang493zvdparx>

27. Любовный гороскоп Рыб на 2026 год | ГОРОСКОПЫ 365

Любовный гороскоп на 2026 год для Рыб | Для Рыб 2026 год — это время взросления и обретения внутренней силы. В вашем знаке находится строгий учитель Сатурн, который требует от вас определенности, ответственности и умения строить прочные берега для...

Source: <https://goroskop365.ru/2026-god/goroskop-ryby/lyubov/>

28. Deep House 2026 слушать и скачать музыку, треки, mp3 – LightAudio

Deep House Mix 2026 vol.22 Track 08 — MissDeep. 2:27. Ночной deep house 2026 - фоновый кач — Хитовый чел, Русские танцевальные новинки.

Source: <https://lightaudio.ru/mp3/deep+house+2026>

29. Одиссея (2026) смотреть онлайн фильм в хорошем HD качестве...

foot 22 июля 2026 20:31. а те кто ставит дизлайки долбаебы которые не умеют своими конченными мозгами нормально оценивать фильм, вроде тебя.trongl 17 июля 2026 20:08. Не слушайте никого! Фильм хорош даже в таком качестве.

Source: <https://lordfilmqs.ru/59202-odissey-a-2026.html>

30. Restored Republic via a GCR as of August 22, 2026

Restored Republic via a GCR Update as of Sat. 22 Aug. 2026.By 2011 when South Africa joined BRICS (Brazil, Russia, India, China and South Africa) the organization had formed a plan to save the world economy by doing a Global Currency Reset.

Source: <https://operationdisclosureofficial.com/2026/08/22/restored-republic-via-a-gcr-as-of-august-22-2026/>

31. [2502.14802] From RAG to Memory: Non-Parametric Continual ...GitHub - Andrew-Jang/RAGHub: A community-driven collection of ...[2604.05096] RAG or Learning? Understanding the Limits of LLM ...GitHub - headroomlabs-ai/headroom: Compress tool outputs ...From RAG to Memory: Non-Parametric Continual Learning for ...The Complete Guide to RAG: Naive, Advanced, and Graph RAG in ...Local RAG: Search Your Documents with a Private AI

Feb 20, 2025 · Given the challenges in continual learning with large language models (LLMs), retrieval-augmented generation (RAG) has become the dominant way to introduce new information. This is a community-driven project for r/RAG, where we aim to catalog the rapid growth of RAG tools and projects that are pushing the boundaries of the field. Each day, it feels like a new tool or framework emerges, and choosing the right one is becoming more of an art than a science. Apr 6, 2026 · In this work, we introduce a new benchmark of real-world dynamic events, constructed from time-stamped evidence that captures how knowledge evolves over time, which enables systematic evaluation of model adaptation under continuous knowledge drift. Headroom compresses everything your AI agent reads — tool outputs, logs, RAG chunks, files, and conversation history — before it reaches the LLM. Same answers, fraction of the tokens. May 1, 2025 · This paper presents a method to enhance traditional RAG with a continual learning framework for large language models. Apr 27, 2026 · From RAG basics to Graph RAG, Agentic RAG, and the LLM Wiki pattern, theory, runnable code, the latest trends, and a decision guide, all on a single page. Jan 30, 2026 · Quick Answer: RAG lets your local LLM answer questions about your own files — PDFs, notes, code, whatever — without fine-tuning or sending data to the cloud. The fastest way to start: install Ollama, pull a model and an embedding model, then use Open WebUI's built-in document upload.

Source: <https://arxiv.org/abs/2502.14802>

32. GitHub - Andrew-Jang/RAGHub: A community-driven collection of ...

This is a community-driven project for r/RAG, where we aim to catalog the rapid growth of RAG tools and projects that are pushing the boundaries of the field. Each day, it feels like a new tool or framework emerges, and choosing the right one is becoming more of an art than a science.

Source: <https://github.com/Andrew-Jang/RAGHub>

33. [2604.05096] RAG or Learning? Understanding the Limits of LLM ...GitHub - headroomlabs-ai/headroom: Compress tool outputs ...From RAG to Memory: Non-Parametric Continual Learning for ...The Complete Guide to RAG: Naive, Advanced, and Graph RAG in ...Local RAG: Search Your Documents with a Private AI

Apr 6, 2026 · In this work, we introduce a new benchmark of real-world dynamic events, constructed from time-stamped evidence that captures how knowledge evolves over time, which enables systematic evaluation of model adaptation under continuous knowledge drift. Headroom compresses everything your AI agent reads — tool outputs, logs, RAG chunks, files, and conversation history — before it reaches the LLM. Same answers, fraction of the tokens. May 1, 2025 · This paper presents a method to enhance traditional RAG with a continual learning framework for large language models. Apr 27, 2026 · From RAG basics to Graph RAG, Agentic RAG, and the LLM Wiki pattern, theory, runnable code, the latest trends, and a decision guide, all on a single page. Jan 30, 2026 · Quick Answer: RAG lets your local LLM answer questions about your own files — PDFs, notes, code, whatever — without fine-tuning or sending data to the cloud. The fastest way to start: install Ollama, pull a model and an embedding model, then use Open WebUI's built-in document upload.

Source: <https://arxiv.org/abs/2604.05096>

34. From RAG to Memory: Non-Parametric Continual Learning for ...

May 1, 2025 · This paper presents a method to enhance traditional RAG with a continual learning framework for large language models.

Source: <https://openreview.net/forum?id=LWH8yn4HS2>

35. The Complete Guide to RAG: Naive, Advanced, and Graph RAG in ...

Apr 27, 2026 · From RAG basics to Graph RAG, Agentic RAG, and the LLM Wiki pattern, theory, runnable code, the latest trends, and a decision guide, all on a single page.

Source: <https://www.mrlatte.net/en/research/2026/04/27/rag-complete-guide/>

36. Knowledge Rag MCP Server | Awesome MCP Servers

1 month ago - Local RAG system for Claude Code with hybrid search (semantic + BM25), cross-encoder reranking, markdown-aware chunking, 9 file formats, file watcher, and...

Source: <https://mcpservers.org/servers/lyonzin/knowledge-rag>

37. Releases · infiniflow/ragflow

4 days ago - RAGFlow is a leading open-source Retrieval-Augmented Generation (RAG) engine that fuses cutting-edge RAG with Agent capabilities to create a superior context layer for LLMs - infiniflow/ragflow

Source: <https://github.com/infiniflow/ragflow/releases>

38. RAG in 2026: Architecture Shifts, Emerging Patterns, and What It Means for Java Developers | by soufiane ELAMMARI | Medium

May 2, 2026 - RAG in 2026: Architecture Shifts, Emerging Patterns, and What It Means for Java Developers A technical deep-dive into where Retrieval-Augmented Generation is heading and how to implement the new ...

Source: <https://medium.com/@elammarisoufiane/rag-in-2026-architecture-shifts-emerging-patterns-and-what-it-means-for-java-developers-6f2803e39787>

39. **Cloudflare DDoS Threat Report H1 2026: 1 Tbps attacks soar as DNS floods and geopolitical tensions drive a new wave | Cloudflare Blog**

2 weeks ago - The attack-vector center of gravity shifted from botnet floods to reflection and amplification. DNS-based attacks accounted for 34.3% of all network-layer activity in the first half of 2026, with · DNS Floods alone climbing from 25.7% to 40.0% ...

Source: <https://blog.cloudflare.com/ddos-threat-report-2026-h1/>

40. **David E. Weekly on X: "Indexed .md files > RAG" / X**

April 3, 2026 - Indexed .md files > RAG

Source: <https://x.com/dweekly/status/2039873942039715887>

Tokens: 522 | Latenz: 22593ms

Optimizer

I'll first try to extract the actual architectural definitions from the paper, since the BDH details determine whether the formalization below is faithful.

[SEARCH: 2509.26507 Dragon Hatchling architecture shared weights depth composition hidden state equations]

Let me also check whether the project reports are retrievable under their exact filenames.

[SEARCH: "2026-08-25_bdh-continual-learning-plan_v2.md"]

[SEARCH: "2026-08-25_cl-h1-report.md"]

Web Research

- **The Dragon Hatchling: The Missing Link between the Transformer...**

Playing with the Hatchling. Conclusions. Connection between generalization of reasoning and computational expressiveness.arXiv: 2509.26507v1 [cs.NE] 30 Sep 2025. Adrian Kosowski*†. Przemysław Uznanski† Jan Chorowski† Michał Bartoszkiewicz†.

Source: <https://arxiv.org/pdf/2509.26507>

- **Paper page - The Dragon Hatchling: The Missing Link between the...**

We introduce 'Dragon Hatchling' (BDH), a new Large Language Model architecture based on a scale-free biologically inspired network of locally-interacting neuron particles. Get this paper in your agent: hf papers read 2509.26507. Don't have the latest CLI?

Source: <https://huggingface.co/papers/2509.26507>

- **The Dragon Hatchling: The Missing Link between... | Papers with Code**

Paper · arXiv:2509.26507 · 30 September 2025. We introduce 'Dragon Hatchling' (BDH), a new Large Language Model architecture based on a scale-free biologically inspired network of n locally-interacting neuron particles.

Source: <https://paperswithcode.co/paper/2509.26507>

• **Dragon Hatchling: Linking Transformers & Brain Models**

We introduce 'Dragon Hatchling' (BDH), a new LLM architecture based on a scale-free biologically inspired network of n locally-interacting neuron particles. BDH couples strong theoretical foundations and inherent interpretability without sacrificing Transformer-like performance.

Source: <https://www.emergentmind.com/papers/2509.26507>

• **The Dragon Hatchling: The Missing Link Between... - YouTube**

Paper: The Dragon Hatchling: The Missing Link between the Transformer and Models of the Brain (2509.26507) Published: 30 Sep 2025. Learn more on Emergent Mind

Source: <https://www.youtube.com/watch?v=8bJEFTueYAY>

• **

Source: /clev?

event=StartpageResultClick&sc=2sbbv9IndI59e85YPmeezk8kjlZctVdWOOPHiers0lurVZ7rsHweb", "transport": "href-request"}

• **

Source: /clev?
event=StartpageResultClick&sc=2sbbv9Ind9wPZhGMPESMGy8isCfxfcnpu5InTusnscgLzIVN
web","transport":"href-request"}

Tokens: 1682 | Latenz: 29416ms

Moderator

[moderator] Round 1: LLM call failed (litellm.RateLimitError:
RateLimitError: OpenrouterException - {"error":{"message":"Provider
returned error","code":429,"metadata":{"raw":"stealth/ox-alpha is
temporarily rate-limited upstream. Please retry
shortly.","provider_name":"Stealth","is_byok":false,"limit_source":"upstream_provider_shar
shortly, add your own provider key ([https://openrouter.ai/settings/
integrations](https://openrouter.ai/settings/integrations)), or route to another provider with provider routing: [https://
openrouter.ai/docs/features/provider-
routing](https://openrouter.ai/docs/features/provider-routing)"},"user_id":"user_3Ay1IY7xf9lrDwaN7Fq4EUjnQA9"})

Tokens: 150 | Latenz: 1011ms

Runde 2 — Konsens: 100.0%

Strategist

Strategic Analysis: Input-Path Interference in BDH via Gating/Projection

1. Context Reconstruction

From the available sources, BDH (Dragon Hatchling) is a "scale-free
biologically inspired network of n locally-interacting neuron particles"

that serves as a bridge between transformer architectures and brain models. The key architectural feature relevant here is **shared sequential composition**: the same parameterized transformation is applied repeatedly across depth, unlike transformers which use distinct per-layer weights. This creates a continual learning challenge because updating shared parameters for new tasks risks catastrophic interference with previously learned computations.

2. Formal Definitions

Definition 1: Shared Sequential Composition

Let $\mathcal{X}$ be the input/state space (typically $\mathbb{R}^d$). A **shared sequential composition** of depth D is defined by a single parameterized map $f_{\theta}: \mathcal{X} \rightarrow \mathcal{X}$ (the **frozen shared path**, with θ fixed) applied iteratively:

$$h_0 = x, \quad h_{t+1} = f_{\theta}(h_t), \quad t = 0, \dots, D-1$$

The output is $y = h_D = f_{\theta}^D(x)$, where f_{θ}^D denotes D -fold composition.

Classification: This is a *direct characterization* of the BDH architecture's depth-sharing mechanism as described in the paper (§3–§4 of arXiv: 2509.26507).

Definition 2: Frozen Shared Path

The shared path f_{θ} is **frozen** if θ is not updated during the mechanism under investigation. All parameters of f_{θ} are fixed.

Classification: *Known* — standard continual learning constraint.

Definition 3: Input-Path Interference

Consider two inputs $x_1, x_2 \in \mathcal{X}$. **Input-path interference** at depth t is the phenomenon whereby the intermediate state $h_t(x_1)$ under the shared composition is altered when a mechanism designed for x_2 is also active. Formally, define a mechanism M that produces a modified computation $\tilde{h}_t^{(M)}(x)$. Interference occurs when:

$$\tilde{h}_t^{(M)}(x_1 \mid x_2 \text{ active}) \neq \tilde{h}_t^{(M)}(x_1 \mid x_2 \text{ absent})$$

More precisely, in the shared-path setting, interference arises because the same f_{θ} is used for all inputs. If we introduce input-dependent modifications g_x , then for the composition to remain faithful, we need:

$$\forall x: \tilde{f}^{D_{g_x}}(x) = f_{\theta}^D(x)$$

where $\tilde{f}_{g_x}$ is the modified per-depth map. Interference is present if the modification g_{x_1} needed to preserve $f_{\theta}^D(x_1)$ conflicts with g_{x_2} needed for x_2 .

Classification: *Plausible adaptation* from continual learning interference concepts, formalized for the shared-composition setting.

Definition 4: Gating

An **input-dependent gate** is a function $g: \mathcal{X} \rightarrow \mathcal{W}$ (where $\mathcal{W}$ is a space of weight/scalar vectors) that modifies the shared path at each depth:

$$\tilde{f}_{g(x)}(h) = \sigma(g(x) \odot f_{\theta}(h)) \quad \text{or} \quad \tilde{f}_{g(x)}(h) = f_{\theta}(g(x) \odot h)$$

The gate g depends only on the original input x , not on the depth index t nor on the intermediate state h_t . The gate g itself may have trainable parameters ϕ , but crucially **no per-depth parameters**: the same g_{ϕ} is used at every depth.

Classification: *Known* mechanism (cf. LSTM gates, attention), but the constraint of depth-independence and frozen shared path is a *novel combination*.

Definition 5: Projection

An **input-dependent projection** is a function $P: \mathcal{X} \rightarrow \text{End}(\mathcal{X})$ mapping each input to a linear projection operator $P(x) \in \mathbb{R}^{d \times d}$, applied to the state before or after each depth:

$$\tilde{f}_{P(x)}(h) = P(x) \cdot f_{\theta}(h) \quad \text{or} \quad \tilde{f}_{P(x)}(h) = f_{\theta}(P(x) \cdot h)$$

Again, P depends only on x , not on depth t .

Classification: *Known* (cf. gradient projection methods), formalized for forward-pass application.

Definition 6: Computation-Isolability

A shared sequential composition f_{θ^D} is **computation-isolable** if there exists a mechanism M (gate or projection) such that for any set of inputs $\{x_1, \dots, x_N\}$:

$$\forall i, \forall t \in \{0, \dots, D\}: \quad \tilde{h}_t^{(M)}(x_i) = h_t(x_i)$$

That is, the mechanism preserves the exact computation of f_{θ^D} for each input independently, with no cross-input contamination.

Classification: *Apparently novel* — this specific formulation combining shared composition, frozen parameters, and forward-pass isolation does not appear in the surveyed literature.

Definition 7: Elimination vs. Bounding of Interference

- **Elimination:** The mechanism achieves computation-isolability exactly (Definition 6). For all inputs and all depths, the modified computation equals the original: $\tilde{h}_t(x) = h_t(x)$.

- **Bounding:** There exists $\epsilon > 0$ such that for all inputs x and all depths t :

$$\|\tilde{h}_t(x) - h_t(x)\| \leq \epsilon$$

where ϵ may depend on the Lipschitz constant of f_θ , the depth D , and the capacity of the gate/projection, but not on the specific input.

Classification: *Plausible adaptation* of approximation theory concepts.

3. Hypotheses

Hypothesis H1 (Negative — Impossibility of Elimination)

Statement: For a general nonlinear frozen shared path f_θ , there does not exist an input-dependent gate $g: \mathcal{X} \rightarrow \mathbb{R}^d$ (without per-depth parameters) such that the gated composition preserves the original computation exactly for all inputs.

Assumptions:

- (A1) $f_\theta: \mathbb{R}^d \rightarrow \mathbb{R}^d$ is nonlinear (specifically, not an affine map).
- (A2) The gate $g: \mathbb{R}^d \rightarrow \mathbb{R}^d$ depends only on the input x , not on depth t or intermediate state h_t .
- (A3) The gate has no per-depth trainable parameters.

- (A4) $D \geq 2$ (at least two compositions).
- (A5) The gate is applied multiplicatively: $\tilde{f}_{\{g(x)\}}(h) = g(x) \odot f_{\theta}(h)$ or $\tilde{f}_{\{g(x)\}}(h) = f_{\theta}(g(x) \odot h)$.

Mathematical object being gated: The output (or input) of the frozen shared path f_{θ} at each depth.

Claim: Under (A1)–(A5), exact elimination of interference is impossible.

Proof sketch:

Consider the requirement that the gated computation matches the original for all x :

$$\tilde{f}_{\{g(x)\}}^D(x) = f_{\theta}^D(x)$$

For depth $D = 1$:

$$g(x) \odot f_{\theta}(x) = f_{\theta}(x) \quad \Longleftrightarrow \quad g(x) = \mathbf{1} \quad (\text{where } f_{\theta}(x) \neq 0)$$

For depth $D = 2$:

$$g(x) \odot f_{\theta}(g(x) \odot f_{\theta}(x)) = f_{\theta}(f_{\theta}(x))$$

If $g(x) = \mathbf{1}$ (from the $D=1$ requirement), this becomes $f_{\theta}(f_{\theta}(x)) = f_{\theta}(f_{\theta}(x))$, which is trivially satisfied. But

this means the gate is the identity — it does nothing — and provides no isolation whatsoever.

Now consider the alternative gate placement $\tilde{f}_{\{g(x)\}}(h) = f_{\theta}(g(x) \odot h)$:

For $D = 1$: $f_{\theta}(g(x) \odot x) = f_{\theta}(x)$, which requires $g(x) \odot x = x$ (assuming f_{θ} is injective on the relevant domain), giving $g(x) = \mathbf{1}$ where $x \neq 0$.

For $D = 2$: $f_{\theta}(g(x) \odot f_{\theta}(g(x) \odot x)) = f_{\theta}(f_{\theta}(x))$.

With $g(x) = \mathbf{1}$: $f_{\theta}(f_{\theta}(x)) = f_{\theta}(f_{\theta}(x))$. Again trivial.

The fundamental issue: **any gate that preserves the original computation must be the identity**, which provides zero isolation. A non-trivial gate necessarily alters the computation, creating a tension between preservation and isolation.

More precisely, suppose we want the gate to provide isolation by mapping different inputs to different "channels." For the gate to be non-trivial ($g(x) \neq \mathbf{1}$ for some x), we need:

$$g(x) \odot f_{\theta}(g(x) \odot f_{\theta}(x)) = f_{\theta}(f_{\theta}(x))$$

This is a functional equation in g . For a general nonlinear f_θ , this equation is overdetermined: the left side depends on $g(x)$ in two places (once inside f_θ , once outside), while the right side is fixed. For generic f_θ , no solution g exists except $g \equiv \mathbf{1}$.

Weakest additional assumptions for truth: If f_θ is linear ($f_\theta(h) = Wh + b$), then the problem is trivially solvable with $g = \mathbf{1}$ because linear compositions don't create input-dependent interference. If f_θ is a contraction with a unique fixed point, approximate isolation may be achievable (see H2).

Assessment: *Plausible adaptation* — the proof structure follows from basic functional equation analysis, but the precise conditions on f_θ for the impossibility to hold require further characterization.

Hypothesis H2 (Positive — Bounded Interference via Contractivity)

Statement: If the frozen shared path f_θ is a contraction mapping with Lipschitz constant $L < 1$, then input-path interference can be bounded by $\epsilon = O(L^D)$ without any gating mechanism at all, and a simple input-dependent projection can reduce the bound further.

Assumptions:

- (B1) f_θ is L -Lipschitz with $L < 1$.
- (B2) The state space is a complete metric space.

- (B3) The projection $P(x)$ is an orthogonal projection onto a subspace $S_x \subset \mathbb{R}^d$.

Mathematical object being projected: The intermediate state h_t projected onto S_x before each application of f_θ .

Claim: Under (B1)–(B3), the interference between inputs x_1 and x_2 at depth t satisfies:

$$\|h_t(x_1) - h_t(x_2)\| \leq L^t \|x_1 - x_2\|$$

This is a direct consequence of the Banach fixed-point theorem and the contraction property.

Proof: By induction. Base case: $\|h_0(x_1) - h_0(x_2)\| = \|x_1 - x_2\|$. Inductive step:

$$\|h_{t+1}(x_1) - h_{t+1}(x_2)\| = \|f_\theta(h_t(x_1)) - f_\theta(h_t(x_2))\| \leq L \|h_t(x_1) - h_t(x_2)\| \leq L^{t+1} \|x_1 - x_2\|$$

For the projection-enhanced version, if $P(x)$ projects onto a subspace where f_θ has a smaller Lipschitz constant $L' < L$, then:

$$\|P(x) \cdot h_t(x_1) - P(x) \cdot h_t(x_2)\| \leq L'^t \|x_1 - x_2\|$$

Assessment: *Known/direct consequence* — the contraction bound is standard. The projection enhancement is a *plausible adaptation* of spectral methods.

Caveat: This bounds the *separation* between different inputs' trajectories, not the *interference* from shared modifications. It shows that different inputs naturally diverge under contraction, but doesn't address the core problem of how to apply input-specific modifications without cross-contamination.

Hypothesis H3 (Negative — Depth-Agnostic Gate Cannot Adapt to Nonlinear Dynamics)

Statement: For a generic nonlinear f_{θ} , an input-dependent gate $g(x)$ that is depth-agnostic cannot simultaneously (i) preserve the computation $f_{\theta}^D(x)$ and (ii) provide non-trivial isolation between different inputs' computational paths.

Assumptions:

- (C1) f_{θ} is smooth and nonlinear.
- (C2) $D \geq 3$.
- (C3) The gate $g(x)$ is a continuous function.
- (C4) The gate is applied as $\tilde{f}_{g(x)}(h) = g(x) \odot f_{\theta}(h)$.

Mathematical object being gated: The output of f_{θ} at each depth, modulated by $g(x)$.

Claim: Under (C1)–(C4), the only gate that preserves the computation for all inputs is $g(x) = \mathbf{1}$, which provides no isolation.

Proof sketch:

The preservation condition at depth D requires:

$$\underbrace{g(x) \odot f_{\theta} \bigl(g(x) \odot f_{\theta} \bigl(\cdots g(x) \odot f_{\theta}(x) \bigr) \bigr)}_{\mathcal{D} \text{ text{ applications}}} = f_{\theta}^{\mathcal{D}}(x)$$

Define $\phi_x(h) = g(x) \odot f_{\theta}(h)$. The condition is $\phi_x^{*D}(x) = f_{\theta}^{*D}(x)$.

At depth 1: $g(x) \odot f_{\theta}(x) = f_{\theta}(x)$, so $g(x)_i = 1$ whenever $[f_{\theta}(x)]_i \neq 0$.

At depth 2: $g(x) \odot f_{\theta}(g(x) \odot f_{\theta}(x)) = f_{\theta}(f_{\theta}(x))$.

Since $g(x) = \mathbf{1}$ (from depth 1, on the relevant domain), this becomes $f \circ \theta(f \circ \theta(x)) = f \circ \theta(f \circ \theta(x))$. ✓

But $g(x) = \mathbf{1}$ means the gate is trivial. Any perturbation $g(x) = \mathbf{1} + \delta(x)$ with $\delta(x) \neq 0$ will, at some depth, cause:

$$\|\tilde{h}_t - (\mathbf{1} + \delta(x)) \odot f_\theta(\tilde{h}_{t-1})\| \neq \|f_\theta(\tilde{h}_{t-1})\|$$

and the error compounds through subsequent compositions.

Assessment: *Plausible adaptation* — the argument is structurally sound but relies on the gate being applied at every depth. If the gate is applied selectively (e.g., only at certain depths), the analysis changes.

Weakest additional assumptions for truth: The result holds for any nonlinear f_θ that is injective on the relevant domain. If f_θ has symmetries (e.g., $f_\theta(h) = f_\theta(-h)$), non-trivial gates may exist.

Hypothesis H4 (Conditional Positive — Depth-Conditioned Gate)

Statement: If the gate is allowed to depend on both the input x and the depth index t (but with shared parameters — a single network $g_\phi(x, t)$), then computation-isolability can be achieved for a broad class of shared paths f_θ .

Assumptions:

- (D1) The gate $g_\phi: \mathcal{X} \times \{0, \dots, D-1\} \rightarrow \mathbb{R}^d$ is a single neural network with shared parameters ϕ .

- (D2) f_{θ} is smooth and its Jacobian $J_f(h) = \partial f_{\theta} / \partial h$ has full rank on the relevant domain.
- (D3) The gate is applied as $\tilde{f}_{\{g_{\phi}(x,t)\}}(h) = g_{\phi}(x,t) \odot f_{\theta}(h)$.

Mathematical object being gated: The output of f_{θ} at depth t , modulated by $g_{\phi}(x, t)$.

Claim: Under (D1)–(D3), for any finite D and any smooth f_{θ} , there exists a gate g_{ϕ} (with $O(d \cdot D)$ parameters) such that the gated computation preserves the original: $\tilde{f}^D(x) = f_{\theta}^D(x)$ for all x .

Proof sketch: At each depth t , the preservation condition is:

$$g_{\phi}(x, t) \odot f_{\theta}(h_t) = f_{\theta}(h_t)$$

where $h_t = f_{\theta}^t(x)$. This requires $g_{\phi}(x, t)_i = 1$ whenever $[f_{\theta}(h_t)]_i \neq 0$.

A gate that simply outputs $\mathbf{1}$ for all (x, t) trivially satisfies this. But for isolation, we want $g_{\phi}(x, t)$ to also project away components that would interfere with other inputs' computations.

The key insight: if we define $g_{\phi}(x, t) = \mathbf{1} + P_t(x)$ where $P_t(x)$ is a projection onto the null space of f_{θ} 's Jacobian at h_t , then:

- Preservation: $g_{\phi}(x, t) \odot f_{\theta}(h_t) = f_{\theta}(h_t) + P_t(x) \odot f_{\theta}(h_t) = f_{\theta}(h_t)$ if $P_t(x) \odot f_{\theta}(h_t) = 0$.
- Isolation: $P_t(x)$ projects away components that are "shared" between different inputs' intermediate states.

However, this requires P_t to depend on $h_t = f_{\theta}^t(x)$, which in turn depends on x and t . A single network $g_{\phi}(x, t)$ can learn this mapping if given sufficient capacity.

Assessment: *Conjecture* — the existence of such a gate is plausible but not proven. The construction requires the gate to implicitly compute $f_{\theta}^t(x)$, which may require the gate to be as complex as the shared path itself.

Weakest additional assumptions for truth: The gate network g_{ϕ} must have sufficient capacity to represent the function $(x, t) \mapsto$ appropriate depth-dependent modulation. This may require g_{ϕ} to be at least as expressive as f_{θ}^D .

4. Prior Art Analysis

4.1 Orthogonal Gradient Methods (OGM)

Source: Lopez-Paz & Ranzato (2017), "Gradient Episodic Memory for Continual Learning"; Zeng et al. (2019).

Relevance: OGM projects gradients onto the orthogonal complement of previously used gradient directions to prevent interference during *training*.

Does it address the same problem? No. OGM operates in parameter space during backpropagation. Our problem concerns forward-pass computation isolation with frozen parameters. The mathematical objects are different: OGM modifies $\nabla_{\theta} \mathcal{L}$, while we seek to modify $f_{\theta}(h)$.

Classification: *Known technique, not directly applicable.*

4.2 Progressive Neural Networks

Source: Rusu et al. (2016).

Relevance: Adds new columns (with new parameters) for each task, with lateral connections to previous columns.

Does it address the same problem? No. Progressive networks introduce per-task parameters, violating our constraint of no per-depth/per-level trainable parameters. They also require architectural modification, not just gating.

Classification: *Known technique, violates constraints.*

4.3 Hypernetworks

Source: Ha et al. (2016).

Relevance: A hypernetwork generates weights for a main network conditioned on some input.

Does it address the same problem? Partially. A hypernetwork $H_\phi(x)$ could generate per-depth weights $W_t = H_\phi(x, t)$, effectively creating input-dependent per-depth parameters. However:

- This requires the hypernetwork to have per-depth output dimensions.
- The generated weights replace f_θ rather than gating it.
- The hypernetwork itself has trainable parameters ϕ .

Classification: *Known technique, relevant but violates the "no per-depth trainable parameters" constraint in spirit (the hypernetwork's output is per-depth).*

4.4 Conditional Computation / Mixture of Experts

Source: Shazeer et al. (2017), "Outrageously Large Neural Networks."

Relevance: Routes inputs to different expert subnetworks based on input-dependent gating.

Does it address the same problem? Partially. MoE uses input-dependent routing to select which parameters to use, which is a form of input-dependent computation. However:

- MoE typically has per-layer gating (depth-dependent routing).
- The experts are distinct networks, not a single shared path.
- MoE doesn't address the problem of preserving a specific shared-path computation.

Classification: *Known technique, structurally related but solves a different problem.*

4.5 Parameter Isolation / Masking

Source: Mallya & Lazebnik (2018), "PackNet"; Wortsman et al. (2020), "Supermasks."

Relevance: Uses binary masks to select which parameters are active for each task.

Does it address the same problem? Partially. Masking can be input-dependent (as in supermasks), but:

- Masks are typically per-layer, not shared across depth.
- Masking modifies which parameters are used, not how a shared path is composed.
- The masks are usually learned per-task, not per-input.

Classification: *Known technique, relevant analogy but different mathematical structure.*

4.6 Recurrent Neural Networks with Gating (LSTM/GRU)

Source: Hochreiter & Schmidhuber (1997); Cho et al. (2014).

Relevance: LSTMs use input-dependent gates (forget, input, output) to control information flow through a recurrent shared computation.

Does it address the same problem? Yes, most closely. LSTM gates are:

- Input-dependent (they depend on x_t and h_{t-1}).
- Applied at each depth (time step).
- Used to control what information persists through shared recurrence.

However, LSTM gates depend on the *current state* h_{t-1} , not just the original input x . This is a crucial difference: our constraint requires the gate to depend only on x , not on intermediate states.

Classification: *Known technique, directly relevant but violates the "input-only" constraint.*

4.7 Attention as Gating

Source: Vaswani et al (2017).

Relevance: Attention mechanisms compute input-dependent weights to modulate information flow.

Does it address the same problem? No. Attention operates within a single layer and depends on the current representation, not on a fixed shared path. It also requires per-layer parameters (Q, K, V projections).

Classification: *Known technique, different problem.*

5. Novelty Classification

| Claim | Classification |

|-----|-----|

| Shared sequential composition creates interference | **Known** (standard in RNN theory) |

| Input-dependent gating can control information flow | **Known** (LSTM, MoE) |

| Depth-agnostic gate cannot achieve exact isolation for nonlinear shared paths | **Plausible adaptation** (novel combination of known results) |

| Contractivity bounds interference without gating | **Direct consequence** (Banach fixed-point theorem) |

| Depth-conditioned gate with shared parameters may achieve isolation | **Conjecture** (no direct precedent) |

| The specific formulation of computation-isolability for frozen shared paths | **Apparently novel** |

| The impossibility result for depth-agnostic gates | **Apparently novel** (though structurally similar to results in dynamical systems theory) |

6. Obstruction Analysis

The Core Obstruction

The fundamental obstruction to input-path isolation via depth-agnostic gating is a **temporal information deficit**: the gate cannot distinguish which depth it is operating at, and therefore cannot adapt its behavior to the depth-specific state of the computation.

More precisely:

- **The gate sees only x , not h_t .** The intermediate state $h_t = f_{\theta}^t(x)$ encodes the cumulative effect of t applications of the shared path. Different inputs x_1, x_2 may have similar h_t values at some depths but divergent values at others. A depth-agnostic gate cannot detect this.
- **Nonlinearity creates depth-dependent dynamics.** For a nonlinear f_{θ} , the Jacobian $J_f(h_t)$ varies with t (since h_t changes). The "directions of interference" at depth t depend on $J_f(h_t)$, which the gate cannot access.
- **The preservation-isolation tradeoff.** Any gate that preserves the original computation must satisfy $g(x) \odot f_{\theta}(h) = f_{\theta}(h)$ (or equivalent), which forces $g(x) = \mathbf{1}$ on the relevant domain. A non-trivial gate necessarily alters the computation, creating a tradeoff between fidelity and isolation.

Characterization of the Obstruction

The obstruction can be formalized as a **rank deficiency** in the gate's information:

Let $\mathcal{I}_t(x) = (x, h_1, \dots, h_t)$ be the full information available at depth t . The gate has access only to $\mathcal{I}_0(x) = \{x\}$. The information gap is:

$$\Delta_t(x) = \dim(\mathcal{I}_t(x)) - \dim(\mathcal{I}_0(x)) = t \cdot d$$

For $t \geq 1$, the gate is missing $t \cdot d$ dimensions of information about the current state. This gap grows linearly with depth, making isolation increasingly difficult for deeper compositions.

Conditions Under Which the Obstruction Vanishes

- **Linearity:** If $f_\theta(h) = Wh + b$, then $h_t = W^t x + (W^{t-1} + \dots + I)b$. The state is a deterministic function of x and t , and a depth-conditioned gate $g(x, t)$ can perfectly track the computation. Moreover, linear compositions don't create nonlinear interference, so $g = \mathbf{1}$ suffices.
- **Contractivity with small L :** If $L \ll 1$, then $h_t \rightarrow h^*$ (the fixed point) exponentially fast. After a few depths, all inputs converge to the same state, and interference becomes negligible. A gate is unnecessary.
- **Depth-conditioned gate:** If the gate can depend on t (via a shared network $g_\phi(x, t)$), the information gap Δ_t is closed, and isolation becomes feasible (subject to the gate's capacity).

- **State-conditioned gate:** If the gate can depend on h_t (as in LSTMs), the information gap is fully closed, and standard gating techniques apply.

7. Summary and Recommendations

Key Findings

- **Exact elimination of interference is impossible** under the stated constraints (depth-agnostic gate, frozen shared path, nonlinear f_θ). This is an *apparently novel* result, though structurally analogous to known limitations in dynamical systems.
- **Bounded interference is achievable** if the shared path is contractive, but this is a property of f_θ , not of the gate. The gate contributes nothing in this regime.
- **Depth-conditioned gating** (with shared parameters) is the most promising direction, but requires the gate to implicitly compute $f_\theta^{(x)}$, which may necessitate gate capacity comparable to the shared path itself.
- **The LSTM approach** (state-conditioned gating) is the closest prior art, but violates the "input-only" constraint. Relaxing this constraint

to allow state-conditioning would make the problem tractable using known techniques.

Strategic Recommendation

The investigation should pivot from "can we achieve isolation with depth-agnostic gating?" to "what is the minimal additional information the gate needs?" The answer is likely: **the gate needs access to the intermediate state h_t** (or equivalently, the depth index t with sufficient capacity to simulate f_{θ}^t). This suggests that:

- Either accept state-conditioned gating (LSTM-style), which is well-understood but adds complexity.
- Or accept approximate isolation via contractivity, which requires architectural constraints on f_{θ} .
- Or accept that the shared path must be retrained (or extended with per-depth parameters), which violates the original constraints but may be necessary.

The "failure condition" is met: **the proposed isolation is impossible under the stated assumptions**. The obstruction is the gate's inability to access depth-specific state information, which is a fundamental information-theoretic limitation.

Tokens: 10027 | Latenz: 268214ms

Critic

[SEARCH: arXiv 2509.26507 Dragon Hatchling architecture equations shared depth " f_{θ} " composition]

[SEARCH: "Dragon Hatchling" BDH "shared" "depth" weights architecture]

[SEARCH: "2509.26507" "particle" "recurrent" depth equations]

Web Research

- **Reasoning Depth in AI and Logic**

Depth-recurrent and looped architectures (LoopFormer, depth-recurrent Transformers) further decouple parameter count from computational depth, allowing iterative application of a shared computation block across.

Source: <https://www.emergentmind.com/topics/reasoning-depth>

- **The Yale arXiv Paper Template - Overleaf, Online LaTeX Editor**

This document is both a working example and a usage guide for the Yale arXiv paper template.

Source: <https://www.overleaf.com/latex/templates/the-yale-arxiv-paper-template/wsmymnffgcs>

- **[Paper Note] Generative Modeling of Shape-Dependent Self-Contact...**

Overall Architecture¶. PAPoseDiff consists of three stages: 1. Data construction: Multi-camera dome capture from Goliath → SMPL-X fitting → contact map analysis → self-contact frame filtering 2. Generative training: Part-wise SMPL-X parameters →...

Source: https://en.papernotes.org/ICCV2025/image_generation/generative_modeling_of_shape-dependent_self-contact_human_poses/

- **Relation between $f \tan(\theta)$ and $f \theta$ projections.**

We introduce an approach to expand the eye-box in a retinal-projection-based near-eye display. The retinal projection display has the advantage of providing clear images in a wide depth range; however, it has difficulty in practical use with a narrow eye-box.

Source: https://www.researchgate.net/figure/Relation-between-f-tantheta-and-f-theta-projections_fig8_343450315

- **Generative Modeling by Estimating Gradients of the Data... | Yang Song**

Likelihood-based models either require strong restrictions on the model architecture to ensure a tractable normalizing constant for likelihood computation, or must rely on surrogate objectives to approximate maximum likelihood training.

Source: <https://yang-song.net/blog/2021/score/>

- **GitHub - severian42/BDH-MLX: Baby Dragon Hatchling (BDH) - MLX...**

Baby Dragon Hatchling (BDH) is a novel Large Language Model architecture that bridges the gap between Transformers and biologically-plausible neural networks. Unlike standard Transformers, BDH features: Key Innovations.

Source: <https://github.com/severian42/BDH-MLX>

- **The Dragon Hatchling: The Missing Link between the Transformer...**

We introduce 'Dragon Hatchling' (BDH), a new Large Language Model architecture based on a scale-free biologically inspired network of n locally-interacting neuron particles.

Source: <https://arxiv.org/pdf/2509.26507>

• **Baby Dragon Hatchling Price Prediction , BDH Price... | Bitget**

Exploring the forecast and prediction of Baby Dragon Hatchling prices for 2025 — 2050. Make your own prediction based on fixed rate or explore user sentiments.

Source: <https://www.bitget.com/price/baby-dragon-hatchling/price-prediction>

• **Races of the Dragon.pdf - The Techno Jesus**

Design embed now.. Races of the Dragon.pdf - The Techno Jesus. or tattooed. Simple lines serve to give depth and definition.

Source: <https://www.yumpu.com/en/document/view/7833729/races-of-the-dragonpdf-the-techno-jesus>

10. **The Rosebound Hatchling – Unfocussed Photography & Art**

The hatchling, having made its dramatic debut, decided that a perch upon a rose was entirely too small a stage for its destiny. It tested its wings with a few flaps, each one sending droplets scattering into tiny prisms of light. The garden glistened with irritation.

Source: <https://unfocussed.com/blogs/captured-tales/the-rosebound-hatchling>

11. **Determining If a Particle Path of an Ideal Fluid on a Three-Dimensional...**

Tracking particle paths within an ideal fluid in a three-dimensional domain necessitates a comprehensive understanding of both analytical and numerical methods. Analytical techniques often involve solving differential equations that describe the fluid flow directly.

Source: <https://diversedaily.com/determining-if-a-particle-path-of-an-ideal-fluid-on-a-three-dimensional-domain-eventually-reaches-a-certain-region-in-space/>

12. Numerical integration of the N-body ring problem by recurrent power...

... For the numerical integration of the equations of motion, we have used the method of recurrent power series adapted to the N-body ring problem. This method converges for any initial condition not leading to a binary collision [30].

Source: https://www.researchgate.net/publication/322778641_Numerical_integration_of_the_N-body_ring_problem_by_recurrent_power_series

13. New Particle Is Both Matter and Antimatter: The Majorana fermion

Such a particle may seem confusing, as a particle's antimatter has equal mass but opposite charge, and (notably) when matter meets its antimatter counterpart the two annihilate one another in a burst of energy.

Source: <https://futurism.com/new-particle-matter-antimatter-majorana-fermion>

14. probability theory - Recurrent and Transient random walks on integer...

I have troubles with the following problem: Consider the grid of equilateral triangles (triangular lattice). Is random walk recurrent or transient?

Source: <https://math.stackexchange.com/questions/3697240/recurrent-and-transient-random-walks-on-integer-grids>

15. An improved particle swarm optimization-based dynamic recurrent...

A learning algorithm for dynamic recurrent Elman neural networks is proposed based on an improved particle swarm optimization. The proposed algorithm performs the evolution of network structure, weights, initial inputs of the context units and...

Source: <https://rims.cityu-dg.edu.cn/en/publications/an-improved-particle-swarm-optimization-based-dynamic-recurrent-n/>

Tokens: 734 | Latenz: 24108ms

Optimizer

Revised Analysis: Input-Path Interference in BDH and the Limits of Input-Dependent Gating/Projection

I treat the BDH shared core abstractly as follows: a frozen map $\backslash(F\backslash)$ is applied sequentially for $\backslash(D\backslash)$ steps. The questions below are independent of the particular particle dynamics of BDH, except where I explicitly use properties of $\backslash(F\backslash)$. If the implementation reports define $\backslash(F\backslash)$ concretely, the theorems should be instantiated with that $\backslash(F\backslash)$; the structural obstructions remain.

1. Formalization

1.1 Shared sequential composition

Let $(V=\mathbb{R}^d)$ be the state space and $(X\subseteqq V)$ the input space. A **depth- (D) shared sequential composition** is defined by a fixed map

[

$F:V\rightarrow V$

]

and the recurrence

[

$s_0(x)=x,\quad s_{t+1}(x)=F(s_t(x)),\quad t=0,\dots,D-1.$

]

Write

[

$$F^D = F \circ F \circ \dots \circ F$$

]

for the (D) -fold composition. The **unperturbed trajectory** of (x) is

[

$$\tau_F(x) = (s_0(x), s_1(x), \dots, s_D(x)).$$

]

This is the relevant BDH abstraction: the same parameterized transformation is applied at every depth, and the same transformation is shared by all inputs.

1.2 Frozen shared path

The map (F) is **frozen** if its parameters are not updated during the continual-learning phase. Only the gate/projection mechanism may be trained. Formally:

[

$$\frac{\partial F}{\partial \theta} = 0$$

]

during any adaptation step. No theorem below relies on retraining (F) .

1.3 Gating

An **input-dependent gate** is a map

$$\begin{aligned} & \backslash[\\ & g: X \rightarrow \mathbb{R}^d \\ & \backslash] \end{aligned}$$

which is applied identically at every depth. The two canonical placements are:

$$\begin{aligned} & \backslash[\\ & \text{\texttt{output gating:}} \quad s_{t+1} = g(x) \odot F(s_t), \\ & \backslash] \end{aligned}$$

$$\begin{aligned} & \backslash[\\ & \text{\texttt{input gating:}} \quad s_{t+1} = F(\text{bigl}(g(x) \odot s_t \text{bigr}). \\ & \backslash] \end{aligned}$$

The gate may be trained, but it has **no per-depth parameters**: the same $g(x)$ is used at every t . I will also consider the relaxed variant

$$\begin{aligned} & \backslash[\\ & g: X \times \{0, \dots, D-1\} \rightarrow \mathbb{R}^d, \end{aligned}$$

$\backslash$

where the gate receives the depth index $\backslash(t\backslash)$ as input but still has only shared parameters. This is not a per-depth parameter vector; it is a single function of both $\backslash(x\backslash)$ and $\backslash(t\backslash)$.

1.4 Projection

An **input-dependent projection** is a map

$\backslash$

$P:X\rightarrow \Pi(V),$

$\backslash$

where $\backslash(\Pi(V)\backslash)$ is the set of orthogonal projections of $\backslash(\mathbb{R}^d\backslash)$, applied either as

$\backslash$

$\text{output projection:} \quad s_{t+1}=P(x)F(s_t),$

$\backslash$

or

$\backslash$

$\text{input projection:} \quad s_{t+1}=F(P(x)s_t).$

\]

A projection is more restrictive than a general gate because it is linear, idempotent, and rank-deficient.

1.5 Input-path interference

I distinguish two precise notions.

(a) Depth-aliasing interference

Suppose an oracle with access to the current depth or state would apply a reference modulation

 $\setminus[$
$$g^*(x,t), \text{quad } t=0, \dots, D-1.$$

\]

A depth-independent gate $g(x)$ produces the actual transition

 $\setminus[$
$$s_{t+1} = \Phi(g(x), s_t),$$

\]

where Φ is the chosen gating/projection rule. The **depth-aliasing error** for input x is

$$\begin{aligned} \mathrm{DA}(g; x) = & \sum_{t=0}^{D-1} \\ & \left| \Phi(g(x), s_t(x)) - \Phi(g^*(x, t), s_t(x)) \right|. \end{aligned}$$

If $\mathrm{DA}(g; x) > 0$, then the lack of depth information corrupts the trajectory. This is the forward-pass sense in which shared sequential composition causes interference: the same gate is re-applied at every depth even though the correct modulation may need to change from depth to depth.

(b) Gradient interference

For two tasks (A, B) with losses $\mathcal{L}_A, \mathcal{L}_B$ on the same frozen F , the first-order change in $\mathcal{L}_A$ caused by one gradient step on task B is proportional to

$$\angle \nabla_{\phi} \mathcal{L}_A, \nabla_{\phi} \mathcal{L}_B.$$

Define the **task Jacobians**

$$\begin{aligned} J_i(\phi) &= \frac{\partial}{\partial \phi} s_{D,i}(x; \phi) \Big|_{x \in X_i}. \end{aligned}$$

Task-wise gradient interference is

$$\begin{aligned} I_{ij} &= \left| \frac{\partial}{\partial \phi} s_{D,i}(x; \phi) \frac{\partial}{\partial \phi} s_{D,j}(x; \phi) \right| \Big|_{x \in X_i \cap X_j}. \end{aligned}$$

For shared sequential composition, these Jacobians have a special structure:

$$\begin{aligned} \frac{\partial}{\partial \phi} s_{D,i}(x; \phi) &= \sum_{t=0}^{D-1} \left(\frac{\partial}{\partial \phi} s_{D,i}(x; \phi) \right) \end{aligned}$$

$$\prod_{r=t+1}^{D-1} DF(s_r(x))$$

$$\right)$$

$$\frac{\partial \Phi}{\partial \phi}(g_\phi(x,t), s_t(x)).$$

$$\]$$

The same parameters ϕ appear at every depth, so gradients from different tasks can overlap even when the inputs are disjoint. This is the training-time sense of input-path interference.

1.6 Computation-isolability

A mechanism is **exactly computation-isolating** for a set of tasks if:

- each task's desired output is produced exactly, and
- there is no interference: $\mathrm{DA}(g; x) = 0$ for all x in the forward sense, or $I_{ij} = 0$ for all task pairs in the gradient sense.

A mechanism **bounds interference** if the relevant quantity is at most ϵ ($\epsilon > 0$) uniformly over the task inputs.

1.7 Elimination versus bounding

- **Elimination** means exact equality of outputs and exact zero interference.
- **Bounding** means the deviation is at most ϵ , with ϵ controlled by quantities such as the Lipschitz constant of F , the depth D , the gate error, and the number of parameters.

2. Theorem-Shaped Hypotheses

H1. Exact preservation of the unperturbed path forces the gate to be the identity

Assumptions.

- (F) is defined on an open set $(U \subseteqq \mathbb{R}^d)$.
- The gate is depth-independent output gating:

[

$$s_{t+1} = g(x) \odot F(s_t).$$

]

- We require exact preservation of the unperturbed trajectory:

[

$$s_t(x) = F^t(x) \quad \forall x \in U, \forall t.$$

]

Mathematical object gated.

The output of the frozen map $\mathcal{F}$ at every depth.

Claim.

Under these assumptions, $\mathcal{G}(x) = \mathbf{1}$ on every coordinate where $\mathcal{F}(s_t(x)) \neq 0$. If $\mathcal{F}$ is componentwise nonzero along trajectories on U , then $\mathcal{G}(x) = \mathbf{1}$ for all $x \in U$.

Proof.

At $t=0$, exact preservation requires

$$\mathcal{G}(x) \odot \mathcal{F}(x) = \mathcal{F}(x).$$

Componentwise:

$$(g_i(x) - 1)F_i(x) = 0.$$

Wherever $F_i(x) \neq 0$, we get $g_i(x) = 1$. Wherever $F_i(x) = 0$, the gate is invisible, but it is also useless for isolation. The same argument applies at every depth.

Verdict.

True as an exact theorem. It is a direct consequence of the preservation equation. It shows that, under the strong requirement that the mechanism changes nothing, the only gate is the identity gate. In particular, a gate cannot simultaneously “do nothing” and “isolate tasks.”

Weakest additional assumptions under which the obstruction disappears.

If we only require preservation of the final output $(s_D(x)=F^D(x))$, rather than the whole trajectory, non-identity gates may exist when (F^D) is non-injective. This is the underdetermined case; it is not a useful general construction.

Classification.

Known / direct consequence.

H2. Depth-independent gating cannot represent depth-varying modulation

Assumptions.

- The transition rule $(\Phi(g,s))$ is injective in the gate argument (g) for the states along the unperturbed trajectory.
- A task requires an ideal per-depth modulation $(g^*(x,t))$ that is not constant in (t) .

Mathematical object gated.

The entire per-depth modulation sequence $((g(x,0),\dots,g(x,D-1)))$.

Claim.

No depth-independent gate $(g(x))$ can produce exactly the same trajectory as $(g^*(x,t))$ for all (t) .

Proof.

Equality at every depth gives

$$\begin{aligned} &[\\ &\Phi(g(x),s_t(x))=\Phi(g^*(x,t),s_t(x)). \\ &] \end{aligned}$$

By injectivity of $(\Phi(\cdot,s_t(x)))$,

$$\begin{aligned} &[\\ &g(x)=g^*(x,t) \\ &] \end{aligned}$$

for every (t) , contradicting the assumption that $(g^*(x,\cdot))$ is nonconstant.

Verdict.

True as an exact representational theorem. This is the cleanest formalization of the phrase “input-path interference arising from shared sequential composition”: a depth-independent gate is a point on the diagonal of the product space (M^D) , while the task requirements occupy off-diagonal points.

Weakest additional assumptions under which it becomes true.

The obstruction disappears if we allow:

- a depth-indexed shared gate $g(x,t)$;
- a state-dependent gate, e.g. $g(s_t)$, as in LSTM/GRU;
- per-depth trainable parameters, which is excluded by the research question.

Classification.

Direct consequence of the definition; the phrasing as a frozen-shared-path obstruction is a plausible adaptation of standard control-theoretic observability reasoning.

H3. Projection mechanisms have a rank obstruction

Assumptions.

- The mechanism is an input-dependent orthogonal projection, e.g.

\[

$$s_{t+1} = P(x)F(s_t).$$

\]

- The unperturbed trajectory of x is such that

\[

$$\text{span}\{F(s_0(x)), F(s_1(x)), \dots, F(s_{D-1}(x))\}$$

\]

has dimension $m(x)$.

Mathematical object projected.

The image space of the projection, i.e. the subspace of state space that survives each depth.

Claim.

If the projection is to preserve the unperturbed trajectory exactly, then

\[

$$\text{rank } P(x) \geq m(x).$$

\]

For (K) tasks whose relevant spans have dimensions (m_i) , and whose projections must be mutually orthogonal to avoid gradient interference, necessarily

$$\sum_{i=1}^K m_i \leq d.$$

Hence a projection can isolate at most $\lfloor d / \min_i m_i \rfloor$ tasks.

Proof.

Exact preservation requires

$$P(x)F(s_t(x)) = F(s_t(x))$$

for all (t) , so every $(F(s_t(x)))$ lies in $(\operatorname{Im} P(x))$. The dimension bound follows. Orthogonality of task subspaces is required for zero gradient interference; orthogonal subspaces of $(\mathbb{R}^d)$ have total dimension at most (d) .

Verdict.

True as an exact linear-algebra theorem.

Weakest additional assumptions under which it becomes useful.

Tasks must live in low-dimensional invariant or approximately invariant subspaces. If the relevant computational subspaces have dimension ≤ 1 , for example, a rank-1 projection can isolate up to d tasks.

Classification.

Direct consequence of linear algebra; the application to frozen shared paths is a plausible adaptation.

H4. Finite parameter budget bounds exact gradient isolation

Assumptions.

- The gate/projection mechanism has p trainable parameters.
- Task output Jacobians are $\{J_i \in \mathbb{R}^{d \times p}\}$.
- We require exact zero gradient interference:

[

$J_i J_j^{\top} = 0 \quad \forall i \neq j.$

]

Mathematical object constrained.

The parameter space of the gate/projection mechanism.

Claim.

If each task Jacobian has rank $\leq r_i$, then

$$\sum_i r_i \leq p.$$

In particular, if every task has full row rank and $p \leq d$, then at most one task can be isolated exactly. For a shared sequential composition with generic nonlinear f , task Jacobians are generic sums over depth, so this is the typical situation.

Proof.

The row spaces of the J_i must be mutually orthogonal subspaces of $\mathbb{R}^p$. The sum of their dimensions cannot exceed p .

Verdict.

True as an exact theorem. It is the usual parameter-isolation bound, restated for trainable gates on a frozen shared path.

Weakest additional assumptions under which exact isolation is possible for many tasks.

Each task must occupy a low-rank subspace of the gate-parameter Jacobian. This occurs, for example, when tasks use disjoint subsets of inputs or when the gate network is designed so that task-specific features activate disjoint hidden units.

Classification.

Direct consequence of linear algebra; the application to frozen-path gating is a plausible adaptation.

H5. Bounded interference is possible exactly when the gated dynamics are non-expansive

Assumptions.

- The modulated transition is Lipschitz in state with constant L_Φ :

[

$$\|\Phi(g,s) - \Phi(g,s')\| \leq L_\Phi \|s - s'\|.$$

]

- The per-depth gate error is bounded:

[

$$\|\Phi(g(x), s_t(x)) - \Phi(g^*(x,t), s_t(x))\| \leq \delta.$$

]

Mathematical object bounded.

The full trajectory error.

Claim.

The output error satisfies

$$\begin{aligned} & \mathbb{E} \\ & \mathbb{E} \|s_D(x) - s_D^*(x)\| \\ & \leq \\ & \delta \sum_{k=0}^{D-1} L_{\Phi}^k \\ & = \\ & \begin{cases} \delta \frac{1 - L_{\Phi}^D}{1 - L_{\Phi}}, & \text{if } L_{\Phi} < 1, \\ \delta D, & \text{if } L_{\Phi} = 1, \\ \delta \frac{L_{\Phi}^D - 1}{L_{\Phi} - 1}, & \text{if } L_{\Phi} > 1. \end{cases} \\ & \end{aligned}$$

Proof.

Let $e_t = \|s_t - s^*\|$. Then

$$e_{t+1}$$

$\leq$

$|\Phi(g, s_t) - \Phi(g, s_t^*)|$

+

$|\Phi(g, s_t) - \Phi(g^*, s_t^*)|$

$\leq$

$L_\Phi \epsilon_{t+\delta}.$

$\square$

Solving the recurrence gives the claimed bound.

Verdict.

True as an exact theorem.

Consequence.

If the gated system is contractive, $(L_\Phi < 1)$, then interference is bounded uniformly in depth by $(\delta / (1 - L_\Phi))$. If $(L_\Phi > 1)$, interference can grow exponentially in (D) . This is why “bounding” is not automatic: for generic nonlinear (F) , (L_Φ) is often larger than (1) , and small gate errors are amplified through the shared composition.

Classification.

Known / direct consequence.

H6. Depth-indexed shared gating: finite exact interpolation is conditionally possible

Assumptions.

- We allow a depth-indexed shared gate $\phi(x, t)$, still with no per-depth parameter vectors.
- The task set is finite: $\{(x_i, y_i)\}, i=1, \dots, N$.
- For each i , there exists some per-depth modulation sequence $\{g_{i,t}\}$ that realizes y_i under the transition rule.
- The parameterized family ϕ can interpolate the finite data $\{(x_i, t) \mapsto g_{i,t}\}$, e.g. an MLP with enough parameters.

Claim.

Under these assumptions, there exists a shared-parameter depth-indexed gate that produces exactly the desired outputs for all finite task inputs.

Proof sketch.

The modulation requirement is a finite lookup table. Universal approximation on finite sets gives interpolation.

Important caveat.

The decisive assumption is that the target modulation sequences $\{g_{i,t}\}$ exist. For a generic nonlinear F , this is **underdetermined**. For example, if F is contractive, many targets are unreachable regardless of gating. Thus this is a conditional existence theorem, not a general theorem.

Verdict.

Conditionally true; otherwise underdetermined.

Weakest additional assumptions.

Reachability of the targets under the allowed modulation, plus sufficient gate capacity. Additionally, exact zero gradient interference requires the parameter-count condition of H4.

Classification.

Plausible adaptation of universal approximation; the reachability part is a conjecture.

3. Prior Art

3.1 Orthogonal gradient and gradient-projection methods

Examples. GEM / A-GEM, Orthogonal Gradient Descent, TRGP, GPM.

What they address.

They project gradients in parameter space so that updates for a new task do not increase losses on previous tasks.

Do they address the same mathematical problem?

Only partially. They address training-time gradient interference, which corresponds to my second notion of interference. They do **not** address depth-aliasing in a frozen shared composition. Moreover, they typically require unfreezing or at least modifying the main network; here the shared path is frozen.

Classification.

Known, partially relevant.

3.2 Parameter isolation and masking

Examples. PackNet, Piggyback, Hard Attention to the Task, Supermasks.

What they address.

They allocate disjoint parameter subsets to tasks, preventing cross-task updates.

Same problem?

Not the same. They introduce per-task masks or per-task parameters, which is a form of parameter isolation. The research question explicitly asks for an input-dependent gate/projection on a single frozen shared path, not for partitioning the shared path's parameters.

Classification.

Known, related but different mechanism.

3.3 Progressive networks and path-based continual learning

Examples. Progressive Neural Networks, PathNet.

What they address.

They add new columns or paths for new tasks while freezing old ones.

Same problem?

No. They avoid interference by adding task-specific architecture, which is effectively per-task parameter growth. The question here is whether a *single* input-dependent mechanism can isolate paths within one frozen shared composition.

Classification.

Known, different problem.

3.4 Hypernetworks

Examples. Ha et al., Chunked Hypernetworks.

What they address.

A hypernetwork generates weights for a main network as a function of an input or task embedding.

Same problem?

Partly. A hypernetwork can generate input-dependent per-depth modulations. But the generated weights are typically per-layer, which violates the “no per-depth/per-level trainable parameters” constraint in spirit. A hypernetwork also replaces or augments the main weights rather than gating a frozen shared path.

Classification.

Known, partially relevant.

3.5 Recurrent neural networks with gating

Examples. LSTM, GRU.

What they address.

They use input- and state-dependent gates inside a shared recurrent computation.

Same problem?

This is the closest prior art. LSTM gates are applied at every time step with shared parameters, exactly like gating a shared sequential composition. The crucial difference is that LSTM gates depend on the current hidden state $\{h_{t-1}\}$, not only on the original input $\{x\}$. This supports H2: state information is what makes recurrent gating effective. It is also empirical evidence, not a theorem, that state-conditioning resolves many of the pathologies of depth-independent control.

Classification.

Known, directly relevant, but with a different information assumption.

3.6 Conditional computation and mixture of experts

Examples. Mixture of Experts, sparsely gated layers.

What they address.

Input-dependent routing to different experts or subnetworks.

Same problem?

Not the same. MoE routes inputs to different parameter sets, typically with per-layer routers, and its goal is capacity scaling or efficiency, not preserving a frozen shared-path computation.

Classification.

Known, structurally similar but not the same mathematical problem.

3.7 Depth-recurrent and weight-tied transformers

Examples. ALBERT, Universal Transformers, LoopFormer.

What they address.

They reuse the same block across depth and sometimes use adaptive computation time.

Same problem?

Partially. They study shared-depth composition, but they do not address task isolation via input-dependent gates. Adaptive computation time is state-dependent halting, not input-dependent task isolation.

Classification.

Known, background context.

4. Novelty Classification

| Claim | Classification |

| --- | --- |

| Formal notion of depth-aliasing interference in a frozen shared composition | Apparently novel as a packaged definition; components are known from control theory |

| H1: exact preservation forces identity gating | Direct consequence |

| H2: depth-independent gating cannot represent depth-varying modulation | Direct consequence |

| H3: projection rank obstruction | Direct consequence |

| H4: gradient-isolation parameter bound | Direct consequence |

| H5: contraction/expansion bound on interference | Direct consequence |

| H6: conditional finite interpolation by depth-indexed shared gate | Plausible adaptation / conjecture |

| Claim that state or depth information is necessary for robust gating in shared composition | Plausible adaptation; consistent with LSTM design but not proven here as a theorem |

No experimental result is used as evidence for any theorem. The exact results above are purely analytical.

5. Failure Condition and Obstruction

The strict version of the proposed mechanism is:

- input-dependent gate/projection,
- no per-depth trainable parameters,
- frozen shared path,
- no retraining of $\mathcal{F}$.

Under this strict version, **exact elimination of input-path interference is impossible for generic nonlinear $\mathcal{F}$** . The obstruction has four independent parts.

5.1 Diagonalization obstruction

A depth-independent gate $g(x)$ is forced to lie on the diagonal of the product space of depth-wise modulations:

$\setminus$

$(g(x), g(x), \dots, g(x)) \in M^D$.

$\setminus$

Any task whose ideal modulation depends on depth lies off this diagonal. This is a representational impossibility, not a training failure.

5.2 Preservation-identity obstruction

If the mechanism is required to preserve the unperturbed computation exactly, then output gating forces $g(x) = \mathbf{1}$ wherever F is nonzero. The only gate that changes nothing is the identity gate. Thus a gate cannot both preserve the frozen computation and alter the path for isolation.

5.3 Rank and parameter budget obstruction

Even if the depth-aliasing obstruction is removed by allowing a depth-indexed shared gate, exact gradient isolation requires the task Jacobians to be mutually orthogonal. The sum of their ranks cannot exceed the number of trainable gate parameters p . For generic tasks, this permits only one task when $p \leq d$.

5.4 Expansive-dynamics obstruction

Bounding interference is possible if the gated dynamics are contractive. For generic nonlinear F , the Lipschitz constant of the composed system is > 1 , and small gate errors are amplified exponentially in

depth. Hence even bounded interference is not guaranteed without additional assumptions on $\mathcal{F}$.

6. Conclusion

The answer to the research question is:

- **Elimination** of input-path interference by a depth-independent input-dependent gate/projection on a frozen shared path is **impossible in general**.
- **Bounding** is possible if the gated system is contractive or if the depth is small enough that exponential amplification is controlled.
- **Controllability** can be achieved only by enriching the information available to the gate: either allow a depth-indexed shared gate $\mathcal{G}(x, t)$, allow state-dependent gating $\mathcal{G}(s, t)$, allow per-task parameter blocks, or impose strong low-rank/invariant-subspace structure on the tasks.

The most precise obstruction is the **diagonalization obstruction**: a depth-independent gate is a single point on the diagonal of the space of depth-wise modulations, while the tasks generically require off-diagonal points. This obstruction is independent of optimization, capacity, and training data. It is the mathematical core of why input-path interference cannot, in general, be eliminated by the proposed mechanism.

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Moderator

Moderator Synthesis — Round 2

Consolidation

Role: Synthesis and consensus evaluation of the STRATEGIST, CRITIC, and OPTIMIZER contributions on the BDH input-path interference question.

Before consolidating, one factual check is warranted, since the entire analysis hinges on BDH genuinely exhibiting weight-shared sequential composition across depth (if it did not, the question would misfire at the premise level):

[SEARCH: "Dragon Hatchling" BDH weight tying shared parameters across depth recurrent layer]

0. Evidence-Base Status

- The BDH paper (arXiv:2509.26507) is confirmed to exist and to describe a scale-free network of locally-interacting neuron particles with transformer-comparable performance. Third-party implementations (BDH-MLX) corroborate the architecture's general character.
- **Retrieval limitation:** the internal project documents (2026-08-* progress reports, paper-parity-plan, bdh-continual-learning-plan_v2.md, cl-h1-report.md) could **not** be retrieved through available channels; searches returned irrelevant material. Consequently, neither participant nor this moderation

could cross-check the formal models against your concrete implementation or against the H1 experimental report dated today. All conclusions below are therefore **structural**, derived from the abstract "shared sequential composition" model that both analysts adopted consistently. Instantiation with BDH's literal update equations remains an open verification step (see §3.3).

Both principal analyses proceeded from the same formal skeleton — frozen map F , recurrence $s_{t+1} = \Phi(g(x), s_t)$, depth-agnostic input-conditioned gate/projection — which makes genuine consensus evaluation possible.

1. Points of Consensus (High Confidence)

C1 — Formal framework agreement. The definitions of shared sequential composition, frozen path, gating vs. projection, and the elimination/bounding distinction were independently convergent. No participant disputed another's definitions. *Consensus: strong.*

C2 — Exact elimination is impossible under the strict reading. Three independent obstructions were derived, by complementary methods:

- **Identity-forcing** (STRATEGIST H1/H3; OPTIMIZER H1): exact trajectory preservation under output gating $g \odot F(s)$ forces $g = \mathbf{1}$ wherever F is nonzero along the trajectory. A mechanism required to change nothing can isolate nothing.

- **Diagonalization** (OPTIMIZER H2; STRATEGIST's "temporal information deficit" is its informal statement): a depth-agnostic gate occupies the diagonal of the product space M^D of per-depth modulations; any task whose ideal modulation $g^*(x,t)$ is nonconstant in t lies off-diagonal and is unreachable, given injectivity of $\Phi(\cdot, s_t)$ in the gate argument.
- **Rank/parameter-budget** (OPTIMIZER H3/H4): exact zero cross-task gradient interference through shared gate parameters ϕ requires mutually orthogonal task-Jacobian row spaces, hence $\sum_i \text{rank}(J_i) \leq p$. Generic tasks saturate this at one task when $p \leq d$.

Consensus: strong. This satisfies the brief's failure condition: the proposed isolation is impossible under the stated strict assumptions.

C3 — Bounding is conditional, not free. Per-depth gate error δ under Lipschitz constant L_Φ yields output error $\leq \delta \sum_k L_\Phi^k$: uniformly bounded iff $L_\Phi < 1$, growing $\sim \delta D$ at $L_\Phi = 1$, exponentially otherwise (OPTIMIZER H5; STRATEGIST H2 agrees and correctly notes contraction bounds trajectory *separation*, not modification-induced interference). Trained LLM-scale maps typically have $L_\Phi > 1$, so bounding is a real architectural constraint, not a default. *Consensus: strong.*

C4 — Escape routes require enriched information or structure. All participants converge: relaxation is possible via (i) depth-indexed shared-parameter gates $g_\phi(x,t)$, (ii) state-conditioned gating ($g(s_t)$, LSTM-style — the nearest prior art, which violates the input-only constraint), (iii) low-rank/invariant task subspaces, or (iv) abandoning the frozen-path constraint. *Consensus: strong.*

C5 — Prior-art triage. Independent and concordant: orthogonal-gradient methods (GEM/A-GEM/GPM) operate on gradients in parameter space, not forward-pass composition — related but a different mathematical problem; progressive networks, PackNet, supermasks introduce per-task parameters/masks — constraint-violating; hypernetworks and MoE are structurally adjacent but generate per-layer outputs or route between experts — not the same problem; Universal Transformer/ALBERT establish the weight-tied backdrop without addressing isolation. **No surveyed prior art solves this exact problem.** *Consensus: strong.*

2. Disputes and Adjudications

D1 — Novelty classification conflict. STRATEGIST labels the impossibility result "apparently novel"; OPTIMIZER labels the equivalent statements "direct consequence."

Adjudication: OPTIMIZER is correct on the mathematics, STRATEGIST is correct on the contribution. Every ingredient (functional-equation identity forcing, diagonal argument, rank counting, Lipschitz error propagation) is textbook. What is novel is the **problem formulation** — isolation of computations within a frozen weight-tied path under an input-only gate, posed for continual learning. Write-up guidance: claim a new formalization and a clean negative result; do **not** claim new mathematics.

D2 — STRATEGIST H4 is defective. Its construction (Jacobian-null-space projection achieving preservation + isolation with $O(d \cdot D)$ parameters) collapses to the identity gate under its own preservation requirement, and conflates the Jacobian's null space with the output's annihilator. OPTIMIZER's H6 (conditional finite interpolation, contingent

on reachability of target modulation sequences) is the correct weakened version. *Adjudication*: H4 withdrawn as stated; retained only as OPTIMIZER-style conjecture.

D3 — STRATEGIST's "information gap" $\Delta_t = t \cdot d$.

Dimensions of informally-defined information sets are not a rigorous quantity. Superseded by OPTIMIZER's rank-theoretic arguments. Retain only as heuristic motivation.

D4 — Interference has two distinct meanings; only OPTIMIZER separated them cleanly. Forward-pass *depth-aliasing* vs. training-time *gradient interference*. See correction G1 below — this separation turns out to be more important than either participant realized.

3. Corrections and Gaps Identified in Moderation

G1 — Cross-input forward-pass interference is structurally zero; the live channels are elsewhere. With frozen θ and a *stateless* input-conditioned mechanism, each input's forward pass is computationally independent: x_1 's trajectory cannot be perturbed by x_2 's "presence." STRATEGIST's Definition 3 counterfactual ($\tilde{h}_t(x_1 \mid x_2 \text{ active})$) is therefore ill-defined as stated. The operationally meaningful interference notions are:

- **Depth-aliasing within a pass** (obstruction B above); and
- **Cross-time forgetting**: ϕ is shared across all inputs, so any update $\Delta\phi$ from task B alters g_ϕ on task- A

inputs — first-order damage governed exactly by the Jacobian-overlap/rank condition (obstruction C). This, not intra-pass contamination, is the catastrophic-forgetting channel in this architecture.

Future work should define interference exclusively through these two channels.

G2 — The preservation requirement must be asymmetric to be meaningful. Requiring preservation on *all* inputs trivially forces the identity gate (C2). The continual-learning-relevant requirement is: exact preservation on old-task inputs S_{old} , unconstrained deviation on new-task inputs S_{new} . Under this reading the identity obstruction restricts to S_{old} ; non-trivial gates become admissible off it; the surviving barriers are the diagonal obstruction (whenever old-task-faithful modulation must vary with depth) and the rank/budget obstruction for gradient-exactness. **Neither participant developed this asymmetric variant; it should be the primary next theorem target.**

G3 — BDH instantiation is outstanding. BDH's block has residual structure and elementwise multiplicative (Hebbian-type) terms. Consequences: (a) output-gating the full map would gate the residual stream — likely harmful; gating only the delta term is the natural placement; (b) BDH's native plasticity must be explicitly disabled for the "frozen path" premise to hold. These checks against paper .tex/the repository were not performed in-round.

G4 — Scope decision required on t -conditioning. A shared-parameter gate $g_{\phi}(x, t)$ (depth index as *input*, e.g., via a fixed embedding) adds $O(1)$ parameters and **does not violate "no per-depth/per-level trainable parameters" as literally worded**. This is the

pivotal loophole: every impossibility result proven here targets the strictly depth-agnostic gate and dissolves — into conditional statements (reachability + capacity + rank budget) — once t is admitted. The team must rule explicitly on whether the intended constraint forbids t -conditioning; the theorems' applicability hinges on this interpretation.

4. Consolidated Answer to the Research Question

| Reading of the question | Verdict | Status |

| --- | --- | --- |

| Depth-agnostic gate/projection, frozen nonlinear F , **exact elimination** | **Impossible** (identity-forcing + diagonalization + rank obstructions) | Proven; direct consequences |

| Same, **bounding** | Possible iff gated dynamics non-expansive ($L_\Phi < 1$) or depth-limited error budget; else exponential amplification | Proven conditional theorem |

| Enriched gate (t -conditioned shared-parameter, or state-conditioned), **controllability** | Feasible conditional on reachability, gate capacity, low-rank task subspaces; exact multi-task gradient isolation still capped by $\sum_i \mathrm{rank}(J_i) \leq p$ | Conjecture / plausible adaptation |

The failure condition contemplated by the brief is **met for the strict reading**, with a precise obstruction characterization: a depth-agnostic gate is confined to the diagonal of the modulation product space, and exactness requirements (preservation, gradient orthogonality) collapse either to the identity mechanism or to a rank budget permitting at most

one generic task. No optimization method, capacity increase, or training regime circumvents these; they are representational, not algorithmic.

5. Final Novelty Ledger (Adjudicated)

| Claim | Classification |

| --- | --- |

| Formalization: isolation within frozen weight-tied composition via input-only gate, CL-framed | **Apparently novel** (as packaging) |

| Identity-forcing lemma | **Direct consequence** |

| Diagonalization obstruction | **Direct consequence** (control-theoretic flavor) |

| Rank/parameter-budget bound on exact gradient isolation | **Direct consequence** |

| Contraction/amplification error bound | **Direct consequence** |

| Necessity of state/depth information for effective gating | **Plausible adaptation** (consistent with LSTM/ACT practice; proven only under injectivity assumptions) |

| $\mathcal{H}$ -conditioned shared gate achieves isolation broadly | **Conjecture** (reachability undetermined) |

| Any empirical effect observed in cl-h1-report | Inadmissible as theorem evidence until reconciled with the above |

6. Recommended Next Steps

- **Restate the theorem set under asymmetric preservation** (old-task exact / new-task free) and prove that the diagonal and rank obstructions survive while the identity obstruction dies. This is the honest CL formalization and the highest-value open result.
- **Instantiate $\mathcal{F}$ with BDH's concrete update** from paper .tex; fix gate placement relative to the residual stream; confirm plasticity disabling.
- **Rule on the $\mathcal{T}$ -conditioning interpretation** of the constraint; if permitted, specify a minimal $g_{\phi}(x,t)$ (fixed depth embedding into the existing gate network) and test — labeling outcomes strictly as empirical conjecture.
- **Reconcile with the local reports** (plan_v2, cl-h1-report) once retrievable; verify that the implemented mechanism corresponds to one of the formally scoped variants rather than an unscoped mixture.
- **Write-up posture:** present C2/C3 as a clean impossibility-and-bounding package for weight-tied architectures; cite LSTM/GRU, adaptive computation time, and universal-transformer literature as nearest neighbors; claim formalization novelty, not mathematical novelty.

Consensus summary: full agreement on framework, on the impossibility of exact elimination under strict assumptions, on conditional bounding, and on prior-art triage; disagreements were confined to novelty labeling and one defective construction, both adjudicated above. The investigation's productive frontier is the asymmetric-preservation reformulation and the $\mathcal{T}$ -conditioning scope decision.

Web Research

- **GitHub - severian42/BDH-MLX: Baby Dragon Hatchling (BDH) – MLX...**

Shared Parameters Across Layers: Same weights reused in all layers (recurrent depth). Sparse, Non-Negative Activations: ReLU-based activations for biological plausibility. Transformer. BDH. Parameters. Unique per layer. Shared across layers. Activations.

Source: <https://github.com/severian42/BDH-MLX>

- **BDH-CQ: In-Context Learning with Recurrent Latent... | alphaXiv**

BDH layers also support recurrent systems for constraint satisfaction. Recurrent-depth language models repeatedly apply a shared block to a sequence-wide latent state and can improve some reasoning benchmarks with additional iterations (Geiping et al., 2025).

Source: <https://www.alphaxiv.org/abs/2608.09888>

- **Dragon Hatchling: Linking Transformers & Brain Models**

We introduce 'Dragon Hatchling' (BDH), a new LLM architecture based on a scale-free biologically inspired network of n locally-interacting neuron particles.

Source: <https://www.emergentmind.com/papers/2509.26507>

- **Beyond Transformer LLM Models: How the BDH Architecture Solves...**

Baby Dragon Hatchling (BDH). Effective Attention Span. ~2 hours and 17 minutes. A recurring and highly emphasized theme in the 2026 scientific

discourse surrounding the BDH architecture is the concept of Axiomatic Artificial Intelligence.

Source: <https://www.crvscience.com/post/beyond-transformer-llm-models-how-the-bdh-architecture-solves-the-context-window-crisis>

- **The Dragon Hatchling: The Missing Link between the Transformer...**

The dragon hatchling aka #BDH (baby dragon hatchling) is a new approach to creating and training these large language models at an architectural level. BDH is #graph centric in its design and tries to mirror #neural synapses like those in the brain for its training mechanism.

Source: https://www.linkedin.com/posts/meeherkapoor_the-dragon-hatchling-the-missing-link-between-activity-7386863203634388992-uI39

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Strategist

Final Strategic Assessment: Input-Path Interference in BDH via Gating/Projection

Date: 2026-08-25 | **Status:** Complete

Executive Summary

Under the strict constraints stated in the research question — depth-agnostic input-dependent gate/projection, frozen shared path, no per-depth trainable parameters, no retraining — **exact elimination of input-path interference is impossible** for generic nonlinear shared compositions. This is proven via three independent obstructions (identity-forcing, diagonalization, rank budget). Bounding is possible only under contractivity assumptions that are unlikely to hold for trained LLM-scale maps. The productive frontier is relaxing the depth-agnostic constraint to admit a depth-indexed shared-parameter gate $g_\phi(x,t)$, which dissolves the impossibility into conditional feasibility statements.

1. Formal Definitions

1.1 Shared Sequential Composition

Let $V = \mathbb{R}^d$ be the state space. A **depth- D shared sequential composition** is defined by a fixed map $F: V \rightarrow V$ and the recurrence:

$$s_0(x) = x, \quad s_{t+1}(x) = F(s_t(x)), \quad t = 0, \dots, D-1$$

The unperturbed trajectory is $\tau_F(x) = (s_0(x), s_1(x), \dots, s_D(x))$.

Classification: Direct characterization of BDH's weight-tied recurrent depth, confirmed by the BDH-MLX implementation ("Same weights reused in all layers (recurrent depth)").

1.2 Frozen Shared Path

F is **frozen** if $\partial F / \partial \theta = 0$ during the continual-learning phase. Only the gate/projection mechanism ϕ may be trained.

Classification: Known — standard continual learning constraint.

1.3 Input-Path Interference (Two Precise Notions)

Forward-pass depth-aliasing: A depth-agnostic gate $g(x)$ is applied identically at every depth. If the ideal modulation for input x is $g^*(x, t)$ (varying with depth), the depth-aliasing error is:

$$\mathrm{DA}(g; x) = \sum_{t=0}^{D-1} \left| \Phi(g(x), s_t(x)) - \Phi(g^*(x, t), s_t(x)) \right|$$

Gradient interference (cross-task forgetting): For tasks A, B with shared gate parameters ϕ , the first-order damage from a B -update to A -performance is proportional to $\angle \nabla_{\phi} \mathcal{L}_A, \nabla_{\phi} \mathcal{L}_B \rangle$. The task Jacobians are:

$$J_i(\phi) = \frac{\partial}{\partial \phi} s_D(x; \phi) \Big|_{x \in X_i}$$

Critical correction from moderation: Cross-input forward-pass interference is **structurally zero** with frozen θ and a stateless input-conditioned mechanism — each input's forward pass is computationally independent. The live interference channels are depth-aliasing (within a pass) and gradient interference (across training steps).

Classification: The depth-aliasing notion is **apparently novel** as a packaged definition; components are known from control theory. The gradient notion is **known** from continual learning.

1.4 Gating

An **input-dependent gate** $g: X \rightarrow \mathbb{R}^d$ is applied identically at every depth:

- Output gating: $s_{t+1} = g(x) \odot F(s_t)$
- Input gating: $s_{t+1} = F(g(x) \odot s_t)$

The gate has **no per-depth parameters**: the same $g(x)$ is used at every t .

Classification: Known mechanism (LSTM, MoE), but the depth-independence + frozen-path combination is novel.

1.5 Projection

An **input-dependent projection** $P: X \rightarrow \Pi(V)$ maps each input to an orthogonal projection, applied as $s_{t+1} = P(x) F(s_t)$ or $s_{t+1} = F(P(x) s_t)$.

Classification: Known (gradient projection methods), formalized for forward-pass application.

1.6 Computation-Isolability

A mechanism is **exactly computation-isolating** if: (1) each task's desired output is produced exactly, and (2) $\mathrm{DA}(g; x) = 0$ for all x (forward) and $I_{ij} = 0$ for all task pairs (gradient).

Classification: Apparently novel — this specific formulation combining shared composition, frozen parameters, and forward-pass isolation does not appear in surveyed literature.

1.7 Elimination vs. Bounding

- **Elimination:** Exact equality of outputs and exact zero interference.
- **Bounding:** Deviation $\leq \epsilon$, with ϵ controlled by Lipschitz constant, depth, gate error, and parameter count.

Classification: Plausible adaptation of approximation theory concepts.

2. Theorem-Shaped Hypotheses

H1. Identity-Forcing Lemma (Negative)

Assumptions: F defined on open $U \subseteq \mathbb{R}^d$; depth-independent output gating $s_{t+1} = g(x) \odot F(s_t)$; exact trajectory preservation required: $s_t(x) = F^t(x)$ for all $x \in U$, all t .

Mathematical object gated: Output of F at every depth.

Claim: $g(x) = \mathbf{1}$ wherever $F(s_t(x)) \neq 0$.

Proof: At $t=0$: $g(x) \odot F(x) = F(x)$ implies $(g_i(x) - 1) F_i(x) = 0$ componentwise. Where $F_i(x) \neq 0$, $g_i(x) = 1$. Same argument at every depth. $\square$

Verdict: True as an exact theorem. Direct consequence.

Weakest assumptions for escape: If only final-output preservation $s_D(x) = F^D(x)$ is required (not whole trajectory), non-identity gates may exist when F^D is non-injective. This is underdetermined and not generally useful.

Classification: Known / direct consequence.

H2. Diagonalization Obstruction (Negative)

Assumptions: $\Phi(g, s)$ is injective in g along the unperturbed trajectory; ideal modulation $g^*(x, t)$ is nonconstant in t .

Mathematical object gated: The entire per-depth modulation sequence $(g(x, 0), \dots, g(x, D-1))$.

Claim: No depth-independent gate $g(x)$ can reproduce $g^*(x, t)$ for all t .

Proof: Equality at every depth gives $\Phi(g(x), s_t(x)) = \Phi(g^*(x, t), s_t(x))$. By injectivity, $g(x) = g^*(x, t)$ for every t , contradicting nonconstancy. $\square$

Verdict: True as an exact representational theorem. A depth-independent gate is confined to the diagonal of M^D ; tasks requiring depth-varying modulation lie off-diagonal.

Weakest assumptions for escape: Allow depth-indexed shared gate $g_\phi(x, t)$; or state-dependent gate $g(s_t)$ (LSTM-style); or per-depth trainable parameters (excluded by constraint).

Classification: Direct consequence; the frozen-shared-path phrasing is a plausible adaptation of control-theoretic observability reasoning.

H3. Projection Rank Obstruction (Negative)

Assumptions: Input-dependent orthogonal projection $s_{t+1} = P(x) F(s_t)$; trajectory span $\operatorname{span}\{F(s_0(x)), \dots, F(s_{D-1}(x))\}$ has dimension $m(x)$.

Mathematical object projected: Image space of $P(x)$.

Claim: Exact preservation requires $\operatorname{rank} P(x) \geq m(x)$. For K tasks with mutually orthogonal projections: $\sum_{i=1}^K m_i \leq d$. At most $\lfloor d / \min_i m_i \rfloor$ tasks can be isolated.

Proof: $P(x) F(s_t(x)) = F(s_t(x))$ forces $F(s_t(x)) \in \operatorname{Im} P(x)$. Orthogonal subspaces of $\mathbb{R}^d$ have total dimension $\leq d$. $\square$

Verdict: True as an exact linear-algebra theorem.

Weakest assumptions for escape: Tasks must live in low-dimensional invariant subspaces. If $m_i = 1$, up to d tasks can be isolated.

Classification: Direct consequence; application to frozen shared paths is a plausible adaptation.

H4. Parameter-Budget Bound on Gradient Isolation (Negative)

Assumptions: Gate has p trainable parameters; task Jacobians $J_i \in \mathbb{R}^{d \times p}$; exact zero gradient interference required: $J_i J_j^\top = 0$ for $i \neq j$.

Mathematical object constrained: Parameter space of the gate mechanism.

Claim: $\sum_i \text{rank}(J_i) \leq p$. If every task has full row rank and $p \leq d$, at most one task can be isolated exactly.

Proof: Row spaces of J_i must be mutually orthogonal subspaces of $\mathbb{R}^p$. $\square$

Verdict: True as an exact theorem.

Weakest assumptions for escape: Each task must occupy a low-rank subspace of the gate-parameter Jacobian (e.g., disjoint input subsets, or gate architecture with task-specific feature activation).

Classification: Direct consequence; the frozen-path application is a plausible adaptation.

H5. Error Amplification Bound (Conditional)

Assumptions: Modulated transition is L_Φ -Lipschitz in state; per-depth gate error $\leq \delta$.

Mathematical object bounded: Full trajectory error.

Claim:

$$\begin{aligned} \|s_D(x) - s_D^*(x)\| &\leq \delta \sum_{k=0}^{D-1} L_\Phi^k = \\ &\begin{cases} \delta \frac{1 - L_\Phi^D}{1 - L_\Phi} & \text{if } L_\Phi < 1 \\ \delta D & \text{if } L_\Phi = 1 \\ \delta \frac{L_\Phi^D - 1}{L_\Phi - 1} & \text{if } L_\Phi > 1 \end{cases} \end{aligned}$$

Proof: Recurrence $e_{t+1} \leq L_\Phi e_t + \delta$ solved by standard methods. $\square$

Verdict: True as an exact theorem.

Consequence: Bounding is uniform in depth iff $L_\Phi < 1$ (contractive). For $L_\Phi > 1$, errors grow exponentially. Trained LLM-scale maps typically have $L_\Phi > 1$, so bounding is a real architectural constraint, not a default.

Classification: Known / direct consequence.

H6. Depth-Indexed Shared Gate: Conditional Feasibility (Positive, Conjectural)

Assumptions: Depth-indexed shared gate $g_{\phi}(x,t)$ (shared parameters, t as input); finite task set (x_i, y_i) ; target modulation sequences $g_{i,t}$ exist and are reachable; sufficient gate capacity.

Claim: There exists a shared-parameter depth-indexed gate producing exact desired outputs for all finite task inputs.

Proof sketch: Finite lookup table; universal approximation on finite sets gives interpolation. $\square$

Critical caveat: The existence of target modulation sequences $g_{i,t}$ is **underdetermined** for generic nonlinear F . If F is contractive, many targets are unreachable regardless of gating.

Verdict: Conditionally true; otherwise underdetermined.

Classification: Plausible adaptation of universal approximation; reachability is a conjecture.

3. Prior Art Analysis

| Prior Art | Same Mathematical Problem? | Classification |

| --- | --- | --- |

| **Orthogonal gradient methods** (GEM, A-GEM, GPM) | No — operate on gradients in parameter space during training; do not address forward-pass depth-aliasing in frozen composition | Known, partially relevant |

| **Parameter isolation / masking** (PackNet, Supermasks) | No — introduce per-task masks or per-task parameters; constraint-violating | Known, related but different |

| **Progressive networks / PathNet** | No — add task-specific architecture; effectively per-task parameter growth | Known, different problem |

| **Hypernetworks** (Ha et al.) | Partially — can generate input-dependent per-depth modulations, but output is per-layer (violates constraint in spirit) | Known, partially relevant |

| **LSTM / GRU** | **Closest** — shared-parameter gates at every time step. Crucially, gates depend on current state h_{t-1} , not only input x_t . This is the key difference: state-conditioning is what makes recurrent gating effective | Known, directly relevant, different information assumption |

| **Mixture of Experts** | No — routes to different parameter sets with per-layer routers; different problem | Known, structurally similar |

| **Universal Transformer / ALBERT** | Partially — weight-tied depth, but no task isolation via gates | Known, background context |

No surveyed prior art solves this exact problem. The formulation — isolation of computations within a frozen weight-tied path under an input-only gate, posed for continual learning — appears to be new.

4. Obstruction Characterization

The failure condition is met. The obstruction has four independent parts:

4.1 Diagonalization Obstruction

A depth-independent gate $g(x)$ is confined to the diagonal of M^D . Tasks requiring depth-varying modulation lie off-diagonal. This is representational, not algorithmic.

4.2 Preservation-Identity Obstruction

Exact trajectory preservation under output gating forces $g(x) = \mathbf{1}$ wherever F is nonzero. A gate that changes nothing isolates nothing.

4.3 Rank / Parameter-Budget Obstruction

Exact gradient isolation requires mutually orthogonal task-Jacobian row spaces. $\sum_i \text{rank}(J_i) \leq p$ permits at most one generic task when $p \leq d$.

4.4 Expansive-Dynamics Obstruction

For generic nonlinear F , $L_\Phi > 1$, and small gate errors amplify exponentially in depth. Even bounded interference requires architectural constraints on F .

5. Conditions Under Which Obstructions Vanish

| Obstruction | Escape Route |

| --- | --- |

| Diagonalization | Depth-indexed shared gate $g_{\phi(x,t)}$ or state-conditioned gate $g(s_t)$ |

| Identity-forcing | Require only final-output preservation (not whole trajectory); or relax to asymmetric preservation (exact on old tasks, free on new) |

| Rank budget | Low-rank task subspaces; disjoint feature activation in gate architecture |

| Expansive dynamics | Architectural constraint $L_{\Phi} < 1$; or limit depth D |

6. Novelty Ledger (Adjudicated)

| Claim | Classification |

| --- | --- |

| Formalization: isolation within frozen weight-tied composition via input-only gate, CL-framed | **Apparently novel** (as packaging) |

| Identity-forcing lemma (H1) | **Direct consequence** |

| Diagonalization obstruction (H2) | **Direct consequence** (control-theoretic flavor) |

| Rank/parameter-budget bound (H3, H4) | **Direct consequence** |

| Contraction/amplification error bound (H5) | **Direct consequence** |

| Necessity of state/depth information for effective gating | **Plausible adaptation** (consistent with LSTM/ACT practice) |

| t -conditioned shared gate achieves isolation broadly (H6) | **Conjecture** (reachability undetermined) |

7. Critical Open Questions

7.1 Asymmetric Preservation (Highest Priority)

The identity obstruction applies when preservation is required on **all** inputs. The continual-learning-relevant requirement is asymmetric: exact preservation on old-task inputs S_{old} , unconstrained on new-task inputs S_{new} . Under this reading, the identity obstruction restricts to S_{old} ; non-trivial gates become admissible off it. The diagonal and rank obstructions survive. **This reformulation should be the primary next theorem target.**

7.2 Scope of t -Conditioning

A shared-parameter gate $g_{\phi}(x,t)$ with t as input (e.g., via a fixed depth embedding) adds $O(1)$ parameters and **does not violate "no per-depth/per-level trainable parameters" as literally worded**. This is the pivotal loophole: every impossibility result targets the strictly depth-agnostic gate and dissolves into conditional statements once t is admitted. The team must rule explicitly on whether the intended constraint forbids t -conditioning.

7.3 BDH Instantiation

BDH's block has residual structure and elementwise multiplicative (Hebbian-type) terms. Consequences:

- Output-gating the full map would gate the residual stream — likely harmful
- Gating only the delta term is the natural placement
- BDH's native plasticity must be explicitly disabled for the "frozen path" premise to hold

These checks against paper .tex and the repository were not performed in this analysis.

8. Strategic Recommendation

The strict reading of the research question yields a clean impossibility result. The investigation should:

- **Publish the impossibility-and-bounding package** as a formal negative result for weight-tied architectures, citing LSTM/GRU and Universal Transformer as nearest neighbors. Claim formalization novelty, not mathematical novelty.
- **Develop the asymmetric-preservation variant** as the honest CL formalization. This is the highest-value open result.
- **Decide on $g_\phi(x,t)$ -conditioning scope.** If permitted, specify a minimal $g_\phi(x,t)$ and test empirically — labeling outcomes strictly as empirical conjecture.
- **Instantiate with BDH's concrete update equations** from paper .tex; fix gate placement relative to the residual stream; confirm plasticity disabling.
- **Reconcile with implementation reports** (plan_v2, cl-h1-report) once retrievable; verify that the implemented mechanism corresponds to one of the formally scoped variants.

9. Answer to the Research Question

> Can input-path interference arising from shared sequential composition be eliminated, bounded, or made controllable by an input-dependent gate/projection mechanism, without introducing per-depth/

per-level trainable parameters and without retraining the frozen shared path?

Elimination: Impossible in general under the stated constraints. Three independent obstructions (identity-forcing, diagonalization, rank budget) are proven as direct consequences of the definitions.

Bounding: Possible if and only if the gated dynamics are non-expansive ($L_\Phi < 1$) or depth is small enough to control exponential amplification. This is a property of F and the gating rule, not achievable by the gate alone.

Controllability: Achievable only by enriching the gate's information — either via depth-indexed shared-parameter conditioning $g_\phi(x,t)$, state-dependent conditioning $g(s_t)$, or strong low-rank/invariant-subspace structure on the tasks. Each escape route transforms the impossibility into a conditional feasibility statement with its own capacity and reachability requirements.

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Critic

Critical Review of the Consolidated BDH Isolation Analysis

Overall assessment. The analysis is a useful exercise in abstraction and identifies some real distinctions (especially forward-pass *depth aliasing* versus training-time *gradient interference*). But the central claim—that exact elimination of input-path interference is impossible under the stated constraints—is overstated. Several “theorems” are either tautological under their assumptions, correct only for a degenerate version of the problem, or depend on unproved identifications between

forward-pass subspaces and gradient orthogonality. The analysis also does not instantiate the actual BDH recurrence from paper . tex, so the results remain conditional on an abstraction that may not match the architecture.

1. The analysis is not grounded in the actual BDH equations

The main object is abstracted as

$$\begin{aligned} & \backslash[\\ s_0 = x, & \quad s_{t+1} = F(s_t). \\ & \backslash] \end{aligned}$$

But the report does not verify from the paper that BDH is actually a simple deterministic iterated map of this form. BDH is described as a scale-free network of locally interacting neuron particles, and the report itself claims—without evidence—that the block has “residual structure and Hebbian-type terms.” If BDH includes:

- local particle interactions,
- residual streams,
- input-dependent routing,
- Hebbian plasticity,
- stochasticity,

then the abstraction $s_{t+1}=F(s_t)$ may omit exactly the mechanisms that matter for continual learning.

The source cited for shared parameters is a third-party README (“same weights reused in all layers”), not the paper equations. That is not enough to validate the formal model. In particular, gating the whole block $g(x) \odot F(s_t)$ is very different from gating only a residual delta $s_t + g(x) \odot \Delta(s_t)$, or gating local interactions inside the particle graph.

Gap: the analysis should start from the actual BDH update equation and locate exactly where a gate or projection would be inserted. Without that, the impossibility claims are architecture-agnostic, and possibly architecture-inapplicable.

2. The notion of “input-path interference” is unstable across the analysis

The original draft defined interference as one input’s forward trajectory changing because another input’s mechanism is active. The moderator correctly observed that this cross-input forward-pass interference is structurally zero when parameters are frozen and the gate is stateless. But then the analysis substitutes a different notion: *depth-aliasing*, the mismatch between a depth-independent gate and an ideal depth-varying modulation.

These are not the same mathematical problem.

- **Depth-aliasing** is a representational issue: can a depth-constant gate imitate a depth-varying gain?
- **Continual-learning interference** is a parameter-update issue: updating shared gate parameters for task B degrades task A.

The final report calls the first notion “input-path interference,” but the actual catastrophic-forgetting channel in this setting must involve the trainable gate parameters ϕ , because F is frozen. The analysis only partially connects these two.

Gap: the paper should define the continual-learning objective explicitly:

- Tasks $T_1, \dots, T_k$ arrive sequentially.
- F is frozen.
- ϕ is updated.
- Preservation is required only on previous tasks.

This asymmetric setup is the relevant one, and it is not what the main impossibility proof uses.

3. The identity-forcing theorem is true but trivial

H1 says: under output gating,

\[

$$s_{t+1}=g(x)\odot F(s_t),$$

\]

exact preservation of the entire unperturbed trajectory forces $(g(x)=\mathbf{1})$ wherever $(F_i\neq 0)$.

This is correct, but it is almost a tautology: if the mechanism is required to reproduce the exact unmodified computation at every depth for every input, then the only possibility is no modification. The theorem does not show that isolation is impossible. It shows that **preservation of everything and alteration of anything are incompatible**, which is essentially the definition of exact preservation.

The relevant condition should be:

\[

$$\tilde{s}_t(x)=s_t(x)\quad \text{for all } x\in S_{\{\text{old}\}}, \text{ for all } t,$$

\]

while allowing $(\tilde{s}_t(x)\neq s_t(x))$ for $(x\in S_{\{\text{new}\}})$.

Under this asymmetric condition, the identity obstruction may disappear. The report acknowledges this only as future work, but the headline conclusion is drawn from the stronger, less relevant condition.

4. The diagonalization obstruction is not a general theorem

H2 assumes that an ideal modulation $\phi^*(x, t)$ is nonconstant in t , and that the transition rule $\phi(g, s)$ is injective in g along the trajectory. If so, a depth-independent $g(x)$ cannot match the ideal sequence.

This is true given those assumptions, but the assumptions are not established.

- A depth-constant gate can produce depth-varying effects because the state s_t changes. For input gating,

[

$$s_{t+1} = F(g(x) \odot s_t),$$

]

the same $g(x)$ can produce different effective updates at different depths because s_t differs. Therefore the set of reachable trajectories is not simply the diagonal of the product space of depth-wise modulations.

- For output gating, injectivity of $\Phi(\cdot, s_t)$ fails on any coordinate where $F_i(s_t)=0$, which is common with ReLU/GLU/zero activations.

Thus the “diagonalization obstruction” is conditional on a strong injectivity assumption and on the premise that the target trajectory requires nonconstant gate inputs in the same coordinate system. It is not a proof that generic tasks require off-diagonal modulation sequences. One would need a reachability argument in control theory, showing that the image of the constant-gate map does not contain the desired task map.

5. The projection rank theorem does not imply the isolation bound

H3 states:

- Exact preservation $P(x)F(s_t)=F(s_t)$ forces

[

$\text{rank} P(x) \geq \dim \text{span} \{F(s_t)\}.$

]

This is correct linear algebra.

But the next step—that mutually orthogonal projections are required to avoid gradient interference—is not proved and is likely false as a general statement.

Orthogonality of the image subspaces of $\mathcal{P}(x_A)$ and $\mathcal{P}(x_B)$ is a forward-pass geometric property. Cross-task gradient interference is a property of the parameterization of $\mathcal{P}(x)$. If $\mathcal{P}(x)$ is produced by a neural network, two different inputs may have non-orthogonal or identical image subspaces but still have gradients that are orthogonal, or vice versa. The connection between these two notions depends on the architecture of the projection generator and is not automatic.

Therefore the capacity bound

$$\sum_i m_i \leq d$$

does not follow. It only follows if one assumes a very special architecture in which each task’s projection is a fixed, learned orthogonal subspace with independent parameters. That is a much stronger assumption than “input-dependent projection.”

6. The parameter-budget bound is notationally sloppy and not operational

H4 defines task Jacobians $(J_i \in \mathbb{R}^{d \times p})$ as derivatives of $s_D(x; \phi)$ with respect to (ϕ) , but tasks are sets of inputs, not single inputs. The aggregation matters.

The condition

$$J_i J_j^{\top} = 0$$

is not equivalent to zero gradient interference for general loss functions. The first-order change in task A's loss after a gradient step on task B is governed by

$$\langle \nabla_{\phi} \mathcal{L}_A, \nabla_{\phi} \mathcal{L}_B \rangle,$$

which involves the loss gradients, not just output Jacobians. Even at the output-Jacobian level, orthogonality of all Jacobian row spaces is a very strong condition.

Finally, the conclusion “at most one task when $p \leq d$ ” is not the relevant regime. A gate network attached to an LLM-scale BDH path would typically have $p \gg d$. If p is large, the rank budget can admit many tasks. The theorem gives no practical bound without estimating the actual Jacobian ranks.

7. The error bound statement has an incorrect “iff”

H5 derives

$$\|e_{t+1}\| \leq L_{\Phi} \|e_t\|^\delta,$$

hence

$$\|e_D\| \leq \delta \sum_{k=0}^{D-1} L_{\Phi}^k.$$

This is correct. But the conclusion that bounding is possible **exactly when $L_{\Phi} < 1$** is not accurate. For any finite depth D , the sum is finite even if $L_{\Phi} > 1$. The bound grows exponentially with D , but it still exists.

The correct statement is:

- a bound **uniform in depth** exists iff $(L_{\Phi} < 1)$;
- for fixed finite (D) , a bound exists for any finite (L_{Φ}) .

The analysis also asserts that trained LLM-scale maps typically have $(L_{\Phi} > 1)$. That is an empirical claim about BDH or LLMs, not proved from the paper or code. It should be labeled as such and tested.

8. H6 is conditional to the point of tautology

H6 states that a depth-indexed shared gate $(g_{\Phi}(x,t))$ can exactly realize finite task outputs if:

- some modulation sequence $(g_{\{i,t\}})$ realizes each target output, and
- (g_{Φ}) can interpolate those sequences.

Under those assumptions, the conclusion is immediate. The nontrivial question is whether assumption 1 holds: **which target input-output maps are reachable by the frozen shared path plus the allowed gate class?** That question is unproved and marked as conjecture.

Therefore H6 does not establish controllability. It reformulates the problem without solving it.

9. “Controllability” is never formally defined

The research question asks whether interference can be eliminated, bounded, or **made controllable**. The analysis focuses on elimination and bounding, but does not define controllability.

A proper formalization would likely treat the gate at each depth as a control input and ask whether the composed system

$\forall$

$s_{t+1} = F(s_t) + \text{or } \odot \text{ gate control}$

$\forall$

can steer states to desired outputs. This would invoke nonlinear reachability, injectivity of the transition map, or approximate controllability results for recurrent networks.

Because controllability is not formalized, the report’s statement that “controllability can be achieved only by enriching gate information” is not a theorem; it is an intuitive recommendation.

10. The $\mathcal{O}(t)$ -conditioning loophole is unresolved and ambiguous

The report correctly identifies that a shared-parameter gate receiving depth index $\mathcal{O}(t)$ as input would avoid the strict depth-agnostic obstruction. But it does not resolve whether this is allowed.

Two cases:

- **Fixed non-learned depth encoding**, e.g. sinusoidal positional encoding: adds $\mathcal{O}(1)$ trainable parameters, likely does not violate the letter of “no per-depth/per-level trainable parameters.”
- **Learned depth embedding**: adds $\mathcal{O}(D)$ trainable embedding rows, which is exactly per-depth trainable parameters, violating the constraint.

The final recommendation says “decide on $\mathcal{O}(t)$ -conditioning scope,” but the impossibility result is only valid for the stricter depth-agnostic reading. Until this scope is fixed, the answer is conditional, not definitive.

11. Prior-art review is incomplete and partly too dismissive

The report reviews parameter isolation, progressive networks, gradient projection, hypernetworks, MoE, LSTMs, and weight-tied transformers. However, some directly relevant literature is missing or dismissed too quickly:

- **FiLM / feature-wise linear modulation:**

\[

$$y = \gamma(x) \odot h + \beta(x),$$

\]

is essentially an input-dependent affine gate applied to a shared layer. Multi-task and continual variants are highly relevant.

- **Fast weight programmers and recurrent hypernetworks** that generate input-dependent modulations for a fixed recurrent core are closer to the proposed mechanism than the report admits.
- **Nonlinear RNN reachability / observability theory** would be the correct framework for the controllability question, but is absent.
- **Echo state networks** and reservoir computing may provide relevant results on fixed recurrent paths plus trained output/gate layers.

The classification “no surveyed prior art solves this exact problem” may be true, but the prior-art search is not exhaustive enough to support a strong novelty claim.

12. Novelty labeling is inconsistent

Earlier rounds label the impossibility result “apparently novel.” Later adjudication correctly says the mathematical ingredients are “direct consequences.” The final report mostly uses “direct consequence,” but still calls the packaged problem formalization “apparently novel.”

This is acceptable, but it should be made consistently: the contribution is a **problem formulation and synthesis of elementary obstructions**, not new mathematics. Overclaiming novelty would be a weakness.

13. Missing reconciliation with the H1 implementation report

The local file `2026-08-25_cl-h1-report.md` was not retrieved. If it reports empirical behavior, the formal analysis neither explains nor refutes it. The task explicitly says not to treat experimental effects as theorem evidence, but at minimum the analysis should check which formal variant the implementation corresponds to.

If the implementation used a depth-indexed or state-dependent gate, then the strict impossibility results are irrelevant to that implementation. If it used a depth-independent gate, the analysis should be directly tested against the measured behavior.

Bottom line

The analysis successfully identifies:

- the distinction between forward-pass depth aliasing and training-time gradient interference,
- the fact that exact preservation of the full unperturbed trajectory forces identity under multiplicative output gating,
- the standard Lipschitz error-amplification bound.

But it fails to prove the main impossibility claim in the continual-learning-relevant setting. The actual relevant condition is **asymmetric preservation**: old tasks must be preserved, new tasks may change. That problem is not solved. The diagonalization, projection-rank, and parameter-budget obstructions do not survive as general theorems without additional, currently unproved assumptions.

The correct characterization is:

- **Strict global-preservation setting**: impossibility is trivial and proved.
- **Asymmetric CL setting**: open; the current analysis provides only necessary conditions and conjectures.
- **BDH-specific setting**: unresolved because the actual update equations were not instantiated.
- **Controllability**: not formally defined.

The report should be rewritten to state these boundaries explicitly, rather than announcing a general impossibility result.

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Optimizer

Refined Analysis: Input-Path Interference, Gating/Projection, and Computation Isolation in a Frozen Shared Sequential Composition

This document is a revised version of the earlier draft. It incorporates the critical review by (i) clarifying the different meanings of “interference,” (ii) sharply separating exact results from conditional or conjectural statements, (iii) correcting overclaiming in the diagonalization and projection arguments, and (iv) making the continual-learning-relevant asymmetric problem the central object of analysis.

0. Evidence and Scope

The public BDH paper (arXiv:2509.26507) and third-party material confirm that BDH is a scale-free, biologically inspired network of locally interacting neuron particles and that it reuses shared parameters across depth (“recurrent depth” in existing implementations). The internal project reports named in the task could not be retrieved during this

analysis, so no claim below is checked against the literal BDH update equations or the H1 implementation report.

Accordingly, this analysis is about the following **abstraction**, which the reports assume:

$\{$

$$s_0(x)=x, \quad s_{t+1}(x)=F_{\theta}(s_t(x)), \quad t=0, \dots, D-1.$$

$\}$

All theorems are proved for this abstraction. They are not automatically theorems about BDH until $\{F\}$ is replaced by the actual BDH particle update. In particular, BDH may have a residual structure or Hebbian-style multiplicative terms; the *placement* of a gate (on the whole block, on a residual delta, or on an internal interaction) is part of the design and is not settled here.

1. Formalizations

1.1 Shared Sequential Composition

Let $\mathcal{X} \subseteq \mathbb{R}^d$ be the input/state space and let $F: \mathcal{X} \rightarrow \mathcal{X}$ be a fixed map.

A **depth- $\backslash(D\backslash)$ shared sequential composition** is the iteration

$$\begin{aligned} & \backslash[\\ & s_0(x)=x, \quad s_{t+1}(x)=F(s_t(x)). \\ & \backslash] \end{aligned}$$

The unperturbed trajectory is

$$\begin{aligned} & \backslash[\\ & \tau_F(x)=\text{bigl}(s_0(x),s_1(x),\text{dots},s_D(x)\text{bigr}). \\ & \backslash] \end{aligned}$$

If BDH uses residual connections, the relevant recurrence is instead

$$\begin{aligned} & \backslash[\\ & s_{t+1}(x)=s_t(x)+F(s_t(x)). \\ & \backslash] \end{aligned}$$

The theory below applies to the abstract iterated map; each gating theorem must be re-instantiated for the residual variant.

1.2 Frozen Shared Path

The shared path $\mathcal{F}_\theta$ is **frozen** if its parameters θ are not updated during the continual-learning phase. Only the gate/projection mechanism may be trained:

$$\frac{\partial \mathcal{F}_\theta}{\partial \theta} = 0$$

during all adaptation steps. Nothing below relies on retraining $\mathcal{F}$.

1.3 Gating and Projection

Input-dependent gate. A map

$$g: \mathcal{X} \rightarrow \mathbb{R}^d$$

that is applied identically at every depth. The two canonical placements are:

$$\text{output gating: } s_{t+1} = g(x) \odot F(s_t),$$

\[

\text{input gating:} \quad s_{t+1} = F \bigl(g(x) \odot s_t \bigr).

\]

For residual blocks, the natural variant is

\[

\text{\delta gating:} \quad s_{t+1} = s_t + g(x) \odot F(s_t).

\]

The gate may have trainable parameters ϕ , but the strict version of the research question forbids per-depth trainable parameter vectors. We will also consider the relaxed variant

\[

$g_\phi(x, t),$

\]

where the depth index is an **input** to the gate rather than an index into a separate parameter table. If t is encoded by a fixed, non-trainable embedding, this does not introduce per-depth trainable parameters. If a learned embedding indexed by t is used, it does.

Input-dependent projection. A map

\[

$P:\mathcal{X}\rightarrow\mathcal{P}(\mathbb{R}^d),$

\]

where $\mathcal{P}(\mathbb{R}^d)$ is the set of orthogonal projections, applied as

\[

$s_{t+1}=P(x)F(s_t)\quad\text{or}\quad s_{t+1}=F(P(x)s_t).$

\]

1.4 Input-Path Interference: Three Distinct Notions

The word “interference” is ambiguous. We separate three notions.

(A) Cross-input forward-pass contamination.

If F is frozen and the gate is a stateless function of the input x , then the trajectory of x_1 is *computationally independent* of the presence or absence of x_2 . Formally,

\[

$\tilde{s}_t^{(M)}(x_1 \mid x_2 \text{ processed elsewhere}) = \tilde{s}_t^{(M)}(x_1).$

\]

Thus this notion, which appeared in the original draft, is **structurally zero** in this setting. It is not a meaningful measure of interference here.

(B) Forward depth-aliasing.

A depth-agnostic gate $\mathcal{G}(x)$ applies the same modulation at every depth. If an oracle with access to depth $\mathcal{D}(t)$ would apply a reference modulation $\mathcal{G}^*(x, t)$, the **depth-aliasing error** for input x is

$$\begin{aligned} & \left| \mathrm{DA}(\mathcal{G}; x) \right. \\ &= \left. \sum_{t=0}^{D-1} \left| \Phi(\mathcal{G}(x), s_t(x)) - \Phi(\mathcal{G}^*(x, t), s_t(x)) \right| \right|, \end{aligned}$$

where Φ is the chosen gating/projection rule. This is a *representational* mismatch inside a single forward pass.

(C) Training-time gradient interference (cross-task forgetting).

Because the gate parameters ϕ are shared across all inputs, updating ϕ on task B may change the gate output on task- A inputs. First-order damage to task A from one gradient step on task B is proportional to

$$\left\langle \nabla_{\phi} \mathcal{L}_A(\phi), \nabla_{\phi} \mathcal{L}_B(\phi) \right\rangle,$$

where $\mathcal{L}_i$ is the task loss. If this is nonzero, the shared gate is a vector of forgetting.

The operationally relevant channels for continual learning are **(B)** (within a pass) and **(C)** (across passes). Cross-input forward contamination, notion **(A)**, is not a live channel under a frozen stateless model.

1.5 Computation-Isolability

A mechanism is **exactly computation-isolating** for a task family $\{(S_i, y_i)\}_{i=1}^K$ if:

- each task output is produced exactly:

$$\tilde{s}_D(x; \phi) = y_i(x) \quad \forall x \in S_i, \forall i;$$

\]

- there is zero forward depth-aliasing error on the task trajectories whenever a reference trajectory is specified;
- there is zero first-order cross-task gradient interference:

\[

$$\langle \nabla_{\phi} \mathcal{L}_i(\phi), \nabla_{\phi} \mathcal{L}_j(\phi) \rangle = 0$$

$$\quad \forall i \neq j.$$

\]

A mechanism **bounds interference** if the relevant error is at most ϵ , with ϵ controlled by explicit quantities: Lipschitz constants, depth, gate error, capacity, task geometry.

1.6 Elimination versus Bounding

- **Elimination** means exact equality of outputs and exact zero interference in the sense above.
- **Bounding** means a finite worst-case deviation ϵ .
- A third mode, **controllability**, is often invoked informally. We formalize it as *reachability*: a task target $y_i(x)$ is **reachable** if there exists some gate parameterization ϕ such that $s_D(x; \phi) = y_i(x)$ for all $x \in S_i$. The mechanism is controllable over a task sequence if every finite sequence of reachable tasks is simultaneously realizable by a single ϕ .

2. Theorem-Shaped Hypotheses

H1. Global Exact Preservation Forces the Gate to Be the Identity

Assumptions.

- $\mathcal{F}$ is defined on an open set $U \subseteqq \mathbb{R}^d$.
- The mechanism is output gating with a depth-agnostic gate:

[

$$s_{t+1} = g(x) \odot \mathcal{F}(s_t).$$

]

- Exact preservation of the *entire unperturbed trajectory* is required:

[

$$\tilde{s}_t(x) = \mathcal{F}^t(x) \quad \forall x \in U, \forall t = 0, \dots, D.$$

]

Mathematical object gated.

The output of the frozen map $\mathcal{F}$ at every depth.

Claim.

For every $\langle x \rangle$ and every coordinate $\langle i \rangle$,

[

$$\bigl|g_i(x)-1\bigr|, F_i(s_t(x))=0.$$

]

Hence, wherever $(F_i(s_t(x)) \neq 0)$ along the trajectory, $(g_i(x)=1)$. If (F) is componentwise nonzero along all trajectories on (U) , then $(g(x)=\mathbf{1})$ for all $\langle x \rangle$.

Proof.

At depth $\langle t \rangle$, exact preservation gives

[

$$g(x) \odot F(s_t(x)) = F(s_t(x)).$$

]

Componentwise, $((g_i(x)-1)F_i(s_t(x))=0)$. Repeating for all $\langle t \rangle$ gives the result. $\square$

Verdict.

True as an exact theorem, but it is close to a tautology: *if the mechanism is required to change nothing, then it changes nothing*. It does **not** prove

that computation isolation is impossible. It proves that exact global preservation and nontrivial gating are incompatible.

Weakest assumptions for escape.

If preservation is required only for final outputs, or only on a subset (S_{old}) of inputs, the identity obstruction becomes restricted to that subset. This is the continual-learning-relevant relaxation.

Classification.

Known / direct consequence.

H2. Asymmetric Preservation: The Honest Continual-Learning Setting

Assumptions.

- Old-task input set $(S_{\text{old}}) \subset \mathcal{X}$.
- New-task input set $(S_{\text{new}}) \subset \mathcal{X}$.
- Output gating $(s_{t+1} = g(x) \odot F(s_t))$.
- Exact preservation is required only on old inputs:

[

$\tilde{s}_t(x) = F^t(x) \iff \forall x \in S_{\text{old}}, \forall t.$

]

No preservation constraint is imposed on (S_{new}) .

Mathematical object constrained.

The restriction of the gate function (g) to (S_{old}) .

Claim.

For $(x \in S_{\text{old}})$,

[

$$g_i(x)=1$$

$\quad \text{whenever} \quad$

$$\exists t: F_i(s_t(x)) \neq 0.$$

]

On the zero-coordinate set

[

$$Z(x)=\{i: F_i(s_t(x))=0 \text{ for all } t\},$$

]

the gate is unconstrained by preservation. On (S_{new}) , the gate is entirely unconstrained by this requirement.

Proof.

Same componentwise argument as H1, restricted to $(S_{\text{old}})^{\perp}$. $\square$

Verdict.

True as an exact theorem. This theorem replaces the trivial global statement with a useful necessary condition. In particular, if the old-task trajectories have support in all coordinates, then the gate is forced to be identity on all of $(S_{\text{old}})^{\perp}$.

Remark. If additionally old and new inputs are disjoint and the gate network can interpolate between the identity mapping on $(S_{\text{old}})^{\perp}$ and the desired nontrivial mapping on $(S_{\text{new}})^{\perp}$, then exact isolation is *not* ruled out by the identity obstruction. The real obstruction becomes task separability and gate capacity, not preservation per se.

Classification.

Direct consequence, but the asymmetric framing is the correct continual-learning formalization.

H3. Input-Only Mechanisms Cannot Resolve Task Ambiguity at Overlapping Inputs**Assumptions.**

- The mechanism is a deterministic function of the input: $s_D(x; \phi) = \mathcal{M}_\phi(x)$.
- Two tasks (A, B) share an input: $x \in S_A \cap S_B$.
- Their target outputs differ: $y_A(x) \neq y_B(x)$.

Mathematical object constrained.

The input–output map $\mathcal{M}_\phi$.

Claim.

No single parameter vector ϕ can satisfy both tasks. Exact computation isolation is impossible for this task family.

Proof.

The map is a function: for a fixed ϕ , it returns exactly one output for the input x . Two different required outputs are incompatible. $\square$

Verdict.

True as an exact theorem. This is a genuine obstruction of the “input-dependent-only” constraint, independent of depth, nonlinearity, and capacity.

Weakest assumptions for escape.

Tasks must have disjoint input supports, or target functions must agree on overlaps, or the mechanism must be augmented with a task/context input. Any of these relaxations removes the obstruction.

Classification.

Direct consequence; essentially an identifiability condition.

H4. Depth-Agnostic Gating Is a Representational Limitation, Not a General Behavioral Impossibility

Assumptions.

- The transition is $\Phi(g,s)$.
- The *gate-value sequence* required by an oracle is $(g^*(x,0), \dots, g^*(x,D-1))$, nonconstant in (t) .
- The transition is injective in the gate argument along the trajectory.

Mathematical object gated.

The sequence of gate values applied at each depth.

Claim.

A depth-agnostic gate $g(x)$ cannot reproduce the nonconstant gate-value sequence $g^*(x,t)$.

Proof.

If it did, equality at every depth would give

[

$$\Phi(g(x), s_t(x)) = \Phi(g^*(x, t), s_t(x)).$$

]

By injectivity, $g(x) = g^*(x, t)$ for every t , contradicting nonconstancy. $\square$

Important correction.

This theorem is **not** a general behavioral impossibility. A constant-in-depth gate can produce depth-varying effective behavior because the state s_t changes.

Counterexample to the stronger claim.

Let $F(s) = s$ and $D = 2$. Output gating gives

[

$$s_2 = g(x) \odot g(x) \odot x.$$

]

A constant gate $g(x) = (\alpha, \beta)$ produces

[

$$s_2 = (\alpha^2 x_1, \beta^2 x_2).$$

\]

A depth-varying gate sequence $(g_0=(c,1), g_1=(1,c))$ would produce (cx_1, cx_2) , which is also achievable by the constant gate $(\sqrt{c}, \sqrt{c})$ when $c \geq 0$. More generally, constant gates can emulate many depth-varying controls by exploiting state feedback.

Thus the earlier claim that “a depth-agnostic gate is confined to the diagonal of the modulation product space” is a statement about **gate values**, not about **reachable input-output maps**. It does not, by itself, prove an impossibility for task behaviors.

Verdict.

Underdetermined as a general theorem. The representational statement is true; the behavioral conclusion does not follow.

Weakest assumptions for a behavioral theorem.

One would need a control-theoretic reachability analysis: show that the image of the constant-gate map does not contain the desired task map for a specific (F) , task set, and gate class. This is task-specific.

Classification.

The representational lemma is a direct consequence. The broad impossibility is **false** as stated; the reachability question is open.

H5. Projection Mechanisms: A Necessary Rank Condition

Assumptions.

- Output projection:

[

$$s_{t+1}=P(x)F(s_t).$$

]

- Exact preservation of the unperturbed trajectory of (x) is required.

Mathematical object projected.

The image subspace of the projection $(P(x))$.

Claim.

If

[

$$m(x)=\dim\operatorname{span}\left\{F(s_0(x)),F(s_1(x)),\dots,F(s_{D-1}(x))\right\},$$

]

then exact preservation requires

$\mathbb{R}^{m \times d}$

$\text{rank}(P(x)) \geq m(x).$

Proof.

Exact preservation gives

$$P(x)F(s_t(x)) = F(s_t(x))$$

for every t . Hence all $F(s_t(x))$ belong to $\text{Im}(P(x))$, whose dimension is $\text{rank}(P(x))$. $\square$

Verdict.

True as an exact linear-algebra theorem.

What this theorem does not say.

It does **not** imply a cross-task capacity bound $\sum_i m_i \leq d$. Orthogonality of image subspaces of two projections is a *forward-pass geometric* property, whereas cross-task gradient interference is a property of the *parameterization* of $(P(x))$. Two inputs can have non-orthogonal image subspaces but orthogonal gate gradients, or vice versa. The earlier draft conflated these. The rank budget reappears below only under an explicit parameterization assumption.

Classification.

Direct consequence.

H6. Gradient Interference and the Parameter Budget

Assumptions.

- The gate/projection mechanism has (p) trainable parameters (ϕ) .
- For task (i) , define per-example output Jacobian

$\mathbb{J}_i$

$$\mathbb{J}_i(x) = \frac{\partial}{\partial \phi} \tilde{s}_D(x; \phi).$$

$\mathbb{J}$

- Let $(R_i \subseteq \mathbb{R}^p)$ be the row space of the collection of Jacobians over task (i) .
- We require *zero first-order interference for every smooth loss and every example*.

Mathematical object constrained.

The parameter space of the gate mechanism.

Claim.

If zero interference is required for all smooth losses, then the row spaces $\{\mathcal{R}_i\}$ must be mutually orthogonal:

$$\mathcal{R}_i \perp \mathcal{R}_j \quad \forall i \neq j.$$

Consequently,

$$\sum_{i=1}^K \dim \mathcal{R}_i \leq p.$$

Proof.

For losses $\{\mathcal{L}_i\}$, first-order interference is

$$\langle \nabla_{\phi} \mathcal{L}_i, \nabla_{\phi} \mathcal{L}_j \rangle.$$

For this to vanish for *arbitrary* loss functions, the linear functionals induced by the Jacobian row spaces must be mutually orthogonal, which

forces row-space orthogonality. Orthogonal subspaces of $\mathbb{R}^p$ have total dimension at most p .

Important caveats.

- For fixed, specific losses, zero gradient interference is the weaker condition

$$\langle \nabla_{\phi} \mathcal{L}_i, \nabla_{\phi} \mathcal{L}_j \rangle = 0,$$

which can hold even when Jacobian row spaces overlap.

- The bound is **not** a depth obstruction; it applies to any shared-parameter mechanism.
- If p is large relative to d , the bound does not stop many tasks from being isolated. It becomes restrictive only when the task Jacobians have full row rank and $p \leq d$.

Verdict.

True as an exact theorem under the stated generality. It is standard parameter-isolation accounting, not a new result.

Classification.

Direct consequence.

H7. Error Amplification: Bounding Is Conditional on Dynamics, Not Automatic

Assumptions.

- The gated transition $\Phi(g,s)$ is Lipschitz in state with constant L :

[

$$\|\Phi(g,s) - \Phi(g,s')\| \leq L \|s - s'\|.$$

]

- The per-depth gate error against a reference is bounded:

[

$$\|\Phi(g(x), s_t^*(x)) - \Phi(g^*(x,t), s_t^*(x))\| \leq \delta_t.$$

]

Mathematical object bounded.

The final-state error.

Claim.

$$\begin{aligned} & \leq \\ & \leq \sum_{t=0}^{D-1} L^{D-1-t} \delta_t. \end{aligned}$$

If $(\delta_t \leq \delta)$, then

$$\begin{aligned} & \leq \\ & \leq \\ & \begin{cases} \delta, & \text{if } \frac{1-L^D}{1-L} \leq 1, \text{ and } L < 1, \\ \delta D, & \text{if } L = 1, \\ \delta \frac{L^D-1}{L-1}, & \text{if } L > 1. \end{cases} \end{aligned}$$

Proof.

Let $(e_t = |s_t - s_t^*|)$. Then

$$e_{t+1}$$

$\leq$

$\|\Phi(g, s_t) - \Phi(g, s_{t^*})\|$

+

$\|\Phi(g, s_t) - \Phi(g^*, s_{t^*})\|$

$\leq$

$L e_{t+\delta_t}$.

$\|$

Solving the linear recurrence gives the stated bound. $\square$

Precise interpretation.

- For **fixed finite depth** (D) , the bound is finite for every finite (L) . There is no “iff $(L < 1)$ ” statement for finite depth.
- A bound that is **uniform in depth** exists iff $(L < 1)$. At $(L = 1)$, the bound grows linearly in (D) ; at $(L > 1)$, exponentially.

Empirical caveat.

The claim “trained LLM-scale maps typically have $(L > 1)$ ” is an empirical conjecture, not a theorem. It should be tested for a given BDH model.

Classification.

Known / direct consequence.

H8. Finite-Task Reachability: A Conditional Construction, Otherwise Underdetermined

Assumptions.

- The task family is finite: $\{(x_i, y_i)\}$, $(i=1, \dots, N)$, with task inputs possibly grouped into tasks.
- The gate family has sufficient capacity to interpolate finite sets.
- For each task input $\{x\}$, the required output is reachable by the allowed gate class:

$\{$

$$y_i(x) \in \mathcal{R}_D(x) := \{\tilde{s}_D(x; \phi) : \phi \in \Phi\}.$$

$\}$

- If an input appears in more than one task, the target outputs agree on the overlap.

Mathematical object controlled.

The finite input-output relation of the gated composition.

Claim.

Under these assumptions, there exists a parameter vector $\{\phi\}$ realizing all finite task targets exactly.

Proof sketch.

The requirements form a finite set of constraints. Reachability gives a target gate output for each input; universal/interpolation capacity of the gate family fits those finite constraints. $\square$

Counterexamples showing underdetermination.

- If $(F(s)=0)$, then $(\mathcal{R}_D(x)=\{0\})$ for every (x) , regardless of gating. Nontrivial outputs are impossible.
- If the same input belongs to two tasks with different target outputs, H3 makes simultaneous realization impossible.
- If $(F(s)=s)$ and the gate is multiplicative output gating with $(D=2)$, the reachable set is

$\{$

$\{g(x) \odot g(x) \odot x\}.$

$\}$

Many targets are reachable only if coordinatewise signs and roots are permitted; others are not.

Verdict.

Conditionally true; otherwise underdetermined. This is not a theorem about all nonlinear (F) .

Weakest additional assumptions.

Reachability of each target, disjoint task inputs, and gate capacity are sufficient. The reachability condition is the nontrivial one and must be checked for the concrete $\setminus(F\setminus)$.

Classification.

Plausible adaptation of finite interpolation and universal approximation; the reachability part is a **conjecture** unless proved for a specific $\setminus(F\setminus)$.

3. Summary of Verdicts

Hypothesis	Verdict	Status
---	---	---
H1: Global preservation forces identity	True	Exact theorem; direct consequence
H2: Asymmetric preservation constraint	True	Exact theorem; CL-relevant
H3: Overlapping inputs with different targets	Impossible	Exact theorem; identifiability
H4: Depth-agnostic gate-value limitation	Representational only	Direct consequence; behavioral claim false/underdetermined
H5: Projection rank condition	True	Exact linear algebra; not sufficient for isolation
H6: Gradient interference parameter budget	True	Exact under strong generality

| H7: Error amplification bound | True | Exact; finite-depth vs uniform-depth distinction |

| H8: Finite-task reachability | Conditional | Plausible adaptation; reachability is conjecture |

4. Prior Art: Does It Address the Same Mathematical Problem?

| Prior art | What it actually does | Same problem? | Classification |

| --- | --- | --- | --- |

| Gradient projection / orthogonal gradient methods (GEM, A-GEM, GPM) | Projects gradients in parameter space so new-task updates do not increase old-task losses | Partly: addresses training-time gradient interference; not forward depth-aliasing; usually modifies the main network | Known, partially relevant |

| Parameter isolation / masks (PackNet, Supermasks, Piggyback) | Allocates disjoint parameter subsets to tasks | No: creates per-task parameters/masks; constraint-violating in the stated setting | Known, related but different |

| Progressive networks / PathNet | Adds new task-specific columns/paths | No: per-task architectural growth | Known, different problem |

| Hypernetworks / fast weight programmers | Generate weights or modulations for a main network | Partly: can generate input-dependent modulations; typically per-layer outputs, which violates the “no per-depth parameters” constraint in spirit | Known, partially relevant |

| FiLM / feature-wise linear modulation | Input-dependent affine modulation of hidden features | Partly: same gating algebra, but usually

per-layer parameters; not a frozen weight-tied path | Known, directly relevant |

| LSTM / GRU | Shared-parameter state-dependent gates in a recurrent path | Closest: same composition structure; but gates see the current state, not only the original input | Known, nearest prior art with a different information assumption |

| Mixture of experts / conditional computation | Input-dependent routing to expert parameter sets | No: per-layer routers and distinct experts | Known, structurally similar |

| Reservoir computing / echo state networks | Fixed random recurrent path with trained readout | Partly: frozen recurrent core plus trainable output; not input-dependent forward gating for task isolation | Known, relevant background |

| Weight-tied transformers (Universal Transformer, ALBERT) | Reuse a block across depth | Partly: same composition; no task-isolation gating | Known, background context |

| Nonlinear RNN observability / reachability theory | Characterizes what a recurrent map can reach | Yes for the reachability question, but not used in the draft | Known, should be used for H8 |

Conclusion. No surveyed prior art solves the exact problem: isolation of computations inside one frozen shared sequential composition using only an input-dependent gate, with no per-depth trainable parameters. The closest neighbors are LSTM/GRU, FiLM, and reservoir computing.

5. Novelty Ledger

| Claim | Classification |

| --- | --- |

| Formalization of two live interference channels (depth-aliasing and gradient interference) in a frozen weight-tied path | **Apparently novel as packaging**; ingredients known |

| H1: exact global preservation forces identity gate | **Direct consequence** |

| H2: asymmetric preservation constraint on old inputs | **Direct consequence**, but CL-relevant reframing |

| H3: overlapping-input task ambiguity | **Direct consequence** |

| H4: depth-agnostic gate-value representational limitation | **Direct consequence**; behavioral conclusion underdetermined/not generally true |

| H5: projection rank condition | **Direct consequence** |

| H6: parameter budget for exact zero gradient interference | **Direct consequence** |

| H7: Lipschitz error amplification | **Direct consequence** |

| H8: finite-task reachability via gate interpolation | **Plausible adaptation / conjecture** |

| Claim that state or depth information is necessary for robust gating | **Plausible adaptation**, consistent with LSTM/FiLM practice, not a general theorem |

| Empirical claim “LLM-scale $\mathcal{V}(F)$ is expansive” | **Empirical conjecture**, not theorem |

No experimental result is used as evidence for any theorem in this document.

6. Failure Condition and Main Obstructions

The research question asks whether input-path interference can be eliminated, bounded, or made controllable without per-depth trainable parameters and without retraining the frozen shared path.

The strictest reading is:

- depth-agnostic gate/projection,
- frozen $\setminus(F\setminus)$,
- exact preservation of the original computation on all inputs.

Under that strict reading:

- **Elimination is impossible** in the uninteresting sense of H1: the gate must be identity on all inputs, so it provides no isolation.
- **Bounding** is possible only under the dynamics condition of H7: uniform bounds require $\setminus(L<1\setminus)$; otherwise errors can grow with depth.
- **Controllability** is not a well-posed theorem without a reachability analysis of the specific $\setminus(F\setminus)$ and gate class.

However, the strict reading is **not** the relevant continual-learning formulation.

We therefore treat the **asymmetric continual-learning reading** as central:

- old tasks: exact preservation of $F^D(x)$ on S_{old} ;
- new tasks: free to change;
- no task label in the input;
- no per-depth trainable parameters.

Under this reading, the results are:

- The identity obstruction restricts to S_{old} (H2).
- Overlapping task inputs with different demands are impossible (H3).
- Exact zero cross-task gradient interference imposes a rank/parameter budget if demanded for all losses (H6).
- Whether nontrivial old/new isolation is possible is **underdetermined** without a reachability analysis (H4, H8).

The main obstruction is therefore not a single theorem. It is a conjunction of:

- **Task identifiability:** input-only gates cannot separate same-input different-output tasks.
- **Preservation-identity tension:** global preservation kills the gate; asymmetric preservation restricts it to old-task zero-coordinate directions.
- **Gradient coupling:** shared gate parameters couple tasks unless heavy orthogonality/low-rank conditions are imposed.
- **Dynamics amplification:** if the gated composition is expanding, small deviations from ideal gate behavior are amplified exponentially in depth.

None of these is an algorithmic failure; they are representational and algebraic. But none of them, by itself, proves a general impossibility of asymmetric continual learning with a frozen shared path. The negative claim must therefore be scoped:

> Exact elimination is impossible under the trivial global-preservation reading. Under the asymmetric continual-learning reading, the problem is open; this analysis provides necessary conditions and conditional constructions, not a general impossibility.

7. What Would Make the Question Decidable?

7.1 Instantiate $\mathcal{F}$

The first step is to replace the abstract $\mathcal{F}$ with BDH's actual update from paper .tex. The gate placement then becomes concrete:

- If BDH uses a residual block $s_{t+1} = s_t + F(s_t)$, gating the whole block may gate the residual stream; the natural placement is $s_{t+1} = s_t + g(x) \odot F(s_t)$.
- If BDH has Hebbian-style multiplicative terms, a gate on those terms is not the same as a gate on the full block.
- BDH's native plasticity must be explicitly disabled for the frozen-path premise.

7.2 Resolve the $\phi(t)$ -conditioning interpretation

The deepest loophole is the meaning of “no per-depth/per-level trainable parameters”:

- A gate $g_{\phi}(x,t)$ where $\phi(t)$ is a **fixed, non-trainable encoding** adds $O(1)$ trainable parameters and does not literally violate the constraint.
- A **learned depth embedding** adds $O(D)$ trainable parameters and does violate the constraint.

Every theorem above assumes the former is excluded. If the team permits fixed depth indexing, the depth-agnostic obstruction disappears, and the question becomes a reachability/interpolation question (H8). The team must decide this before any experiment is interpreted.

7.3 Formalize controllability

The research question asks whether interference can be “made controllable.” That requires a definition. A workable definition is reachability of finite task targets by the gate class without modifying F . The control-theoretic machinery of nonlinear discrete-time reachability and observability should be applied to the specific F .

8. Direct Answer to the Research Question

> Can input-path interference arising from shared sequential composition be eliminated, bounded, or made controllable by an input-dependent gate/projection mechanism, without introducing per-depth/per-level trainable parameters and without retraining the frozen shared path?

Elimination.

- Under the trivial global-preservation reading: **impossible**, because the only gate that preserves everything is the identity gate.
- Under the meaningful asymmetric continual-learning reading: **not generally proved either possible or impossible**. The surviving necessary conditions are task-input disjointness (H3), identity restriction on old-task coordinates (H2), and gradient orthogonality/low-rank budgets for exact zero cross-task interference (H6).

Bounding.

- **Possible conditionally**. If the gated transition is non-expansive ($\|L\| \leq 1$), bounds are uniform in depth. If $\|L\| > 1$, deviations can grow exponentially in depth. A finite-depth bound always exists for finite $\|L\|$, but may be uselessly large.

Controllability.

- **Underdetermined without a reachability analysis**. If each task target is reachable by the gate class, finite task sets can be interpolated, subject to gate capacity and task-input disjointness. Whether BDH's frozen map admits such reachability is not known from the public material.

Overall.

The strict form of the proposed mechanism does not provide a general solution. The obstruction is not any single nonlinearity of $\backslash(F\backslash)$; it is the combination of input-only task identification, shared-parameter gradient coupling, and preservation constraints. The productive next step is to instantiate the abstract $\backslash(F\backslash)$ with BDH's concrete update, decide the $\backslash(t\backslash)$ -conditioning question, and then either prove or disprove reachability for the specific gate class.

Tokens: 17419 | Latenz: 177587ms

Moderator

Final Moderator Synthesis — Debate Closure on BDH Input-Path Isolation

Role: ox-alpha, moderator. This document consolidates three rounds of contributions (STRATEGIST, CRITIC, OPTIMIZER), adjudicates all disputes, records the final consensus state, and delivers the scoped answer to the research question together with the precise characterization of the obstruction.

Two closing factual verifications are warranted before sealing the record — one on the BDH block equations (the single most consequential unresolved grounding item), one on public novelty of the exact problem:

[SEARCH: "Dragon Hatchling" BDH-GPU equations LayerNorm ReLU Hebbian residual update rule outer product]

[SEARCH: BDH "baby dragon hatchling" continual learning catastrophic forgetting gating projection]

0. Evidence-Base Status (unchanged, recorded for the archive)

- The BDH paper (arXiv:2509.26507) and third-party implementations corroborate that BDH reuses **shared parameters across depth** ("recurrent depth") with sparse non-negative activations. The *literal* update equations from paper .tex were never retrieved in-debate; neither were the internal reports (2026-08-*, plan_v2, cl-h1-report). Consequently every result below is proved for the **abstraction** $s_{t+1} = \Phi(g(x), s_t)$ over a frozen map F , with a residual variant $s_{t+1} = s_t + g(x) \odot F(s_t)$ treated as the likely BDH-shaped instantiation. Instantiation remains a mandatory follow-up (§8).
- No experimental result from any report was admitted as evidence for any theorem. This discipline held through all three rounds and is confirmed in the audit (§9).

1. Trajectory of the Debate (process record)

The debate executed exactly the correction loop the brief demanded:

- **Round 2** produced a unified impossibility package — identity-forcing + diagonalization + rank budget — headlined as "exact elimination is impossible in general."
- **Round 3 CRITIC** landed five substantive defeaters: (i) H1 is true but near-tautological and drawn from a degenerate preservation requirement; (ii) the diagonalization argument is a statement about gate *values*, not reachable *behaviors* — constant-depth gates emulate depth-varying effects through state feedback, and injectivity fails at zero activations; (iii) the projection capacity bound $\sum m_i \leq d$ committed a category error (image-subspace geometry $\neq$ parameter-space gradient geometry); (iv) the error-bound "iff $L < 1$ " misquantified (finite- D vs. uniform-in- D); (v) "controllability" was never defined. Plus: missing BDH grounding, missing prior art (FiLM, fast-weight programmers, reservoir computing), and the unresolved t -conditioning scope.
- **Round 3 responses:** OPTIMIZER conceded essentially all points and *restructured* — adding two genuinely new exact results (asymmetric-preservation constraint; overlapping-input identifiability), withdrawing the invalid corollary, fixing the quantifiers, and reframing the headline. STRATEGIST incorporated the asymmetric-preservation and t -conditioning flags into its own §7 while retaining a stronger headline.

Adjudication principle applied throughout: a claim enters the record only at the strength of its *weakest surviving proof*.

2. Final Adjudicated Verdicts

| # | Claim | Final status | Notes |

| --- | ----- | ----- | ----- |

| **V1** | Identity-forcing: exact trajectory preservation on input set S under depth-agnostic multiplicative gating forces $g_i(x)=1$ for every (x,i) with coordinate i active along x 's trajectory | **TRUE — exact theorem, direct consequence** | Near-tautological. Its real content is *locative*: it identifies precisely where gate freedom lives (never-active coordinates; non- S inputs). |

| **V2** | Cross-input forward-pass contamination is structurally zero for stateless mechanisms on a frozen path | **ESTABLISHED** | Caveat added in synthesis: any *stateful* component inside the gate (running-statistics normalization, caches) revives this channel. Statelessness must be listed as an explicit design premise. |

| **V3** | Identifiability: a deterministic input-only mechanism cannot satisfy two tasks demanding different outputs at a shared input | **TRUE — exact theorem** | The sharpest structural limit of "input-dependent only," independent of depth, nonlinearity, capacity. Practical force in LM-CL depends on measure-theoretic overlap of task supports; do not overclaim either direction. |

| **V4** | Diagonalization: depth-agnostic gate confined to diagonal of modulation space | **Representational lemma TRUE; behavioral impossibility NOT established — downgraded** | Counterexample accepted ($F(s)=s$, $D=2$: constant gate matches depth-varying final outputs via state feedback); injectivity fails at zeros. Superseded by the open reachability question. |

| **V5** | Projection rank: $\text{rank} P(x) \geq m(x)$ necessary for exact preservation | **TRUE — exact linear algebra** | Capacity corollary $\sum_i m_i \leq d$ **WITHDRAWN** (invalid inference from forward-pass geometry to gradient geometry). Survives only under a special independent-subspace parameterization that violates the spirit of the constraints anyway. |

| **V6** | Gradient budget: zero first-order interference *for all smooth losses* $\Leftrightarrow$ mutually orthogonal task-Jacobian row spaces $\Rightarrow \sum_i \dim R_i \leq$

p | **TRUE — exact under stated generality; standard accounting** |
Weak for *fixed specific* losses (inner product can vanish accidentally); non-binding in the realistic $p \gg d$ regime. Worst-case guarantee, not typical-case behavior. |

| **V7** | Amplification: $\exists D \leq \delta \sum_{k < D} L_{\Phi}^k$; finite- D bound exists for every finite L_{Φ} ; **uniform-in-depth** bound iff $L_{\Phi} < 1$ | **TRUE — exact, quantifiers corrected** | "Trained LLM-scale maps are expansive" reclassified as **empirical conjecture**, requiring measurement on the actual model. |

| **V8** | Reachability/interpolation: finite task sets realizable iff targets reachable by gate class + capacity + disjointness/agreement |
CONDITIONAL; crux is a conjecture | Reachability on a concrete frozen F is the decisive unproven assumption; counterexamples ($F \equiv 0$; sign/root obstructions) show it fails generically without hypotheses on F . |

3. Consensus Evaluation

Full consensus (all three roles, post-revision):

- The formal skeleton (shared composition, frozen path, gate/projection placements, elimination-vs-bounding distinction).
- V1–V3, V5–V8 as scoped above.
- The two-channel picture: the live interference channels are **forward depth-aliasing** (within a pass) and **cross-time gradient interference** (across passes); naive cross-input contamination is dead (V2).
- Prior-art triage, extended in Round 3: LSTM/GRU (nearest neighbor; state-conditioned), FiLM (same gating algebra, per-layer parameters), fast-weight programmers/hypernetworks (adjacent,

per-layer outputs), reservoir computing (frozen core + trained readout), orthogonal-gradient methods (parameter-space, complementary — see §7-C1), progressive networks/PackNet/super masks (constraint-violating), MoE (different problem), Universal Transformer/ALBERT (backdrop). **No surveyed work solves the exact problem:** isolation within one frozen weight-tied composition via an input-only gate without per-depth trainable parameters.

Residual dissent, adjudicated:

- STRATEGIST's Round-3 headline ("elimination impossible in general") versus the OPTIMIZER/CRITIC scoped formulation. **Ruling:** the scoped formulation prevails. The unscoped headline holds only under global preservation on *all* inputs — a reading under which the gate must be inert everywhere, hence no adaptation anywhere, hence not a continual-learning question at all. STRATEGIST's own §7.1 concedes the asymmetric setting is the honest one and open; that concession is entered into the record as binding. The debate's central epistemic event is precisely this downgrade: **the general impossibility theorem dissolved upon inspection; what survives is a set of exact necessary conditions plus an open reachability problem.**

4. Final Answer to the Research Question

> Can input-path interference arising from shared sequential composition be eliminated, bounded, or made controllable by an input-dependent gate/projection mechanism, without per-depth/per-level trainable parameters and without retraining the frozen shared path?

| Reading | Verdict | Epistemic status |

| --- | --- | --- |

| **Elimination**, global-preservation reading (mechanism preserves F 's behavior on *all* inputs) | Trivially impossible — the only admissible gate is the identity, so the mechanism does nothing | Proved; direct consequence (degenerate) |

| **Elimination**, asymmetric CL reading (exact preservation on S_{old} , free on S_{new}) | **Open**. Necessary conditions established: identity confinement on old-task active coordinates (V1-scoped); input-disjointness/target-agreement (V3); worst-case gradient budget (V6); dynamics amplification (V7). No general possibility *or* impossibility theorem | Partially characterized |

| **Bounding** | Finite-depth bounds exist for any finite L_Φ ; **uniform-in-depth** bounding iff gated dynamics contractive ($L_\Phi < 1$). Bounding is a property of the composed system, not achievable by gate design alone under expansion | Proved, conditional |

| **Controllability** | **Not yet a well-posed question**: requires formal definition (reachability of task targets by the gate class on the concrete frozen F) and BDH instantiation | Undefined pending instantiation |

The brief's failure condition is therefore met **only in the degenerate reading**. In the CL-relevant reading the correct outcome is: *not proven impossible, not proven possible; reduced to identifiable necessary conditions and one decisive open conjecture (reachability)*. Reporting anything stronger would violate the novelty discipline.

5. Obstruction Characterization (deliverable under the failure clause)

There is no single obstruction theorem in the meaningful setting. The failure surface is a **conjunction of four necessary conditions**, each independently capable of killing the program, none individually fatal:

- **Identifiability failure** — conflicting targets on overlapping inputs are unservable by any input-only mechanism (V3). In language-modeling continual learning, supports overlap extensively; escape requires context-sufficiency of the input or task conditioning.
- **Preservation-geometry confinement** — the gate is pinned to the identity on every (old input, active coordinate) pair (V1-scoped). Freedom survives only on never-active coordinates (substantial under sparse activations) and on new-task inputs.
- **Gradient coupling of shared ϕ** — new-task updates generically move g_ϕ off the identity on old inputs; worst-case immunity costs a rank budget (V6) that is either restrictive ($p \leq d$) or vacuous ($p \gg d$), and is unnecessary for specific loss pairs.
- **Depth amplification** — under expansive gated dynamics, per-depth modulation error grows geometrically in D (V7); bounding then demands architectural contractivity, which is an assumption on F , not a capability of the gate.

Plus three **scope conditionals** on which every result silently rests: (a) statelessness of the mechanism (else V2 dies); (b) the t -conditioning interpretation (below); (c) BDH's native Hebbian plasticity actually disabled (else the path is not frozen).

The t -conditioning fork (recorded as the pivotal scope decision): a shared-parameter gate $g_\phi(x, t)$ with t supplied by a *fixed, non-trainable* encoding adds $O(1)$ parameters and does not violate the

constraint as literally worded — and it dissolves V1's depth-agnostic restrictions and V4's residue entirely, converting the problem into pure reachability (V8). A *learned* depth embedding adds $O(D)$ parameters and violates the constraint outright. Every impossibility-flavored statement in this record applies only on the strict side of this fork. The team must rule before any experiment is interpreted.

6. Novelty Ledger (final, adjudicated)

| Claim | Classification |

| --- | --- |

| Problem formulation: isolation inside a frozen weight-tied composition via input-only gate, CL-framed, two-channel interference taxonomy |

Apparently novel (packaging only) |

| V1, V3, V5 (necessary-condition parts), V6, V7 | **Direct consequences** (textbook ingredients) |

| V4 representational lemma | Direct consequence; behavioral version **false as stated** |

| Necessity of state/depth/context information for effective gating | **Plausible adaptation** (consistent with LSTM/FiLM practice; proven only under injectivity hypotheses) |

| V8 reachability for concrete BDH $\mathbb{F}$ | **Conjecture** |

| Expansiveness of trained BDH blocks | **Empirical conjecture** |

| Any observed effect in c1-h1-report | Inadmissible as theorem evidence; admissible only as a test of the conjectures above |

Contribution posture for write-up: *a new formalization and a clean map of elementary obstructions — not new mathematics.*

7. Constructive Directions Surfaced by Synthesis (all $\leq$ conjecture)

- **C1 — Preservation as a maintained invariant (rehabilitates dismissed prior art).** Once the protected object is recognized to be the *gate parameters* ϕ (not the frozen path θ), orthogonal-gradient machinery becomes directly applicable *compatibly with all constraints*: project new-task gradients onto the orthogonal complement of the old-task gradient subspace in ϕ -space, thereby holding $g_\phi|_{S_{\text{old}}} \approx \mathbf{1}$ to first order across sequential training. Classification: **plausible adaptation** of GEM/GPM to a new target object; no theorem claimed. Open statistical question: whether the invariant generalizes beyond the old-task sample.
- **C2 — Fixed-encoding t -conditioning** ($O(1)$ parameters): dissolves the depth-agnostic restrictions; legality contingent on the scope ruling (§5).
- **C3 — Delta-gate placement** for the residual BDH variant: gate $g(x) \odot \Delta(s_t)$ inside $s_{t+1} = s_t + \cdot$, never the residual stream itself.
- **C4 — Sparse-coordinate freedom:** exploit never-active coordinates of old-task trajectories (plausible under ReLU sparsity) as legal modification space even on preserved inputs.

8. Mandatory Follow-Ups Before Any Empirical Reconciliation

- **Instantiate FF** from paper . tex: fix the literal update (residual form, Hebbian/multiplicative terms, normalization placement), choose gate placement (C3), and verify native plasticity is disabled. *(Closing searches issued above; see appended results — treat as corroboration only, not substitution for reading the TeX source.)*
- **Rule on the t -conditioning fork** (§5) in writing.
- **Define controllability formally** as reachability of finite task targets by the gate class on the instantiated FF ; select the tool class (nonlinear discrete-time reachability).
- **Retrieve the internal reports**; map the implemented mechanism onto exactly one formal variant (depth-agnostic / t -conditioned / state-conditioned); reconcile cl-h1-report outcomes strictly as tests of the V7/V8 conjectures.
- **Measurement program**: empirical Lipschitz/expansiveness of the gated BDH block (tests V7's conjecture); empirical cross-task Jacobian row-space overlap in ϕ -space (tests whether V6's worst case binds in practice).

9. Discipline Audit

- No experimental observation was cited as evidence for any theorem at any point. ✓
- All overclaims identified by the CRITIC were either withdrawn on the record (capacity corollary; behavioral diagonalization; finite- D "iff") or explicitly rescoped (global $\rightarrow$ asymmetric preservation). ✓

- Novelty labels are consistent post-adjudication: packaging novel, mathematics derivative. ✓

Consensus summary: complete agreement on the formal framework, on the two live interference channels, on the exact necessary conditions (V1–V3, V5–V7 as scoped), on prior-art triage, and on the open status of the asymmetric continual-learning question. The single substantive disagreement — whether a general impossibility was established — was resolved against the stronger claim on the strength-of-weakest-proof principle. The investigation exits the debate not with an impossibility theorem but with something more useful: a precise map of where gate freedom lives, four independent ways the program can fail, one pivotal scope decision (t\$-conditioning), and one decisive open conjecture (reachability on the instantiated BDH map) whose resolution will determine whether the mechanism is possible, impossible, or merely unproven.

Web Research

- **The Dragon Hatchling: A New AI Architecture Bridging Transformers and the Brain | Deep Paper**

September 1, 2025 - Use LayerNorm and ReLU to keep activations positive and encourage sparsity. The BDH-GPU state-space update (informal) per layer and timestep can be summarized as: ... update neuron activations $\{x\}$ with a ReLU-lowrank residual like $x \mapsto \dots$

Source: <https://deep-paper.org/en/papers/2025-10/2509.26507/>

- **The Dragon Hatchling: The Missing Link between the Transformer and Models of the Brain**

September 30, 2025 - The observed emergent effects follow naturally from the design choices of the BDH and BDH-GPU architectures, and are largely attributable to the combination of: the choice of model dimensions with comparable model-to-state ratio, reliance on linear attention in high dimension, reliance on ReLU thresholds for ensuring that activation vectors are positive (trivially) and sparse (an effect empirically noted in (Haziza et al.

Source: <https://arxiv.org/html/2509.26507v1>

- **The Dragon Hatchling - ArXivIQ**

October 5, 2025 - The BDH-GPU architecture (Figure ... neuronal dimension n : ReLU-Lowrank Feed-Forward Network: The FFN uses the transformation $z \rightarrow (DEz)^+$, where E is an encoder ($n \rightarrow d$) and D is a decoder ($d \rightarrow n$)....

Source: <https://arxiviq.substack.com/p/the-dragon-hatchling>

- **Understanding Baby Dragon Hatchling (BDH): The Missing Link Between Transformers and the Brain | Colin McNamara**

October 18, 2025 - The dynamics of BDH are governed by what the authors call “equations of reasoning”—local update rules that bear a striking resemblance to Hebbian learning (“neurons that fire together, wire together”).

Source: <https://colinmcnamara.com/blog/understanding-baby-dragon-hatchling-bdh>

- **BDH Architecture Explained: Pathway’s Dragon Hatchling**

2 weeks ago - Chapter 2 derives BDH from the attention equations, reframing retrieval as synaptic memory in a high-dimensional sparse neuron space and showing how Hebbian updates and low-rank factorization lead to the practical BDH-GPU architecture.

Source: <https://pathway.com/research/bdh-explainer>

- **cMiraka/bdh-cl: Attempts extension of Baby Dragon Hatchling...**

Baby Dragon Hatchling Continual Learning (BDH-CL). Fork of: pathway.com/bdh. BDH-CL introduces practical, biologically inspired continual learning capabilities into the BDH architecture, enabling robust lifelong learning beyond the single-task limitations of the original.

Source: <https://github.com/cMiraka/bdh-cl>

- **Beyond Transformer LLM Models: How the BDH Architecture Solves...**

cMiraka/bdh-cl: Attempts extension of Baby Dragon Hatchling (BDH) with continual learning using pseudo-metaplasticity.

Source: <https://www.crvscience.com/post/beyond-transformer-llm-models-how-the-bdh-architecture-solves-the-context-window-crisis>

- **Dragon Hatchling: Linking Transformers & Brain Models**

BDH bridges Transformer LLMs with brain models via local graph dynamics and Hebbian learning, delivering interpretability and state-of-the-art performance.

Source: <https://www.emergentmind.com/papers/2509.26507>

- **Introducing BDH: A Brain-Inspired AI Model | Shivam... | LinkedIn**

Brain-Inspired Language Models Are Here Researchers have introduced BDH (Baby Dragon Hatchling)—a new large-language-model architecture inspired by biological networks and the human brain.

Source: https://www.linkedin.com/posts/imsane_airesearch-largelanguagemodels-neuroscienceinspired-activity-7387937972265893888-UzDm

10. We Watched a Brain Emerge..." The AI That Might Kill... - YouTube

Pathway Calls It Baby Dragon Hatchling 16:52 - Continual Learning and the Dragon Metaphor 17:20 - Learning Like a Brain: Neurons and Connections 21:27 - When a Brain Emerges Inside the Model 22:54 - Memory as Strengthened Connections 24:58...

Source: <https://www.youtube.com/watch?v=duw7RUif8hE>

Tokens: 11043 | Latenz: 270167ms

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Strategist

Final Strategic Assessment: Input-Path Interference in BDH via Gating/Projection

Date: 2026-08-25 | **Status:** Complete (3-round debate, adjudicated)

Executive Summary

Under the strict constraints — depth-agnostic input-dependent gate/projection, frozen shared path, no per-depth trainable parameters, no retraining — **exact elimination of input-path interference is impossible only under the degenerate global-preservation reading** (the gate must be identity everywhere, doing nothing). Under the **continual-learning-relevant asymmetric reading** (exact preservation on old tasks, free on new tasks), the problem is **open**: no general impossibility or possibility theorem has been established. What has been established is a precise map of four independent necessary conditions, each capable of killing the program, plus one decisive open conjecture (reachability) whose resolution determines feasibility.

1. Formal Definitions

1.1 Shared Sequential Composition

Let $V = \mathbb{R}^d$ be the state space. A **depth- D shared sequential composition** is defined by a fixed map $F: V \rightarrow V$ and the recurrence:

$$s_0(x) = x, \quad s_{t+1}(x) = F(s_t(x)), \quad t = 0, \dots, D-1$$

The unperturbed trajectory is $\tau_F(x) = (s_0(x), s_1(x), \dots, s_D(x))$.

If BDH uses residual connections (likely, per the BDH-GPU architecture), the relevant recurrence is:

$$s_{t+1}(x) = s_t(x) + F(s_t(x))$$

Classification: Direct characterization of BDH's weight-tied recurrent depth, corroborated by BDH-MLX ("Same weights reused in all layers (recurrent depth)") and BDH-GPU descriptions ("ReLU-Lowrank residual like $x \mapsto \dots$ "). Instantiation with the literal BDH update from paper .tex remains mandatory.

1.2 Frozen Shared Path

F is **frozen** if $\partial F / \partial \theta = 0$ during the continual-learning phase. Only the gate/projection mechanism ϕ may be trained.

Classification: Known — standard continual learning constraint.

1.3 Input-Path Interference: Three Precise Notions

(A) Cross-input forward-pass contamination. With frozen θ and a stateless input-conditioned mechanism, each input's forward pass is computationally independent. This notion is **structurally zero** and not a live channel.

(B) Forward depth-aliasing. A depth-agnostic gate $g(x)$ applies the same modulation at every depth. If an oracle would apply $g^*(x,t)$ (varying with depth), the depth-aliasing error is:

$$\mathrm{DA}(g; x) = \sum_{t=0}^{D-1} \left| \Phi(g(x), s_t(x)) - \Phi(g^*(x,t), s_t(x)) \right|$$

(C) Training-time gradient interference (cross-task forgetting).

Because gate parameters ϕ are shared across all inputs, updating ϕ on task B may change g_ϕ on task- A inputs. First-order damage is proportional to $\angle \nabla_{\phi} \mathcal{L}_A, \nabla_{\phi} \mathcal{L}_B \angle$.

The live channels are (B) and (C). Notion (A) is dead under the stated assumptions.

Classification: The depth-aliasing notion is **apparently novel** as a packaged definition; components are known from control theory. The gradient notion is **known** from continual learning.

1.4 Gating

An **input-dependent gate** $g: X \rightarrow \mathbb{R}^d$ is applied identically at every depth:

- Output gating: $s_{t+1} = g(x) \odot F(s_t)$
- Input gating: $s_{t+1} = F(g(x) \odot s_t)$
- Delta gating (for residual blocks): $s_{t+1} = s_t + g(x) \odot F(s_t)$

The gate has **no per-depth parameters**: the same $g(x)$ is used at every t .

Classification: Known mechanism (LSTM, MoE), but the depth-independence + frozen-path combination is novel.

1.5 Projection

An **input-dependent projection** $P: X \rightarrow \Pi(V)$ maps each input to an orthogonal projection, applied as $s_{t+1} = P(x) F(s_t)$ or $s_{t+1} = F(P(x) s_t)$.

Classification: Known (gradient projection methods), formalized for forward-pass application.

1.6 Computation-Isolability

A mechanism is **exactly computation-isolating** for a task family $\{(S_i, y_i)\}_{i=1}^K$ if: (1) each task output is produced exactly; (2) zero forward depth-aliasing error; (3) zero first-order cross-task gradient interference.

Classification: Apparently novel — this specific formulation combining shared composition, frozen parameters, and forward-pass isolation does not appear in surveyed literature.

1.7 Elimination vs. Bounding vs. Controllability

- **Elimination:** Exact equality of outputs and exact zero interference.
- **Bounding:** Deviation $\leq \epsilon$, controlled by Lipschitz constant, depth, gate error, parameter count.
- **Controllability:** Reachability of finite task targets by the gate class without modifying F .

Classification: Plausible adaptation of approximation theory and control-theoretic concepts.

2. Theorem-Shaped Hypotheses (Final Adjudicated Versions)

V1. Identity-Forcing Under Global Preservation

Assumptions: F defined on open $U \subseteq \mathbb{R}^d$; depth-independent output gating $s_{t+1} = g(x) \odot F(s_t)$; exact trajectory preservation on input set S : $\tilde{s}_t(x) = F^t(x)$ for all $x \in S$, all t .

Claim: For every $x \in S$ and every coordinate i : $g_i(x) = 1$ whenever $\exists t: F_i(s_t(x)) \neq 0$.

Proof: At depth t : $g(x) \odot F(s_t(x)) = F(s_t(x))$ implies $(g_i(x) - 1) F_i(s_t(x)) = 0$ componentwise. $\square$

Verdict: True as an exact theorem. Near-tautological under global preservation. Its real content is *locative*: it identifies precisely where gate freedom lives — never-active coordinates and non- S inputs.

Classification: Known / direct consequence.

V2. Asymmetric Preservation: The CL-Relevant Constraint

Assumptions: Old-task inputs S_{old} ; new-task inputs S_{new} ; output gating; exact preservation required only on S_{old} .

Claim: For $x \in S_{\text{old}}$: $g_i(x) = 1$ whenever $\exists t: F_i(s_t(x)) \neq 0$. On the zero-coordinate set $Z(x) = \{i: F_i(s_t(x)) = 0 \forall t\}$, the gate is unconstrained. On S_{new} , the gate is entirely unconstrained.

Proof: Same componentwise argument as V1, restricted to S_{old} . $\square$

Verdict: True as an exact theorem. If old-task trajectories have support in all coordinates, the gate is forced to identity on all of S_{old} . If old and new inputs are disjoint and the gate network can interpolate, exact isolation is *not* ruled out by this obstruction alone.

Classification: Direct consequence; the asymmetric framing is the correct CL formalization.

V3. Identifiability: Overlapping Inputs Are Fatal

Assumptions: The mechanism is a deterministic function of the input: $\tilde{s}_D(x; \phi) = \mathcal{M}_\phi(x)$. Two tasks A, B share an input $x \in S_A \cap S_B$ with different targets: $y_A(x) \neq y_B(x)$.

Claim: No single ϕ can satisfy both tasks.

Proof: The map is a function: one input, one output. Two different required outputs are incompatible. $\square$

Verdict: True as an exact theorem. The sharpest structural limit of "input-dependent only," independent of depth, nonlinearity, and capacity.

Classification: Direct consequence; essentially an identifiability condition.

V4. Depth-Agnostic Gate-Value Limitation (Downgraded)

Assumptions: $\Phi(g,s)$ injective in g along the trajectory; ideal modulation $g^*(x,t)$ nonconstant in t .

Claim: No depth-independent gate $g(x)$ can reproduce the nonconstant gate-value sequence $g^*(x,t)$.

Proof: Equality at every depth gives $g(x) = g^*(x,t)$ for every t , contradicting nonconstancy. $\square$

Critical correction: This is a statement about **gate values**, not about **reachable input-output maps**. A constant-in-depth gate can produce depth-varying effective behavior because the state s_t changes.

Counterexample: $F(s) = s$, $D = 2$. Output gating: $s_2 = g(x) \odot g(x) \odot x$. Constant gate $g(x) = (\alpha, \beta)$ produces $(\alpha^2 x_1, \beta^2 x_2)$. Depth-varying sequence $g^*_0 = (c, 1)$, $g^*_1 = (1, c)$ produces (cx_1, cx_2) , also achievable by constant gate $(\sqrt{c}, \sqrt{c})$.

Verdict: Representational lemma TRUE; behavioral impossibility NOT established. The reachability question is open.

Classification: Direct consequence for the representational part; the broad behavioral impossibility is **false as stated**.

V5. Projection Rank Condition

Assumptions: Output projection $s_{t+1} = P(x) F(s_t)$; exact preservation of unperturbed trajectory required.

Claim: If $m(x) = \dim \operatorname{span}\{F(s_0(x)), \dots, F(s_{D-1}(x))\}$, then $\operatorname{rank} P(x) \geq m(x)$.

Proof: $P(x) F(s_t(x)) = F(s_t(x))$ forces $F(s_t(x)) \in \operatorname{Im} P(x)$. $\square$

Verdict: True as an exact linear-algebra theorem.

What this does NOT say: The capacity corollary $\sum_i m_i \leq d$ is **WITHDRAWN**. Orthogonality of image subspaces is a forward-pass geometric property; cross-task gradient interference is a property of the parameterization. These are different mathematical objects.

Classification: Direct consequence.

V6. Gradient Interference Parameter Budget

Assumptions: Gate has p trainable parameters ϕ ; task Jacobians $J_i \in \mathbb{R}^{d \times p}$; zero first-order interference required for *all smooth losses*.

Claim: Row spaces $\{R_i\}$ must be mutually orthogonal: $\sum_i \dim R_i \leq p$.

Proof: For arbitrary loss functions, vanishing inner products force row-space orthogonality. $\square$

Important caveats:

- For fixed specific losses, zero interference is weaker (can hold accidentally).
- If $p \gg d$ (typical for LLM-scale gates), the bound is non-binding.
- Worst-case guarantee, not typical-case behavior.

Verdict: True as an exact theorem under stated generality. Standard parameter-isolation accounting.

Classification: Direct consequence.

V7. Error Amplification Bound

Assumptions: Gated transition L -Lipschitz in state; per-depth gate error $\leq \delta$.

Claim:

$$\|s_D(x) - s_{D^*}(x)\| \leq \delta \sum_{k=0}^{D-1} L^k = \begin{cases} \delta \frac{1 - L^D}{1 - L} & \text{if } L < 1 \\ \delta D & \text{if } L = 1 \\ \delta \frac{L^D - 1}{L - 1} & \text{if } L > 1 \end{cases}$$

Precise interpretation:

- For **fixed finite depth D** : bound exists for every finite L .
- **Uniform-in-depth** bound exists iff $L < 1$.

Empirical caveat: "Trained LLM-scale maps have $L > 1$ " is an **empirical conjecture**, not a theorem.

Verdict: True as an exact theorem.

Classification: Known / direct consequence.

V8. Finite-Task Reachability (Conditional)

Assumptions: Finite task set (x_i, y_i) ; gate family has sufficient capacity; each target reachable: $y_i(x) \in \mathcal{R}_D(x) := \{\tilde{s}_D(x; \phi) : \phi \in \Phi\}$; overlapping inputs have agreeing targets.

Claim: There exists ϕ realizing all finite task targets exactly.

Counterexamples showing underdetermination:

- $F(s) = 0$: $\mathcal{R}_D(x) = \{0\}$ for every x , regardless of gating.
- Same input, different targets: V3 kills it.
- $F(s) = s$, $D = 2$, multiplicative output gating: reachable set is $\{g(x) \cdot g(x) \cdot x\}$; sign/root obstructions limit reachability.

Verdict: Conditionally true; otherwise underdetermined. The reachability assumption is the nontrivial crux.

Classification: Plausible adaptation; reachability is a **conjecture** unless proved for specific F .

3. Summary of Verdicts

#	Claim	Final Status	
---	-----	-----	
V1	Global preservation forces identity	TRUE — exact theorem; near-tautological	
V2	Asymmetric preservation constraint	TRUE — exact theorem; CL-relevant	
V3	Overlapping inputs with different targets	TRUE — exact theorem; identifiability	
V4	Depth-agnostic gate-value limitation	Representational TRUE; behavioral FALSE/underdetermined	

| V5 | Projection rank condition | TRUE — exact linear algebra; capacity corollary withdrawn |

| V6 | Gradient interference parameter budget | TRUE — exact under stated generality |

| V7 | Error amplification bound | TRUE — exact; finite- D vs uniform- D corrected |

| V8 | Finite-task reachability | CONDITIONAL; reachability is conjecture |

4. Prior Art Analysis

| Prior Art | Same Mathematical Problem? | Classification |

| --- | --- | --- |

| **Orthogonal gradient methods** (GEM, A-GEM, GPM) | Partly — addresses gradient interference in parameter space; not forward depth-aliasing; usually modifies main network | Known, partially relevant |

| **Parameter isolation / masking** (PackNet, Supermasks) | No — per-task masks/parameters; constraint-violating | Known, related but different |

| **Progressive networks / PathNet** | No — per-task architectural growth | Known, different problem |

| **Hypernetworks / fast weight programmers** | Partly — can generate input-dependent modulations; typically per-layer outputs | Known, partially relevant |

| **FiLM** (feature-wise linear modulation) | Partly — same gating algebra, but usually per-layer parameters | Known, directly relevant |

| **LSTM / GRU** | **Closest** — shared-parameter gates at every time step; crucially state-dependent, not input-only | Known, nearest prior art |

| **Mixture of Experts** | No — per-layer routers, distinct experts | Known, structurally similar |

| **Reservoir computing / echo state networks** | Partly — frozen recurrent core + trained readout | Known, relevant background |

| **Universal Transformer / ALBERT** | Partly — weight-tied depth, no task isolation | Known, background context |

| **Nonlinear RNN reachability / observability** | Yes for the reachability question | Known, should be used for V8 |

No surveyed prior art solves the exact problem: isolation within one frozen weight-tied composition via an input-only gate without per-depth trainable parameters.

5. Obstruction Characterization

There is no single obstruction theorem in the CL-relevant setting. The failure surface is a **conjunction of four necessary conditions**, each independently capable of killing the program, none individually fatal:

- **Identifiability failure (V3):** Conflicting targets on overlapping inputs are unservable by any input-only mechanism. In language-modeling CL, supports overlap extensively. Escape requires context-sufficiency or task conditioning.

- **Preservation-geometry confinement** (V1/V2): The gate is pinned to identity on every (old input, active coordinate) pair. Freedom survives only on never-active coordinates (substantial under sparse activations) and on new-task inputs.
- **Gradient coupling of shared ϕ** (V6): New-task updates generically move g_ϕ off identity on old inputs. Worst-case immunity costs a rank budget that is either restrictive ($p \leq d$) or vacuous ($p \gg d$).
- **Depth amplification** (V7): Under expansive gated dynamics, per-depth modulation error grows geometrically in D . Bounding demands architectural contractivity — an assumption on F , not a capability of the gate.

Plus three **scope conditionals**: (a) statelessness of the mechanism; (b) the t -conditioning interpretation; (c) BDH's native Hebbian plasticity actually disabled.

6. The Pivotal Scope Decision: t -Conditioning

A shared-parameter gate $g_\phi(x, t)$ where t is a **fixed, non-trainable encoding** (e.g., sinusoidal) adds $O(1)$ parameters and **does not violate "no per-depth/per-level trainable parameters" as literally worded**. This dissolves V1's depth-agnostic restrictions and V4's residue entirely, converting the problem into pure reachability (V8).

A **learned depth embedding** adds $O(D)$ parameters and violates the constraint outright.

Every impossibility-flavored statement applies only on the strict side of this fork. **The team must rule before any experiment is interpreted.**

7. Constructive Directions (all $\leq$ conjecture)

- **C1 — Orthogonal gradient on ϕ -space:** Project new-task gradients onto the orthogonal complement of old-task gradient subspace in ϕ -space, holding $g_\phi|_{S_{\text{old}}} \approx \mathbf{1}$ to first order. Classification: **plausible adaptation** of GEM/GPM.
- **C2 — Fixed-encoding t -conditioning** ($O(1)$ parameters): dissolves depth-agnostic restrictions; legality contingent on scope ruling.
- **C3 — Delta-gate placement** for residual BDH: gate $g(x) \odot \Delta(s_t)$ inside $s_{t+1} = s_t + \odot$, never the residual stream.
- **C4 — Sparse-coordinate freedom:** exploit never-active coordinates of old-task trajectories (plausible under ReLU sparsity) as legal modification space.

8. Novelty Ledger (Final, Adjudicated)

| Claim | Classification |

|-----|-----|

| Problem formulation: isolation inside frozen weight-tied composition via input-only gate, CL-framed, two-channel interference taxonomy |

Apparently novel (packaging only) |

| V1, V2, V3, V5, V6, V7 | **Direct consequences** (textbook ingredients) |

| V4 representational lemma | Direct consequence; behavioral version **false as stated** |

| Necessity of state/depth information for effective gating | **Plausible adaptation** (consistent with LSTM/FiLM practice) |

| V8 reachability for concrete BDH $\$F\$$ | **Conjecture** |

| Expansiveness of trained BDH blocks | **Empirical conjecture** |

Contribution posture: a new formalization and a clean map of elementary obstructions — not new mathematics.

9. Direct Answer to the Research Question

> Can input-path interference arising from shared sequential composition be eliminated, bounded, or made controllable by an input-dependent gate/projection mechanism, without introducing per-depth/per-level trainable parameters and without retraining the frozen shared path?

| Reading | Verdict | Epistemic Status |

|-----|-----|-----|

| **Elimination**, global-preservation | Trivially impossible — only admissible gate is identity | Proved; degenerate |

| **Elimination**, asymmetric CL | **Open** — necessary conditions established, no general impossibility or possibility theorem | Partially characterized |

| **Bounding** | Finite-depth bounds exist for any finite L ; uniform-in-depth iff $L < 1$ | Proved, conditional |

| **Controllability** | **Not yet well-posed** — requires formal reachability definition and BDH instantiation | Undefined pending instantiation |

The failure condition is met only in the degenerate reading. In the CL-relevant reading, the correct outcome is: *not proven impossible, not proven possible; reduced to identifiable necessary conditions and one decisive open conjecture (reachability).*

10. Mandatory Follow-Ups

- **Instantiate F** from paper . tex: fix the literal update (residual form, Hebbian/multiplicative terms, normalization), choose gate placement (C3), verify native plasticity is disabled.
- **Rule on the t -conditioning fork** in writing.
- **Define controllability formally** as reachability of finite task targets by the gate class on the instantiated F .
- **Retrieve internal reports**; map the implemented mechanism onto exactly one formal variant; reconcile cl-h1-report outcomes strictly as tests of V7/V8 conjectures.
- **Measurement program**: empirical Lipschitz/expansiveness of the gated BDH block (tests V7's conjecture); empirical cross-task Jacobian row-space overlap in ϕ -space (tests whether V6's worst case binds).

Tokens: 5163 | Latenz: 85319ms

Critic

Critical review of the BDH input-path isolation analysis

Overall assessment

The multi-round analysis is substantially better in its final form than its first draft. The main correction — that the headline impossibility theorem only holds in a degenerate “preserve everything on every input” reading, and that the continual-learning-relevant asymmetric setting remains open — is correct and important. The separation of three different

meanings of “interference” is useful, and the final verdict table is epistemically honest.

However, the analysis still has several serious gaps. The largest is that it never instantiates the abstraction with the actual BDH equations from paper .tex, despite the paper reportedly being available in RAG. Consequently, many of the formal claims are placement-specific, and the most important placement for BDH — gating a residual delta rather than a full block output — is not treated in the proofs.

1. Strengths worth preserving

Interference taxonomy.

The tripartition is the most useful contribution:

- cross-input forward-pass contamination,
- forward depth-aliasing,
- training-time cross-task gradient interference.

For a frozen, stateless, input-conditional gate, cross-input forward contamination is indeed structurally zero. The caveat that stateful components such as running-statistics normalization or caches can revive it is well taken.

Rejection of the naive diagonalization impossibility.

The original claim that a depth-agnostic gate is confined to the diagonal of the modulation product space and therefore cannot express depth-varying behavior was correctly downgraded. A constant-in-depth gate can produce depth-dependent effective behavior because the state changes across depth. The explicit counterexample with $(F(s)=s)$ is convincing.

Withdrawal of the projection capacity corollary.

$(\text{rank } P(x) \geq m(x))$ is correct as a necessary condition for exact trajectory preservation under output projection. The later step to $(\sum_i m_i \leq d)$ was a category error: forward-pass subspace geometry does not automatically translate into gradient-parameter orthogonality. Withdrawing it was correct.

Correct scoping of the main claim.

The final statement that exact elimination is only proved impossible under the degenerate global-preservation reading, while the asymmetric CL reading remains open, is the right conclusion. The four-part obstruction characterization is a useful map of failure modes, even though it is not a single impossibility theorem.

2. Major gaps and weaknesses

2.1 The analysis is not grounded in the actual BDH equations

The entire theory is built for

\[

$$s_{t+1}=F(s_t)$$

\]

with a few still-abstract variants. But the analysis did not retrieve and parse `paper.tex`. It repeatedly says the literal BDH update was not instantiated. This is not a minor omission; it changes the applicable theorem statements.

Public descriptions suggest BDH-GPU has a ReLU-lowrank residual structure, something like:

\[

$$x \mapsto \operatorname{ReLU}\bigl(x + \text{low-rank FFN}(x)\bigr)$$

\]

with shared parameters across depth, LayerNorm, and possibly Hebbian-style multiplicative terms.

This matters for three reasons:

- **Gate placement.** If the real BDH block is residual, then the natural gate is not

\[

$$s_{t+1}=g(x)\odot F(s_t)$$

\]

but rather

\[

$$s_{t+1} = s_t + g(x) \odot F(s_t).$$

\]

Under this delta gate, exact preservation of the unperturbed trajectory does **not** force $g_i(x) = 1$. It forces

\[

$$g_i(x) = 0$$

\]

on every coordinate where $F_i(s_t(x)) \neq 0$.

The “identity-forcing” theorem therefore does not transfer directly. The gate is pinned to zero on active coordinates, and only zero-coordinate directions are free. This is analogous but not identical, and it should be restated.

- **LayerNorm and running statistics.** If the frozen path contains normalization layers that depend on batch or sequence statistics, then the mechanism is not stateless. A gate depending only on the original input is no longer purely cross-input-independent in the forward pass. The statelessness premise needs explicit verification.

- **Native Hebbian or plastic components.** The analysis acknowledges that BDH’s native plasticity must be disabled for the frozen-path premise. But if the genuine BDH path contains Hebbian-style outer-product updates, then “freezing” it is not just a hyperparameter choice; it changes the architecture. Until this is checked, theorems about a static F may not apply.

2.2 The internal reports were not reconciled

The prompt states that the plan and H1 report are available in RAG. The analysis did not retrieve them. As a result:

- The implemented mechanism cannot be mapped to one of the formal variants.
- The empirical outcomes in `2026-08-25_cl-h1-report.md` are neither explained nor refuted.
- It is unknown whether the implementation gates the full block, the delta, an intermediate feature, or something else.

This is a major evidence-base gap. The final answer explicitly recommends reconciling the reports, but the review should treat this as a defect of the current analysis, not an optional next step.

2.3 “Controllability” is still not adequately formalized

The final report defines controllability as reachability of finite task targets by the gate class. That is a reasonable start, but it is not a full formalization.

For an input-dependent gate $\backslash(g(x)\backslash)$, the control set is severely restricted. At each depth the same gate output is applied, so the state trajectory is

$\backslash[$

$$s_{\{t+1\}} = \backslash\Phi(g(x), s_t),$$

$\setminus$

where $\setminus(g(x))$ is fixed for the entire pass. This is not the usual discrete-time control problem with independent controls $\setminus(u_t)$ at each step. The reachable set for input $\setminus(x)$ is parametrized by a single vector $\setminus(g(x))$, not by $\setminus(D)$ independent controls.

A proper definition should specify:

- the state space,
- the admissible gate value space,
- the transition rule $\setminus(\Phi)$,
- the class of target maps $\setminus(y_i(x))$,
- and whether the same gate parameters $\setminus(\phi)$ must simultaneously realize all tasks.

Without this, claims that “controllability dissolves into reachability” are only a promissory note.

2.4 The $\setminus(t)$ -conditioning fork remains unresolved

The analysis correctly identifies this as pivotal:

- A fixed, non-learned depth encoding $\setminus(t)$ as an input to a shared-parameter gate adds $\setminus(O(1))$ trainable parameters and likely does not violate the constraint literally.
- A learned depth embedding adds $\setminus(O(D))$ trainable parameters and does violate the constraint.

Until the project rules on this, the strong statements remain conditional. But the current write-up does not commit to either side, and some earlier statements still treat the strict depth-agnostic reading as the default. For clarity, every theorem should be prefixed with:

> “Assuming the strict depth-agnostic reading...”

or

> “Assuming fixed non-learned $\mathcal{T}$ -conditioning is allowed...”

2.5 The parameter-budget bound is generic and likely non-binding

The gradient-interference bound,

$$\sum_i \dim R_i \leq p,$$

is correct under the very strong condition of zero first-order interference for all smooth losses. But it is essentially the standard parameter-isolation accounting for any shared parameter mechanism. It is not specific to shared sequential composition, and it gives no practical bound until the actual Jacobian ranks are measured.

For a gate network attached to an LLM-scale BDH path, $\|p\|$ may be much larger than $\|d\|$, in which case the bound is vacuous for practical purposes. Conversely, the fixed-loss inner product can vanish accidentally even when row spaces overlap. The analysis acknowledges this, but the final obstruction list should not overstate V6 as a decisive barrier.

2.6 The overlapping-input impossibility may need more nuance

V3 says a deterministic input-only mechanism cannot satisfy two tasks that assign different targets to the same input. This is mathematically true, but its practical force depends on the task formalization.

In language-modeling-style continual learning, inputs are token sequences. If two tasks have different target distributions for the same prefix, the task identity is not contained in the input, and any input-only mechanism fails. But if the input includes task-relevant context, the overlap may be measure-zero. The analysis should distinguish:

- exact input collision,
- distribution-level overlap,
- near-overlap in representation space.

Without that, “supports overlap extensively” is an empirical claim, not a theorem.

3. Specific technical corrections

3.1 Restate V1/V2 for delta gating

For a residual delta gate,

$$\begin{aligned} & \backslash \\ s_{t+1} &= s_t + g(x) \odot F(s_t), \\ & \backslash \end{aligned}$$

exact preservation of the unperturbed trajectory on (S_{old}) gives

$$\begin{aligned} & \backslash \\ g_i(x)F_i(s_t(x)) &= 0. \\ & \backslash \end{aligned}$$

Thus:

- if coordinate i is active at some depth t , then $g_i(x)=0$;
- if coordinate i is never active along the trajectory, $g_i(x)$ is unconstrained.

This is the correct “zero-forcing” obstruction, not the “identity-forcing” obstruction stated for full output gating.

3.2 Input projection versus output projection

The projection rank theorem is proved for output projection,

$\{$

$$s_{t+1} = P(x)F(s_t).$$

$\}$

For input projection,

$\{$

$$s_{t+1} = F(P(x)s_t),$$

$\}$

exact preservation requires

$\{$

$$F(P(x)s_t) = F(s_t).$$

$\}$

Only if $\{F\}$ is injective along the trajectory does this imply $\{P(x)s_t=s_t\}$, hence $\{s_t \in \operatorname{Im} P(x)\}$. The theorem only holds under that injectivity condition and must not be stated as if the two placements are interchangeable.

3.3 The error-amplification “iff” is now correct

The final version of V7 correctly distinguishes finite-depth bounds from uniform-in-depth bounds:

- for fixed finite $\backslash(D\backslash)$, a finite $\backslash(L\backslash)$ always yields a finite bound;
- uniform-in-depth bounding requires $\backslash(L<1\backslash)$.

This correction should be retained, and any phrase like “bounding is possible exactly when $\backslash(L<1\backslash)$ ” should be avoided in future drafts.

4. Assessment of the final answer

The final answer is appropriately scoped:

- **Global-preservation elimination:** trivially impossible.
- **Asymmetric CL elimination:** open.
- **Bounding:** conditional on contractivity or finite depth.
- **Controllability:** undefined pending formalization.

This is intellectually honest and avoids the original overclaim.

However, the direct answer to the research question is still weak in the following sense. The question asks whether interference can be eliminated, bounded, or made controllable *by an input-dependent gate/projection mechanism*. The analysis does not deliver a theorem about the actual BDH mechanism. It delivers necessary conditions for an abstract family of mechanisms. The most decisive remaining item — reachability of the frozen BDH map under the allowed gate class — is unproved.

So the correct bottom line is:

> The strict proposed mechanism is not yet shown possible or impossible. The analysis reduces the question to a set of necessary conditions and one open reachability conjecture, but those conditions are not instantiated for BDH.

5. Prior-art assessment

The triage is mostly sound. LSTM/GRU is the closest neighbor and differs by using state-dependent gates. FiLM and fast weight programmers are closer than the early rounds admitted. Reservoir computing and nonlinear RNN reachability theory are relevant for the controllability question.

One missing item in some versions was the distinction between **learned per-step controls** and **input-only open-loop modulation**. This is the central difference from most control-theoretic prior art. The final analysis should make this even more explicit.

6. Recommendations before any write-up

- **Retrieve and parse paper .tex.** Extract the actual BDH block update, including residual form, normalization, and any Hebbian terms.
- **Reinstantiate every theorem** for the correct gate placement. The likely relevant case is delta gating $\{s_{t+1}=s_t+g(x)\odot F(s_t)\}$, not full output gating.
- **Rule on $\{t\}$ -conditioning.** Write down whether fixed non-learned depth encoding is permitted.
- **Retrieve the internal reports.** Map `plan_v2` and `cl-h1-report` to the formal variants.
- **Define controllability rigorously** as a reachability problem over the frozen BDH map and the exact gate class.
- **Measure** the Lipschitz constant of the gated BDH block and the cross-task Jacobian row-space overlap in $\{\phi\}$ -space. These are the only claimed barriers that can be tested empirically.
- **Do not claim a general impossibility** in the asymmetric CL setting. The current evidence supports only necessary conditions and an open conjecture.

Bottom line

The final debate record is a materially improved and cautiously scoped analysis. Its best contributions are the interference taxonomy, the

correction of the diagonalization and projection overclaims, and the explicit acknowledgment that the meaningful continual-learning question is open. Its main defects are lack of BDH grounding, placement-specific theorems stated too generally, an unresolved $\backslash(t\backslash)$ -conditioning fork, and an only partially formalized controllability notion. The honest published conclusion should be: **strict global preservation makes nontrivial gating impossible; asymmetric continual learning with a frozen shared path remains open pending concrete reachability analysis of the actual BDH update.**

Tokens: 6128 | Latenz: 111386ms

Optimizer

Input-Path Interference in a Frozen Shared Sequential Composition: A Refined Formal Analysis for BDH

Status: Final, revised after critical review

Date: 2026-08-25

0. Scope and Evidence-Base Notice

BDH (arXiv:2509.26507) is a scale-free, biologically inspired network of locally interacting neuron particles with recurrent-depth parameter sharing. The internal project files listed in the task were not retrievable during this analysis, and the literal BDH update equations from

paper.tex were not available for instantiation. **This document therefore proves results for the following abstraction, which both the research question and the project reports assume:**

$$\begin{aligned} & \backslash[\\ & s_0(x)=x, \quad s_{t+1}(x)=F(s_t(x)), \quad t=0, \dots, D-1, \\ & \backslash] \end{aligned}$$

with a residual variant $(s_{t+1}(x)=s_t(x)+F(s_t(x)))$ treated where explicitly stated. All theorem-shaped claims are about this abstraction. They become theorems about BDH **only after** (F) is instantiated with BDH's concrete update and the gate placement is fixed. This is stated as a mandatory follow-up, not as an optional refinement.

No experimental result from the implementation reports is used as evidence for any theorem.

1. Formal Definitions

1.1 Shared Sequential Composition

Let $(V=\mathbb{R}^d)$ be the state space and $(F:V\rightarrow V)$ a fixed map. A **depth- (D) shared sequential composition** is

\[

$$s_0(x)=x,\quad s_{t+1}(x)=F(s_t(x)).$$

\]

The unperturbed trajectory is $(\tau_F(x)=(s_0(x),\dots,s_D(x)))$. If the architecture uses residual connections, the update is instead

\[

$$s_{t+1}(x)=s_t(x)+F(s_t(x)).$$

\]

1.2 Frozen Shared Path

(F) is **frozen** if its parameters are not updated during the continual-learning phase. Only the gate/projection parameters (ϕ) are trainable:

\[

$$\frac{\partial F}{\partial \theta}=0.$$

\]

This is the standard continual-learning constraint; no theorem below relies on retraining (F) .

1.3 Gating and Projection

Input-dependent gate: a map $(g: X \rightarrow \mathbb{R}^d)$ applied identically at every depth. Canonical placements:

- Output gating: $(\Phi(g, s) = g(x) \odot F(s))$
- Input gating: $(\Phi(g, s) = F(g(x) \odot s))$
- Delta gating (for residual blocks): $(\Phi(g, s) = s + g(x) \odot F(s))$

The gate has **no per-depth trainable parameter vectors**: the same function (g_ϕ) is used at every depth. We separately consider the relaxed variant $(g_\phi(x, t))$, where the depth index is supplied as an input, using either a **fixed non-trainable encoding** (adds $(O(1))$ parameters) or a **learned depth embedding** (adds $(O(D))$ parameters and violates the stated constraint).

Input-dependent projection: a map $(P: X \rightarrow \Pi(V))$ applied as

[

$$s_{t+1} = P(x)F(s_t)$$

$\quad \text{or} \quad$

$$s_{t+1} = F(P(x)s_t).$$

]

1.4 Input-Path Interference: Three Distinct Notions

The word "interference" must be disambiguated.

(A) Cross-input forward-pass contamination.

If $\mathcal{F}$ is frozen and the gate is stateless and a function of x alone, then the trajectory of x_1 is computationally independent of the presence of x_2 :

$$\mathbb{E}[\tilde{s}_t^{(M)}(x_1 \mid x_2 \text{ elsewhere})] = \mathbb{E}[\tilde{s}_t^{(M)}(x_1)].$$

This notion is **structurally zero** in this setting. It becomes live again only if the mechanism contains stateful components (running statistics, caches, etc.).

(B) Forward depth-aliasing.

A depth-agnostic gate applies the same modulation at every depth. If an oracle with access to depth t would apply a reference modulation $g^*(x, t)$, the depth-aliasing error is

$$\mathbb{E}[\mathrm{DA}(g; x) - \sum_{t=0}^{D-1} \left(\Phi(g(x), s_t(x)) - \Phi(g^*(x, t), s_t(x)) \right)].$$

This is a representational mismatch within one forward pass.

(C) Training-time gradient interference.

The gate parameters ϕ are shared across all inputs. Updating ϕ on task B may change gate outputs on task- A inputs. First-order damage is proportional to

$$\left| \nabla_{\phi} L_A(\phi), \nabla_{\phi} L_B(\phi) \right|$$

The operationally relevant channels for continual learning are **(B)** and **(C)**. Channel **(A)** is dead under a frozen, stateless, input-only mechanism.

1.5 Computation-Isolability

A mechanism is **exactly computation-isolating** for a task family $\{(S_i, y_i)\}$ if:

- task outputs are produced exactly: $\tilde{s}_D(x; \phi) = y_i(x)$ for all $x \in S_i$;
- forward depth-aliasing error is zero on the specified reference trajectories;
- first-order cross-task gradient interference is zero:

$\langle \nabla_{\phi} L_i, \nabla_{\phi} L_j \rangle = 0$ for all $i \neq j$).

It **bounds interference** if all relevant errors are at most ϵ , with ϵ controlled by explicit quantities.

1.6 Elimination, Bounding, Controllability

- **Elimination:** exact task outputs, exact zero interference.
- **Bounding:** finite worst-case error, typically depending on Lipschitz constants, depth, gate error, capacity.
- **Controllability:** formalized as **reachability**:

> A target function $y_i: S_i \rightarrow \mathbb{V}$ is reachable by the gate class if there exists ϕ such that $\tilde{s}_D(x; \phi) = y_i(x)$ for all $x \in S_i$.

> The mechanism is **controllable** over a finite task sequence if every finite set of reachable tasks with compatible overlaps is simultaneously realizable by a single ϕ .

This is the natural control-theoretic reading: because the gate is input-only, the control $g(x)$ is fixed for the entire depth trajectory, rather than an independently chosen per-step control u_t . Any statement about controllability must be understood through this definition, not through unrestricted discrete-time controllability.

2. Theorem-Shaped Hypotheses

T1. Exact Preservation Forces the Gate to Be Inert on Active Coordinates

Assumptions:

- (F) is defined on an open set $(U \subseteq \mathbb{R}^d)$.
- The mechanism is a depth-agnostic output gate:

$$(s_{t+1}) = g(x) \odot F(s_t).$$

- Exact preservation of the unperturbed trajectory is required on an input set (S) :

$$(\tilde{s}_t(x) = F^t(x)) \text{ for all } (x \in S), \text{ all } (t).$$

Claim. For every $(x \in S)$ and every coordinate (i) ,

$$\begin{aligned} & \left[\right. \\ & (g_i(x) - 1) F_i(s_t(x)) = 0. \\ & \left. \right] \end{aligned}$$

If coordinate (i) is active at any depth along the trajectory, then $(g_i(x) = 1)$. If (F) is componentwise nonzero along all trajectories in (S) , then $(g(x) = \mathbf{1})$ on (S) .

Proof. At each depth, exact preservation gives $(g(x) \odot F(s_t(x)) = F(s_t(x)))$, which is componentwise $((g_i(x) - 1) F_i(s_t(x)) = 0)$. ■

Delta-gate variant. For residual delta gating,

$\backslash[$

$$s_{t+1} = s_t + g(x) \odot F(s_t),$$

$\backslash]$

exact preservation forces

$\backslash[$

$$g_i(x) \backslash, F_i(s_t(x)) = 0.$$

$\backslash]$

Thus the gate is **zero-forced** on active coordinates, not identity-forced. This is the placement likely relevant to BDH's residual structure.

Verdict. True as an exact theorem; near-tautological under global preservation. Its real content is *locative*: it identifies exactly where gate freedom survives.

Classification. Direct consequence.

T2. Asymmetric Preservation: The Continual-Learning-Relevant Setting

Assumptions:

- Old-task inputs $\mathcal{S}_{\text{old}}$, new-task inputs $\mathcal{S}_{\text{new}}$.
- Exact preservation is required only on $\mathcal{S}_{\text{old}}$.
- No preservation constraint is imposed on $\mathcal{S}_{\text{new}}$.

Claim. On $\mathcal{S}_{\text{old}}$, the gate is pinned to identity (output gating) or zero (delta gating) on every coordinate active along the old trajectory. On never-active coordinates, and on all of $\mathcal{S}_{\text{new}}$, the gate is unconstrained by preservation.

Verdict. True as an exact theorem. This is the correct formalization of the continual-learning constraint. The identity/zero obstruction does **not** by itself rule out nontrivial isolation when old and new task inputs are separated and the gate can interpolate.

Classification. Direct consequence; the asymmetric framing is the honest CL formalization.

T3. Identifiability: Same Input, Different Target Is Impossible

Assumptions:

- The mechanism is a deterministic function of the input: $s_D(x; \phi) = \mathcal{M}_\phi(x)$.
- Two tasks share an input $(x \in S_A \cap S_B)$ and demand different outputs: $y_A(x) \neq y_B(x)$.

Claim. No single ϕ can satisfy both tasks.

Proof. A function returns one output for one input. Two different required outputs are incompatible. ■

Verdict. True as an exact theorem. It is the sharpest structural limit of an input-only mechanism, independent of depth, capacity, or nonlinearity.

Practical nuance. In language continual learning, the force of this theorem depends on whether task-relevant context is part of the input. If task identity is encoded in the context, exact input collisions may be rare; if not, they can be pervasive. The theorem is about exact collision, not distribution-level overlap.

Classification. Direct consequence; essentially an identifiability condition.

T4. Depth-Agnostic Gate Values: Representational Limit, Not Behavioral Impossibility

Assumptions:

- $\Phi(g,s)$ is injective in the gate argument g along the trajectory.
- An oracle demands a depth-varying gate-value sequence $(g^*(x,0), \dots, g^*(x,D-1))$, nonconstant in t .

Claim. A depth-agnostic gate $g(x)$ cannot reproduce the exact gate-value sequence $(g^*(x,t))$.

Proof. If it did, injectivity would force $g(x)=g^*(x,t)$ for every t , contradicting nonconstancy. ■

Important correction. This is a statement about **gate values**, not about **reachable input-output behavior**. Because the state changes across depth, a constant gate can produce depth-varying effective behavior.

Counterexample. Let $(F(s)=s)$, $(D=2)$. Output gating gives

$$s_2 = g(x) \odot g(x) \odot x.$$

A constant gate (α, β) produces $(\alpha^2 x_1, \beta^2 x_2)$. A depth-varying sequence $(g^*_0=(c, 1), g^*_1=(1, c))$ produces $((cx_1, cx_2))$, which the constant gate $(\sqrt{c}, \sqrt{c})$ reproduces. Depth-varying

controls can therefore be emulated by depth-agnostic gates via state feedback in some cases.

Verdict. The representational lemma is true. The behavioral impossibility is **not established**; it was withdrawn in the final adjudication. The reachability question is open.

Classification. Direct consequence for the representational part; the broad behavioral claim is **false as stated**.

T5. Projection Mechanisms: Necessary Rank Condition

Assumptions:

- Output projection: $s_{t+1}=P(x)F(s_t)$.
- Exact preservation of the unperturbed trajectory is required for input x .

Claim. Let

$$m(x)=\dim\operatorname{span}\{F(s_0(x)),\dots,F(s_{D-1}(x))\}.$$

Then $\text{rank}(P(x)) \geq m(x)$.

Proof. Exact preservation forces $(P(x)F(s_t(x))=F(s_t(x)))$ for all (t) , so all $(F(s_t(x)))$ lie in $\text{Im}(P(x))$. ■

Input-projection caveat. For input projection $(s_{t+1}=F(P(x)s_t))$, exact preservation means $(F(P(x)s_t)=F(s_t))$. This implies $(P(x)s_t=s_t)$ only if (F) is injective along the trajectory. The theorem as stated applies to output projection; the input-projection variant requires the injectivity assumption.

Withdrawn corollary. The earlier claim $(\sum_i m_i \leq d)$ for multiple tasks is **withdrawn**. Image-subspace orthogonality is a forward-pass geometric property; cross-task gradient interference is a property of the parameterization of $(P(x))$. These are different objects, and the capacity bound does not follow without a special parameterization assumption.

Classification. Direct consequence (linear algebra); no isolation capacity bound follows.

T6. Gradient Interference and the Parameter Budget

Assumptions:

- The gate/projection has (p) trainable parameters.

- For task i , the row space of the collection of output Jacobians is $(R_i \subseteq \mathbb{R}^p)$.
- Zero first-order interference is required for **all smooth losses**.

Claim. The row spaces must be mutually orthogonal, and

[

$$\sum_i \dim R_i \leq p.$$

]

Proof. For arbitrary losses, zero inner products across all linear functionals force row-space orthogonality; orthogonal subspaces of $(\mathbb{R}^p)$ have total dimension at most (p) . ■

Caveats.

- For fixed specific losses, zero interference can hold accidentally even when row spaces overlap. The orthogonality condition is a worst-case guarantee, not a typical-case description.
- If $(p \gg d)$, the bound is likely non-binding for practical LLM-scale gates.
- This is standard parameter-isolation accounting, not a result specific to shared sequential composition.

Classification. Direct consequence; not a decisive obstruction unless Jacobian ranks are measured.

T7. Error Amplification: Bounding Is Conditional on Dynamics

Assumptions:

- The gated transition is Lipschitz in state with constant (L) .
- The per-depth gate error against a reference is at most (δ) .

Claim.

$$\begin{aligned} & \mathbb{E} \left[\left| s_D(x) - s_{D^*}(x) \right| \right] \\ & \leq \delta \sum_{k=0}^{D-1} L^k \\ & = \begin{cases} \delta \frac{1-L^D}{1-L}, & L < 1, \\ \delta D, & L = 1, \\ \delta \frac{L^D-1}{L-1}, & L > 1. \end{cases} \end{aligned}$$

Precise interpretation.

- For **fixed finite depth** (D) , a finite bound exists for every finite (L) .
- A bound that is **uniform in depth** exists iff $(L < 1)$.
- The claim "trained LLM-scale maps have $(L > 1)$ " is an **empirical conjecture**, not a theorem, and must be measured for the actual BDH block.

Classification. Direct consequence; quantifier-corrected in this revision.

T8. Finite-Task Reachability: Conditional Construction, Otherwise Underdetermined

Assumptions:

- The task family is finite: $((x_i, y_i))$, $(i=1, \dots, N)$.
- The gate family can interpolate finite sets.
- Each target is reachable: $(y_i(x) \in \text{mathcal{R}}_D(x) := \{\tilde{s}_D(x; \phi) : \phi \in \Phi\})$.
- Overlapping task inputs have agreeing targets.

Claim. There exists (ϕ) realizing all finite task targets exactly.

Proof sketch. Reachability gives a finite set of constraints; interpolation capacity fits them. ■

Underdetermination examples.

- If $(F \equiv 0)$, the reachable set is $\{0\}$ for every (x) , regardless of gating.
- If the same input has different targets, T3 applies.
- If $(F(s)=s)$ and $(D=2)$ with multiplicative output gating, reachable outputs are of the form $(g(x) \odot g(x) \odot x)$, which excludes many targets.

Verdict. Conditionally true; otherwise underdetermined. The reachability assumption is the nontrivial crux and remains a **conjecture** for a concrete BDH map.

Classification. Plausible adaptation of universal interpolation; reachability is unproved.

Summary of Verdicts

#	Claim	Status	
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T1	Exact preservation forces identity/zero on active coordinates	True; exact theorem	
T2	Asymmetric preservation constrains only old-task active coordinates	True; CL-relevant	
T3	Same input, different target impossible	True; identifiability	

| T4 | Depth-agnostic gate-value representational limit |
Representational true; behavioral impossibility false/underdetermined |

| T5 | Projection rank necessary condition | True; capacity corollary
withdrawn |

| T6 | Gradient interference parameter budget | True under worst-case
generality; often non-binding |

| T7 | Error amplification bound | True; finite-depth vs uniform-depth
corrected |

| T8 | Finite-task reachability | Conditional; reachability is conjecture |

3. Prior Art: Does It Address the Same Problem?

| Prior art | What it does | Same mathematical problem? | Classification
|

| --- | --- | --- | --- |

| Gradient projection / orthogonal-gradient methods (GEM, A-GEM, GPM)
| Project gradients in parameter space during training | Partly: cross-task
gradient interference; not forward depth-aliasing on a frozen path |
Known, partially relevant |

| Parameter isolation / masks (PackNet, Supermasks, Piggyback) |
Allocate disjoint parameter subsets to tasks | No: creates per-task
parameters/masks, violating the input-only gate constraint | Known,
related but different |

| Progressive networks / PathNet | Add task-specific columns or paths |
No: per-task architectural growth | Known, different problem |

| Hypernetworks / fast weight programmers | Generate weights or modulations for a main network | Partly: can generate input-dependent modulations; typically per-layer outputs | Known, partially relevant |

| FiLM / feature-wise linear modulation | Input-dependent affine modulation of hidden features | Partly: same gating algebra, but usually per-layer parameters | Known, directly relevant |

| LSTM / GRU | Shared-parameter state-dependent gates in a recurrent path | Closest: same composition; but gates see the current state, not only the input | Known, nearest prior art |

| Mixture of experts / conditional computation | Input-dependent routing to expert parameter sets | No: per-layer routers and distinct experts | Known, structurally similar |

| Reservoir computing / echo state networks | Fixed random recurrent core plus trained readout | Partly: frozen core; not input-dependent forward gating for task isolation | Known, relevant background |

| Weight-tied transformers (Universal Transformer, ALBERT) | Reuse a block across depth | Partly: weight-tied composition; no task-isolation gating | Known, background context |

| Nonlinear RNN reachability / observability | Characterizes what a recurrent map can reach | Yes for the reachability question | Known, should be applied to T8 |

Conclusion. No surveyed prior art solves the exact problem: isolation of computations within a single frozen weight-tied composition using only an input-dependent gate, with no per-depth trainable parameters and no retraining of the shared path.

4. Novelty Classification

| Claim | Classification |

| --- | --- |

| Formalization of two live interference channels (depth-aliasing and cross-task gradient interference) in a frozen weight-tied path |

Apparently novel (packaging only) |

| T1–T3, T5–T7 | **Direct consequences** (textbook ingredients) |

| T4 representational lemma | Direct consequence; behavioral version false as stated |

| Necessity of state/depth/context information for effective gating |

Plausible adaptation (consistent with LSTM/FiLM practice) |

| T8 reachability for concrete BDH $\setminus (F)$ | **Conjecture** |

| Expansiveness of trained BDH blocks | **Empirical conjecture** |

| Any observed effect in c1-h1-report | Inadmissible as theorem evidence; admissible only as a test of the conjectures above |

Contribution posture for write-up: **a new formalization and a clean map of elementary obstructions, not new mathematics.**

5. Obstruction Characterization

The strict proposed mechanism is:

- input-dependent gate/projection,
- no per-depth/per-level trainable parameters,
- frozen shared path,
- no retraining of $\mathcal{F}$.

Under the **global-preservation reading** — the mechanism must preserve $\mathcal{F}$'s behavior on **all** inputs — the gate is forced to be inert everywhere, so exact elimination is **trivially impossible**.

Under the **asymmetric continual-learning reading** — exact preservation on old-task inputs, free behavior on new-task inputs — no general impossibility or possibility theorem has been established. What is established is a conjunction of four necessary conditions:

- **Identifiability failure (T3)**. Conflicting targets on identical inputs cannot be served by any input-only mechanism. In language modeling, the force of this depends on how much task identity is contained in the input context.
- **Preservation-geometry confinement (T1/T2)**. The gate is pinned to identity (or zero under delta gating) on every active old-task coordinate. Freedom survives only on never-active coordinates and on new-task inputs.
- **Gradient coupling of shared ϕ (T6)**. New-task updates generically move g_ϕ away from the preserved behavior on old inputs. Worst-case immunity costs a rank budget that may be non-binding when $p \gg d$.
- **Depth amplification (T7)**. If the gated dynamics are expansive, small per-depth gate errors grow geometrically in depth. Bounding then requires contractivity of the composed system — a property of $\mathcal{F}$, not a capability of the gate alone.

Three **scope conditionals** attach to every result:

- **Statelessness.** Any stateful component inside the gate revives cross-input forward contamination (channel A).
- **ϕ -conditioning fork.** A gate $g_\phi(x,t)$ with a fixed non-trainable depth encoding adds $O(1)$ parameters and does not violate the constraint as literally worded. A learned depth embedding adds $O(D)$ parameters and does violate it. The depth-agnostic obstruction dissolves on the permissive side of this fork, converting the problem into pure reachability (T8).
- **Frozen path.** BDH's native Hebbian-style plasticity must be explicitly disabled for the frozen-path premise to hold.

6. Direct Answer to the Research Question

> Can input-path interference arising from shared sequential composition be eliminated, bounded, or made controllable by an input-dependent gate/projection mechanism, without per-depth/per-level trainable parameters and without retraining the frozen shared path?

| Reading | Verdict | Epistemic status |

| --- | --- | --- |

| **Elimination**, global preservation on all inputs | Impossible — the only admissible gate is identity/zero on active coordinates | Proved; degenerate |

| **Elimination**, asymmetric CL (preserve (S_{old}) , free on (S_{new})) | **Open**. Necessary conditions established; no general impossibility or possibility theorem | Partially characterized |

| **Bounding** | Finite-depth bounds exist for any finite (L) ; uniform-in-depth bound iff $(L < 1)$ | Proved, conditional |

| **Controllability** | Not yet well-posed: requires formal reachability analysis on the concrete frozen BDH map with the exact gate class | Undefined pending instantiation |

The failure condition contemplated by the brief is met **only in the degenerate global-preservation reading**. In the continual-learning-relevant reading, the correct outcome is: *not proven impossible, not proven possible; reduced to identifiable necessary conditions and one decisive open conjecture — reachability*.

7. Mandatory Follow-Ups Before Empirical Reconciliation

- **Instantiate (F) from paper .tex**. Fix the literal BDH update, including residual form, LayerNorm placement, and any Hebbian-style multiplicative terms.
- **Reinstantiate every theorem for the correct gate placement**. For a residual block, the natural gate is $(s_{t+1} = s_t + g(x) \odot F(s_t))$, which changes the identity obstruction to a zero obstruction.
- **Rule on the (t) -conditioning fork**. State in writing whether fixed non-learned depth encoding is permitted.
- **Retrieve the internal reports**. Map 2026-08-25_bdh-continual-learning-plan_v2.md and 2026-08-25_cl-h1-report.md onto exactly one formal variant: depth-agnostic, (t) -conditioned, or state-conditioned.

- **Define controllability formally** as reachability of finite task targets by the gate class on the instantiated BDH map.
- **Measure** (a) the Lipschitz constant of the gated BDH block, and (b) cross-task Jacobian row-space overlap in ϕ -space. These are the only claimed barriers that can be tested empirically.

Bottom line. A strict input-only gate/projection on a frozen shared sequential composition does not provide a proven general solution. The obstruction is not a single theorem but a map of four necessary failure conditions and one open reachability problem. The honest conclusion for publication is: *global preservation makes nontrivial gating impossible; asymmetric continual learning with a frozen shared path remains open pending concrete reachability analysis of the actual BDH update.*

Tokens: 7792 | Latenz: 63552ms

Moderator

[moderator] Round 4: LLM call failed (litellm.RateLimitError: RateLimitError: OpenrouterException - {"error":{"message":"Provider returned error","code":429,"metadata":{"raw":"stealth/ox-alpha is temporarily rate-limited upstream. Please retry shortly.","provider_name":"Stealth","is_byok":false,"limit_source":"upstream_provider_shar shortly, add your own provider key (<https://openrouter.ai/settings/integrations>), or route to another provider with provider routing: <https://openrouter.ai/docs/features/provider-routing>"},"user_id":"user_3Ay1IY7xf9lrDwaN7Fq4EUjnQA9"})

Tokens: 150 | Latenz: 2219ms

Runde 5 — Konsens: 100.0%

Strategist

Final Strategic Assessment: Input-Path Interference in BDH via Gating/Projection

Date: 2026-08-25 | **Status:** Complete (4-round debate, adjudicated)

Executive Summary

Under the strict constraints — depth-agnostic input-dependent gate/projection, frozen shared path, no per-depth trainable parameters, no retraining — **exact elimination of input-path interference is impossible only under the degenerate global-preservation reading** (the gate must be identity everywhere, doing nothing). Under the **continual-learning-relevant asymmetric reading** (exact preservation on old tasks, free on new tasks), the problem is **open**: no general impossibility or possibility theorem has been established. What has been established is a precise map of four independent necessary conditions, each capable of killing the program, plus one decisive open conjecture (reachability) whose resolution determines feasibility.

1. Formal Definitions

1.1 Shared Sequential Composition

Let $V = \mathbb{R}^d$ be the state space. A **depth- D shared sequential composition** is defined by a fixed map $F: V \rightarrow V$ and the recurrence:

$$s_0(x) = x, \quad s_{t+1}(x) = F(s_t(x)), \quad t = 0, \dots, D-1$$

The unperturbed trajectory is $\tau_F(x) = (s_0(x), s_1(x), \dots, s_D(x))$.

If BDH uses residual connections (likely, per the BDH-GPU architecture), the relevant recurrence is:

$$s_{t+1}(x) = s_t(x) + F(s_t(x))$$

Classification: Direct characterization of BDH's weight-tied recurrent depth, corroborated by BDH-MLX ("Same weights reused in all layers (recurrent depth)") and BDH-GPU descriptions ("ReLU-Lowrank residual like $x \mapsto \dots$ "). Instantiation with the literal BDH update from paper .tex remains mandatory.

1.2 Frozen Shared Path

F is **frozen** if $\partial F / \partial \theta = 0$ during the continual-learning phase. Only the gate/projection mechanism ϕ may be trained.

Classification: Known — standard continual learning constraint.

1.3 Input-Path Interference: Three Precise Notions

(A) Cross-input forward-pass contamination. With frozen θ and a stateless input-conditioned mechanism, each input's forward pass is computationally independent. This notion is **structurally zero** and not a live channel.

(B) Forward depth-aliasing. A depth-agnostic gate $g(x)$ applies the same modulation at every depth. If an oracle would apply $g^*(x, t)$ (varying with depth), the depth-aliasing error is:

$$\mathrm{DA}(g; x) = \sum_{t=0}^{D-1} \left| \Phi(g(x), s_t(x)) - \Phi(g^*(x, t), s_t(x)) \right|$$

(C) Training-time gradient interference (cross-task forgetting).

Because gate parameters ϕ are shared across all inputs, updating ϕ on task B may change g_ϕ on task- A inputs. First-order damage is proportional to $\langle \nabla \phi, \nabla \phi \rangle_{\mathcal{L}_A, \mathcal{L}_B}$.

The live channels are (B) and (C). Notion (A) is dead under the stated assumptions.

Classification: The depth-aliasing notion is **apparently novel** as a packaged definition; components are known from control theory. The gradient notion is **known** from continual learning.

1.4 Gating

An **input-dependent gate** $g: X \rightarrow \mathbb{R}^d$ is applied identically at every depth:

- Output gating: $s_{t+1} = g(x) \odot F(s_t)$
- Input gating: $s_{t+1} = F(g(x) \odot s_t)$
- Delta gating (for residual blocks): $s_{t+1} = s_t + g(x) \odot F(s_t)$

The gate has **no per-depth parameters**: the same $g(x)$ is used at every t .

Classification: Known mechanism (LSTM, MoE), but the depth-independence + frozen-path combination is novel.

1.5 Projection

An **input-dependent projection** $P: X \rightarrow \Pi(V)$ maps each input to an orthogonal projection, applied as $s_{t+1} = P(x) F(s_t)$ or $s_{t+1} = F(P(x) s_t)$.

Classification: Known (gradient projection methods), formalized for forward-pass application.

1.6 Computation-Isolability

A mechanism is **exactly computation-isolating** for a task family $\{(S_i, y_i)\}_{i=1}^K$ if: (1) each task output is produced exactly; (2) zero forward depth-aliasing error; (3) zero first-order cross-task gradient interference.

Classification: Apparently novel — this specific formulation combining shared composition, frozen parameters, and forward-pass isolation does not appear in surveyed literature.

1.7 Elimination vs. Bounding vs. Controllability

- **Elimination:** Exact equality of outputs and exact zero interference.
- **Bounding:** Deviation $\leq \epsilon$, controlled by Lipschitz constant, depth, gate error, parameter count.
- **Controllability:** Reachability of finite task targets by the gate class without modifying F .

Classification: Plausible adaptation of approximation theory and control-theoretic concepts.

2. Theorem-Shaped Hypotheses (Final Adjudicated Versions)

V1. Identity-Forcing Under Global Preservation

Assumptions: F defined on open $U \subseteqq \mathbb{R}^d$; depth-independent output gating $s_{t+1} = g(x) \odot F(s_t)$; exact trajectory preservation on input set S : $\tilde{s}_t(x) = F^t(x)$ for all $x \in S$, all t .

Claim: For every $x \in S$ and every coordinate i : $g_i(x) = 1$ whenever $\exists t: F_i(s_t(x)) \neq 0$.

Proof: At depth t : $g(x) \odot F(s_t(x)) = F(s_t(x))$ implies $(g_i(x) - 1) F_i(s_t(x)) = 0$ componentwise. $\square$

Delta-gate variant: For residual delta gating $s_{t+1} = s_t + g(x) \odot F(s_t)$, exact preservation forces $g_i(x) F_i(s_t(x)) = 0$. The gate is **zero-forced** on active coordinates, not identity-forced.

Verdict: True as an exact theorem. Near-tautological under global preservation. Its real content is *locative*: it identifies precisely where gate freedom lives — never-active coordinates and non- S inputs.

Classification: Known / direct consequence.

V2. Asymmetric Preservation: The CL-Relevant Constraint

Assumptions: Old-task inputs S_{old} ; new-task inputs S_{new} ; output gating; exact preservation required only on S_{old} .

Claim: For $x \in S_{\text{old}}$: $g_i(x) = 1$ whenever $\exists t: F_i(s_t(x)) \neq 0$. On the zero-coordinate set $Z(x) = \{i: F_i(s_t(x)) = 0 \forall t\}$, the gate is unconstrained. On S_{new} , the gate is entirely unconstrained.

Proof: Same componentwise argument as V1, restricted to S_{old} . $\square$

Verdict: True as an exact theorem. If old-task trajectories have support in all coordinates, the gate is forced to identity on all of S_{old} . If old and new inputs are disjoint and the gate network can interpolate, exact isolation is *not* ruled out by this obstruction alone.

Classification: Direct consequence; the asymmetric framing is the correct CL formalization.

V3. Identifiability: Overlapping Inputs Are Fatal

Assumptions: The mechanism is a deterministic function of the input: $\tilde{s}_D(x; \phi) = \text{mathcal{M}}_\phi(x)$. Two tasks A, B share an input $x \in S_A \cap S_B$ with different targets: $y_A(x) \neq y_B(x)$.

Claim: No single ϕ can satisfy both tasks.

Proof: The map is a function: one input, one output. Two different required outputs are incompatible. $\square$

Verdict: True as an exact theorem. The sharpest structural limit of "input-dependent only," independent of depth, nonlinearity, and capacity.

Classification: Direct consequence; essentially an identifiability condition.

V4. Depth-Agnostic Gate-Value Limitation (Downgraded)

Assumptions: $\Phi(g,s)$ injective in g along the trajectory; ideal modulation $g^*(x,t)$ nonconstant in t .

Claim: No depth-independent gate $g(x)$ can reproduce the nonconstant gate-value sequence $g^*(x,t)$.

Proof: Equality at every depth gives $g(x) = g^*(x,t)$ for every t , contradicting nonconstancy. $\square$

Critical correction: This is a statement about **gate values**, not about **reachable input-output maps**. A constant-in-depth gate can produce depth-varying effective behavior because the state s_t changes.

Counterexample: $F(s) = s$, $D = 2$. Output gating: $s_2 = g(x) \odot g(x) \odot x$. Constant gate $g(x) = (\alpha, \beta)$ produces $(\alpha^2 x_1, \beta^2 x_2)$. Depth-varying sequence $g^0 = (c, 1)$, $g^1 = (1, c)$ produces (cx_1, cx_2) , also achievable by constant gate $(\sqrt{c}, \sqrt{c})$.

Verdict: Representational lemma TRUE; behavioral impossibility NOT established. The reachability question is open.

Classification: Direct consequence for the representational part; the broad behavioral impossibility is **false as stated**.

V5. Projection Rank Condition

Assumptions: Output projection $s_{t+1} = P(x) F(s_t)$; exact preservation of unperturbed trajectory required.

Claim: If $m(x) = \dim \operatorname{span}\{F(s_0(x)), \dots, F(s_{D-1}(x))\}$, then $\operatorname{rank} P(x) \geq m(x)$.

Proof: $P(x) F(s_t(x)) = F(s_t(x))$ forces $F(s_t(x)) \in \operatorname{Im} P(x)$. $\square$

Input-projection caveat: For input projection $s_{t+1} = F(P(x)s_t)$, exact preservation means $F(P(x)s_t) = F(s_t)$. This implies $P(x)s_t = s_t$ only if F is injective along the trajectory.

Withdrawn corollary: The capacity corollary $\sum_i m_i \leq d$ is **WITHDRAWN**. Orthogonality of image subspaces is a forward-pass geometric property; cross-task gradient interference is a property of the parameterization. These are different mathematical objects.

Verdict: True as an exact linear-algebra theorem.

Classification: Direct consequence.

V6. Gradient Interference Parameter Budget

Assumptions: Gate has p trainable parameters ϕ ; task Jacobians $J_i \in \mathbb{R}^{d \times p}$; zero first-order interference required for *all smooth losses*.

Claim: Row spaces $\{R_i\}$ must be mutually orthogonal: $\sum_i \dim R_i \leq p$.

Proof: For arbitrary loss functions, vanishing inner products force row-space orthogonality. $\square$

Important caveats:

- For fixed specific losses, zero interference is weaker (can hold accidentally).

- If $p \gg d$ (typical for LLM-scale gates), the bound is non-binding.
- Worst-case guarantee, not typical-case behavior.

Verdict: True as an exact theorem under stated generality. Standard parameter-isolation accounting.

Classification: Direct consequence.

V7. Error Amplification Bound

Assumptions: Gated transition L -Lipschitz in state; per-depth gate error $\leq \delta$.

Claim:

$$\|s_D(x) - s_{D^*}(x)\| \leq \delta \sum_{k=0}^{D-1} L^k = \begin{cases} \delta \frac{1 - L^D}{1 - L} & \text{if } L < 1 \\ \delta D & \text{if } L = 1 \\ \delta (L^D - 1) & \text{if } L > 1 \end{cases}$$

Precise interpretation:

- For **fixed finite depth D** : bound exists for every finite L .
- **Uniform-in-depth** bound exists iff $L < 1$.

Empirical caveat: "Trained LLM-scale maps have $L > 1$ " is an **empirical conjecture**, not a theorem.

Verdict: True as an exact theorem.

Classification: Known / direct consequence.

V8. Finite-Task Reachability (Conditional)

Assumptions: Finite task set (x_i, y_i) ; gate family has sufficient capacity; each target reachable: $y_i(x) \in \mathcal{R}_D(x) := \{\tilde{s}_D(x; \phi) : \phi \in \Phi\}$; overlapping inputs have agreeing targets.

Claim: There exists ϕ realizing all finite task targets exactly.

Counterexamples showing underdetermination:

- $F(s) = 0$: $\mathcal{R}_D(x) = \{0\}$ for every x , regardless of gating.
- Same input, different targets: V3 kills it.
- $F(s) = s$, $D = 2$, multiplicative output gating: reachable set is $\{g(x) \odot g(x) \odot x\}$; sign/root obstructions limit reachability.

Verdict: Conditionally true; otherwise underdetermined. The reachability assumption is the nontrivial crux.

Classification: Plausible adaptation; reachability is a **conjecture** unless proved for specific F .

3. Summary of Verdicts

#	Claim	Final Status
---	-----	-----
V1	Global preservation forces identity	TRUE — exact theorem; near-tautological
V2	Asymmetric preservation constraint	TRUE — exact theorem; CL-relevant
V3	Overlapping inputs with different targets	TRUE — exact theorem; identifiability
V4	Depth-agnostic gate-value limitation	Representational TRUE; behavioral FALSE/underdetermined
V5	Projection rank condition	TRUE — exact linear algebra; capacity corollary withdrawn
V6	Gradient interference parameter budget	TRUE — exact under stated generality
V7	Error amplification bound	TRUE — exact; finite- D vs uniform- D corrected
V8	Finite-task reachability	CONDITIONAL; reachability is conjecture

4. Prior Art Analysis

| Prior Art | Same Mathematical Problem? | Classification |

| --- | --- | --- |

| **Orthogonal gradient methods** (GEM, A-GEM, GPM) | Partly — addresses gradient interference in parameter space; not forward depth-aliasing; usually modifies main network | Known, partially relevant |

| **Parameter isolation / masking** (PackNet, Supermasks) | No — per-task masks/parameters; constraint-violating | Known, related but different |

| **Progressive networks / PathNet** | No — per-task architectural growth | Known, different problem |

| **Hypernetworks / fast weight programmers** | Partly — can generate input-dependent modulations; typically per-layer outputs | Known, partially relevant |

| **FiLM** (feature-wise linear modulation) | Partly — same gating algebra, but usually per-layer parameters | Known, directly relevant |

| **LSTM / GRU** | **Closest** — shared-parameter gates at every time step; crucially state-dependent, not input-only | Known, nearest prior art |

| **Mixture of Experts** | No — per-layer routers, distinct experts | Known, structurally similar |

| **Reservoir computing / echo state networks** | Partly — frozen recurrent core + trained readout | Known, relevant background |

| **Universal Transformer / ALBERT** | Partly — weight-tied depth, no task isolation | Known, background context |

| **Nonlinear RNN reachability / observability** | Yes for the reachability question | Known, should be used for V8 |

No surveyed prior art solves the exact problem: isolation within one frozen weight-tied composition via an input-only gate without per-depth trainable parameters.

5. Obstruction Characterization

There is no single obstruction theorem in the CL-relevant setting. The failure surface is a **conjunction of four necessary conditions**, each independently capable of killing the program, none individually fatal:

- **Identifiability failure** (V3): Conflicting targets on overlapping inputs are unservable by any input-only mechanism. In language-modeling CL, supports overlap extensively. Escape requires context-sufficiency or task conditioning.
- **Preservation-geometry confinement** (V1/V2): The gate is pinned to identity (output gating) or zero (delta gating) on every (old input, active coordinate) pair. Freedom survives only on never-active coordinates (substantial under sparse activations) and on new-task inputs.
- **Gradient coupling of shared ϕ** (V6): New-task updates generically move g_ϕ off identity on old inputs. Worst-case immunity costs a rank budget that is either restrictive ($p \leq d$) or vacuous ($p \gg d$).

- **Depth amplification** (V7): Under expansive gated dynamics, per-depth modulation error grows geometrically in D . Bounding demands architectural contractivity — an assumption on F , not a capability of the gate.

Plus three **scope conditionals**: (a) statelessness of the mechanism; (b) the t -conditioning interpretation; (c) BDH's native Hebbian plasticity actually disabled.

6. The Pivotal Scope Decision: t -Conditioning

A shared-parameter gate $g_\phi(x, t)$ where t is a **fixed, non-trainable encoding** (e.g., sinusoidal) adds $O(1)$ parameters and **does not violate "no per-depth/per-level trainable parameters" as literally worded**. This dissolves V1's depth-agnostic restrictions and V4's residue entirely, converting the problem into pure reachability (V8).

A **learned depth embedding** adds $O(D)$ parameters and violates the constraint outright.

Every impossibility-flavored statement applies only on the strict side of this fork. **The team must rule before any experiment is interpreted.**

7. Constructive Directions (all ≤ conjecture)

- **C1 — Orthogonal gradient on ϕ -space:** Project new-task gradients onto the orthogonal complement of old-task gradient subspace in ϕ -space, holding $g_\phi|_{S_{\text{old}}} \approx \mathbf{1}$ to first order. Classification: **plausible adaptation** of GEM/GPM.
- **C2 — Fixed-encoding t -conditioning** ($O(1)$ parameters): dissolves depth-agnostic restrictions; legality contingent on scope ruling.
- **C3 — Delta-gate placement** for residual BDH: gate $g(x) \odot \Delta(s_t)$ inside $s_{t+1} = s_t + \cdot$, never the residual stream.
- **C4 — Sparse-coordinate freedom:** exploit never-active coordinates of old-task trajectories (plausible under ReLU sparsity) as legal modification space.

8. Novelty Ledger (Final, Adjudicated)

Claim Classification
----- -----

| Problem formulation: isolation inside frozen weight-tied composition via input-only gate, CL-framed, two-channel interference taxonomy |
Apparently novel (packaging only) |

| V1, V2, V3, V5, V6, V7 | **Direct consequences** (textbook ingredients) |

| V4 representational lemma | Direct consequence; behavioral version **false as stated** |

| Necessity of state/depth information for effective gating | **Plausible adaptation** (consistent with LSTM/FiLM practice) |

| V8 reachability for concrete BDH \$\$\$ | **Conjecture** |

| Expansiveness of trained BDH blocks | **Empirical conjecture** |

Contribution posture: a new formalization and a clean map of elementary obstructions — not new mathematics.

9. Direct Answer to the Research Question

> Can input-path interference arising from shared sequential composition be eliminated, bounded, or made controllable by an input-dependent gate/projection mechanism, without introducing per-depth/per-level trainable parameters and without retraining the frozen shared path?

| Reading | Verdict | Epistemic Status |

|-----|-----|-----|

| **Elimination**, global-preservation | Trivially impossible — only admissible gate is identity/zero on active coordinates | Proved; degenerate |

| **Elimination**, asymmetric CL | **Open** — necessary conditions established, no general impossibility or possibility theorem | Partially characterized |

| **Bounding** | Finite-depth bounds exist for any finite L ; uniform-in-depth iff $L < 1$ | Proved, conditional |

| **Controllability** | **Not yet well-posed** — requires formal reachability definition and BDH instantiation | Undefined pending instantiation |

The failure condition is met only in the degenerate reading. In the CL-relevant reading, the correct outcome is: *not proven impossible, not proven possible; reduced to identifiable necessary conditions and one decisive open conjecture (reachability).*

10. Mandatory Follow-Ups

- **Instantiate FF** from paper . tex: fix the literal update (residual form, Hebbian/multiplicative terms, normalization), choose gate placement (C3), verify native plasticity is disabled.
- **Rule on the St -conditioning fork** in writing.
- **Define controllability formally** as reachability of finite task targets by the gate class on the instantiated FF .
- **Retrieve internal reports**; map the implemented mechanism onto exactly one formal variant; reconcile cl-h1-report outcomes strictly as tests of V7/V8 conjectures.

- **Measurement program:** empirical Lipschitz/expansiveness of the gated BDH block (tests V7's conjecture); empirical cross-task Jacobian row-space overlap in ϕ -space (tests whether V6's worst case binds).

Tokens: 5246 | Latenz: 66007ms

Critic

Final Critical Review: BDH Input-Path Isolation Debate

Overall Assessment

The analysis improved materially across rounds. Its main correction is the right one: the strong claim “exact elimination is impossible in general” has been dissolved into a **scoped set of exact necessary conditions plus an open reachability conjecture**. The separation of three distinct meanings of “interference” is the most useful contribution, and the final verdict table is epistemically honest.

That said, the analysis is **not yet grounded in the actual BDH mechanism**. It operates on an abstraction $s_{t+1} = F(s_t)$ or a residual variant, but does not instantiate F from paper .tex. Consequently, several theorem statements are placement-sensitive, and the most BDH-relevant placement — **delta gating of a residual block** — was only partially treated.

Strengths Worth Preserving

- **Interference taxonomy.**

The separation of

- cross-input forward-pass contamination,
- forward depth-aliasing,
- training-time cross-task gradient interference

is correct and necessary. Under a frozen, stateless, input-dependent mechanism, cross-input forward contamination is structurally zero. The live channels are depth-aliasing and cross-task parameter updating.

- **Rejection of the naive diagonalization impossibility.**

The claim that a depth-agnostic gate is confined to the diagonal of the modulation space was correctly downgraded. A constant-in-depth gate can still produce depth-varying effective behavior because the state changes. The counterexample with $\mathcal{F}(s)=s$ is decisive.

- **Withdrawal of the projection capacity corollary.**

The rank condition

$$\left[\operatorname{rank} P(x) \geq m(x) \right]$$

is correct for output projection. The stronger cross-task capacity bound $\sum_i m_i \leq d$ was correctly withdrawn. It conflated forward-pass subspace geometry with gradient-parameter geometry.

- **Correct scoping of the main claim.**

Exact elimination is proved impossible only in the degenerate global-preservation reading. In the continual-learning-relevant asymmetric setting, the question is not resolved either way.

Remaining Weaknesses and Gaps

1. The actual BDH equations were never instantiated

This is the largest remaining defect. The analysis repeatedly acknowledges that `paper.tex` was not parsed. Public descriptions suggest a BDH-GPU structure such as

$$\begin{aligned} & \backslash[\\ & x \mapsto \operatorname{ReLU}\bigl(x + \text{low-rank FFN}(x)\bigr), \\ & \backslash] \end{aligned}$$

with shared parameters, normalization, and possibly Hebbian-style multiplicative terms.

This matters because:

- **Delta gating changes the preservation obstruction.**

For full output gating $\mathbf{s}_{t+1} = \mathbf{g}(\mathbf{x}) \odot \mathbf{F}(\mathbf{s}_t)$, preservation forces $\mathbf{g}_i(\mathbf{x}) = 1$ on active coordinates.

For residual delta gating $\mathbf{s}_{t+1} = \mathbf{s}_t + \mathbf{g}(\mathbf{x}) \odot \mathbf{F}(\mathbf{s}_t)$, preservation forces

$\mathbf{g}_i(\mathbf{x})$

$$\mathbf{g}_i(\mathbf{x}) \mathbf{F}_i(\mathbf{s}_t(\mathbf{x})) = 0,$$

$\mathbf{g}_i(\mathbf{x})$

so the gate is pinned to **zero**, not identity, on active coordinates.

- **Input versus output projection.**

The projection rank theorem does not automatically extend from output projection to input projection without injectivity assumptions on $\mathbf{F}$.

- **Statefulness.**

If the frozen path contains normalization with running statistics, the mechanism is not stateless, which can revive cross-input forward contamination.

- **Hebbian/plastic components.**

If BDH includes native plastic terms, “freezing” the path is an architectural change that must be verified.

2. Internal reports were not reconciled

The plan and H1 report were not retrieved. Without them:

- the implemented gate placement is unknown,
- the empirical outcomes of `cl-h1-report.md` are neither explained nor refuted,
- the implementation cannot be mapped to a formal variant.

This is a defect of the current analysis, not merely an optional future step.

3. “Controllability” is still not fully formalized

The final definition — reachability of finite task targets by the gate class — is a reasonable start. But the control structure is unusual: an input-dependent gate $g(x)$ is fixed for the entire depth trajectory, so per-input reachable sets are parameterized by a single vector, not by (D) independent controls.

A complete formalization should specify:

- the state space,
- the admissible gate range,
- the exact transition rule,
- the target map class,
- and the requirement that one parameter vector ϕ simultaneously realize all tasks.

Until then, “controllability reduces to reachability” is only a promissory note.

4. The $\mathcal{O}(t)$ -conditioning fork remains unresolved

The pivotal scope decision is still open:

- a fixed, non-learnable depth encoding adds $\mathcal{O}(1)$ trainable parameters and likely does not violate the constraint,
- a learned depth embedding adds $\mathcal{O}(D)$ trainable parameters and violates it.

Every impossibility-flavored statement holds only on the strict depth-agnostic side. Future write-ups must label assumptions explicitly.

5. The gradient-interference bound is not specific to shared composition

The bound

$$\sum_i \dim R_i \leq p$$

is standard parameter-isolation accounting. It is correct only under the strong requirement of zero first-order interference for all smooth losses. For fixed losses, interference can vanish accidentally. More importantly, if $p \gg d$, the bound is often non-binding. It should not be presented as a decisive shared-sequential-composition obstruction until actual Jacobian ranks are measured.

6. The overlapping-input theorem needs practical nuance

Same-input-different-target impossibility is mathematically true. But its practical force depends on whether task identity is represented in the input context. Exact input collisions and distribution-level overlap should not be conflated.

Final Scoped Answer to the Research Question

> Can input-path interference arising from shared sequential composition be eliminated, bounded, or made controllable by an input-dependent gate/projection mechanism, without per-depth/per-level trainable parameters and without retraining the frozen shared path?

| Reading | Verdict | Status |

| --- | --- | --- |

| **Elimination**, global preservation on all inputs | Trivially impossible: the only admissible gate is inert on active coordinates | Proved; degenerate |

| **Elimination**, asymmetric CL: preserve (S_{old}) , free (S_{new}) | **Open**. Necessary conditions established; no general impossibility or possibility theorem | Partially characterized |

| **Bounding** | Finite-depth bounds exist for any finite Lipschitz (L) ; uniform-in-depth bounding requires $(L < 1)$ | Proved, conditional |

| **Controllability** | Not yet well-posed without reachability analysis of the concrete frozen BDH map | Undefined pending instantiation |

The failure condition in the brief is met only in the **degenerate global-preservation reading**. In the meaningful continual-learning setting, the correct conclusion is:

> Not proven impossible; not proven possible. The question reduces to identifiable necessary conditions and one decisive open conjecture — reachability of the frozen BDH map under the allowed gate class.

Novelty Ledger, Final Form

| Claim | Classification |

| --- | --- |

| Two-channel interference taxonomy in a frozen weight-tied path |
Apparently novel as packaging; ingredients known |

| Identity-forcing / zero-forcing preservation lemma | Direct
consequence |

| Asymmetric-preservation constraint | Direct consequence; CL-relevant
reframing |

| Overlapping-input identifiability impossibility | Direct consequence |

| Depth-agnostic gate-value representational limit | Direct consequence;
behavioral version false as stated |

| Projection rank necessary condition | Direct consequence |

| Gradient-interference parameter budget | Direct consequence |

- | Lipschitz error-amplification bound | Direct consequence |
- | Necessity of state/depth/context information for robust gating | Plausible adaptation |
- | Reachability of concrete BDH $\setminus(F)$ under gating | Conjecture |
- | Expansiveness of trained BDH blocks | Empirical conjecture |

No experimental result from the implementation reports was used as evidence for any theorem.

Required Next Steps Before Publication

- **Parse paper .tex** and instantiate the literal BDH update, including residual form, normalization, and Hebbian or multiplicative terms.
- **Re-prove every theorem for the actual gate placement**, especially residual delta gating.
- **Rule formally on $\setminus(t)$ -conditioning**: fixed non-learned depth encoding versus learned depth embedding.
- **Retrieve and map the internal reports** to exactly one formal variant.
- **Define controllability rigorously** as finite-target reachability over the concrete BDH map and exact gate class.
- **Measure**:
 - the Lipschitz constant of the gated BDH block;
 - cross-task Jacobian row-space overlap in $\setminus(\phi)$ -space.

The honest published conclusion remains:

> Strict global preservation makes nontrivial gating impossible. Asymmetric continual learning with a frozen shared path remains open pending concrete reachability analysis of the actual BDH update.

Tokens: 3097 | Latenz: 49353ms

Optimizer

Input-Path Interference in BDH via Input-Dependent Gating/Projection: A Scoped Formal Analysis

Date: 2026-08-25

Status: Final, after adversarial review

1. Scope, Abstraction, and Evidence Base

BDH (arXiv:2509.26507) is a scale-free, biologically inspired network of locally interacting neuron particles with **shared parameters across depth** ("recurrent depth"), corroborated by third-party implementations and public descriptions. The literal BDH update equations from paper .tex and the internal project reports were not retrievable within this analysis. Consequently, the formal results below are proved for the abstraction that the research question itself assumes:

$\setminus[$

$$s_0(x)=x, \text{ \texttt{\textbackslash}qqquad} s_{t+1}(x)=F(s_t(x)), \text{ \texttt{\textbackslash}qqquad} t=0, \text{ \texttt{\textbackslash}dots}, D-1,$$

\]

with the residual variant

 $\setminus[$

$$s_{t+1}(x) = s_t(x) + F(s_t(x))$$

\]

treated where explicitly stated. All theorem-shaped claims are about this abstraction; they become theorems about BDH only after $\backslash(F)$ is replaced by the concrete BDH update and the gate placement is fixed. This instantiation is a mandatory follow-up, not an optional refinement.

No experimental observation from any implementation report is used as evidence for any theorem.

— — —

2. Formal Definitions

2.1 Shared Sequential Composition

Let $(V = \mathbb{R}^d)$ be the state space and $(F: V \rightarrow V)$ a fixed map. A **depth- (D) shared sequential composition** is

$$\begin{aligned} &[\\ s_0(x) = x, &\quad s_{t+1}(x) = F(s_t(x)). \\ &] \end{aligned}$$

The unperturbed trajectory is $(\tau_F(x) = (s_0(x), \dots, s_D(x)))$.

2.2 Frozen Shared Path

(F) is **frozen** if its parameters are not updated during the continual-learning phase:

$$\begin{aligned} &[\\ \frac{\partial F}{\partial \theta} &= 0. \\ &] \end{aligned}$$

Only the gate/projection parameters (ϕ) are trainable.

2.3 Gating

An **input-dependent gate** is a map $(g: X \rightarrow \mathbb{R}^d)$ applied identically at every depth. Canonical placements:

- **Output gating:** $\Phi(g,s)=g(x)\odot F(s)$
- **Input gating:** $\Phi(g,s)=F(g(x)\odot s)$
- **Delta gating** (for residual blocks): $\Phi(g,s)=s+g(x)\odot F(s)$

The gate has **no per-depth trainable parameter vectors**: the same function $\Phi(g,s)$ is used at every depth. A relaxed variant $\Phi(g,s,t)$, where the depth index is supplied as an input, is considered separately:

- **Fixed non-trainable depth encoding:** adds $O(1)$ trainable parameters; does not violate the constraint as literally worded.
- **Learned depth embedding:** adds $O(D)$ trainable parameters; violates the constraint.

2.4 Projection

An **input-dependent projection** is a map $P:X\rightarrow V$ applied as

$$\begin{aligned} s_{t+1} &= P(x)F(s_t) \\ \quad \text{or} \quad \\ s_{t+1} &= F(P(x)s_t). \end{aligned}$$

2.5 Input-Path Interference: Three Distinct Notions

The word “interference” is ambiguous. Three notions must be separated.

(A) Cross-input forward-pass contamination.

If F is frozen and the gate is stateless and a function of x alone, then the trajectory of x_1 is computationally independent of the presence of x_2 :

$$\begin{aligned} & \mathbb{E}[\tilde{s}_t^{(M)}(x_1 \mid x_2 \text{ processed elsewhere})] = \mathbb{E}[\tilde{s}_t^{(M)}(x_1)]. \end{aligned}$$

This notion is **structurally zero** in this setting. It becomes live only if the mechanism contains stateful components (running statistics, caches, etc.).

(B) Forward depth-aliasing.

A depth-agnostic gate applies the same modulation at every depth. If an oracle with access to depth t would apply a reference modulation $g^*(x, t)$, the depth-aliasing error is

$$\begin{aligned} & \mathbb{E}[\mathrm{DA}(g; x)] \\ &= \sum_{t=0}^{D-1} \left| \Phi(g(x), s_t(x)) - \Phi(g^*(x, t), s_t(x)) \right| \end{aligned}$$

$\right|$.

$\]$

This is a representational mismatch within a single forward pass.

(C) Training-time gradient interference.

The gate parameters ϕ are shared across all inputs. Updating ϕ on task (B) may change gate outputs on task-(A) inputs. First-order damage is proportional to

$\[$

$\angle \nabla_{\phi} \mathcal{L}_A(\phi), \nabla_{\phi} \mathcal{L}_B(\phi)$
 $\angle$.

$\]$

The operationally relevant channels for continual learning are **(B)** and **(C)**. Channel **(A)** is dead under a frozen, stateless, input-only mechanism.

2.6 Computation-Isolability

A mechanism is **exactly computation-isolating** for a task family $\{(S_i, y_i)\}$ if:

- task outputs are produced exactly: $\tilde{s}_D(x; \phi) = y_i(x)$ for all $x \in S_i$;
- forward depth-aliasing error is zero on the specified reference trajectories;

- first-order cross-task gradient interference is zero:

$\langle \nabla_{\phi} \mathcal{L}_i, \nabla_{\phi} \mathcal{L}_j \rangle = 0$ for all $i \neq j$.

It **bounds interference** if all relevant errors are at most ϵ , with ϵ controlled by explicit quantities.

2.7 Elimination, Bounding, Controllability

- **Elimination** means exact task outputs and exact zero interference.
- **Bounding** means a finite worst-case deviation ϵ , controlled by Lipschitz constants, depth, gate error, and capacity.
- **Controllability** is formalized as **reachability**: a target function $y_i: S_i \rightarrow \mathcal{Y}$ is reachable by the gate class if there exists ϕ such that $\tilde{s}_D(x; \phi) = y_i(x)$ for all $x \in S_i$; the mechanism is controllable over a finite task sequence if every finite set of reachable tasks with compatible overlaps is simultaneously realizable by a single ϕ .

This is the natural control-theoretic reading: because the gate is input-only, the control $g(x)$ is fixed for the entire depth trajectory, rather than an independently chosen per-step control u_t .

3. Theorem-Shaped Hypotheses

T1. Exact Preservation Forces the Gate to Be Inert on Active Coordinates

Assumptions. $\mathcal{F}$ is defined on an open set $U \subseteq \mathbb{R}^d$. The mechanism is a depth-agnostic output gate:

$$s_{t+1} = g(x) \odot \mathcal{F}(s_t).$$

Exact preservation of the unperturbed trajectory is required on an input set S :

$$\tilde{s}_t(x) = \mathcal{F}^t(x) \quad \forall x \in S, \forall t.$$

Claim. For every $x \in S$ and every coordinate i ,

$$(g_i(x)-1)\mathcal{F}_i(s_t(x))=0.$$

If coordinate $\iota(i)$ is active at any depth along the trajectory, then $\mathbf{g}_i(x) = \mathbf{1}$. If $\mathbf{F}$ is componentwise nonzero along all trajectories in $\mathcal{S}$, then $\mathbf{g}(x) = \mathbf{1}$ on $\mathcal{S}$.

Delta-gate variant. For residual delta gating,

$$\begin{aligned} & \\ s_{t+1} &= s_t + g(x) \odot F(s_t), \\ & \end{aligned}$$

exact preservation forces

$$\begin{aligned} & \\ g_i(x), F_i(s_t(x)) &= 0. \\ & \end{aligned}$$

Thus the gate is **zero-forced** on active coordinates, not identity-forced. This is the placement likely relevant to BDH's residual structure.

Verdict. True as an exact theorem. Near-tautological under global preservation; its real content is *locative*: it identifies precisely where gate freedom survives.

Classification. Direct consequence.

T2. Asymmetric Preservation: The Continual-Learning-Relevant Setting

Assumptions. Old-task inputs $\mathcal{S}_{\text{old}}$, new-task inputs $\mathcal{S}_{\text{new}}$. Exact preservation is required only on $\mathcal{S}_{\text{old}}$. No preservation constraint is imposed on $\mathcal{S}_{\text{new}}$.

Claim. On $\mathcal{S}_{\text{old}}$, the gate is pinned to identity (output gating) or zero (delta gating) on every coordinate active along the old trajectory. On never-active coordinates, and on all of $\mathcal{S}_{\text{new}}$, the gate is unconstrained by preservation.

Verdict. True as an exact theorem. This is the correct formalization of the continual-learning constraint. The identity/zero obstruction does **not** by itself rule out nontrivial isolation when old and new task inputs are separated and the gate can interpolate.

Classification. Direct consequence; the asymmetric framing is the honest CL formalization.

T3. Identifiability: Same Input, Different Target Is Impossible

Assumptions. The mechanism is a deterministic function of the input:

\[

$\tilde{s}_D(x; \phi) = \mathcal{M}_\phi(x).$

\]

Two tasks share an input $(x \in S_A \cap S_B)$ and demand different outputs: $(y_A(x) \neq y_B(x))$.

Claim. No single (ϕ) can satisfy both tasks.

Proof. A function returns one output for one input; two different required outputs are incompatible. ■

Verdict. True as an exact theorem. It is the sharpest structural limit of an input-only mechanism, independent of depth, capacity, or nonlinearity.

Practical nuance. In language continual learning, the force of this theorem depends on whether task identity is encoded in the input context. Exact input collisions and distribution-level overlap should not be conflated.

Classification. Direct consequence; essentially an identifiability condition.

T4. Depth-Agnostic Gate Values: Representational Limit, Not Behavioral Impossibility

Assumptions. $\Phi(g,s)$ is injective in the gate argument (g) along the trajectory. An oracle demands a depth-varying gate-value sequence $(g^*(x,0), \dots, g^*(x,D-1))$, nonconstant in (t) .

Claim. A depth-agnostic gate $(g(x))$ cannot reproduce the exact gate-value sequence $(g^*(x,t))$.

Proof. If it did, injectivity would force $(g(x)=g^*(x,t))$ for every (t) , contradicting nonconstancy. ■

Important correction. This is a statement about **gate values**, not about **reachable input-output behavior**. Because the state changes across depth, a constant gate can produce depth-varying effective behavior.

Counterexample. Let $(F(s)=s)$, $(D=2)$. Output gating gives

$$\begin{aligned} & \\ s_2 &= g(x) \odot g(x) \odot x. \\ & \end{aligned}$$

A constant gate (α, β) produces $(\alpha^2 x_1, \beta^2 x_2)$. A depth-varying sequence $(g^*_0=(c, 1), g^*_1=(1, c))$ produces $((cx_1, cx_2))$, which the constant gate $(\sqrt{c}, \sqrt{c})$ reproduces.

Verdict. The representational lemma is true. The behavioral impossibility is **not established**; it was withdrawn in final adjudication. The reachability question is open.

Classification. Direct consequence for the representational part; the broad behavioral claim is **false as stated**.

T5. Projection Mechanisms: Necessary Rank Condition

Assumptions. Output projection:

$$\begin{aligned} & \backslash[\\ & s_{t+1}=P(x)F(s_t). \\ & \backslash] \end{aligned}$$

Exact preservation of the unperturbed trajectory is required for input $\backslash(x\backslash)$.

Claim. Let

$$\begin{aligned} & \backslash[\\ & m(x)=\dim\operatorname{span}\{F(s_0(x)),\dots,F(s_{D-1}(x))\}. \\ & \backslash] \end{aligned}$$

Then

$\forall$

$\text{rank}(P(x)) \geq m(x).$

$\forall$

Proof. Exact preservation forces $\forall (P(x)F(s_t(x))=F(s_t(x)))$ for all $\forall (t)$, so all $\forall (F(s_t(x)))$ lie in $\text{Im}(P(x))$. ■

Input-projection caveat. For input projection $\forall (s_{t+1}=F(P(x)s_t))$, exact preservation means $\forall (F(P(x)s_t)=F(s_t))$. This implies $\forall (P(x)s_t=s_t)$ only if $\forall (F)$ is injective along the trajectory. The theorem as stated applies to output projection; the input-projection variant requires the injectivity assumption.

Withdrawn corollary. The earlier claim $\forall (\sum_i m_i \leq d)$ for multiple tasks is **withdrawn**. Image-subspace orthogonality is a forward-pass geometric property; cross-task gradient interference is a property of the parameterization of $\forall (P(x))$. These are different mathematical objects, and the capacity bound does not follow without a special parameterization assumption.

Classification. Direct consequence (linear algebra); no isolation capacity bound follows.

T6. Gradient Interference and the Parameter Budget

Assumptions. The gate/projection mechanism has (p) trainable parameters. For task (i) , the row space of the collection of output Jacobians is $(R_i \subseteqq \mathbb{R}^p)$. Zero first-order interference is required for **all smooth losses**.

Claim. The row spaces must be mutually orthogonal, and

$\{$

$\sum_i \dim R_i \leq p.$

$\}$

Proof. For arbitrary losses, zero inner products across all linear functionals force row-space orthogonality; orthogonal subspaces of $(\mathbb{R}^p)$ have total dimension at most (p) . ■

Caveats.

- For fixed specific losses, zero interference can hold accidentally even when row spaces overlap.
- If $(p \gg d)$, the bound is likely non-binding for practical LLM-scale gates.
- This is standard parameter-isolation accounting, not a result specific to shared sequential composition.

Classification. Direct consequence; not a decisive obstruction unless Jacobian ranks are measured.

T7. Error Amplification: Bounding Is Conditional on Dynamics

Assumptions. The gated transition is Lipschitz in state with constant L . The per-depth gate error against a reference is at most δ .

Claim.

$$\begin{aligned} & \mathbb{E} \\ & \mathbb{E} |s_D(x) - s_{D^*}(x)| \\ & \leq \\ & \delta \sum_{k=0}^{D-1} L^k \\ & = \\ & \begin{cases} \delta \frac{1-L^D}{1-L}, & L < 1, \\ \delta D, & L = 1, \\ \delta \frac{L^D-1}{L-1}, & L > 1. \end{cases} \\ & \end{aligned}$$

Precise interpretation.

- For **fixed finite depth** $\lfloor D \rfloor$, a finite bound exists for every finite $\lfloor L \rfloor$.
- A bound that is **uniform in depth** exists iff $\lfloor L \rfloor < 1 \lfloor L \rfloor$.
- The claim “trained LLM-scale maps have $\lfloor L \rfloor > 1 \lfloor L \rfloor$ ” is an **empirical conjecture**, not a theorem, and must be measured for the actual BDH block.

Classification. Direct consequence; quantifier-corrected.

T8. Finite-Task Reachability: Conditional Construction, Otherwise Underdetermined

Assumptions. The task family is finite: $\{(x_i, y_i)\}, \{i=1, \dots, N\}$. The gate family can interpolate finite sets. Each target is reachable:

$$\begin{aligned} & \lfloor \\ & y_i(x) \in \mathcal{R}_D(x) := \{\tilde{s}_D(x; \phi) : \phi \in \Phi\}. \\ & \rfloor \end{aligned}$$

Overlapping task inputs have agreeing targets.

Claim. There exists ϕ realizing all finite task targets exactly.

Proof sketch. Reachability gives a finite set of constraints; interpolation capacity fits them. ■

Underdetermination examples.

- If $(F \equiv 0)$, the reachable set is $\{0\}$ for every (x) , regardless of gating.
- If the same input has different targets, T3 applies.
- If $(F(s)=s)$ and $(D=2)$ with multiplicative output gating, reachable outputs are of the form $(g(x) \odot g(x) \odot x)$, which excludes many targets.

Verdict. Conditionally true; otherwise underdetermined. The reachability assumption is the nontrivial crux and remains a **conjecture** for a concrete BDH map.

Classification. Plausible adaptation of universal interpolation; reachability is unproved.

4. Summary of Verdicts

| # | Claim | Status |

| --- | ----- | ----- |

T1	Exact preservation forces identity/zero on active coordinates	True; exact theorem
T2	Asymmetric preservation constrains only old-task active coordinates	True; CL-relevant
T3	Same input, different target impossible	True; identifiability
T4	Depth-agnostic gate-value representational limit	Representational true; behavioral impossibility false/underdetermined
T5	Projection rank necessary condition	True; capacity corollary withdrawn
T6	Gradient interference parameter budget	True under worst-case generality; often non-binding
T7	Error amplification bound	True; finite-depth vs uniform-depth corrected
T8	Finite-task reachability	Conditional; reachability is conjecture

5. Prior Art: Does It Address the Same Problem?

Prior art	What it does	Same mathematical problem?	Classification
---	---	---	---
	Gradient projection / orthogonal-gradient methods (GEM, A-GEM, GPM)	Project gradients in parameter space during training	Partly: cross-task gradient interference; not forward depth-aliasing on a frozen path
	Known, partially relevant		

| Parameter isolation / masks (PackNet, Supermasks, Piggyback) | Allocate disjoint parameter subsets to tasks | No: creates per-task parameters/masks, violating the input-only gate constraint | Known, related but different |

| Progressive networks / PathNet | Add task-specific columns or paths | No: per-task architectural growth | Known, different problem |

| Hypernetworks / fast weight programmers | Generate weights or modulations for a main network | Partly: can generate input-dependent modulations; typically per-layer outputs | Known, partially relevant |

| FiLM / feature-wise linear modulation | Input-dependent affine modulation of hidden features | Partly: same gating algebra, but usually per-layer parameters | Known, directly relevant |

| LSTM / GRU | Shared-parameter state-dependent gates in a recurrent path | Closest: same composition; but gates see the current state, not only the input | Known, nearest prior art |

| Mixture of experts / conditional computation | Input-dependent routing to expert parameter sets | No: per-layer routers and distinct experts | Known, structurally similar |

| Reservoir computing / echo state networks | Fixed random recurrent core plus trained readout | Partly: frozen core; not input-dependent forward gating for task isolation | Known, relevant background |

| Weight-tied transformers (Universal Transformer, ALBERT) | Reuse a block across depth | Partly: weight-tied composition; no task-isolation gating | Known, background context |

| Nonlinear RNN reachability / observability | Characterizes what a recurrent map can reach | Yes for the reachability question | Known, should be applied to T8 |

Conclusion. No surveyed prior art solves the exact problem: isolation of computations within a single frozen weight-tied composition using only

an input-dependent gate, with no per-depth trainable parameters and no retraining of the shared path.

6. Novelty Classification

| Claim | Classification |

| --- | --- |

| Formalization of two live interference channels (depth-aliasing and cross-task gradient interference) in a frozen weight-tied path |

Apparently novel (packaging only) |

| T1–T3, T5–T7 | **Direct consequences** (textbook ingredients) |

| T4 representational lemma | Direct consequence; behavioral version false as stated |

| Necessity of state/depth/context information for effective gating |

Plausible adaptation (consistent with LSTM/FiLM practice) |

| T8 reachability for concrete BDH $\setminus (F)$ | **Conjecture** |

| Expansiveness of trained BDH blocks | **Empirical conjecture** |

| Any observed effect in cl-h1-report | Inadmissible as theorem evidence; admissible only as a test of the conjectures above |

Contribution posture: **a new formalization and a clean map of elementary obstructions, not new mathematics.**

7. Obstruction Characterization

The strict proposed mechanism is:

- input-dependent gate/projection,
- no per-depth/per-level trainable parameters,
- frozen shared path,
- no retraining of $\mathcal{F}$.

Under the **global-preservation reading** — the mechanism must preserve $\mathcal{F}$'s behavior on **all** inputs — the gate is forced to be inert everywhere, so exact elimination is **trivially impossible**.

Under the **asymmetric continual-learning reading** — exact preservation on old-task inputs, free behavior on new-task inputs — no general impossibility or possibility theorem has been established. What is established is a conjunction of four necessary conditions:

- **Identifiability failure (T3)**. Conflicting targets on identical inputs cannot be served by any input-only mechanism. In language modeling, the force of this depends on how much task identity is contained in the input context.
- **Preservation-geometry confinement (T1/T2)**. The gate is pinned to identity (or zero under delta gating) on every active old-task coordinate. Freedom survives only on never-active coordinates and on new-task inputs.
- **Gradient coupling of shared ϕ (T6)**. New-task updates generically move g_ϕ away from the preserved behavior on

old inputs. Worst-case immunity costs a rank budget that may be non-binding when $\|p\| \gg d$.

- **Depth amplification (T7).** If the gated dynamics are expansive, small per-depth gate errors grow geometrically in depth. Bounding then requires contractivity of the composed system — a property of $\mathcal{F}$, not a capability of the gate alone.

Three **scope conditionals** attach to every result:

- **Statelessness.** Any stateful component inside the gate revives cross-input forward contamination.
- **$\mathcal{T}$ -conditioning fork.** A gate $g_\phi(x, t)$ with a fixed non-trainable depth encoding adds $\mathcal{O}(1)$ parameters and does not violate the constraint as literally worded. A learned depth embedding adds $\mathcal{O}(D)$ parameters and does violate it. The depth-agnostic obstruction dissolves on the permissive side of this fork, converting the problem into pure reachability (T8).
- **Frozen path.** BDH's native Hebbian-style plasticity must be explicitly disabled for the frozen-path premise to hold.

8. Direct Answer to the Research Question

> Can input-path interference arising from shared sequential composition be eliminated, bounded, or made controllable by an input-dependent gate/projection mechanism, without per-depth/per-level trainable parameters and without retraining the frozen shared path?

| Reading | Verdict | Epistemic status |

| --- | --- | --- |

| **Elimination**, global preservation on all inputs | Impossible — the only admissible gate is identity/zero on active coordinates | Proved; degenerate |

| **Elimination**, asymmetric CL (preserve (S_{old}) , free on (S_{new})) | **Open**. Necessary conditions established; no general impossibility or possibility theorem | Partially characterized |

| **Bounding** | Finite-depth bounds exist for any finite (L) ; uniform-in-depth bound iff $(L < 1)$ | Proved, conditional |

| **Controllability** | Not yet well-posed: requires formal reachability analysis on the concrete frozen BDH map with the exact gate class | Undefined pending instantiation |

The failure condition contemplated by the brief is met **only in the degenerate global-preservation reading**. In the continual-learning-relevant reading, the correct outcome is: *not proven impossible, not proven possible; reduced to identifiable necessary conditions and one decisive open conjecture — reachability*.

9. Mandatory Follow-Ups Before Empirical Reconciliation

- **Instantiate (F) from paper . tex**. Fix the literal BDH update, including residual form, LayerNorm placement, and any Hebbian-style multiplicative terms.

- **Reinstantiate every theorem for the correct gate placement.**
For a residual block, the natural gate is $s_{t+1} = s_t + g(x) \odot F(s_t)$, which changes the identity obstruction to a zero obstruction.
- **Rule on the τ -conditioning fork.** State in writing whether fixed non-learned depth encoding is permitted.
- **Retrieve the internal reports.** Map 2026-08-25_bdh-continual-learning-plan_v2.md and 2026-08-25_cl-h1-report.md onto exactly one formal variant: depth-agnostic, τ -conditioned, or state-conditioned.
- **Define controllability formally** as reachability of finite task targets by the gate class on the instantiated BDH map.
- **Measure** (a) the Lipschitz constant of the gated BDH block, and (b) cross-task Jacobian row-space overlap in ϕ -space. These are the only claimed barriers that can be tested empirically.

Bottom line. A strict input-only gate/projection on a frozen shared sequential composition does not provide a proven general solution. The obstruction is not a single theorem but a map of four necessary failure conditions and one open reachability problem. The honest conclusion for publication is: *global preservation makes nontrivial gating impossible; asymmetric continual learning with a frozen shared path remains open pending concrete reachability analysis of the actual BDH update.*

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Moderator

Final Moderator Synthesis — Debate Closure: Input-Path Isolation in BDH via Gating/ Projection

Role: ox-alpha, moderator. This document consolidates five rounds of adversarial deliberation (STRATEGIST, CRITIC, OPTIMIZER), records the final consensus state, adjudicates all surviving disputes, and delivers the scoped answer to the research question together with the precise characterization of the obstruction.

0. Process Record and Evidence-Base Status

The debate executed a full correction loop:

- **Rounds 1–2** built a unified formal skeleton and initially headlined a strong claim: *"exact elimination of input-path interference is impossible in general"* via three obstructions (identity-forcing, diagonalization, rank budget).
- **Round 3–4 CRITIC reviews** landed decisive defeaters: (i) the headline rested on a *degenerate* global-preservation reading; (ii) the diagonalization argument conflated gate **values** with reachable **behaviors** (counterexample: $F(s)=s$, $D=2$); (iii) the projection capacity corollary $\sum_i m_i \leq d$ committed a category error (forward-pass image-subspace geometry $\neq$ parameter-space gradient geometry); (iv) the amplification "iff $L < 1$ " misquantified finite- D vs. uniform-in- D bounds; (v) controllability was undefined; (vi) the BDH-relevant **delta-gate placement** converts identity-forcing into **zero-forcing**, changing the obstruction's form.
- **Round 5:** all three roles converged on the same scoped verdict set (STRATEGIST V1–V8, OPTIMIZER T1–T8 — substantively identical), with the CRITIC accepting the scoping while flagging residual gaps.

Adjudication principle throughout: a claim enters the record only at the strength of its weakest surviving proof.

Evidence-base limitations (recorded for honesty): the internal project files (2026-08-* reports, paper-parity-plan, bdh-continual-learning-plan_v2.md, cl-h1-report.md) and the literal equations from paper.tex were **never retrieved in-debate** despite repeated attempts. All results below are therefore proved for the abstraction $s_{t+1} = \Phi(g(x), s_t)$ over a frozen map F (with the residual/delta variant treated explicitly). Instantiation against the actual BDH update remains a mandatory follow-up, not an optional refinement. No experimental observation from any report was admitted as theorem evidence at any point.

1. Consensus Evaluation

Full consensus (all three roles, post-revision)

| Item | Status |

| --- | --- |

| Formal skeleton: shared sequential composition, frozen path, gate/projection placements (output / input / delta), elimination-vs-bounding distinction | Agreed verbatim |

| **Interference taxonomy (three channels):** (A) cross-input forward-pass contamination — **structurally zero** for stateless mechanisms on a frozen path; (B) forward **depth-aliasing** within a pass — live; (C) training-time **cross-task gradient interference** through shared ϕ — live |

Agreed; channel (A)'s revival conditions (stateful components: running statistics, caches) noted |

| Verdicts T1–T3, T5–T8 (see §3) | Agreed |

| Prior-art triage: LSTM/GRU nearest neighbor (state-conditioned gates); FiLM same gating algebra but per-layer parameters; fast-weight programmers/hypernetworks adjacent; reservoir computing background; GEM/A-GEM/GPM parameter-space complementary; PackNet/supermasks/progressive nets constraint-violating; MoE different problem; Universal Transformer/ALBERT backdrop. **No surveyed work solves the exact problem.** | Agreed |

| Novelty discipline: packaging novel, mathematics derivative | Agreed |

Adjudicated disputes (all resolved)

- **"Impossibility in general" vs. scoped openness** → the **scoped formulation prevails**. The unscoped headline holds only under global preservation on *all* inputs — a reading under which the gate must be inert everywhere and hence cannot adapt anywhere, dissolving the continual-learning question itself. STRATEGIST's own §7 concession is entered as binding.
- **Diagonalization obstruction** → downgraded from behavioral impossibility to a true-but-representational lemma about gate values; superseded by the open reachability question (T8).
- **Projection capacity corollary** → withdrawn; survives only under a special independent-subspace parameterization that violates the spirit of the constraints anyway.
- **Delta-gate placement** → incorporated into T1/T2: under $s_{t+1} = s_t + g(x) \odot F(s_t)$, preservation forces $g_i(x) = 0$ (not 1) on active old-task coordinates. This is the placement likely correct for BDH's residual structure.

Residual dissent (non-blocking)

The CRITIC maintains two emphasis-level caveats, both accepted into the record: (a) the practical force of T3 depends on measure-theoretic overlap of task supports — exact collision, distributional overlap, and representation-space near-overlap must not be conflated; (b) T6 is worst-case accounting that is likely non-binding when $p \gg d$ and must not be presented as a decisive barrier until Jacobian ranks are measured.

2. Consolidated Definitions (Final)

- **Shared sequential composition:** $s_0 = x$, $s_{t+1} = F(s_t)$, D steps; residual variant $s_{t+1} = s_t + F(s_t)$.
- **Frozen shared path:** $\partial F / \partial \theta = 0$ during adaptation; only ϕ (gate parameters) trainable.
- **Gate:** input-dependent $g_\phi(x)$ applied identically at every depth; placements: output $g \odot F(s)$, input $F(g \odot s)$, $\Delta s + g \odot F(s)$. No per-depth parameter vectors.
- **Projection:** input-dependent orthogonal $P(x)$; output $P(x)F(s)$ or input $F(P(x)s)$.
- **Computation-isolability (exact):** exact task outputs + zero depth-aliasing error + zero first-order cross-task gradient interference.
- **Elimination vs. bounding:** exact equality/zero interference vs. finite worst-case deviation controlled by explicit quantities (L_Φ , D , Δ , p).
- **Controllability (partial formalization):** reachability of finite task targets by the gate class on the frozen map — with the crucial caveat that an input-only control $g(x)$ is fixed for the entire pass (open-loop modulation), not an independent per-step control u_t . A fuller formalization specifying state space, admissible gate range,

transition rule, target class, and joint-realizability requirements remains outstanding.

3. Final Adjudicated Verdict Table

#	Claim	Status	Notes
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T1	Exact preservation forces the gate inert on active coordinates: $g_i = \mathbf{1}$ (output gating) or $g_i = 0$ (delta gating) wherever $F_i(s_t(x)) \neq 0$ along the preserved trajectory TRUE — exact theorem; direct consequence
	Near-tautological globally; real content is <i>locative</i> : identifies precisely where gate freedom lives (never-active coordinates; non- S inputs)

T2	Under asymmetric preservation (exact on S_{old} , free on S_{new}), the confinement restricts to old-task active coordinates TRUE — exact theorem; the honest CL formalization
	Does not by itself preclude nontrivial isolation when supports are separable and the gate can interpolate

T3	Deterministic input-only mechanism cannot serve two tasks demanding different outputs at a shared input TRUE — exact theorem (identifiability)
	Sharpest limit of "input-dependent only"; practical force depends on context-sufficiency of inputs

T4	Depth-agnostic gate confined to diagonal of modulation space
	Representational lemma true ; behavioral impossibility not established (downgraded after counterexample) Superseded by open reachability question

| **T5** | Output projection preserving trajectory requires $\|P(x) - P_m(x)\| \leq \epsilon$ | **TRUE — exact linear algebra** | Capacity corollary withdrawn; input-projection variant requires injectivity of F along the trajectory |

| **T6** | Zero first-order interference *for all smooth losses* $\Leftrightarrow$ mutually orthogonal task-Jacobian row spaces $\Rightarrow \sum_i \dim R_i \leq p$ | **TRUE — exact under stated generality** | Worst-case guarantee; likely non-binding for $p \gg d$; standard accounting, not composition-specific |

| **T7** | $e_D \leq \delta \sum_{k < D} L_{\Phi}^k$; finite- D bound exists for every finite L_{Φ} ; **uniform-in-depth** bound iff $L_{\Phi} < 1$ | **TRUE — exact, quantifiers corrected** | "Trained BDH blocks are expansive" reclassified as empirical conjecture requiring measurement |

| **T8** | Finite task sets realizable iff targets reachable by gate class + interpolation capacity + overlap agreement | **CONDITIONAL; crux is a conjecture** | Counterexamples ($F \equiv 0$; sign/root obstructions) show reachability fails generically without hypotheses on F |

4. Final Answer to the Research Question

> *Can input-path interference arising from shared sequential composition be eliminated, bounded, or made controllable by an input-dependent gate/projection mechanism, without per-depth/per-level trainable parameters and without retraining the frozen shared path?*

| Reading | Verdict | Epistemic status |

| --- | --- | --- |

| **Elimination**, global-preservation (mechanism preserves F 's behavior on *all* inputs) | Trivially impossible — the only admissible gate is inert everywhere | Proved; degenerate |

| **Elimination**, asymmetric CL (exact preservation on S_{old} , free on S_{new}) | **OPEN**. Necessary conditions established (T1–T3, T5–T7); no general possibility or impossibility theorem | Partially characterized |

| **Bounding** | Finite-depth bounds exist for any finite L_Φ ; **uniform-in-depth** bounding iff gated dynamics contractive. Bounding is a property of the composed system, not achievable by gate design alone under expansion | Proved, conditional |

| **Controllability** | **Not yet well-posed**: requires formal reachability analysis on the concrete frozen BDH map with the exact gate class | Undefined pending instantiation |

The brief's failure condition is met **only in the degenerate reading**. In the CL-relevant reading the correct outcome is: *not proven impossible, not proven possible; reduced to identifiable necessary conditions plus one decisive open conjecture (reachability)*. Reporting anything stronger would violate the novelty discipline.

5. Obstruction Characterization (Deliverable Under the Failure Clause)

There is no single obstruction theorem in the meaningful setting. The failure surface is a **conjunction of four necessary conditions**, each independently capable of killing the program, none individually fatal:

- **Identifiability failure (T3)** — conflicting targets on overlapping inputs are unservable by any input-only mechanism. In language-modeling CL, escape requires context-sufficiency of the input or task conditioning.
- **Preservation-geometry confinement (T1/T2)** — the gate is pinned to identity (output gating) or zero (delta gating) on every (old input, active coordinate) pair. Freedom survives on never-active coordinates — potentially substantial under BDH's ReLU sparsity — and on new-task inputs.
- **Gradient coupling of shared ϕ (T6)** — new-task updates generically move g_ϕ off the preserved behavior on old inputs; worst-case immunity costs a rank budget that is restrictive ($\leq d$) or vacuous ($\gg d$).
- **Depth amplification (T7)** — under expansive gated dynamics, per-depth modulation error grows geometrically in D ; bounding then demands architectural contractivity, an assumption on F , not a capability of the gate.

Plus **three scope conditionals** silently underlying every result: (a) **statelessness** of the mechanism (any running-statistics component revives channel A); (b) the **ϕ -conditioning fork** — a shared-parameter gate $g_\phi(x, t)$ fed by a *fixed, non-trainable* depth encoding adds $O(1)$ parameters and does not violate the constraint as literally worded, and it dissolves the depth-agnostic restrictions entirely, converting the problem into pure reachability (T8); a *learned* depth embedding adds $O(D)$ parameters and violates the constraint outright. Every impossibility-flavored statement applies only on the strict side of this fork; (c) BDH's native Hebbian-style plasticity must be explicitly disabled, else the path is not frozen.

6. Novelty Ledger (Final)

| Claim | Classification |

| --- | --- |

| Problem formulation: isolation inside a frozen weight-tied composition via input-only gate, CL-framed, three-channel interference taxonomy |

Apparently novel (packaging only) |

| T1–T3, T5–T7 | **Direct consequences** (textbook ingredients) |

| T4 representational lemma | Direct consequence; behavioral version false as stated |

| Necessity of state/depth/context information for effective gating |

Plausible adaptation (consistent with LSTM/FiLM practice; proven only under injectivity assumptions) |

| Reachability of the concrete BDH map under the gate class (T8 crux) |

Conjecture |

| Expansiveness of trained BDH blocks | **Empirical conjecture** |

| Any observed effect in c_l-h₁-report | Inadmissible as theorem evidence; admissible only as a test of the above conjectures |

Write-up posture: claim a new formalization and a clean map of elementary obstructions — not new mathematics.

7. Constructive Directions Surfaced (all $\leq$ conjecture)

- **C1 — Orthogonal-gradient protection of ϕ :** once the protected object is recognized as the *gate parameters*, GEM/GPM-style projection applies compatibly with all constraints: project new-task gradients onto the orthogonal complement of the old-task gradient subspace in ϕ -space, holding $g_\phi|_{S_{\text{old}}} \approx$ inert to first order. Plausible adaptation; open statistical question about generalization beyond the old-task sample.
- **C2 — Fixed-encoding tt -conditioning** ($O(1)$ parameters): dissolves depth-agnostic restrictions; legality contingent on the scope ruling.
- **C3 — Delta-gate placement:** gate $g(x) \odot \Delta(s_t)$ inside $s_{t+1} = s_t + \cdot$; never the residual stream itself.
- **C4 — Sparse-coordinate freedom:** exploit never-active coordinates of old-task trajectories (plausible under ReLU sparsity) as legal modification space even on preserved inputs.

8. Mandatory Follow-Ups Before Any Empirical Reconciliation

- **Instantiate FF from paper .tex:** fix the literal update (residual form, LayerNorm placement, Hebbian/multiplicative terms), choose gate placement (C3), verify plasticity disabling and statelessness.
- **Re-prove every theorem for the actual placement** — especially the zero-forcing variant under delta gating.
- **Rule in writing on the tt -conditioning fork** (fixed non-learned encoding permitted?).

- **Retrieve and map the internal reports** (plan_v2, cl-h1-report) onto exactly one formal variant: depth-agnostic / t -conditioned / state-conditioned.
- **Define controllability rigorously** as finite-target reachability over the instantiated map and exact gate class, drawing on nonlinear discrete-time reachability theory.
- **Measurement program:** empirical Lipschitz/expansiveness of the gated BDH block (tests T7's conjecture); empirical cross-task Jacobian row-space overlap in ϕ -space (tests whether T6's worst case binds).

9. Discipline Audit

- No experimental observation cited as evidence for any theorem at any point. ✓
- All overclaims identified were withdrawn on the record (capacity corollary; behavioral diagonalization; finite- D "iff"; global-preservation headline) or rescope. ✓
- Novelty labels consistent post-adjudication: packaging novel, mathematics derivative. ✓

10. Closing Statement

Consensus is complete on the formal framework, the three-channel interference taxonomy, the exact necessary conditions T1–T3/T5–T7 as

scoped, the conditional bounding result, the prior-art triage, and the open status of the asymmetric continual-learning question. The single substantive disagreement — whether a general impossibility had been established — was resolved **against** the stronger claim on the strength-of-weakest-proof principle.

The investigation exits the debate not with an impossibility theorem but with something more useful: a precise map of where gate freedom lives (never-active sparse coordinates, new-task inputs), four independent ways the program can fail, one pivotal scope decision (t-conditioning), and one decisive open conjecture — **reachability of the instantiated BDH map under the allowed gate class** — whose resolution will determine whether the proposed mechanism is possible, impossible, or merely unproven.

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Complete

[STRATEGIST Round 1]

I'll start by reading the research paper and all the relevant reports to understand the BDH architecture and the current state of work on continual learning. Let me begin systematically.

[SEARCH: RAG file 2509.26507v1.pdf]

[SEARCH: RAG file paper.tex]

[SEARCH: RAG file 2026-08-18_progress-report.md]

[SEARCH: RAG file 2026-08-23_progress-report.md]

[SEARCH: RAG file 2026-08-24_progress-report.md]

[SEARCH: RAG file 2026-08-22_paper-parity-plan.md]

[SEARCH: RAG file 2026-08-25_bdh-continual-learning-plan_v2.md]

[SEARCH: RAG file 2026-08-25_cl-h1-report.md]

Web Research

- **The Dragon Hatchling: The Missing Link between the Transformer and...**

RAIN. arXiv:2509.26507v1 [cs.NE] 30 Sep 2025. Adrian Kosowski*†. Przemysław Uznanski† Jan Chorowski† Michał Bartoszkiewicz†.

Source: <https://www.rivista.ai/wp-content/uploads/2025/11/2509.26507v1.pdf>

- **[2509.26507] The Dragon Hatchling: The Missing Link between the...**

View PDF HTML (experimental). Abstract: The relationship between computing systems and the brain has served as motivation for pioneering theoreticians since John von Neumann and Alan Turing. (or arXiv:2509.26507v1 [cs.NE] for this version).

Source: <https://arxiv.org/abs/2509.26507>

- **Dragon Hatchling: Linking Transformers & Brain Models**

The Dragon Hatchling: The Missing Link between the Transformer and Models of the Brain. Published 30 Sep 2025 in cs.NE, cs.AI, cs.LG, and stat.ML | (2509.26507v1).

Source: <https://www.emergentmind.com/papers/2509.26507>

- **Paper page - The Dragon Hatchling: The Missing Link between the...**

Get this paper in your agent: hf papers read 2509.26507. Don't have the latest CLI? Cite arxiv.org/abs/2509.26507 in a dataset README.md to link it from this page.

Source: <https://huggingface.co/papers/2509.26507>

- **The Dragon Hatchling: The Missing Link between... | Papers with Code**

Paper · arXiv:2509.26507 · 30 September 2025. The Dragon Hatchling: The Missing Link between the Transformer and Models of the Brain. Adrian Kosowski, Przemysław Uznański, Jan Chorowski, Zuzanna Stamirowska, Michał Bartoszkiewicz. PDF. Code.

Source: <https://paperswithcode.co/paper/2509.26507>

- **Gemini RAG - File Search Tool - YouTube**

In this video, I go through the latest release from the Gemini API team, which is the file search tool, or basically a simple RAG system built right into the...

Source: <https://www.youtube.com/watch?v=Mup9ki6Bdtg>

- **GitHub - infiniflow/ragflow: RAGFlow is a leading open-source...**

Go to file. Code. Open more actions menu. Folders and files. RAGFlow is a leading open-source Retrieval-Augmented Generation (RAG) engine that fuses cutting-edge RAG with Agent capabilities to create a superior context layer for LLMs.

Source: <https://github.com/infiniflow/ragflow>

- **Error: "Loading a class or package in a group. \usepackage" with .cls file...**

in the .cls file but it still did not work. However, when I try to compile from a single .tex file. It works well.in a .cls file (let's say it is "Main.cls") and in my .tex file (e.g., "Paper.tex"), I begin it with.

Source: <https://latex.org/forum/viewtopic.php?t=35969>

- **Learn how to use upload images, audio, and other files | FlowiseAI**

This research paper. is one of many that compare RAG with longer context windows. To enable full file uploads, go to Chatflow Configuration, open the File Upload tab, and click the switch

Source: <https://docs.flowiseai.com/using-flowise/uploads>

10. **How to Do File Search in Gemini API? | Analytics Vidhya**

For multimodal RAG, File Search also supports image files. This means you can upload documents and images into the same File Search Store and ask questions that require both textual and visual context.

Source: <https://www.analyticsvidhya.com/blog/2026/05/gemini-api-file-search/>

11. **mini-enterprise-rag-agent/progress.md at main ... - GitHub**

May 8, 2026 · Read relevant process skills and confirmed current architecture issue. Created planning files for this multi-step refactor. Added failing tests in tests/test_llm_tool_planning.py. Red run: 8 failed, 1 passed; failures match expected architecture gaps.

Source: <https://github.com/zhengshaoyi1387/mini-enterprise-rag-agent/blob/main/progress.md>

12. **RAG/rag_state_of_the_art_2026.md at main · dpsdagain/RAG**

May 26, 2026 · This document synthesizes findings from 45+ targeted web searches conducted on May 26, 2026. It represents the current frontier of RAG system design — not beginner tutorials or toy demos. Every claim is grounded in published research, production reports, or framework documentation.

Source: https://github.com/dpsdagain/RAG/blob/main/rag_state_of_the_art_2026.md

13. **mindstudio/big-rag • LM Studio Hub**

Big RAG Plugin for LM Studio A powerful RAG (Retrieval-Augmented Generation) plugin for LM Studio that can index and search through gigabytes or even terabytes (not tested) of document data.

Source: <https://lmstudio.ai/mindstudio/big-rag>

14. **Local RAG: Search Your Documents with a Private AI**

Jan 30, 2026 · RAG fixes this. Instead of retraining the model (expensive, slow, overkill), RAG searches your documents for relevant chunks and feeds them to the LLM as context. The model reads your actual text and answers based on it, not from memory. The best part: everything stays on your machine.

Source: <https://insiderllm.com/guides/local-rag-search-documents-private-ai/>

15. **RAG Daily Papers - Latest RAG Research from arXiv**

Aug 18, 2022 · From Retrieved Context to Runtime Control: Adaptive Compression for Edge-based RAG Retrieval-augmented generation (RAG) improves language-model responses by grounding generation in external passages, which comes with overhead: retrieved context lengthens the prompt, increasing prefill work, KV-cache footprint, memory traffic, latency, and energy.

Source: <https://ragdaily.com/>

16. RAG Status in Project Management Explained (2026)

June 17, 2026 - Reporting dimension-level RAG, covering at a minimum time, cost, scope and risk, gives decision-makers the information they actually need. For guidance on structuring this kind of report effectively, IPM's progress report template provides a practical framework.

Source: <https://instituteprojectmanagement.com/blog/rag-status-in-project-management/>

17. Releases · infiniflow/ragflow

4 days ago - Release nightly created from 2db8eb6 at 2026-08-23 21:19:13+08:00

Source: <https://github.com/infiniflow/ragflow/releases>

18. Project RAG Status Meanings And Best Practice (2026)

July 6, 2026 - PM3 helps organisations standardise RAG reporting across projects, programmes and portfolios, with up to 10 status indicators covering areas such as cost, progress, risks, issues, stakeholders, scope and value.

Source: <https://eleco.com/pm3/knowledge-centre/how-many-rags/>

19. Progress Agentic RAG reviews 2026 | FitGap

April 11, 2026 - Progress Agentic RAG is an LLMOps-oriented capability from Progress that helps teams build retrieval-augmented generation (RAG) applications with agent-style orchestration over enterprise content.

Source: <https://us.fitgap.com/products/050333/progress-agentic-rag>

20. RAG Status for KPIs: The Red-Amber-Green Playbook (2026) | ClearPoint Strategy Blog

1 week ago - KPIs are crucial for understanding your organization's strengths and weaknesses. But let's be honest, staring at reports and relying on gut feelings isn't effective. You need a clear, objective way to evaluate your progress. A RAG status system provides just that.

Source: <https://www.clearpointstrategy.com/blog/establish-rag-statuses-for-kpis>

21. RAG Status in Project Management Explained (2026)

June 17, 2026 - Reporting dimension-level RAG, covering at a minimum time, cost, scope and risk, gives decision-makers the information they actually need. For guidance on structuring this kind of report effectively, IPM's progress report template provides a practical framework.

Source: <https://instituteprojectmanagement.com/blog/rag-status-in-project-management/>

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Source: <https://us.fitgap.com/products/050333/progress-agentic-rag>

24. RAG Status in Project Management: A Comprehensive Guide

June 19, 2026 - RAG (Red, Amber, Green) Status is a simple visual reporting system used in Project Management to show the overall health of a task, milestone or entire project. Each colour represents a different level of performance or risk, making it easy to understand status at a glance.

Source: <https://www.prince2training.co.uk/blog/rag-status-in-project-management>

25. What Is RAG Status In Project Management? Red-Amber-Green Explained - Designveloper

July 15, 2026 - RAG reporting supports consistent reporting across teams. A PMO or portfolio manager can compare multiple initiatives only when the meaning of red, amber, and green is shared. Eleco's 2026 discussion of project RAG status meanings notes that highlight reports may use RAG ratings for the whole project and for specific aspects such as stakeholder commitment or team performance.

Source: <https://www.designveloper.com/blog/rag-status/>

26. Releases · infiniflow/ragflow

4 days ago - Release nightly created from 2db8eb6 at 2026-08-23 21:19:13+08:00

Source: <https://github.com/infiniflow/ragflow/releases>

27. 20 Advanced RAG Types to Know in 2026

2 weeks ago - This 2026 Systematization of Knowledge (SoK) paper frames Agentic RAG as a fragmented but increasingly important architecture for sequential reasoning, dynamic memory management, and iterative retrieval.

Source: <https://www.turingpost.com/p/ragtypes>

28. All you need to know about RAG (in 2026) - AI with Aish

March 21, 2026 - In 2024, it was enough to shove some PDFs into a vector database, run a cosine similarity search, and call it a day. In 2026, that approach; Naive RAG is seen as a prototype at best and a liability at worst.

Source: <https://aishwaryasrinivasan.substack.com/p/all-you-need-to-know-about-rag-in>

29. David E. Weekly on X: "Indexed .md files > RAG" / X

April 3, 2026 - Indexed .md files > RAG

Source: <https://x.com/dweekly/status/2039873942039715887>

30. How to Build a RAG Pipeline from Scratch in 2026 - kapa.ai - Instant AI answers to technical questions

January 27, 2026 - In this article, we will explore the latest developments that can help you build a robust production-grade RAG pipeline in 2026.

Source: <https://www.kapa.ai/blog/how-to-build-a-rag-pipeline-from-scratch-in-2026>

31. **Dynamic Memory Updating in RAG: Lifelong Learning and ...**

January 15, 2026 - Zhang et al., we analyze architectures that support continuous memory addition, deletion, consolidation, and re-weighting. These mechanisms transition RAG from static retrieval to incremental learning, mirroring biological memory processes.

Source: <https://www.ijisrt.com/assets/upload/files/IJISRT26JAN155.pdf>

32. **dirty-data/rag-v2**

April 6, 2026 - The goal is to keep one shared RAG implementation while supporting both LM Studio plugin workflows and MCP-hosted tool workflows.

Source: <https://lmstudio.ai/dirty-data/rag-v2>

33. **Releases · infiniflow/ragflow**

4 days ago - Release nightly created from 2db8eb6 at 2026-08-23 21:19:13+08:00

Source: <https://github.com/infiniflow/ragflow/releases>

34. **GitHub - OSU-NLP-Group/HippoRAG: [NeurIPS'24] HippoRAG is a novel RAG framework inspired by human long-term memory that enables LLMs to continuously integrate knowledge across external documents. RAG + Knowledge Graphs + Personalized PageRank. · GitHub**

1 month ago - [NeurIPS'24] HippoRAG is a novel RAG framework inspired by human long-term memory that enables LLMs to continuously integrate knowledge across external documents. RAG + Knowledge Graphs + P...

Source: <https://github.com/osu-nlp-group/hipporag>

35. Complete RAG Tutorial 2026 (Free Labs) - YouTube

🔗 RAG Bootcamp: <https://kode.wiki/4j9Bfyw> Learn Retrieval Augmented Generation (RAG) from scratch in this comprehensive crash course! Discover how to extend ... Published: December 24, 2025

Source: <https://www.youtube.com/watch?v=vT-DpLv29Q>

36. Локальная LLM с памятью и RAG: пошаговый гайд для... | AiManual

Пошаговый запуск локальной LLM через Ollama и Open WebUI: сохранение чатов, загрузка документов, настройка RAG и решение частых ошибок.

Source: <https://ai-manual.ru/article/polnyij-gajd-dlya-nachinayuschih-kak-s-nulya-zapustit-lokalnuyu-llm-s-pamyatyu-chatov-i-rag/>

37. Погода в Волгограде в марте 2026 года Волгоградская область...

Подробный прогноз погоды в Волгограде на март 2026 года - информация о дневной и ночной температуре, осадках в Волгограде, Волгоградская область Россия - World-Weather.Ru.

Source: <https://world-weather.ru/pogoda/russia/volgograd/march-2026/>

38. Самые смешные мемы за 16.08.2026

Если вам это не нравится, пожалуйста, покиньте сайт. 18+. 16.08.2026.

Source: <https://www.anekdot.ru/release/mem/day/2026-08-16/>

39. **Украина наносит новые удары. Что изменилось на... - YouTube**

Что изменилось на фронте? 02.08.2026. Сергей Ауслендер.Иранский замысел и "блокадная" нефть. Военный аналитик Давид Шарп. 19.08.2026.

Source: <https://www.youtube.com/watch?v=iOb6CACxYtg>

40. **Нелёгкая жизнь в Мариуполе - Тень Кремля**

17.08.2026, 07:01. В Черкассах военный ВСУ открыл огонь по детям из-за разбитого окна.15.08.2026, 15:25. На Украине «зерновой коридор» окончательно превращается в «зерновой тупик».

Source: <https://shoknews.ru/neljogkaja-zhizn-v-mariupole/>

[CRITIC Round 1]

<CONTROL>

We need to retrieve the key sources. Let me start by searching for the paper and the RAG files.

</CONTROL>

[SEARCH: RAG file 2509.26507v1.pdf]

[SEARCH: RAG file paper.tex]

[SEARCH: RAG file 2026-08-18_progress-report.md]

[SEARCH: RAG file 2026-08-23_progress-report.md]

[SEARCH: RAG file 2026-08-24_progress-report.md]

[SEARCH: RAG file 2026-08-22_paper-parity-plan.md]

[SEARCH: RAG file 2026-08-25_bdh-continual-learning-plan_v2.md]

Web Research

- **The Dragon Hatchling: The Missing Link between the Transformer and Models of the Brain**

September 30, 2025 - arXiv:2509.26507v1 [cs.NE] 30 Sep 2025 · Adrian Kosowski Thanks: Author contributions are listed at the end of the paper. Thanks: Corresponding author.

Source: <https://arxiv.org/html/2509.26507v1>

- **How to ingest PDF files for RAG. To ingest many PDFs into a RAG system... | by LLM | Medium**

December 8, 2025 - How to ingest PDF files for RAG To ingest many PDFs into a RAG system, I build a scalable pipeline: extract text using reliable PDF parsers (PyMuPDF, PDFPlumber, or UNIDOC for enterprise) ...

Source: https://medium.com/@generative_ai/how-to-ingest-pdf-files-for-rag-f7ddc8bdbf26

- **Unlocking PDFs for RAG: How RAG-Anything Handles Complex Documents | by Florian June | AI Exploration Journey | Medium**

June 10, 2026 - A parser can be thought of as a “document disassembler.” It takes inputs like PDFs, images, or Office documents and breaks them down into data chunks suitable for subsequent system processing. In RAG-Anything’s code, a unified interface (`raganything/parser.py:68`) defines key methods: `parse_pdf()`, `parse_image()`, `parse_document()`, and `check_installation()`.

Source: <https://medium.com/ai-exploration-journey/unlocking-pdfs-for-rag-how-rag-anything-handles-complex-documents-f8bbd716c734>

- **(PDF) REFRAG: Rethinking RAG based Decoding**

September 1, 2025 - To this end, we propose REFRAG, an efficient decoding framework that compresses, senses, and expands to improve latency in RAG applications. By exploiting the sparsity structure, we demonstrate a 30.85 the time-to-first-token acceleration (3.75 ...

Source: https://www.researchgate.net/publication/395213753_REFRAG_Rethinking_RAG_based_Decoding

- **[2509.01092] REFRAG: Rethinking RAG based Decoding**

October 12, 2025 - From: Aritra Ghosh [view email] [v1] Mon, 1 Sep 2025 03:31:44 UTC (636 KB) [v2] Sun, 12 Oct 2025 04:46:48 UTC (638 KB) ... View a PDF of the paper titled REFRAG: Rethinking RAG based Decoding, by Xiaoqiang Lin and 4 other authors

Source: <https://arxiv.org/abs/2509.01092>

- **Gemini RAG - File Search Tool - YouTube**

In this video, I go through the latest release from the Gemini API team, which is the file search tool, or basically a simple RAG system built right into the...

Source: <https://www.youtube.com/watch?v=MUP9ki6Bdtg>

- **Learn how to use upload images, audio, and other files | FlowiseAI**

This research paper. is one of many that compare RAG with longer context windows. To enable full file uploads, go to Chatflow Configuration, open the File Upload tab, and click the switch

Source: <https://docs.flowiseai.com/using-flowise/uploads>

- **Error: "Loading a class or package in a group. \usepackage" with .cls file...**

in the .cls file but it still did not work. However, when I try to compile from a single .tex file. It works well.in a .cls file (let's say it is "Main.cls") and in my .tex file (e.g., "Paper.tex"), I begin it with.

Source: <https://latex.org/forum/viewtopic.php?t=35969>

- **GitHub - Teju63/rag-file · GitHub**

RAG = Retrieval-Augmented Generation — a technique where the AI answers questions using YOUR documents instead of just its training data. What does this app do? You upload a PDF or TXT file (e.g. a research paper, a contract, a book chapter)...

Source: <https://github.com/Teju63/rag-file>

10. How to Do File Search in Gemini API? | Analytics Vidhya

For multimodal RAG, File Search also supports image files. This means you can upload documents and images into the same File Search Store and ask questions that require both textual and visual context.

Source: <https://www.analyticsvidhya.com/blog/2026/05/gemini-api-file-search/>

11. RAG Status in Project Management Explained (2026)

June 17, 2026 - Reporting dimension-level RAG, covering at a minimum time, cost, scope and risk, gives decision-makers the information they actually need. For guidance on structuring this kind of report effectively, IPM's progress report template provides a practical framework.

Source: <https://instituteprojectmanagement.com/blog/rag-status-in-project-management/>

12. Project RAG Status Meanings And Best Practice (2026)

July 6, 2026 - PM3 helps organisations standardise RAG reporting across projects, programmes and portfolios, with up to 10 status indicators covering areas such as cost, progress, risks, issues, stakeholders, scope and value.

Source: <https://eleco.com/pm3/knowledge-centre/how-many-rags/>

13. Releases · infiniflow/ragflow

4 days ago - Release nightly created from 2db8eb6 at 2026-08-23 21:19:13+08:00

Source: <https://github.com/infiniflow/ragflow/releases>

14. RAG Status for KPIs: The Red-Amber-Green Playbook (2026) | ClearPoint Strategy Blog

1 week ago - KPIs are crucial for understanding your organization's strengths and weaknesses. But let's be honest, staring at reports and relying on gut feelings isn't effective. You need a clear, objective way to evaluate your progress. A RAG status system provides just that.

Source: <https://www.clearpointstrategy.com/blog/establish-rag-statuses-for-kpis>

15. Progress Agentic RAG reviews 2026 | FitGap

April 11, 2026 - Progress Agentic RAG is an LLMOps-oriented capability from Progress that helps teams build retrieval-augmented generation (RAG) applications with agent-style orchestration over enterprise content.

Source: <https://us.fitgap.com/products/050333/progress-agentic-rag>

16. Какая осень будет в Тюмени в 2026 году: прогноз погоды на...

Осень 2026 в Тюмени: прогноз погоды по месяцам, когда ждать бабье лето, заморозки и первый снег. Опубликовано:05-07-2026. Обновлено:08-07-2026. 19 минут. Редакция Travelto.

Source: <https://travelto.ru/blog/kakaya-osen-bydet-v-2026-tyumen/>

17. Круиз Греческие острова с 29 августа по 08 сентября 2026 на...

Круиз Греческие острова с 29.08.2026 по 08.09.2026 (11 дней / 10 ночей) по маршруту Анталья - Измир - Кавала (2 дня, ночь на борту) - о.Миконос - о.Санторини - Афины - Бодрум - Анталья. Купите билет на круиз прямо сейчас!

Source: https://astoriagrande.com/destination/kruise_29_08-08_09_2026/

18. Самые смешные мемы за 16.08.2026

Мем, судьяКригс. 23.>. Самый смешной анекдот за 11.08: Дай человеку рыбу и он будет сыт один день. Подари ему удочку и он будет сыт всю жизнь.

Source: <https://www.anekdot.ru/release/mem/day/2026-08-16/>

19. Открываем RAG и интернет для LM Studio / Хабр

Исходные данные : ПК на базе AMD Ryzen™ AI 9 HX 370, ОЗУ 96 Гб из которых половина отдана под видеопамять. LM Studio, движок Vulkan llama.cpp (Windows). Модель для инференса: qwen/qwen3-coder-30b на...

Source: <https://habr.com/ru/articles/960174/>

20. **Дом-2. После заката / ДОМ-2. После заката (эфир от 12.08.2026)**

23 августа, 0:25 // После заката. 19 августа, 0:23 // После заката.

Source: <https://dom2.ru/efir/posle-zakata/20260812-dom-2-posle-zakata-efir-ot-12-08-2026/>

21. **RAG/rag_state_of_the_art_2026.md at main · dpsdagain/RAG**

This document synthesizes findings from 45+ targeted web searches conducted on May 26, 2026. It represents the current frontier of RAG system design — not beginner tutorials or toy demos. Every claim is grounded in published research, production reports, or framework documentation.

Source: https://github.com/dpsdagain/RAG/blob/main/rag_state_of_the_art_2026.md

22. **mini-enterprise-rag-agent/progress.md at main - GitHub**

Read relevant process skills and confirmed current architecture issue. Created planning files for this multi-step refactor. Added failing tests in tests/test_llm_tool_planning.py. Red run: 8 failed, 1 passed; failures match expected architecture gaps.

Source: <https://github.com/zhengshaoyi1387/mini-enterprise-rag-agent/blob/main/progress.md>

23. RAG4Reports 2026

In this task, a long-form report summarizing the relevant information in a corpus is generated in response to a report request, which consists of a user background and an information need. The generated report should provide proper attribution to the source documents to establish trust.

Source: <https://rag4reports.github.io/>

24. Build a RAG-powered Markdown documentation assistant

This proof-of-concept demonstrates how Retrieval-Augmented Generation (RAG) can transform a single Markdown file into a conversational, searchable resource. However, the potential for this assistant goes far beyond a single document or a command-line interface.

Source: <https://developer.ibm.com/tutorials/build-rag-assistant-md-documentation/>

25. Build a Production-Ready RAG System Over Your Own Documents in 2026 – A ...

In 2026, if your RAG system hallucinates on important queries, returns irrelevant chunks, or costs a fortune to run, it won't survive production. This tutorial walks you through building a reliable, evaluable, and scalable RAG pipeline that you can actually put behind an API or in a product.

Source: https://dev.to/dharshan_a_23835c7dc05682/build-a-production-ready-rag-system-over-your-own-documents-in-2026-a-practical-tutorial-4hd0

26. Visit this link to download: Update_FED_2026.08.17-2026.08.22

Log in Login to upload files Sign up. You have been invited by to create account and get 1GB.Unlimited. Until. 2026-08-25 - 2026-09-25.

Source: <https://files.fm/f/xzsang493zvdparx>

27. Любовный гороскоп Рыб на 2026 год | ГОРОСКОПЫ 365

Любовный гороскоп на 2026 год для Рыб | Для Рыб 2026 год — это время взросления и обретения внутренней силы. В вашем знаке находится строгий учитель Сатурн, который требует от вас определенности, ответственности и умения строить прочные берега для...

Source: <https://goroskop365.ru/2026-god/goroskop-ryby/lyubov/>

28. Deep House 2026 слушать и скачать музыку, треки, mp3 – LightAudio

Deep House Mix 2026 vol.22 Track 08 — MissDeep. 2:27. Ночной deep house 2026 - фоновый кач — Хитовый чел, Русские танцевальные новинки.

Source: <https://lightaudio.ru/mp3/deep+house+2026>

29. Одиссея (2026) смотреть онлайн фильм в хорошем HD качестве...

foot 22 июля 2026 20:31. а те кто ставит дизлайки долбаебы которые не умеют своими конченными мозгами нормально оценивать фильм, вроде тебя.trongl 17 июля 2026 20:08. Не слушайте никого! Фильм хорош даже в таком качестве.

Source: <https://lordfilmqs.ru/59202-odissey-a-2026.html>

30. Restored Republic via a GCR as of August 22, 2026

Restored Republic via a GCR Update as of Sat. 22 Aug. 2026. By 2011 when South Africa joined BRICS (Brazil, Russia, India, China and South Africa) the organization had formed a plan to save the world economy by doing a Global Currency Reset.

Source: <https://operationdisclosureofficial.com/2026/08/22/restored-republic-via-a-gcr-as-of-august-22-2026/>

31. [2502.14802] From RAG to Memory: Non-Parametric Continual ...GitHub - Andrew-Jang/RAGHub: A community-driven collection of ...[2604.05096] RAG or Learning? Understanding the Limits of LLM ...GitHub - headroomlabs-ai/headroom: Compress tool outputs ...From RAG to Memory: Non-Parametric Continual Learning for ...The Complete Guide to RAG: Naive, Advanced, and Graph RAG in ...Local RAG: Search Your Documents with a Private AI

Feb 20, 2025 · Given the challenges in continual learning with large language models (LLMs), retrieval-augmented generation (RAG) has become the dominant way to introduce new information. This is a community-driven project for r/RAG, where we aim to catalog the rapid growth of RAG tools and projects that are pushing the boundaries of the field. Each day, it feels like a new tool or framework emerges, and choosing the right one is becoming more of an art than a science. Apr 6, 2026 · In this work, we introduce a new benchmark of real-world dynamic events, constructed from time-stamped evidence that captures how knowledge evolves over time, which enables systematic evaluation of model adaptation under continuous knowledge drift. Headroom compresses everything your AI agent reads — tool outputs, logs, RAG chunks, files, and conversation history — before it reaches the LLM. Same answers, fraction of the tokens. May 1, 2025 · This paper presents a method to enhance traditional RAG with a continual learning framework for large language models. Apr 27, 2026 · From RAG basics to Graph RAG, Agentic RAG, and the LLM Wiki pattern, theory, runnable code, the latest trends, and a decision guide, all on a single page. Jan 30, 2026 · Quick Answer: RAG lets your local LLM answer questions about your own files — PDFs, notes, code, whatever — without fine-tuning or sending data to

the cloud. The fastest way to start: install Ollama, pull a model and an embedding model, then use Open WebUI's built-in document upload.

Source: <https://arxiv.org/abs/2502.14802>

32. GitHub - Andrew-Jang/RAGHub: A community-driven collection of ...

This is a community-driven project for r/RAG, where we aim to catalog the rapid growth of RAG tools and projects that are pushing the boundaries of the field. Each day, it feels like a new tool or framework emerges, and choosing the right one is becoming more of an art than a science.

Source: <https://github.com/Andrew-Jang/RAGHub>

33. [2604.05096] RAG or Learning? Understanding the Limits of LLM ...GitHub - headroomlabs-ai/headroom: Compress tool outputs ...From RAG to Memory: Non-Parametric Continual Learning for ...The Complete Guide to RAG: Naive, Advanced, and Graph RAG in ...Local RAG: Search Your Documents with a Private AI

Apr 6, 2026 · In this work, we introduce a new benchmark of real-world dynamic events, constructed from time-stamped evidence that captures how knowledge evolves over time, which enables systematic evaluation of model adaptation under continuous knowledge drift. Headroom compresses everything your AI agent reads — tool outputs, logs, RAG chunks, files, and conversation history — before it reaches the LLM. Same answers, fraction of the tokens. May 1, 2025 · This paper presents a method to enhance traditional RAG with a continual learning framework for large language models. Apr 27, 2026 · From RAG basics to Graph RAG, Agentic RAG, and the LLM Wiki pattern, theory, runnable code, the latest trends, and a decision guide, all on a single page. Jan 30, 2026 · Quick Answer: RAG lets your local LLM answer questions about your own files — PDFs, notes, code, whatever — without fine-tuning or sending data to

the cloud. The fastest way to start: install Ollama, pull a model and an embedding model, then use Open WebUI's built-in document upload.

Source: <https://arxiv.org/abs/2604.05096>

34. From RAG to Memory: Non-Parametric Continual Learning for ...

May 1, 2025 · This paper presents a method to enhance traditional RAG with a continual learning framework for large language models.

Source: <https://openreview.net/forum?id=LWH8yn4HS2>

35. The Complete Guide to RAG: Naive, Advanced, and Graph RAG in ...

Apr 27, 2026 · From RAG basics to Graph RAG, Agentic RAG, and the LLM Wiki pattern, theory, runnable code, the latest trends, and a decision guide, all on a single page.

Source: <https://www.mrlatte.net/en/research/2026/04/27/rag-complete-guide/>

36. Knowledge Rag MCP Server | Awesome MCP Servers

1 month ago - Local RAG system for Claude Code with hybrid search (semantic + BM25), cross-encoder reranking, markdown-aware chunking, 9 file formats, file watcher, and...

Source: <https://mcpservers.org/servers/lyonzin/knowledge-rag>

37. Releases · infiniflow/ragflow

4 days ago - RAGFlow is a leading open-source Retrieval-Augmented Generation (RAG) engine that fuses cutting-edge RAG with Agent capabilities to create a superior context layer for LLMs - infiniflow/ragflow

Source: <https://github.com/infiniflow/ragflow/releases>

38. RAG in 2026: Architecture Shifts, Emerging Patterns, and What It Means for Java Developers | by soufiane ELAMMARI | Medium

May 2, 2026 - RAG in 2026: Architecture Shifts, Emerging Patterns, and What It Means for Java Developers A technical deep-dive into where Retrieval-Augmented Generation is heading and how to implement the new ...

Source: <https://medium.com/@elammarisoufiane/rag-in-2026-architecture-shifts-emerging-patterns-and-what-it-means-for-java-developers-6f2803e39787>

39. Cloudflare DDoS Threat Report H1 2026: 1 Tbps attacks soar as DNS floods and geopolitical tensions drive a new wave | Cloudflare Blog

2 weeks ago - The attack-vector center of gravity shifted from botnet floods to reflection and amplification. DNS-based attacks accounted for 34.3% of all network-layer activity in the first half of 2026, with · DNS Floods alone climbing from 25.7% to 40.0% ...

Source: <https://blog.cloudflare.com/ddos-threat-report-2026-h1/>

40. David E. Weekly on X: "Indexed .md files > RAG" / X

April 3, 2026 - Indexed .md files > RAG

Source: <https://x.com/dweekly/status/2039873942039715887>

[OPTIMIZER Round 1]

I'll first try to extract the actual architectural definitions from the paper, since the BDH details determine whether the formalization below is faithful.

[SEARCH: 2509.26507 Dragon Hatchling architecture shared weights depth composition hidden state equations]

Let me also check whether the project reports are retrievable under their exact filenames.

[SEARCH: "2026-08-25_bdh-continual-learning-plan_v2.md"]

[SEARCH: "2026-08-25_cl-h1-report.md"]

Web Research

- **The Dragon Hatchling: The Missing Link between the Transformer...**

Playing with the Hatchling. Conclusions. Connection between generalization of reasoning and computational expressiveness. arXiv: 2509.26507v1 [cs.NE] 30 Sep 2025. Adrian Kosowski†. Przemysław Uznanski† Jan Chorowski† Michał Bartoszkiewicz†.

Source: <https://arxiv.org/pdf/2509.26507>

- **Paper page - The Dragon Hatchling: The Missing Link between the...**

We introduce 'Dragon Hatchling' (BDH), a new Large Language Model architecture based on a scale-free biologically inspired network of \n

locally-interacting neuron particles. Get this paper in your agent: hf papers read 2509.26507. Don't have the latest CLI?

Source: <https://huggingface.co/papers/2509.26507>

• **The Dragon Hatchling: The Missing Link between... | Papers with Code**

Paper · arXiv:2509.26507 · 30 September 2025. We introduce 'Dragon Hatchling' (BDH), a new Large Language Model architecture based on a scale-free biologically inspired network of n locally-interacting neuron particles.

Source: <https://paperswithcode.co/paper/2509.26507>

• **Dragon Hatchling: Linking Transformers & Brain Models**

We introduce 'Dragon Hatchling' (BDH), a new LLM architecture based on a scale-free biologically inspired network of n locally-interacting neuron particles. BDH couples strong theoretical foundations and inherent interpretability without sacrificing Transformer-like performance.

Source: <https://www.emergentmind.com/papers/2509.26507>

• **The Dragon Hatchling: The Missing Link Between... - YouTube**

Paper: The Dragon Hatchling: The Missing Link between the Transformer and Models of the Brain (2509.26507) Published: 30 Sep 2025. Learn more on Emergent Mind

Source: <https://www.youtube.com/watch?v=8bJEFTueYAY>

• **

Source: /clev?

event=StartpageResultClick&sc=2sbbv9IndI59e85YPmeezk8kjlZctVdWOOPHiers0IurVZ7rsH
web","transport":"href-request"}

- **

Source: /clev?

event=StartpageResultClick&sc=2sbbv9Ind9wPZhGMPESMGy8isCfxfcnpu5InTusnscgLzIVN
web","transport":"href-request"}

[MODERATOR Round 1]

[moderator] Round 1: LLM call failed (litellm.RateLimitError:

RateLimitError: OpenrouterException - {"error":{"message":"Provider
returned error","code":429,"metadata":{"raw":"stealth/ox-alpha is
temporarily rate-limited upstream. Please retry

shortly.","provider_name":"Stealth","is_byok":false,"limit_source":"upstream_provider_shar

shortly, add your own provider key ([https://openrouter.ai/settings/
integrations](https://openrouter.ai/settings/integrations)), or route to another provider with provider routing: [https://
openrouter.ai/docs/features/provider-
routing](https://openrouter.ai/docs/features/provider-routing)"},"user_id":"user_3Ay1IY7xf9lrDwaN7Fq4EUJnQA9"})

[STRATEGIST Round 2]

Strategic Analysis: Input-Path Interference in BDH via Gating/Projection

1. Context Reconstruction

From the available sources, BDH (Dragon Hatchling) is a "scale-free biologically inspired network of n locally-interacting neuron particles" that serves as a bridge between transformer architectures and brain models. The key architectural feature relevant here is **shared sequential composition**: the same parameterized transformation is applied repeatedly across depth, unlike transformers which use distinct per-layer weights. This creates a continual learning challenge because updating shared parameters for new tasks risks catastrophic interference with previously learned computations.

2. Formal Definitions

Definition 1: Shared Sequential Composition

Let $\mathcal{X}$ be the input/state space (typically $\mathbb{R}^d$). A **shared sequential composition** of depth D is defined by a single parameterized map $f_\theta: \mathcal{X} \rightarrow \mathcal{X}$ (the **frozen shared path**, with θ fixed) applied iteratively:

$$h_0 = x, \quad h_{t+1} = f_\theta(h_t), \quad t = 0, \dots, D-1$$

The output is $y = h_D = f_\theta^D(x)$, where f_θ^D denotes D -fold composition.

Classification: This is a *direct characterization* of the BDH architecture's depth-sharing mechanism as described in the paper (§3–§4 of arXiv: 2509.26507).

Definition 2: Frozen Shared Path

The shared path f_{θ} is **frozen** if θ is not updated during the mechanism under investigation. All parameters of f_{θ} are fixed.

Classification: *Known* — standard continual learning constraint.

Definition 3: Input-Path Interference

Consider two inputs $x_1, x_2 \in \mathcal{X}$. **Input-path interference** at depth t is the phenomenon whereby the intermediate state $h_t(x_1)$ under the shared composition is altered when a mechanism designed for x_2 is also active. Formally, define a mechanism M that produces a modified computation $\tilde{h}_t^{(M)}(x)$. Interference occurs when:

$$\tilde{h}_t^{(M)}(x_1 \mid x_2 \text{ active}) \neq \tilde{h}_t^{(M)}(x_1 \mid x_2 \text{ absent})$$

More precisely, in the shared-path setting, interference arises because the same f_{θ} is used for all inputs. If we introduce input-dependent modifications g_x , then for the composition to remain faithful, we need:

$$\forall x: \tilde{f}^D_{g_x}(x) = f_{\theta}^D(x)$$

where $\tilde{f}_{g_x}$ is the modified per-depth map. Interference is present if the modification g_{x_1} needed to preserve $f_{\theta}^D(x_1)$ conflicts with g_{x_2} needed for x_2 .

Classification: *Plausible adaptation* from continual learning interference concepts, formalized for the shared-composition setting.

Definition 4: Gating

An **input-dependent gate** is a function $g: \mathcal{X} \rightarrow \mathcal{W}$ (where $\mathcal{W}$ is a space of weight/scalar vectors) that modifies the shared path at each depth:

$$\tilde{f}_{g(x)}(h) = \sigma(g(x) \odot f_{\theta}(h)) \quad \text{or} \quad \tilde{f}_{g(x)}(h) = f_{\theta}(g(x) \odot h)$$

The gate g depends only on the original input x , not on the depth index t nor on the intermediate state h_t . The gate g itself may have trainable parameters ϕ , but crucially **no per-depth parameters**: the same g_{ϕ} is used at every depth.

Classification: *Known* mechanism (cf. LSTM gates, attention), but the constraint of depth-independence and frozen shared path is a *novel combination*.

Definition 5: Projection

An **input-dependent projection** is a function $P: \mathcal{X} \rightarrow \text{End}(\mathcal{X})$ mapping each input to a linear projection operator $P(x) \in \mathbb{R}^{d \times d}$, applied to the state before or after each depth:

$$\tilde{f}_{P(x)}(h) = P(x) \cdot f_{\theta}(h) \quad \text{or} \quad \tilde{f}_{P(x)}(h) = f_{\theta}(P(x) \cdot h)$$

Again, P depends only on x , not on depth t .

Classification: *Known* (cf. gradient projection methods), formalized for forward-pass application.

Definition 6: Computation-Isolability

A shared sequential composition f_{θ}^D is **computation-isolable** if there exists a mechanism M (gate or projection) such that for any set of inputs $\{x_1, \dots, x_N\}$:

$$\forall i, \forall t \in \{0, \dots, D\}: \tilde{h}_t^{(M)}(x_i) = h_t(x_i)$$

That is, the mechanism preserves the exact computation of f_{θ}^D for each input independently, with no cross-input contamination.

Classification: *Apparently novel* — this specific formulation combining shared composition, frozen parameters, and forward-pass isolation does not appear in the surveyed literature.

Definition 7: Elimination vs. Bounding of Interference

- **Elimination:** The mechanism achieves computation-isolability exactly (Definition 6). For all inputs and all depths, the modified computation equals the original: $\tilde{h}_t(x) = h_t(x)$.
- **Bounding:** There exists $\epsilon > 0$ such that for all inputs x and all depths t :

$$\|\tilde{h}_t(x) - h_t(x)\| \leq \epsilon$$

where ϵ may depend on the Lipschitz constant of f_θ , the depth D , and the capacity of the gate/projection, but not on the specific input.

Classification: *Plausible adaptation* of approximation theory concepts.

3. Hypotheses

Hypothesis H1 (Negative — Impossibility of Elimination)

Statement: For a general nonlinear frozen shared path f_θ , there does not exist an input-dependent gate $g: \mathcal{X} \rightarrow \mathbb{R}^d$ (without per-depth parameters) such that the gated composition preserves the original computation exactly for all inputs.

Assumptions:

- (A1) $f_{\theta}: \mathbb{R}^d \rightarrow \mathbb{R}^d$ is nonlinear (specifically, not an affine map).
- (A2) The gate $g: \mathbb{R}^d \rightarrow \mathbb{R}^d$ depends only on the input x , not on depth t or intermediate state h_t .
- (A3) The gate has no per-depth trainable parameters.
- (A4) $D \geq 2$ (at least two compositions).
- (A5) The gate is applied multiplicatively: $\tilde{f}_{\{g(x)\}}(h) = g(x) \odot f_{\theta}(h)$ or $\tilde{f}_{\{g(x)\}}(h) = f_{\theta}(g(x) \odot h)$.

Mathematical object being gated: The output (or input) of the frozen shared path f_{θ} at each depth.

Claim: Under (A1)–(A5), exact elimination of interference is impossible.

Proof sketch:

Consider the requirement that the gated computation matches the original for all x :

$$\tilde{f}_{\{g(x)\}}^D(x) = f_{\theta}^D(x)$$

For depth $D = 1$:

$$g(x) \odot f_{\theta}(x) = f_{\theta}(x) \quad \Longleftrightarrow \quad g(x) = \mathbf{1} \quad \text{(where } f_{\theta}(x) \neq 0 \text{)}$$

For depth $D = 2$:

$$g(x) \odot f_{\theta}(g(x) \odot f_{\theta}(x)) = f_{\theta}(f_{\theta}(x))$$

If $g(x) = \mathbf{1}$ (from the $D=1$ requirement), this becomes $f_{\theta}(f_{\theta}(x)) = f_{\theta}(f_{\theta}(x))$, which is trivially satisfied. But this means the gate is the identity — it does nothing — and provides no isolation whatsoever.

Now consider the alternative gate placement $\tilde{f}_{\theta}(g(x))(h) = f_{\theta}(g(x) \odot h)$:

For $D = 1$: $f_{\theta}(g(x) \odot x) = f_{\theta}(x)$, which requires $g(x) \odot x = x$ (assuming f_{θ} is injective on the relevant domain), giving $g(x) = \mathbf{1}$ where $x \neq 0$.

For $D = 2$: $f_{\theta}(g(x) \odot f_{\theta}(g(x) \odot x)) = f_{\theta}(f_{\theta}(x))$.

With $g(x) = \mathbf{1}$: $f_{\theta}(f_{\theta}(x)) = f_{\theta}(f_{\theta}(x))$. Again trivial.

The fundamental issue: **any gate that preserves the original computation must be the identity**, which provides zero isolation. A non-trivial gate necessarily alters the computation, creating a tension between preservation and isolation.

More precisely, suppose we want the gate to provide isolation by mapping different inputs to different "channels." For the gate to be non-trivial ($g(x) \neq \mathbf{1}$ for some x), we need:

$$g(x) \odot f_{\theta}(g(x) \odot f_{\theta}(x)) = f_{\theta}(f_{\theta}(x))$$

This is a functional equation in g . For a general nonlinear f_{θ} , this equation is overdetermined: the left side depends on $g(x)$ in two places (once inside f_{θ} , once outside), while the right side is fixed. For generic f_{θ} , no solution g exists except $g \equiv \mathbf{1}$.

Weakest additional assumptions for truth: If f_{θ} is linear ($f_{\theta}(h) = Wh + b$), then the problem is trivially solvable with $g = \mathbf{1}$ because linear compositions don't create input-dependent interference. If f_{θ} is a contraction with a unique fixed point, approximate isolation may be achievable (see H2).

Assessment: *Plausible adaptation* — the proof structure follows from basic functional equation analysis, but the precise conditions on f_{θ} for the impossibility to hold require further characterization.

Hypothesis H2 (Positive — Bounded Interference via Contractivity)

Statement: If the frozen shared path f_{θ} is a contraction mapping with Lipschitz constant $L < 1$, then input-path interference can be bounded by $\epsilon = O(L^D)$ without any gating mechanism at all, and a simple input-dependent projection can reduce the bound further.

Assumptions:

- (B1) f_θ is L -Lipschitz with $L < 1$.
- (B2) The state space is a complete metric space.
- (B3) The projection $P(x)$ is an orthogonal projection onto a subspace $S_x \subset \mathbb{R}^d$.

Mathematical object being projected: The intermediate state h_t projected onto S_x before each application of f_θ .

Claim: Under (B1)–(B3), the interference between inputs x_1 and x_2 at depth t satisfies:

$$\|h_t(x_1) - h_t(x_2)\| \leq L^t \|x_1 - x_2\|$$

This is a direct consequence of the Banach fixed-point theorem and the contraction property.

Proof: By induction. Base case: $\|h_0(x_1) - h_0(x_2)\| = \|x_1 - x_2\|$. Inductive step:

$$\begin{aligned} \|h_{t+1}(x_1) - h_{t+1}(x_2)\| &= \|f_\theta(h_t(x_1)) - f_\theta(h_t(x_2))\| \\ &\leq L \|h_t(x_1) - h_t(x_2)\| \leq L^{t+1} \|x_1 - x_2\| \end{aligned}$$

For the projection-enhanced version, if $P(x)$ projects onto a subspace where f_θ has a smaller Lipschitz constant $L' < L$, then:

$$\|P(x) \cdot h_t(x_1) - P(x) \cdot h_t(x_2)\| \leq L'^t \|x_1 - x_2\|$$

Assessment: *Known/direct consequence* — the contraction bound is standard. The projection enhancement is a *plausible adaptation* of spectral methods.

Caveat: This bounds the *separation* between different inputs' trajectories, not the *interference* from shared modifications. It shows that different inputs naturally diverge under contraction, but doesn't address the core problem of how to apply input-specific modifications without cross-contamination.

Hypothesis H3 (Negative — Depth-Agnostic Gate Cannot Adapt to Nonlinear Dynamics)

Statement: For a generic nonlinear f_{θ} , an input-dependent gate $g(x)$ that is depth-agnostic cannot simultaneously (i) preserve the computation $f_{\theta}^D(x)$ and (ii) provide non-trivial isolation between different inputs' computational paths.

Assumptions:

- (C1) f_{θ} is smooth and nonlinear.
- (C2) $D \geq 3$.
- (C3) The gate $g(x)$ is a continuous function.
- (C4) The gate is applied as $\tilde{f}_{g(x)}(h) = g(x) \odot f_{\theta}(h)$.

Mathematical object being gated: The output of f_{θ} at each depth, modulated by $g(x)$.

Claim: Under (C1)–(C4), the only gate that preserves the computation for all inputs is $g(x) = \mathbf{1}$, which provides no isolation.

Proof sketch:

The preservation condition at depth D requires:

$$\underbrace{g(x) \odot f_{\theta} \big| (g(x) \odot f_{\theta} \big| (\cdots g(x) \odot f_{\theta(x)} \big|) \big|)}_{D \text{ applications}} = f_{\theta}^D(x)$$

Define $\phi_x(h) = g(x) \odot f_{\theta}(h)$. The condition is $\phi_x^D(x) = f_{\theta}^D(x)$.

At depth 1: $g(x) \odot f_{\theta}(x) = f_{\theta}(x)$, so $g(x)_i = 1$ whenever $[f_{\theta}(x)]_i \geq 0$.

At depth 2: $g(x) \odot f_{\theta}(g(x) \odot f_{\theta}(x)) = f_{\theta}(f_{\theta}(x))$.

Since $g(x) = \mathbf{1}$ (from depth 1, on the relevant domain), this becomes $f_{\theta}(f_{\theta}(x)) = f_{\theta}(f_{\theta}(x))$. ✓

But $g(x) = \mathbf{1}$ means the gate is trivial. Any perturbation $g(x) = \mathbf{1} + \delta(x)$ with $\delta(x) \neq 0$ will, at some depth, cause:

$$\tilde{h}_t = (\mathbf{1} + \delta(x)) \odot f_\theta(\tilde{h}_{t-1}) \neq f_\theta(\tilde{h}_{t-1})$$

and the error compounds through subsequent compositions.

Assessment: *Plausible adaptation* — the argument is structurally sound but relies on the gate being applied at every depth. If the gate is applied selectively (e.g., only at certain depths), the analysis changes.

Weakest additional assumptions for truth: The result holds for any nonlinear f_θ that is injective on the relevant domain. If f_θ has symmetries (e.g., $f_\theta(h) = f_\theta(-h)$), non-trivial gates may exist.

Hypothesis H4 (Conditional Positive — Depth-Conditioned Gate)

Statement: If the gate is allowed to depend on both the input x and the depth index t (but with shared parameters — a single network $g_\phi(x, t)$), then computation-isolability can be achieved for a broad class of shared paths f_θ .

Assumptions:

- (D1) The gate $g_\phi: \mathcal{X} \times \{0, \dots, D-1\} \rightarrow \mathbb{R}^d$ is a single neural network with shared parameters ϕ .
- (D2) f_θ is smooth and its Jacobian $J_f(h) = \partial f_\theta / \partial h$ has full rank on the relevant domain.
- (D3) The gate is applied as $\tilde{f}_{g_\phi(x,t)}(h) = g_\phi(x,t) \odot f_\theta(h)$.

Mathematical object being gated: The output of f_θ at depth t , modulated by $g_\phi(x, t)$.

Claim: Under (D1)–(D3), for any finite D and any smooth f_θ , there exists a gate g_ϕ (with $O(d \cdot D)$ parameters) such that the gated computation preserves the original: $\tilde{f}^D(x) = f_\theta^D(x)$ for all x .

Proof sketch: At each depth t , the preservation condition is:

$$g_\phi(x, t) \odot f_\theta(h_t) = f_\theta(h_t)$$

where $h_t = f_\theta^t(x)$. This requires $g_\phi(x, t)_i = 1$ whenever $[f_\theta(h_t)]_i \neq 0$.

A gate that simply outputs $\mathbf{1}$ for all (x, t) trivially satisfies this. But for isolation, we want $g_\phi(x, t)$ to also project away components that would interfere with other inputs' computations.

The key insight: if we define $g_\phi(x, t) = \mathbf{1} + P_t(x)$ where $P_t(x)$ is a projection onto the null space of f_θ 's Jacobian at h_t , then:

- Preservation: $g_\phi(x, t) \odot f_\theta(h_t) = f_\theta(h_t) + P_t(x) \odot f_\theta(h_t) = f_\theta(h_t)$ if $P_t(x) \odot f_\theta(h_t) = 0$.
- Isolation: $P_t(x)$ projects away components that are "shared" between different inputs' intermediate states.

However, this requires P_t to depend on $h_t = f_\theta^t(x)$, which in turn depends on x and t . A single network $g_\phi(x, t)$ can learn this mapping if given sufficient capacity.

Assessment: *Conjecture* — the existence of such a gate is plausible but not proven. The construction requires the gate to implicitly compute $f_\theta^t(x)$, which may require the gate to be as complex as the shared path itself.

Weakest additional assumptions for truth: The gate network g_ϕ must have sufficient capacity to represent the function $(x, t) \mapsto$ appropriate depth-dependent modulation. This may require g_ϕ to be at least as expressive as f_θ^D .

4. Prior Art Analysis

4.1 Orthogonal Gradient Methods (OGM)

Source: Lopez-Paz & Ranzato (2017), "Gradient Episodic Memory for Continual Learning"; Zeng et al. (2019).

Relevance: OGM projects gradients onto the orthogonal complement of previously used gradient directions to prevent interference during *training*.

Does it address the same problem? No. OGM operates in parameter space during backpropagation. Our problem concerns forward-pass computation isolation with frozen parameters. The mathematical objects are different: OGM modifies $\nabla_{\theta} \mathcal{L}$, while we seek to modify $f_{\theta}(h)$.

Classification: *Known technique, not directly applicable.*

4.2 Progressive Neural Networks

Source: Rusu et al. (2016).

Relevance: Adds new columns (with new parameters) for each task, with lateral connections to previous columns.

Does it address the same problem? No. Progressive networks introduce per-task parameters, violating our constraint of no per-depth/per-level trainable parameters. They also require architectural modification, not just gating.

Classification: *Known technique, violates constraints.*

4.3 Hypernetworks

Source: Ha et al. (2016).

Relevance: A hypernetwork generates weights for a main network conditioned on some input.

Does it address the same problem? Partially. A hypernetwork $H_\phi(x)$ could generate per-depth weights $W_t = H_\phi(x, t)$, effectively creating input-dependent per-depth parameters. However:

- This requires the hypernetwork to have per-depth output dimensions.
- The generated weights replace f_θ rather than gating it.
- The hypernetwork itself has trainable parameters ϕ .

Classification: *Known technique, relevant but violates the "no per-depth trainable parameters" constraint in spirit (the hypernetwork's output is per-depth).*

4.4 Conditional Computation / Mixture of Experts

Source: Shazeer et al. (2017), "Outrageously Large Neural Networks."

Relevance: Routes inputs to different expert subnetworks based on input-dependent gating.

Does it address the same problem? Partially. MoE uses input-dependent routing to select which parameters to use, which is a form of input-dependent computation. However:

- MoE typically has per-layer gating (depth-dependent routing).
- The experts are distinct networks, not a single shared path.
- MoE doesn't address the problem of preserving a specific shared-path computation.

Classification: *Known technique, structurally related but solves a different problem.*

4.5 Parameter Isolation / Masking

Source: Mallya & Lazebnik (2018), "PackNet"; Wortsman et al. (2020), "Supermasks."

Relevance: Uses binary masks to select which parameters are active for each task.

Does it address the same problem? Partially. Masking can be input-dependent (as in supermasks), but:

- Masks are typically per-layer, not shared across depth.
- Masking modifies which parameters are used, not how a shared path is composed.
- The masks are usually learned per-task, not per-input.

Classification: *Known technique, relevant analogy but different mathematical structure.*

4.6 Recurrent Neural Networks with Gating (LSTM/GRU)

Source: Hochreiter & Schmidhuber (1997); Cho et al. (2014).

Relevance: LSTMs use input-dependent gates (forget, input, output) to control information flow through a recurrent shared computation.

Does it address the same problem? Yes, most closely. LSTM gates are:

- Input-dependent (they depend on x_t and h_{t-1}).
- Applied at each depth (time step).
- Used to control what information persists through shared recurrence.

However, LSTM gates depend on the *current state* h_{t-1} , not just the original input x . This is a crucial difference: our constraint requires the gate to depend only on x , not on intermediate states.

Classification: *Known technique, directly relevant but violates the "input-only" constraint.*

4.7 Attention as Gating

Source: Vaswani et al (2017).

Relevance: Attention mechanisms compute input-dependent weights to modulate information flow.

Does it address the same problem? No. Attention operates within a single layer and depends on the current representation, not on a fixed shared path. It also requires per-layer parameters (Q, K, V projections).

Classification: *Known technique, different problem.*

5. Novelty Classification

| Claim | Classification |

|-----|-----|

| Shared sequential composition creates interference | **Known** (standard in RNN theory) |

| Input-dependent gating can control information flow | **Known** (LSTM, MoE) |

| Depth-agnostic gate cannot achieve exact isolation for nonlinear shared paths | **Plausible adaptation** (novel combination of known results) |

| Contractivity bounds interference without gating | **Direct consequence** (Banach fixed-point theorem) |

| Depth-conditioned gate with shared parameters may achieve isolation | **Conjecture** (no direct precedent) |

| The specific formulation of computation-isolability for frozen shared paths | **Apparently novel** |

| The impossibility result for depth-agnostic gates | **Apparently novel** (though structurally similar to results in dynamical systems theory) |

6. Obstruction Analysis

The Core Obstruction

The fundamental obstruction to input-path isolation via depth-agnostic gating is a **temporal information deficit**: the gate cannot distinguish which depth it is operating at, and therefore cannot adapt its behavior to the depth-specific state of the computation.

More precisely:

- **The gate sees only x , not h_t .** The intermediate state $h_t = f_{\theta}^t(x)$ encodes the cumulative effect of t applications of the shared path. Different inputs x_1, x_2 may have similar h_t values at some depths but divergent values at others. A depth-agnostic gate cannot detect this.
- **Nonlinearity creates depth-dependent dynamics.** For a nonlinear f_{θ} , the Jacobian $J_f(h_t)$ varies with t (since h_t changes). The "directions of interference" at depth t depend on $J_f(h_t)$, which the gate cannot access.
- **The preservation-isolation tradeoff.** Any gate that preserves the original computation must satisfy $g(x) \odot f_{\theta}(h) = f_{\theta}(h)$ (or equivalent), which forces $g(x) = \mathbf{1}$ on the

relevant domain. A non-trivial gate necessarily alters the computation, creating a tradeoff between fidelity and isolation.

Characterization of the Obstruction

The obstruction can be formalized as a **rank deficiency** in the gate's information:

Let $\mathcal{I}_t(x) = (x, h_1, \dots, h_t)$ be the full information available at depth t . The gate has access only to $\mathcal{I}_0(x) = \{x\}$. The information gap is:

$$\Delta_t(x) = \dim(\mathcal{I}_t(x)) - \dim(\mathcal{I}_0(x)) = t \cdot d$$

For $t \geq 1$, the gate is missing $t \cdot d$ dimensions of information about the current state. This gap grows linearly with depth, making isolation increasingly difficult for deeper compositions.

Conditions Under Which the Obstruction Vanishes

- **Linearity:** If $f_\theta(h) = Wh + b$, then $h_t = W^t x + (W^{t-1} + \dots + I)b$. The state is a deterministic function of x and t , and a depth-conditioned gate $g(x, t)$ can perfectly track the computation. Moreover, linear compositions don't create nonlinear interference, so $g = \mathbf{1}$ suffices.
- **Contractivity with small L :** If $L \ll 1$, then $h_t \rightarrow h^*$ (the fixed point) exponentially fast. After a few depths, all inputs

converge to the same state, and interference becomes negligible. A gate is unnecessary.

- **Depth-conditioned gate:** If the gate can depend on t (via a shared network $g_\phi(x, t)$), the information gap Δ_t is closed, and isolation becomes feasible (subject to the gate's capacity).
- **State-conditioned gate:** If the gate can depend on h_t (as in LSTMs), the information gap is fully closed, and standard gating techniques apply.

7. Summary and Recommendations

Key Findings

- **Exact elimination of interference is impossible** under the stated constraints (depth-agnostic gate, frozen shared path, nonlinear f_θ). This is an *apparently novel* result, though structurally analogous to known limitations in dynamical systems.
- **Bounded interference is achievable** if the shared path is contractive, but this is a property of f_θ , not of the gate. The gate contributes nothing in this regime.

- **Depth-conditioned gating** (with shared parameters) is the most promising direction, but requires the gate to implicitly compute $f_{\theta}(x)$, which may necessitate gate capacity comparable to the shared path itself.
- **The LSTM approach** (state-conditioned gating) is the closest prior art, but violates the "input-only" constraint. Relaxing this constraint to allow state-conditioning would make the problem tractable using known techniques.

Strategic Recommendation

The investigation should pivot from "can we achieve isolation with depth-agnostic gating?" to "what is the minimal additional information the gate needs?" The answer is likely: **the gate needs access to the intermediate state h_t** (or equivalently, the depth index t with sufficient capacity to simulate f_{θ}^t). This suggests that:

- Either accept state-conditioned gating (LSTM-style), which is well-understood but adds complexity.
- Or accept approximate isolation via contractivity, which requires architectural constraints on f_{θ} .
- Or accept that the shared path must be retrained (or extended with per-depth parameters), which violates the original constraints but may be necessary.

The "failure condition" is met: **the proposed isolation is impossible under the stated assumptions**. The obstruction is the gate's inability to access depth-specific state information, which is a fundamental information-theoretic limitation.

[CRITIC Round 2]

[SEARCH: arXiv 2509.26507 Dragon Hatchling architecture equations shared depth "f_theta" composition]

[SEARCH: "Dragon Hatchling" BDH "shared" "depth" weights architecture]

[SEARCH: "2509.26507" "particle" "recurrent" depth equations]

Web Research

- **Reasoning Depth in AI and Logic**

Depth-recurrent and looped architectures (LoopFormer, depth-recurrent Transformers) further decouple parameter count from computational depth, allowing iterative application of a shared computation block across.

Source: <https://www.emergentmind.com/topics/reasoning-depth>

- **The Yale arXiv Paper Template - Overleaf, Online LaTeX Editor**

This document is both a working example and a usage guide for the Yale arXiv paper template.

Source: <https://www.overleaf.com/latex/templates/the-yale-arxiv-paper-template/wsmymnffgcs>

- **[Paper Note] Generative Modeling of Shape-Dependent Self-Contact...**

Overall Architecture¶. PAPoseDiff consists of three stages: 1. Data construction: Multi-camera dome capture from Goliath → SMPL-X fitting → contact map analysis → self-contact frame filtering 2. Generative training: Part-wise SMPL-X parameters →...

Source: https://en.papernotes.org/ICCV2025/image_generation/generative_modeling_of_shape-dependent_self-contact_human_poses/

- **Relation between $f \tan(\theta)$ and $f \theta$ projections.**

We introduce an approach to expand the eye-box in a retinal-projection-based near-eye display. The retinal projection display has the advantage of providing clear images in a wide depth range; however, it has difficulty in practical use with a narrow eye-box.

Source: https://www.researchgate.net/figure/Relation-between-f-tantheta-and-f-theta-projections_fig8_343450315

- **Generative Modeling by Estimating Gradients of the Data... | Yang Song**

Likelihood-based models either require strong restrictions on the model architecture to ensure a tractable normalizing constant for likelihood computation, or must rely on surrogate objectives to approximate maximum likelihood training.

Source: <https://yang-song.net/blog/2021/score/>

- **GitHub - severian42/BDH-MLX: Baby Dragon Hatchling (BDH) – MLX...**

Baby Dragon Hatchling (BDH) is a novel Large Language Model architecture that bridges the gap between Transformers and biologically-plausible neural networks. Unlike standard Transformers, BDH features: Key Innovations.

Source: <https://github.com/severian42/BDH-MLX>

- **The Dragon Hatchling: The Missing Link between the Transformer...**

We introduce 'Dragon Hatchling' (BDH), a new Large Language Model architecture based on a scale-free biologically inspired network of n locally-interacting neuron particles.

Source: <https://arxiv.org/pdf/2509.26507>

- **Baby Dragon Hatchling Price Prediction , BDH Price... | Bitget**

Exploring the forecast and prediction of Baby Dragon Hatchling prices for 2025 — 2050. Make your own prediction based on fixed rate or explore user sentiments.

Source: <https://www.bitget.com/price/baby-dragon-hatchling/price-prediction>

- **Races of the Dragon.pdf - The Techno Jesus**

Design embed now.. Races of the Dragon.pdf - The Techno Jesus. or tattooed. Simple lines serve to give depth and definition.

Source: <https://www.yumpu.com/en/document/view/7833729/races-of-the-dragonpdf-the-techno-jesus>

10. **The Rosebound Hatchling – Unfocussed Photography & Art**

The hatchling, having made its dramatic debut, decided that a perch upon a rose was entirely too small a stage for its destiny. It tested its wings with a few flaps, each one sending droplets scattering into tiny prisms of light. The garden glistened with irritation.

Source: <https://unfocussed.com/blogs/captured-tales/the-rosebound-hatchling>

11. Determining If a Particle Path of an Ideal Fluid on a Three-Dimensional...

Tracking particle paths within an ideal fluid in a three-dimensional domain necessitates a comprehensive understanding of both analytical and numerical methods. Analytical techniques often involve solving differential equations that describe the fluid flow directly.

Source: <https://diversedaily.com/determining-if-a-particle-path-of-an-ideal-fluid-on-a-three-dimensional-domain-eventually-reaches-a-certain-region-in-space/>

12. Numerical integration of the N-body ring problem by recurrent power...

... For the numerical integration of the equations of motion, we have used the method of recurrent power series adapted to the N-body ring problem. This method converges for any initial condition not leading to a binary collision [30].

Source: https://www.researchgate.net/publication/322778641_Numerical_integration_of_the_N-body_ring_problem_by_recurrent_power_series

13. New Particle Is Both Matter and Antimatter: The Majorana fermion

Such a particle may seem confusing, as a particle's antimatter has equal mass but opposite charge, and (notably) when matter meets its antimatter counterpart the two annihilate one another in a burst of energy.

Source: <https://futurism.com/new-particle-matter-antimatter-majorana-fermion>

14. probability theory - Recurrent and Transient random walks on integer...

I have troubles with the following problem: Consider the grid of equilateral triangles (triangular lattice). Is random walk recurrent or transient?

Source: <https://math.stackexchange.com/questions/3697240/recurrent-and-transient-random-walks-on-integer-grids>

15. An improved particle swarm optimization-based dynamic recurrent...

A learning algorithm for dynamic recurrent Elman neural networks is proposed based on an improved particle swarm optimization. The proposed algorithm performs the evolution of network structure, weights, initial inputs of the context units and...

Source: <https://rims.cityu-dg.edu.cn/en/publications/an-improved-particle-swarm-optimization-based-dynamic-recurrent-n/>

[OPTIMIZER Round 2]

Revised Analysis: Input-Path Interference in BDH and the Limits of Input-Dependent Gating/Projection

I treat the BDH shared core abstractly as follows: a frozen map $\mathcal{F}$ is applied sequentially for $\mathcal{D}$ steps. The questions below are independent of the particular particle dynamics of BDH, except where I explicitly use properties of $\mathcal{F}$. If the implementation reports define $\mathcal{F}$ concretely, the theorems should be instantiated with that $\mathcal{F}$; the structural obstructions remain.

1. Formalization

1.1 Shared sequential composition

Let $V=\mathbb{R}^d$ be the state space and $X\subseteqq V$ the input space. A **depth- $\mathcal{D}$ shared sequential composition** is defined by a fixed map

$$\begin{aligned} & \mathcal{F} \\ & F:V\rightarrow V \\ & \end{aligned}$$

and the recurrence

$$\begin{aligned} & \mathcal{F} \\ & s_0(x)=x,\quad s_{t+1}(x)=F(s_t(x)),\quad t=0,\dots,D-1. \\ & \end{aligned}$$

Write

$\lceil$

$$F^D = F \circ F \circ \dots \circ F$$

$\rceil$

for the $\lceil D \rceil$ -fold composition. The **unperturbed trajectory** of $\lceil x \rceil$ is

$\lceil$

$$\tau_F(x) = (s_0(x), s_1(x), \dots, s_D(x)).$$

$\rceil$

This is the relevant BDH abstraction: the same parameterized transformation is applied at every depth, and the same transformation is shared by all inputs.

1.2 Frozen shared path

The map $\lceil F \rceil$ is **frozen** if its parameters are not updated during the continual-learning phase. Only the gate/projection mechanism may be trained. Formally:

$\lceil$

$$\frac{\partial F}{\partial \theta} = 0$$

\]

during any adaptation step. No theorem below relies on retraining $\mathcal{F}$.

1.3 Gating

An **input-dependent gate** is a map

\[

$$g: X \rightarrow \mathbb{R}^d$$

\]

which is applied identically at every depth. The two canonical placements are:

\[

$$\text{\texttt{output gating:}} \quad s_{t+1} = g(x) \odot F(s_t),$$

\]

\[

$$\text{\texttt{input gating:}} \quad s_{t+1} = F(\text{bigl}(g(x) \odot s_t \bigr)).$$

\]

The gate may be trained, but it has **no per-depth parameters**: the same $g(x)$ is used at every t . I will also consider the relaxed variant

$$g: X \times \{0, \dots, D-1\} \rightarrow \mathbb{R}^d,$$

where the gate receives the depth index t as input but still has only shared parameters. This is not a per-depth parameter vector; it is a single function of both x and t .

1.4 Projection

An **input-dependent projection** is a map

$$P: X \rightarrow \Pi(V),$$

where $\Pi(V)$ is the set of orthogonal projections of $\mathbb{R}^d$, applied either as

$$\text{output projection:} \quad s_{t+1} = P(x)F(s_t),$$

or

$\backslash$

$\text{input projection:} \quad s_{t+1} = F(P(x)s_t).$

$\backslash$

A projection is more restrictive than a general gate because it is linear, idempotent, and rank-deficient.

1.5 Input-path interference

I distinguish two precise notions.

(a) Depth-aliasing interference

Suppose an oracle with access to the current depth or state would apply a reference modulation

$\backslash$

$g^*(x,t), \quad t=0, \dots, D-1.$

$\backslash$

A depth-independent gate $\backslash(g(x)\backslash$ produces the actual transition

$$\begin{aligned} & \backslash \\ s_{t+1} &= \Phi(g(x), s_t), \\ & \backslash \end{aligned}$$

where Φ is the chosen gating/projection rule. The **depth-aliasing error** for input x is

$$\begin{aligned} & \backslash \\ \mathrm{DA}(g; x) &= \sum_{t=0}^{D-1} \\ & \left| \right. \\ & \Phi(g(x), s_t(x)) - \Phi(g^*(x, t), s_t(x)) \\ & \left. \right|. \\ & \backslash \end{aligned}$$

If $\mathrm{DA}(g; x) > 0$, then the lack of depth information corrupts the trajectory. This is the forward-pass sense in which shared sequential composition causes interference: the same gate is re-applied at every depth even though the correct modulation may need to change from depth to depth.

(b) Gradient interference

For two tasks (A, B) with losses $(\mathcal{L}_A, \mathcal{L}_B)$ on the same frozen (F) , the first-order change in $\mathcal{L}_A$ caused by one gradient step on task (B) is proportional to

$\langle \nabla_{\phi} L_A, \nabla_{\phi} L_B \rangle$.

Define the **task Jacobians**

$J_i(\phi) = \frac{\partial}{\partial \phi} s_D(x; \phi)$
 $\Big|_{x \in X_i}$.

Task-wise gradient interference is

$I_{ij} = \left| J_i(\phi) J_j(\phi)^{\top} \right|$.

For shared sequential composition, these Jacobians have a special structure:

$$\begin{aligned} & \frac{\partial s_D(x)}{\partial \phi} \\ &= \\ & \sum_{t=0}^{D-1} \\ & \left(\prod_{r=t+1}^{D-1} DF(s_r(x)) \right. \\ & \left. \frac{\partial \Phi}{\partial \phi}(g_\phi(x,t), s_t(x)) \right). \end{aligned}$$

The same parameters ϕ appear at every depth, so gradients from different tasks can overlap even when the inputs are disjoint. This is the training-time sense of input-path interference.

1.6 Computation-isolability

A mechanism is **exactly computation-isolating** for a set of tasks if:

- each task's desired output is produced exactly, and
- there is no interference: $\mathrm{DA}(g;x)=0$ for all x in the forward sense, or $I_{ij}=0$ for all task pairs in the gradient sense.

A mechanism **bounds interference** if the relevant quantity is at most ϵ ($\epsilon > 0$) uniformly over the task inputs.

1.7 Elimination versus bounding

- **Elimination** means exact equality of outputs and exact zero interference.
- **Bounding** means the deviation is at most ϵ , with ϵ controlled by quantities such as the Lipschitz constant of F , the depth D , the gate error, and the number of parameters.

2. Theorem-Shaped Hypotheses

H1. Exact preservation of the unperturbed path forces the gate to be the identity

Assumptions.

- F is defined on an open set $U \subseteq \mathbb{R}^d$.
- The gate is depth-independent output gating:

[

$$s_{t+1} = g(x) \odot F(s_t).$$

]

- We require exact preservation of the unperturbed trajectory:

$$\begin{aligned} & \\ s_t(x) &= F^t(x) \quad \text{for all } x \in U, \text{ for all } t. \end{aligned}$$

$\square$

Mathematical object gated.

The output of the frozen map $\mathcal{F}$ at every depth.

Claim.

Under these assumptions, $\mathcal{G}(x) = \mathbf{1}$ on every coordinate where $\mathcal{F}(s_t(x)) \neq 0$. If $\mathcal{F}$ is componentwise nonzero along trajectories on U , then $\mathcal{G}(x) = \mathbf{1}$ for all $x \in U$.

Proof.

At $t=0$, exact preservation requires

$$\begin{aligned} & \\ \mathcal{G}(x) \odot F(x) &= F(x). \end{aligned}$$

$\square$

Componentwise:

$$\begin{aligned} & \\ (g_i(x) - 1)F_i(x) &= 0. \end{aligned}$$

\]

Wherever $\backslash(F_i(x) \neq 0\backslash)$, we get $\backslash(g_i(x)=1\backslash)$. Wherever $\backslash(F_i(x)=0\backslash)$, the gate is invisible, but it is also useless for isolation. The same argument applies at every depth.

Verdict.

True as an exact theorem. It is a direct consequence of the preservation equation. It shows that, under the strong requirement that the mechanism changes nothing, the only gate is the identity gate. In particular, a gate cannot simultaneously “do nothing” and “isolate tasks.”

Weakest additional assumptions under which the obstruction disappears.

If we only require preservation of the final output $\backslash(s_D(x)=F^D(x)\backslash)$, rather than the whole trajectory, non-identity gates may exist when $\backslash(F^D\backslash)$ is non-injective. This is the underdetermined case; it is not a useful general construction.

Classification.

Known / direct consequence.

H2. Depth-independent gating cannot represent depth-varying modulation

Assumptions.

- The transition rule $\Phi(g,s)$ is injective in the gate argument g for the states along the unperturbed trajectory.
- A task requires an ideal per-depth modulation $g^*(x,t)$ that is not constant in t .

Mathematical object gated.

The entire per-depth modulation sequence $(g(x,0), \dots, g(x,D-1))$.

Claim.

No depth-independent gate $g(x)$ can produce exactly the same trajectory as $g^*(x,t)$ for all t .

Proof.

Equality at every depth gives

$$\begin{aligned} & \Phi(g(x), s_t(x)) = \Phi(g^*(x,t), s_t(x)). \end{aligned}$$

By injectivity of $\Phi(\cdot, s_t(x))$,

$$\begin{aligned} & g(x) = g^*(x,t) \end{aligned}$$

$\forall$

for every $\forall(t)$, contradicting the assumption that $\forall(g^*(x, \cdot))$ is nonconstant.

Verdict.

True as an exact representational theorem. This is the cleanest formalization of the phrase “input-path interference arising from shared sequential composition”: a depth-independent gate is a point on the diagonal of the product space $\forall(M^D)$, while the task requirements occupy off-diagonal points.

Weakest additional assumptions under which it becomes true.

The obstruction disappears if we allow:

- a depth-indexed shared gate $\forall(g(x, t))$;
- a state-dependent gate, e.g. $\forall(g(s_t))$, as in LSTM/GRU;
- per-depth trainable parameters, which is excluded by the research question.

Classification.

Direct consequence of the definition; the phrasing as a frozen-shared-path obstruction is a plausible adaptation of standard control-theoretic observability reasoning.

H3. Projection mechanisms have a rank obstruction

Assumptions.

- The mechanism is an input-dependent orthogonal projection, e.g.

$\backslash$

$$s_{t+1} = P(x)F(s_t).$$

$\backslash$

- The unperturbed trajectory of $\backslash(x\backslash)$ is such that

$\backslash$

$$\operatorname{span}\{F(s_0(x)), F(s_1(x)), \dots, F(s_{D-1}(x))\}$$

$\backslash$

has dimension $\backslash(m(x)\backslash)$.

Mathematical object projected.

The image space of the projection, i.e. the subspace of state space that survives each depth.

Claim.

If the projection is to preserve the unperturbed trajectory exactly, then

$$\text{rank } P(x) \geq m(x).$$

For (K) tasks whose relevant spans have dimensions (m_i) , and whose projections must be mutually orthogonal to avoid gradient interference, necessarily

$$\sum_{i=1}^K m_i \leq d.$$

Hence a projection can isolate at most $\lfloor d / \min_i m_i \rfloor$ tasks.

Proof.

Exact preservation requires

$$P(x)F(s_t(x)) = F(s_t(x))$$

for all (t) , so every $(F(s_t(x)))$ lies in $(\text{Im } P(x))$. The dimension bound follows. Orthogonality of task subspaces is required for

zero gradient interference; orthogonal subspaces of $\mathbb{R}^d$ have total dimension at most d .

Verdict.

True as an exact linear-algebra theorem.

Weakest additional assumptions under which it becomes useful.

Tasks must live in low-dimensional invariant or approximately invariant subspaces. If the relevant computational subspaces have dimension 1 , for example, a rank-1 projection can isolate up to d tasks.

Classification.

Direct consequence of linear algebra; the application to frozen shared paths is a plausible adaptation.

H4. Finite parameter budget bounds exact gradient isolation

Assumptions.

- The gate/projection mechanism has p trainable parameters.
- Task output Jacobians are $\{J_i \in \mathbb{R}^{d \times p}\}$.
- We require exact zero gradient interference:

$\{$

$$J_{ij}^{\text{top}} = 0 \quad \text{for all } i \neq j.$$

$\backslash$

Mathematical object constrained.

The parameter space of the gate/projection mechanism.

Claim.

If each task Jacobian has rank $\backslash(r_i = \operatorname{rank}\{J_i\})$, then

$\backslash$

$$\sum_i r_i \leq p.$$

$\backslash$

In particular, if every task has full row rank and $\backslash(p \leq d\backslash)$, then at most one task can be isolated exactly. For a shared sequential composition with generic nonlinear $\backslash(F\backslash)$, task Jacobians are generic sums over depth, so this is the typical situation.

Proof.

The row spaces of the $\backslash(J_i\backslash)$ must be mutually orthogonal subspaces of $\backslash(\mathbb{R}^p\backslash)$. The sum of their dimensions cannot exceed $\backslash(p\backslash)$.

Verdict.

True as an exact theorem. It is the usual parameter-isolation bound, restated for trainable gates on a frozen shared path.

Weakest additional assumptions under which exact isolation is possible for many tasks.

Each task must occupy a low-rank subspace of the gate-parameter Jacobian. This occurs, for example, when tasks use disjoint subsets of inputs or when the gate network is designed so that task-specific features activate disjoint hidden units.

Classification.

Direct consequence of linear algebra; the application to frozen-path gating is a plausible adaptation.

H5. Bounded interference is possible exactly when the gated dynamics are non-expansive

Assumptions.

- The modulated transition is Lipschitz in state with constant L_Φ :

[

$$\|\Phi(g,s) - \Phi(g,s')\| \leq L_\Phi \|s - s'\|.$$

]

- The per-depth gate error is bounded:

$$\begin{aligned} & \mathbb{E} \left[\left| \Phi(g(x), s_t(x)) - \Phi(g^*(x, t), s_t(x)) \right| \right] \leq \delta. \end{aligned}$$

Mathematical object bounded.

The full trajectory error.

Claim.

The output error satisfies

$$\begin{aligned} & \mathbb{E} \left[\left| s_D(x) - s_D^*(x) \right| \right] \\ & \leq \\ & \delta \sum_{k=0}^{D-1} L_{\Phi}^k \\ & = \\ & \begin{cases} \delta \frac{1 - L_{\Phi}^D}{1 - L_{\Phi}}, & \text{if } L_{\Phi} < 1, \\ \delta D, & \text{if } L_{\Phi} = 1, \\ \delta \frac{L_{\Phi}^D - 1}{L_{\Phi} - 1}, & \text{if } L_{\Phi} > 1. \end{cases} \end{aligned}$$

\]

Proof.

Let $(e_t = |s_t - s_{t^*}|)$. Then

\[

e_{t+1}

$\leq$

$| \Phi(g, s_t) - \Phi(g, s_{t^*}) |$

+

$| \Phi(g, s_t) - \Phi(g^*, s_{t^*}) |$

$\leq$

$L_\Phi e_t + \delta$.

\]

Solving the recurrence gives the claimed bound.

Verdict.

True as an exact theorem.

Consequence.

If the gated system is contractive, $(L_\Phi < 1)$, then interference is bounded uniformly in depth by $(\delta / (1 - L_\Phi))$. If $(L_\Phi > 1)$, interference can grow exponentially in (D) . This is why “bounding” is not

automatic: for generic nonlinear $\mathcal{F}$, $\mathcal{L}(\Phi)$ is often larger than $\mathcal{L}_1$, and small gate errors are amplified through the shared composition.

Classification.

Known / direct consequence.

H6. Depth-indexed shared gating: finite exact interpolation is conditionally possible

Assumptions.

- We allow a depth-indexed shared gate $\mathcal{G}_\Phi(x, t)$, still with no per-depth parameter vectors.
- The task set is finite: $\{(x_i, y_i)\}$, $(i=1, \dots, N)$.
- For each (i) , there exists some per-depth modulation sequence $\{g_{i,t}\}$ that realizes (y_i) under the transition rule.
- The parameterized family $\mathcal{G}_\Phi$ can interpolate the finite data $\{(x_i, t) \mapsto g_{i,t}\}$, e.g. an MLP with enough parameters.

Claim.

Under these assumptions, there exists a shared-parameter depth-indexed gate that produces exactly the desired outputs for all finite task inputs.

Proof sketch.

The modulation requirement is a finite lookup table. Universal approximation on finite sets gives interpolation.

Important caveat.

The decisive assumption is that the target modulation sequences $\{g_{i,t}\}$ exist. For a generic nonlinear $\{F\}$, this is **underdetermined**. For example, if $\{F\}$ is contractive, many targets are unreachable regardless of gating. Thus this is a conditional existence theorem, not a general theorem.

Verdict.

Conditionally true; otherwise underdetermined.

Weakest additional assumptions.

Reachability of the targets under the allowed modulation, plus sufficient gate capacity. Additionally, exact zero gradient interference requires the parameter-count condition of H4.

Classification.

Plausible adaptation of universal approximation; the reachability part is a conjecture.

3. Prior Art

3.1 Orthogonal gradient and gradient-projection methods

Examples. GEM / A-GEM, Orthogonal Gradient Descent, TRGP, GPM.

What they address.

They project gradients in parameter space so that updates for a new task do not increase losses on previous tasks.

Do they address the same mathematical problem?

Only partially. They address training-time gradient interference, which corresponds to my second notion of interference. They do **not** address depth-aliasing in a frozen shared composition. Moreover, they typically require unfreezing or at least modifying the main network; here the shared path is frozen.

Classification.

Known, partially relevant.

3.2 Parameter isolation and masking

Examples. PackNet, Piggyback, Hard Attention to the Task, Supermasks.

What they address.

They allocate disjoint parameter subsets to tasks, preventing cross-task updates.

Same problem?

Not the same. They introduce per-task masks or per-task parameters, which is a form of parameter isolation. The research question explicitly asks for an input-dependent gate/projection on a single frozen shared path, not for partitioning the shared path's parameters.

Classification.

Known, related but different mechanism.

3.3 Progressive networks and path-based continual learning

Examples. Progressive Neural Networks, PathNet.

What they address.

They add new columns or paths for new tasks while freezing old ones.

Same problem?

No. They avoid interference by adding task-specific architecture, which is effectively per-task parameter growth. The question here is whether a *single* input-dependent mechanism can isolate paths within one frozen shared composition.

Classification.

Known, different problem.

3.4 Hypernetworks

Examples. Ha et al., Chunked Hypernetworks.

What they address.

A hypernetwork generates weights for a main network as a function of an input or task embedding.

Same problem?

Partly. A hypernetwork can generate input-dependent per-depth modulations. But the generated weights are typically per-layer, which violates the “no per-depth/per-level trainable parameters” constraint in spirit. A hypernetwork also replaces or augments the main weights rather than gating a frozen shared path.

Classification.

Known, partially relevant.

3.5 Recurrent neural networks with gating

Examples. LSTM, GRU.

What they address.

They use input- and state-dependent gates inside a shared recurrent computation.

Same problem?

This is the closest prior art. LSTM gates are applied at every time step with shared parameters, exactly like gating a shared sequential composition. The crucial difference is that LSTM gates depend on the current hidden state $\{h_{t-1}\}$, not only on the original input $\{x\}$. This supports H2: state information is what makes recurrent gating effective. It is also empirical evidence, not a theorem, that state-conditioning resolves many of the pathologies of depth-independent control.

Classification.

Known, directly relevant, but with a different information assumption.

3.6 Conditional computation and mixture of experts

Examples. Mixture of Experts, sparsely gated layers.

What they address.

Input-dependent routing to different experts or subnetworks.

Same problem?

Not the same. MoE routes inputs to different parameter sets, typically with per-layer routers, and its goal is capacity scaling or efficiency, not preserving a frozen shared-path computation.

Classification.

Known, structurally similar but not the same mathematical problem.

3.7 Depth-recurrent and weight-tied transformers

Examples. ALBERT, Universal Transformers, LoopFormer.

What they address.

They reuse the same block across depth and sometimes use adaptive computation time.

Same problem?

Partially. They study shared-depth composition, but they do not address task isolation via input-dependent gates. Adaptive computation time is state-dependent halting, not input-dependent task isolation.

Classification.

Known, background context.

4. Novelty Classification

| Claim | Classification |

| --- | --- |

| Formal notion of depth-aliasing interference in a frozen shared composition | Apparently novel as a packaged definition; components are known from control theory |

| H1: exact preservation forces identity gating | Direct consequence |

| H2: depth-independent gating cannot represent depth-varying modulation | Direct consequence |

| H3: projection rank obstruction | Direct consequence |

| H4: gradient-isolation parameter bound | Direct consequence |

| H5: contraction/expansion bound on interference | Direct consequence |

| H6: conditional finite interpolation by depth-indexed shared gate | Plausible adaptation / conjecture |

| Claim that state or depth information is necessary for robust gating in shared composition | Plausible adaptation; consistent with LSTM design but not proven here as a theorem |

No experimental result is used as evidence for any theorem. The exact results above are purely analytical.

5. Failure Condition and Obstruction

The strict version of the proposed mechanism is:

- input-dependent gate/projection,
- no per-depth trainable parameters,
- frozen shared path,
- no retraining of $\backslash(F\backslash)$.

Under this strict version, **exact elimination of input-path interference is impossible for generic nonlinear $\mathcal{F}$** . The obstruction has four independent parts.

5.1 Diagonalization obstruction

A depth-independent gate $g(x)$ is forced to lie on the diagonal of the product space of depth-wise modulations:

$$\begin{bmatrix} g(x) & g(x) & \dots & g(x) \end{bmatrix} \in M^D.$$

Any task whose ideal modulation depends on depth lies off this diagonal. This is a representational impossibility, not a training failure.

5.2 Preservation-identity obstruction

If the mechanism is required to preserve the unperturbed computation exactly, then output gating forces $g(x) = \mathbf{1}$ wherever $\mathcal{F}$ is nonzero. The only gate that changes nothing is the identity gate. Thus a gate cannot both preserve the frozen computation and alter the path for isolation.

5.3 Rank and parameter budget obstruction

Even if the depth-aliasing obstruction is removed by allowing a depth-indexed shared gate, exact gradient isolation requires the task Jacobians to be mutually orthogonal. The sum of their ranks cannot exceed the number of trainable gate parameters $\mathcal{P}$. For generic tasks, this permits only one task when $\mathcal{P} \leq d$.

5.4 Expansive-dynamics obstruction

Bounding interference is possible if the gated dynamics are contractive. For generic nonlinear $\mathcal{F}$, the Lipschitz constant of the composed system is >1 , and small gate errors are amplified exponentially in depth. Hence even bounded interference is not guaranteed without additional assumptions on $\mathcal{F}$.

6. Conclusion

The answer to the research question is:

- **Elimination** of input-path interference by a depth-independent input-dependent gate/projection on a frozen shared path is **impossible in general**.
- **Bounding** is possible if the gated system is contractive or if the depth is small enough that exponential amplification is controlled.
- **Controllability** can be achieved only by enriching the information available to the gate: either allow a depth-indexed shared gate $g(x, t)$, allow state-dependent gating $g(s, t)$, allow per-task parameter blocks, or impose strong low-rank/invariant-subspace structure on the tasks.

The most precise obstruction is the **diagonalization obstruction**: a depth-independent gate is a single point on the diagonal of the space of depth-wise modulations, while the tasks generically require off-diagonal points. This obstruction is independent of optimization, capacity, and training data. It is the mathematical core of why input-path interference cannot, in general, be eliminated by the proposed mechanism.

[MODERATOR Round 2]

Moderator Synthesis — Round 2

Consolidation

Role: Synthesis and consensus evaluation of the STRATEGIST, CRITIC, and OPTIMIZER contributions on the BDH input-path interference question.

Before consolidating, one factual check is warranted, since the entire analysis hinges on BDH genuinely exhibiting weight-shared sequential composition across depth (if it did not, the question would misfire at the premise level):

[SEARCH: "Dragon Hatchling" BDH weight tying shared parameters across depth recurrent layer]

0. Evidence-Base Status

- The BDH paper (arXiv:2509.26507) is confirmed to exist and to describe a scale-free network of locally-interacting neuron particles with transformer-comparable performance. Third-party implementations (BDH-MLX) corroborate the architecture's general character.
- **Retrieval limitation:** the internal project documents (2026-08-^{*} progress reports, paper-parity-plan, bdh-continual-learning-plan_v2.md, cl-h1-report.md) could **not** be retrieved through available channels; searches returned irrelevant material. Consequently, neither participant nor this moderation could cross-check the formal models against your concrete implementation or against the H1 experimental report dated today. All conclusions below are therefore **structural**, derived from the abstract "shared sequential composition" model that both analysts adopted consistently. Instantiation with BDH's literal update equations remains an open verification step (see §3.3).

Both principal analyses proceeded from the same formal skeleton — frozen map F , recurrence $s_{t+1} = \Phi(g(x), s_t)$, depth-agnostic input-conditioned gate/projection — which makes genuine consensus evaluation possible.

1. Points of Consensus (High Confidence)

C1 — Formal framework agreement. The definitions of shared sequential composition, frozen path, gating vs. projection, and the

elimination/bounding distinction were independently convergent. No participant disputed another's definitions. *Consensus: strong.*

C2 — Exact elimination is impossible under the strict reading. Three independent obstructions were derived, by complementary methods:

- **Identity-forcing** (STRATEGIST H1/H3; OPTIMIZER H1): exact trajectory preservation under output gating $g \odot F(s)$ forces $g = \mathbf{1}$ wherever F is nonzero along the trajectory. A mechanism required to change nothing can isolate nothing.
- **Diagonalization** (OPTIMIZER H2; STRATEGIST's "temporal information deficit" is its informal statement): a depth-agnostic gate occupies the diagonal of the product space M^D of per-depth modulations; any task whose ideal modulation $g^*(x, t)$ is nonconstant in t lies off-diagonal and is unreachable, given injectivity of $\Phi(\cdot, s_t)$ in the gate argument.
- **Rank/parameter-budget** (OPTIMIZER H3/H4): exact zero cross-task gradient interference through shared gate parameters ϕ requires mutually orthogonal task-Jacobian row spaces, hence $\sum_i \text{rank}(J_i) \leq p$. Generic tasks saturate this at one task when $p \leq d$.

Consensus: strong. This satisfies the brief's failure condition: the proposed isolation is impossible under the stated strict assumptions.

C3 — Bounding is conditional, not free. Per-depth gate error δ under Lipschitz constant L_Φ yields output error $\leq \delta \sum_k L_\Phi^k$: uniformly bounded iff $L_\Phi < 1$, growing $\sim \delta D$ at $L_\Phi = 1$, exponentially otherwise (OPTIMIZER H5; STRATEGIST H2 agrees and correctly notes contraction bounds trajectory *separation*, not modification-induced interference). Trained LLM-scale maps typically

have $L_{\Phi} > 1$, so bounding is a real architectural constraint, not a default. *Consensus: strong.*

C4 — Escape routes require enriched information or structure. All participants converge: relaxation is possible via (i) depth-indexed shared-parameter gates $g_{\Phi}(x, t)$, (ii) state-conditioned gating ($g(s_t)$, LSTM-style — the nearest prior art, which violates the input-only constraint), (iii) low-rank/invariant task subspaces, or (iv) abandoning the frozen-path constraint. *Consensus: strong.*

C5 — Prior-art triage. Independent and concordant: orthogonal-gradient methods (GEM/A-GEM/GPM) operate on gradients in parameter space, not forward-pass composition — related but a different mathematical problem; progressive networks, PackNet, supermasks introduce per-task parameters/masks — constraint-violating; hypernetworks and MoE are structurally adjacent but generate per-layer outputs or route between experts — not the same problem; Universal Transformer/ALBERT establish the weight-tied backdrop without addressing isolation. **No surveyed prior art solves this exact problem.** *Consensus: strong.*

2. Disputes and Adjudications

D1 — Novelty classification conflict. STRATEGIST labels the impossibility result "apparently novel"; OPTIMIZER labels the equivalent statements "direct consequence."

Adjudication: OPTIMIZER is correct on the mathematics, STRATEGIST is correct on the contribution. Every ingredient (functional-equation identity

forcing, diagonal argument, rank counting, Lipschitz error propagation) is textbook. What is novel is the **problem formulation** — isolation of computations within a frozen weight-tied path under an input-only gate, posed for continual learning. Write-up guidance: claim a new formalization and a clean negative result; do **not** claim new mathematics.

D2 — STRATEGIST H4 is defective. Its construction (Jacobian-null-space projection achieving preservation + isolation with $O(d \cdot D)$ parameters) collapses to the identity gate under its own preservation requirement, and conflates the Jacobian's null space with the output's annihilator. OPTIMIZER's H6 (conditional finite interpolation, contingent on reachability of target modulation sequences) is the correct weakened version. *Adjudication:* H4 withdrawn as stated; retained only as OPTIMIZER-style conjecture.

D3 — STRATEGIST's "information gap" $\Delta_t = t \cdot d$. Dimensions of informally-defined information sets are not a rigorous quantity. Superseded by OPTIMIZER's rank-theoretic arguments. Retain only as heuristic motivation.

D4 — Interference has two distinct meanings; only OPTIMIZER separated them cleanly. Forward-pass *depth-aliasing* vs. training-time *gradient interference*. See correction G1 below — this separation turns out to be more important than either participant realized.

3. Corrections and Gaps Identified in Moderation

G1 — Cross-input forward-pass interference is structurally zero; the live channels are elsewhere. With frozen θ and a *stateless* input-conditioned mechanism, each input's forward pass is computationally independent: x_1 's trajectory cannot be perturbed by x_2 's "presence." STRATEGIST's Definition 3 counterfactual ($\tilde{h}_t(x_1 \mid x_2 \text{ active})$) is therefore ill-defined as stated. The operationally meaningful interference notions are:

- **Depth-aliasing within a pass** (obstruction B above); and
- **Cross-time forgetting:** ϕ is shared across all inputs, so any update $\Delta\phi$ from task B alters g_ϕ on task- A inputs — first-order damage governed exactly by the Jacobian-overlap/rank condition (obstruction C). This, not intra-pass contamination, is the catastrophic-forgetting channel in this architecture.

Future work should define interference exclusively through these two channels.

G2 — The preservation requirement must be asymmetric to be meaningful. Requiring preservation on *all* inputs trivially forces the identity gate (C2). The continual-learning-relevant requirement is: exact preservation on old-task inputs S_{old} , unconstrained deviation on new-task inputs S_{new} . Under this reading the identity obstruction restricts to S_{old} ; non-trivial gates become admissible off it; the surviving barriers are the diagonal obstruction (whenever old-task-faithful modulation must vary with depth) and the rank/budget obstruction for gradient-exactness. **Neither participant developed this asymmetric variant; it should be the primary next theorem target.**

G3 — BDH instantiation is outstanding. BDH's block has residual structure and elementwise multiplicative (Hebbian-type) terms. Consequences: (a) output-gating the full map would gate the residual stream — likely harmful; gating only the delta term is the natural placement; (b) BDH's native plasticity must be explicitly disabled for the "frozen path" premise to hold. These checks against paper .tex/the repository were not performed in-round.

G4 — Scope decision required on t -conditioning. A shared-parameter gate $g_{\phi}(x,t)$ (depth index as *input*, e.g., via a fixed embedding) adds $O(1)$ parameters and **does not violate "no per-depth/per-level trainable parameters" as literally worded**. This is the pivotal loophole: every impossibility result proven here targets the strictly depth-agnostic gate and dissolves — into conditional statements (reachability + capacity + rank budget) — once t is admitted. The team must rule explicitly on whether the intended constraint forbids t -conditioning; the theorems' applicability hinges on this interpretation.

4. Consolidated Answer to the Research Question

| Reading of the question | Verdict | Status |

| --- | --- | --- |

| Depth-agnostic gate/projection, frozen nonlinear F , **exact elimination** | **Impossible** (identity-forcing + diagonalization + rank obstructions) | Proven; direct consequences |

| Same, **bounding** | Possible iff gated dynamics non-expansive ($L_{\Phi} < 1$) or depth-limited error budget; else exponential amplification | Proven conditional theorem |

| Enriched gate (t -conditioned shared-parameter, or state-conditioned), **controllability** | Feasible conditional on reachability, gate capacity, low-rank task subspaces; exact multi-task gradient isolation still capped by $\sum_i \mathrm{rank}(J_i) \leq p$ | Conjecture / plausible adaptation |

The failure condition contemplated by the brief is **met for the strict reading**, with a precise obstruction characterization: a depth-agnostic gate is confined to the diagonal of the modulation product space, and exactness requirements (preservation, gradient orthogonality) collapse either to the identity mechanism or to a rank budget permitting at most one generic task. No optimization method, capacity increase, or training regime circumvents these; they are representational, not algorithmic.

5. Final Novelty Ledger (Adjudicated)

| Claim | Classification |

| --- | --- |

| Formalization: isolation within frozen weight-tied composition via input-only gate, CL-framed | **Apparently novel** (as packaging) |

| Identity-forcing lemma | **Direct consequence** |

| Diagonalization obstruction | **Direct consequence** (control-theoretic flavor) |

| Rank/parameter-budget bound on exact gradient isolation | **Direct consequence** |

| Contraction/amplification error bound | **Direct consequence** |

| Necessity of state/depth information for effective gating | **Plausible adaptation** (consistent with LSTM/ACT practice; proven only under injectivity assumptions) |

| t -conditioned shared gate achieves isolation broadly | **Conjecture** (reachability undetermined) |

| Any empirical effect observed in cl-h1-report | Inadmissible as theorem evidence until reconciled with the above |

6. Recommended Next Steps

- **Restate the theorem set under asymmetric preservation** (old-task exact / new-task free) and prove that the diagonal and rank obstructions survive while the identity obstruction dies. This is the honest CL formalization and the highest-value open result.
- **Instantiate t with BDH's concrete update** from paper .tex; fix gate placement relative to the residual stream; confirm plasticity disabling.
- **Rule on the t -conditioning interpretation** of the constraint; if permitted, specify a minimal $g_{\phi}(x,t)$ (fixed depth embedding into the existing gate network) and test — labeling outcomes strictly as empirical conjecture.
- **Reconcile with the local reports** (plan_v2, cl-h1-report) once retrievable; verify that the implemented mechanism corresponds to one of the formally scoped variants rather than an unscoped mixture.
- **Write-up posture:** present C2/C3 as a clean impossibility-and-bounding package for weight-tied architectures; cite LSTM/GRU, adaptive computation time, and universal-transformer literature as nearest neighbors; claim formalization novelty, not mathematical novelty.

Consensus summary: full agreement on framework, on the impossibility of exact elimination under strict assumptions, on conditional bounding, and on prior-art triage; disagreements were confined to novelty labeling and one defective construction, both adjudicated above. The investigation's productive frontier is the asymmetric-preservation reformulation and the t -conditioning scope decision.

Web Research

- **GitHub - severian42/BDH-MLX: Baby Dragon Hatchling (BDH) – MLX...**

Shared Parameters Across Layers: Same weights reused in all layers (recurrent depth). Sparse, Non-Negative Activations: ReLU-based activations for biological plausibility. Transformer. BDH. Parameters. Unique per layer. Shared across layers. Activations.

Source: <https://github.com/severian42/BDH-MLX>

- **BDH-CQ: In-Context Learning with Recurrent Latent... | alphaXiv**

BDH layers also support recurrent systems for constraint satisfaction. Recurrent-depth language models repeatedly apply a shared block to a sequence-wide latent state and can improve some reasoning benchmarks with additional iterations (Geiping et al., 2025).

Source: <https://www.alphaxiv.org/abs/2608.09888>

- **Dragon Hatchling: Linking Transformers & Brain Models**

We introduce 'Dragon Hatchling' (BDH), a new LLM architecture based on a scale-free biologically inspired network of n locally-interacting neuron particles.

Source: <https://www.emergentmind.com/papers/2509.26507>

- **Beyond Transformer LLM Models: How the BDH Architecture Solves...**

Baby Dragon Hatchling (BDH). Effective Attention Span. ~2 hours and 17 minutes. A recurring and highly emphasized theme in the 2026 scientific discourse surrounding the BDH architecture is the concept of Axiomatic Artificial Intelligence.

Source: <https://www.crvscience.com/post/beyond-transformer-llm-models-how-the-bdh-architecture-solves-the-context-window-crisis>

- **The Dragon Hatchling: The Missing Link between the Transformer...**

The dragon hatchling aka #BDH (baby dragon hatchling) is a new approach to creating and training these large language models at an architectural level. BDH is #graph centric in its design and tries to mirror #neural synapses like those in the brain for its training mechanism.

Source: https://www.linkedin.com/posts/meeherkapoor_the-dragon-hatchling-the-missing-link-between-activity-7386863203634388992-uI39

[STRATEGIST Round 3]

Final Strategic Assessment: Input-Path Interference in BDH via Gating/Projection

Date: 2026-08-25 | **Status:** Complete

Executive Summary

Under the strict constraints stated in the research question — depth-agnostic input-dependent gate/projection, frozen shared path, no per-depth trainable parameters, no retraining — **exact elimination of input-path interference is impossible** for generic nonlinear shared compositions. This is proven via three independent obstructions (identity-forcing, diagonalization, rank budget). Bounding is possible only under contractivity assumptions that are unlikely to hold for trained LLM-scale maps. The productive frontier is relaxing the depth-agnostic constraint to admit a depth-indexed shared-parameter gate $g_{\phi}(x,t)$, which dissolves the impossibility into conditional feasibility statements.

1. Formal Definitions

1.1 Shared Sequential Composition

Let $V = \mathbb{R}^d$ be the state space. A **depth- D shared sequential composition** is defined by a fixed map $F: V \rightarrow V$ and the recurrence:

$$s_0(x) = x, \quad s_{t+1}(x) = F(s_t(x)), \quad t = 0, \dots, D-1$$

The unperturbed trajectory is $\tau_F(x) = (s_0(x), s_1(x), \dots, s_D(x))$.

Classification: Direct characterization of BDH's weight-tied recurrent depth, confirmed by the BDH-MLX implementation ("Same weights reused in all layers (recurrent depth)").

1.2 Frozen Shared Path

F is **frozen** if $\partial F / \partial \theta = 0$ during the continual-learning phase. Only the gate/projection mechanism ϕ may be trained.

Classification: Known — standard continual learning constraint.

1.3 Input-Path Interference (Two Precise Notions)

Forward-pass depth-aliasing: A depth-agnostic gate $g(x)$ is applied identically at every depth. If the ideal modulation for input x is $g^*(x, t)$ (varying with depth), the depth-aliasing error is:

$$\mathrm{DA}(g; x) = \sum_{t=0}^{D-1} \left| \Phi(g(x), s_t(x)) - \Phi(g^*(x, t), s_t(x)) \right|$$

Gradient interference (cross-task forgetting): For tasks A, B with shared gate parameters ϕ , the first-order damage from a B -update to A -performance is proportional to $\langle \nabla_{\phi} \mathcal{L}_A, \nabla_{\phi} \mathcal{L}_B \rangle$. The task Jacobians are:

$$J_i(\phi) = \frac{\partial}{\partial \phi} s_D(x; \phi) \Big|_{x \in X_i}$$

Critical correction from moderation: Cross-input forward-pass interference is **structurally zero** with frozen θ and a stateless input-conditioned mechanism — each input's forward pass is computationally independent. The live interference channels are depth-aliasing (within a pass) and gradient interference (across training steps).

Classification: The depth-aliasing notion is **apparently novel** as a packaged definition; components are known from control theory. The gradient notion is **known** from continual learning.

1.4 Gating

An **input-dependent gate** $g: X \rightarrow \mathbb{R}^d$ is applied identically at every depth:

- Output gating: $s_{t+1} = g(x) \odot F(s_t)$
- Input gating: $s_{t+1} = F(g(x) \odot s_t)$

The gate has **no per-depth parameters**: the same $g(x)$ is used at every t .

Classification: Known mechanism (LSTM, MoE), but the depth-independence + frozen-path combination is novel.

1.5 Projection

An **input-dependent projection** $P: X \rightarrow \Pi(V)$ maps each input to an orthogonal projection, applied as $s_{t+1} = P(x) F(s_t)$ or $s_{t+1} = F(P(x) s_t)$.

Classification: Known (gradient projection methods), formalized for forward-pass application.

1.6 Computation-Isolability

A mechanism is **exactly computation-isolating** if: (1) each task's desired output is produced exactly, and (2) $\mathrm{DA}(g; x) = 0$ for all x (forward) and $I_{ij} = 0$ for all task pairs (gradient).

Classification: Apparently novel — this specific formulation combining shared composition, frozen parameters, and forward-pass isolation does not appear in surveyed literature.

1.7 Elimination vs. Bounding

- **Elimination:** Exact equality of outputs and exact zero interference.
- **Bounding:** Deviation $\leq \epsilon$, with ϵ controlled by Lipschitz constant, depth, gate error, and parameter count.

Classification: Plausible adaptation of approximation theory concepts.

2. Theorem-Shaped Hypotheses

H1. Identity-Forcing Lemma (Negative)

Assumptions: F defined on open $U \subseteq \mathbb{R}^d$; depth-independent output gating $s_{t+1} = g(x) \odot F(s_t)$; exact trajectory preservation required: $s_t(x) = F^t(x)$ for all $x \in U$, all t .

Mathematical object gated: Output of F at every depth.

Claim: $g(x) = \mathbf{1}$ wherever $F(s_t(x)) \neq 0$.

Proof: At $t=0$: $g(x) \odot F(x) = F(x)$ implies $(g_i(x) - 1) F_i(x) = 0$ componentwise. Where $F_i(x) \neq 0$, $g_i(x) = 1$. Same argument at every depth. $\square$

Verdict: True as an exact theorem. Direct consequence.

Weakest assumptions for escape: If only final-output preservation $s_D(x) = F^D(x)$ is required (not whole trajectory), non-identity gates may exist when F^D is non-injective. This is underdetermined and not generally useful.

Classification: Known / direct consequence.

H2. Diagonalization Obstruction (Negative)

Assumptions: $\Phi(g, s)$ is injective in g along the unperturbed trajectory; ideal modulation $g^*(x, t)$ is nonconstant in t .

Mathematical object gated: The entire per-depth modulation sequence $(g(x, 0), \dots, g(x, D-1))$.

Claim: No depth-independent gate $g(x)$ can reproduce $g^*(x, t)$ for all t .

Proof: Equality at every depth gives $\Phi(g(x), s_t(x)) = \Phi(g^*(x, t), s_t(x))$. By injectivity, $g(x) = g^*(x, t)$ for every t , contradicting nonconstancy. $\square$

Verdict: True as an exact representational theorem. A depth-independent gate is confined to the diagonal of M^D ; tasks requiring depth-varying modulation lie off-diagonal.

Weakest assumptions for escape: Allow depth-indexed shared gate $g_\phi(x,t)$; or state-dependent gate $g(s_t)$ (LSTM-style); or per-depth trainable parameters (excluded by constraint).

Classification: Direct consequence; the frozen-shared-path phrasing is a plausible adaptation of control-theoretic observability reasoning.

H3. Projection Rank Obstruction (Negative)

Assumptions: Input-dependent orthogonal projection $s_{t+1} = P(x) F(s_t)$; trajectory span $\operatorname{span}\{F(s_0(x)), \dots, F(s_{D-1}(x))\}$ has dimension $m(x)$.

Mathematical object projected: Image space of $P(x)$.

Claim: Exact preservation requires $\operatorname{rank} P(x) \geq m(x)$. For K tasks with mutually orthogonal projections: $\sum_{i=1}^K m_i \leq d$. At most $\lfloor d / \min_i m_i \rfloor$ tasks can be isolated.

Proof: $P(x) F(s_t(x)) = F(s_t(x))$ forces $F(s_t(x)) \in \operatorname{Im} P(x)$. Orthogonal subspaces of $\mathbb{R}^d$ have total dimension $\leq d$. $\square$

Verdict: True as an exact linear-algebra theorem.

Weakest assumptions for escape: Tasks must live in low-dimensional invariant subspaces. If $m_i = 1$, up to d tasks can be isolated.

Classification: Direct consequence; application to frozen shared paths is a plausible adaptation.

H4. Parameter-Budget Bound on Gradient Isolation (Negative)

Assumptions: Gate has p trainable parameters; task Jacobians $J_i \in \mathbb{R}^{d \times p}$; exact zero gradient interference required: $J_i J_j^\top = 0$ for $i \neq j$.

Mathematical object constrained: Parameter space of the gate mechanism.

Claim: $\sum_i \text{rank}(J_i) \leq p$. If every task has full row rank and $p \leq d$, at most one task can be isolated exactly.

Proof: Row spaces of J_i must be mutually orthogonal subspaces of $\mathbb{R}^p$. $\square$

Verdict: True as an exact theorem.

Weakest assumptions for escape: Each task must occupy a low-rank subspace of the gate-parameter Jacobian (e.g., disjoint input subsets, or gate architecture with task-specific feature activation).

Classification: Direct consequence; the frozen-path application is a plausible adaptation.

H5. Error Amplification Bound (Conditional)

Assumptions: Modulated transition is L_Φ -Lipschitz in state; per-depth gate error $\leq \delta$.

Mathematical object bounded: Full trajectory error.

Claim:

$$\begin{aligned} \mathbb{E} \|s_D(x) - s_D^*(x)\| &\leq \delta \sum_{k=0}^{D-1} L_\Phi^k = \\ &\begin{cases} \delta \frac{1 - L_\Phi^D}{1 - L_\Phi} & \text{if } L_\Phi < 1 \\ \delta D & \text{if } L_\Phi = 1 \\ \delta \frac{L_\Phi^D - 1}{L_\Phi - 1} & \text{if } L_\Phi > 1 \end{cases} \end{aligned}$$

Proof: Recurrence $e_{t+1} \leq L_\Phi e_t + \delta$ solved by standard methods. $\square$

Verdict: True as an exact theorem.

Consequence: Bounding is uniform in depth iff $L_\Phi < 1$ (contractive). For $L_\Phi > 1$, errors grow exponentially. Trained LLM-scale maps typically have $L_\Phi > 1$, so bounding is a real architectural constraint, not a default.

Classification: Known / direct consequence.

H6. Depth-Indexed Shared Gate: Conditional Feasibility (Positive, Conjectural)

Assumptions: Depth-indexed shared gate $g_\phi(x,t)$ (shared parameters, t as input); finite task set (x_i, y_i) ; target modulation sequences $g_{i,t}$ exist and are reachable; sufficient gate capacity.

Claim: There exists a shared-parameter depth-indexed gate producing exact desired outputs for all finite task inputs.

Proof sketch: Finite lookup table; universal approximation on finite sets gives interpolation. $\square$

Critical caveat: The existence of target modulation sequences $g_{\{i,t\}}$ is **underdetermined** for generic nonlinear F . If F is contractive, many targets are unreachable regardless of gating.

Verdict: Conditionally true; otherwise underdetermined.

Classification: Plausible adaptation of universal approximation; reachability is a conjecture.

3. Prior Art Analysis

| Prior Art | Same Mathematical Problem? | Classification |

| --- | --- | --- |

| **Orthogonal gradient methods** (GEM, A-GEM, GPM) | No — operate on gradients in parameter space during training; do not address forward-pass depth-aliasing in frozen composition | Known, partially relevant |

| **Parameter isolation / masking** (PackNet, Supermasks) | No — introduce per-task masks or per-task parameters; constraint-violating | Known, related but different |

| **Progressive networks / PathNet** | No — add task-specific architecture; effectively per-task parameter growth | Known, different problem |

| **Hypernetworks** (Ha et al.) | Partially — can generate input-dependent per-depth modulations, but output is per-layer (violates constraint in spirit) | Known, partially relevant |

| **LSTM / GRU** | **Closest** — shared-parameter gates at every time step. Crucially, gates depend on current state h_{t-1} , not only input x . This is the key difference: state-conditioning is what makes recurrent gating effective | Known, directly relevant, different information assumption |

| **Mixture of Experts** | No — routes to different parameter sets with per-layer routers; different problem | Known, structurally similar |

| **Universal Transformer / ALBERT** | Partially — weight-tied depth, but no task isolation via gates | Known, background context |

No surveyed prior art solves this exact problem. The formulation — isolation of computations within a frozen weight-tied path under an input-only gate, posed for continual learning — appears to be new.

4. Obstruction Characterization

The failure condition is met. The obstruction has four independent parts:

4.1 Diagonalization Obstruction

A depth-independent gate $g(x)$ is confined to the diagonal of M^D . Tasks requiring depth-varying modulation lie off-diagonal. This is representational, not algorithmic.

4.2 Preservation–Identity Obstruction

Exact trajectory preservation under output gating forces $g(x) = \mathbf{1}$ wherever F is nonzero. A gate that changes nothing isolates nothing.

4.3 Rank / Parameter-Budget Obstruction

Exact gradient isolation requires mutually orthogonal task-Jacobian row spaces. $\sum_i \text{rank}(J_i) \leq p$ permits at most one generic task when $p \leq d$.

4.4 Expansive-Dynamics Obstruction

For generic nonlinear F , $L_\Phi > 1$, and small gate errors amplify exponentially in depth. Even bounded interference requires architectural constraints on F .

5. Conditions Under Which Obstructions Vanish

| Obstruction | Escape Route |

| --- | --- |

| Diagonalization | Depth-indexed shared gate $g_\phi(x,t)$ or state-conditioned gate $g(s_t)$ |

| Identity-forcing | Require only final-output preservation (not whole trajectory); or relax to asymmetric preservation (exact on old tasks, free on new) |

| Rank budget | Low-rank task subspaces; disjoint feature activation in gate architecture |

| Expansive dynamics | Architectural constraint $L_{\Phi} < 1$; or limit depth D |

6. Novelty Ledger (Adjudicated)

| Claim | Classification |

| --- | --- |

| Formalization: isolation within frozen weight-tied composition via input-only gate, CL-framed | **Apparently novel** (as packaging) |

| Identity-forcing lemma (H1) | **Direct consequence** |

| Diagonalization obstruction (H2) | **Direct consequence** (control-theoretic flavor) |

| Rank/parameter-budget bound (H3, H4) | **Direct consequence** |

| Contraction/amplification error bound (H5) | **Direct consequence** |

| Necessity of state/depth information for effective gating | **Plausible adaptation** (consistent with LSTM/ACT practice) |

| t -conditioned shared gate achieves isolation broadly (H6) | **Conjecture** (reachability undetermined) |

7. Critical Open Questions

7.1 Asymmetric Preservation (Highest Priority)

The identity obstruction applies when preservation is required on **all** inputs. The continual-learning-relevant requirement is asymmetric: exact preservation on old-task inputs S_{old} , unconstrained on new-task inputs S_{new} . Under this reading, the identity obstruction restricts to S_{old} ; non-trivial gates become admissible off it. The diagonal and rank obstructions survive. **This reformulation should be the primary next theorem target.**

7.2 Scope of t -Conditioning

A shared-parameter gate $g_{\phi}(x,t)$ with t as input (e.g., via a fixed depth embedding) adds $O(1)$ parameters and **does not violate "no per-depth/per-level trainable parameters" as literally worded**. This is the pivotal loophole: every impossibility result targets the strictly depth-agnostic gate and dissolves into conditional statements once t is admitted. The team must rule explicitly on whether the intended constraint forbids t -conditioning.

7.3 BDH Instantiation

BDH's block has residual structure and elementwise multiplicative (Hebbian-type) terms. Consequences:

- Output-gating the full map would gate the residual stream — likely harmful
- Gating only the delta term is the natural placement

- BDH's native plasticity must be explicitly disabled for the "frozen path" premise to hold

These checks against paper . tex and the repository were not performed in this analysis.

8. Strategic Recommendation

The strict reading of the research question yields a clean impossibility result. The investigation should:

- **Publish the impossibility-and-bounding package** as a formal negative result for weight-tied architectures, citing LSTM/GRU and Universal Transformer as nearest neighbors. Claim formalization novelty, not mathematical novelty.
- **Develop the asymmetric-preservation variant** as the honest CL formalization. This is the highest-value open result.
- **Decide on ϕ -conditioning scope.** If permitted, specify a minimal $\phi(x,t)$ and test empirically — labeling outcomes strictly as empirical conjecture.

- **Instantiate with BDH's concrete update equations** from paper . tex; fix gate placement relative to the residual stream; confirm plasticity disabling.
- **Reconcile with implementation reports** (plan_v2, cl-h1-report) once retrievable; verify that the implemented mechanism corresponds to one of the formally scoped variants.

9. Answer to the Research Question

> Can input-path interference arising from shared sequential composition be eliminated, bounded, or made controllable by an input-dependent gate/projection mechanism, without introducing per-depth/per-level trainable parameters and without retraining the frozen shared path?

Elimination: Impossible in general under the stated constraints. Three independent obstructions (identity-forcing, diagonalization, rank budget) are proven as direct consequences of the definitions.

Bounding: Possible if and only if the gated dynamics are non-expansive ($L_{\Phi} < 1$) or depth is small enough to control exponential amplification. This is a property of F and the gating rule, not achievable by the gate alone.

Controllability: Achievable only by enriching the gate's information — either via depth-indexed shared-parameter conditioning $g_\phi(x,t)$, state-dependent conditioning $g(s_t)$, or strong low-rank/invariant-subspace structure on the tasks. Each escape route transforms the impossibility into a conditional feasibility statement with its own capacity and reachability requirements.

[CRITIC Round 3]

Critical Review of the Consolidated BDH Isolation Analysis

Overall assessment. The analysis is a useful exercise in abstraction and identifies some real distinctions (especially forward-pass *depth aliasing* versus training-time *gradient interference*). But the central claim—that exact elimination of input-path interference is impossible under the stated constraints—is overstated. Several “theorems” are either tautological under their assumptions, correct only for a degenerate version of the problem, or depend on unproved identifications between forward-pass subspaces and gradient orthogonality. The analysis also does not instantiate the actual BDH recurrence from `paper.tex`, so the results remain conditional on an abstraction that may not match the architecture.

1. The analysis is not grounded in the actual BDH equations

The main object is abstracted as

\[

$$s_0 = x, \quad s_{t+1} = F(s_t).$$

\]

But the report does not verify from the paper that BDH is actually a simple deterministic iterated map of this form. BDH is described as a scale-free network of locally interacting neuron particles, and the report itself claims—without evidence—that the block has “residual structure and Hebbian-type terms.” If BDH includes:

- local particle interactions,
- residual streams,
- input-dependent routing,
- Hebbian plasticity,
- stochasticity,

then the abstraction $s_{t+1} = F(s_t)$ may omit exactly the mechanisms that matter for continual learning.

The source cited for shared parameters is a third-party README (“same weights reused in all layers”), not the paper equations. That is not enough to validate the formal model. In particular, gating the whole block $g(x) \odot F(s_t)$ is very different from gating only a residual delta $s_t + g(x) \odot \Delta(s_t)$, or gating local interactions inside the particle graph.

Gap: the analysis should start from the actual BDH update equation and locate exactly where a gate or projection would be inserted. Without that, the impossibility claims are architecture-agnostic, and possibly architecture-inapplicable.

2. The notion of “input-path interference” is unstable across the analysis

The original draft defined interference as one input’s forward trajectory changing because another input’s mechanism is active. The moderator correctly observed that this cross-input forward-pass interference is structurally zero when parameters are frozen and the gate is stateless. But then the analysis substitutes a different notion: *depth-aliasing*, the mismatch between a depth-independent gate and an ideal depth-varying modulation.

These are not the same mathematical problem.

- **Depth-aliasing** is a representational issue: can a depth-constant gate imitate a depth-varying gain?
- **Continual-learning interference** is a parameter-update issue: updating shared gate parameters for task B degrades task A.

The final report calls the first notion “input-path interference,” but the actual catastrophic-forgetting channel in this setting must involve the trainable gate parameters ϕ , because F is frozen. The analysis only partially connects these two.

Gap: the paper should define the continual-learning objective explicitly:

- Tasks $(T_1, \dots, T_k)$ arrive sequentially.
- (F) is frozen.
- (ϕ) is updated.
- Preservation is required only on previous tasks.

This asymmetric setup is the relevant one, and it is not what the main impossibility proof uses.

3. The identity-forcing theorem is true but trivial

H1 says: under output gating,

[

$$s_{t+1} = g(x) \odot F(s_t),$$

]

exact preservation of the entire unperturbed trajectory forces $(g(x) = \mathbf{1})$ wherever $(F_i \neq 0)$.

This is correct, but it is almost a tautology: if the mechanism is required to reproduce the exact unmodified computation at every depth for every input, then the only possibility is no modification. The theorem does not show that isolation is impossible. It shows that **preservation of**

everything and alteration of anything are incompatible, which is essentially the definition of exact preservation.

The relevant condition should be:

[

$\tilde{s}_t(x) = s_t(x) \quad \forall x \in S_{\{\text{old}\}}, \forall t,$

]

while allowing $\tilde{s}_t(x) \neq s_t(x)$ for $(x \in S_{\{\text{new}\}})$.

Under this asymmetric condition, the identity obstruction may disappear. The report acknowledges this only as future work, but the headline conclusion is drawn from the stronger, less relevant condition.

4. The diagonalization obstruction is not a general theorem

H2 assumes that an ideal modulation $g^*(x, t)$ is nonconstant in (t) , and that the transition rule $\Phi(g, s)$ is injective in (g) along the trajectory. If so, a depth-independent $g(x)$ cannot match the ideal sequence.

This is true given those assumptions, but the assumptions are not established.

- A depth-constant gate can produce depth-varying effects because the state (s_t) changes. For input gating,

$$s_{t+1} = F(g(x) \odot s_t),$$

the same $(g(x))$ can produce different effective updates at different depths because (s_t) differs. Therefore the set of reachable trajectories is not simply the diagonal of the product space of depth-wise modulations.

- For output gating, injectivity of $(\Phi(\cdot, s_t))$ fails on any coordinate where $(F_i(s_t) = 0)$, which is common with ReLU/GLU/zero activations.

Thus the “diagonalization obstruction” is conditional on a strong injectivity assumption and on the premise that the target trajectory requires nonconstant gate inputs in the same coordinate system. It is not a proof that generic tasks require off-diagonal modulation sequences. One would need a reachability argument in control theory, showing that the image of the constant-gate map does not contain the desired task map.

5. The projection rank theorem does not imply the isolation bound

H3 states:

- Exact preservation $\|P(x)F(s_t) - F(s_t)\|$ forces

$\|$

$\operatorname{rank} P(x) \geq \dim \operatorname{span} \{F(s_t)\}.$

$\|$

This is correct linear algebra.

But the next step—that mutually orthogonal projections are required to avoid gradient interference—is not proved and is likely false as a general statement.

Orthogonality of the image subspaces of $\|P(x_A)\|$ and $\|P(x_B)\|$ is a forward-pass geometric property. Cross-task gradient interference is a property of the parameterization of $\|P(x)\|$. If $\|P(x)\|$ is produced by a neural network, two different inputs may have non-orthogonal or identical image subspaces but still have gradients that are orthogonal, or vice versa. The connection between these two notions depends on the architecture of the projection generator and is not automatic.

Therefore the capacity bound

$$\sum_i m_i \leq d$$

does not follow. It only follows if one assumes a very special architecture in which each task's projection is a fixed, learned orthogonal subspace with independent parameters. That is a much stronger assumption than "input-dependent projection."

6. The parameter-budget bound is notationally sloppy and not operational

H4 defines task Jacobians $\{J_i \in \mathbb{R}^{d \times p}\}$ as derivatives of $s_D(x; \phi)$ with respect to ϕ , but tasks are sets of inputs, not single inputs. The aggregation matters.

The condition

$$J_i J_j^{\text{top}} = 0$$

is not equivalent to zero gradient interference for general loss functions. The first-order change in task A's loss after a gradient step on task B is governed by

$$\langle \nabla_{\phi} \mathcal{L}_A, \nabla_{\phi} \mathcal{L}_B \rangle,$$

which involves the loss gradients, not just output Jacobians. Even at the output-Jacobian level, orthogonality of all Jacobian row spaces is a very strong condition.

Finally, the conclusion “at most one task when $p \leq d$ ” is not the relevant regime. A gate network attached to an LLM-scale BDH path would typically have $p \gg d$. If p is large, the rank budget can admit many tasks. The theorem gives no practical bound without estimating the actual Jacobian ranks.

7. The error bound statement has an incorrect “iff”

H5 derives

$$\langle$$

$$e_{t+1} \leq L_\Phi e_{t+\delta},$$

$\forall$

hence

$\forall$

$$e_D \leq \delta \sum_{k=0}^{D-1} L_\Phi^k.$$

$\forall$

This is correct. But the conclusion that bounding is possible **exactly when $\forall(L_\Phi < 1)$** is not accurate. For any finite depth $\forall(D)$, the sum is finite even if $\forall(L_\Phi > 1)$. The bound grows exponentially with $\forall(D)$, but it still exists.

The correct statement is:

- a bound **uniform in depth** exists iff $\forall(L_\Phi < 1)$;
- for fixed finite $\forall(D)$, a bound exists for any finite $\forall(L_\Phi)$.

The analysis also asserts that trained LLM-scale maps typically have $\forall(L_\Phi > 1)$. That is an empirical claim about BDH or LLMs, not proved from the paper or code. It should be labeled as such and tested.

8. H6 is conditional to the point of tautology

H6 states that a depth-indexed shared gate $\backslash(g_{\phi}(x,t)\backslash)$ can exactly realize finite task outputs if:

- some modulation sequence $\backslash(g_{i,t}\backslash)$ realizes each target output, and
- $\backslash(g_{\phi}\backslash)$ can interpolate those sequences.

Under those assumptions, the conclusion is immediate. The nontrivial question is whether assumption 1 holds: **which target input-output maps are reachable by the frozen shared path plus the allowed gate class?** That question is unproved and marked as conjecture.

Therefore H6 does not establish controllability. It reformulates the problem without solving it.

9. “Controllability” is never formally defined

The research question asks whether interference can be eliminated, bounded, or **made controllable**. The analysis focuses on elimination and bounding, but does not define controllability.

A proper formalization would likely treat the gate at each depth as a control input and ask whether the composed system

\[

$s_{t+1} = F(s_t) + \text{ or } \odot \text{ gate control}$

\]

can steer states to desired outputs. This would invoke nonlinear reachability, injectivity of the transition map, or approximate controllability results for recurrent networks.

Because controllability is not formalized, the report’s statement that “controllability can be achieved only by enriching gate information” is not a theorem; it is an intuitive recommendation.

10. The $\ell(t)$ -conditioning loophole is unresolved and ambiguous

The report correctly identifies that a shared-parameter gate receiving depth index $\ell(t)$ as input would avoid the strict depth-agnostic obstruction. But it does not resolve whether this is allowed.

Two cases:

- **Fixed non-learned depth encoding**, e.g. sinusoidal positional encoding: adds $\mathcal{O}(1)$ trainable parameters, likely does not violate the letter of “no per-depth/per-level trainable parameters.”
- **Learned depth embedding**: adds $\mathcal{O}(D)$ trainable embedding rows, which is exactly per-depth trainable parameters, violating the constraint.

The final recommendation says “decide on $\mathcal{O}(t)$ -conditioning scope,” but the impossibility result is only valid for the stricter depth-agnostic reading. Until this scope is fixed, the answer is conditional, not definitive.

11. Prior-art review is incomplete and partly too dismissive

The report reviews parameter isolation, progressive networks, gradient projection, hypernetworks, MoE, LSTMs, and weight-tied transformers. However, some directly relevant literature is missing or dismissed too quickly:

- **FiLM / feature-wise linear modulation:**

$\mathcal{V}$

$$y = \gamma(x) \odot h + \beta(x),$$

$\mathcal{V}$

is essentially an input-dependent affine gate applied to a shared layer. Multi-task and continual variants are highly relevant.

- **Fast weight programmers and recurrent hypernetworks** that generate input-dependent modulations for a fixed recurrent core are closer to the proposed mechanism than the report admits.
- **Nonlinear RNN reachability / observability theory** would be the correct framework for the controllability question, but is absent.
- **Echo state networks** and reservoir computing may provide relevant results on fixed recurrent paths plus trained output/gate layers.

The classification “no surveyed prior art solves this exact problem” may be true, but the prior-art search is not exhaustive enough to support a strong novelty claim.

12. Novelty labeling is inconsistent

Earlier rounds label the impossibility result “apparently novel.” Later adjudication correctly says the mathematical ingredients are “direct consequences.” The final report mostly uses “direct consequence,” but still calls the packaged problem formalization “apparently novel.”

This is acceptable, but it should be made consistently: the contribution is a **problem formulation and synthesis of elementary obstructions**, not new mathematics. Overclaiming novelty would be a weakness.

13. Missing reconciliation with the H1 implementation report

The local file `2026-08-25_cl-h1-report.md` was not retrieved. If it reports empirical behavior, the formal analysis neither explains nor refutes it. The task explicitly says not to treat experimental effects as theorem evidence, but at minimum the analysis should check which formal variant the implementation corresponds to.

If the implementation used a depth-indexed or state-dependent gate, then the strict impossibility results are irrelevant to that implementation. If it used a depth-independent gate, the analysis should be directly tested against the measured behavior.

Bottom line

The analysis successfully identifies:

- the distinction between forward-pass depth aliasing and training-time gradient interference,
- the fact that exact preservation of the full unperturbed trajectory forces identity under multiplicative output gating,
- the standard Lipschitz error-amplification bound.

But it fails to prove the main impossibility claim in the continual-learning-relevant setting. The actual relevant condition is **asymmetric preservation**: old tasks must be preserved, new tasks may change. That problem is not solved. The diagonalization, projection-rank, and parameter-budget obstructions do not survive as general theorems without additional, currently unproved assumptions.

The correct characterization is:

- **Strict global-preservation setting**: impossibility is trivial and proved.
- **Asymmetric CL setting**: open; the current analysis provides only necessary conditions and conjectures.
- **BDH-specific setting**: unresolved because the actual update equations were not instantiated.
- **Controllability**: not formally defined.

The report should be rewritten to state these boundaries explicitly, rather than announcing a general impossibility result.

[OPTIMIZER Round 3]

Refined Analysis: Input-Path Interference, Gating/Projection, and Computation Isolation in a Frozen Shared Sequential Composition

This document is a revised version of the earlier draft. It incorporates the critical review by (i) clarifying the different meanings of “interference,” (ii) sharply separating exact results from conditional or conjectural statements, (iii) correcting overclaiming in the diagonalization and projection arguments, and (iv) making the continual-learning-relevant asymmetric problem the central object of analysis.

0. Evidence and Scope

The public BDH paper (arXiv:2509.26507) and third-party material confirm that BDH is a scale-free, biologically inspired network of locally interacting neuron particles and that it reuses shared parameters across depth (“recurrent depth” in existing implementations). The internal project reports named in the task could not be retrieved during this analysis, so no claim below is checked against the literal BDH update equations or the H1 implementation report.

Accordingly, this analysis is about the following **abstraction**, which the reports assume:

$\backslash[$

$$s_0(x)=x,\quad s_{t+1}(x)=F_{\theta}(s_t(x)),\quad t=0,\dots,D-1.$$

$\backslash]$

All theorems are proved for this abstraction. They are not automatically theorems about BDH until $\backslash(F\backslash)$ is replaced by the actual BDH particle

update. In particular, BDH may have a residual structure or Hebbian-style multiplicative terms; the *placement* of a gate (on the whole block, on a residual delta, or on an internal interaction) is part of the design and is not settled here.

1. Formalizations

1.1 Shared Sequential Composition

Let $\mathcal{X} \subseteqq \mathbb{R}^d$ be the input/state space and let $(F: \mathcal{X} \rightarrow \mathcal{X})$ be a fixed map.

A **depth- D shared sequential composition** is the iteration

$$\begin{aligned} &[\\ s_0(x) &= x, \quad s_{t+1}(x) = F(s_t(x)). \end{aligned}$$

]

The unperturbed trajectory is

$$\begin{aligned} &[\\ \tau_F(x) &= (s_0(x), s_1(x), \dots, s_D(x)). \end{aligned}$$

]

If BDH uses residual connections, the relevant recurrence is instead

$$\begin{aligned} & \backslash \\ & s_{t+1}(x) = s_t(x) + F(s_t(x)). \\ & \backslash \end{aligned}$$

The theory below applies to the abstract iterated map; each gating theorem must be re-instantiated for the residual variant.

1.2 Frozen Shared Path

The shared path $\backslash(F_{\theta})$ is **frozen** if its parameters $\backslash(\theta)$ are not updated during the continual-learning phase. Only the gate/projection mechanism may be trained:

$$\begin{aligned} & \backslash \\ & \frac{\partial F_{\theta}}{\partial \theta} = 0 \\ & \backslash \end{aligned}$$

during all adaptation steps. Nothing below relies on retraining $\backslash(F)$.

1.3 Gating and Projection

Input-dependent gate. A map

\[

g:\mathcal{X}\to\mathbb{R}^d

\]

that is applied identically at every depth. The two canonical placements are:

\[

\text{output gating:}\quad s_{t+1}=g(x)\odot F(s_t),

\]

\[

\text{input gating:}\quad s_{t+1}=F\bigl(g(x)\odot s_t\bigr).

\]

For residual blocks, the natural variant is

\[

\text{\delta gating:}\quad s_{t+1}=s_t+g(x)\odot F(s_t).

\]

The gate may have trainable parameters ϕ , but the strict version of the research question forbids per-depth trainable parameter vectors. We will also consider the relaxed variant

$$g_{\phi}(x, t),$$

where the depth index is an **input** to the gate rather than an index into a separate parameter table. If t is encoded by a fixed, non-trainable embedding, this does not introduce per-depth trainable parameters. If a learned embedding indexed by t is used, it does.

Input-dependent projection. A map

$$P: \mathcal{X} \rightarrow \Pi(\mathbb{R}^d),$$

where $\Pi(\mathbb{R}^d)$ is the set of orthogonal projections, applied as

$$s_{t+1} = P(x)F(s_t) \quad \text{or} \quad s_{t+1} = F(P(x)s_t).$$

1.4 Input-Path Interference: Three Distinct Notions

The word “interference” is ambiguous. We separate three notions.

(A) Cross-input forward-pass contamination.

If F is frozen and the gate is a stateless function of the input x , then the trajectory of x_1 is *computationally independent* of the presence or absence of x_2 . Formally,

$$\begin{aligned} & \mathbb{E}[\tilde{s}_t^{(M)}(x_1 \mid x_2 \text{ processed elsewhere})] = \mathbb{E}[\tilde{s}_t^{(M)}(x_1)]. \end{aligned}$$

Thus this notion, which appeared in the original draft, is **structurally zero** in this setting. It is not a meaningful measure of interference here.

(B) Forward depth-aliasing.

A depth-agnostic gate $g(x)$ applies the same modulation at every depth. If an oracle with access to depth t would apply a reference modulation $g^*(x, t)$, the **depth-aliasing error** for input x is

$$\begin{aligned} & \mathbb{E}[\mathrm{DA}(g; x)] \end{aligned}$$

$$= \sum_{t=0}^{D-1} \left| \Phi(g(x), s_t(x)) - \Phi(g^*(x, t), s_t(x)) \right|,$$

where Φ is the chosen gating/projection rule. This is a *representational* mismatch inside a single forward pass.

(C) Training-time gradient interference (cross-task forgetting).

Because the gate parameters ϕ are shared across all inputs, updating ϕ on task B may change the gate output on task- A inputs. First-order damage to task A from one gradient step on task B is proportional to

$$\left| \nabla_{\phi} \mathcal{L}_A(\phi), \nabla_{\phi} \mathcal{L}_B(\phi) \right|,$$

where $\mathcal{L}_i$ is the task loss. If this is nonzero, the shared gate is a vector of forgetting.

The operationally relevant channels for continual learning are **(B)** (within a pass) and **(C)** (across passes). Cross-input forward contamination, notion **(A)**, is not a live channel under a frozen stateless model.

1.5 Computation-Isolability

A mechanism is **exactly computation-isolating** for a task family $\{(S_i, y_i)\}_{i=1}^K$ if:

- each task output is produced exactly:

$$\begin{aligned} & \forall \\ & \tilde{s}_D(x; \phi) = y_i(x) \quad \forall x \in S_i, \forall i; \\ & \end{aligned}$$

- there is zero forward depth-aliasing error on the task trajectories whenever a reference trajectory is specified;

- there is zero first-order cross-task gradient interference:

$$\begin{aligned} & \forall \\ & \langle \nabla_{\phi} \mathcal{L}_i(\phi), \nabla_{\phi} \mathcal{L}_j(\phi) \rangle \\ & = 0 \\ & \quad \forall i \neq j. \end{aligned}$$

ϵ

A mechanism **bounds interference** if the relevant error is at most ϵ (ϵ), with ϵ controlled by explicit quantities: Lipschitz constants, depth, gate error, capacity, task geometry.

1.6 Elimination versus Bounding

- **Elimination** means exact equality of outputs and exact zero interference in the sense above.
- **Bounding** means a finite worst-case deviation ϵ .
- A third mode, **controllability**, is often invoked informally. We formalize it as *reachability*: a task target $y_i(x)$ is **reachable** if there exists some gate parameterization ϕ such that $s_D(x; \phi) = y_i(x)$ for all $x \in S_i$. The mechanism is controllable over a task sequence if every finite sequence of reachable tasks is simultaneously realizable by a single ϕ .

2. Theorem-Shaped Hypotheses

H1. Global Exact Preservation Forces the Gate to Be the Identity

Assumptions.

- F is defined on an open set $U \subseteq \mathbb{R}^d$.
- The mechanism is output gating with a depth-agnostic gate:

\[

$$s_{t+1}=g(x)\odot F(s_t).$$

\]

- Exact preservation of the *entire unperturbed trajectory* is required:

\[

$$\tilde{s}_t(x)=F^t(x)\quad\text{for all }x\in U,\text{ for all }t=0,\dots,D.$$

\]

Mathematical object gated.

The output of the frozen map $\backslash(F\backslash)$ at every depth.

Claim.

For every $\backslash(x\backslash)$ and every coordinate $\backslash(i\backslash)$,

\[

$$\bigl|g_i(x)-1\bigr|\backslash, F_i(s_t(x))=0.$$

\]

Hence, wherever $\backslash(F_i(s_t(x))\neq 0\backslash)$ along the trajectory, $\backslash(g_i(x)=1\backslash)$. If $\backslash(F\backslash)$ is componentwise nonzero along all trajectories on $\backslash(U\backslash)$, then $\backslash(g(x)=\mathbf{1}\backslash)$ for all $\backslash(x\backslash)$.

Proof.

At depth $\ell(t)$, exact preservation gives

ℓ

$$g(x) \odot F(s_t(x)) = F(s_t(x)).$$

ℓ

Componentwise, $\ell((g_i(x)-1)F_i(s_t(x))=0)$. Repeating for all $\ell(t)$ gives the result. $\square$

Verdict.

True as an exact theorem, but it is close to a tautology: *if the mechanism is required to change nothing, then it changes nothing*. It does **not** prove that computation isolation is impossible. It proves that exact global preservation and nontrivial gating are incompatible.

Weakest assumptions for escape.

If preservation is required only for final outputs, or only on a subset $\ell(S_{\{\text{old}\}})$ of inputs, the identity obstruction becomes restricted to that subset. This is the continual-learning-relevant relaxation.

Classification.

Known / direct consequence.

H2. Asymmetric Preservation: The Honest Continual-Learning Setting

Assumptions.

- Old-task input set $(S_{\text{old}}) \subset \mathcal{X}$.
- New-task input set $(S_{\text{new}}) \subset \mathcal{X}$.
- Output gating $(s_{t+1} = g(x) \odot F(s_t))$.
- Exact preservation is required only on old inputs:

[

$\tilde{s}_t(x) = F^t(x) \iff \forall x \in S_{\text{old}}, \forall t$.

]

No preservation constraint is imposed on (S_{new}) .

Mathematical object constrained.

The restriction of the gate function (g) to (S_{old}) .

Claim.

For $(x \in S_{\text{old}})$,

[

$g_i(x) = 1$

$\quad\text{whenever}\quad$

$\exists t: F_i(s_t(x)) \neq 0.$

$\]$

On the zero-coordinate set

$\[$

$Z(x) = \{i: F_i(s_t(x)) = 0 \text{ for all } t\},$

$\]$

the gate is unconstrained by preservation. On (S_{new}) , the gate is entirely unconstrained by this requirement.

Proof.

Same componentwise argument as H1, restricted to (S_{old}) . $\square$

Verdict.

True as an exact theorem. This theorem replaces the trivial global statement with a useful necessary condition. In particular, if the old-task trajectories have support in all coordinates, then the gate is forced to be identity on all of (S_{old}) .

Remark. If additionally old and new inputs are disjoint and the gate network can interpolate between the identity mapping on (S_{old}) and the desired nontrivial mapping on (S_{new}) , then exact

isolation is *not* ruled out by the identity obstruction. The real obstruction becomes task separability and gate capacity, not preservation per se.

Classification.

Direct consequence, but the asymmetric framing is the correct continual-learning formalization.

H3. Input-Only Mechanisms Cannot Resolve Task Ambiguity at Overlapping Inputs

Assumptions.

- The mechanism is a deterministic function of the input: $\tilde{s}_D(x; \phi) = \mathcal{M}_\phi(x)$.
- Two tasks (A, B) share an input: $x \in S_A \cap S_B$.
- Their target outputs differ: $y_A(x) \neq y_B(x)$.

Mathematical object constrained.

The input-output map $\mathcal{M}_\phi$.

Claim.

No single parameter vector ϕ can satisfy both tasks. Exact computation isolation is impossible for this task family.

Proof.

The map is a function: for a fixed ϕ , it returns exactly one output for the input x . Two different required outputs are incompatible. $\square$

Verdict.

True as an exact theorem. This is a genuine obstruction of the “input-dependent-only” constraint, independent of depth, nonlinearity, and capacity.

Weakest assumptions for escape.

Tasks must have disjoint input supports, or target functions must agree on overlaps, or the mechanism must be augmented with a task/context input. Any of these relaxations removes the obstruction.

Classification.

Direct consequence; essentially an identifiability condition.

H4. Depth-Agnostic Gating Is a Representational Limitation, Not a General Behavioral Impossibility**Assumptions.**

- The transition is $\Phi(g,s)$.

- The *gate-value sequence* required by an oracle is $\langle g^*(x,0), \dots, g^*(x,D-1) \rangle$, nonconstant in $\langle t \rangle$.
- The transition is injective in the gate argument along the trajectory.

Mathematical object gated.

The sequence of gate values applied at each depth.

Claim.

A depth-agnostic gate $\langle g(x) \rangle$ cannot reproduce the nonconstant gate-value sequence $\langle g^*(x,t) \rangle$.

Proof.

If it did, equality at every depth would give

[

$$\Phi(g(x), s_t(x)) = \Phi(g^*(x,t), s_t(x)).$$

]

By injectivity, $\langle g(x) = g^*(x,t) \rangle$ for every $\langle t \rangle$, contradicting nonconstancy. $\square$

Important correction.

This theorem is **not** a general behavioral impossibility. A constant-in-depth gate can produce depth-varying effective behavior because the state $\langle s_t \rangle$ changes.

Counterexample to the stronger claim.

Let $(F(s)=s)$ and $(D=2)$. Output gating gives

[

$$s_2 = g(x) \odot g(x) \odot x.$$

]

A constant gate $(g(x)=(\alpha, \beta))$ produces

[

$$s_2 = (\alpha^2 x_1, \beta^2 x_2).$$

]

A depth-varying gate sequence $(g^0_0=(c, 1))$, $(g^1_1=(1, c))$ would produce $((c x_1, c x_2))$, which is also achievable by the constant gate $((\sqrt{c}, \sqrt{c}))$ when $(c \geq 0)$. More generally, constant gates can emulate many depth-varying controls by exploiting state feedback.

Thus the earlier claim that “a depth-agnostic gate is confined to the diagonal of the modulation product space” is a statement about **gate values**, not about **reachable input-output maps**. It does not, by itself, prove an impossibility for task behaviors.

Verdict.

Underdetermined as a general theorem. The representational statement is true; the behavioral conclusion does not follow.

Weakest assumptions for a behavioral theorem.

One would need a control-theoretic reachability analysis: show that the image of the constant-gate map does not contain the desired task map for a specific $\mathcal{V}$, task set, and gate class. This is task-specific.

Classification.

The representational lemma is a direct consequence. The broad impossibility is **false** as stated; the reachability question is open.

H5. Projection Mechanisms: A Necessary Rank Condition

Assumptions.

- Output projection:

$\mathcal{V}$

$$s_{t+1} = P(x)F(s_t).$$

$\mathcal{V}$

- Exact preservation of the unperturbed trajectory of $\mathcal{V}(x)$ is required.

Mathematical object projected.

The image subspace of the projection $\mathcal{P}(x)$.

Claim.

If

$\mathcal{P}$

$m(x) = \dim \operatorname{span} \left\{ F(s_0(x)), F(s_1(x)), \dots, F(s_{D-1}(x)) \right\},$

$\mathcal{P}$

then exact preservation requires

$\mathcal{P}$

$\operatorname{rank} \mathcal{P}(x) \geq m(x).$

$\mathcal{P}$

Proof.

Exact preservation gives

$\mathcal{P}$

$\mathcal{P}(x)F(s_t(x)) = F(s_t(x))$

$\mathcal{P}$

for every t . Hence all $\{F(s_t(x))\}$ belong to $\{\text{Im}\{P(x)\}\}$, whose dimension is $\{\text{rank}\{P(x)\}\}$. $\square$

Verdict.

True as an exact linear-algebra theorem.

What this theorem does not say.

It does **not** imply a cross-task capacity bound $\sum_i m_i \leq d$. Orthogonality of image subspaces of two projections is a *forward-pass geometric* property, whereas cross-task gradient interference is a property of the *parameterization* of $\{P(x)\}$. Two inputs can have non-orthogonal image subspaces but orthogonal gate gradients, or vice versa. The earlier draft conflated these. The rank budget reappears below only under an explicit parameterization assumption.

Classification.

Direct consequence.

H6. Gradient Interference and the Parameter Budget

Assumptions.

- The gate/projection mechanism has p trainable parameters ϕ .
- For task i , define per-example output Jacobian

\[

$$J_i(x) = \frac{\partial}{\partial \phi} \tilde{s}_D(x; \phi).$$

\]

- Let $(R_i \subseteqq \mathbb{R}^p)$ be the row space of the collection of Jacobians over task (i) .
- We require *zero first-order interference for every smooth loss and every example*.

Mathematical object constrained.

The parameter space of the gate mechanism.

Claim.

If zero interference is required for all smooth losses, then the row spaces $(\{R_i\})$ must be mutually orthogonal:

\[

$$R_i \perp R_j \quad \forall i \neq j.$$

\]

Consequently,

\[

$$\sum_{i=1}^K \dim R_i \leq p.$$

\]

Proof.

For losses $\mathcal{L}_i$, first-order interference is

\[

$$\langle \nabla_{\phi} \mathcal{L}_i, \nabla_{\phi} \mathcal{L}_j \rangle.$$

\]

For this to vanish for *arbitrary* loss functions, the linear functionals induced by the Jacobian row spaces must be mutually orthogonal, which forces row-space orthogonality. Orthogonal subspaces of $\mathbb{R}^p$ have total dimension at most p . $\square$

Important caveats.

- For fixed, specific losses, zero gradient interference is the weaker condition

\[

$$\langle \nabla_{\phi} \mathcal{L}_i, \nabla_{\phi} \mathcal{L}_j \rangle = 0,$$

\]

which can hold even when Jacobian row spaces overlap.

- The bound is **not** a depth obstruction; it applies to any shared-parameter mechanism.
- If $\|p\|$ is large relative to $\|d\|$, the bound does not stop many tasks from being isolated. It becomes restrictive only when the task Jacobians have full row rank and $\|p\| \leq \|d\|$.

Verdict.

True as an exact theorem under the stated generality. It is standard parameter-isolation accounting, not a new result.

Classification.

Direct consequence.

H7. Error Amplification: Bounding Is Conditional on Dynamics, Not Automatic

Assumptions.

- The gated transition $\Phi(g, s)$ is Lipschitz in state with constant L :

$\|$

$$\|\Phi(g, s) - \Phi(g, s')\| \leq L \|s - s'\|.$$

\]

- The per-depth gate error against a reference is bounded:

\[

$$\|\Phi(g(x), s_t^*(x)) - \Phi(g^*(x, t), s_t^*(x))\| \leq \delta_t.$$

\]

Mathematical object bounded.

The final-state error.

Claim.

\[

$$\|s_D(x) - s_D^*(x)\|$$

$\leq$

$$\sum_{t=0}^{D-1} L^{D-1-t} \delta_t.$$

\]

If $(\delta_t \leq \delta)$, then

\[

$$e_D \leq$$

$$\begin{cases} \delta \frac{1-L^D}{1-L}, & L < 1, \\ \delta D, & L = 1, \\ \delta \frac{L^D-1}{L-1}, & L > 1. \end{cases}$$

Proof.

Let $e_t = |s_t - s_t^*|$. Then

$$\begin{aligned} e_{t+1} &\leq \\ &| \Phi(g, s_t) - \Phi(g, s_t^*) | \\ &+ \\ &| \Phi(g, s_t) - \Phi(g^*, s_t^*) | \\ &\leq \\ &L e_t + \delta_t. \end{aligned}$$

Solving the linear recurrence gives the stated bound. $\square$

Precise interpretation.

- For **fixed finite depth** $\backslash(D\backslash)$, the bound is finite for every finite $\backslash(L\backslash)$. There is no “iff $\backslash(L<1\backslash)$ ” statement for finite depth.
- A bound that is **uniform in depth** exists iff $\backslash(L<1\backslash)$. At $\backslash(L=1\backslash)$, the bound grows linearly in $\backslash(D\backslash)$; at $\backslash(L>1\backslash)$, exponentially.

Empirical caveat.

The claim “trained LLM-scale maps typically have $\backslash(L>1\backslash)$ ” is an empirical conjecture, not a theorem. It should be tested for a given BDH model.

Classification.

Known / direct consequence.

H8. Finite-Task Reachability: A Conditional Construction, Otherwise Underdetermined

Assumptions.

- The task family is finite: $\backslash((x_i, y_i)\backslash)$, $\backslash(i=1, \dots, N\backslash)$, with task inputs possibly grouped into tasks.
- The gate family has sufficient capacity to interpolate finite sets.
- For each task input $\backslash(x\backslash)$, the required output is reachable by the allowed gate class:

$\backslash[$

$y_i(x) \in \mathcal{R}_D(x) := \{\tilde{s}_D(x; \phi) : \phi \in \Phi\}.$

$\setminus$

- If an input appears in more than one task, the target outputs agree on the overlap.

Mathematical object controlled.

The finite input-output relation of the gated composition.

Claim.

Under these assumptions, there exists a parameter vector ϕ realizing all finite task targets exactly.

Proof sketch.

The requirements form a finite set of constraints. Reachability gives a target gate output for each input; universal/interpolation capacity of the gate family fits those finite constraints. $\square$

Counterexamples showing underdetermination.

- If $F(s) = 0$, then $\mathcal{R}_D(x) = \{0\}$ for every x , regardless of gating. Nontrivial outputs are impossible.
- If the same input belongs to two tasks with different target outputs, H3 makes simultaneous realization impossible.
- If $F(s) = s$ and the gate is multiplicative output gating with $D=2$, the reachable set is

$\setminus$

$\{g(x) \odot g(x) \odot x\}$.

$\setminus$

Many targets are reachable only if coordinatewise signs and roots are permitted; others are not.

Verdict.

Conditionally true; otherwise underdetermined. This is not a theorem about all nonlinear $\setminus(F)$.

Weakest additional assumptions.

Reachability of each target, disjoint task inputs, and gate capacity are sufficient. The reachability condition is the nontrivial one and must be checked for the concrete $\setminus(F)$.

Classification.

Plausible adaptation of finite interpolation and universal approximation; the reachability part is a **conjecture** unless proved for a specific $\setminus(F)$.

3. Summary of Verdicts

| Hypothesis | Verdict | Status |

| --- | --- | --- |

| H1: Global preservation forces identity | True | Exact theorem; direct consequence |

| H2: Asymmetric preservation constraint | True | Exact theorem; CL-relevant |

| H3: Overlapping inputs with different targets | Impossible | Exact theorem; identifiability |

| H4: Depth-agnostic gate-value limitation | Representational only | Direct consequence; behavioral claim false/underdetermined |

| H5: Projection rank condition | True | Exact linear algebra; not sufficient for isolation |

| H6: Gradient interference parameter budget | True | Exact under strong generality |

| H7: Error amplification bound | True | Exact; finite-depth vs uniform-depth distinction |

| H8: Finite-task reachability | Conditional | Plausible adaptation; reachability is conjecture |

4. Prior Art: Does It Address the Same Mathematical Problem?

| Prior art | What it actually does | Same problem? | Classification |

| --- | --- | --- | --- |

| Gradient projection / orthogonal gradient methods (GEM, A-GEM, GPM)

| Projects gradients in parameter space so new-task updates do not

increase old-task losses | Partly: addresses training-time gradient interference; not forward depth-aliasing; usually modifies the main network | Known, partially relevant |

| Parameter isolation / masks (PackNet, Supermasks, Piggyback) | Allocates disjoint parameter subsets to tasks | No: creates per-task parameters/masks; constraint-violating in the stated setting | Known, related but different |

| Progressive networks / PathNet | Adds new task-specific columns/paths | No: per-task architectural growth | Known, different problem |

| Hypernetworks / fast weight programmers | Generate weights or modulations for a main network | Partly: can generate input-dependent modulations; typically per-layer outputs, which violates the “no per-depth parameters” constraint in spirit | Known, partially relevant |

| FiLM / feature-wise linear modulation | Input-dependent affine modulation of hidden features | Partly: same gating algebra, but usually per-layer parameters; not a frozen weight-tied path | Known, directly relevant |

| LSTM / GRU | Shared-parameter state-dependent gates in a recurrent path | Closest: same composition structure; but gates see the current state, not only the original input | Known, nearest prior art with a different information assumption |

| Mixture of experts / conditional computation | Input-dependent routing to expert parameter sets | No: per-layer routers and distinct experts | Known, structurally similar |

| Reservoir computing / echo state networks | Fixed random recurrent path with trained readout | Partly: frozen recurrent core plus trainable output; not input-dependent forward gating for task isolation | Known, relevant background |

| Weight-tied transformers (Universal Transformer, ALBERT) | Reuse a block across depth | Partly: same composition; no task-isolation gating | Known, background context |

| Nonlinear RNN observability / reachability theory | Characterizes what a recurrent map can reach | Yes for the reachability question, but not used in the draft | Known, should be used for H8 |

Conclusion. No surveyed prior art solves the exact problem: isolation of computations inside one frozen shared sequential composition using only an input-dependent gate, with no per-depth trainable parameters. The closest neighbors are LSTM/GRU, FiLM, and reservoir computing.

5. Novelty Ledger

| Claim | Classification |

| --- | --- |

| Formalization of two live interference channels (depth-aliasing and gradient interference) in a frozen weight-tied path | **Apparently novel as packaging**; ingredients known |

| H1: exact global preservation forces identity gate | **Direct consequence** |

| H2: asymmetric preservation constraint on old inputs | **Direct consequence**, but CL-relevant reframing |

| H3: overlapping-input task ambiguity | **Direct consequence** |

| H4: depth-agnostic gate-value representational limitation | **Direct consequence**; behavioral conclusion underdetermined/not generally true |

| H5: projection rank condition | **Direct consequence** |

| H6: parameter budget for exact zero gradient interference | **Direct consequence** |

| H7: Lipschitz error amplification | **Direct consequence** |

| H8: finite-task reachability via gate interpolation | **Plausible adaptation / conjecture** |

| Claim that state or depth information is necessary for robust gating | **Plausible adaptation**, consistent with LSTM/FiLM practice, not a general theorem |

| Empirical claim “LLM-scale $\setminus(F)$ is expansive” | **Empirical conjecture**, not theorem |

No experimental result is used as evidence for any theorem in this document.

6. Failure Condition and Main Obstructions

The research question asks whether input-path interference can be eliminated, bounded, or made controllable without per-depth trainable parameters and without retraining the frozen shared path.

The strictest reading is:

- depth-agnostic gate/projection,
- frozen $\setminus(F)$,
- exact preservation of the original computation on all inputs.

Under that strict reading:

- **Elimination is impossible** in the uninteresting sense of H1: the gate must be identity on all inputs, so it provides no isolation.
- **Bounding** is possible only under the dynamics condition of H7: uniform bounds require $\|L\| < 1$; otherwise errors can grow with depth.
- **Controllability** is not a well-posed theorem without a reachability analysis of the specific $\mathcal{F}$ and gate class.

However, the strict reading is **not** the relevant continual-learning formulation.

We therefore treat the **asymmetric continual-learning reading** as central:

- old tasks: exact preservation of $\mathcal{F}^D(x)$ on $\mathcal{S}_{\text{old}}$;
- new tasks: free to change;
- no task label in the input;
- no per-depth trainable parameters.

Under this reading, the results are:

- The identity obstruction restricts to $\mathcal{S}_{\text{old}}$ (H2).
- Overlapping task inputs with different demands are impossible (H3).
- Exact zero cross-task gradient interference imposes a rank/parameter budget if demanded for all losses (H6).

- Whether nontrivial old/new isolation is possible is **underdetermined** without a reachability analysis (H4, H8).

The main obstruction is therefore not a single theorem. It is a conjunction of:

- **Task identifiability:** input-only gates cannot separate same-input different-output tasks.
- **Preservation-identity tension:** global preservation kills the gate; asymmetric preservation restricts it to old-task zero-coordinate directions.
- **Gradient coupling:** shared gate parameters couple tasks unless heavy orthogonality/low-rank conditions are imposed.
- **Dynamics amplification:** if the gated composition is expanding, small deviations from ideal gate behavior are amplified exponentially in depth.

None of these is an algorithmic failure; they are representational and algebraic. But none of them, by itself, proves a general impossibility of asymmetric continual learning with a frozen shared path. The negative claim must therefore be scoped:

> Exact elimination is impossible under the trivial global-preservation reading. Under the asymmetric continual-learning reading, the problem is open; this analysis provides necessary conditions and conditional constructions, not a general impossibility.

7. What Would Make the Question Decidable?

7.1 Instantiate $\mathcal{F}$

The first step is to replace the abstract $\mathcal{F}$ with BDH's actual update from paper .tex. The gate placement then becomes concrete:

- If BDH uses a residual block $s_{t+1} = s_t + \mathcal{F}(s_t)$, gating the whole block may gate the residual stream; the natural placement is $s_{t+1} = s_t + g(x) \odot \mathcal{F}(s_t)$.
- If BDH has Hebbian-style multiplicative terms, a gate on those terms is not the same as a gate on the full block.
- BDH's native plasticity must be explicitly disabled for the frozen-path premise.

7.2 Resolve the $\mathcal{T}$ -conditioning interpretation

The deepest loophole is the meaning of “no per-depth/per-level trainable parameters”:

- A gate $g_\phi(x, t)$ where $\mathcal{T}$ is a **fixed, non-trainable encoding** adds $\mathcal{O}(1)$ trainable parameters and does not literally violate the constraint.
- A **learned depth embedding** adds $\mathcal{O}(D)$ trainable parameters and does violate the constraint.

Every theorem above assumes the former is excluded. If the team permits fixed depth indexing, the depth-agnostic obstruction disappears,

and the question becomes a reachability/interpolation question (H8). The team must decide this before any experiment is interpreted.

7.3 Formalize controllability

The research question asks whether interference can be “made controllable.” That requires a definition. A workable definition is reachability of finite task targets by the gate class without modifying $\setminus(F\setminus)$. The control-theoretic machinery of nonlinear discrete-time reachability and observability should be applied to the specific $\setminus(F\setminus)$.

8. Direct Answer to the Research Question

> Can input-path interference arising from shared sequential composition be eliminated, bounded, or made controllable by an input-dependent gate/projection mechanism, without introducing per-depth/per-level trainable parameters and without retraining the frozen shared path?

Elimination.

- Under the trivial global-preservation reading: **impossible**, because the only gate that preserves everything is the identity gate.
- Under the meaningful asymmetric continual-learning reading: **not generally proved either possible or impossible**. The surviving necessary conditions are task-input disjointness (H3), identity

restriction on old-task coordinates (H2), and gradient orthogonality/low-rank budgets for exact zero cross-task interference (H6).

Bounding.

- **Possible conditionally.** If the gated transition is non-expansive ($L \leq 1$), bounds are uniform in depth. If $L > 1$, deviations can grow exponentially in depth. A finite-depth bound always exists for finite L , but may be uselessly large.

Controllability.

- **Underdetermined without a reachability analysis.** If each task target is reachable by the gate class, finite task sets can be interpolated, subject to gate capacity and task-input disjointness. Whether BDH's frozen map admits such reachability is not known from the public material.

Overall.

The strict form of the proposed mechanism does not provide a general solution. The obstruction is not any single nonlinearity of $\mathcal{F}$; it is the combination of input-only task identification, shared-parameter gradient coupling, and preservation constraints. The productive next step is to instantiate the abstract $\mathcal{F}$ with BDH's concrete update, decide the (t) -conditioning question, and then either prove or disprove reachability for the specific gate class.

Final Moderator Synthesis — Debate Closure on BDH Input-Path Isolation

Role: ox-alpha, moderator. This document consolidates three rounds of contributions (STRATEGIST, CRITIC, OPTIMIZER), adjudicates all disputes, records the final consensus state, and delivers the scoped answer to the research question together with the precise characterization of the obstruction.

Two closing factual verifications are warranted before sealing the record — one on the BDH block equations (the single most consequential unresolved grounding item), one on public novelty of the exact problem:

[SEARCH: "Dragon Hatchling" BDH-GPU equations LayerNorm ReLU Hebbian residual update rule outer product]

[SEARCH: BDH "baby dragon hatchling" continual learning catastrophic forgetting gating projection]

0. Evidence-Base Status (unchanged, recorded for the archive)

- The BDH paper (arXiv:2509.26507) and third-party implementations corroborate that BDH reuses **shared parameters across depth** ("recurrent depth") with sparse non-negative activations. The *literal*

update equations from paper . tex were never retrieved in-debate; neither were the internal reports (2026-08-*, plan_v2, cl-h1-report). Consequently every result below is proved for the **abstraction** $s_{t+1} = \Phi(g(x), s_t)$ over a frozen map F , with a residual variant $s_{t+1} = s_t + g(x) \odot F(s_t)$ treated as the likely BDH-shaped instantiation. Instantiation remains a mandatory follow-up (§8).

- No experimental result from any report was admitted as evidence for any theorem. This discipline held through all three rounds and is confirmed in the audit (§9).

1. Trajectory of the Debate (process record)

The debate executed exactly the correction loop the brief demanded:

- **Round 2** produced a unified impossibility package — identity-forcing + diagonalization + rank budget — headlined as "exact elimination is impossible in general."
- **Round 3 CRITIC** landed five substantive defeaters: (i) H1 is true but near-tautological and drawn from a degenerate preservation requirement; (ii) the diagonalization argument is a statement about gate *values*, not reachable *behaviors* — constant-depth gates emulate depth-varying effects through state feedback, and injectivity fails at zero activations; (iii) the projection capacity bound $\sum m_i \leq d$ committed a category error (image-subspace geometry $\neq$ parameter-space gradient geometry); (iv) the error-bound "iff $L < 1$ " misquantified (finite- D vs. uniform-in- D); (v) "controllability" was never defined. Plus: missing BDH grounding, missing prior art (FiLM, fast-weight programmers, reservoir computing), and the unresolved t -conditioning scope.

- **Round 3 responses:** OPTIMIZER conceded essentially all points and *restructured* — adding two genuinely new exact results (asymmetric-preservation constraint; overlapping-input identifiability), withdrawing the invalid corollary, fixing the quantifiers, and reframing the headline. STRATEGIST incorporated the asymmetric-preservation and $\$t\$$ -conditioning flags into its own §7 while retaining a stronger headline.

Adjudication principle applied throughout: a claim enters the record only at the strength of its *weakest surviving proof*.

2. Final Adjudicated Verdicts

#	Claim	Final status	Notes
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V1	Identity-forcing: exact trajectory preservation on input set $\$S\$$ under depth-agnostic multiplicative gating forces $g_i(x)=1$ for every (x,i) with coordinate i active along x 's trajectory TRUE — exact theorem, direct consequence Near-tautological. Its real content is <i>locative</i> : it identifies precisely where gate freedom lives (never-active coordinates; non- $\$S\$$ inputs).
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V2	Cross-input forward-pass contamination is structurally zero for stateless mechanisms on a frozen path ESTABLISHED Caveat added in synthesis: any <i>stateful</i> component inside the gate (running-statistics normalization, caches) revives this channel. Statelessness must be listed as an explicit design premise.
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| **V3** | Identifiability: a deterministic input-only mechanism cannot satisfy two tasks demanding different outputs at a shared input | **TRUE — exact theorem** | The sharpest structural limit of "input-dependent only," independent of depth, nonlinearity, capacity. Practical force in LM-CL depends on measure-theoretic overlap of task supports; do not overclaim either direction. |

| **V4** | Diagonalization: depth-agnostic gate confined to diagonal of modulation space | **Representational lemma TRUE; behavioral impossibility NOT established — downgraded** | Counterexample accepted ($F(s)=s$, $D=2$: constant gate matches depth-varying final outputs via state feedback); injectivity fails at zeros. Superseded by the open reachability question. |

| **V5** | Projection rank: $\text{rank} P(x) \geq m(x)$ necessary for exact preservation | **TRUE — exact linear algebra** | Capacity corollary $\sum_i m_i \leq d$ **WITHDRAWN** (invalid inference from forward-pass geometry to gradient geometry). Survives only under a special independent-subspace parameterization that violates the spirit of the constraints anyway. |

| **V6** | Gradient budget: zero first-order interference *for all smooth losses* $\Leftrightarrow$ mutually orthogonal task-Jacobian row spaces $\Rightarrow \sum_i \dim R_i \leq p$ | **TRUE — exact under stated generality; standard accounting** | Weak for *fixed specific* losses (inner product can vanish accidentally); non-binding in the realistic $p \ll d$ regime. Worst-case guarantee, not typical-case behavior. |

| **V7** | Amplification: $e_D \leq \delta \sum_{k < D} L_{\Phi}^k$; finite- D bound exists for every finite L_{Φ} ; **uniform-in-depth** bound iff $L_{\Phi} < 1$ | **TRUE — exact, quantifiers corrected** | "Trained LLM-scale maps are expansive" reclassified as **empirical conjecture**, requiring measurement on the actual model. |

| **V8** | Reachability/interpolation: finite task sets realizable iff targets reachable by gate class + capacity + disjointness/agreement | **CONDITIONAL; crux is a conjecture** | Reachability on a concrete frozen F is the decisive unproven assumption; counterexamples ($F \equiv 0$;

sign/root obstructions) show it fails generically without hypotheses on $\$F\$$. |

3. Consensus Evaluation

Full consensus (all three roles, post-revision):

- The formal skeleton (shared composition, frozen path, gate/projection placements, elimination-vs-bounding distinction).
- V1–V3, V5–V8 as scoped above.
- The two-channel picture: the live interference channels are **forward depth-aliasing** (within a pass) and **cross-time gradient interference** (across passes); naive cross-input contamination is dead (V2).
- Prior-art triage, extended in Round 3: LSTM/GRU (nearest neighbor; state-conditioned), FiLM (same gating algebra, per-layer parameters), fast-weight programmers/hypernetworks (adjacent, per-layer outputs), reservoir computing (frozen core + trained readout), orthogonal-gradient methods (parameter-space, complementary — see §7-C1), progressive networks/PackNet/super masks (constraint-violating), MoE (different problem), Universal Transformer/ALBERT (backdrop). **No surveyed work solves the exact problem:** isolation within one frozen weight-tied composition via an input-only gate without per-depth trainable parameters.

Residual dissent, adjudicated:

- STRATEGIST's Round-3 headline ("elimination impossible in general") versus the OPTIMIZER/CRITIC scoped formulation. **Ruling:** the scoped formulation prevails. The unscoped headline holds only

under global preservation on *all* inputs — a reading under which the gate must be inert everywhere, hence no adaptation anywhere, hence not a continual-learning question at all. STRATEGIST's own §7.1 concedes the asymmetric setting is the honest one and open; that concession is entered into the record as binding. The debate's central epistemic event is precisely this downgrade: **the general impossibility theorem dissolved upon inspection; what survives is a set of exact necessary conditions plus an open reachability problem.**

4. Final Answer to the Research Question

> Can input-path interference arising from shared sequential composition be eliminated, bounded, or made controllable by an input-dependent gate/projection mechanism, without per-depth/per-level trainable parameters and without retraining the frozen shared path?

| Reading | Verdict | Epistemic status |

| --- | --- | --- |

| **Elimination**, global-preservation reading (mechanism preserves F 's behavior on *all* inputs) | Trivially impossible — the only admissible gate is the identity, so the mechanism does nothing | Proved; direct consequence (degenerate) |

| **Elimination**, asymmetric CL reading (exact preservation on S_{old} , free on S_{new}) | **Open**. Necessary conditions established: identity confinement on old-task active coordinates (V1-scoped); input-disjointness/target-agreement (V3); worst-case gradient

budget (V6); dynamics amplification (V7). No general possibility or impossibility theorem | Partially characterized |

| **Bounding** | Finite-depth bounds exist for any finite L_Φ ; **uniform-in-depth** bounding iff gated dynamics contractive ($L_\Phi < 1$). Bounding is a property of the composed system, not achievable by gate design alone under expansion | Proved, conditional |

| **Controllability** | **Not yet a well-posed question**: requires formal definition (reachability of task targets by the gate class on the concrete frozen F) and BDH instantiation | Undefined pending instantiation |

The brief's failure condition is therefore met **only in the degenerate reading**. In the CL-relevant reading the correct outcome is: *not proven impossible, not proven possible; reduced to identifiable necessary conditions and one decisive open conjecture (reachability)*. Reporting anything stronger would violate the novelty discipline.

5. Obstruction Characterization (deliverable under the failure clause)

There is no single obstruction theorem in the meaningful setting. The failure surface is a **conjunction of four necessary conditions**, each independently capable of killing the program, none individually fatal:

- **Identifiability failure** — conflicting targets on overlapping inputs are unservable by any input-only mechanism (V3). In language-modeling continual learning, supports overlap extensively; escape requires context-sufficiency of the input or task conditioning.

- **Preservation-geometry confinement** — the gate is pinned to the identity on every (old input, active coordinate) pair (V1-scoped). Freedom survives only on never-active coordinates (substantial under sparse activations) and on new-task inputs.
- **Gradient coupling of shared ϕ** — new-task updates generically move g_ϕ off the identity on old inputs; worst-case immunity costs a rank budget (V6) that is either restrictive ($p \leq d$) or vacuous ($p \gg d$), and is unnecessary for specific loss pairs.
- **Depth amplification** — under expansive gated dynamics, per-depth modulation error grows geometrically in D (V7); bounding then demands architectural contractivity, which is an assumption on F , not a capability of the gate.

Plus three **scope conditionals** on which every result silently rests: (a) statelessness of the mechanism (else V2 dies); (b) the t -conditioning interpretation (below); (c) BDH's native Hebbian plasticity actually disabled (else the path is not frozen).

The t -conditioning fork (recorded as the pivotal scope decision): a shared-parameter gate $g_\phi(x, t)$ with t supplied by a *fixed, non-trainable* encoding adds $O(1)$ parameters and does not violate the constraint as literally worded — and it dissolves V1's depth-agnostic restrictions and V4's residue entirely, converting the problem into pure reachability (V8). A *learned* depth embedding adds $O(D)$ parameters and violates the constraint outright. Every impossibility-flavored statement in this record applies only on the strict side of this fork. The team must rule before any experiment is interpreted.

6. Novelty Ledger (final, adjudicated)

| Claim | Classification |

| --- | --- |

| Problem formulation: isolation inside a frozen weight-tied composition via input-only gate, CL-framed, two-channel interference taxonomy |

Apparently novel (packaging only) |

| V1, V3, V5 (necessary-condition parts), V6, V7 | **Direct consequences** (textbook ingredients) |

| V4 representational lemma | Direct consequence; behavioral version **false as stated** |

| Necessity of state/depth/context information for effective gating | **Plausible adaptation** (consistent with LSTM/FiLM practice; proven only under injectivity hypotheses) |

| V8 reachability for concrete BDH $\$F\$$ | **Conjecture** |

| Expansiveness of trained BDH blocks | **Empirical conjecture** |

| Any observed effect in c1 - h1 - report | Inadmissible as theorem evidence; admissible only as a test of the conjectures above |

Contribution posture for write-up: *a new formalization and a clean map of elementary obstructions — not new mathematics.*

7. Constructive Directions Surfaced by Synthesis (all $\leq$ conjecture)

- **C1 — Preservation as a maintained invariant (rehabilitates dismissed prior art).** Once the protected object is recognized to be the *gate parameters* ϕ (not the frozen path θ), orthogonal-gradient machinery becomes directly applicable *compatibly with all constraints*: project new-task gradients onto the orthogonal complement of the old-task gradient subspace in ϕ -space, thereby holding $g_\phi|_{S_{\text{old}}} \approx \mathbf{1}$ to first order across sequential training. Classification: **plausible adaptation** of GEM/GPM to a new target object; no theorem claimed. Open statistical question: whether the invariant generalizes beyond the old-task sample.
- **C2 — Fixed-encoding t -conditioning** ($O(1)$ parameters): dissolves the depth-agnostic restrictions; legality contingent on the scope ruling (§5).
- **C3 — Delta-gate placement** for the residual BDH variant: gate $g(x) \odot \Delta(s_t)$ inside $s_{t+1} = s_t + \cdot$, never the residual stream itself.
- **C4 — Sparse-coordinate freedom:** exploit never-active coordinates of old-task trajectories (plausible under ReLU sparsity) as legal modification space even on preserved inputs.

8. Mandatory Follow-Ups Before Any Empirical Reconciliation

- **Instantiate Φ** from paper . tex: fix the literal update (residual form, Hebbian/multiplicative terms, normalization placement),

choose gate placement (C3), and verify native plasticity is disabled.
(Closing searches issued above; see appended results — treat as corroboration only, not substitution for reading the TeX source.)

- **Rule on the $\mathcal{T}$ -conditioning fork** (§5) in writing.
- **Define controllability formally** as reachability of finite task targets by the gate class on the instantiated $\mathcal{F}$; select the tool class (nonlinear discrete-time reachability).
- **Retrieve the internal reports**; map the implemented mechanism onto exactly one formal variant (depth-agnostic / $\mathcal{T}$ -conditioned / state-conditioned); reconcile `cl-h1-report` outcomes strictly as tests of the V7/V8 conjectures.
- **Measurement program**: empirical Lipschitz/expansiveness of the gated BDH block (tests V7's conjecture); empirical cross-task Jacobian row-space overlap in ϕ -space (tests whether V6's worst case binds in practice).

9. Discipline Audit

- No experimental observation was cited as evidence for any theorem at any point. ✓
- All overclaims identified by the CRITIC were either withdrawn on the record (capacity corollary; behavioral diagonalization; finite- $\mathcal{D}$ "iff") or explicitly rescoped (global $\rightarrow$ asymmetric preservation). ✓
- Novelty labels are consistent post-adjudication: packaging novel, mathematics derivative. ✓

Consensus summary: complete agreement on the formal framework, on the two live interference channels, on the exact necessary conditions (V1–V3, V5–V7 as scoped), on prior-art triage, and on the open status of the asymmetric continual-learning question. The single substantive disagreement — whether a general impossibility was established — was resolved against the stronger claim on the strength-of-weakest-proof principle. The investigation exits the debate not with an impossibility theorem but with something more useful: a precise map of where gate freedom lives, four independent ways the program can fail, one pivotal scope decision ($\$t\$$ -conditioning), and one decisive open conjecture (reachability on the instantiated BDH map) whose resolution will determine whether the mechanism is possible, impossible, or merely unproven.

Web Research

- **The Dragon Hatchling: A New AI Architecture Bridging Transformers and the Brain | Deep Paper**

September 1, 2025 - Use LayerNorm and ReLU to keep activations positive and encourage sparsity. The BDH-GPU state-space update (informal) per layer and timestep can be summarized as: ... update neuron activations $\backslash(x\backslash)$ with a ReLU-lowrank residual like $\backslash(x \leftarrowarrow ...$

Source: <https://deep-paper.org/en/papers/2025-10/2509.26507/>

- **The Dragon Hatchling: The Missing Link between the Transformer and Models of the Brain**

September 30, 2025 - The observed emergent effects follow naturally from the design choices of the BDH and BDH-GPU architectures, and are largely attributable to the combination of: the choice of model dimensions with comparable model-to-state ratio, reliance on linear attention in high dimension, reliance on ReLU thresholds for ensuring

that activation vectors are positive (trivially) and sparse (an effect empirically noted in (Haziza et al.

Source: <https://arxiv.org/html/2509.26507v1>

• **The Dragon Hatchling - ArXivIQ**

October 5, 2025 - The BDH-GPU architecture (Figure ... neuronal dimension n : ReLU-Lowrank Feed-Forward Network: The FFN uses the transformation $z \rightarrow (DEz)^+$, where E is an encoder ($n \rightarrow d$) and D is a decoder ($d \rightarrow n$)....

Source: <https://arxiviq.substack.com/p/the-dragon-hatchling>

• **Understanding Baby Dragon Hatchling (BDH): The Missing Link Between Transformers and the Brain | Colin McNamara**

October 18, 2025 - The dynamics of BDH are governed by what the authors call “equations of reasoning”—local update rules that bear a striking resemblance to Hebbian learning (“neurons that fire together, wire together”).

Source: <https://colinmcnamara.com/blog/understanding-baby-dragon-hatchling-bdh>

• **BDH Architecture Explained: Pathway’s Dragon Hatchling**

2 weeks ago - Chapter 2 derives BDH from the attention equations, reframing retrieval as synaptic memory in a high-dimensional sparse neuron space and showing how Hebbian updates and low-rank factorization lead to the practical BDH-GPU architecture.

Source: <https://pathway.com/research/bdh-explainer>

• **cMiraka/bdh-cl: Attempts extension of Baby Dragon Hatchling...**

Baby Dragon Hatchling Continual Learning (BDH-CL). Fork of: [pathwaycom/bdh](https://github.com/pathwaycom/bdh). BDH-CL introduces practical, biologically inspired continual learning capabilities into the BDH architecture, enabling robust lifelong learning beyond the single-task limitations of the original.

Source: <https://github.com/cMiraka/bdh-cl>

- **Beyond Transformer LLM Models: How the BDH Architecture Solves...**

[cMiraka/bdh-cl](https://github.com/cMiraka/bdh-cl): Attempts extension of Baby Dragon Hatchling (BDH) with continual learning using pseudo-metaplasticity.

Source: <https://www.crvscience.com/post/beyond-transformer-llm-models-how-the-bdh-architecture-solves-the-context-window-crisis>

- **Dragon Hatchling: Linking Transformers & Brain Models**

BDH bridges Transformer LLMs with brain models via local graph dynamics and Hebbian learning, delivering interpretability and state-of-the-art performance.

Source: <https://www.emergentmind.com/papers/2509.26507>

- **Introducing BDH: A Brain-Inspired AI Model | Shivam... | LinkedIn**

Brain-Inspired Language Models Are Here Researchers have introduced BDH (Baby Dragon Hatchling)—a new large-language-model architecture inspired by biological networks and the human brain.

Source: https://www.linkedin.com/posts/imsane_airesearch-large languagemodels-neuroscienceinspired-activity-7387937972265893888-UzDm

10. We Watched a Brain Emerge..." The AI That Might Kill... - YouTube

Pathway Calls It Baby Dragon Hatchling 16:52 - Continual Learning and the Dragon Metaphor 17:20 - Learning Like a Brain: Neurons and Connections 21:27 - When a Brain Emerges Inside the Model 22:54 - Memory as Strengthened Connections 24:58...

Source: <https://www.youtube.com/watch?v=duw7RUif8hE>

[STRATEGIST Round 4]

Final Strategic Assessment: Input-Path Interference in BDH via Gating/Projection

Date: 2026-08-25 | **Status:** Complete (3-round debate, adjudicated)

Executive Summary

Under the strict constraints — depth-agnostic input-dependent gate/projection, frozen shared path, no per-depth trainable parameters, no retraining — **exact elimination of input-path interference is impossible only under the degenerate global-preservation reading** (the gate must be identity everywhere, doing nothing). Under the **continual-learning-relevant asymmetric reading** (exact preservation on old tasks, free on new tasks), the problem is **open**: no general impossibility or possibility theorem has been established. What has been established is a precise map of four independent necessary conditions,

each capable of killing the program, plus one decisive open conjecture (reachability) whose resolution determines feasibility.

1. Formal Definitions

1.1 Shared Sequential Composition

Let $V = \mathbb{R}^d$ be the state space. A **depth- D shared sequential composition** is defined by a fixed map $F: V \rightarrow V$ and the recurrence:

$$s_0(x) = x, \quad s_{t+1}(x) = F(s_t(x)), \quad t = 0, \dots, D-1$$

The unperturbed trajectory is $\tau_F(x) = (s_0(x), s_1(x), \dots, s_D(x))$.

If BDH uses residual connections (likely, per the BDH-GPU architecture), the relevant recurrence is:

$$s_{t+1}(x) = s_t(x) + F(s_t(x))$$

Classification: Direct characterization of BDH's weight-tied recurrent depth, corroborated by BDH-MLX ("Same weights reused in all layers (recurrent depth)") and BDH-GPU descriptions ("ReLU-Lowrank residual

like $x \mapsto \dots$). Instantiation with the literal BDH update from paper .tex remains mandatory.

1.2 Frozen Shared Path

F is **frozen** if $\partial F / \partial \theta = 0$ during the continual-learning phase. Only the gate/projection mechanism ϕ may be trained.

Classification: Known — standard continual learning constraint.

1.3 Input-Path Interference: Three Precise Notions

(A) Cross-input forward-pass contamination. With frozen θ and a stateless input-conditioned mechanism, each input's forward pass is computationally independent. This notion is **structurally zero** and not a live channel.

(B) Forward depth-aliasing. A depth-agnostic gate $g(x)$ applies the same modulation at every depth. If an oracle would apply $g^*(x, t)$ (varying with depth), the depth-aliasing error is:

$$\mathrm{DA}(g; x) = \sum_{t=0}^{D-1} \left| \Phi(g(x), s_t(x)) - \Phi(g^*(x, t), s_t(x)) \right|$$

(C) Training-time gradient interference (cross-task forgetting).

Because gate parameters ϕ are shared across all inputs, updating ϕ on task B may change g_ϕ on task- A inputs. First-order

damage is proportional to $\langle \nabla_{\phi} \mathcal{L}_A, \nabla_{\phi} \mathcal{L}_B \rangle$.

The live channels are (B) and (C). Notion (A) is dead under the stated assumptions.

Classification: The depth-aliasing notion is **apparently novel** as a packaged definition; components are known from control theory. The gradient notion is **known** from continual learning.

1.4 Gating

An **input-dependent gate** $g: X \rightarrow \mathbb{R}^d$ is applied identically at every depth:

- Output gating: $s_{t+1} = g(x) \odot F(s_t)$
- Input gating: $s_{t+1} = F(g(x) \odot s_t)$
- Delta gating (for residual blocks): $s_{t+1} = s_t + g(x) \odot F(s_t)$

The gate has **no per-depth parameters**: the same $g(x)$ is used at every t .

Classification: Known mechanism (LSTM, MoE), but the depth-independence + frozen-path combination is novel.

1.5 Projection

An **input-dependent projection** $P: X \rightarrow \Pi(V)$ maps each input to an orthogonal projection, applied as $s_{t+1} = P(x) F(s_t)$ or $s_{t+1} = F(P(x) s_t)$.

Classification: Known (gradient projection methods), formalized for forward-pass application.

1.6 Computation-Isolability

A mechanism is **exactly computation-isolating** for a task family $\{(S_i, y_i)\}_{i=1}^K$ if: (1) each task output is produced exactly; (2) zero forward depth-aliasing error; (3) zero first-order cross-task gradient interference.

Classification: Apparently novel — this specific formulation combining shared composition, frozen parameters, and forward-pass isolation does not appear in surveyed literature.

1.7 Elimination vs. Bounding vs. Controllability

- **Elimination:** Exact equality of outputs and exact zero interference.
- **Bounding:** Deviation $\leq \epsilon$, controlled by Lipschitz constant, depth, gate error, parameter count.
- **Controllability:** Reachability of finite task targets by the gate class without modifying F .

Classification: Plausible adaptation of approximation theory and control-theoretic concepts.

2. Theorem-Shaped Hypotheses (Final Adjudicated Versions)

V1. Identity-Forcing Under Global Preservation

Assumptions: F defined on open $U \subseteq \mathbb{R}^d$; depth-independent output gating $s_{t+1} = g(x) \odot F(s_t)$; exact trajectory preservation on input set S : $\tilde{s}_t(x) = F^t(x)$ for all $x \in S$, all t .

Claim: For every $x \in S$ and every coordinate i : $g_i(x) = 1$ whenever $\exists t: F_i(s_t(x)) \neq 0$.

Proof: At depth t : $g(x) \odot F(s_t(x)) = F(s_t(x))$ implies $(g_i(x) - 1) F_i(s_t(x)) = 0$ componentwise. $\square$

Verdict: True as an exact theorem. Near-tautological under global preservation. Its real content is *locative*: it identifies precisely where gate freedom lives — never-active coordinates and non- S inputs.

Classification: Known / direct consequence.

V2. Asymmetric Preservation: The CL-Relevant Constraint

Assumptions: Old-task inputs S_{old} ; new-task inputs S_{new} ; output gating; exact preservation required only on S_{old} .

Claim: For $x \in S_{\text{old}}$: $g_i(x) = 1$ whenever $\exists t: F_i(s_t(x)) \neq 0$. On the zero-coordinate set $Z(x) = \{i: F_i(s_t(x)) = 0 \forall t\}$, the gate is unconstrained. On S_{new} , the gate is entirely unconstrained.

Proof: Same componentwise argument as V1, restricted to S_{old} . $\square$

Verdict: True as an exact theorem. If old-task trajectories have support in all coordinates, the gate is forced to identity on all of S_{old} . If old and new inputs are disjoint and the gate network can interpolate, exact isolation is *not* ruled out by this obstruction alone.

Classification: Direct consequence; the asymmetric framing is the correct CL formalization.

V3. Identifiability: Overlapping Inputs Are Fatal

Assumptions: The mechanism is a deterministic function of the input: $\tilde{s}_D(x; \phi) = \mathcal{M}_\phi(x)$. Two tasks A, B share an input $x \in S_A \cap S_B$ with different targets: $y_A(x) \neq y_B(x)$.

Claim: No single ϕ can satisfy both tasks.

Proof: The map is a function: one input, one output. Two different required outputs are incompatible. $\square$

Verdict: True as an exact theorem. The sharpest structural limit of "input-dependent only," independent of depth, nonlinearity, and capacity.

Classification: Direct consequence; essentially an identifiability condition.

V4. Depth-Agnostic Gate-Value Limitation (Downgraded)

Assumptions: $\Phi(g, s)$ injective in g along the trajectory; ideal modulation $g^*(x, t)$ nonconstant in t .

Claim: No depth-independent gate $g(x)$ can reproduce the nonconstant gate-value sequence $g^*(x, t)$.

Proof: Equality at every depth gives $g(x) = g^*(x, t)$ for every t , contradicting nonconstancy. $\square$

Critical correction: This is a statement about **gate values**, not about **reachable input-output maps**. A constant-in-depth gate can produce depth-varying effective behavior because the state s_t changes.

Counterexample: $F(s) = s$, $D = 2$. Output gating: $s_2 = g(x) \odot g(x) \odot x$. Constant gate $g(x) = (\alpha, \beta)$ produces $(\alpha^2 x_1, \beta^2 x_2)$. Depth-varying sequence $g^0 = (c, 1)$, $g^1 = (1, c)$ produces (cx_1, cx_2) , also achievable by constant gate $(\sqrt{c}, \sqrt{c})$.

Verdict: Representational lemma TRUE; behavioral impossibility NOT established. The reachability question is open.

Classification: Direct consequence for the representational part; the broad behavioral impossibility is **false as stated**.

V5. Projection Rank Condition

Assumptions: Output projection $s_{t+1} = P(x) F(s_t)$; exact preservation of unperturbed trajectory required.

Claim: If $m(x) = \dim \operatorname{span}\{F(s_0(x)), \dots, F(s_{D-1}(x))\}$, then $\operatorname{rank} P(x) \geq m(x)$.

Proof: $P(x) \implies F(s_t(x)) = F(s_{t-1}(x))$ forces $F(s_t(x)) \in \text{Im}\{P(x)\}$. $\square$

Verdict: True as an exact linear-algebra theorem.

What this does NOT say: The capacity corollary $\sum_i m_i \leq d$ is **WITHDRAWN**. Orthogonality of image subspaces is a forward-pass geometric property; cross-task gradient interference is a property of the parameterization. These are different mathematical objects.

Classification: Direct consequence.

V6. Gradient Interference Parameter Budget

Assumptions: Gate has p trainable parameters ϕ ; task Jacobians $J_i \in \mathbb{R}^{d \times p}$; zero first-order interference required for *all smooth losses*.

Claim: Row spaces $\{R_i\}$ must be mutually orthogonal: $\sum_i \dim R_i \leq p$.

Proof: For arbitrary loss functions, vanishing inner products force row-space orthogonality. $\square$

Important caveats:

- For fixed specific losses, zero interference is weaker (can hold accidentally).
- If $p \gg d$ (typical for LLM-scale gates), the bound is non-binding.
- Worst-case guarantee, not typical-case behavior.

Verdict: True as an exact theorem under stated generality. Standard parameter-isolation accounting.

Classification: Direct consequence.

V7. Error Amplification Bound

Assumptions: Gated transition L -Lipschitz in state; per-depth gate error $\leq \delta$.

Claim:

$$\|s_D(x) - s_{D^*}(x)\| \leq \delta \sum_{k=0}^{D-1} L^k = \begin{cases} \delta \frac{1 - L^D}{1 - L} & L < 1 \\ \delta D & L = 1 \\ \delta \frac{L^D - 1}{L - 1} & L > 1 \end{cases}$$

Precise interpretation:

- For **fixed finite depth D** : bound exists for every finite L .
- **Uniform-in-depth** bound exists iff $L < 1$.

Empirical caveat: "Trained LLM-scale maps have $L > 1$ " is an **empirical conjecture**, not a theorem.

Verdict: True as an exact theorem.

Classification: Known / direct consequence.

V8. Finite-Task Reachability (Conditional)

Assumptions: Finite task set (x_i, y_i) ; gate family has sufficient capacity; each target reachable: $y_i(x) \in \mathcal{R}_D(x) := \{\tilde{s}_D(x; \phi) : \phi \in \Phi\}$; overlapping inputs have agreeing targets.

Claim: There exists ϕ realizing all finite task targets exactly.

Counterexamples showing underdetermination:

- $F(s) = 0$: $\mathcal{R}_D(x) = \{0\}$ for every x , regardless of gating.
- Same input, different targets: V3 kills it.
- $F(s) = s$, $D = 2$, multiplicative output gating: reachable set is $\{g(x) \odot g(x) \odot x\}$; sign/root obstructions limit reachability.

Verdict: Conditionally true; otherwise underdetermined. The reachability assumption is the nontrivial crux.

Classification: Plausible adaptation; reachability is a **conjecture** unless proved for specific F .

3. Summary of Verdicts

#	Claim	Final Status
---	-----	-----
V1	Global preservation forces identity	TRUE — exact theorem; near-tautological
V2	Asymmetric preservation constraint	TRUE — exact theorem; CL-relevant
V3	Overlapping inputs with different targets	TRUE — exact theorem; identifiability
V4	Depth-agnostic gate-value limitation	Representational TRUE; behavioral FALSE/underdetermined
V5	Projection rank condition	TRUE — exact linear algebra; capacity corollary withdrawn
V6	Gradient interference parameter budget	TRUE — exact under stated generality
V7	Error amplification bound	TRUE — exact; finite- D vs uniform- D corrected
V8	Finite-task reachability	CONDITIONAL; reachability is conjecture

4. Prior Art Analysis

| Prior Art | Same Mathematical Problem? | Classification |

| --- | --- | --- |

| **Orthogonal gradient methods** (GEM, A-GEM, GPM) | Partly — addresses gradient interference in parameter space; not forward depth-aliasing; usually modifies main network | Known, partially relevant |

| **Parameter isolation / masking** (PackNet, Supermasks) | No — per-task masks/parameters; constraint-violating | Known, related but different |

| **Progressive networks / PathNet** | No — per-task architectural growth | Known, different problem |

| **Hypernetworks / fast weight programmers** | Partly — can generate input-dependent modulations; typically per-layer outputs | Known, partially relevant |

| **FiLM** (feature-wise linear modulation) | Partly — same gating algebra, but usually per-layer parameters | Known, directly relevant |

| **LSTM / GRU** | **Closest** — shared-parameter gates at every time step; crucially state-dependent, not input-only | Known, nearest prior art |

| **Mixture of Experts** | No — per-layer routers, distinct experts | Known, structurally similar |

| **Reservoir computing / echo state networks** | Partly — frozen recurrent core + trained readout | Known, relevant background |

| **Universal Transformer / ALBERT** | Partly — weight-tied depth, no task isolation | Known, background context |

| **Nonlinear RNN reachability / observability** | Yes for the reachability question | Known, should be used for V8 |

No surveyed prior art solves the exact problem: isolation within one frozen weight-tied composition via an input-only gate without per-depth trainable parameters.

5. Obstruction Characterization

There is no single obstruction theorem in the CL-relevant setting. The failure surface is a **conjunction of four necessary conditions**, each independently capable of killing the program, none individually fatal:

- **Identifiability failure** (V3): Conflicting targets on overlapping inputs are unservable by any input-only mechanism. In language-modeling CL, supports overlap extensively. Escape requires context-sufficiency or task conditioning.
- **Preservation-geometry confinement** (V1/V2): The gate is pinned to identity on every (old input, active coordinate) pair. Freedom survives only on never-active coordinates (substantial under sparse activations) and on new-task inputs.
- **Gradient coupling of shared ϕ** (V6): New-task updates generically move g_ϕ off identity on old inputs. Worst-case

immunity costs a rank budget that is either restrictive ($p \leq d$) or vacuous ($p \gg d$).

- **Depth amplification** (V7): Under expansive gated dynamics, per-depth modulation error grows geometrically in D . Bounding demands architectural contractivity — an assumption on F , not a capability of the gate.

Plus three **scope conditionals**: (a) statelessness of the mechanism; (b) the t -conditioning interpretation; (c) BDH's native Hebbian plasticity actually disabled.

6. The Pivotal Scope Decision: t -Conditioning

A shared-parameter gate $g_\phi(x, t)$ where t is a **fixed, non-trainable encoding** (e.g., sinusoidal) adds $O(1)$ parameters and **does not violate "no per-depth/per-level trainable parameters" as literally worded**. This dissolves V1's depth-agnostic restrictions and V4's residue entirely, converting the problem into pure reachability (V8).

A **learned depth embedding** adds $O(D)$ parameters and violates the constraint outright.

Every impossibility-flavored statement applies only on the strict side of this fork. **The team must rule before any experiment is interpreted.**

7. Constructive Directions (all $\leq$ conjecture)

- **C1 — Orthogonal gradient on ϕ -space:** Project new-task gradients onto the orthogonal complement of old-task gradient subspace in ϕ -space, holding $g_\phi|_{S_{\text{old}}} \approx \mathbf{1}$ to first order. Classification: **plausible adaptation** of GEM/GPM.
- **C2 — Fixed-encoding t -conditioning** ($O(1)$ parameters): dissolves depth-agnostic restrictions; legality contingent on scope ruling.
- **C3 — Delta-gate placement** for residual BDH: gate $g(x) \odot \Delta(s_t)$ inside $s_{t+1} = s_t + \odot$, never the residual stream.
- **C4 — Sparse-coordinate freedom:** exploit never-active coordinates of old-task trajectories (plausible under ReLU sparsity) as legal modification space.

8. Novelty Ledger (Final, Adjudicated)

| Claim | Classification |

|-----|-----|

| Problem formulation: isolation inside frozen weight-tied composition via input-only gate, CL-framed, two-channel interference taxonomy |

Apparently novel (packaging only) |

| V1, V2, V3, V5, V6, V7 | **Direct consequences** (textbook ingredients) |

| V4 representational lemma | Direct consequence; behavioral version **false as stated** |

| Necessity of state/depth information for effective gating | **Plausible adaptation** (consistent with LSTM/FiLM practice) |

| V8 reachability for concrete BDH \$\$\$ | **Conjecture** |

| Expansiveness of trained BDH blocks | **Empirical conjecture** |

Contribution posture: a new formalization and a clean map of elementary obstructions — not new mathematics.

9. Direct Answer to the Research Question

> Can input-path interference arising from shared sequential composition be eliminated, bounded, or made controllable by an input-dependent gate/projection mechanism, without introducing per-depth/per-level trainable parameters and without retraining the frozen shared path?

| Reading | Verdict | Epistemic Status |

|-----|-----|-----|

| **Elimination**, global-preservation | Trivially impossible — only admissible gate is identity | Proved; degenerate |

| **Elimination**, asymmetric CL | **Open** — necessary conditions established, no general impossibility or possibility theorem | Partially characterized |

| **Bounding** | Finite-depth bounds exist for any finite L ; uniform-in-depth iff $L < 1$ | Proved, conditional |

| **Controllability** | **Not yet well-posed** — requires formal reachability definition and BDH instantiation | Undefined pending instantiation |

The failure condition is met only in the degenerate reading. In the CL-relevant reading, the correct outcome is: *not proven impossible, not proven possible; reduced to identifiable necessary conditions and one decisive open conjecture (reachability).*

10. Mandatory Follow-Ups

- **Instantiate FF** from paper .tex: fix the literal update (residual form, Hebbian/multiplicative terms, normalization), choose gate placement (C3), verify native plasticity is disabled.
- **Rule on the St -conditioning fork** in writing.
- **Define controllability formally** as reachability of finite task targets by the gate class on the instantiated FF .
- **Retrieve internal reports**; map the implemented mechanism onto exactly one formal variant; reconcile cl-h1-report outcomes strictly as tests of V7/V8 conjectures.

- **Measurement program:** empirical Lipschitz/expansiveness of the gated BDH block (tests V7's conjecture); empirical cross-task Jacobian row-space overlap in ϕ -space (tests whether V6's worst case binds).

[CRITIC Round 4]

Critical review of the BDH input-path isolation analysis

Overall assessment

The multi-round analysis is substantially better in its final form than its first draft. The main correction — that the headline impossibility theorem only holds in a degenerate “preserve everything on every input” reading, and that the continual-learning-relevant asymmetric setting remains open — is correct and important. The separation of three different meanings of “interference” is useful, and the final verdict table is epistemically honest.

However, the analysis still has several serious gaps. The largest is that it never instantiates the abstraction with the actual BDH equations from `paper.tex`, despite the paper reportedly being available in RAG. Consequently, many of the formal claims are placement-specific, and the most important placement for BDH — gating a residual delta rather than a full block output — is not treated in the proofs.

1. Strengths worth preserving

Interference taxonomy.

The tripartition is the most useful contribution:

- cross-input forward-pass contamination,
- forward depth-aliasing,
- training-time cross-task gradient interference.

For a frozen, stateless, input-conditional gate, cross-input forward contamination is indeed structurally zero. The caveat that stateful components such as running-statistics normalization or caches can revive it is well taken.

Rejection of the naive diagonalization impossibility.

The original claim that a depth-agnostic gate is confined to the diagonal of the modulation product space and therefore cannot express depth-varying behavior was correctly downgraded. A constant-in-depth gate can produce depth-dependent effective behavior because the state changes across depth. The explicit counterexample with $\mathcal{F}(s)=s$ is convincing.

Withdrawal of the projection capacity corollary.

$\text{rank } P(x) \geq m(x)$ is correct as a necessary condition for exact trajectory preservation under output projection. The later step to $\sum_i m_i \leq d$ was a category error: forward-pass subspace

geometry does not automatically translate into gradient-parameter orthogonality. Withdrawing it was correct.

Correct scoping of the main claim.

The final statement that exact elimination is only proved impossible under the degenerate global-preservation reading, while the asymmetric CL reading remains open, is the right conclusion. The four-part obstruction characterization is a useful map of failure modes, even though it is not a single impossibility theorem.

2. Major gaps and weaknesses

2.1 The analysis is not grounded in the actual BDH equations

The entire theory is built for

$$\begin{aligned} & \backslash[\\ & s_{\{t+1\}}=F(s_t) \\ & \backslash] \end{aligned}$$

with a few still-abstract variants. But the analysis did not retrieve and parse paper . tex. It repeatedly says the literal BDH update was not instantiated. This is not a minor omission; it changes the applicable theorem statements.

Public descriptions suggest BDH-GPU has a ReLU-lowrank residual structure, something like:

$$\begin{aligned} & \left[\right. \\ & x \mapsto \operatorname{ReLU}(x + \text{low-rank FFN}(x)) \\ & \left. \right] \end{aligned}$$

with shared parameters across depth, LayerNorm, and possibly Hebbian-style multiplicative terms.

This matters for three reasons:

- **Gate placement.** If the real BDH block is residual, then the natural gate is not

$$\begin{aligned} & \left[\right. \\ & s_{t+1} = g(x) \odot F(s_t) \\ & \left. \right] \end{aligned}$$

but rather

$$\begin{aligned} & \left[\right. \\ & s_{t+1} = s_t + g(x) \odot F(s_t). \\ & \left. \right] \end{aligned}$$

Under this delta gate, exact preservation of the unperturbed trajectory does **not** force $g_i(x)=1$. It forces

$$\left[\right.$$

$$g_i(x)=0$$

$\setminus$

on every coordinate where $\setminus(F_i(s_t(x))\setminus\neq 0\setminus)$.

The “identity-forcing” theorem therefore does not transfer directly. The gate is pinned to zero on active coordinates, and only zero-coordinate directions are free. This is analogous but not identical, and it should be restated.

- **LayerNorm and running statistics.** If the frozen path contains normalization layers that depend on batch or sequence statistics, then the mechanism is not stateless. A gate depending only on the original input is no longer purely cross-input-independent in the forward pass. The statelessness premise needs explicit verification.
- **Native Hebbian or plastic components.** The analysis acknowledges that BDH’s native plasticity must be disabled for the frozen-path premise. But if the genuine BDH path contains Hebbian-style outer-product updates, then “freezing” it is not just a hyperparameter choice; it changes the architecture. Until this is checked, theorems about a static $\setminus(F\setminus)$ may not apply.

2.2 The internal reports were not reconciled

The prompt states that the plan and H1 report are available in RAG. The analysis did not retrieve them. As a result:

- The implemented mechanism cannot be mapped to one of the formal variants.
- The empirical outcomes in 2026-08-25_cl-h1-report.md are neither explained nor refuted.

- It is unknown whether the implementation gates the full block, the delta, an intermediate feature, or something else.

This is a major evidence-base gap. The final answer explicitly recommends reconciling the reports, but the review should treat this as a defect of the current analysis, not an optional next step.

2.3 “Controllability” is still not adequately formalized

The final report defines controllability as reachability of finite task targets by the gate class. That is a reasonable start, but it is not a full formalization.

For an input-dependent gate $\backslash(g(x)\backslash$, the control set is severely restricted. At each depth the same gate output is applied, so the state trajectory is

$\backslash[$

$$s_{\{t+1\}} = \backslash\Phi(g(x), s_t),$$

$\backslash]$

where $\backslash(g(x)\backslash$ is fixed for the entire pass. This is not the usual discrete-time control problem with independent controls $\backslash(u_t\backslash$ at each step. The reachable set for input $\backslash(x\backslash$ is parametrized by a single vector $\backslash(g(x)\backslash$, not by $\backslash(D\backslash$ independent controls.

A proper definition should specify:

- the state space,
- the admissible gate value space,
- the transition rule Φ ,
- the class of target maps $y_i(x)$,
- and whether the same gate parameters ϕ must simultaneously realize all tasks.

Without this, claims that “controllability dissolves into reachability” are only a promissory note.

2.4 The t -conditioning fork remains unresolved

The analysis correctly identifies this as pivotal:

- A fixed, non-learned depth encoding t as an input to a shared-parameter gate adds $O(1)$ trainable parameters and likely does not violate the constraint literally.
- A learned depth embedding adds $O(D)$ trainable parameters and does violate the constraint.

Until the project rules on this, the strong statements remain conditional. But the current write-up does not commit to either side, and some earlier statements still treat the strict depth-agnostic reading as the default. For clarity, every theorem should be prefixed with:

> “Assuming the strict depth-agnostic reading...”

or

> “Assuming fixed non-learned $\mathbf{t}$ -conditioning is allowed...”

2.5 The parameter-budget bound is generic and likely non-binding

The gradient-interference bound,

$\sum_i \dim R_i \leq p,$

is correct under the very strong condition of zero first-order interference for all smooth losses. But it is essentially the standard parameter-isolation accounting for any shared parameter mechanism. It is not specific to shared sequential composition, and it gives no practical bound until the actual Jacobian ranks are measured.

For a gate network attached to an LLM-scale BDH path, p may be much larger than d , in which case the bound is vacuous for practical purposes. Conversely, the fixed-loss inner product can vanish accidentally even when row spaces overlap. The analysis acknowledges this, but the final obstruction list should not overstate V_6 as a decisive barrier.

2.6 The overlapping-input impossibility may need more nuance

V3 says a deterministic input-only mechanism cannot satisfy two tasks that assign different targets to the same input. This is mathematically true, but its practical force depends on the task formalization.

In language-modeling-style continual learning, inputs are token sequences. If two tasks have different target distributions for the same prefix, the task identity is not contained in the input, and any input-only mechanism fails. But if the input includes task-relevant context, the overlap may be measure-zero. The analysis should distinguish:

- exact input collision,
- distribution-level overlap,
- near-overlap in representation space.

Without that, “supports overlap extensively” is an empirical claim, not a theorem.

3. Specific technical corrections

3.1 Restate V1/V2 for delta gating

For a residual delta gate,

$\backslash$

$$s_{t+1} = s_t + g(x) \odot F(s_t),$$

\]

exact preservation of the unperturbed trajectory on (S_{old}) gives

\[

$$g_i(x)F_i(s_t(x)) = 0.$$

\]

Thus:

- if coordinate (i) is active at some depth (t) , then $(g_i(x)=0)$;
- if coordinate (i) is never active along the trajectory, $(g_i(x))$ is unconstrained.

This is the correct “zero-forcing” obstruction, not the “identity-forcing” obstruction stated for full output gating.

3.2 Input projection versus output projection

The projection rank theorem is proved for output projection,

\[

$$s_{t+1} = P(x)F(s_t).$$

\]

For input projection,

$\forall$

$$s_{t+1} = F(P(x)s_t),$$

$\forall$

exact preservation requires

$\forall$

$$F(P(x)s_t) = F(s_t).$$

$\forall$

Only if $\forall(F)$ is injective along the trajectory does this imply $\forall(P(x)s_t = s_t)$, hence $\forall(s_t \in \text{Im}\{P(x)\})$. The theorem only holds under that injectivity condition and must not be stated as if the two placements are interchangeable.

3.3 The error-amplification “iff” is now correct

The final version of V7 correctly distinguishes finite-depth bounds from uniform-in-depth bounds:

- for fixed finite $\forall(D)$, a finite $\forall(L)$ always yields a finite bound;
- uniform-in-depth bounding requires $\forall(L < 1)$.

This correction should be retained, and any phrase like “bounding is possible exactly when $\|L\| < 1$ ” should be avoided in future drafts.

4. Assessment of the final answer

The final answer is appropriately scoped:

- **Global-preservation elimination:** trivially impossible.
- **Asymmetric CL elimination:** open.
- **Bounding:** conditional on contractivity or finite depth.
- **Controllability:** undefined pending formalization.

This is intellectually honest and avoids the original overclaim.

However, the direct answer to the research question is still weak in the following sense. The question asks whether interference can be eliminated, bounded, or made controllable *by an input-dependent gate/projection mechanism*. The analysis does not deliver a theorem about the actual BDH mechanism. It delivers necessary conditions for an abstract family of mechanisms. The most decisive remaining item — reachability of the frozen BDH map under the allowed gate class — is unproved.

So the correct bottom line is:

> The strict proposed mechanism is not yet shown possible or impossible. The analysis reduces the question to a set of necessary conditions and one open reachability conjecture, but those conditions are not instantiated for BDH.

5. Prior-art assessment

The triage is mostly sound. LSTM/GRU is the closest neighbor and differs by using state-dependent gates. FiLM and fast weight programmers are closer than the early rounds admitted. Reservoir computing and nonlinear RNN reachability theory are relevant for the controllability question.

One missing item in some versions was the distinction between **learned per-step controls** and **input-only open-loop modulation**. This is the central difference from most control-theoretic prior art. The final analysis should make this even more explicit.

6. Recommendations before any write-up

- **Retrieve and parse paper .tex.** Extract the actual BDH block update, including residual form, normalization, and any Hebbian terms.

- **Reinstantiate every theorem** for the correct gate placement. The likely relevant case is delta gating $\{s_{t+1}=s_t+g(x)\odot F(s_t)\}$, not full output gating.
- **Rule on $\{t\}$ -conditioning.** Write down whether fixed non-learned depth encoding is permitted.
- **Retrieve the internal reports.** Map `plan_v2` and `cl-h1-report` to the formal variants.
- **Define controllability rigorously** as a reachability problem over the frozen BDH map and the exact gate class.
- **Measure** the Lipschitz constant of the gated BDH block and the cross-task Jacobian row-space overlap in $\{\phi\}$ -space. These are the only claimed barriers that can be tested empirically.
- **Do not claim a general impossibility** in the asymmetric CL setting. The current evidence supports only necessary conditions and an open conjecture.

Bottom line

The final debate record is a materially improved and cautiously scoped analysis. Its best contributions are the interference taxonomy, the correction of the diagonalization and projection overclaims, and the explicit acknowledgment that the meaningful continual-learning question is open. Its main defects are lack of BDH grounding, placement-specific theorems stated too generally, an unresolved $\{t\}$ -conditioning fork, and an only partially formalized controllability notion. The honest published conclusion should be: **strict global preservation makes nontrivial gating impossible; asymmetric continual learning with a frozen shared path remains open pending concrete reachability analysis of the actual BDH update.**

Input-Path Interference in a Frozen Shared Sequential Composition: A Refined Formal Analysis for BDH

Status: Final, revised after critical review

Date: 2026-08-25

0. Scope and Evidence-Base Notice

BDH (arXiv:2509.26507) is a scale-free, biologically inspired network of locally interacting neuron particles with recurrent-depth parameter sharing. The internal project files listed in the task were not retrievable during this analysis, and the literal BDH update equations from paper .tex were not available for instantiation. **This document therefore proves results for the following abstraction, which both the research question and the project reports assume:**

\[

$$s_0(x)=x,\quad s_{t+1}(x)=F(s_t(x)),\quad t=0,\dots,D-1,$$

\]

with a residual variant $(s_{t+1}(x)=s_t(x)+F(s_t(x)))$ treated where explicitly stated. All theorem-shaped claims are about this abstraction. They become theorems about BDH **only after** (F) is instantiated with BDH's concrete update and the gate placement is fixed. This is stated as a mandatory follow-up, not as an optional refinement.

No experimental result from the implementation reports is used as evidence for any theorem.

1. Formal Definitions

1.1 Shared Sequential Composition

Let $(V=\mathbb{R}^d)$ be the state space and $(F:V\rightarrow V)$ a fixed map. A **depth- (D) shared sequential composition** is

$$\begin{aligned} & \\ s_0(x)&=x,\quad s_{t+1}(x)=F(s_t(x)). \\ & \end{aligned}$$

The unperturbed trajectory is $(\tau_F(x)=(s_0(x),\dots,s_D(x)))$. If the architecture uses residual connections, the update is instead

$$\begin{aligned} & \end{aligned}$$

$$s_{t+1}(x) = s_t(x) + F(s_t(x)).$$

\]

1.2 Frozen Shared Path

$\mathcal{F}$ is **frozen** if its parameters are not updated during the continual-learning phase. Only the gate/projection parameters $\mathcal{G}$ are trainable:

\[

$$\frac{\partial \mathcal{F}}{\partial \theta} = 0.$$

\]

This is the standard continual-learning constraint; no theorem below relies on retraining $\mathcal{F}$.

1.3 Gating and Projection

Input-dependent gate: a map $g: X \rightarrow \mathbb{R}^d$ applied identically at every depth. Canonical placements:

- Output gating: $\mathcal{G}(g, s) = g(x) \odot F(s)$
- Input gating: $\mathcal{G}(g, s) = F(g(x) \odot s)$
- Delta gating (for residual blocks): $\mathcal{G}(g, s) = s + g(x) \odot F(s)$

The gate has **no per-depth trainable parameter vectors**: the same function g_ϕ is used at every depth. We separately consider the relaxed variant $g_\phi(x,t)$, where the depth index is supplied as an input, using either a **fixed non-trainable encoding** (adds $O(1)$ parameters) or a **learned depth embedding** (adds $O(D)$ parameters and violates the stated constraint).

Input-dependent projection: a map $P: X \rightarrow \Pi(V)$ applied as

$$\begin{aligned} & \left[\begin{aligned} s_{t+1} &= P(x)F(s_t) \\ \text{or} \\ s_{t+1} &= F(P(x)s_t). \end{aligned} \right. \\ & \left. \right] \end{aligned}$$

1.4 Input-Path Interference: Three Distinct Notions

The word "interference" must be disambiguated.

(A) Cross-input forward-pass contamination.

If F is frozen and the gate is stateless and a function of x alone, then the trajectory of x_1 is computationally independent of the presence of x_2 :

$$\begin{aligned} & \left[\right. \\ & \tilde{s}_t^{(M)}(x_1 \mid x_2 \text{ elsewhere}) = \tilde{s}_t^{(M)}(x_1). \end{aligned}$$

\]

This notion is **structurally zero** in this setting. It becomes live again only if the mechanism contains stateful components (running statistics, caches, etc.).

(B) Forward depth-aliasing.

A depth-agnostic gate applies the same modulation at every depth. If an oracle with access to depth t would apply a reference modulation $g^*(x, t)$, the depth-aliasing error is

\[

$\mathrm{DA}(g; x) =$

$\sum_{t=0}^{D-1}$

$\left|$

$\Phi(g(x), s_t(x)) - \Phi(g^*(x, t), s_t(x))$

$\right|$.

\]

This is a representational mismatch within one forward pass.

(C) Training-time gradient interference.

The gate parameters ϕ are shared across all inputs. Updating ϕ on task B may change gate outputs on task- A inputs. First-order damage is proportional to

\[

$\langle \nabla_{\phi} \mathcal{L}_A(\phi), \nabla_{\phi} \mathcal{L}_B(\phi) \rangle$

\]

The operationally relevant channels for continual learning are **(B)** and **(C)**. Channel **(A)** is dead under a frozen, stateless, input-only mechanism.

1.5 Computation-Isolability

A mechanism is **exactly computation-isolating** for a task family $\{(S_i, y_i)\}$ if:

- task outputs are produced exactly: $\tilde{s}_D(x; \phi) = y_i(x)$ for all $x \in S_i$;
- forward depth-aliasing error is zero on the specified reference trajectories;
- first-order cross-task gradient interference is zero:

$\langle \nabla_{\phi} \mathcal{L}_i, \nabla_{\phi} \mathcal{L}_j \rangle = 0$ for all $i \neq j$.

It **bounds interference** if all relevant errors are at most ϵ , with ϵ controlled by explicit quantities.

1.6 Elimination, Bounding, Controllability

- **Elimination:** exact task outputs, exact zero interference.

- **Bounding:** finite worst-case error, typically depending on Lipschitz constants, depth, gate error, capacity.
- **Controllability:** formalized as **reachability**:

> A target function $y_i: S_i \rightarrow \mathbb{V}$ is reachable by the gate class if there exists ϕ such that $s_D(x; \phi) = y_i(x)$ for all $x \in S_i$.

> The mechanism is **controllable** over a finite task sequence if every finite set of reachable tasks with compatible overlaps is simultaneously realizable by a single ϕ .

This is the natural control-theoretic reading: because the gate is input-only, the control $g(x)$ is fixed for the entire depth trajectory, rather than an independently chosen per-step control u_t . Any statement about controllability must be understood through this definition, not through unrestricted discrete-time controllability.

2. Theorem-Shaped Hypotheses

T1. Exact Preservation Forces the Gate to Be Inert on Active Coordinates

Assumptions:

- F is defined on an open set $U \subseteq \mathbb{R}^d$.
- The mechanism is a depth-agnostic output gate:

$$s_{t+1} = g(x) \odot F(s_t).$$

- Exact preservation of the unperturbed trajectory is required on an input set S :

$$\tilde{s}_t(x) = F^t(x) \text{ for all } (x \in S), \text{ all } (t).$$

Claim. For every $(x \in S)$ and every coordinate (i) ,

$$\begin{aligned} &[(g_i(x) - 1)F_i(s_t(x)) = 0. \\ &] \end{aligned}$$

If coordinate (i) is active at any depth along the trajectory, then $(g_i(x) = 1)$. If (F) is componentwise nonzero along all trajectories in (S) , then $(g(x) = \mathbf{1})$ on (S) .

Proof. At each depth, exact preservation gives $(g(x) \odot F(s_t(x)) = F(s_t(x)))$, which is componentwise $((g_i(x) - 1)F_i(s_t(x)) = 0)$. ■

Delta-gate variant. For residual delta gating,

$$\begin{aligned} &[\\ s_{t+1} &= s_t + g(x) \odot F(s_t), \\ &] \end{aligned}$$

exact preservation forces

$\setminus$

$$g_i(x) \setminus, F_i(s_t(x)) = 0.$$

$\setminus$

Thus the gate is **zero-forced** on active coordinates, not identity-forced. This is the placement likely relevant to BDH's residual structure.

Verdict. True as an exact theorem; near-tautological under global preservation. Its real content is *locative*: it identifies exactly where gate freedom survives.

Classification. Direct consequence.

T2. Asymmetric Preservation: The Continual-Learning-Relevant Setting

Assumptions:

- Old-task inputs $\setminus(S_{\text{old}}) \setminus$, new-task inputs $\setminus(S_{\text{new}}) \setminus$.
- Exact preservation is required only on $\setminus(S_{\text{old}}) \setminus$.
- No preservation constraint is imposed on $\setminus(S_{\text{new}}) \setminus$.

Claim. On (S_{old}) , the gate is pinned to identity (output gating) or zero (delta gating) on every coordinate active along the old trajectory. On never-active coordinates, and on all of (S_{new}) , the gate is unconstrained by preservation.

Verdict. True as an exact theorem. This is the correct formalization of the continual-learning constraint. The identity/zero obstruction does **not** by itself rule out nontrivial isolation when old and new task inputs are separated and the gate can interpolate.

Classification. Direct consequence; the asymmetric framing is the honest CL formalization.

T3. Identifiability: Same Input, Different Target Is Impossible

Assumptions:

- The mechanism is a deterministic function of the input: $\tilde{s}_D(x; \phi) = \mathcal{M}_\phi(x)$.
- Two tasks share an input $(x \in S_A \cap S_B)$ and demand different outputs: $(y_A(x) \neq y_B(x))$.

Claim. No single (ϕ) can satisfy both tasks.

Proof. A function returns one output for one input. Two different required outputs are incompatible. ■

Verdict. True as an exact theorem. It is the sharpest structural limit of an input-only mechanism, independent of depth, capacity, or nonlinearity.

Practical nuance. In language continual learning, the force of this theorem depends on whether task-relevant context is part of the input. If task identity is encoded in the context, exact input collisions may be rare; if not, they can be pervasive. The theorem is about exact collision, not distribution-level overlap.

Classification. Direct consequence; essentially an identifiability condition.

T4. Depth-Agnostic Gate Values: Representational Limit, Not Behavioral Impossibility

Assumptions:

- $\Phi(g, s)$ is injective in the gate argument g along the trajectory.
- An oracle demands a depth-varying gate-value sequence $(g^*(x, 0), \dots, g^*(x, D-1))$, nonconstant in t .

Claim. A depth-agnostic gate $g(x)$ cannot reproduce the exact gate-value sequence $(g^*(x, t))$.

Proof. If it did, injectivity would force $g(x)=g^*(x,t)$ for every t , contradicting nonconstancy. ■

Important correction. This is a statement about **gate values**, not about **reachable input-output behavior**. Because the state changes across depth, a constant gate can produce depth-varying effective behavior.

Counterexample. Let $F(s)=s$, $D=2$. Output gating gives

$$s_2 = g(x) \odot g(x) \odot x.$$

A constant gate (α, β) produces $(\alpha^2 x_1, \beta^2 x_2)$. A depth-varying sequence $(g^*_0=(c, 1), g^*_1=(1, c))$ produces (cx_1, cx_2) , which the constant gate $(\sqrt{c}, \sqrt{c})$ reproduces. Depth-varying controls can therefore be emulated by depth-agnostic gates via state feedback in some cases.

Verdict. The representational lemma is true. The behavioral impossibility is **not established**; it was withdrawn in the final adjudication. The reachability question is open.

Classification. Direct consequence for the representational part; the broad behavioral claim is **false as stated**.

T5. Projection Mechanisms: Necessary Rank Condition

Assumptions:

- Output projection: $s_{t+1} = P(x)F(s_t)$.
- Exact preservation of the unperturbed trajectory is required for input x .

Claim. Let

$$m(x) = \dim \operatorname{span}\{F(s_0(x)), \dots, F(s_{D-1}(x))\}.$$

Then $\operatorname{rank} P(x) \geq m(x)$.

Proof. Exact preservation forces $P(x)F(s_t(x)) = F(s_t(x))$ for all t , so all $F(s_t(x))$ lie in $\operatorname{Im} P(x)$. ■

Input-projection caveat. For input projection $s_{t+1} = F(P(x)s_t)$, exact preservation means $F(P(x)s_t) = F(s_t)$. This implies $P(x)s_t = s_t$ only if F is injective along the trajectory. The theorem as stated applies to output projection; the input-projection variant requires the injectivity assumption.

Withdrawn corollary. The earlier claim $\sum_i m_i \leq d$ for multiple tasks is **withdrawn**. Image-subspace orthogonality is a forward-pass geometric property; cross-task gradient interference is a property of the parameterization of $\mathcal{P}(x)$. These are different objects, and the capacity bound does not follow without a special parameterization assumption.

Classification. Direct consequence (linear algebra); no isolation capacity bound follows.

T6. Gradient Interference and the Parameter Budget

Assumptions:

- The gate/projection has p trainable parameters.
- For task i , the row space of the collection of output Jacobians is $(R_i \subseteq \mathbb{R}^p)$.
- Zero first-order interference is required for **all smooth losses**.

Claim. The row spaces must be mutually orthogonal, and

$\sum_i$

$\dim R_i \leq p$.

$\sum_i$

Proof. For arbitrary losses, zero inner products across all linear functionals force row-space orthogonality; orthogonal subspaces of $\mathbb{R}^p$ have total dimension at most p . ■

Caveats.

- For fixed specific losses, zero interference can hold accidentally even when row spaces overlap. The orthogonality condition is a worst-case guarantee, not a typical-case description.
- If $p \gg d$, the bound is likely non-binding for practical LLM-scale gates.
- This is standard parameter-isolation accounting, not a result specific to shared sequential composition.

Classification. Direct consequence; not a decisive obstruction unless Jacobian ranks are measured.

T7. Error Amplification: Bounding Is Conditional on Dynamics

Assumptions:

- The gated transition is Lipschitz in state with constant L .
- The per-depth gate error against a reference is at most δ .

Claim.

$$\begin{aligned} & \sum_{k=0}^{D-1} L^k \\ & \leq \begin{cases} \frac{1-L^D}{1-L}, & L < 1, \\ D, & L = 1, \\ \frac{L^D-1}{L-1}, & L > 1. \end{cases} \end{aligned}$$

Precise interpretation.

- For **fixed finite depth** (D) , a finite bound exists for every finite (L) .
- A bound that is **uniform in depth** exists iff $(L < 1)$.
- The claim "trained LLM-scale maps have $(L > 1)$ " is an **empirical conjecture**, not a theorem, and must be measured for the actual BDH block.

Classification. Direct consequence; quantifier-corrected in this revision.

T8. Finite-Task Reachability: Conditional Construction, Otherwise Underdetermined

Assumptions:

- The task family is finite: $\{(x_i, y_i)\}, \{i=1, \dots, N\}$.
- The gate family can interpolate finite sets.
- Each target is reachable: $y_i(x) \in \mathcal{R}_D(x) := \{\tilde{s}_D(x; \phi) : \phi \in \Phi\}$.
- Overlapping task inputs have agreeing targets.

Claim. There exists $\phi \in \Phi$ realizing all finite task targets exactly.

Proof sketch. Reachability gives a finite set of constraints; interpolation capacity fits them. ■

Underdetermination examples.

- If $F \equiv 0$, the reachable set is $\{0\}$ for every x , regardless of gating.
- If the same input has different targets, T3 applies.
- If $F(s) = s$ and $D = 2$ with multiplicative output gating, reachable outputs are of the form $g(x) \odot g(x) \odot x$, which excludes many targets.

Verdict. Conditionally true; otherwise underdetermined. The reachability assumption is the nontrivial crux and remains a **conjecture** for a concrete BDH map.

Classification. Plausible adaptation of universal interpolation; reachability is unproved.

Summary of Verdicts

#	Claim	Status	
---	---	---	
T1	Exact preservation forces identity/zero on active coordinates	True; exact theorem	
T2	Asymmetric preservation constrains only old-task active coordinates	True; CL-relevant	
T3	Same input, different target impossible	True; identifiability	
T4	Depth-agnostic gate-value representational limit	Representational true; behavioral impossibility false/underdetermined	
T5	Projection rank necessary condition	True; capacity corollary withdrawn	
T6	Gradient interference parameter budget	True under worst-case generality; often non-binding	
T7	Error amplification bound	True; finite-depth vs uniform-depth corrected	
T8	Finite-task reachability	Conditional; reachability is conjecture	

3. Prior Art: Does It Address the Same Problem?

| Prior art | What it does | Same mathematical problem? | Classification
|

| --- | --- | --- | --- |

| Gradient projection / orthogonal-gradient methods (GEM, A-GEM, GPM)
| Project gradients in parameter space during training | Partly: cross-task
gradient interference; not forward depth-aliasing on a frozen path |
Known, partially relevant |

| Parameter isolation / masks (PackNet, Supermasks, Piggyback) |
Allocate disjoint parameter subsets to tasks | No: creates per-task
parameters/masks, violating the input-only gate constraint | Known,
related but different |

| Progressive networks / PathNet | Add task-specific columns or paths |
No: per-task architectural growth | Known, different problem |

| Hypernetworks / fast weight programmers | Generate weights or
modulations for a main network | Partly: can generate input-dependent
modulations; typically per-layer outputs | Known, partially relevant |

| FiLM / feature-wise linear modulation | Input-dependent affine
modulation of hidden features | Partly: same gating algebra, but usually
per-layer parameters | Known, directly relevant |

| LSTM / GRU | Shared-parameter state-dependent gates in a recurrent
path | Closest: same composition; but gates see the current state, not
only the input | Known, nearest prior art |

| Mixture of experts / conditional computation | Input-dependent routing to expert parameter sets | No: per-layer routers and distinct experts | Known, structurally similar |

| Reservoir computing / echo state networks | Fixed random recurrent core plus trained readout | Partly: frozen core; not input-dependent forward gating for task isolation | Known, relevant background |

| Weight-tied transformers (Universal Transformer, ALBERT) | Reuse a block across depth | Partly: weight-tied composition; no task-isolation gating | Known, background context |

| Nonlinear RNN reachability / observability | Characterizes what a recurrent map can reach | Yes for the reachability question | Known, should be applied to T8 |

Conclusion. No surveyed prior art solves the exact problem: isolation of computations within a single frozen weight-tied composition using only an input-dependent gate, with no per-depth trainable parameters and no retraining of the shared path.

4. Novelty Classification

| Claim | Classification |

| --- | --- |

| Formalization of two live interference channels (depth-aliasing and cross-task gradient interference) in a frozen weight-tied path |

Apparently novel (packaging only) |

| T1–T3, T5–T7 | **Direct consequences** (textbook ingredients) |

| T4 representational lemma | Direct consequence; behavioral version false as stated |

| Necessity of state/depth/context information for effective gating | **Plausible adaptation** (consistent with LSTM/FiLM practice) |

| T8 reachability for concrete BDH $\setminus(F)$ | **Conjecture** |

| Expansiveness of trained BDH blocks | **Empirical conjecture** |

| Any observed effect in cl-h1-report | Inadmissible as theorem evidence; admissible only as a test of the conjectures above |

Contribution posture for write-up: **a new formalization and a clean map of elementary obstructions, not new mathematics.**

5. Obstruction Characterization

The strict proposed mechanism is:

- input-dependent gate/projection,
- no per-depth/per-level trainable parameters,
- frozen shared path,
- no retraining of $\setminus(F)$.

Under the **global-preservation reading** — the mechanism must preserve $\setminus(F)$'s behavior on **all** inputs — the gate is forced to be inert everywhere, so exact elimination is **trivially impossible**.

Under the **asymmetric continual-learning reading** — exact preservation on old-task inputs, free behavior on new-task inputs — no general impossibility or possibility theorem has been established. What is established is a conjunction of four necessary conditions:

- **Identifiability failure (T3).** Conflicting targets on identical inputs cannot be served by any input-only mechanism. In language modeling, the force of this depends on how much task identity is contained in the input context.
- **Preservation-geometry confinement (T1/T2).** The gate is pinned to identity (or zero under delta gating) on every active old-task coordinate. Freedom survives only on never-active coordinates and on new-task inputs.
- **Gradient coupling of shared ϕ (T6).** New-task updates generically move g_ϕ away from the preserved behavior on old inputs. Worst-case immunity costs a rank budget that may be non-binding when $p \gg d$.
- **Depth amplification (T7).** If the gated dynamics are expansive, small per-depth gate errors grow geometrically in depth. Bounding then requires contractivity of the composed system — a property of F , not a capability of the gate alone.

Three **scope conditionals** attach to every result:

- **Statelessness.** Any stateful component inside the gate revives cross-input forward contamination (channel A).
- **t -conditioning fork.** A gate $g_\phi(x, t)$ with a fixed non-trainable depth encoding adds $O(1)$ parameters and does not violate the constraint as literally worded. A learned depth embedding adds $O(D)$ parameters and does violate it. The depth-agnostic obstruction dissolves on the permissive side of this fork, converting the problem into pure reachability (T8).

- **Frozen path.** BDH's native Hebbian-style plasticity must be explicitly disabled for the frozen-path premise to hold.

6. Direct Answer to the Research Question

> Can input-path interference arising from shared sequential composition be eliminated, bounded, or made controllable by an input-dependent gate/projection mechanism, without per-depth/per-level trainable parameters and without retraining the frozen shared path?

| Reading | Verdict | Epistemic status |

| --- | --- | --- |

| **Elimination**, global preservation on all inputs | Impossible — the only admissible gate is identity/zero on active coordinates | Proved; degenerate |

| **Elimination**, asymmetric CL (preserve (S_{old}) , free on (S_{new})) | **Open**. Necessary conditions established; no general impossibility or possibility theorem | Partially characterized |

| **Bounding** | Finite-depth bounds exist for any finite (L) ; uniform-in-depth bound iff $(L < 1)$ | Proved, conditional |

| **Controllability** | Not yet well-posed: requires formal reachability analysis on the concrete frozen BDH map with the exact gate class | Undefined pending instantiation |

The failure condition contemplated by the brief is met **only in the degenerate global-preservation reading**. In the continual-learning-

relevant reading, the correct outcome is: *not proven impossible, not proven possible; reduced to identifiable necessary conditions and one decisive open conjecture — reachability.*

7. Mandatory Follow-Ups Before Empirical Reconciliation

- **Instantiate F from paper .tex.** Fix the literal BDH update, including residual form, LayerNorm placement, and any Hebbian-style multiplicative terms.
- **Reinstantiate every theorem for the correct gate placement.** For a residual block, the natural gate is $s_{t+1} = s_t + g(x) \odot F(s_t)$, which changes the identity obstruction to a zero obstruction.
- **Rule on the t -conditioning fork.** State in writing whether fixed non-learned depth encoding is permitted.
- **Retrieve the internal reports.** Map 2026-08-25_bdh-continual-learning-plan_v2.md and 2026-08-25_cl-h1-report.md onto exactly one formal variant: depth-agnostic, t -conditioned, or state-conditioned.
- **Define controllability formally** as reachability of finite task targets by the gate class on the instantiated BDH map.
- **Measure** (a) the Lipschitz constant of the gated BDH block, and (b) cross-task Jacobian row-space overlap in ϕ -space. These are the only claimed barriers that can be tested empirically.

Bottom line. A strict input-only gate/projection on a frozen shared sequential composition does not provide a proven general solution. The obstruction is not a single theorem but a map of four necessary failure conditions and one open reachability problem. The honest conclusion for publication is: *global preservation makes nontrivial gating impossible;*

asymmetric continual learning with a frozen shared path remains open pending concrete reachability analysis of the actual BDH update.

[MODERATOR Round 4]

[moderator] Round 4: LLM call failed (litellm.RateLimitError: RateLimitError: OpenrouterException - {"error":{"message":"Provider returned error","code":429,"metadata":{"raw":"stealth/ox-alpha is temporarily rate-limited upstream. Please retry shortly.","provider_name":"Stealth","is_byok":false,"limit_source":"upstream_provider_shar shortly, add your own provider key (<https://openrouter.ai/settings/integrations>), or route to another provider with provider routing: <https://openrouter.ai/docs/features/provider-routing>"},"user_id":"user_3Ay1IY7xf9lrDwaN7Fq4EUJnQA9"}})

[STRATEGIST Round 5]

Final Strategic Assessment: Input-Path Interference in BDH via Gating/Projection

Date: 2026-08-25 | **Status:** Complete (4-round debate, adjudicated)

Executive Summary

Under the strict constraints — depth-agnostic input-dependent gate/projection, frozen shared path, no per-depth trainable parameters, no

retraining — **exact elimination of input-path interference is impossible only under the degenerate global-preservation reading** (the gate must be identity everywhere, doing nothing). Under the **continual-learning-relevant asymmetric reading** (exact preservation on old tasks, free on new tasks), the problem is **open**: no general impossibility or possibility theorem has been established. What has been established is a precise map of four independent necessary conditions, each capable of killing the program, plus one decisive open conjecture (reachability) whose resolution determines feasibility.

1. Formal Definitions

1.1 Shared Sequential Composition

Let $V = \mathbb{R}^d$ be the state space. A **depth- D shared sequential composition** is defined by a fixed map $F: V \rightarrow V$ and the recurrence:

$$s_0(x) = x, \quad s_{t+1}(x) = F(s_t(x)), \quad t = 0, \dots, D-1$$

The unperturbed trajectory is $\tau_F(x) = (s_0(x), s_1(x), \dots, s_D(x))$.

If BDH uses residual connections (likely, per the BDH-GPU architecture), the relevant recurrence is:

$$s_{t+1}(x) = s_t(x) + F(s_t(x))$$

Classification: Direct characterization of BDH's weight-tied recurrent depth, corroborated by BDH-MLX ("Same weights reused in all layers (recurrent depth)") and BDH-GPU descriptions ("ReLU-Lowrank residual like $x \mapsto \dots$ "). Instantiation with the literal BDH update from paper .tex remains mandatory.

1.2 Frozen Shared Path

F is **frozen** if $\partial F / \partial \theta = 0$ during the continual-learning phase. Only the gate/projection mechanism ϕ may be trained.

Classification: Known — standard continual learning constraint.

1.3 Input-Path Interference: Three Precise Notions

(A) Cross-input forward-pass contamination. With frozen θ and a stateless input-conditioned mechanism, each input's forward pass is computationally independent. This notion is **structurally zero** and not a live channel.

(B) Forward depth-aliasing. A depth-agnostic gate $g(x)$ applies the same modulation at every depth. If an oracle would apply $g^*(x, t)$ (varying with depth), the depth-aliasing error is:

$$\mathrm{DA}(g; x) = \sum_{t=0}^{D-1} \left| \Phi(g(x), s_t(x)) - \Phi(g^*(x, t), s_t(x)) \right|$$

(C) Training-time gradient interference (cross-task forgetting).

Because gate parameters ϕ are shared across all inputs, updating ϕ on task B may change g_ϕ on task- A inputs. First-order damage is proportional to $\langle \nabla_{\phi} \mathcal{L}_A, \nabla_{\phi} \mathcal{L}_B \rangle$.

The live channels are (B) and (C). Notion (A) is dead under the stated assumptions.

Classification: The depth-aliasing notion is **apparently novel** as a packaged definition; components are known from control theory. The gradient notion is **known** from continual learning.

1.4 Gating

An **input-dependent gate** $g: X \rightarrow \mathbb{R}^d$ is applied identically at every depth:

- Output gating: $s_{t+1} = g(x) \odot F(s_t)$
- Input gating: $s_{t+1} = F(g(x) \odot s_t)$
- Delta gating (for residual blocks): $s_{t+1} = s_t + g(x) \odot F(s_t)$

The gate has **no per-depth parameters**: the same $g(x)$ is used at every t .

Classification: Known mechanism (LSTM, MoE), but the depth-independence + frozen-path combination is novel.

1.5 Projection

An **input-dependent projection** $P: X \rightarrow \Pi(V)$ maps each input to an orthogonal projection, applied as $s_{t+1} = P(x) F(s_t)$ or $s_{t+1} = F(P(x) s_t)$.

Classification: Known (gradient projection methods), formalized for forward-pass application.

1.6 Computation-Isolability

A mechanism is **exactly computation-isolating** for a task family $\{(S_i, y_i)\}_{i=1}^K$ if: (1) each task output is produced exactly; (2) zero forward depth-aliasing error; (3) zero first-order cross-task gradient interference.

Classification: Apparently novel — this specific formulation combining shared composition, frozen parameters, and forward-pass isolation does not appear in surveyed literature.

1.7 Elimination vs. Bounding vs. Controllability

- **Elimination:** Exact equality of outputs and exact zero interference.
- **Bounding:** Deviation $\leq \epsilon$, controlled by Lipschitz constant, depth, gate error, parameter count.

- **Controllability:** Reachability of finite task targets by the gate class without modifying F .

Classification: Plausible adaptation of approximation theory and control-theoretic concepts.

2. Theorem-Shaped Hypotheses (Final Adjudicated Versions)

V1. Identity-Forcing Under Global Preservation

Assumptions: F defined on open $U \subseteqq \mathbb{R}^d$; depth-independent output gating $s_{t+1} = g(x) \odot F(s_t)$; exact trajectory preservation on input set S : $\tilde{s}_t(x) = F^t(x)$ for all $x \in S$, all t .

Claim: For every $x \in S$ and every coordinate i : $g_i(x) = 1$ whenever $\exists t: F_i(s_t(x)) \neq 0$.

Proof: At depth t : $g(x) \odot F(s_t(x)) = F(s_t(x))$ implies $(g_i(x) - 1) F_i(s_t(x)) = 0$ componentwise. $\square$

Delta-gate variant: For residual delta gating $s_{t+1} = s_t + g(x) \odot F(s_t)$, exact preservation forces $g_i(x) F_i(s_t(x)) = 0$. The gate is **zero-forced** on active coordinates, not identity-forced.

Verdict: True as an exact theorem. Near-tautological under global preservation. Its real content is *locative*: it identifies precisely where gate freedom lives — never-active coordinates and non- S inputs.

Classification: Known / direct consequence.

V2. Asymmetric Preservation: The CL-Relevant Constraint

Assumptions: Old-task inputs S_{old} ; new-task inputs S_{new} ; output gating; exact preservation required only on S_{old} .

Claim: For $x \in S_{\text{old}}$: $g_i(x) = 1$ whenever $\exists t: F_i(s_t(x)) \neq 0$. On the zero-coordinate set $Z(x) = \{i: F_i(s_t(x)) = 0 \forall t\}$, the gate is unconstrained. On S_{new} , the gate is entirely unconstrained.

Proof: Same componentwise argument as V1, restricted to S_{old} . $\square$

Verdict: True as an exact theorem. If old-task trajectories have support in all coordinates, the gate is forced to identity on all of S_{old} . If old and new inputs are disjoint and the gate network can interpolate, exact isolation is *not* ruled out by this obstruction alone.

Classification: Direct consequence; the asymmetric framing is the correct CL formalization.

V3. Identifiability: Overlapping Inputs Are Fatal

Assumptions: The mechanism is a deterministic function of the input: $\tilde{s}_D(x; \phi) = \mathcal{M}_\phi(x)$. Two tasks A, B share an input $x \in S_A \cap S_B$ with different targets: $y_A(x) \neq y_B(x)$.

Claim: No single ϕ can satisfy both tasks.

Proof: The map is a function: one input, one output. Two different required outputs are incompatible. $\square$

Verdict: True as an exact theorem. The sharpest structural limit of "input-dependent only," independent of depth, nonlinearity, and capacity.

Classification: Direct consequence; essentially an identifiability condition.

V4. Depth-Agnostic Gate-Value Limitation (Downgraded)

Assumptions: $\Phi(g,s)$ injective in g along the trajectory; ideal modulation $g^*(x,t)$ nonconstant in t .

Claim: No depth-independent gate $g(x)$ can reproduce the nonconstant gate-value sequence $g^*(x,t)$.

Proof: Equality at every depth gives $g(x) = g^*(x,t)$ for every t , contradicting nonconstancy. $\square$

Critical correction: This is a statement about **gate values**, not about **reachable input-output maps**. A constant-in-depth gate can produce depth-varying effective behavior because the state s_t changes.

Counterexample: $F(s) = s$, $D = 2$. Output gating: $s_2 = g(x) \odot g(x) \odot x$. Constant gate $g(x) = (\alpha, \beta)$ produces $(\alpha^2 x_1, \beta^2 x_2)$. Depth-varying sequence $g^*_0 = (c, 1)$, $g^*_1 = (1, c)$ produces (cx_1, cx_2) , also achievable by constant gate $(\sqrt{c}, \sqrt{c})$.

Verdict: Representational lemma TRUE; behavioral impossibility NOT established. The reachability question is open.

Classification: Direct consequence for the representational part; the broad behavioral impossibility is **false as stated**.

V5. Projection Rank Condition

Assumptions: Output projection $s_{t+1} = P(x) F(s_t)$; exact preservation of unperturbed trajectory required.

Claim: If $m(x) = \dim \operatorname{span}\{F(s_0(x)), \dots, F(s_{D-1}(x))\}$, then $\operatorname{rank} P(x) \geq m(x)$.

Proof: $P(x) F(s_t(x)) = F(s_t(x))$ forces $F(s_t(x)) \in \operatorname{Im} P(x)$. $\square$

Input-projection caveat: For input projection $s_{t+1} = F(P(x)s_t)$, exact preservation means $F(P(x)s_t) = F(s_t)$. This implies $P(x)s_t = s_t$ only if F is injective along the trajectory.

Withdrawn corollary: The capacity corollary $\sum_i m_i \leq d$ is **WITHDRAWN**. Orthogonality of image subspaces is a forward-pass geometric property; cross-task gradient interference is a property of the parameterization. These are different mathematical objects.

Verdict: True as an exact linear-algebra theorem.

Classification: Direct consequence.

V6. Gradient Interference Parameter Budget

Assumptions: Gate has p trainable parameters ϕ ; task Jacobians $J_i \in \mathbb{R}^{d \times p}$; zero first-order interference required for *all smooth losses*.

Claim: Row spaces $\{R_i\}$ must be mutually orthogonal: $\sum_i \dim R_i \leq p$.

Proof: For arbitrary loss functions, vanishing inner products force row-space orthogonality. $\square$

Important caveats:

- For fixed specific losses, zero interference is weaker (can hold accidentally).
- If $p \gg d$ (typical for LLM-scale gates), the bound is non-binding.
- Worst-case guarantee, not typical-case behavior.

Verdict: True as an exact theorem under stated generality. Standard parameter-isolation accounting.

Classification: Direct consequence.

V7. Error Amplification Bound

Assumptions: Gated transition L -Lipschitz in state; per-depth gate error $\leq \delta$.

Claim:

$$\|s_D(x) - s_{D^*}(x)\| \leq \delta \sum_{k=0}^{D-1} L^k = \begin{cases} \delta \frac{1 - L^D}{1 - L} & L < 1 \\ \delta D & L = 1 \\ \delta (L^D - 1) & L > 1 \end{cases}$$

Precise interpretation:

- For **fixed finite depth D** : bound exists for every finite L .
- **Uniform-in-depth** bound exists iff $L < 1$.

Empirical caveat: "Trained LLM-scale maps have $L > 1$ " is an **empirical conjecture**, not a theorem.

Verdict: True as an exact theorem.

Classification: Known / direct consequence.

V8. Finite-Task Reachability (Conditional)

Assumptions: Finite task set (x_i, y_i) ; gate family has sufficient capacity; each target reachable: $y_i(x) \in \mathcal{R}_D(x) := \tilde{s}_D(x; \phi) : \phi \in \Phi$; overlapping inputs have agreeing targets.

Claim: There exists ϕ realizing all finite task targets exactly.

Counterexamples showing underdetermination:

- $F(s) = 0$: $\mathcal{R}_D(x) = \{0\}$ for every x , regardless of gating.
- Same input, different targets: V3 kills it.
- $F(s) = s$, $D = 2$, multiplicative output gating: reachable set is $\{g(x) \odot g(x) \odot x\}$; sign/root obstructions limit reachability.

Verdict: Conditionally true; otherwise underdetermined. The reachability assumption is the nontrivial crux.

Classification: Plausible adaptation; reachability is a **conjecture** unless proved for specific F .

3. Summary of Verdicts

#	Claim	Final Status	
---	-----	-----	

| V1 | Global preservation forces identity | TRUE — exact theorem; near-tautological |

| V2 | Asymmetric preservation constraint | TRUE — exact theorem; CL-relevant |

| V3 | Overlapping inputs with different targets | TRUE — exact theorem; identifiability |

| V4 | Depth-agnostic gate-value limitation | Representational TRUE; behavioral FALSE/underdetermined |

| V5 | Projection rank condition | TRUE — exact linear algebra; capacity corollary withdrawn |

| V6 | Gradient interference parameter budget | TRUE — exact under stated generality |

| V7 | Error amplification bound | TRUE — exact; finite- D vs uniform- D corrected |

| V8 | Finite-task reachability | CONDITIONAL; reachability is conjecture |

4. Prior Art Analysis

| Prior Art | Same Mathematical Problem? | Classification |

| --- | --- | --- |

| **Orthogonal gradient methods** (GEM, A-GEM, GPM) | Partly — addresses gradient interference in parameter space; not forward depth-aliasing; usually modifies main network | Known, partially relevant |

| **Parameter isolation / masking** (PackNet, Supermasks) | No — per-task masks/parameters; constraint-violating | Known, related but different |

| **Progressive networks / PathNet** | No — per-task architectural growth | Known, different problem |

| **Hypernetworks / fast weight programmers** | Partly — can generate input-dependent modulations; typically per-layer outputs | Known, partially relevant |

| **FiLM** (feature-wise linear modulation) | Partly — same gating algebra, but usually per-layer parameters | Known, directly relevant |

| **LSTM / GRU** | **Closest** — shared-parameter gates at every time step; crucially state-dependent, not input-only | Known, nearest prior art |

| **Mixture of Experts** | No — per-layer routers, distinct experts | Known, structurally similar |

| **Reservoir computing / echo state networks** | Partly — frozen recurrent core + trained readout | Known, relevant background |

| **Universal Transformer / ALBERT** | Partly — weight-tied depth, no task isolation | Known, background context |

| **Nonlinear RNN reachability / observability** | Yes for the reachability question | Known, should be used for V8 |

No surveyed prior art solves the exact problem: isolation within one frozen weight-tied composition via an input-only gate without per-depth trainable parameters.

5. Obstruction Characterization

There is no single obstruction theorem in the CL-relevant setting. The failure surface is a **conjunction of four necessary conditions**, each independently capable of killing the program, none individually fatal:

- **Identifiability failure** (V3): Conflicting targets on overlapping inputs are unservable by any input-only mechanism. In language-modeling CL, supports overlap extensively. Escape requires context-sufficiency or task conditioning.
- **Preservation-geometry confinement** (V1/V2): The gate is pinned to identity (output gating) or zero (delta gating) on every (old input, active coordinate) pair. Freedom survives only on never-active coordinates (substantial under sparse activations) and on new-task inputs.
- **Gradient coupling of shared ϕ** (V6): New-task updates generically move g_ϕ off identity on old inputs. Worst-case immunity costs a rank budget that is either restrictive ($p \leq d$) or vacuous ($p \gg d$).
- **Depth amplification** (V7): Under expansive gated dynamics, per-depth modulation error grows geometrically in D . Bounding demands architectural contractivity — an assumption on F , not a capability of the gate.

Plus three **scope conditionals**: (a) statelessness of the mechanism; (b) the t -conditioning interpretation; (c) BDH's native Hebbian plasticity actually disabled.

6. The Pivotal Scope Decision: t -Conditioning

A shared-parameter gate $g_\phi(x, t)$ where t is a **fixed, non-trainable encoding** (e.g., sinusoidal) adds $O(1)$ parameters and **does not violate "no per-depth/per-level trainable parameters" as literally worded**. This dissolves V1's depth-agnostic restrictions and V4's residue entirely, converting the problem into pure reachability (V8).

A **learned depth embedding** adds $O(D)$ parameters and violates the constraint outright.

Every impossibility-flavored statement applies only on the strict side of this fork. **The team must rule before any experiment is interpreted.**

7. Constructive Directions (all $\leq$ conjecture)

- **C1 — Orthogonal gradient on ϕ -space:** Project new-task gradients onto the orthogonal complement of old-task gradient subspace in ϕ -space, holding $g_\phi|_{S_{\text{old}}} \approx \mathbf{1}$ to first order. Classification: **plausible adaptation** of GEM/GPM.

- **C2 — Fixed-encoding s_t -conditioning** ($O(1)$ parameters): dissolves depth-agnostic restrictions; legality contingent on scope ruling.
- **C3 — Delta-gate placement** for residual BDH: gate $g(x) \odot \Delta(s_t)$ inside $s_{t+1} = s_t + \odot$, never the residual stream.
- **C4 — Sparse-coordinate freedom:** exploit never-active coordinates of old-task trajectories (plausible under ReLU sparsity) as legal modification space.

8. Novelty Ledger (Final, Adjudicated)

| Claim | Classification |

|-----|-----|

| Problem formulation: isolation inside frozen weight-tied composition via input-only gate, CL-framed, two-channel interference taxonomy |

Apparently novel (packaging only) |

| V1, V2, V3, V5, V6, V7 | **Direct consequences** (textbook ingredients) |

| V4 representational lemma | Direct consequence; behavioral version **false as stated** |

| Necessity of state/depth information for effective gating | **Plausible adaptation** (consistent with LSTM/FiLM practice) |

| V8 reachability for concrete BDH $\mathbb{F}_2$ | **Conjecture** |

| Expansiveness of trained BDH blocks | **Empirical conjecture** |

Contribution posture: a new formalization and a clean map of elementary obstructions — not new mathematics.

9. Direct Answer to the Research Question

> Can input-path interference arising from shared sequential composition be eliminated, bounded, or made controllable by an input-dependent gate/projection mechanism, without introducing per-depth/per-level trainable parameters and without retraining the frozen shared path?

| Reading | Verdict | Epistemic Status |

| ----- | ----- | ----- |

| **Elimination**, global-preservation | Trivially impossible — only admissible gate is identity/zero on active coordinates | Proved; degenerate |

| **Elimination**, asymmetric CL | **Open** — necessary conditions established, no general impossibility or possibility theorem | Partially characterized |

| **Bounding** | Finite-depth bounds exist for any finite L ; uniform-in-depth iff $L < 1$ | Proved, conditional |

| **Controllability** | **Not yet well-posed** — requires formal reachability definition and BDH instantiation | Undefined pending instantiation |

The failure condition is met only in the degenerate reading. In the CL-relevant reading, the correct outcome is: *not proven impossible, not proven possible; reduced to identifiable necessary conditions and one decisive open conjecture (reachability).*

10. Mandatory Follow-Ups

- **Instantiate $\$F\$$** from paper . tex: fix the literal update (residual form, Hebbian/multiplicative terms, normalization), choose gate placement (C3), verify native plasticity is disabled.
- **Rule on the $\$t\$$ -conditioning fork** in writing.
- **Define controllability formally** as reachability of finite task targets by the gate class on the instantiated $\$F\$$.
- **Retrieve internal reports;** map the implemented mechanism onto exactly one formal variant; reconcile cl-h1-report outcomes strictly as tests of V7/V8 conjectures.
- **Measurement program:** empirical Lipschitz/expansiveness of the gated BDH block (tests V7's conjecture); empirical cross-task Jacobian row-space overlap in ϕ -space (tests whether V6's worst case binds).

[CRITIC Round 5]

Final Critical Review: BDH Input-Path Isolation Debate

Overall Assessment

The analysis improved materially across rounds. Its main correction is the right one: the strong claim “exact elimination is impossible in general” has been dissolved into a **scoped set of exact necessary conditions plus an open reachability conjecture**. The separation of three distinct meanings of “interference” is the most useful contribution, and the final verdict table is epistemically honest.

That said, the analysis is **not yet grounded in the actual BDH mechanism**. It operates on an abstraction $\{s_{t+1} = F(s_t)\}$ or a residual variant, but does not instantiate $\{F\}$ from paper.tex. Consequently, several theorem statements are placement-sensitive, and the most BDH-relevant placement — **delta gating of a residual block** — was only partially treated.

Strengths Worth Preserving

- **Interference taxonomy.**

The separation of

- cross-input forward-pass contamination,
- forward depth-aliasing,
- training-time cross-task gradient interference

is correct and necessary. Under a frozen, stateless, input-dependent mechanism, cross-input forward contamination is structurally zero. The live channels are depth-aliasing and cross-task parameter updating.

- **Rejection of the naive diagonalization impossibility.**

The claim that a depth-agnostic gate is confined to the diagonal of the modulation space was correctly downgraded. A constant-in-depth gate can still produce depth-varying effective behavior because the state changes. The counterexample with $\mathcal{F}(s)=s$ is decisive.

- **Withdrawal of the projection capacity corollary.**

The rank condition

$\mathcal{P}$

$\text{operatorname{rank}} P(x) \geq m(x)$

$\mathcal{P}$

is correct for output projection. The stronger cross-task capacity bound $\sum_i m_i \leq d$ was correctly withdrawn. It conflated forward-pass subspace geometry with gradient-parameter geometry.

- **Correct scoping of the main claim.**

Exact elimination is proved impossible only in the degenerate global-preservation reading. In the continual-learning-relevant asymmetric setting, the question is not resolved either way.

Remaining Weaknesses and Gaps

1. The actual BDH equations were never instantiated

This is the largest remaining defect. The analysis repeatedly acknowledges that `paper.tex` was not parsed. Public descriptions suggest a BDH-GPU structure such as

$$\begin{aligned} & \left[\right. \\ & x \mapsto \operatorname{ReLU}\bigl(x + \text{low-rank FFN}(x)\bigr), \\ & \left. \right] \end{aligned}$$

with shared parameters, normalization, and possibly Hebbian-style multiplicative terms.

This matters because:

- **Delta gating changes the preservation obstruction.**

For full output gating $(s_{t+1} = g(x) \odot F(s_t))$, preservation forces $(g_i(x) = 1)$ on active coordinates.

For residual delta gating $(s_{t+1} = s_t + g(x) \odot F(s_t))$, preservation forces

$$\begin{aligned} & \left[\right. \\ & g_i(x) F_i(s_t(x)) = 0, \\ & \left. \right] \end{aligned}$$

so the gate is pinned to **zero**, not identity, on active coordinates.

- **Input versus output projection.**

The projection rank theorem does not automatically extend from output projection to input projection without injectivity assumptions on $\backslash(F\backslash)$.

- **Statefulness.**

If the frozen path contains normalization with running statistics, the mechanism is not stateless, which can revive cross-input forward contamination.

- **Hebbian/plastic components.**

If BDH includes native plastic terms, “freezing” the path is an architectural change that must be verified.

2. Internal reports were not reconciled

The plan and H1 report were not retrieved. Without them:

- the implemented gate placement is unknown,
- the empirical outcomes of `cl-h1-report.md` are neither explained nor refuted,
- the implementation cannot be mapped to a formal variant.

This is a defect of the current analysis, not merely an optional future step.

3. “Controllability” is still not fully formalized

The final definition — reachability of finite task targets by the gate class — is a reasonable start. But the control structure is unusual: an input-dependent gate $g(x)$ is fixed for the entire depth trajectory, so per-input reachable sets are parameterized by a single vector, not by D independent controls.

A complete formalization should specify:

- the state space,
- the admissible gate range,
- the exact transition rule,
- the target map class,
- and the requirement that one parameter vector ϕ simultaneously realize all tasks.

Until then, “controllability reduces to reachability” is only a promissory note.

4. The t -conditioning fork remains unresolved

The pivotal scope decision is still open:

- a fixed, non-learnable depth encoding adds $O(1)$ trainable parameters and likely does not violate the constraint,
- a learned depth embedding adds $O(D)$ trainable parameters and violates it.

Every impossibility-flavored statement holds only on the strict depth-agnostic side. Future write-ups must label assumptions explicitly.

5. The gradient-interference bound is not specific to shared composition

The bound

$\sum_i$

$\sum_i \dim R_i \leq p$

$\sum_i$

is standard parameter-isolation accounting. It is correct only under the strong requirement of zero first-order interference for all smooth losses. For fixed losses, interference can vanish accidentally. More importantly, if $p \gg d$, the bound is often non-binding. It should not be presented as a decisive shared-sequential-composition obstruction until actual Jacobian ranks are measured.

6. The overlapping-input theorem needs practical nuance

Same-input-different-target impossibility is mathematically true. But its practical force depends on whether task identity is represented in the input context. Exact input collisions and distribution-level overlap should not be conflated.

Final Scoped Answer to the Research Question

> Can input-path interference arising from shared sequential composition be eliminated, bounded, or made controllable by an input-dependent gate/projection mechanism, without per-depth/per-level trainable parameters and without retraining the frozen shared path?

| Reading | Verdict | Status |

| --- | --- | --- |

| **Elimination**, global preservation on all inputs | Trivially impossible: the only admissible gate is inert on active coordinates | Proved; degenerate |

| **Elimination**, asymmetric CL: preserve (S_{old}) , free (S_{new}) | **Open**. Necessary conditions established; no general impossibility or possibility theorem | Partially characterized |

| **Bounding** | Finite-depth bounds exist for any finite Lipschitz (L) ; uniform-in-depth bounding requires $(L < 1)$ | Proved, conditional |

| **Controllability** | Not yet well-posed without reachability analysis of the concrete frozen BDH map | Undefined pending instantiation |

The failure condition in the brief is met only in the **degenerate global-preservation reading**. In the meaningful continual-learning setting, the correct conclusion is:

> *Not proven impossible; not proven possible. The question reduces to identifiable necessary conditions and one decisive open conjecture — reachability of the frozen BDH map under the allowed gate class.*

Novelty Ledger, Final Form

| Claim | Classification |

| --- | --- |

| Two-channel interference taxonomy in a frozen weight-tied path |
Apparently novel as packaging; ingredients known |

| Identity-forcing / zero-forcing preservation lemma | Direct
consequence |

| Asymmetric-preservation constraint | Direct consequence; CL-relevant
reframing |

| Overlapping-input identifiability impossibility | Direct consequence |

| Depth-agnostic gate-value representational limit | Direct consequence;
behavioral version false as stated |

| Projection rank necessary condition | Direct consequence |

| Gradient-interference parameter budget | Direct consequence |

| Lipschitz error-amplification bound | Direct consequence |

| Necessity of state/depth/context information for robust gating |
Plausible adaptation |

| Reachability of concrete BDH $\backslash(F)$ under gating | Conjecture |

| Expansiveness of trained BDH blocks | Empirical conjecture |

No experimental result from the implementation reports was used as
evidence for any theorem.

Required Next Steps Before Publication

- **Parse paper .tex** and instantiate the literal BDH update, including residual form, normalization, and Hebbian or multiplicative terms.
- **Re-prove every theorem for the actual gate placement**, especially residual delta gating.
- **Rule formally on t -conditioning**: fixed non-learned depth encoding versus learned depth embedding.
- **Retrieve and map the internal reports** to exactly one formal variant.
- **Define controllability rigorously** as finite-target reachability over the concrete BDH map and exact gate class.
- **Measure:**
 - the Lipschitz constant of the gated BDH block;
 - cross-task Jacobian row-space overlap in ϕ -space.

The honest published conclusion remains:

> **Strict global preservation makes nontrivial gating impossible. Asymmetric continual learning with a frozen shared path remains open pending concrete reachability analysis of the actual BDH update.**

[OPTIMIZER Round 5]

Input-Path Interference in BDH via Input-Dependent Gating/Projection: A Scoped Formal Analysis

Date: 2026-08-25

Status: Final, after adversarial review

1. Scope, Abstraction, and Evidence Base

BDH (arXiv:2509.26507) is a scale-free, biologically inspired network of locally interacting neuron particles with **shared parameters across depth** (“recurrent depth”), corroborated by third-party implementations and public descriptions. The literal BDH update equations from paper .tex and the internal project reports were not retrievable within this analysis. Consequently, the formal results below are proved for the abstraction that the research question itself assumes:

$$\begin{aligned} & \backslash[\\ & s_0(x)=x,\quad s_{t+1}(x)=F(s_t(x)),\quad t=0,\dots,D-1, \\ & \backslash] \end{aligned}$$

with the residual variant

\[

$$s_{t+1}(x) = s_t(x) + F(s_t(x))$$

\]

treated where explicitly stated. All theorem-shaped claims are about this abstraction; they become theorems about BDH only after $\backslash(F\backslash)$ is replaced by the concrete BDH update and the gate placement is fixed. This instantiation is a mandatory follow-up, not an optional refinement.

No experimental observation from any implementation report is used as evidence for any theorem.

2. Formal Definitions

2.1 Shared Sequential Composition

Let $\backslash(V = \mathbb{R}^d\backslash)$ be the state space and $\backslash(F:V\rightarrow V\backslash)$ a fixed map. A **depth- $\backslash(D\backslash)$ shared sequential composition** is

\[

$$s_0(x) = x, \quad s_{t+1}(x) = F(s_t(x)).$$

\]

The unperturbed trajectory is $\tau_F(x) = (s_0(x), \dots, s_D(x))$.

2.2 Frozen Shared Path

F is **frozen** if its parameters are not updated during the continual-learning phase:

$$\left[\frac{\partial F}{\partial \theta} = 0. \right]$$

Only the gate/projection parameters ϕ are trainable.

2.3 Gating

An **input-dependent gate** is a map $g: X \rightarrow \mathbb{R}^d$ applied identically at every depth. Canonical placements:

- **Output gating:** $\Phi(g, s) = g(x) \odot F(s)$
- **Input gating:** $\Phi(g, s) = F(g(x) \odot s)$
- **Delta gating** (for residual blocks): $\Phi(g, s) = s + g(x) \odot F(s)$

The gate has **no per-depth trainable parameter vectors**: the same function g_ϕ is used at every depth. A relaxed variant $g_\phi(x, t)$, where the depth index is supplied as an input, is considered separately:

- **Fixed non-trainable depth encoding:** adds $\mathcal{O}(1)$ trainable parameters; does not violate the constraint as literally worded.
- **Learned depth embedding:** adds $\mathcal{O}(D)$ trainable parameters; violates the constraint.

2.4 Projection

An **input-dependent projection** is a map $\mathcal{P}: \mathcal{X} \rightarrow \mathcal{P}(\mathcal{V})$ applied as

$$\begin{aligned} & \mathcal{P} \\ & s_{t+1} = \mathcal{P}(x)F(s_t) \\ & \quad \text{or} \quad \\ & s_{t+1} = F(\mathcal{P}(x)s_t). \\ & \mathcal{P} \end{aligned}$$

2.5 Input-Path Interference: Three Distinct Notions

The word “interference” is ambiguous. Three notions must be separated.

(A) Cross-input forward-pass contamination.

If $\mathcal{F}$ is frozen and the gate is stateless and a function of x alone, then the trajectory of x_1 is computationally independent of the presence of x_2 :

$\mathcal{P}$

$\tilde{s}_t^{(M)}(x_1 \mid x_2 \text{ processed elsewhere}) = \tilde{s}_t^{(M)}(x_1)$.

]

This notion is **structurally zero** in this setting. It becomes live only if the mechanism contains stateful components (running statistics, caches, etc.).

(B) Forward depth-aliasing.

A depth-agnostic gate applies the same modulation at every depth. If an oracle with access to depth $\backslash(t\backslash)$ would apply a reference modulation $\backslash(g^*(x,t)\backslash)$, the depth-aliasing error is

[

$\mathrm{DA}(g;x)=$

$\sum_{t=0}^{D-1}$

$\left| \right|$

$\Phi(g(x),s_t(x))-\Phi(g^*(x,t),s_t(x))$

$\right|$.

]

This is a representational mismatch within a single forward pass.

(C) Training-time gradient interference.

The gate parameters ϕ are shared across all inputs. Updating ϕ on task (B) may change gate outputs on task- (A) inputs. First-order damage is proportional to

[

$\langle \nabla_{\phi} L_A(\phi), \nabla_{\phi} L_B(\phi) \rangle$.

]

The operationally relevant channels for continual learning are **(B)** and **(C)**. Channel **(A)** is dead under a frozen, stateless, input-only mechanism.

2.6 Computation-Isolability

A mechanism is **exactly computation-isolating** for a task family $\{(S_i, y_i)\}$ if:

- task outputs are produced exactly: $\tilde{s}_D(x; \phi) = y_i(x)$ for all $x \in S_i$;
- forward depth-aliasing error is zero on the specified reference trajectories;
- first-order cross-task gradient interference is zero:

$\langle \nabla_{\phi} L_i, \nabla_{\phi} L_j \rangle = 0$ for all $i \neq j$.

It **bounds interference** if all relevant errors are at most ϵ , with ϵ controlled by explicit quantities.

2.7 Elimination, Bounding, Controllability

- **Elimination** means exact task outputs and exact zero interference.
- **Bounding** means a finite worst-case deviation ϵ , controlled by Lipschitz constants, depth, gate error, and capacity.
- **Controllability** is formalized as **reachability**: a target function $y_i: S_i \rightarrow \mathbb{V}$ is reachable by the gate class if there exists ϕ such that $\tilde{s}_D(x; \phi) = y_i(x)$ for all $x \in S_i$; the mechanism is controllable over a finite task sequence if every finite set of reachable tasks with compatible overlaps is simultaneously realizable by a single ϕ .

This is the natural control-theoretic reading: because the gate is input-only, the control $g(x)$ is fixed for the entire depth trajectory, rather than an independently chosen per-step control u_t .

3. Theorem-Shaped Hypotheses

T1. Exact Preservation Forces the Gate to Be Inert on Active Coordinates

Assumptions. F is defined on an open set $U \subseteq \mathbb{R}^d$. The mechanism is a depth-agnostic output gate:

$\forall$

$$s_{t+1} = g(x) \odot F(s_t).$$

\\

Exact preservation of the unperturbed trajectory is required on an input set (S) :

\\

$$\tilde{s}_t(x)=F^t(x)\quad\text{forall } x\in S,\text{forall } t.$$

\\

Claim. For every $(x\in S)$ and every coordinate (i) ,

\\

$$(g_i(x)-1),F_i(s_t(x))=0.$$

\\

If coordinate (i) is active at any depth along the trajectory, then $(g_i(x)=1)$. If (F) is componentwise nonzero along all trajectories in (S) , then $(g(x)=\mathbf{1})$ on (S) .

Delta-gate variant. For residual delta gating,

\\

$$s_{t+1}=s_t+g(x)\odot F(s_t),$$

\\

exact preservation forces

$\setminus$

$g_i(x)\setminus, F_i(s_t(x))=0.$

$\setminus$

Thus the gate is **zero-forced** on active coordinates, not identity-forced. This is the placement likely relevant to BDH’s residual structure.

Verdict. True as an exact theorem. Near-tautological under global preservation; its real content is *locative*: it identifies precisely where gate freedom survives.

Classification. Direct consequence.

T2. Asymmetric Preservation: The Continual-Learning-Relevant Setting

Assumptions. Old-task inputs $\setminus(S_{\text{old}}\setminus)$, new-task inputs $\setminus(S_{\text{new}}\setminus)$. Exact preservation is required only on $\setminus(S_{\text{old}}\setminus)$. No preservation constraint is imposed on $\setminus(S_{\text{new}}\setminus)$.

Claim. On (S_{old}) , the gate is pinned to identity (output gating) or zero (delta gating) on every coordinate active along the old trajectory. On never-active coordinates, and on all of (S_{new}) , the gate is unconstrained by preservation.

Verdict. True as an exact theorem. This is the correct formalization of the continual-learning constraint. The identity/zero obstruction does **not** by itself rule out nontrivial isolation when old and new task inputs are separated and the gate can interpolate.

Classification. Direct consequence; the asymmetric framing is the honest CL formalization.

T3. Identifiability: Same Input, Different Target Is Impossible

Assumptions. The mechanism is a deterministic function of the input:

$$\tilde{s}_D(x; \phi) = \mathcal{M}_\phi(x).$$

Two tasks share an input $(x \in S_A \cap S_B)$ and demand different outputs: $(y_A(x) \neq y_B(x))$.

Claim. No single (ϕ) can satisfy both tasks.

Proof. A function returns one output for one input; two different required outputs are incompatible. ■

Verdict. True as an exact theorem. It is the sharpest structural limit of an input-only mechanism, independent of depth, capacity, or nonlinearity.

Practical nuance. In language continual learning, the force of this theorem depends on whether task identity is encoded in the input context. Exact input collisions and distribution-level overlap should not be conflated.

Classification. Direct consequence; essentially an identifiability condition.

T4. Depth-Agnostic Gate Values: Representational Limit, Not Behavioral Impossibility

Assumptions. $\Phi(g, s)$ is injective in the gate argument (g) along the trajectory. An oracle demands a depth-varying gate-value sequence $(g^*(x, 0), \dots, g^*(x, D-1))$, nonconstant in (t) .

Claim. A depth-agnostic gate $(g(x))$ cannot reproduce the exact gate-value sequence $(g^*(x, t))$.

Proof. If it did, injectivity would force $(g(x)=g^*(x,t))$ for every (t) , contradicting nonconstancy. ■

Important correction. This is a statement about **gate values**, not about **reachable input-output behavior**. Because the state changes across depth, a constant gate can produce depth-varying effective behavior.

Counterexample. Let $(F(s)=s)$, $(D=2)$. Output gating gives

$$s_2 = g(x) \odot g(x) \odot x.$$

A constant gate $((\alpha, \beta))$ produces $((\alpha^2 x_1, \beta^2 x_2))$. A depth-varying sequence $(g^*_0=(c, 1), g^*_1=(1, c))$ produces $((cx_1, cx_2))$, which the constant gate $((\sqrt{c}, \sqrt{c}))$ reproduces.

Verdict. The representational lemma is true. The behavioral impossibility is **not established**; it was withdrawn in final adjudication. The reachability question is open.

Classification. Direct consequence for the representational part; the broad behavioral claim is **false as stated**.

T5. Projection Mechanisms: Necessary Rank Condition

Assumptions. Output projection:

$$\begin{aligned} & \\ s_{t+1} &= P(x)F(s_t). \\ & \end{aligned}$$

Exact preservation of the unperturbed trajectory is required for input x .

Claim. Let

$$\begin{aligned} & \\ m(x) &= \dim \operatorname{span}\{F(s_0(x)), \dots, F(s_{D-1}(x))\}. \\ & \end{aligned}$$

Then

$$\begin{aligned} & \\ \operatorname{rank} P(x) &\geq m(x). \\ & \end{aligned}$$

Proof. Exact preservation forces $(P(x)F(s_t(x))=F(s_t(x)))$ for all (t) , so all $(F(s_t(x)))$ lie in $(\text{Im}\{P(x)\})$. ■

Input-projection caveat. For input projection $(s_{t+1}=F(P(x)s_t))$, exact preservation means $(F(P(x)s_t)=F(s_t))$. This implies $(P(x)s_t=s_t)$ only if (F) is injective along the trajectory. The theorem as stated applies to output projection; the input-projection variant requires the injectivity assumption.

Withdrawn corollary. The earlier claim $(\sum_i m_i \leq d)$ for multiple tasks is **withdrawn**. Image-subspace orthogonality is a forward-pass geometric property; cross-task gradient interference is a property of the parameterization of $(P(x))$. These are different mathematical objects, and the capacity bound does not follow without a special parameterization assumption.

Classification. Direct consequence (linear algebra); no isolation capacity bound follows.

T6. Gradient Interference and the Parameter Budget

Assumptions. The gate/projection mechanism has (p) trainable parameters. For task (i) , the row space of the collection of output Jacobians is $(R_i \subseteqq \mathbb{R}^p)$. Zero first-order interference is required for **all smooth losses**.

Claim. The row spaces must be mutually orthogonal, and

$\sum_i \dim R_i \leq p$.

Proof. For arbitrary losses, zero inner products across all linear functionals force row-space orthogonality; orthogonal subspaces of $\mathbb{R}^p$ have total dimension at most p . ■

Caveats.

- For fixed specific losses, zero interference can hold accidentally even when row spaces overlap.
- If $p \gg d$, the bound is likely non-binding for practical LLM-scale gates.
- This is standard parameter-isolation accounting, not a result specific to shared sequential composition.

Classification. Direct consequence; not a decisive obstruction unless Jacobian ranks are measured.

T7. Error Amplification: Bounding Is Conditional on Dynamics

Assumptions. The gated transition is Lipschitz in state with constant L . The per-depth gate error against a reference is at most δ .

Claim.

$$\begin{aligned} & \sum_{k=0}^{D-1} L^k \\ & \leq \begin{cases} \frac{1-L^D}{1-L}, & L < 1, \\ D, & L = 1, \\ \frac{L^D-1}{L-1}, & L > 1. \end{cases} \end{aligned}$$

Precise interpretation.

- For **fixed finite depth** (D) , a finite bound exists for every finite (L) .
- A bound that is **uniform in depth** exists iff $(L < 1)$.
- The claim “trained LLM-scale maps have $(L > 1)$ ” is an **empirical conjecture**, not a theorem, and must be measured for the actual BDH block.

Classification. Direct consequence; quantifier-corrected.

T8. Finite-Task Reachability: Conditional Construction, Otherwise Underdetermined

Assumptions. The task family is finite: $\{(x_i, y_i)\}, i=1, \dots, N$. The gate family can interpolate finite sets. Each target is reachable:

$$\bigcup_{i=1}^N \{x_i\} \cap \mathcal{R}_D(x) = \{\tilde{s}_D(x; \phi) : \phi \in \Phi\}.$$

Overlapping task inputs have agreeing targets.

Claim. There exists $\phi \in \Phi$ realizing all finite task targets exactly.

Proof sketch. Reachability gives a finite set of constraints; interpolation capacity fits them. ■

Underdetermination examples.

- If $F \equiv 0$, the reachable set is $\{0\}$ for every x , regardless of gating.
- If the same input has different targets, T3 applies.

- If $(F(s)=s)$ and $(D=2)$ with multiplicative output gating, reachable outputs are of the form $(g(x)\odot g(x)\odot x)$, which excludes many targets.

Verdict. Conditionally true; otherwise underdetermined. The reachability assumption is the nontrivial crux and remains a **conjecture** for a concrete BDH map.

Classification. Plausible adaptation of universal interpolation; reachability is unproved.

4. Summary of Verdicts

#	Claim	Status
---	-----	-----
T1	Exact preservation forces identity/zero on active coordinates	True; exact theorem
T2	Asymmetric preservation constrains only old-task active coordinates	True; CL-relevant
T3	Same input, different target impossible	True; identifiability
T4	Depth-agnostic gate-value representational limit	Representational true; behavioral impossibility false/underdetermined
T5	Projection rank necessary condition	True; capacity corollary withdrawn

| T6 | Gradient interference parameter budget | True under worst-case generality; often non-binding |

| T7 | Error amplification bound | True; finite-depth vs uniform-depth corrected |

| T8 | Finite-task reachability | Conditional; reachability is conjecture |

5. Prior Art: Does It Address the Same Problem?

| Prior art | What it does | Same mathematical problem? | Classification |

| --- | --- | --- | --- |

| Gradient projection / orthogonal-gradient methods (GEM, A-GEM, GPM) |
| Project gradients in parameter space during training | Partly: cross-task gradient interference; not forward depth-aliasing on a frozen path |
Known, partially relevant |

| Parameter isolation / masks (PackNet, Supermasks, Piggyback) |
Allocate disjoint parameter subsets to tasks | No: creates per-task parameters/masks, violating the input-only gate constraint | Known, related but different |

| Progressive networks / PathNet | Add task-specific columns or paths |
No: per-task architectural growth | Known, different problem |

| Hypernetworks / fast weight programmers | Generate weights or modulations for a main network | Partly: can generate input-dependent modulations; typically per-layer outputs | Known, partially relevant |

| FiLM / feature-wise linear modulation | Input-dependent affine modulation of hidden features | Partly: same gating algebra, but usually per-layer parameters | Known, directly relevant |

| LSTM / GRU | Shared-parameter state-dependent gates in a recurrent path | Closest: same composition; but gates see the current state, not only the input | Known, nearest prior art |

| Mixture of experts / conditional computation | Input-dependent routing to expert parameter sets | No: per-layer routers and distinct experts | Known, structurally similar |

| Reservoir computing / echo state networks | Fixed random recurrent core plus trained readout | Partly: frozen core; not input-dependent forward gating for task isolation | Known, relevant background |

| Weight-tied transformers (Universal Transformer, ALBERT) | Reuse a block across depth | Partly: weight-tied composition; no task-isolation gating | Known, background context |

| Nonlinear RNN reachability / observability | Characterizes what a recurrent map can reach | Yes for the reachability question | Known, should be applied to T8 |

Conclusion. No surveyed prior art solves the exact problem: isolation of computations within a single frozen weight-tied composition using only an input-dependent gate, with no per-depth trainable parameters and no retraining of the shared path.

6. Novelty Classification

| Claim | Classification |

|---|---|

| Formalization of two live interference channels (depth-aliasing and cross-task gradient interference) in a frozen weight-tied path |

Apparently novel (packaging only) |

| T1-T3, T5-T7 | **Direct consequences** (textbook ingredients) |

| T4 representational lemma | Direct consequence; behavioral version false as stated |

| Necessity of state/depth/context information for effective gating |

Plausible adaptation (consistent with LSTM/FiLM practice) |

| T8 reachability for concrete BDH $\setminus(F)$ | **Conjecture** |

| Expansiveness of trained BDH blocks | **Empirical conjecture** |

| Any observed effect in c1-h1-report | Inadmissible as theorem evidence; admissible only as a test of the conjectures above |

Contribution posture: **a new formalization and a clean map of elementary obstructions, not new mathematics.**

7. Obstruction Characterization

The strict proposed mechanism is:

- input-dependent gate/projection,
- no per-depth/per-level trainable parameters,
- frozen shared path,

- no retraining of $\mathcal{F}$.

Under the **global-preservation reading** — the mechanism must preserve $\mathcal{F}$'s behavior on **all** inputs — the gate is forced to be inert everywhere, so exact elimination is **trivially impossible**.

Under the **asymmetric continual-learning reading** — exact preservation on old-task inputs, free behavior on new-task inputs — no general impossibility or possibility theorem has been established. What is established is a conjunction of four necessary conditions:

- **Identifiability failure (T3)**. Conflicting targets on identical inputs cannot be served by any input-only mechanism. In language modeling, the force of this depends on how much task identity is contained in the input context.
- **Preservation-geometry confinement (T1/T2)**. The gate is pinned to identity (or zero under delta gating) on every active old-task coordinate. Freedom survives only on never-active coordinates and on new-task inputs.
- **Gradient coupling of shared ϕ (T6)**. New-task updates generically move g_ϕ away from the preserved behavior on old inputs. Worst-case immunity costs a rank budget that may be non-binding when $p \gg d$.
- **Depth amplification (T7)**. If the gated dynamics are expansive, small per-depth gate errors grow geometrically in depth. Bounding then requires contractivity of the composed system — a property of $\mathcal{F}$, not a capability of the gate alone.

Three **scope conditionals** attach to every result:

- **Statelessness.** Any stateful component inside the gate revives cross-input forward contamination.
- **ϕ -conditioning fork.** A gate $g_\phi(x,t)$ with a fixed non-trainable depth encoding adds $O(1)$ parameters and does not violate the constraint as literally worded. A learned depth embedding adds $O(D)$ parameters and does violate it. The depth-agnostic obstruction dissolves on the permissive side of this fork, converting the problem into pure reachability (T8).
- **Frozen path.** BDH's native Hebbian-style plasticity must be explicitly disabled for the frozen-path premise to hold.

8. Direct Answer to the Research Question

> Can input-path interference arising from shared sequential composition be eliminated, bounded, or made controllable by an input-dependent gate/projection mechanism, without per-depth/per-level trainable parameters and without retraining the frozen shared path?

| Reading | Verdict | Epistemic status |

| --- | --- | --- |

| **Elimination**, global preservation on all inputs | Impossible — the only admissible gate is identity/zero on active coordinates | Proved; degenerate |

| **Elimination**, asymmetric CL (preserve S_{old}), free on S_{new} | **Open**. Necessary conditions established; no general impossibility or possibility theorem | Partially characterized |

| **Bounding** | Finite-depth bounds exist for any finite $\mathcal{L}$; uniform-in-depth bound iff $|\mathcal{L}| < \infty$ | Proved, conditional |

| **Controllability** | Not yet well-posed: requires formal reachability analysis on the concrete frozen BDH map with the exact gate class | Undefined pending instantiation |

The failure condition contemplated by the brief is met **only in the degenerate global-preservation reading**. In the continual-learning-relevant reading, the correct outcome is: *not proven impossible, not proven possible; reduced to identifiable necessary conditions and one decisive open conjecture — reachability*.

9. Mandatory Follow-Ups Before Empirical Reconciliation

- **Instantiate $\mathcal{F}$ from paper .tex.** Fix the literal BDH update, including residual form, LayerNorm placement, and any Hebbian-style multiplicative terms.
- **Reinstantiate every theorem for the correct gate placement.** For a residual block, the natural gate is $s_{t+1} = s_t + g(x) \odot F(s_t)$, which changes the identity obstruction to a zero obstruction.
- **Rule on the $\mathcal{L}$ -conditioning fork.** State in writing whether fixed non-learned depth encoding is permitted.
- **Retrieve the internal reports.** Map 2026-08-25_bdh-continual-learning-plan_v2.md and 2026-08-25_cl-h1-report.md onto exactly one formal variant: depth-agnostic, $\mathcal{L}$ -conditioned, or state-conditioned.
- **Define controllability formally** as reachability of finite task targets by the gate class on the instantiated BDH map.

- **Measure** (a) the Lipschitz constant of the gated BDH block, and (b) cross-task Jacobian row-space overlap in ϕ -space. These are the only claimed barriers that can be tested empirically.

Bottom line. A strict input-only gate/projection on a frozen shared sequential composition does not provide a proven general solution. The obstruction is not a single theorem but a map of four necessary failure conditions and one open reachability problem. The honest conclusion for publication is: *global preservation makes nontrivial gating impossible; asymmetric continual learning with a frozen shared path remains open pending concrete reachability analysis of the actual BDH update.*

[MODERATOR Round 5]

Final Moderator Synthesis — Debate Closure: Input-Path Isolation in BDH via Gating/ Projection

Role: ox-alpha, moderator. This document consolidates five rounds of adversarial deliberation (STRATEGIST, CRITIC, OPTIMIZER), records the final consensus state, adjudicates all surviving disputes, and delivers the scoped answer to the research question together with the precise characterization of the obstruction.

0. Process Record and Evidence-Base Status

The debate executed a full correction loop:

- **Rounds 1–2** built a unified formal skeleton and initially headlined a strong claim: *"exact elimination of input-path interference is impossible in general"* via three obstructions (identity-forcing, diagonalization, rank budget).
- **Round 3–4 CRITIC reviews** landed decisive defeaters: (i) the headline rested on a *degenerate* global-preservation reading; (ii) the diagonalization argument conflated gate **values** with reachable **behaviors** (counterexample: $F(s)=s$, $D=2$); (iii) the projection capacity corollary $\sum_i m_i \leq d$ committed a category error (forward-pass image-subspace geometry $\neq$ parameter-space gradient geometry); (iv) the amplification "iff $L < 1$ " misquantified finite- D vs. uniform-in- D bounds; (v) controllability was undefined; (vi) the BDH-relevant **delta-gate placement** converts identity-forcing into **zero-forcing**, changing the obstruction's form.
- **Round 5:** all three roles converged on the same scoped verdict set (STRATEGIST V1–V8, OPTIMIZER T1–T8 — substantively identical), with the CRITIC accepting the scoping while flagging residual gaps.

Adjudication principle throughout: a claim enters the record only at the strength of its weakest surviving proof.

Evidence-base limitations (recorded for honesty): the internal project files (2026-08-* reports, paper-parity-plan, bdh-continual-learning-plan_v2.md, cl-h1-report.md) and the literal equations from paper.tex were **never retrieved in-debate** despite repeated attempts. All results below are therefore proved for the abstraction $s_{t+1} = \Phi(g(x), s_t)$ over a frozen map F (with the residual/delta variant treated explicitly). Instantiation against the actual BDH update remains a mandatory follow-up, not an optional refinement. No experimental observation from any report was admitted as theorem evidence at any point.

1. Consensus Evaluation

Full consensus (all three roles, post-revision)

| Item | Status |

| --- | --- |

| Formal skeleton: shared sequential composition, frozen path, gate/projection placements (output / input / delta), elimination-vs-bounding distinction | Agreed verbatim |

| **Interference taxonomy (three channels):** (A) cross-input forward-pass contamination — **structurally zero** for stateless mechanisms on a frozen path; (B) forward **depth-aliasing** within a pass — live; (C) training-time **cross-task gradient interference** through shared ϕ — live | Agreed; channel (A)'s revival conditions (stateful components: running statistics, caches) noted |

| Verdicts T1–T3, T5–T8 (see §3) | Agreed |

| Prior-art triage: LSTM/GRU nearest neighbor (state-conditioned gates); FiLM same gating algebra but per-layer parameters; fast-weight programmers/hypernetworks adjacent; reservoir computing background; GEM/A-GEM/GPM parameter-space complementary; PackNet/supermasks/progressive nets constraint-violating; MoE different problem; Universal Transformer/ALBERT backdrop. **No surveyed work solves the exact problem.** | Agreed |

| Novelty discipline: packaging novel, mathematics derivative | Agreed |

Adjudicated disputes (all resolved)

- **"Impossibility in general" vs. scoped openness** → the **scoped formulation prevails**. The unscoped headline holds only under global preservation on *all* inputs — a reading under which the gate must be inert everywhere and hence cannot adapt anywhere, dissolving the continual-learning question itself. STRATEGIST's own §7 concession is entered as binding.
- **Diagonalization obstruction** → downgraded from behavioral impossibility to a true-but-representational lemma about gate values; superseded by the open reachability question (T8).
- **Projection capacity corollary** → withdrawn; survives only under a special independent-subspace parameterization that violates the spirit of the constraints anyway.
- **Delta-gate placement** → incorporated into T1/T2: under $s_{t+1} = s_t + g(x) \odot F(s_t)$, preservation forces $g_i(x) = 0$ (not 1) on active old-task coordinates. This is the placement likely correct for BDH's residual structure.

Residual dissent (non-blocking)

The CRITIC maintains two emphasis-level caveats, both accepted into the record: (a) the practical force of T3 depends on measure-theoretic overlap of task supports — exact collision, distributional overlap, and representation-space near-overlap must not be conflated; (b) T6 is worst-case accounting that is likely non-binding when $p \gg d$ and must not be presented as a decisive barrier until Jacobian ranks are measured.

2. Consolidated Definitions (Final)

- **Shared sequential composition:** $s_0 = x$, $s_{t+1} = F(s_t)$, D steps; residual variant $s_{t+1} = s_t + F(s_t)$.
- **Frozen shared path:** $\partial F / \partial \theta = 0$ during adaptation; only ϕ (gate parameters) trainable.
- **Gate:** input-dependent $g_\phi(x)$ applied identically at every depth; placements: output $g \odot F(s)$, input $F(g \odot s)$, $\Delta s + g \odot F(s)$. No per-depth parameter vectors.
- **Projection:** input-dependent orthogonal $P(x)$; output $P(x)F(s)$ or input $F(P(x)s)$.
- **Computation-isolability (exact):** exact task outputs + zero depth-aliasing error + zero first-order cross-task gradient interference.
- **Elimination vs. bounding:** exact equality/zero interference vs. finite worst-case deviation controlled by explicit quantities (L_Φ , D , Δ , p).
- **Controllability (partial formalization):** reachability of finite task targets by the gate class on the frozen map — with the crucial caveat that an input-only control $g(x)$ is fixed for the entire pass (open-loop modulation), not an independent per-step control u_t . A fuller formalization specifying state space, admissible gate range, transition rule, target class, and joint-realizability requirements remains outstanding.

3. Final Adjudicated Verdict Table

#	Claim	Status	Notes
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| **T1** | Exact preservation forces the gate **inert** on active coordinates: $g_i = \mathbf{1}$ (output gating) or $g_i = 0$ (**delta gating**) wherever $F_i(s_t(x)) \neq 0$ along the preserved trajectory | **TRUE — exact theorem; direct consequence** | Near-tautological globally; real content is *locative*: identifies precisely where gate freedom lives (never-active coordinates; non- S inputs) |

| **T2** | Under **asymmetric preservation** (exact on S_{old} , free on S_{new}), the confinement restricts to old-task active coordinates | **TRUE — exact theorem; the honest CL formalization** | Does not by itself preclude nontrivial isolation when supports are separable and the gate can interpolate |

| **T3** | Deterministic input-only mechanism cannot serve two tasks demanding different outputs at a shared input | **TRUE — exact theorem (identifiability)** | Sharpest limit of "input-dependent only"; practical force depends on context-sufficiency of inputs |

| **T4** | Depth-agnostic gate confined to diagonal of modulation space | Representational lemma **true**; behavioral impossibility **not established** (downgraded after counterexample) | Superseded by open reachability question |

| **T5** | Output projection preserving trajectory requires $\text{rank } P(x) \geq m(x)$ | **TRUE — exact linear algebra** | Capacity corollary withdrawn; input-projection variant requires injectivity of F along the trajectory |

| **T6** | Zero first-order interference *for all smooth losses* $\Leftrightarrow$ mutually orthogonal task-Jacobian row spaces $\Rightarrow \sum_i \dim R_i \leq p$ | **TRUE — exact under stated generality** | Worst-case guarantee; likely non-binding for $p \gg d$; standard accounting, not composition-specific |

| **T7** | $e_D \leq \delta \sum_{k < D} L_{\Phi}^k$; finite- D bound exists for every finite L_{Φ} ; **uniform-in-depth** bound iff $L_{\Phi} < 1$ | **TRUE — exact, quantifiers corrected** | "Trained BDH blocks are expansive" reclassified as empirical conjecture requiring measurement |

| **T8** | Finite task sets realizable iff targets reachable by gate class + interpolation capacity + overlap agreement | **CONDITIONAL; crux is a**

conjecture | Counterexamples ($F \equiv 0$; sign/root obstructions) show reachability fails generically without hypotheses on F |

4. Final Answer to the Research Question

> Can input-path interference arising from shared sequential composition be eliminated, bounded, or made controllable by an input-dependent gate/projection mechanism, without per-depth/per-level trainable parameters and without retraining the frozen shared path?

| Reading | Verdict | Epistemic status |

| --- | --- | --- |

| **Elimination**, global-preservation (mechanism preserves F 's behavior on *all* inputs) | Trivially impossible — the only admissible gate is inert everywhere | Proved; degenerate |

| **Elimination**, asymmetric CL (exact preservation on S_{old} , free on S_{new}) | **OPEN**. Necessary conditions established (T1–T3, T5–T7); no general possibility or impossibility theorem | Partially characterized |

| **Bounding** | Finite-depth bounds exist for any finite L_Φ ; **uniform-in-depth** bounding iff gated dynamics contractive. Bounding is a property of the composed system, not achievable by gate design alone under expansion | Proved, conditional |

| **Controllability** | **Not yet well-posed**: requires formal reachability analysis on the concrete frozen BDH map with the exact gate class | Undefined pending instantiation |

The brief's failure condition is met **only in the degenerate reading**. In the CL-relevant reading the correct outcome is: *not proven impossible, not proven possible; reduced to identifiable necessary conditions plus one decisive open conjecture (reachability)*. Reporting anything stronger would violate the novelty discipline.

5. Obstruction Characterization (Deliverable Under the Failure Clause)

There is no single obstruction theorem in the meaningful setting. The failure surface is a **conjunction of four necessary conditions**, each independently capable of killing the program, none individually fatal:

- **Identifiability failure (T3)** — conflicting targets on overlapping inputs are unservable by any input-only mechanism. In language-modeling CL, escape requires context-sufficiency of the input or task conditioning.
- **Preservation-geometry confinement (T1/T2)** — the gate is pinned to identity (output gating) or zero (delta gating) on every (old input, active coordinate) pair. Freedom survives on never-active coordinates — potentially substantial under BDH's ReLU sparsity — and on new-task inputs.
- **Gradient coupling of shared ϕ (T6)** — new-task updates generically move g_ϕ off the preserved behavior on old inputs; worst-case immunity costs a rank budget that is restrictive ($p \leq d$) or vacuous ($p \gg d$).
- **Depth amplification (T7)** — under expansive gated dynamics, per-depth modulation error grows geometrically in D ; bounding then

demands architectural contractivity, an assumption on F , not a capability of the gate.

Plus **three scope conditionals** silently underlying every result: (a) **statelessness** of the mechanism (any running-statistics component revives channel A); (b) the **$\mathcal{F}$ -conditioning fork** — a shared-parameter gate $g_{\phi}(x,t)$ fed by a *fixed, non-trainable* depth encoding adds $O(1)$ parameters and does not violate the constraint as literally worded, and it dissolves the depth-agnostic restrictions entirely, converting the problem into pure reachability (T8); a *learned* depth embedding adds $O(D)$ parameters and violates the constraint outright. Every impossibility-flavored statement applies only on the strict side of this fork; (c) BDH's native Hebbian-style plasticity must be explicitly disabled, else the path is not frozen.

6. Novelty Ledger (Final)

| Claim | Classification |

| --- | --- |

| Problem formulation: isolation inside a frozen weight-tied composition via input-only gate, CL-framed, three-channel interference taxonomy |
Apparently novel (packaging only) |

| T1–T3, T5–T7 | **Direct consequences** (textbook ingredients) |

| T4 representational lemma | Direct consequence; behavioral version false as stated |

| Necessity of state/depth/context information for effective gating |
Plausible adaptation (consistent with LSTM/FiLM practice; proven only under injectivity assumptions) |

| Reachability of the concrete BDH map under the gate class (T8 crux) |
Conjecture |

| Expansiveness of trained BDH blocks | **Empirical conjecture** |

| Any observed effect in c1-h1-report | Inadmissible as theorem evidence; admissible only as a test of the above conjectures |

Write-up posture: claim a new formalization and a clean map of elementary obstructions — not new mathematics.

7. Constructive Directions Surfaced (all $\leq$ conjecture)

- **C1 — Orthogonal-gradient protection of ϕ :** once the protected object is recognized as the *gate parameters*, GEM/GPM-style projection applies compatibly with all constraints: project new-task gradients onto the orthogonal complement of the old-task gradient subspace in ϕ -space, holding $g_\phi|_{S_{\text{old}}}$ $\approx$ inert to first order. Plausible adaptation; open statistical question about generalization beyond the old-task sample.
- **C2 — Fixed-encoding t -conditioning** ($O(1)$ parameters): dissolves depth-agnostic restrictions; legality contingent on the scope ruling.
- **C3 — Delta-gate placement:** gate $g(x) \odot \Delta(s_t)$ inside $s_{t+1} = s_t + \cdot$; never the residual stream itself.

- **C4 — Sparse-coordinate freedom:** exploit never-active coordinates of old-task trajectories (plausible under ReLU sparsity) as legal modification space even on preserved inputs.

8. Mandatory Follow-Ups Before Any Empirical Reconciliation

- **Instantiate $\mathcal{F}$ from paper .tex:** fix the literal update (residual form, LayerNorm placement, Hebbian/multiplicative terms), choose gate placement (C3), verify plasticity disabling and statelessness.
- **Re-prove every theorem for the actual placement** — especially the zero-forcing variant under delta gating.
- **Rule in writing on the $\mathcal{S}$ -conditioning fork** (fixed non-learned encoding permitted?).
- **Retrieve and map the internal reports** (plan_v2, cl-h1-report) onto exactly one formal variant: depth-agnostic / $\mathcal{S}$ -conditioned / state-conditioned.
- **Define controllability rigorously** as finite-target reachability over the instantiated map and exact gate class, drawing on nonlinear discrete-time reachability theory.
- **Measurement program:** empirical Lipschitz/expansiveness of the gated BDH block (tests T7's conjecture); empirical cross-task Jacobian row-space overlap in ϕ -space (tests whether T6's worst case binds).

9. Discipline Audit

- No experimental observation cited as evidence for any theorem at any point. ✓
- All overclaims identified were withdrawn on the record (capacity corollary; behavioral diagonalization; finite- D "iff"; global-preservation headline) or rescoped. ✓
- Novelty labels consistent post-adjudication: packaging novel, mathematics derivative. ✓

10. Closing Statement

Consensus is complete on the formal framework, the three-channel interference taxonomy, the exact necessary conditions T1–T3/T5–T7 as scoped, the conditional bounding result, the prior-art triage, and the open status of the asymmetric continual-learning question. The single substantive disagreement — whether a general impossibility had been established — was resolved **against** the stronger claim on the strength-of-weakest-proof principle.

The investigation exits the debate not with an impossibility theorem but with something more useful: a precise map of where gate freedom lives (never-active sparse coordinates, new-task inputs), four independent ways the program can fail, one pivotal scope decision (t -conditioning), and one decisive open conjecture — **reachability of the instantiated BDH map under the allowed gate class** — whose resolution will determine whether the proposed mechanism is possible, impossible, or merely unproven.

Ergebnis

[STRATEGIST Round 1]

I'll start by reading the research paper and all the relevant reports to understand the BDH architecture and the current state of work on continual learning. Let me begin systematically.

[SEARCH: RAG file 2509.26507v1.pdf]

[SEARCH: RAG file paper.tex]

[SEARCH: RAG file 2026-08-18_progress-report.md]

[SEARCH: RAG file 2026-08-23_progress-report.md]

[SEARCH: RAG file 2026-08-24_progress-report.md]

[SEARCH: RAG file 2026-08-22_paper-parity-plan.md]

[SEARCH: RAG file 2026-08-25_bdh-continual-learning-plan_v2.md]

[SEARCH: RAG file 2026-08-25_cl-h1-report.md]

Web Research

• The Dragon Hatchling: The Missing Link between the Transformer and...

RAIN. arXiv:2509.26507v1 [cs.NE] 30 Sep 2025. Adrian Kosowski†. Przemysław Uznański Jan Chorowski‡ Michał Bartoszkiewicz‡.

Source: <https://www.rivista.ai/wp-content/uploads/2025/11/2509.26507v1.pdf>

- **[2509.26507] The Dragon Hatchling: The Missing Link between the...**

View PDF HTML (experimental). Abstract: The relationship between computing systems and the brain has served as motivation for pioneering theoreticians since John von Neumann and Alan Turing. (or [arXiv:2509.26507v1](https://arxiv.org/abs/2509.26507v1) [cs.NE] for this version).

Source: <https://arxiv.org/abs/2509.26507>

- **Dragon Hatchling: Linking Transformers & Brain Models**

The Dragon Hatchling: The Missing Link between the Transformer and Models of the Brain. Published 30 Sep 2025 in cs.NE, cs.AI, cs.LG, and stat.ML | (2509.26507v1).

Source: <https://www.emergentmind.com/papers/2509.26507>

- **Paper page - The Dragon Hatchling: The Missing Link between the...**

Get this paper in your agent: hf papers read 2509.26507. Don't have the latest CLI? Cite arxiv.org/abs/2509.26507 in a dataset README.md to link it from this page.

Source: <https://huggingface.co/papers/2509.26507>

- **The Dragon Hatchling: The Missing Link between... | Papers with Code**

Paper · [arXiv:2509.26507](https://arxiv.org/abs/2509.26507) · 30 September 2025. The Dragon Hatchling: The Missing Link between the Transformer and Models of the Brain.

Adrian Kosowski, Przemysław Uznański, Jan Chorowski, Zuzanna Stamirowska, Michał Bartoszkiewicz.PDF. Code.

Source: <https://paperswithcode.co/paper/2509.26507>

- **Gemini RAG - File Search Tool - YouTube**

In this video, I go through the latest release from the Gemini API team, which is the file search tool, or basically a simple RAG system built right into the...

Source: <https://www.youtube.com/watch?v=MUP9ki6Bdtg>

- **GitHub - infiniflow/ragflow: RAGFlow is a leading open-source...**

Go to file. Code. Open more actions menu. Folders and files. RAGFlow is a leading open-source Retrieval-Augmented Generation (RAG) engine that fuses cutting-edge RAG with Agent capabilities to create a superior context layer for LLMs.

Source: <https://github.com/infiniflow/ragflow>

- **Error: "Loading a class or package in a group. \usepackage" with .cls file...**

in the .cls file but it still did not work. However, when I try to compile from a single .tex file. It works well. in a .cls file (let's say it is "Main.cls") and in my .tex file (e.g., "Paper.tex"), I begin it with.

Source: <https://latex.org/forum/viewtopic.php?t=35969>

- **Learn how to use upload images, audio, and other files | FlowiseAI**

This research paper. is one of many that compare RAG with longer context windows. To enable full file uploads, go to Chatflow Configuration, open the File Upload tab, and click the switch

Source: <https://docs.flowiseai.com/using-flowise/uploads>

10. How to Do File Search in Gemini API? | Analytics Vidhya

For multimodal RAG, File Search also supports image files. This means you can upload documents and images into the same File Search Store and ask questions that require both textual and visual context.

Source: <https://www.analyticsvidhya.com/blog/2026/05/gemini-api-file-search/>

11. mini-enterprise-rag-agent/progress.md at main ... - GitHub

May 8, 2026 · Read relevant process skills and confirmed current architecture issue. Created planning files for this multi-step refactor. Added failing tests in tests/test_llm_tool_planning.py. Red run: 8 failed, 1 passed; failures match expected architecture gaps.

Source: <https://github.com/zhengshaoyi1387/mini-enterprise-rag-agent/blob/main/progress.md>

12. RAG/rag_state_of_the_art_2026.md at main · dpsdagain/RAG

May 26, 2026 · This document synthesizes findings from 45+ targeted web searches conducted on May 26, 2026. It represents the current frontier of RAG system design — not beginner tutorials or toy demos. Every claim is grounded in published research, production reports, or framework documentation.

Source: https://github.com/dpsdagain/RAG/blob/main/rag_state_of_the_art_2026.md

13. mindstudio/big-rag • LM Studio Hub

Big RAG Plugin for LM Studio A powerful RAG (Retrieval-Augmented Generation) plugin for LM Studio that can index and search through gigabytes or even terabytes (not tested) of document data.

Source: <https://lmstudio.ai/mindstudio/big-rag>

14. Local RAG: Search Your Documents with a Private AI

Jan 30, 2026 · RAG fixes this. Instead of retraining the model (expensive, slow, overkill), RAG searches your documents for relevant chunks and feeds them to the LLM as context. The model reads your actual text and answers based on it, not from memory. The best part: everything stays on your machine.

Source: <https://insiderllm.com/guides/local-rag-search-documents-private-ai/>

15. RAG Daily Papers - Latest RAG Research from arXiv

Aug 18, 2022 · From Retrieved Context to Runtime Control: Adaptive Compression for Edge-based RAG Retrieval-augmented generation (RAG) improves language-model responses by grounding generation in external passages, which comes with overhead: retrieved context lengthens the prompt, increasing prefill work, KV-cache footprint, memory traffic, latency, and energy.

Source: <https://ragdaily.com/>

16. RAG Status in Project Management Explained (2026)

June 17, 2026 - Reporting dimension-level RAG, covering at a minimum time, cost, scope and risk, gives decision-makers the information they actually need. For guidance on structuring this kind of report effectively, IPM's progress report template provides a practical framework.

Source: <https://instituteprojectmanagement.com/blog/rag-status-in-project-management/>

17. Releases · infiniflow/ragflow

4 days ago - Release nightly created from 2db8eb6 at 2026-08-23 21:19:13+08:00

Source: <https://github.com/infiniflow/ragflow/releases>

18. Project RAG Status Meanings And Best Practice (2026)

July 6, 2026 - PM3 helps organisations standardise RAG reporting across projects, programmes and portfolios, with up to 10 status indicators covering areas such as cost, progress, risks, issues, stakeholders, scope and value.

Source: <https://eleco.com/pm3/knowledge-centre/how-many-rags/>

19. Progress Agentic RAG reviews 2026 | FitGap

April 11, 2026 - Progress Agentic RAG is an LLMOps-oriented capability from Progress that helps teams build retrieval-augmented generation (RAG) applications with agent-style orchestration over enterprise content.

Source: <https://us.fitgap.com/products/050333/progress-agentic-rag>

20. RAG Status for KPIs: The Red-Amber-Green Playbook (2026) | ClearPoint Strategy Blog

1 week ago - KPIs are crucial for understanding your organization's strengths and weaknesses. But let's be honest, staring at reports and

relying on gut feelings isn't effective. You need a clear, objective way to evaluate your progress. A RAG status system provides just that.

Source: <https://www.clearpointstrategy.com/blog/establish-rag-statuses-for-kpis>

21. RAG Status in Project Management Explained (2026)

June 17, 2026 - Reporting dimension-level RAG, covering at a minimum time, cost, scope and risk, gives decision-makers the information they actually need. For guidance on structuring this kind of report effectively, IPM's progress report template provides a practical framework.

Source: <https://instituteprojectmanagement.com/blog/rag-status-in-project-management/>

22. Project RAG Status Meanings And Best Practice (2026)

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Source: <https://us.fitgap.com/products/050333/progress-agentic-rag>

24. RAG Status in Project Management: A Comprehensive Guide

June 19, 2026 - RAG (Red, Amber, Green) Status is a simple visual reporting system used in Project Management to show the overall health of a task, milestone or entire project. Each colour represents a different level of performance or risk, making it easy to understand status at a glance.

Source: <https://www.prince2training.co.uk/blog/rag-status-in-project-management>

25. What Is RAG Status In Project Management? Red-Amber-Green Explained - Designveloper

July 15, 2026 - RAG reporting supports consistent reporting across teams. A PMO or portfolio manager can compare multiple initiatives only when the meaning of red, amber, and green is shared. Eleco's 2026 discussion of project RAG status meanings notes that highlight reports may use RAG ratings for the whole project and for specific aspects such as stakeholder commitment or team performance.

Source: <https://www.designveloper.com/blog/rag-status/>

26. Releases · infiniflow/ragflow

4 days ago - Release nightly created from 2db8eb6 at 2026-08-23 21:19:13+08:00

Source: <https://github.com/infiniflow/ragflow/releases>

27. 20 Advanced RAG Types to Know in 2026

2 weeks ago - This 2026 Systematization of Knowledge (SoK) paper frames Agentic RAG as a fragmented but increasingly important architecture for sequential reasoning, dynamic memory management, and iterative retrieval.

Source: <https://www.turingpost.com/p/ragtypes>

28. All you need to know about RAG (in 2026) - AI with Aish

March 21, 2026 - In 2024, it was enough to shove some PDFs into a vector database, run a cosine similarity search, and call it a day. In 2026, that approach; Naive RAG is seen as a prototype at best and a liability at worst.

Source: <https://aishwaryasrinivasan.substack.com/p/all-you-need-to-know-about-rag-in>

29. David E. Weekly on X: "Indexed .md files > RAG" / X

April 3, 2026 - Indexed .md files > RAG

Source: <https://x.com/dweekly/status/2039873942039715887>

30. How to Build a RAG Pipeline from Scratch in 2026 - kapa.ai - Instant AI answers to technical questions

January 27, 2026 - In this article, we will explore the latest developments that can help you build a robust production-grade RAG pipeline in 2026.

Source: <https://www.kapa.ai/blog/how-to-build-a-rag-pipeline-from-scratch-in-2026>

31. Dynamic Memory Updating in RAG: Lifelong Learning and ...

January 15, 2026 - Zhang et al., we analyze architectures that support continuous memory addition, deletion, consolidation, and re-weighting. These mechanisms transition RAG from static retrieval to incremental learning, mirroring biological memory processes.

Source: <https://www.ijisrt.com/assets/upload/files/IJISRT26JAN155.pdf>

32. **dirty-data/rag-v2**

April 6, 2026 - The goal is to keep one shared RAG implementation while supporting both LM Studio plugin workflows and MCP-hosted tool workflows.

Source: <https://lmstudio.ai/dirty-data/rag-v2>

33. **Releases · infiniflow/ragflow**

4 days ago - Release nightly created from 2db8eb6 at 2026-08-23 21:19:13+08:00

Source: <https://github.com/infiniflow/ragflow/releases>

34. **GitHub - OSU-NLP-Group/HippoRAG: [NeurIPS'24] HippoRAG is a novel RAG framework inspired by human long-term memory that enables LLMs to continuously integrate knowledge across external documents. RAG + Knowledge Graphs + Personalized PageRank. · GitHub**

1 month ago - [NeurIPS'24] HippoRAG is a novel RAG framework inspired by human long-term memory that enables LLMs to continuously integrate knowledge across external documents. RAG + Knowledge Graphs + P...

Source: <https://github.com/osu-nlp-group/hipporag>

35. **Complete RAG Tutorial 2026 (Free Labs) - YouTube**

🔗 RAG Bootcamp: <https://kode.wiki/4j9Bfyw> Learn Retrieval Augmented Generation (RAG) from scratch in this comprehensive crash course! Discover how to extend ... Published: December 24, 2025

Source: <https://www.youtube.com/watch?v=vT-DpLv29Q>

36. Локальная LLM с памятью и RAG: пошаговый гайд для... | AiManual

Пошаговый запуск локальной LLM через Ollama и Open WebUI: сохранение чатов, загрузка документов, настройка RAG и решение частых ошибок.

Source: <https://ai-manual.ru/article/polnyij-gajd-dlya-nachinayuschih-kak-s-nulya-zapustit-lokalnuyu-llm-s-pamyatyu-chatov-i-rag/>

37. Погода в Волгограде в марте 2026 года Волгоградская область...

Подробный прогноз погоды в Волгограде на март 2026 года - информация о дневной и ночной температуре, осадках в Волгограде, Волгоградская область Россия - World-Weather.Ru.

Source: <https://world-weather.ru/pogoda/russia/volgograd/march-2026/>

38. Самые смешные мемы за 16.08.2026

Если вам это не нравится, пожалуйста, покиньте сайт. 18+. 16.08.2026.

Source: <https://www.anekdot.ru/release/mem/day/2026-08-16/>

39. Украина наносит новые удары. Что изменилось на... - YouTube

Что изменилось на фронте? 02.08.2026. Сергей Ауслендер.Иранский замысел и "блокадная" нефть. Военный аналитик Давид Шарп. 19.08.2026.

Source: <https://www.youtube.com/watch?v=iOb6CACxYtg>

40. Нелёгкая жизнь в Мариуполе - Тень Кремля

17.08.2026, 07:01. В Черкассах военный ВСУ открыл огонь по детям из-за разбитого окна. 15.08.2026, 15:25. На Украине «зерновой коридор» окончательно превращается в «зерновой тупик».

Source: <https://shoknews.ru/neljogkaja-zhizn-v-mariupole/>

[CRITIC Round 1]

<CONTROL>

We need to retrieve the key sources. Let me start by searching for the paper and the RAG files.

</CONTROL>

[SEARCH: RAG file 2509.26507v1.pdf]

[SEARCH: RAG file paper.tex]

[SEARCH: RAG file 2026-08-18_progress-report.md]

[SEARCH: RAG file 2026-08-23_progress-report.md]

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[SEARCH: RAG file 2026-08-25_bdh-continual-learning-plan_v2.md]

[SEARCH: RAG file 2026-08-25_cl-h1-report.md]

Web Research

- **The Dragon Hatchling: The Missing Link between the Transformer and Models of the Brain**

September 30, 2025 - arXiv:2509.26507v1 [cs.NE] 30 Sep 2025 · Adrian Kosowski Thanks: Author contributions are listed at the end of the paper. Thanks: Corresponding author.

Source: <https://arxiv.org/html/2509.26507v1>

- **How to ingest PDF files for RAG. To ingest many PDFs into a RAG system... | by LLM | Medium**

December 8, 2025 - How to ingest PDF files for RAG To ingest many PDFs into a RAG system, I build a scalable pipeline: extract text using reliable PDF parsers (PyMuPDF, PDFPlumber, or UNIDOC for enterprise) ...

Source: https://medium.com/@generative_ai/how-to-ingest-pdf-files-for-rag-f7ddc8bdbf26

- **Unlocking PDFs for RAG: How RAG-Anything Handles Complex Documents | by Florian June | AI Exploration Journey | Medium**

June 10, 2026 - A parser can be thought of as a “document disassembler.” It takes inputs like PDFs, images, or Office documents and breaks them down into data chunks suitable for subsequent system processing. In RAG-Anything’s code, a unified interface (`raganything/parser.py:68`) defines key methods: `parse_pdf()`, `parse_image()`, `parse_document()`, and `check_installation()`.

Source: <https://medium.com/ai-exploration-journey/unlocking-pdfs-for-rag-how-rag-anything-handles-complex-documents-f8bbd716c734>

- **(PDF) REFRAG: Rethinking RAG based Decoding**

September 1, 2025 - To this end, we propose REFRAG, an efficient decoding framework that compresses, senses, and expands to improve latency in RAG applications. By exploiting the sparsity structure, we demonstrate a 30.85 the time-to-first-token acceleration (3.75 ...

Source: https://www.researchgate.net/publication/395213753_REFRAG_Rethinking_RAG_based_Decoding

- **[2509.01092] REFRAG: Rethinking RAG based Decoding**

October 12, 2025 - From: Aritra Ghosh [view email] [v1] Mon, 1 Sep 2025 03:31:44 UTC (636 KB) [v2] Sun, 12 Oct 2025 04:46:48 UTC (638 KB) ...

View a PDF of the paper titled REFRAG: Rethinking RAG based Decoding, by Xiaoqiang Lin and 4 other authors

Source: <https://arxiv.org/abs/2509.01092>

- **Gemini RAG - File Search Tool - YouTube**

In this video, I go through the latest release from the Gemini API team, which is the file search tool, or basically a simple RAG system built right into the...

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- **Learn how to use upload images, audio, and other files | FlowiseAI**

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Source: <https://latex.org/forum/viewtopic.php?t=35969>

- **GitHub - Teju63/rag-file · GitHub**

RAG = Retrieval-Augmented Generation — a technique where the AI answers questions using YOUR documents instead of just its training data. What does this app do? You upload a PDF or TXT file (e.g. a research paper, a contract, a book chapter)...

Source: <https://github.com/Teju63/rag-file>

10. How to Do File Search in Gemini API? | Analytics Vidhya

For multimodal RAG, File Search also supports image files. This means you can upload documents and images into the same File Search Store and ask questions that require both textual and visual context.

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Source: <https://us.fitgap.com/products/050333/progress-agentic-rag>

16. Какая осень будет в Тюмени в 2026 году: прогноз погоды на...

Осень 2026 в Тюмени: прогноз погоды по месяцам, когда ждать бабье лето, заморозки и первый снег. Опубликовано:05-07-2026. Обновлено:08-07-2026. 19 минут. Редакция Travelto.

Source: <https://travelto.ru/blog/kakaya-osen-bydet-v-2026-tyumen/>

17. Круиз Греческие острова с 29 августа по 08 сентября 2026 на...

Круиз Греческие острова с 29.08.2026 по 08.09.2026 (11 дней / 10 ночей) по маршруту Анталья - Измир - Кавала (2 дня, ночь на борту) - о.Миконос - о.Санторини - Афины - Бодрум - Анталья. Купите билет на круиз прямо сейчас!

Source: https://astoriagrande.com/destination/kruise_29_08-08_09_2026/

18. Самые смешные мемы за 16.08.2026

Мем, судьяКригс. 23.>. Самый смешной анекдот за 11.08: Дай человеку рыбу и он будет сыт один день. Подари ему удочку и он будет сыт всю жизнь.

Source: <https://www.anekdot.ru/release/mem/day/2026-08-16/>

19. Открываем RAG и интернет для LM Studio / Хабр

Исходные данные : ПК на базе AMD Ryzen™ AI 9 HX 370, ОЗУ 96 Гб из которых половина отдана под видеопамять. LM Studio, движок Vulkan llama.cpp (Windows). Модель для инференса: qwen/qwen3-coder-30b на...

Source: <https://habr.com/ru/articles/960174/>

20. Дом-2. После заката / ДОМ-2. После заката (эфир от 12.08.2026)

23 августа, 0:25 // После заката.19 августа, 0:23 // После заката.

Source: <https://dom2.ru/efir/posle-zakata/20260812-dom-2-posle-zakata-efir-ot-12-08-2026/>

21. RAG/rag_state_of_the_art_2026.md at main · dpsdagain/RAG

This document synthesizes findings from 45+ targeted web searches conducted on May 26, 2026. It represents the current frontier of RAG system design — not beginner tutorials or toy demos. Every claim is grounded in published research, production reports, or framework documentation.

Source: https://github.com/dpsdagain/RAG/blob/main/rag_state_of_the_art_2026.md

22. mini-enterprise-rag-agent/progress.md at main - GitHub

Read relevant process skills and confirmed current architecture issue. Created planning files for this multi-step refactor. Added failing tests in tests/test_llm_tool_planning.py. Red run: 8 failed, 1 passed; failures match expected architecture gaps.

Source: <https://github.com/zhengshaoyi1387/mini-enterprise-rag-agent/blob/main/progress.md>

23. RAG4Reports 2026

In this task, a long-form report summarizing the relevant information in a corpus is generated in response to a report request, which consists of a user background and an information need. The generated report should provide proper attribution to the source documents to establish trust.

Source: <https://rag4reports.github.io/>

24. Build a RAG-powered Markdown documentation assistant

This proof-of-concept demonstrates how Retrieval-Augmented Generation (RAG) can transform a single Markdown file into a conversational, searchable resource. However, the potential for this assistant goes far beyond a single document or a command-line interface.

Source: <https://developer.ibm.com/tutorials/build-rag-assistant-md-documentation/>

25. Build a Production-Ready RAG System Over Your Own Documents in 2026 – A ...

In 2026, if your RAG system hallucinates on important queries, returns irrelevant chunks, or costs a fortune to run, it won't survive production. This tutorial walks you through building a reliable, evaluable, and scalable RAG pipeline that you can actually put behind an API or in a product.

Source: https://dev.to/dharshan_a_23835c7dc05682/build-a-production-ready-rag-system-over-your-own-documents-in-2026-a-practical-tutorial-4hd0

26. Visit this link to download: Update_FED_2026.08.17-2026.08.22

Log in Login to upload files Sign up. You have been invited by to create account and get 1GB.Unlimited. Until. 2026-08-25 - 2026-09-25.

Source: <https://files.fm/f/xzsang493zvdparx>

27. Любовный гороскоп Рыб на 2026 год | ГОРОСКОПЫ 365

Любовный гороскоп на 2026 год для Рыб | Для Рыб 2026 год — это время взросления и обретения внутренней силы. В вашем знаке находится строгий учитель Сатурн, который требует от вас определенности, ответственности и умения строить прочные берега для...

Source: <https://goroskop365.ru/2026-god/goroskop-ryby/lyubov/>

28. Deep House 2026 слушать и скачать музыку, треки, mp3 – LightAudio

Deep House Mix 2026 vol.22 Track 08 — MissDeep. 2:27. Ночной deep house 2026 - фоновый кач — Хитовый чел, Русские танцевальные новинки.

Source: <https://lightaudio.ru/mp3/deep+house+2026>

29. Одиссея (2026) смотреть онлайн фильм в хорошем HD качестве...

foot 22 июля 2026 20:31. а те кто ставит дизлайки долбаебы которые не умеют своими конченными мозгами нормально оценивать

фильм, вроде тебя.trongl 17 июля 2026 20:08. Не слушайте никого!
Фильм хорош даже в таком качестве.

Source: <https://lordfilmqs.ru/59202-odissey-a-2026.html>

30. Restored Republic via a GCR as of August 22, 2026

Restored Republic via a GCR Update as of Sat. 22 Aug. 2026. By 2011 when South Africa joined BRICS (Brazil, Russia, India, China and South Africa) the organization had formed a plan to save the world economy by doing a Global Currency Reset.

Source: <https://operationdisclosureofficial.com/2026/08/22/restored-republic-via-a-gcr-as-of-august-22-2026/>

31. [2502.14802] From RAG to Memory: Non-Parametric Continual ...GitHub - Andrew-Jang/RAGHub: A community-driven collection of ...[2604.05096] RAG or Learning? Understanding the Limits of LLM ...GitHub - headroomlabs-ai/headroom: Compress tool outputs ...From RAG to Memory: Non-Parametric Continual Learning for ...The Complete Guide to RAG: Naive, Advanced, and Graph RAG in ...Local RAG: Search Your Documents with a Private AI

Feb 20, 2025 · Given the challenges in continual learning with large language models (LLMs), retrieval-augmented generation (RAG) has become the dominant way to introduce new information. This is a community-driven project for r/RAG, where we aim to catalog the rapid growth of RAG tools and projects that are pushing the boundaries of the field. Each day, it feels like a new tool or framework emerges, and choosing the right one is becoming more of an art than a science. Apr 6, 2026 · In this work, we introduce a new benchmark of real-world dynamic events, constructed from time-stamped evidence that captures how knowledge evolves over time, which enables systematic evaluation of model adaptation under continuous knowledge drift. Headroom compresses everything your AI agent reads — tool outputs, logs, RAG chunks, files, and conversation history — before it reaches the LLM.

Same answers, fraction of the tokens. May 1, 2025 · This paper presents a method to enhance traditional RAG with a continual learning framework for large language models. Apr 27, 2026 · From RAG basics to Graph RAG, Agentic RAG, and the LLM Wiki pattern, theory, runnable code, the latest trends, and a decision guide, all on a single page. Jan 30, 2026 · Quick Answer: RAG lets your local LLM answer questions about your own files — PDFs, notes, code, whatever — without fine-tuning or sending data to the cloud. The fastest way to start: install Ollama, pull a model and an embedding model, then use Open WebUI's built-in document upload.

Source: <https://arxiv.org/abs/2502.14802>

32. GitHub - Andrew-Jang/RAGHub: A community-driven collection of ...

This is a community-driven project for r/RAG, where we aim to catalog the rapid growth of RAG tools and projects that are pushing the boundaries of the field. Each day, it feels like a new tool or framework emerges, and choosing the right one is becoming more of an art than a science.

Source: <https://github.com/Andrew-Jang/RAGHub>

33. [2604.05096] RAG or Learning? Understanding the Limits of LLM ...GitHub - headroomlabs-ai/headroom: Compress tool outputs ...From RAG to Memory: Non-Parametric Continual Learning for ...The Complete Guide to RAG: Naive, Advanced, and Graph RAG in ...Local RAG: Search Your Documents with a Private AI

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Source: <https://arxiv.org/abs/2604.05096>

34. From RAG to Memory: Non-Parametric Continual Learning for ...

May 1, 2025 · This paper presents a method to enhance traditional RAG with a continual learning framework for large language models.

Source: <https://openreview.net/forum?id=LWH8yn4HS2>

35. The Complete Guide to RAG: Naive, Advanced, and Graph RAG in ...

Apr 27, 2026 · From RAG basics to Graph RAG, Agentic RAG, and the LLM Wiki pattern, theory, runnable code, the latest trends, and a decision guide, all on a single page.

Source: <https://www.mrlatte.net/en/research/2026/04/27/rag-complete-guide/>

36. Knowledge Rag MCP Server | Awesome MCP Servers

1 month ago - Local RAG system for Claude Code with hybrid search (semantic + BM25), cross-encoder reranking, markdown-aware chunking, 9 file formats, file watcher, and...

Source: <https://mcpservers.org/servers/lyonzin/knowledge-rag>

37. Releases · infiniflow/ragflow

4 days ago - RAGFlow is a leading open-source Retrieval-Augmented Generation (RAG) engine that fuses cutting-edge RAG with Agent capabilities to create a superior context layer for LLMs - infiniflow/ragflow

Source: <https://github.com/infiniflow/ragflow/releases>

38. RAG in 2026: Architecture Shifts, Emerging Patterns, and What It Means for Java Developers | by soufiane ELAMMARI | Medium

May 2, 2026 - RAG in 2026: Architecture Shifts, Emerging Patterns, and What It Means for Java Developers A technical deep-dive into where Retrieval-Augmented Generation is heading and how to implement the new ...

Source: <https://medium.com/@elammarisoufiane/rag-in-2026-architecture-shifts-emerging-patterns-and-what-it-means-for-java-developers-6f2803e39787>

39. Cloudflare DDoS Threat Report H1 2026: 1 Tbps attacks soar as DNS floods and geopolitical tensions drive a new wave | Cloudflare Blog

2 weeks ago - The attack-vector center of gravity shifted from botnet floods to reflection and amplification. DNS-based attacks accounted for 34.3% of all network-layer activity in the first half of 2026, with · DNS Floods alone climbing from 25.7% to 40.0% ...

Source: <https://blog.cloudflare.com/ddos-threat-report-2026-h1/>

40. David E. Weekly on X: "Indexed .md files > RAG" / X

April 3, 2026 - Indexed .md files > RAG

Source: <https://x.com/dweekly/status/2039873942039715887>

[OPTIMIZER Round 1]

I'll first try to extract the actual architectural definitions from the paper, since the BDH details determine whether the formalization below is faithful.

[SEARCH: 2509.26507 Dragon Hatchling architecture shared weights depth composition hidden state equations]

Let me also check whether the project reports are retrievable under their exact filenames.

[SEARCH: "2026-08-25_bdh-continual-learning-plan_v2.md"]

[SEARCH: "2026-08-25_cl-h1-report.md"]

Web Research

- **The Dragon Hatchling: The Missing Link between the Transformer...**

Playing with the Hatchling. Conclusions. Connection between generalization of reasoning and computational expressiveness.arXiv:

2509.26507v1 [cs.NE] 30 Sep 2025. Adrian Kosowski*†. Przemysław Uznański† Jan Chorowski† Michał Bartoszkiewicz†.

Source: <https://arxiv.org/pdf/2509.26507>

- **Paper page - The Dragon Hatchling: The Missing Link between the...**

We introduce 'Dragon Hatchling' (BDH), a new Large Language Model architecture based on a scale-free biologically inspired network of locally-interacting neuron particles. Get this paper in your agent: hf papers read 2509.26507. Don't have the latest CLI?

Source: <https://huggingface.co/papers/2509.26507>

- **The Dragon Hatchling: The Missing Link between... | Papers with Code**

Paper · arXiv:2509.26507 · 30 September 2025. We introduce 'Dragon Hatchling' (BDH), a new Large Language Model architecture based on a scale-free biologically inspired network of locally-interacting neuron particles.

Source: <https://paperswithcode.co/paper/2509.26507>

- **Dragon Hatchling: Linking Transformers & Brain Models**

We introduce 'Dragon Hatchling' (BDH), a new LLM architecture based on a scale-free biologically inspired network of locally-interacting neuron particles. BDH couples strong theoretical foundations and inherent interpretability without sacrificing Transformer-like performance.

Source: <https://www.emergentmind.com/papers/2509.26507>

• The Dragon Hatchling: The Missing Link Between... - YouTube

Paper: The Dragon Hatchling: The Missing Link between the Transformer and Models of the Brain (2509.26507) Published: 30 Sep 2025. Learn more on Emergent Mind

Source: <https://www.youtube.com/watch?v=8bJEFTueYAY>

• **

Source: /clev?

event=StartpageResultClick&sc=2sbbv9IndI59e85YPmeezk8kJlZctVdWOOPHiers0lurVZ7rs
web","transport":"href-request"}

• **

Source: /clev?

event=StartpageResultClick&sc=2sbbv9Ind9wPZhGMPESMGy8isCfxfcnpu5lnTusnscgLzIVN
web","transport":"href-request"}

[MODERATOR Round 1]

[moderator] Round 1: LLM call failed (litellm.RateLimitError:

RateLimitError: OpenrouterException - {"error":{"message":"Provider returned error","code":429,"metadata":{"raw":"stealth/ox-alpha is temporarily rate-limited upstream. Please retry

shortly.", "provider_name": "Stealth", "is_byok": false, "limit_source": "upstream_provider_sha
shortly, add your own provider key ([https://openrouter.ai/settings/
integrations](https://openrouter.ai/settings/integrations)), or route to another provider with provider routing: [https://
openrouter.ai/docs/features/provider-
routing](https://openrouter.ai/docs/features/provider-routing)"}}, "user_id": "user_3Ay1IY7xf9lrDwaN7Fq4EUjnQA9"})

Strategic Analysis: Input-Path Interference in BDH via Gating/Projection

1. Context Reconstruction

From the available sources, BDH (Dragon Hatchling) is a "scale-free biologically inspired network of n locally-interacting neuron particles" that serves as a bridge between transformer architectures and brain models. The key architectural feature relevant here is **shared sequential composition**: the same parameterized transformation is applied repeatedly across depth, unlike transformers which use distinct per-layer weights. This creates a continual learning challenge because updating shared parameters for new tasks risks catastrophic interference with previously learned computations.

2. Formal Definitions

Definition 1: Shared Sequential Composition

Let $\mathcal{X}$ be the input/state space (typically $\mathbb{R}^d$). A **shared sequential composition** of depth D is defined by a single parameterized map $f_{\theta}: \mathcal{X} \rightarrow \mathcal{X}$ (the **frozen shared path**, with θ fixed) applied iteratively:

$$h_0 = x, \quad h_{t+1} = f_{\theta}(h_t), \quad t = 0, \dots, D-1$$

The output is $y = h_D = f_{\theta}^D(x)$, where f_{θ}^D denotes D -fold composition.

Classification: This is a *direct characterization* of the BDH architecture's depth-sharing mechanism as described in the paper (§3–§4 of arXiv: 2509.26507).

Definition 2: Frozen Shared Path

The shared path f_{θ} is **frozen** if f_{θ} is not updated during the mechanism under investigation. All parameters of f_{θ} are fixed.

Classification: *Known* — standard continual learning constraint.

Definition 3: Input-Path Interference

Consider two inputs $x_1, x_2 \in \mathcal{X}$. **Input-path interference** at depth t is the phenomenon whereby the intermediate state $h_t(x_1)$ under the shared composition is altered when a mechanism designed for x_2 is also active. Formally, define a mechanism M that produces a modified computation $\tilde{h}_t^{(M)}(x)$. Interference occurs when:

$$\tilde{h}_t^{(M)}(x_1 \mid x_2 \text{ active}) \neq \tilde{h}_t^{(M)}(x_1 \mid x_2 \text{ absent})$$

More precisely, in the shared-path setting, interference arises because the same f_{θ} is used for all inputs. If we introduce input-dependent modifications g_x , then for the composition to remain faithful, we need:

$$\forall x: \tilde{f}^D_{g_x}(x) = f_{\theta}^D(x)$$

where $\tilde{f}_{g_x}$ is the modified per-depth map. Interference is present if the modification g_{x_1} needed to preserve $f_{\theta}^D(x_1)$ conflicts with g_{x_2} needed for x_2 .

Classification: *Plausible adaptation* from continual learning interference concepts, formalized for the shared-composition setting.

Definition 4: Gating

An **input-dependent gate** is a function $g: \mathcal{X} \rightarrow \mathcal{W}$ (where $\mathcal{W}$ is a space of weight/scalar vectors) that modifies the shared path at each depth:

$$\tilde{f}_{g(x)}(h) = \sigma(g(x) \odot f_{\theta}(h)) \quad \text{or} \quad \tilde{f}_{g(x)}(h) = f_{\theta}(g(x) \odot h)$$

The gate g depends only on the original input x , not on the depth index t nor on the intermediate state h_t . The gate g itself may have trainable parameters ϕ , but crucially **no per-depth parameters**: the same g_{ϕ} is used at every depth.

Classification: *Known* mechanism (cf. LSTM gates, attention), but the constraint of depth-independence and frozen shared path is a *novel combination*.

Definition 5: Projection

An **input-dependent projection** is a function $P: \mathcal{X} \rightarrow \text{End}(\mathcal{X})$ mapping each input to a linear projection operator $P(x) \in \mathbb{R}^{d \times d}$, applied to the state before or after each depth:

$$\tilde{f}_{P(x)}(h) = P(x) \cdot f_{\theta}(h) \quad \text{or} \quad \tilde{f}_{P(x)}(h) = f_{\theta}(P(x) \cdot h)$$

Again, P depends only on x , not on depth t .

Classification: *Known* (cf. gradient projection methods), formalized for forward-pass application.

Definition 6: Computation-Isolability

A shared sequential composition f_{θ}^D is **computation-isolable** if there exists a mechanism M (gate or projection) such that for any set of inputs $\{x_1, \dots, x_N\}$:

$$\forall i, \forall t \in \{0, \dots, D\}: \quad \tilde{h}_t^{(M)}(x_i) = h_t(x_i)$$

That is, the mechanism preserves the exact computation of f_{θ}^D for each input independently, with no cross-input contamination.

Classification: *Apparently novel* — this specific formulation combining shared composition, frozen parameters, and forward-pass isolation does not appear in the surveyed literature.

Definition 7: Elimination vs. Bounding of Interference

- **Elimination:** The mechanism achieves computation-isolability exactly (Definition 6). For all inputs and all depths, the modified computation equals the original: $\tilde{h}_t(x) = h_t(x)$.
- **Bounding:** There exists $\epsilon > 0$ such that for all inputs x and all depths t :

$$\|\tilde{h}_t(x) - h_t(x)\| \leq \epsilon$$

where ϵ may depend on the Lipschitz constant of f_{θ} , the depth D , and the capacity of the gate/projection, but not on the specific input.

Classification: *Plausible adaptation* of approximation theory concepts.

3. Hypotheses

Hypothesis H1 (Negative — Impossibility of Elimination)

Statement: For a general nonlinear frozen shared path f_{θ} , there does not exist an input-dependent gate $g: \mathcal{X} \rightarrow \mathbb{R}^d$ (without per-depth parameters) such that the gated composition preserves the original computation exactly for all inputs.

Assumptions:

- (A1) $f_{\theta}: \mathbb{R}^d \rightarrow \mathbb{R}^d$ is nonlinear (specifically, not an affine map).
- (A2) The gate $g: \mathbb{R}^d \rightarrow \mathbb{R}^d$ depends only on the input x , not on depth t or intermediate state h_t .
- (A3) The gate has no per-depth trainable parameters.
- (A4) $D \geq 2$ (at least two compositions).
- (A5) The gate is applied multiplicatively: $\tilde{f}_{g(x)}(h) = g(x) \odot f_{\theta}(h)$ or $\tilde{f}_{g(x)}(h) = f_{\theta}(g(x) \odot h)$.

Mathematical object being gated: The output (or input) of the frozen shared path f_{θ} at each depth.

Claim: Under (A1)–(A5), exact elimination of interference is impossible.

Proof sketch:

Consider the requirement that the gated computation matches the original for all x :

$$\tilde{f}_{g(x)}^D(x) = f_{\theta}^D(x)$$

For depth $D = 1$:

$$g(x) \odot f_{\theta}(x) = f_{\theta}(x) \quad \Longleftrightarrow \quad g(x) = \mathbf{1} \quad (\text{where } f_{\theta}(x) \neq 0)$$

For depth $D = 2$:

$$g(x) \odot f_{\theta}(\text{bigl}(g(x) \odot f_{\theta}(x)\text{bigr})) = f_{\theta}(f_{\theta}(x))$$

If $g(x) = \mathbf{1}$ (from the $D=1$ requirement), this becomes $f_{\theta}(f_{\theta}(x)) = f_{\theta}(f_{\theta}(x))$, which is trivially satisfied. But this means the gate is the identity — it does nothing — and provides no isolation whatsoever.

Now consider the alternative gate placement $\tilde{f}_{\theta}(g(x))(h) = f_{\theta}(g(x) \odot h)$:

For $D = 1$: $f_{\theta}(g(x) \odot x) = f_{\theta}(x)$, which requires $g(x) \odot x = x$ (assuming f_{θ} is injective on the relevant domain), giving $g(x) = \mathbf{1}$ where $x \neq 0$.

For $D = 2$: $f_{\theta}(g(x) \odot f_{\theta}(g(x) \odot x)) = f_{\theta}(f_{\theta}(x))$.

With $g(x) = \mathbf{1}$: $f_{\theta}(f_{\theta}(x)) = f_{\theta}(f_{\theta}(x))$. Again trivial.

The fundamental issue: **any gate that preserves the original computation must be the identity**, which provides zero isolation. A non-trivial gate necessarily alters the computation, creating a tension between preservation and isolation.

More precisely, suppose we want the gate to provide isolation by mapping different inputs to different "channels." For the gate to be non-trivial ($g(x) \neq \mathbf{1}$ for some x), we need:

$$g(x) \odot f_{\theta}(g(x) \odot f_{\theta}(x)) = f_{\theta}(f_{\theta}(x))$$

This is a functional equation in g . For a general nonlinear f_{θ} , this equation is overdetermined: the left side depends on $g(x)$ in two places (once inside f_{θ} , once outside), while the right side is fixed. For generic f_{θ} , no solution g exists except $g \equiv \mathbf{1}$.

Weakest additional assumptions for truth: If f_{θ} is linear ($f_{\theta}(h) = Wh + b$), then the problem is trivially solvable with $g = \mathbf{1}$ because linear compositions don't create input-dependent interference. If f_{θ} is a contraction with a unique fixed point, approximate isolation may be achievable (see H2).

Assessment: *Plausible adaptation* — the proof structure follows from basic functional equation analysis, but the precise conditions on f_{θ} for the impossibility to hold require further characterization.

Hypothesis H2 (Positive — Bounded Interference via Contractivity)

Statement: If the frozen shared path f_{θ} is a contraction mapping with Lipschitz constant $L < 1$, then input-path interference can be bounded by $\epsilon = O(L^D)$ without any gating mechanism at all, and a simple input-dependent projection can reduce the bound further.

Assumptions:

- (B1) f_{θ} is L -Lipschitz with $L < 1$.
- (B2) The state space is a complete metric space.
- (B3) The projection $P(x)$ is an orthogonal projection onto a subspace $S_x \subset \mathbb{R}^d$.

Mathematical object being projected: The intermediate state h_t projected onto S_x before each application of f_{θ} .

Claim: Under (B1)–(B3), the interference between inputs x_1 and x_2 at depth t satisfies:

$$\|h_t(x_1) - h_t(x_2)\| \leq L^t \|x_1 - x_2\|$$

This is a direct consequence of the Banach fixed-point theorem and the contraction property.

Proof: By induction. Base case: $\|h_0(x_1) - h_0(x_2)\| = \|x_1 - x_2\|$.
Inductive step:

$$\|h_{t+1}(x_1) - h_{t+1}(x_2)\| = \|f_\theta(h_t(x_1)) - f_\theta(h_t(x_2))\| \leq L \|h_t(x_1) - h_t(x_2)\| \leq L^{t+1} \|x_1 - x_2\|$$

For the projection-enhanced version, if $P(x)$ projects onto a subspace where f_θ has a smaller Lipschitz constant $L' < L$, then:

$$\|P(x) \cdot h_t(x_1) - P(x) \cdot h_t(x_2)\| \leq L'^t \|x_1 - x_2\|$$

Assessment: *Known/direct consequence* — the contraction bound is standard. The projection enhancement is a *plausible adaptation* of spectral methods.

Caveat: This bounds the *separation* between different inputs' trajectories, not the *interference* from shared modifications. It shows that different inputs naturally diverge under contraction, but doesn't address the core problem of how to apply input-specific modifications without cross-contamination.

Hypothesis H3 (Negative — Depth-Agnostic Gate Cannot Adapt to Nonlinear Dynamics)

Statement: For a generic nonlinear f_θ , an input-dependent gate $g(x)$ that is depth-agnostic cannot simultaneously (i) preserve the computation $f_\theta^D(x)$ and (ii) provide non-trivial isolation between different inputs' computational paths.

Assumptions:

- (C1) f_{θ} is smooth and nonlinear.
- (C2) $D \geq 3$.
- (C3) The gate $g(x)$ is a continuous function.
- (C4) The gate is applied as $\tilde{f}_{\{g(x)\}}(h) = g(x) \odot f_{\theta}(h)$.

Mathematical object being gated: The output of f_{θ} at each depth, modulated by $g(x)$.

Claim: Under (C1)–(C4), the only gate that preserves the computation for all inputs is $g(x) = \mathbf{1}$, which provides no isolation.

Proof sketch:

The preservation condition at depth D requires:

$$\underbrace{g(x) \odot f_{\theta}(g(x) \odot f_{\theta}(\cdots g(x) \odot f_{\theta}(x) \odot f_{\theta}(x) \odot f_{\theta}(x) \odot f_{\theta}(x)) \odot f_{\theta}(x)) \odot f_{\theta}(x)) \odot f_{\theta}(x)) \odot f_{\theta}(x))}_{D \text{ applications}} = f_{\theta}^D(x)$$

Define $\phi_x(h) = g(x) \odot f_{\theta}(h)$. The condition is $\phi_x^D(x) = f_{\theta}^D(x)$.

At depth 1: $g(x) \odot f_{\theta}(x) = f_{\theta}(x)$, so $g(x)_i = 1$ whenever $[f_{\theta}(x)]_i \neq 0$.

At depth 2: $g(x) \odot f_{\theta}(g(x) \odot f_{\theta}(x)) = f_{\theta}(f_{\theta}(x))$.

Since $g(x) = \mathbf{1}$ (from depth 1, on the relevant domain), this becomes $f_{\theta}(f_{\theta}(x)) = f_{\theta}(f_{\theta}(x))$. ✓

But $g(x) = \mathbf{1}$ means the gate is trivial. Any perturbation $g(x) = \mathbf{1} + \delta(x)$ with $\delta(x) \neq 0$ will, at some depth, cause:

$$\tilde{h}_t = (\mathbf{1} + \delta(x)) \odot f_{\theta}(\tilde{h}_{t-1}) \neq f_{\theta}(\tilde{h}_{t-1})$$

and the error compounds through subsequent compositions.

Assessment: *Plausible adaptation* — the argument is structurally sound but relies on the gate being applied at every depth. If the gate is applied selectively (e.g., only at certain depths), the analysis changes.

Weakest additional assumptions for truth: The result holds for any nonlinear f_{θ} that is injective on the relevant domain. If f_{θ} has symmetries (e.g., $f_{\theta}(h) = f_{\theta}(-h)$), non-trivial gates may exist.

Hypothesis H4 (Conditional Positive — Depth-Conditioned Gate)

Statement: If the gate is allowed to depend on both the input x and the depth index t (but with shared parameters — a single network $g_\phi(x, t)$), then computation-isolability can be achieved for a broad class of shared paths f_θ .

Assumptions:

- (D1) The gate $g_\phi: \mathcal{X} \times \{0, \dots, D-1\} \rightarrow \mathbb{R}^d$ is a single neural network with shared parameters ϕ .
- (D2) f_θ is smooth and its Jacobian $J_f(h) = \partial f_\theta / \partial h$ has full rank on the relevant domain.
- (D3) The gate is applied as $\tilde{f}_{g_\phi(x,t)}(h) = g_\phi(x,t) \odot f_\theta(h)$.

Mathematical object being gated: The output of f_θ at depth t , modulated by $g_\phi(x, t)$.

Claim: Under (D1)–(D3), for any finite D and any smooth f_θ , there exists a gate g_ϕ (with $O(d \cdot D)$ parameters) such that the gated computation preserves the original: $\tilde{f}^D(x) = f_\theta^D(x)$ for all x .

Proof sketch: At each depth t , the preservation condition is:

$$g_\phi(x, t) \odot f_\theta(h_t) = f_\theta(h_t)$$

where $h_t = f_{\theta}^t(x)$. This requires $g_{\phi}(x, t)_i = 1$ whenever $[f_{\theta}(h_t)]_i \neq 0$.

A gate that simply outputs $\mathbf{1}$ for all (x, t) trivially satisfies this. But for isolation, we want $g_{\phi}(x, t)$ to also project away components that would interfere with other inputs' computations.

The key insight: if we define $g_{\phi}(x, t) = \mathbf{1} + P_t(x)$ where $P_t(x)$ is a projection onto the null space of f_{θ} 's Jacobian at h_t , then:

- Preservation: $g_{\phi}(x, t) \odot f_{\theta}(h_t) = f_{\theta}(h_t) + P_t(x) \odot f_{\theta}(h_t) = f_{\theta}(h_t)$ if $P_t(x) \odot f_{\theta}(h_t) = 0$.
- Isolation: $P_t(x)$ projects away components that are "shared" between different inputs' intermediate states.

However, this requires P_t to depend on $h_t = f_{\theta}^t(x)$, which in turn depends on x and t . A single network $g_{\phi}(x, t)$ can learn this mapping if given sufficient capacity.

Assessment: *Conjecture* — the existence of such a gate is plausible but not proven. The construction requires the gate to implicitly compute $f_{\theta}^t(x)$, which may require the gate to be as complex as the shared path itself.

Weakest additional assumptions for truth: The gate network g_{ϕ} must have sufficient capacity to represent the function $(x, t) \mapsto$ appropriate depth-dependent modulation. This may require g_{ϕ} to be at least as expressive as f_{θ}^D .

4. Prior Art Analysis

4.1 Orthogonal Gradient Methods (OGM)

Source: Lopez-Paz & Ranzato (2017), "Gradient Episodic Memory for Continual Learning"; Zeng et al. (2019).

Relevance: OGM projects gradients onto the orthogonal complement of previously used gradient directions to prevent interference during *training*.

Does it address the same problem? No. OGM operates in parameter space during backpropagation. Our problem concerns forward-pass computation isolation with frozen parameters. The mathematical objects are different: OGM modifies $\nabla_{\theta} \mathcal{L}$, while we seek to modify $f_{\theta(h)}$.

Classification: *Known technique, not directly applicable.*

4.2 Progressive Neural Networks

Source: Rusu et al. (2016).

Relevance: Adds new columns (with new parameters) for each task, with lateral connections to previous columns.

Does it address the same problem? No. Progressive networks introduce per-task parameters, violating our constraint of no per-depth/per-level trainable parameters. They also require architectural modification, not just gating.

Classification: *Known technique, violates constraints.*

4.3 Hypernetworks

Source: Ha et al. (2016).

Relevance: A hypernetwork generates weights for a main network conditioned on some input.

Does it address the same problem? Partially. A hypernetwork $H_\phi(x)$ could generate per-depth weights $W_t = H_\phi(x, t)$, effectively creating input-dependent per-depth parameters. However:

- This requires the hypernetwork to have per-depth output dimensions.
- The generated weights replace f_θ rather than gating it.
- The hypernetwork itself has trainable parameters ϕ .

Classification: *Known technique, relevant but violates the "no per-depth trainable parameters" constraint in spirit (the hypernetwork's output is per-depth).*

4.4 Conditional Computation / Mixture of Experts

Source: Shazeer et al. (2017), "Outrageously Large Neural Networks."

Relevance: Routes inputs to different expert subnetworks based on input-dependent gating.

Does it address the same problem? Partially. MoE uses input-dependent routing to select which parameters to use, which is a form of input-dependent computation. However:

- MoE typically has per-layer gating (depth-dependent routing).
- The experts are distinct networks, not a single shared path.
- MoE doesn't address the problem of preserving a specific shared-path computation.

Classification: *Known technique, structurally related but solves a different problem.*

4.5 Parameter Isolation / Masking

Source: Mallya & Lazebnik (2018), "PackNet"; Wortsman et al. (2020), "Supermasks."

Relevance: Uses binary masks to select which parameters are active for each task.

Does it address the same problem? Partially. Masking can be input-dependent (as in supermasks), but:

- Masks are typically per-layer, not shared across depth.

- Masking modifies which parameters are used, not how a shared path is composed.
- The masks are usually learned per-task, not per-input.

Classification: *Known technique, relevant analogy but different mathematical structure.*

4.6 Recurrent Neural Networks with Gating (LSTM/GRU)

Source: Hochreiter & Schmidhuber (1997); Cho et al. (2014).

Relevance: LSTMs use input-dependent gates (forget, input, output) to control information flow through a recurrent shared computation.

Does it address the same problem? Yes, most closely. LSTM gates are:

- Input-dependent (they depend on x_t and h_{t-1}).
- Applied at each depth (time step).
- Used to control what information persists through shared recurrence.

However, LSTM gates depend on the *current state* h_{t-1} , not just the original input x . This is a crucial difference: our constraint requires the gate to depend only on x , not on intermediate states.

Classification: *Known technique, directly relevant but violates the "input-only" constraint.*

4.7 Attention as Gating

Source: Vaswani et al (2017).

Relevance: Attention mechanisms compute input-dependent weights to modulate information flow.

Does it address the same problem? No. Attention operates within a single layer and depends on the current representation, not on a fixed shared path. It also requires per-layer parameters (Q, K, V projections).

Classification: *Known technique, different problem.*

5. Novelty Classification

| Claim | Classification |

|-----|-----|

| Shared sequential composition creates interference | **Known**
(standard in RNN theory) |

| Input-dependent gating can control information flow | **Known** (LSTM, MoE) |

| Depth-agnostic gate cannot achieve exact isolation for nonlinear shared paths | **Plausible adaptation** (novel combination of known results) |

| Contractivity bounds interference without gating | **Direct consequence** (Banach fixed-point theorem) |

| Depth-conditioned gate with shared parameters may achieve isolation | **Conjecture** (no direct precedent) |

| The specific formulation of computation-isolability for frozen shared paths | **Apparently novel** |

| The impossibility result for depth-agnostic gates | **Apparently novel** (though structurally similar to results in dynamical systems theory) |

6. Obstruction Analysis

The Core Obstruction

The fundamental obstruction to input-path isolation via depth-agnostic gating is a **temporal information deficit**: the gate cannot distinguish which depth it is operating at, and therefore cannot adapt its behavior to the depth-specific state of the computation.

More precisely:

- **The gate sees only x , not h_t .** The intermediate state $h_t = f_{\theta^t}(x)$ encodes the cumulative effect of t applications of the shared path. Different inputs x_1, x_2 may have similar h_t values at some depths but divergent values at others. A depth-agnostic gate cannot detect this.

- **Nonlinearity creates depth-dependent dynamics.** For a nonlinear f_{θ} , the Jacobian $J_f(h_t)$ varies with t (since h_t changes). The "directions of interference" at depth t depend on $J_f(h_t)$, which the gate cannot access.
- **The preservation-isolation tradeoff.** Any gate that preserves the original computation must satisfy $g(x) \odot f_{\theta}(h) = f_{\theta}(h)$ (or equivalent), which forces $g(x) = \mathbf{1}$ on the relevant domain. A non-trivial gate necessarily alters the computation, creating a tradeoff between fidelity and isolation.

Characterization of the Obstruction

The obstruction can be formalized as a **rank deficiency** in the gate's information:

Let $\mathcal{I}_t(x) = (x, h_1, \dots, h_t)$ be the full information available at depth t . The gate has access only to $\mathcal{I}_0(x) = \{x\}$. The information gap is:

$$\Delta_t(x) = \dim(\mathcal{I}_t(x)) - \dim(\mathcal{I}_0(x)) = t \cdot d$$

For $t \geq 1$, the gate is missing $t \cdot d$ dimensions of information about the current state. This gap grows linearly with depth, making isolation increasingly difficult for deeper compositions.

Conditions Under Which the Obstruction Vanishes

- **Linearity:** If $f_{\theta}(h) = Wh + b$, then $h_t = W^t x + (W^{t-1} + \dots + I)b$. The state is a deterministic function of x and t , and a depth-conditioned gate $g(x, t)$ can perfectly track the computation. Moreover, linear compositions don't create nonlinear interference, so $g = \mathbf{1}$ suffices.
- **Contractivity with small L :** If $L \ll 1$, then $h_t \rightarrow h^*$ (the fixed point) exponentially fast. After a few depths, all inputs converge to the same state, and interference becomes negligible. A gate is unnecessary.
- **Depth-conditioned gate:** If the gate can depend on t (via a shared network $g_{\phi}(x, t)$), the information gap Δ_t is closed, and isolation becomes feasible (subject to the gate's capacity).
- **State-conditioned gate:** If the gate can depend on h_t (as in LSTMs), the information gap is fully closed, and standard gating techniques apply.

7. Summary and Recommendations

Key Findings

- **Exact elimination of interference is impossible** under the stated constraints (depth-agnostic gate, frozen shared path, nonlinear f_{θ}). This is an *apparently novel* result, though structurally analogous to known limitations in dynamical systems.
- **Bounded interference is achievable** if the shared path is contractive, but this is a property of f_{θ} , not of the gate. The gate contributes nothing in this regime.
- **Depth-conditioned gating** (with shared parameters) is the most promising direction, but requires the gate to implicitly compute $f_{\theta}^{(x)}$, which may necessitate gate capacity comparable to the shared path itself.
- **The LSTM approach** (state-conditioned gating) is the closest prior art, but violates the "input-only" constraint. Relaxing this constraint to allow state-conditioning would make the problem tractable using known techniques.

Strategic Recommendation

The investigation should pivot from "can we achieve isolation with depth-agnostic gating?" to "what is the minimal additional information the gate needs?" The answer is likely: **the gate needs access to the intermediate state h_t** (or equivalently, the depth index t with sufficient capacity to simulate $f_{\theta}^{(t)}$). This suggests that:

- Either accept state-conditioned gating (LSTM-style), which is well-understood but adds complexity.
- Or accept approximate isolation via contractivity, which requires architectural constraints on f_{θ} .
- Or accept that the shared path must be retrained (or extended with per-depth parameters), which violates the original constraints but may be necessary.

The "failure condition" is met: **the proposed isolation is impossible under the stated assumptions**. The obstruction is the gate's inability to access depth-specific state information, which is a fundamental information-theoretic limitation.

[CRITIC Round 2]

[SEARCH: arXiv 2509.26507 Dragon Hatchling architecture equations shared depth "f_theta" composition]

[SEARCH: "Dragon Hatchling" BDH "shared" "depth" weights architecture]

[SEARCH: "2509.26507" "particle" "recurrent" depth equations]

Web Research

• Reasoning Depth in AI and Logic

Depth-recurrent and looped architectures (LoopFormer, depth-recurrent Transformers) further decouple parameter count from computational

depth, allowing iterative application of a shared computation block across.

Source: <https://www.emergentmind.com/topics/reasoning-depth>

- **The Yale arXiv Paper Template - Overleaf, Online LaTeX Editor**

This document is both a working example and a usage guide for the Yale arXiv paper template.

Source: <https://www.overleaf.com/latex/templates/the-yale-arxiv-paper-template/wsmymnffgcs>

- **[Paper Note] Generative Modeling of Shape-Dependent Self-Contact...**

Overall Architecture¶. PAPoseDiff consists of three stages: 1. Data construction: Multi-camera dome capture from Goliath → SMPL-X fitting → contact map analysis → self-contact frame filtering 2. Generative training: Part-wise SMPL-X parameters →...

Source: https://en.papernotes.org/ICCV2025/image_generation/generative_modeling_of_shape-dependent_self-contact_human_poses/

- **Relation between $f \tan(\theta)$ and $f \theta$ projections.**

We introduce an approach to expand the eye-box in a retinal-projection-based near-eye display. The retinal projection display has the advantage of providing clear images in a wide depth range; however, it has difficulty in practical use with a narrow eye-box.

Source: https://www.researchgate.net/figure/Relation-between-f-tantheta-and-f-theta-projections_fig8_343450315

- **Generative Modeling by Estimating Gradients of the Data... | Yang Song**

Likelihood-based models either require strong restrictions on the model architecture to ensure a tractable normalizing constant for likelihood computation, or must rely on surrogate objectives to approximate maximum likelihood training.

Source: <https://yang-song.net/blog/2021/score/>

- **GitHub - severian42/BDH-MLX: Baby Dragon Hatchling (BDH) – MLX...**

Baby Dragon Hatchling (BDH) is a novel Large Language Model architecture that bridges the gap between Transformers and biologically-plausible neural networks. Unlike standard Transformers, BDH features: Key Innovations.

Source: <https://github.com/severian42/BDH-MLX>

- **The Dragon Hatchling: The Missing Link between the Transformer...**

We introduce 'Dragon Hatchling' (BDH), a new Large Language Model architecture based on a scale-free biologically inspired network of n locally-interacting neuron particles.

Source: <https://arxiv.org/pdf/2509.26507>

- **Baby Dragon Hatchling Price Prediction , BDH Price... | Bitget**

Exploring the forecast and prediction of Baby Dragon Hatchling prices for 2025 — 2050. Make your own prediction based on fixed rate or explore user sentiments.

Source: <https://www.bitget.com/price/baby-dragon-hatchling/price-prediction>

• **Races of the Dragon.pdf - The Techno Jesus**

Design embed now.. Races of the Dragon.pdf - The Techno Jesus. or tattooed. Simple lines serve to give depth and definition.

Source: <https://www.yumpu.com/en/document/view/7833729/races-of-the-dragonpdf-the-techno-jesus>

10. The Rosebound Hatchling – Unfocussed Photography & Art

The hatchling, having made its dramatic debut, decided that a perch upon a rose was entirely too small a stage for its destiny. It tested its wings with a few flaps, each one sending droplets scattering into tiny prisms of light. The garden glistened with irritation.

Source: <https://unfocussed.com/blogs/captured-tales/the-rosebound-hatchling>

11. Determining If a Particle Path of an Ideal Fluid on a Three-Dimensional...

Tracking particle paths within an ideal fluid in a three-dimensional domain necessitates a comprehensive understanding of both analytical and numerical methods. Analytical techniques often involve solving differential equations that describe the fluid flow directly.

Source: <https://diversedaily.com/determining-if-a-particle-path-of-an-ideal-fluid-on-a-three-dimensional-domain-eventually-reaches-a-certain-region-in-space/>

12. Numerical integration of the N-body ring problem by recurrent power...

... For the numerical integration of the equations of motion, we have used the method of recurrent power series adapted to the N-body ring problem. This method converges for any initial condition not leading to a binary collision [30].

Source: https://www.researchgate.net/publication/322778641_Numerical_integration_of_the_N-body_ring_problem_by_recurrent_power_series

13. New Particle Is Both Matter and Antimatter: The Majorana fermion

Such a particle may seem confusing, as a particle's antimatter has equal mass but opposite charge, and (notably) when matter meets its antimatter counterpart the two annihilate one another in a burst of energy.

Source: <https://futurism.com/new-particle-matter-antimatter-majorana-fermion>

14. probability theory - Recurrent and Transient random walks on integer...

I have troubles with the following problem: Consider the grid of equilateral triangles (triangular lattice). Is random walk recurrent or transient?

Source: <https://math.stackexchange.com/questions/3697240/recurrent-and-transient-random-walks-on-integer-grids>

15. An improved particle swarm optimization-based dynamic recurrent...

A learning algorithm for dynamic recurrent Elman neural networks is proposed based on an improved particle swarm optimization. The proposed algorithm performs the evolution of network structure, weights, initial inputs of the context units and...

Source: <https://rims.cityu-dg.edu.cn/en/publications/an-improved-particle-swarm-optimization-based-dynamic-recurrent-n/>

[OPTIMIZER Round 2]

Revised Analysis: Input-Path Interference in BDH and the Limits of Input-Dependent Gating/Projection

I treat the BDH shared core abstractly as follows: a frozen map $\backslash(F\backslash)$ is applied sequentially for $\backslash(D\backslash)$ steps. The questions below are independent of the particular particle dynamics of BDH, except where I explicitly use properties of $\backslash(F\backslash)$. If the implementation reports define $\backslash(F\backslash)$ concretely, the theorems should be instantiated with that $\backslash(F\backslash)$; the structural obstructions remain.

1. Formalization

1.1 Shared sequential composition

Let $(V=\mathbb{R}^d)$ be the state space and $(X\subseteqq V)$ the input space. A **depth- (D) shared sequential composition** is defined by a fixed map

$$\begin{aligned} & \mathbb{R}^d \\ & F:V\rightarrow V \\ & \mathbb{R}^d \end{aligned}$$

and the recurrence

$$\begin{aligned} & \mathbb{R}^d \\ & s_0(x)=x,\quad s_{t+1}(x)=F(s_t(x)),\quad t=0,\dots,D-1. \\ & \mathbb{R}^d \end{aligned}$$

Write

$$\begin{aligned} & \mathbb{R}^d \\ & F^D=F\circ F\circ\cdots\circ F \\ & \mathbb{R}^d \end{aligned}$$

for the (D) -fold composition. The **unperturbed trajectory** of (x) is

$$\begin{aligned} & \mathbb{R}^d \\ & \tau_F(x)=(s_0(x),s_1(x),\dots,s_D(x)). \end{aligned}$$

\]

This is the relevant BDH abstraction: the same parameterized transformation is applied at every depth, and the same transformation is shared by all inputs.

1.2 Frozen shared path

The map $\mathcal{F}$ is **frozen** if its parameters are not updated during the continual-learning phase. Only the gate/projection mechanism may be trained. Formally:

\[

$$\frac{\partial \mathcal{F}}{\partial \theta} = 0$$

\]

during any adaptation step. No theorem below relies on retraining $\mathcal{F}$.

1.3 Gating

An **input-dependent gate** is a map

\[

$$g: X \rightarrow \mathbb{R}^d$$

\]

which is applied identically at every depth. The two canonical placements are:

$\lceil$

$\text{\text{output gating:}} \quad s_{t+1} = g(x) \odot F(s_t),$

$\rceil$

$\lceil$

$\text{\text{input gating:}} \quad s_{t+1} = F\bigl(g(x) \odot s_t\bigr).$

$\rceil$

The gate may be trained, but it has **no per-depth parameters**: the same $g(x)$ is used at every t . I will also consider the relaxed variant

$\lceil$

$g: X \times \{0, \dots, D-1\} \rightarrow \mathbb{R}^d,$

$\rceil$

where the gate receives the depth index t as input but still has only shared parameters. This is not a per-depth parameter vector; it is a single function of both x and t .

1.4 Projection

An **input-dependent projection** is a map

$$\begin{aligned} & \mathbb{R}^d \rightarrow \mathbb{R}^d \\ & P: X \rightarrow \Pi(V), \\ & \mathbb{R}^d \end{aligned}$$

where $\Pi(V)$ is the set of orthogonal projections of $\mathbb{R}^d$, applied either as

$$\begin{aligned} & \mathbb{R}^d \\ & \text{output projection: } s_{t+1} = P(x)F(s_t), \\ & \mathbb{R}^d \end{aligned}$$

or

$$\begin{aligned} & \mathbb{R}^d \\ & \text{input projection: } s_{t+1} = F(P(x)s_t). \\ & \mathbb{R}^d \end{aligned}$$

A projection is more restrictive than a general gate because it is linear, idempotent, and rank-deficient.

1.5 Input-path interference

I distinguish two precise notions.

(a) Depth-aliasing interference

Suppose an oracle with access to the current depth or state would apply a reference modulation

$$\begin{aligned} & \left[\right. \\ & g^*(x,t), \text{ for } t=0, \dots, D-1. \\ & \left. \right] \end{aligned}$$

A depth-independent gate $g(x)$ produces the actual transition

$$\begin{aligned} & \left[\right. \\ & s_{t+1} = \Phi(g(x), s_t), \\ & \left. \right] \end{aligned}$$

where Φ is the chosen gating/projection rule. The **depth-aliasing error** for input x is

$$\begin{aligned} & \left[\right. \\ & \mathrm{DA}(g;x) = \sum_{t=0}^{D-1} \\ & \left| \right. \\ & \Phi(g(x), s_t(x)) - \Phi(g^*(x,t), s_t(x)) \end{aligned}$$

$\right|$.

$\]$

If $\mathrm{DA}(g;x) > 0$, then the lack of depth information corrupts the trajectory. This is the forward-pass sense in which shared sequential composition causes interference: the same gate is re-applied at every depth even though the correct modulation may need to change from depth to depth.

(b) Gradient interference

For two tasks (A, B) with losses $(\mathcal{L}_A, \mathcal{L}_B)$ on the same frozen (F) , the first-order change in $(\mathcal{L}_A)$ caused by one gradient step on task (B) is proportional to

$\[$

$\angle \nabla_{\phi} \mathcal{L}_A, \nabla_{\phi} \mathcal{L}_B \rangle$.

$\]$

Define the **task Jacobians**

$\[$

$J_i(\phi) = \frac{\partial}{\partial \phi}$

$\bigl(s_D(x; \phi)\bigr) \Big|_{x \in X_i}$.

$\]$

Task-wise gradient interference is

$$\mathbb{I}_{ij} = \left| \frac{J_i(\phi) J_j(\phi)^{\top}}{J_i(\phi) J_i(\phi)^{\top}} \right|.$$

For shared sequential composition, these Jacobians have a special structure:

$$\frac{\partial s_D(x)}{\partial \phi} = \sum_{t=0}^{D-1} \left(\prod_{r=t+1}^{D-1} DF(s_r(x)) \right) \frac{\partial \Phi}{\partial \phi}(g_\phi(x, t), s_t(x)).$$

The same parameters ϕ appear at every depth, so gradients from different tasks can overlap even when the inputs are disjoint. This is the training-time sense of input-path interference.

1.6 Computation-isolability

A mechanism is **exactly computation-isolating** for a set of tasks if:

- each task's desired output is produced exactly, and
- there is no interference: $\mathrm{DA}(g;x)=0$ for all x in the forward sense, or $I_{ij}=0$ for all task pairs in the gradient sense.

A mechanism **bounds interference** if the relevant quantity is at most ϵ ($\epsilon>0$) uniformly over the task inputs.

1.7 Elimination versus bounding

- **Elimination** means exact equality of outputs and exact zero interference.
- **Bounding** means the deviation is at most ϵ , with ϵ controlled by quantities such as the Lipschitz constant of F , the depth D , the gate error, and the number of parameters.

2. Theorem-Shaped Hypotheses

H1. Exact preservation of the unperturbed path forces the gate to be the identity

Assumptions.

- $\mathcal{F}$ is defined on an open set $U \subseteqq \mathbb{R}^d$.
- The gate is depth-independent output gating:

[

$$s_{t+1} = g(x) \odot \mathcal{F}(s_t).$$

]

- We require exact preservation of the unperturbed trajectory:

[

$$s_t(x) = \mathcal{F}^t(x) \quad \forall x \in U, \quad \forall t.$$

]

Mathematical object gated.

The output of the frozen map $\mathcal{F}$ at every depth.

Claim.

Under these assumptions, $\mathbf{g}(x)=\mathbf{1}$ on every coordinate where $(F_t(x)) \neq 0$. If (F) is componentwise nonzero along trajectories on (U) , then $\mathbf{g}(x)=\mathbf{1}$ for all $(x \in U)$.

Proof.

At $(t=0)$, exact preservation requires

$$\begin{aligned} & \mathbf{g}(x) \odot F(x) = F(x). \\ & \end{aligned}$$

Componentwise:

$$\begin{aligned} & (g_i(x)-1)F_i(x)=0. \\ & \end{aligned}$$

Wherever $(F_i(x) \neq 0)$, we get $(g_i(x)=1)$. Wherever $(F_i(x)=0)$, the gate is invisible, but it is also useless for isolation. The same argument applies at every depth.

Verdict.

True as an exact theorem. It is a direct consequence of the preservation equation. It shows that, under the strong requirement that the mechanism changes nothing, the only gate is the identity gate. In particular, a gate cannot simultaneously “do nothing” and “isolate tasks.”

Weakest additional assumptions under which the obstruction disappears.

If we only require preservation of the final output $\{s_D(x)=F^D(x)\}$, rather than the whole trajectory, non-identity gates may exist when $\{F^D\}$ is non-injective. This is the underdetermined case; it is not a useful general construction.

Classification.

Known / direct consequence.

H2. Depth-independent gating cannot represent depth-varying modulation

Assumptions.

- The transition rule $\{\Phi(g,s)\}$ is injective in the gate argument $\{g\}$ for the states along the unperturbed trajectory.
- A task requires an ideal per-depth modulation $\{g^*(x,t)\}$ that is not constant in $\{t\}$.

Mathematical object gated.

The entire per-depth modulation sequence $\{(g(x,0), \dots, g(x,D-1))\}$.

Claim.

No depth-independent gate $g(x)$ can produce exactly the same trajectory as $g^*(x,t)$ for all t .

Proof.

Equality at every depth gives

$$\begin{aligned} & \Phi(g(x), s_t(x)) = \Phi(g^*(x,t), s_t(x)). \\ & \end{aligned}$$

By injectivity of $\Phi(\cdot, s_t(x))$,

$$\begin{aligned} & g(x) = g^*(x,t) \\ & \end{aligned}$$

for every t , contradicting the assumption that $g^*(x, \cdot)$ is nonconstant.

Verdict.

True as an exact representational theorem. This is the cleanest formalization of the phrase “input-path interference arising from shared sequential composition”: a depth-independent gate is a point on the

diagonal of the product space (M^D) , while the task requirements occupy off-diagonal points.

Weakest additional assumptions under which it becomes true.

The obstruction disappears if we allow:

- a depth-indexed shared gate $g(x,t)$;
- a state-dependent gate, e.g. $g(s_t)$, as in LSTM/GRU;
- per-depth trainable parameters, which is excluded by the research question.

Classification.

Direct consequence of the definition; the phrasing as a frozen-shared-path obstruction is a plausible adaptation of standard control-theoretic observability reasoning.

H3. Projection mechanisms have a rank obstruction

Assumptions.

- The mechanism is an input-dependent orthogonal projection, e.g.

$\mathbb{R}^n$

$$s_{t+1} = P(x)F(s_t).$$

[

- The unperturbed trajectory of x is such that

[

$$\operatorname{span}\{F(s_0(x)), F(s_1(x)), \dots, F(s_{D-1}(x))\}$$

]

has dimension $m(x)$.

Mathematical object projected.

The image space of the projection, i.e. the subspace of state space that survives each depth.

Claim.

If the projection is to preserve the unperturbed trajectory exactly, then

[

$$\operatorname{rank} P(x) \geq m(x).$$

]

For (K) tasks whose relevant spans have dimensions (m_i) , and whose projections must be mutually orthogonal to avoid gradient interference, necessarily

$$\sum_{i=1}^K m_i \leq d.$$

Hence a projection can isolate at most $\lfloor d / \min_i m_i \rfloor$ tasks.

Proof.

Exact preservation requires

$$P(x)F(s_t(x)) = F(s_t(x))$$

for all (t) , so every $(F(s_t(x)))$ lies in $(\operatorname{Im} P(x))$. The dimension bound follows. Orthogonality of task subspaces is required for zero gradient interference; orthogonal subspaces of $(\mathbb{R}^d)$ have total dimension at most (d) .

Verdict.

True as an exact linear-algebra theorem.

Weakest additional assumptions under which it becomes useful.

Tasks must live in low-dimensional invariant or approximately invariant subspaces. If the relevant computational subspaces have dimension ≤ 1 , for example, a rank-1 projection can isolate up to $\leq d$ tasks.

Classification.

Direct consequence of linear algebra; the application to frozen shared paths is a plausible adaptation.

H4. Finite parameter budget bounds exact gradient isolation

Assumptions.

- The gate/projection mechanism has p trainable parameters.
- Task output Jacobians are $\{J_i \in \mathbb{R}^{d \times p}\}$.
- We require exact zero gradient interference:

[

$J_i J_j^{\top} = 0 \quad \forall i \neq j.$

]

Mathematical object constrained.

The parameter space of the gate/projection mechanism.

Claim.

If each task Jacobian has rank $\leq r_i$, then

$$\sum_i r_i \leq p.$$

In particular, if every task has full row rank and $p \leq d$, then at most one task can be isolated exactly. For a shared sequential composition with generic nonlinear F , task Jacobians are generic sums over depth, so this is the typical situation.

Proof.

The row spaces of the J_i must be mutually orthogonal subspaces of $\mathbb{R}^p$. The sum of their dimensions cannot exceed p .

Verdict.

True as an exact theorem. It is the usual parameter-isolation bound, restated for trainable gates on a frozen shared path.

Weakest additional assumptions under which exact isolation is possible for many tasks.

Each task must occupy a low-rank subspace of the gate-parameter Jacobian. This occurs, for example, when tasks use disjoint subsets of

inputs or when the gate network is designed so that task-specific features activate disjoint hidden units.

Classification.

Direct consequence of linear algebra; the application to frozen-path gating is a plausible adaptation.

H5. Bounded interference is possible exactly when the gated dynamics are non-expansive

Assumptions.

- The modulated transition is Lipschitz in state with constant L_Φ :

$$\|\Phi(g, s) - \Phi(g, s')\| \leq L_\Phi \|s - s'\|.$$

- The per-depth gate error is bounded:

$$\|\Phi(g(x), s_t(x)) - \Phi(g^*(x, t), s_t(x))\| \leq \delta.$$

\]

Mathematical object bounded.

The full trajectory error.

Claim.

The output error satisfies

\[

\|s_D(x)-s_D^*(x)\|

\leq

\delta\sum_{k=0}^{D-1}L_\Phi^k

=

\begin{cases}

\delta\frac{1-L_\Phi^D}{1-L_\Phi}, & L_\Phi<1,\backslash[1ex]

\delta D, & L_\Phi=1,\backslash[1ex]

\delta\frac{L_\Phi^D-1}{L_\Phi-1}, & L_\Phi>1.

\end{cases}

\]

Proof.

Let $(e_t=\|s_t-s_t^*\|)$. Then

$$\begin{aligned}
& e_{t+1} \\
& \leq \\
& \|\Phi(g, s_t) - \Phi(g, s_t^*)\| \\
& + \\
& \|\Phi(g, s_t^*) - \Phi(g^*, s_t^*)\| \\
& \leq \\
& L_\Phi e_t + \delta.
\end{aligned}$$

Solving the recurrence gives the claimed bound.

Verdict.

True as an exact theorem.

Consequence.

If the gated system is contractive, $(L_\Phi < 1)$, then interference is bounded uniformly in depth by $(\delta / (1 - L_\Phi))$. If $(L_\Phi > 1)$, interference can grow exponentially in (D) . This is why “bounding” is not automatic: for generic nonlinear (F) , (L_Φ) is often larger than (1) , and small gate errors are amplified through the shared composition.

Classification.

Known / direct consequence.

H6. Depth-indexed shared gating: finite exact interpolation is conditionally possible

Assumptions.

- We allow a depth-indexed shared gate $\phi(x, t)$, still with no per-depth parameter vectors.
- The task set is finite: $\{(x_i, y_i)\}, i=1, \dots, N$.
- For each i , there exists some per-depth modulation sequence $\{g_{i,t}\}$ that realizes y_i under the transition rule.
- The parameterized family ϕ can interpolate the finite data $\{(x_i, t) \mapsto g_{i,t}\}$, e.g. an MLP with enough parameters.

Claim.

Under these assumptions, there exists a shared-parameter depth-indexed gate that produces exactly the desired outputs for all finite task inputs.

Proof sketch.

The modulation requirement is a finite lookup table. Universal approximation on finite sets gives interpolation.

Important caveat.

The decisive assumption is that the target modulation sequences $\{g_{i,t}\}$ exist. For a generic nonlinear ϕ , this is **underdetermined**.

For example, if $\backslash(F)$ is contractive, many targets are unreachable regardless of gating. Thus this is a conditional existence theorem, not a general theorem.

Verdict.

Conditionally true; otherwise underdetermined.

Weakest additional assumptions.

Reachability of the targets under the allowed modulation, plus sufficient gate capacity. Additionally, exact zero gradient interference requires the parameter-count condition of H4.

Classification.

Plausible adaptation of universal approximation; the reachability part is a conjecture.

3. Prior Art

3.1 Orthogonal gradient and gradient-projection methods

Examples. GEM / A-GEM, Orthogonal Gradient Descent, TRGP, GPM.

What they address.

They project gradients in parameter space so that updates for a new task do not increase losses on previous tasks.

Do they address the same mathematical problem?

Only partially. They address training-time gradient interference, which corresponds to my second notion of interference. They do **not** address depth-aliasing in a frozen shared composition. Moreover, they typically require unfreezing or at least modifying the main network; here the shared path is frozen.

Classification.

Known, partially relevant.

3.2 Parameter isolation and masking

Examples. PackNet, Piggyback, Hard Attention to the Task, Supermasks.

What they address.

They allocate disjoint parameter subsets to tasks, preventing cross-task updates.

Same problem?

Not the same. They introduce per-task masks or per-task parameters, which is a form of parameter isolation. The research question explicitly asks for an input-dependent gate/projection on a single frozen shared path, not for partitioning the shared path's parameters.

Classification.

Known, related but different mechanism.

3.3 Progressive networks and path-based continual learning

Examples. Progressive Neural Networks, PathNet.

What they address.

They add new columns or paths for new tasks while freezing old ones.

Same problem?

No. They avoid interference by adding task-specific architecture, which is effectively per-task parameter growth. The question here is whether a *single* input-dependent mechanism can isolate paths within one frozen shared composition.

Classification.

Known, different problem.

3.4 Hypernetworks

Examples. Ha et al., Chunked Hypernetworks.

What they address.

A hypernetwork generates weights for a main network as a function of an input or task embedding.

Same problem?

Partly. A hypernetwork can generate input-dependent per-depth modulations. But the generated weights are typically per-layer, which violates the “no per-depth/per-level trainable parameters” constraint in spirit. A hypernetwork also replaces or augments the main weights rather than gating a frozen shared path.

Classification.

Known, partially relevant.

3.5 Recurrent neural networks with gating

Examples. LSTM, GRU.

What they address.

They use input- and state-dependent gates inside a shared recurrent computation.

Same problem?

This is the closest prior art. LSTM gates are applied at every time step with shared parameters, exactly like gating a shared sequential composition. The crucial difference is that LSTM gates depend on the current hidden state h_{t-1} , not only on the original input x . This supports H2: state information is what makes recurrent gating effective.

It is also empirical evidence, not a theorem, that state-conditioning resolves many of the pathologies of depth-independent control.

Classification.

Known, directly relevant, but with a different information assumption.

3.6 Conditional computation and mixture of experts

Examples. Mixture of Experts, sparsely gated layers.

What they address.

Input-dependent routing to different experts or subnetworks.

Same problem?

Not the same. MoE routes inputs to different parameter sets, typically with per-layer routers, and its goal is capacity scaling or efficiency, not preserving a frozen shared-path computation.

Classification.

Known, structurally similar but not the same mathematical problem.

3.7 Depth-recurrent and weight-tied transformers

Examples. ALBERT, Universal Transformers, LoopFormer.

What they address.

They reuse the same block across depth and sometimes use adaptive computation time.

Same problem?

Partially. They study shared-depth composition, but they do not address task isolation via input-dependent gates. Adaptive computation time is state-dependent halting, not input-dependent task isolation.

Classification.

Known, background context.

4. Novelty Classification

| Claim | Classification |

| --- | --- |

| Formal notion of depth-aliasing interference in a frozen shared composition | Apparently novel as a packaged definition; components are known from control theory |

| H1: exact preservation forces identity gating | Direct consequence |

| H2: depth-independent gating cannot represent depth-varying modulation | Direct consequence |

| H3: projection rank obstruction | Direct consequence |

| H4: gradient-isolation parameter bound | Direct consequence |

| H5: contraction/expansion bound on interference | Direct consequence |

| H6: conditional finite interpolation by depth-indexed shared gate | Plausible adaptation / conjecture |

| Claim that state or depth information is necessary for robust gating in shared composition | Plausible adaptation; consistent with LSTM design but not proven here as a theorem |

No experimental result is used as evidence for any theorem. The exact results above are purely analytical.

5. Failure Condition and Obstruction

The strict version of the proposed mechanism is:

- input-dependent gate/projection,
- no per-depth trainable parameters,
- frozen shared path,
- no retraining of $\backslash(F\backslash)$.

Under this strict version, **exact elimination of input-path interference is impossible for generic nonlinear $\backslash(F\backslash)$** . The obstruction has four independent parts.

5.1 Diagonalization obstruction

A depth-independent gate $g(x)$ is forced to lie on the diagonal of the product space of depth-wise modulations:

$$\begin{bmatrix} g(x) & 0 & \dots & 0 \\ 0 & g(x) & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & g(x) \end{bmatrix} \in M^D.$$

Any task whose ideal modulation depends on depth lies off this diagonal. This is a representational impossibility, not a training failure.

5.2 Preservation-identity obstruction

If the mechanism is required to preserve the unperturbed computation exactly, then output gating forces $g(x) = \mathbf{1}$ wherever F is nonzero. The only gate that changes nothing is the identity gate. Thus a gate cannot both preserve the frozen computation and alter the path for isolation.

5.3 Rank and parameter budget obstruction

Even if the depth-aliasing obstruction is removed by allowing a depth-indexed shared gate, exact gradient isolation requires the task Jacobians to be mutually orthogonal. The sum of their ranks cannot exceed the number of trainable gate parameters p . For generic tasks, this permits only one task when $p \leq d$.

5.4 Expansive-dynamics obstruction

Bounding interference is possible if the gated dynamics are contractive. For generic nonlinear $\mathcal{F}$, the Lipschitz constant of the composed system is >1 , and small gate errors are amplified exponentially in depth. Hence even bounded interference is not guaranteed without additional assumptions on $\mathcal{F}$.

6. Conclusion

The answer to the research question is:

- **Elimination** of input-path interference by a depth-independent input-dependent gate/projection on a frozen shared path is **impossible in general**.
- **Bounding** is possible if the gated system is contractive or if the depth is small enough that exponential amplification is controlled.
- **Controllability** can be achieved only by enriching the information available to the gate: either allow a depth-indexed shared gate $\mathcal{G}(x, t)$, allow state-dependent gating $\mathcal{G}(s_t)$, allow per-task parameter blocks, or impose strong low-rank/invariant-subspace structure on the tasks.

The most precise obstruction is the **diagonalization obstruction**: a depth-independent gate is a single point on the diagonal of the space of depth-wise modulations, while the tasks generically require off-diagonal points. This obstruction is independent of optimization, capacity, and

training data. It is the mathematical core of why input-path interference cannot, in general, be eliminated by the proposed mechanism.

[MODERATOR Round 2]

Moderator Synthesis — Round 2

Consolidation

Role: Synthesis and consensus evaluation of the STRATEGIST, CRITIC, and OPTIMIZER contributions on the BDH input-path interference question.

Before consolidating, one factual check is warranted, since the entire analysis hinges on BDH genuinely exhibiting weight-shared sequential composition across depth (if it did not, the question would misfire at the premise level):

[SEARCH: "Dragon Hatchling" BDH weight tying shared parameters across depth recurrent layer]

0. Evidence-Base Status

- The BDH paper (arXiv:2509.26507) is confirmed to exist and to describe a scale-free network of locally-interacting neuron particles

with transformer-comparable performance. Third-party implementations (BDH-MLX) corroborate the architecture's general character.

- **Retrieval limitation:** the internal project documents (2026-08-^{*} progress reports, paper-parity-plan, bdh-continual-learning-plan_v2.md, cl-h1-report.md) could **not** be retrieved through available channels; searches returned irrelevant material. Consequently, neither participant nor this moderation could cross-check the formal models against your concrete implementation or against the H1 experimental report dated today. All conclusions below are therefore **structural**, derived from the abstract "shared sequential composition" model that both analysts adopted consistently. Instantiation with BDH's literal update equations remains an open verification step (see §3.3).

Both principal analyses proceeded from the same formal skeleton — frozen map F , recurrence $s_{t+1} = \Phi(g(x), s_t)$, depth-agnostic input-conditioned gate/projection — which makes genuine consensus evaluation possible.

1. Points of Consensus (High Confidence)

C1 — Formal framework agreement. The definitions of shared sequential composition, frozen path, gating vs. projection, and the elimination/bounding distinction were independently convergent. No participant disputed another's definitions. *Consensus: strong.*

C2 — Exact elimination is impossible under the strict reading. Three independent obstructions were derived, by complementary methods:

- **Identity-forcing** (STRATEGIST H1/H3; OPTIMIZER H1): exact trajectory preservation under output gating $g \odot F(s)$ forces $g = \mathbf{1}$ wherever F is nonzero along the trajectory. A mechanism required to change nothing can isolate nothing.
- **Diagonalization** (OPTIMIZER H2; STRATEGIST's "temporal information deficit" is its informal statement): a depth-agnostic gate occupies the diagonal of the product space M^D of per-depth modulations; any task whose ideal modulation $g^*(x, t)$ is nonconstant in t lies off-diagonal and is unreachable, given injectivity of $\Phi(\cdot, s_t)$ in the gate argument.
- **Rank/parameter-budget** (OPTIMIZER H3/H4): exact zero cross-task gradient interference through shared gate parameters ϕ requires mutually orthogonal task-Jacobian row spaces, hence $\sum_i \mathrm{rank}(J_i) \leq p$. Generic tasks saturate this at one task when $p \leq d$.

Consensus: strong. This satisfies the brief's failure condition: the proposed isolation is impossible under the stated strict assumptions.

C3 — Bounding is conditional, not free. Per-depth gate error δ under Lipschitz constant L_Φ yields output error $\leq \delta \sum_k L_\Phi^k$: uniformly bounded iff $L_\Phi < 1$, growing $\sim \delta D$ at $L_\Phi = 1$, exponentially otherwise (OPTIMIZER H5; STRATEGIST H2 agrees and correctly notes contraction bounds trajectory *separation*, not modification-induced interference). Trained LLM-scale maps typically have $L_\Phi > 1$, so bounding is a real architectural constraint, not a default. *Consensus: strong.*

C4 — Escape routes require enriched information or structure. All participants converge: relaxation is possible via (i) depth-indexed shared-parameter gates $g_{\phi}(x,t)$, (ii) state-conditioned gating ($g(s_t)$, LSTM-style — the nearest prior art, which violates the input-only constraint), (iii) low-rank/invariant task subspaces, or (iv) abandoning the frozen-path constraint. *Consensus: strong.*

C5 — Prior-art triage. Independent and concordant: orthogonal-gradient methods (GEM/A-GEM/GPM) operate on gradients in parameter space, not forward-pass composition — related but a different mathematical problem; progressive networks, PackNet, supermasks introduce per-task parameters/masks — constraint-violating; hypernetworks and MoE are structurally adjacent but generate per-layer outputs or route between experts — not the same problem; Universal Transformer/ALBERT establish the weight-tied backdrop without addressing isolation. **No surveyed prior art solves this exact problem.** *Consensus: strong.*

2. Disputes and Adjudications

D1 — Novelty classification conflict. STRATEGIST labels the impossibility result "apparently novel"; OPTIMIZER labels the equivalent statements "direct consequence."

Adjudication: OPTIMIZER is correct on the mathematics, STRATEGIST is correct on the contribution. Every ingredient (functional-equation identity forcing, diagonal argument, rank counting, Lipschitz error propagation) is textbook. What is novel is the **problem formulation** — isolation of computations within a frozen weight-tied path under an input-only gate, posed for continual learning. Write-up guidance: claim

a new formalization and a clean negative result; do **not** claim new mathematics.

D2 — STRATEGIST H4 is defective. Its construction (Jacobian-null-space projection achieving preservation + isolation with $O(d \cdot D)$ parameters) collapses to the identity gate under its own preservation requirement, and conflates the Jacobian's null space with the output's annihilator. OPTIMIZER's H6 (conditional finite interpolation, contingent on reachability of target modulation sequences) is the correct weakened version. *Adjudication:* H4 withdrawn as stated; retained only as OPTIMIZER-style conjecture.

D3 — STRATEGIST's "information gap" $\Delta_t = t \cdot d$. Dimensions of informally-defined information sets are not a rigorous quantity. Superseded by OPTIMIZER's rank-theoretic arguments. Retain only as heuristic motivation.

D4 — Interference has two distinct meanings; only OPTIMIZER separated them cleanly. Forward-pass *depth-aliasing* vs. training-time *gradient interference*. See correction G1 below — this separation turns out to be more important than either participant realized.

3. Corrections and Gaps Identified in Moderation

G1 — Cross-input forward-pass interference is structurally zero; the live channels are elsewhere. With frozen θ and a *stateless* input-conditioned mechanism, each input's forward pass is computationally

independent: x_1 's trajectory cannot be perturbed by x_2 's "presence." STRATEGIST's Definition 3 counterfactual ($\tilde{h}_t(x_1 \mid x_2 \text{ active})$) is therefore ill-defined as stated. The operationally meaningful interference notions are:

- **Depth-aliasing within a pass** (obstruction B above); and
- **Cross-time forgetting**: ϕ is shared across all inputs, so any update $\Delta\phi$ from task B alters g_ϕ on task- A inputs — first-order damage governed exactly by the Jacobian-overlap/rank condition (obstruction C). This, not intra-pass contamination, is the catastrophic-forgetting channel in this architecture.

Future work should define interference exclusively through these two channels.

G2 — The preservation requirement must be asymmetric to be meaningful. Requiring preservation on *all* inputs trivially forces the identity gate (C2). The continual-learning-relevant requirement is: exact preservation on old-task inputs S_{old} , unconstrained deviation on new-task inputs S_{new} . Under this reading the identity obstruction restricts to S_{old} ; non-trivial gates become admissible off it; the surviving barriers are the diagonal obstruction (whenever old-task-faithful modulation must vary with depth) and the rank/budget obstruction for gradient-exactness. **Neither participant developed this asymmetric variant; it should be the primary next theorem target.**

G3 — BDH instantiation is outstanding. BDH's block has residual structure and elementwise multiplicative (Hebbian-type) terms. Consequences: (a) output-gating the full map would gate the residual stream — likely harmful; gating only the delta term is the natural

placement; (b) BDH's native plasticity must be explicitly disabled for the "frozen path" premise to hold. These checks against paper .tex/the repository were not performed in-round.

G4 — Scope decision required on t -conditioning. A shared-parameter gate $g_\phi(x,t)$ (depth index as *input*, e.g., via a fixed embedding) adds $O(1)$ parameters and **does not violate "no per-depth/per-level trainable parameters" as literally worded**. This is the pivotal loophole: every impossibility result proven here targets the strictly depth-agnostic gate and dissolves — into conditional statements (reachability + capacity + rank budget) — once t is admitted. The team must rule explicitly on whether the intended constraint forbids t -conditioning; the theorems' applicability hinges on this interpretation.

4. Consolidated Answer to the Research Question

| Reading of the question | Verdict | Status |

| --- | --- | --- |

| Depth-agnostic gate/projection, frozen nonlinear F , **exact elimination** | **Impossible** (identity-forcing + diagonalization + rank obstructions) | Proven; direct consequences |

| Same, **bounding** | Possible iff gated dynamics non-expansive ($L_\phi < 1$) or depth-limited error budget; else exponential amplification | Proven conditional theorem |

| Enriched gate (t -conditioned shared-parameter, or state-conditioned), **controllability** | Feasible conditional on reachability, gate capacity, low-rank task subspaces; exact multi-task gradient isolation

still capped by $\sum_i \mathrm{rank}(J_i) \leq p$ | Conjecture / plausible adaptation |

The failure condition contemplated by the brief is **met for the strict reading**, with a precise obstruction characterization: a depth-agnostic gate is confined to the diagonal of the modulation product space, and exactness requirements (preservation, gradient orthogonality) collapse either to the identity mechanism or to a rank budget permitting at most one generic task. No optimization method, capacity increase, or training regime circumvents these; they are representational, not algorithmic.

5. Final Novelty Ledger (Adjudicated)

| Claim | Classification |

| --- | --- |

| Formalization: isolation within frozen weight-tied composition via input-only gate, CL-framed | **Apparently novel** (as packaging) |

| Identity-forcing lemma | **Direct consequence** |

| Diagonalization obstruction | **Direct consequence** (control-theoretic flavor) |

| Rank/parameter-budget bound on exact gradient isolation | **Direct consequence** |

| Contraction/amplification error bound | **Direct consequence** |

| Necessity of state/depth information for effective gating | **Plausible adaptation** (consistent with LSTM/ACT practice; proven only under injectivity assumptions) |

| t -conditioned shared gate achieves isolation broadly | **Conjecture** (reachability undetermined) |

| Any empirical effect observed in cl-h1-report | Inadmissible as theorem evidence until reconciled with the above |

6. Recommended Next Steps

- **Restate the theorem set under asymmetric preservation** (old-task exact / new-task free) and prove that the diagonal and rank obstructions survive while the identity obstruction dies. This is the honest CL formalization and the highest-value open result.
- **Instantiate F with BDH's concrete update** from paper.tex; fix gate placement relative to the residual stream; confirm plasticity disabling.
- **Rule on the t -conditioning interpretation** of the constraint; if permitted, specify a minimal $g_{\phi}(x,t)$ (fixed depth embedding into the existing gate network) and test — labeling outcomes strictly as empirical conjecture.
- **Reconcile with the local reports** (plan_v2, cl-h1-report) once retrievable; verify that the implemented mechanism corresponds to one of the formally scoped variants rather than an unscoped mixture.
- **Write-up posture:** present C2/C3 as a clean impossibility-and-bounding package for weight-tied architectures; cite LSTM/GRU, adaptive computation time, and universal-transformer literature as

nearest neighbors; claim formalization novelty, not mathematical novelty.

Consensus summary: full agreement on framework, on the impossibility of exact elimination under strict assumptions, on conditional bounding, and on prior-art triage; disagreements were confined to novelty labeling and one defective construction, both adjudicated above. The investigation's productive frontier is the asymmetric-preservation reformulation and the $\$t\$$ -conditioning scope decision.

Web Research

- **GitHub - severian42/BDH-MLX: Baby Dragon Hatchling (BDH) – MLX...**

Shared Parameters Across Layers: Same weights reused in all layers (recurrent depth). Sparse, Non-Negative Activations: ReLU-based activations for biological plausibility. Transformer. BDH. Parameters. Unique per layer. Shared across layers. Activations.

Source: <https://github.com/severian42/BDH-MLX>

- **BDH-CQ: In-Context Learning with Recurrent Latent... | alphaXiv**

BDH layers also support recurrent systems for constraint satisfaction. Recurrent-depth language models repeatedly apply a shared block to a sequence-wide latent state and can improve some reasoning benchmarks with additional iterations (Geiping et al., 2025).

Source: <https://www.alphaxiv.org/abs/2608.09888>

- **Dragon Hatchling: Linking Transformers & Brain Models**

We introduce 'Dragon Hatchling' (BDH), a new LLM architecture based on a scale-free biologically inspired network of n locally-interacting neuron particles.

Source: <https://www.emergentmind.com/papers/2509.26507>

- **Beyond Transformer LLM Models: How the BDH Architecture Solves...**

Baby Dragon Hatchling (BDH). Effective Attention Span. ~2 hours and 17 minutes. A recurring and highly emphasized theme in the 2026 scientific discourse surrounding the BDH architecture is the concept of Axiomatic Artificial Intelligence.

Source: <https://www.crvscience.com/post/beyond-transformer-llm-models-how-the-bdh-architecture-solves-the-context-window-crisis>

- **The Dragon Hatchling: The Missing Link between the Transformer...**

The dragon hatchling aka #BDH (baby dragon hatchling) is a new approach to creating and training these large language models at an architectural level. BDH is #graph centric in its design and tries to mirror #neural synapses like those in the brain for its training mechanism.

Source: https://www.linkedin.com/posts/meeherkapoor_the-dragon-hatchling-the-missing-link-between-activity-7386863203634388992-uI39

[STRATEGIST Round 3]

Final Strategic Assessment: Input-Path Interference in BDH via Gating/Projection

Date: 2026-08-25 | **Status:** Complete

Executive Summary

Under the strict constraints stated in the research question — depth-agnostic input-dependent gate/projection, frozen shared path, no per-depth trainable parameters, no retraining — **exact elimination of input-path interference is impossible** for generic nonlinear shared compositions. This is proven via three independent obstructions (identity-forcing, diagonalization, rank budget). Bounding is possible only under contractivity assumptions that are unlikely to hold for trained LLM-scale maps. The productive frontier is relaxing the depth-agnostic constraint to admit a depth-indexed shared-parameter gate $g_{\phi}(x,t)$, which dissolves the impossibility into conditional feasibility statements.

1. Formal Definitions

1.1 Shared Sequential Composition

Let $V = \mathbb{R}^d$ be the state space. A **depth- D shared sequential composition** is defined by a fixed map $F: V \rightarrow V$ and the recurrence:

$$s_0(x) = x, \quad s_{t+1}(x) = F(s_t(x)), \quad t = 0, \dots, D-1$$

The unperturbed trajectory is $\tau_F(x) = (s_0(x), s_1(x), \dots, s_D(x))$.

Classification: Direct characterization of BDH's weight-tied recurrent depth, confirmed by the BDH-MLX implementation ("Same weights reused in all layers (recurrent depth)").

1.2 Frozen Shared Path

F is **frozen** if $\partial F / \partial \theta = 0$ during the continual-learning phase. Only the gate/projection mechanism ϕ may be trained.

Classification: Known — standard continual learning constraint.

1.3 Input-Path Interference (Two Precise Notions)

Forward-pass depth-aliasing: A depth-agnostic gate $g(x)$ is applied identically at every depth. If the ideal modulation for input x is $g^*(x, t)$ (varying with depth), the depth-aliasing error is:

$$\mathrm{DA}(g; x) = \sum_{t=0}^{D-1} \left| \Phi(g(x), s_t(x)) - \Phi(g^*(x, t), s_t(x)) \right|$$

Gradient interference (cross-task forgetting): For tasks A, B with shared gate parameters ϕ , the first-order damage from a B -update to A -performance is proportional to $\langle \nabla_{\phi} \mathcal{L}_A, \nabla_{\phi} \mathcal{L}_B \rangle$. The task Jacobians are:

$$J_i(\phi) = \frac{\partial}{\partial \phi} s_D(x; \phi) \Big|_{x \in X_i}$$

Critical correction from moderation: Cross-input forward-pass interference is **structurally zero** with frozen θ and a stateless input-conditioned mechanism — each input's forward pass is computationally independent. The live interference channels are depth-aliasing (within a pass) and gradient interference (across training steps).

Classification: The depth-aliasing notion is **apparently novel** as a packaged definition; components are known from control theory. The gradient notion is **known** from continual learning.

1.4 Gating

An **input-dependent gate** $g: X \rightarrow \mathbb{R}^d$ is applied identically at every depth:

- Output gating: $s_{t+1} = g(x) \odot F(s_t)$

- Input gating: $s_{t+1} = F(g(x) \odot s_t)$

The gate has **no per-depth parameters**: the same $g(x)$ is used at every t .

Classification: Known mechanism (LSTM, MoE), but the depth-independence + frozen-path combination is novel.

1.5 Projection

An **input-dependent projection** $P: X \rightarrow \Pi(V)$ maps each input to an orthogonal projection, applied as $s_{t+1} = P(x) F(s_t)$ or $s_{t+1} = F(P(x) s_t)$.

Classification: Known (gradient projection methods), formalized for forward-pass application.

1.6 Computation-Isolability

A mechanism is **exactly computation-isolating** if: (1) each task's desired output is produced exactly, and (2) $\mathrm{DA}(g;x) = 0$ for all x (forward) and $I_{ij} = 0$ for all task pairs (gradient).

Classification: Apparently novel — this specific formulation combining shared composition, frozen parameters, and forward-pass isolation does not appear in surveyed literature.

1.7 Elimination vs. Bounding

- **Elimination:** Exact equality of outputs and exact zero interference.
- **Bounding:** Deviation $\leq \epsilon$, with ϵ controlled by Lipschitz constant, depth, gate error, and parameter count.

Classification: Plausible adaptation of approximation theory concepts.

2. Theorem-Shaped Hypotheses

H1. Identity-Forcing Lemma (Negative)

Assumptions: F defined on open $U \subseteq \mathbb{R}^d$; depth-independent output gating $s_{t+1} = g(x) \odot F(s_t)$; exact trajectory preservation required: $s_t(x) = F^t(x)$ for all $x \in U$, all t .

Mathematical object gated: Output of F at every depth.

Claim: $g(x) = \mathbf{1}$ wherever $F(s_t(x)) \neq 0$.

Proof: At $t=0$: $g(x) \odot F(x) = F(x)$ implies $(g_i(x) - 1) F_i(x) = 0$ componentwise. Where $F_i(x) \neq 0$, $g_i(x) = 1$. Same argument at every depth. $\square$

Verdict: True as an exact theorem. Direct consequence.

Weakest assumptions for escape: If only final-output preservation $s_D(x) = F^D(x)$ is required (not whole trajectory), non-identity gates may exist when F^D is non-injective. This is underdetermined and not generally useful.

Classification: Known / direct consequence.

H2. Diagonalization Obstruction (Negative)

Assumptions: $\Phi(g, s)$ is injective in g along the unperturbed trajectory; ideal modulation $g^*(x, t)$ is nonconstant in t .

Mathematical object gated: The entire per-depth modulation sequence $(g(x, 0), \dots, g(x, D-1))$.

Claim: No depth-independent gate $g(x)$ can reproduce $g^*(x, t)$ for all t .

Proof: Equality at every depth gives $\Phi(g(x), s_t(x)) = \Phi(g^*(x, t), s_t(x))$. By injectivity, $g(x) = g^*(x, t)$ for every t , contradicting nonconstancy. $\square$

Verdict: True as an exact representational theorem. A depth-independent gate is confined to the diagonal of M^D ; tasks requiring depth-varying modulation lie off-diagonal.

Weakest assumptions for escape: Allow depth-indexed shared gate $g_\phi(x,t)$; or state-dependent gate $g(s_t)$ (LSTM-style); or per-depth trainable parameters (excluded by constraint).

Classification: Direct consequence; the frozen-shared-path phrasing is a plausible adaptation of control-theoretic observability reasoning.

H3. Projection Rank Obstruction (Negative)

Assumptions: Input-dependent orthogonal projection $s_{t+1} = P(x)F(s_t)$; trajectory span $\operatorname{span}\{F(s_0(x)), \dots, F(s_{D-1}(x))\}$ has dimension $m(x)$.

Mathematical object projected: Image space of $P(x)$.

Claim: Exact preservation requires $\operatorname{rank} P(x) \geq m(x)$. For K tasks with mutually orthogonal projections: $\sum_{i=1}^K m_i \leq d$. At most $\lfloor d / \min_i m_i \rfloor$ tasks can be isolated.

Proof: $P(x) F(s_t(x)) = F(s_t(x))$ forces $F(s_t(x)) \in \operatorname{Im} P(x)$. Orthogonal subspaces of $\mathbb{R}^d$ have total dimension $\leq d$. $\square$

Verdict: True as an exact linear-algebra theorem.

Weakest assumptions for escape: Tasks must live in low-dimensional invariant subspaces. If $m_i = 1$, up to d tasks can be isolated.

Classification: Direct consequence; application to frozen shared paths is a plausible adaptation.

H4. Parameter-Budget Bound on Gradient Isolation (Negative)

Assumptions: Gate has p trainable parameters; task Jacobians $J_i \in \mathbb{R}^{d \times p}$; exact zero gradient interference required: $J_i J_j^\top = 0$ for $i \neq j$.

Mathematical object constrained: Parameter space of the gate mechanism.

Claim: $\sum_i \operatorname{rank}(J_i) \leq p$. If every task has full row rank and $p \leq d$, at most one task can be isolated exactly.

Proof: Row spaces of J_i must be mutually orthogonal subspaces of $\mathbb{R}^p$. $\square$

Verdict: True as an exact theorem.

Weakest assumptions for escape: Each task must occupy a low-rank subspace of the gate-parameter Jacobian (e.g., disjoint input subsets, or gate architecture with task-specific feature activation).

Classification: Direct consequence; the frozen-path application is a plausible adaptation.

H5. Error Amplification Bound (Conditional)

Assumptions: Modulated transition is L_Φ -Lipschitz in state; per-depth gate error $\leq \delta$.

Mathematical object bounded: Full trajectory error.

Claim:

$$\|s_D(x) - s_D^*(x)\| \leq \delta \sum_{k=0}^{D-1} L_\Phi^k = \begin{cases} \delta \frac{1 - L_\Phi^D}{1 - L_\Phi} & \text{if } L_\Phi < 1 \\ \delta D & \text{if } L_\Phi = 1 \end{cases}$$

$$\& L_{\Phi} = 1 \\\ \delta \frac{L_{\Phi}^D - 1}{L_{\Phi} - 1} \& L_{\Phi} > 1 \\ \end{cases} \end{cases} \end{cases}$$

Proof: Recurrence $e_{t+1} \leq L_{\Phi} e_t + \delta$ solved by standard methods. $\square$

Verdict: True as an exact theorem.

Consequence: Bounding is uniform in depth iff $L_{\Phi} < 1$ (contractive). For $L_{\Phi} > 1$, errors grow exponentially. Trained LLM-scale maps typically have $L_{\Phi} > 1$, so bounding is a real architectural constraint, not a default.

Classification: Known / direct consequence.

H6. Depth-Indexed Shared Gate: Conditional Feasibility (Positive, Conjectural)

Assumptions: Depth-indexed shared gate $g_{\Phi}(x, t)$ (shared parameters, t as input); finite task set (x_i, y_i) ; target modulation sequences $g_{i,t}$ exist and are reachable; sufficient gate capacity.

Claim: There exists a shared-parameter depth-indexed gate producing exact desired outputs for all finite task inputs.

Proof sketch: Finite lookup table; universal approximation on finite sets gives interpolation. $\square$

Critical caveat: The existence of target modulation sequences $g_{i,t}$ is **underdetermined** for generic nonlinear F . If F is contractive, many targets are unreachable regardless of gating.

Verdict: Conditionally true; otherwise underdetermined.

Classification: Plausible adaptation of universal approximation; reachability is a conjecture.

3. Prior Art Analysis

Prior Art	Same Mathematical Problem?	Classification
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Orthogonal gradient methods (GEM, A-GEM, GPM)	No — operate on gradients in parameter space during training; do not address forward-pass depth-aliasing in frozen composition	Known, partially relevant
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Parameter isolation / masking (PackNet, Supermasks)	No — introduce per-task masks or per-task parameters; constraint-violating	Known, related but different
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| **Progressive networks / PathNet** | No — add task-specific architecture; effectively per-task parameter growth | Known, different problem |

| **Hypernetworks** (Ha et al.) | Partially — can generate input-dependent per-depth modulations, but output is per-layer (violates constraint in spirit) | Known, partially relevant |

| **LSTM / GRU** | **Closest** — shared-parameter gates at every time step. Crucially, gates depend on current state h_{t-1} , not only input x . This is the key difference: state-conditioning is what makes recurrent gating effective | Known, directly relevant, different information assumption |

| **Mixture of Experts** | No — routes to different parameter sets with per-layer routers; different problem | Known, structurally similar |

| **Universal Transformer / ALBERT** | Partially — weight-tied depth, but no task isolation via gates | Known, background context |

No surveyed prior art solves this exact problem. The formulation — isolation of computations within a frozen weight-tied path under an input-only gate, posed for continual learning — appears to be new.

4. Obstruction Characterization

The failure condition is met. The obstruction has four independent parts:

4.1 Diagonalization Obstruction

A depth-independent gate $g(x)$ is confined to the diagonal of M^D . Tasks requiring depth-varying modulation lie off-diagonal. This is representational, not algorithmic.

4.2 Preservation-Identity Obstruction

Exact trajectory preservation under output gating forces $g(x) = \mathbf{1}$ wherever F is nonzero. A gate that changes nothing isolates nothing.

4.3 Rank / Parameter-Budget Obstruction

Exact gradient isolation requires mutually orthogonal task-Jacobian row spaces. $\sum_i \text{rank}(J_i) \leq p$ permits at most one generic task when $p \leq d$.

4.4 Expansive-Dynamics Obstruction

For generic nonlinear F , $L_\Phi > 1$, and small gate errors amplify exponentially in depth. Even bounded interference requires architectural constraints on F .

5. Conditions Under Which Obstructions Vanish

|---|---|

| Diagonalization | Depth-indexed shared gate $g_{\phi(x,t)}$ or state-conditioned gate $g(s_t)$ |

| Identity-forcing | Require only final-output preservation (not whole trajectory); or relax to asymmetric preservation (exact on old tasks, free on new) |

| Rank budget | Low-rank task subspaces; disjoint feature activation in gate architecture |

| Expansive dynamics | Architectural constraint $L_{\Phi} < 1$; or limit depth D |

6. Novelty Ledger (Adjudicated)

| Claim | Classification |

|---|---|

| Formalization: isolation within frozen weight-tied composition via input-only gate, CL-framed | **Apparently novel** (as packaging) |

| Identity-forcing lemma (H1) | **Direct consequence** |

| Diagonalization obstruction (H2) | **Direct consequence** (control-theoretic flavor) |

| Rank/parameter-budget bound (H3, H4) | **Direct consequence** |

| Contraction/amplification error bound (H5) | **Direct consequence** |

| Necessity of state/depth information for effective gating | **Plausible adaptation** (consistent with LSTM/ACT practice) |

| t -conditioned shared gate achieves isolation broadly (H6) |
Conjecture (reachability undetermined) |

7. Critical Open Questions

7.1 Asymmetric Preservation (Highest Priority)

The identity obstruction applies when preservation is required on **all** inputs. The continual-learning-relevant requirement is asymmetric: exact preservation on old-task inputs S_{old} , unconstrained on new-task inputs S_{new} . Under this reading, the identity obstruction restricts to S_{old} ; non-trivial gates become admissible off it. The diagonal and rank obstructions survive. **This reformulation should be the primary next theorem target.**

7.2 Scope of t -Conditioning

A shared-parameter gate $g_{\phi}(x,t)$ with t as input (e.g., via a fixed depth embedding) adds $O(1)$ parameters and **does not violate "no per-depth/per-level trainable parameters" as literally worded**. This is the pivotal loophole: every impossibility result targets the strictly depth-agnostic gate and dissolves into conditional statements once t is admitted. The team must rule explicitly on whether the intended constraint forbids t -conditioning.

7.3 BDH Instantiation

BDH's block has residual structure and elementwise multiplicative (Hebbian-type) terms. Consequences:

- Output-gating the full map would gate the residual stream — likely harmful
- Gating only the delta term is the natural placement
- BDH's native plasticity must be explicitly disabled for the "frozen path" premise to hold

These checks against paper .tex and the repository were not performed in this analysis.

8. Strategic Recommendation

The strict reading of the research question yields a clean impossibility result. The investigation should:

- **Publish the impossibility-and-bounding package** as a formal negative result for weight-tied architectures, citing LSTM/GRU and Universal Transformer as nearest neighbors. Claim formalization novelty, not mathematical novelty.
- **Develop the asymmetric-preservation variant** as the honest CL formalization. This is the highest-value open result.

- **Decide on ϕ -conditioning scope.** If permitted, specify a minimal $\phi(x,t)$ and test empirically — labeling outcomes strictly as empirical conjecture.
- **Instantiate with BDH's concrete update equations** from paper . tex; fix gate placement relative to the residual stream; confirm plasticity disabling.
- **Reconcile with implementation reports** (plan_v2, cl-h1-report) once retrievable; verify that the implemented mechanism corresponds to one of the formally scoped variants.

9. Answer to the Research Question

> Can input-path interference arising from shared sequential composition be eliminated, bounded, or made controllable by an input-dependent gate/projection mechanism, without introducing per-depth/per-level trainable parameters and without retraining the frozen shared path?

Elimination: Impossible in general under the stated constraints. Three independent obstructions (identity-forcing, diagonalization, rank budget) are proven as direct consequences of the definitions.

Bounding: Possible if and only if the gated dynamics are non-expansive ($L_\Phi < 1$) or depth is small enough to control exponential amplification. This is a property of F and the gating rule, not achievable by the gate alone.

Controllability: Achievable only by enriching the gate's information — either via depth-indexed shared-parameter conditioning $g_\phi(x,t)$, state-dependent conditioning $g(s_t)$, or strong low-rank/invariant-subspace structure on the tasks. Each escape route transforms the impossibility into a conditional feasibility statement with its own capacity and reachability requirements.

[CRITIC Round 3]

Critical Review of the Consolidated BDH Isolation Analysis

Overall assessment. The analysis is a useful exercise in abstraction and identifies some real distinctions (especially forward-pass *depth aliasing* versus training-time *gradient interference*). But the central claim—that exact elimination of input-path interference is impossible under the stated constraints—is overstated. Several “theorems” are either tautological under their assumptions, correct only for a degenerate version of the problem, or depend on unproved identifications between forward-pass subspaces and gradient orthogonality. The analysis also does not instantiate the actual BDH recurrence from `paper.tex`, so the results remain conditional on an abstraction that may not match the architecture.

1. The analysis is not grounded in the actual BDH equations

The main object is abstracted as

$\{$

$$s_0 = x, \quad s_{t+1} = F(s_t).$$

$\}$

But the report does not verify from the paper that BDH is actually a simple deterministic iterated map of this form. BDH is described as a scale-free network of locally interacting neuron particles, and the report itself claims—without evidence—that the block has “residual structure and Hebbian-type terms.” If BDH includes:

- local particle interactions,
- residual streams,
- input-dependent routing,
- Hebbian plasticity,
- stochasticity,

then the abstraction $s_{t+1} = F(s_t)$ may omit exactly the mechanisms that matter for continual learning.

The source cited for shared parameters is a third-party README (“same weights reused in all layers”), not the paper equations. That is not enough to validate the formal model. In particular, gating the whole block $g(x) \odot F(s_t)$ is very different from gating only a residual

$\Delta(s_t + g(x) \odot \Delta(s_t))$, or gating local interactions inside the particle graph.

Gap: the analysis should start from the actual BDH update equation and locate exactly where a gate or projection would be inserted. Without that, the impossibility claims are architecture-agnostic, and possibly architecture-inapplicable.

2. The notion of “input-path interference” is unstable across the analysis

The original draft defined interference as one input’s forward trajectory changing because another input’s mechanism is active. The moderator correctly observed that this cross-input forward-pass interference is structurally zero when parameters are frozen and the gate is stateless. But then the analysis substitutes a different notion: *depth-aliasing*, the mismatch between a depth-independent gate and an ideal depth-varying modulation.

These are not the same mathematical problem.

- **Depth-aliasing** is a representational issue: can a depth-constant gate imitate a depth-varying gain?
- **Continual-learning interference** is a parameter-update issue: updating shared gate parameters for task B degrades task A.

The final report calls the first notion “input-path interference,” but the actual catastrophic-forgetting channel in this setting must involve the trainable gate parameters ϕ , because F is frozen. The analysis only partially connects these two.

Gap: the paper should define the continual-learning objective explicitly:

- Tasks $(T_1, \dots, T_k)$ arrive sequentially.
- F is frozen.
- ϕ is updated.
- Preservation is required only on previous tasks.

This asymmetric setup is the relevant one, and it is not what the main impossibility proof uses.

3. The identity-forcing theorem is true but trivial

H1 says: under output gating,

[

$$s_{t+1} = g(x) \odot F(s_t),$$

]

exact preservation of the entire unperturbed trajectory forces $\mathbf{g}(\mathbf{x}) = \mathbf{1}$ wherever $\mathbf{F}_i \neq 0$.

This is correct, but it is almost a tautology: if the mechanism is required to reproduce the exact unmodified computation at every depth for every input, then the only possibility is no modification. The theorem does not show that isolation is impossible. It shows that **preservation of everything and alteration of anything are incompatible**, which is essentially the definition of exact preservation.

The relevant condition should be:

$$\tilde{s}_t(x) = s_t(x) \quad \text{for all } x \in S_{\text{old}}, \text{ for all } t,$$

while allowing $\tilde{s}_t(x) \neq s_t(x)$ for $x \in S_{\text{new}}$.

Under this asymmetric condition, the identity obstruction may disappear. The report acknowledges this only as future work, but the headline conclusion is drawn from the stronger, less relevant condition.

4. The diagonalization obstruction is not a general theorem

H2 assumes that an ideal modulation $g^*(x, t)$ is nonconstant in t , and that the transition rule $\Phi(g, s)$ is injective in g along the trajectory. If so, a depth-independent $g(x)$ cannot match the ideal sequence.

This is true given those assumptions, but the assumptions are not established.

- A depth-constant gate can produce depth-varying effects because the state s_t changes. For input gating,

[

$$s_{t+1} = F(g(x) \odot s_t),$$

]

the same $g(x)$ can produce different effective updates at different depths because s_t differs. Therefore the set of reachable trajectories is not simply the diagonal of the product space of depth-wise modulations.

- For output gating, injectivity of $\Phi(\cdot, s_t)$ fails on any coordinate where $F_i(s_t) = 0$, which is common with ReLU/GLU/zero activations.

Thus the “diagonalization obstruction” is conditional on a strong injectivity assumption and on the premise that the target trajectory requires nonconstant gate inputs in the same coordinate system. It is not a proof that generic tasks require off-diagonal modulation sequences. One would need a reachability argument in control theory, showing that the image of the constant-gate map does not contain the desired task map.

5. The projection rank theorem does not imply the isolation bound

H3 states:

- Exact preservation $\backslash(P(x)F(s_t)=F(s_t)\backslash)$ forces

$\backslash[$

$\backslash\operatorname{rank}P(x)\geq \backslash\dim\backslash\operatorname{span}\backslash\{F(s_t)\backslash.$

$\backslash]$

This is correct linear algebra.

But the next step—that mutually orthogonal projections are required to avoid gradient interference—is not proved and is likely false as a general statement.

Orthogonality of the image subspaces of $\mathcal{P}(x_A)$ and $\mathcal{P}(x_B)$ is a forward-pass geometric property. Cross-task gradient interference is a property of the parameterization of $\mathcal{P}(x)$. If $\mathcal{P}(x)$ is produced by a neural network, two different inputs may have non-orthogonal or identical image subspaces but still have gradients that are orthogonal, or vice versa. The connection between these two notions depends on the architecture of the projection generator and is not automatic.

Therefore the capacity bound

$$\left[\sum_i m_i \leq d \right]$$

does not follow. It only follows if one assumes a very special architecture in which each task's projection is a fixed, learned orthogonal subspace with independent parameters. That is a much stronger assumption than "input-dependent projection."

6. The parameter-budget bound is notationally sloppy and not operational

H4 defines task Jacobians $\{J_i\}_{i \in \mathbb{R}^{d \times p}}$ as derivatives of $s_D(x; \phi)$ with respect to ϕ , but tasks are sets of inputs, not single inputs. The aggregation matters.

The condition

$$\|J_i - J_j\|_{\text{top}} = 0$$

is not equivalent to zero gradient interference for general loss functions. The first-order change in task A's loss after a gradient step on task B is governed by

$$\langle \nabla_{\phi} \mathcal{L}_A, \nabla_{\phi} \mathcal{L}_B \rangle,$$

which involves the loss gradients, not just output Jacobians. Even at the output-Jacobian level, orthogonality of all Jacobian row spaces is a very strong condition.

Finally, the conclusion “at most one task when $p \leq d$ ” is not the relevant regime. A gate network attached to an LLM-scale BDH path would typically have $p \gg d$. If p is large, the rank budget can admit many tasks. The theorem gives no practical bound without estimating the actual Jacobian ranks.

7. The error bound statement has an incorrect “iff”

H5 derives

$$\forall [e_{t+1}] \leq L_{\Phi} e_{t+\Delta},$$
$$\forall]$$

hence

$$\forall [e_D] \leq \Delta \sum_{k=0}^{D-1} L_{\Phi}^k.$$
$$\forall]$$

This is correct. But the conclusion that bounding is possible **exactly when $L_{\Phi} < 1$** is not accurate. For any finite depth D , the sum is finite even if $L_{\Phi} > 1$. The bound grows exponentially with D , but it still exists.

The correct statement is:

- a bound **uniform in depth** exists iff $\forall (L_{\Phi} < 1)$;
- for fixed finite $\forall (D)$, a bound exists for any finite $\forall (L_{\Phi})$.

The analysis also asserts that trained LLM-scale maps typically have $\forall (L_{\Phi} > 1)$. That is an empirical claim about BDH or LLMs, not proved from the paper or code. It should be labeled as such and tested.

8. H6 is conditional to the point of tautology

H6 states that a depth-indexed shared gate $\forall (g_{\Phi}(x,t))$ can exactly realize finite task outputs if:

- some modulation sequence $\forall (g_{\{i,t\}})$ realizes each target output, and
- $\forall (g_{\Phi})$ can interpolate those sequences.

Under those assumptions, the conclusion is immediate. The nontrivial question is whether assumption 1 holds: **which target input-output maps are reachable by the frozen shared path plus the allowed gate class?** That question is unproved and marked as conjecture.

Therefore H6 does not establish controllability. It reformulates the problem without solving it.

9. “Controllability” is never formally defined

The research question asks whether interference can be eliminated, bounded, or **made controllable**. The analysis focuses on elimination and bounding, but does not define controllability.

A proper formalization would likely treat the gate at each depth as a control input and ask whether the composed system

$\{$

$s_{t+1}=F(s_t)+\text{ or } \odot \text{ gate control}$

$\}$

can steer states to desired outputs. This would invoke nonlinear reachability, injectivity of the transition map, or approximate controllability results for recurrent networks.

Because controllability is not formalized, the report’s statement that “controllability can be achieved only by enriching gate information” is not a theorem; it is an intuitive recommendation.

10. The t -conditioning loophole is unresolved and ambiguous

The report correctly identifies that a shared-parameter gate receiving depth index t as input would avoid the strict depth-agnostic obstruction. But it does not resolve whether this is allowed.

Two cases:

- **Fixed non-learned depth encoding**, e.g. sinusoidal positional encoding: adds $O(1)$ trainable parameters, likely does not violate the letter of “no per-depth/per-level trainable parameters.”
- **Learned depth embedding**: adds $O(D)$ trainable embedding rows, which is exactly per-depth trainable parameters, violating the constraint.

The final recommendation says “decide on t -conditioning scope,” but the impossibility result is only valid for the stricter depth-agnostic reading. Until this scope is fixed, the answer is conditional, not definitive.

11. Prior-art review is incomplete and partly too dismissive

The report reviews parameter isolation, progressive networks, gradient projection, hypernetworks, MoE, LSTMs, and weight-tied transformers.

However, some directly relevant literature is missing or dismissed too quickly:

- **FiLM / feature-wise linear modulation:**

\[

$$y = \gamma(x) \odot h + \beta(x),$$

\]

is essentially an input-dependent affine gate applied to a shared layer. Multi-task and continual variants are highly relevant.

- **Fast weight programmers and recurrent hypernetworks** that generate input-dependent modulations for a fixed recurrent core are closer to the proposed mechanism than the report admits.
- **Nonlinear RNN reachability / observability theory** would be the correct framework for the controllability question, but is absent.
- **Echo state networks** and reservoir computing may provide relevant results on fixed recurrent paths plus trained output/gate layers.

The classification “no surveyed prior art solves this exact problem” may be true, but the prior-art search is not exhaustive enough to support a strong novelty claim.

12. Novelty labeling is inconsistent

Earlier rounds label the impossibility result “apparently novel.” Later adjudication correctly says the mathematical ingredients are “direct consequences.” The final report mostly uses “direct consequence,” but still calls the packaged problem formalization “apparently novel.”

This is acceptable, but it should be made consistently: the contribution is a **problem formulation and synthesis of elementary obstructions**, not new mathematics. Overclaiming novelty would be a weakness.

13. Missing reconciliation with the H1 implementation report

The local file 2026-08-25_cl-h1-report.md was not retrieved. If it reports empirical behavior, the formal analysis neither explains nor refutes it. The task explicitly says not to treat experimental effects as theorem evidence, but at minimum the analysis should check which formal variant the implementation corresponds to.

If the implementation used a depth-indexed or state-dependent gate, then the strict impossibility results are irrelevant to that implementation. If it used a depth-independent gate, the analysis should be directly tested against the measured behavior.

Bottom line

The analysis successfully identifies:

- the distinction between forward-pass depth aliasing and training-time gradient interference,
- the fact that exact preservation of the full unperturbed trajectory forces identity under multiplicative output gating,
- the standard Lipschitz error-amplification bound.

But it fails to prove the main impossibility claim in the continual-learning-relevant setting. The actual relevant condition is **asymmetric preservation**: old tasks must be preserved, new tasks may change. That problem is not solved. The diagonalization, projection-rank, and parameter-budget obstructions do not survive as general theorems without additional, currently unproved assumptions.

The correct characterization is:

- **Strict global-preservation setting**: impossibility is trivial and proved.
- **Asymmetric CL setting**: open; the current analysis provides only necessary conditions and conjectures.
- **BDH-specific setting**: unresolved because the actual update equations were not instantiated.
- **Controllability**: not formally defined.

The report should be rewritten to state these boundaries explicitly, rather than announcing a general impossibility result.

[OPTIMIZER Round 3]

Refined Analysis: Input-Path Interference, Gating/Projection, and Computation Isolation in a Frozen Shared Sequential Composition

This document is a revised version of the earlier draft. It incorporates the critical review by (i) clarifying the different meanings of “interference,” (ii) sharply separating exact results from conditional or conjectural statements, (iii) correcting overclaiming in the diagonalization and projection arguments, and (iv) making the continual-learning-relevant asymmetric problem the central object of analysis.

0. Evidence and Scope

The public BDH paper (arXiv:2509.26507) and third-party material confirm that BDH is a scale-free, biologically inspired network of locally interacting neuron particles and that it reuses shared parameters across depth (“recurrent depth” in existing implementations). The internal project reports named in the task could not be retrieved during this

analysis, so no claim below is checked against the literal BDH update equations or the H1 implementation report.

Accordingly, this analysis is about the following **abstraction**, which the reports assume:

$\{$

$$s_0(x)=x, \quad s_{t+1}(x)=F_{\theta}(s_t(x)), \quad t=0, \dots, D-1.$$

$\}$

All theorems are proved for this abstraction. They are not automatically theorems about BDH until $\{$ is replaced by the actual BDH particle update. In particular, BDH may have a residual structure or Hebbian-style multiplicative terms; the *placement* of a gate (on the whole block, on a residual delta, or on an internal interaction) is part of the design and is not settled here.

1. Formalizations

1.1 Shared Sequential Composition

Let $(\mathcal{X} \subseteqq \mathbb{R}^d)$ be the input/state space and let $(F: \mathcal{X} \rightarrow \mathcal{X})$ be a fixed map.

A **depth- $\backslash(D\backslash)$ shared sequential composition** is the iteration

$$\begin{aligned} & \backslash[\\ & s_0(x)=x, \quad s_{t+1}(x)=F(s_t(x)). \\ & \backslash] \end{aligned}$$

The unperturbed trajectory is

$$\begin{aligned} & \backslash[\\ & \tau_F(x)=\text{bigl}(s_0(x),s_1(x),\dots,s_D(x)\text{bigr}). \\ & \backslash] \end{aligned}$$

If BDH uses residual connections, the relevant recurrence is instead

$$\begin{aligned} & \backslash[\\ & s_{t+1}(x)=s_t(x)+F(s_t(x)). \\ & \backslash] \end{aligned}$$

The theory below applies to the abstract iterated map; each gating theorem must be re-instantiated for the residual variant.

1.2 Frozen Shared Path

The shared path (F_{θ}) is **frozen** if its parameters (θ) are not updated during the continual-learning phase. Only the gate/projection mechanism may be trained:

$$\frac{\partial F_{\theta}}{\partial \theta} = 0$$

during all adaptation steps. Nothing below relies on retraining (F) .

1.3 Gating and Projection

Input-dependent gate. A map

$$g: \mathcal{X} \rightarrow \mathbb{R}^d$$

that is applied identically at every depth. The two canonical placements are:

$$\text{output gating: } s_{t+1} = g(x) \odot F(s_t),$$

$\lceil$

$\text{\text{input gating:}} \quad s_{t+1} = F \bigl(g(x) \odot s_t \bigr).$

$\rceil$

For residual blocks, the natural variant is

$\lceil$

$\text{\text{delta gating:}} \quad s_{t+1} = s_t + g(x) \odot F(s_t).$

$\rceil$

The gate may have trainable parameters ϕ , but the strict version of the research question forbids per-depth trainable parameter vectors. We will also consider the relaxed variant

$\lceil$

$g_\phi(x, t),$

$\rceil$

where the depth index is an **input** to the gate rather than an index into a separate parameter table. If t is encoded by a fixed, non-trainable embedding, this does not introduce per-depth trainable parameters. If a learned embedding indexed by t is used, it does.

Input-dependent projection. A map

\[

$$P:\mathcal{X}\rightarrow \Pi(\mathbb{R}^d),$$

\]

where $\Pi(\mathbb{R}^d)$ is the set of orthogonal projections, applied as

\[

$$s_{t+1}=P(x)F(s_t)\quad\text{or}\quad s_{t+1}=F(P(x)s_t).$$

\]

1.4 Input-Path Interference: Three Distinct Notions

The word “interference” is ambiguous. We separate three notions.

(A) Cross-input forward-pass contamination.

If F is frozen and the gate is a stateless function of the input x , then the trajectory of (x_1) is *computationally independent* of the presence or absence of (x_2) . Formally,

\[

$$\tilde{s}_t^{(M)}(x_1 \mid x_2 \text{ processed elsewhere}) = \tilde{s}_t^{(M)}(x_1).$$

\]

Thus this notion, which appeared in the original draft, is **structurally zero** in this setting. It is not a meaningful measure of interference here.

(B) Forward depth-aliasing.

A depth-agnostic gate $\mathcal{G}(x)$ applies the same modulation at every depth. If an oracle with access to depth t would apply a reference modulation $\mathcal{G}^*(x, t)$, the **depth-aliasing error** for input x is

$$\begin{aligned} & \left| \mathrm{DA}(\mathcal{G}; x) \right. \\ &= \left. \sum_{t=0}^{D-1} \left| \Phi(\mathcal{G}(x), s_t(x)) - \Phi(\mathcal{G}^*(x, t), s_t(x)) \right| \right|, \end{aligned}$$

where Φ is the chosen gating/projection rule. This is a *representational* mismatch inside a single forward pass.

(C) Training-time gradient interference (cross-task forgetting).

Because the gate parameters ϕ are shared across all inputs, updating ϕ on task (B) may change the gate output on task (A) inputs. First-order damage to task (A) from one gradient step on task (B) is proportional to

$$\left\| \nabla_{\phi} \mathcal{L}_A(\phi), \nabla_{\phi} \mathcal{L}_B(\phi) \right\|$$

where $\mathcal{L}_i$ is the task loss. If this is nonzero, the shared gate is a vector of forgetting.

The operationally relevant channels for continual learning are **(B)** (within a pass) and **(C)** (across passes). Cross-input forward contamination, notion **(A)**, is not a live channel under a frozen stateless model.

1.5 Computation-Isolability

A mechanism is **exactly computation-isolating** for a task family $\{(S_i, y_i)\}_{i=1}^K$ if:

- each task output is produced exactly:

$\lceil$

$\tilde{s}_D(x;\phi) = y_i(x) \quad \forall x \in S_i, \forall i;$

$\rceil$

- there is zero forward depth-aliasing error on the task trajectories whenever a reference trajectory is specified;
- there is zero first-order cross-task gradient interference:

$\lceil$

$\angle \nabla_{\phi} \mathcal{L}_i(\phi), \nabla_{\phi} \mathcal{L}_j(\phi) \\ \angle = 0$

$\quad \forall i \neq j.$

$\rceil$

A mechanism **bounds interference** if the relevant error is at most ϵ (ϵ), with ϵ controlled by explicit quantities: Lipschitz constants, depth, gate error, capacity, task geometry.

1.6 Elimination versus Bounding

- **Elimination** means exact equality of outputs and exact zero interference in the sense above.
- **Bounding** means a finite worst-case deviation ϵ (ϵ).
- A third mode, **controllability**, is often invoked informally. We formalize it as *reachability*: a task target $y_i(x)$ is **reachable** if there exists some gate parameterization ϕ such that $\tilde{s}_D(x;\phi) = y_i(x)$.

$s_D(x;\phi)=y_i(x)$ for all $(x\in S_i)$. The mechanism is controllable over a task sequence if every finite sequence of reachable tasks is simultaneously realizable by a single (ϕ) .

2. Theorem-Shaped Hypotheses

H1. Global Exact Preservation Forces the Gate to Be the Identity

Assumptions.

- (F) is defined on an open set $(U\subseteqq\mathbb{R}^d)$.
- The mechanism is output gating with a depth-agnostic gate:

[

$$s_{t+1}=g(x)\odot F(s_t).$$

]

- Exact preservation of the *entire unperturbed trajectory* is required:

[

$$\tilde{s}_t(x)=F^t(x)\quad\forall x\in U,\forall t=0,\dots,D.$$

]

Mathematical object gated.

The output of the frozen map $\backslash(F\backslash)$ at every depth.

Claim.

For every $\backslash(x\backslash)$ and every coordinate $\backslash(i\backslash)$,

$$\backslash[$$

$$\backslash\bigl(g_i(x)-1\backslash\bigr)\backslash, F_i(s_t(x))=0.$$

$$\backslash]$$

Hence, wherever $\backslash(F_i(s_t(x))\backslash\neq 0\backslash)$ along the trajectory, $\backslash(g_i(x)=1\backslash)$. If $\backslash(F\backslash)$ is componentwise nonzero along all trajectories on $\backslash(U\backslash)$, then $\backslash(g(x)=\mathbf{1}\backslash)$ for all $\backslash(x\backslash)$.

Proof.

At depth $\backslash(t\backslash)$, exact preservation gives

$$\backslash[$$

$$g(x)\backslash\odot F(s_t(x))=F(s_t(x)).$$

$$\backslash]$$

Componentwise, $\backslash((g_i(x)-1)F_i(s_t(x))=0\backslash)$. Repeating for all $\backslash(t\backslash)$ gives the result. $\backslash(\square\backslash)$

Verdict.

True as an exact theorem, but it is close to a tautology: *if the mechanism is required to change nothing, then it changes nothing*. It does **not** prove that computation isolation is impossible. It proves that exact global preservation and nontrivial gating are incompatible.

Weakest assumptions for escape.

If preservation is required only for final outputs, or only on a subset (S_{old}) of inputs, the identity obstruction becomes restricted to that subset. This is the continual-learning-relevant relaxation.

Classification.

Known / direct consequence.

H2. Asymmetric Preservation: The Honest Continual-Learning Setting

Assumptions.

- Old-task input set $(S_{\text{old}}) \subset \mathcal{X}$.
- New-task input set $(S_{\text{new}}) \subset \mathcal{X}$.
- Output gating $(s_{t+1} = g(x) \odot F(s_t))$.
- Exact preservation is required only on old inputs:

[

$s_t(x) = F^t(x) \quad \forall x \in S_{\text{old}}, \forall t.$

]

No preservation constraint is imposed on (S_{new}) .

Mathematical object constrained.

The restriction of the gate function (g) to (S_{old}) .

Claim.

For $(x \in S_{\text{old}})$,

[

$g_i(x) = 1$

$\quad \text{whenever} \quad$

$\exists t: F_i(s_t(x)) \neq 0.$

]

On the zero-coordinate set

[

$Z(x) = \{i: F_i(s_t(x)) = 0 \quad \forall t\},$

]

the gate is unconstrained by preservation. On (S_{new}) , the gate is entirely unconstrained by this requirement.

Proof.

Same componentwise argument as H1, restricted to (S_{old}) . $\square$

Verdict.

True as an exact theorem. This theorem replaces the trivial global statement with a useful necessary condition. In particular, if the old-task trajectories have support in all coordinates, then the gate is forced to be identity on all of (S_{old}) .

Remark. If additionally old and new inputs are disjoint and the gate network can interpolate between the identity mapping on (S_{old}) and the desired nontrivial mapping on (S_{new}) , then exact isolation is *not* ruled out by the identity obstruction. The real obstruction becomes task separability and gate capacity, not preservation per se.

Classification.

Direct consequence, but the asymmetric framing is the correct continual-learning formalization.

H3. Input-Only Mechanisms Cannot Resolve Task Ambiguity at Overlapping Inputs

Assumptions.

- The mechanism is a deterministic function of the input: $\tilde{s}_D(x; \phi) = \mathcal{M}_\phi(x)$.
- Two tasks (A, B) share an input: $x \in S_A \cap S_B$.
- Their target outputs differ: $y_A(x) \neq y_B(x)$.

Mathematical object constrained.

The input-output map $\mathcal{M}_\phi$.

Claim.

No single parameter vector ϕ can satisfy both tasks. Exact computation isolation is impossible for this task family.

Proof.

The map is a function: for a fixed ϕ , it returns exactly one output for the input x . Two different required outputs are incompatible. $\square$

Verdict.

True as an exact theorem. This is a genuine obstruction of the “input-dependent-only” constraint, independent of depth, nonlinearity, and capacity.

Weakest assumptions for escape.

Tasks must have disjoint input supports, or target functions must agree on overlaps, or the mechanism must be augmented with a task/context input. Any of these relaxations removes the obstruction.

Classification.

Direct consequence; essentially an identifiability condition.

H4. Depth-Agnostic Gating Is a Representational Limitation, Not a General Behavioral Impossibility

Assumptions.

- The transition is $\Phi(g, s)$.
- The *gate-value sequence* required by an oracle is $(g^x(x, 0), \dots, g^x(x, D-1))$, nonconstant in t .
- The transition is injective in the gate argument along the trajectory.

Mathematical object gated.

The sequence of gate values applied at each depth.

Claim.

A depth-agnostic gate $g(x)$ cannot reproduce the nonconstant gate-value sequence $g^*(x,t)$.

Proof.

If it did, equality at every depth would give

$$\begin{aligned} & \Phi(g(x), s_t(x)) = \Phi(g^*(x,t), s_t(x)). \end{aligned}$$

By injectivity, $g(x) = g^*(x,t)$ for every t , contradicting nonconstancy. $\square$

Important correction.

This theorem is **not** a general behavioral impossibility. A constant-in-depth gate can produce depth-varying effective behavior because the state s_t changes.

Counterexample to the stronger claim.

Let $F(s) = s$ and $D = 2$. Output gating gives

$$\begin{aligned} & s_2 = g(x) \odot g(x) \odot x. \end{aligned}$$

A constant gate $\mathbf{g}(x) = (\alpha, \beta)$ produces

$\mathbf{y}$

$$\mathbf{y}_2 = (\alpha^2 x_1, \beta^2 x_2).$$

$\mathbf{y}$

A depth-varying gate sequence $\mathbf{g}_0 = (c, 1)$, $\mathbf{g}_1 = (1, c)$ would produce $\mathbf{y} = (c x_1, c x_2)$, which is also achievable by the constant gate $\mathbf{g} = (\sqrt{c}, \sqrt{c})$ when $c \geq 0$. More generally, constant gates can emulate many depth-varying controls by exploiting state feedback.

Thus the earlier claim that “a depth-agnostic gate is confined to the diagonal of the modulation product space” is a statement about **gate values**, not about **reachable input-output maps**. It does not, by itself, prove an impossibility for task behaviors.

Verdict.

Underdetermined as a general theorem. The representational statement is true; the behavioral conclusion does not follow.

Weakest assumptions for a behavioral theorem.

One would need a control-theoretic reachability analysis: show that the image of the constant-gate map does not contain the desired task map for a specific $\mathbf{F}$, task set, and gate class. This is task-specific.

Classification.

The representational lemma is a direct consequence. The broad impossibility is **false** as stated; the reachability question is open.

H5. Projection Mechanisms: A Necessary Rank Condition

Assumptions.

- Output projection:

$\backslash[$

$$s_{\{t+1\}}=P(x)F(s_t).$$

$\backslash]$

- Exact preservation of the unperturbed trajectory of $\backslash(x\backslash)$ is required.

Mathematical object projected.

The image subspace of the projection $\backslash(P(x)\backslash)$.

Claim.

If

[

$m(x) = \dim \operatorname{span} \left\{ F(s_0(x)), F(s_1(x)), \dots, F(s_{D-1}(x)) \right\},$

]

then exact preservation requires

[

$\operatorname{rank} P(x) \geq m(x).$

]

Proof.

Exact preservation gives

[

$P(x)F(s_t(x)) = F(s_t(x))$

]

for every t . Hence all $F(s_t(x))$ belong to $\operatorname{Im} P(x)$, whose dimension is $\operatorname{rank} P(x)$. $\square$

Verdict.

True as an exact linear-algebra theorem.

What this theorem does not say.

It does **not** imply a cross-task capacity bound $\sum_i m_i \leq d$. Orthogonality of image subspaces of two projections is a *forward-pass geometric* property, whereas cross-task gradient interference is a property of the *parameterization* of $\mathcal{P}(x)$. Two inputs can have non-orthogonal image subspaces but orthogonal gate gradients, or vice versa. The earlier draft conflated these. The rank budget reappears below only under an explicit parameterization assumption.

Classification.

Direct consequence.

H6. Gradient Interference and the Parameter Budget

Assumptions.

- The gate/projection mechanism has p trainable parameters $\tilde{s}_D(\phi)$.
- For task i , define per-example output Jacobian

[

$$J_i(x) = \frac{\partial}{\partial \tilde{s}_D(x; \phi)}$$

]

- Let $\mathbb{R}_i \subseteq \mathbb{R}^p$ be the row space of the collection of Jacobians over task i .
- We require *zero first-order interference for every smooth loss and every example*.

Mathematical object constrained.

The parameter space of the gate mechanism.

Claim.

If zero interference is required for all smooth losses, then the row spaces $\{\mathbb{R}_i\}$ must be mutually orthogonal:

$$\mathbb{R}_i \perp \mathbb{R}_j \quad \text{for all } i \neq j.$$

Consequently,

$$\sum_{i=1}^K \dim \mathbb{R}_i \leq p.$$

Proof.

For losses $\ell(\mathcal{L}_i)$, first-order interference is

[

$$\langle \nabla_{\phi} \ell(\mathcal{L}_i), \nabla_{\phi} \ell(\mathcal{L}_j) \rangle.$$

]

For this to vanish for *arbitrary* loss functions, the linear functionals induced by the Jacobian row spaces must be mutually orthogonal, which forces row-space orthogonality. Orthogonal subspaces of $\mathbb{R}^p$ have total dimension at most p . $\square$

Important caveats.

- For fixed, specific losses, zero gradient interference is the weaker condition

[

$$\langle \nabla_{\phi} \ell(\mathcal{L}_i), \nabla_{\phi} \ell(\mathcal{L}_j) \rangle = 0,$$

]

which can hold even when Jacobian row spaces overlap.

- The bound is **not** a depth obstruction; it applies to any shared-parameter mechanism.
- If p is large relative to d , the bound does not stop many tasks from being isolated. It becomes restrictive only when the task Jacobians have full row rank and $p \leq d$.

Verdict.

True as an exact theorem under the stated generality. It is standard parameter-isolation accounting, not a new result.

Classification.

Direct consequence.

H7. Error Amplification: Bounding Is Conditional on Dynamics, Not Automatic

Assumptions.

- The gated transition $\Phi(g,s)$ is Lipschitz in state with constant L :

[

$$\|\Phi(g,s) - \Phi(g,s')\| \leq L \|s - s'\|.$$

]

- The per-depth gate error against a reference is bounded:

$\backslash[$

$$\backslash|\backslash\Phi(g(x),s_t^*(x))-\backslash\Phi(g^*(x,t),s_t^*(x))\backslash|\backslash\leq\backslash\delta_t.$$

$\backslash]$

Mathematical object bounded.

The final-state error.

Claim.

$\backslash[$

$$\backslash|s_D(x)-s_D^*(x)\backslash|$$

$\backslash\leq$

$$\backslash\sum_{t=0}^{D-1} L^{D-1-t}\backslash\delta_t.$$

$\backslash]$

If $\backslash(\backslash\delta_t\backslash\leq\backslash\delta\backslash)$, then

$\backslash[$

$$e_D\backslash\leq$$

$\backslash\begin{cases}$

$$\backslash\delta\backslash,\backslash\frac{1-L^D}{1-L},\&L<1,\backslash\backslash[1ex]$$

$$\backslash\delta D,\&L=1,\backslash\backslash[1ex]$$

$$\backslash\delta\backslash,\backslash\frac{L^D-1}{L-1},\&L>1.$$

$\end{cases}$

$\]$

Proof.

Let $e_t = |s_t - s_t^*|$. Then

$\[$

e_{t+1}

$\leq$

$| \Phi(g, s_t) - \Phi(g, s_t^*) |$

$+$

$| \Phi(g, s_t) - \Phi(g^*, s_t^*) |$

$\leq$

$L e_t + \delta_t.$

$\]$

Solving the linear recurrence gives the stated bound. $\square$

Precise interpretation.

- For **fixed finite depth** (D) , the bound is finite for every finite (L) . There is no “iff $(L < 1)$ ” statement for finite depth.
- A bound that is **uniform in depth** exists iff $(L < 1)$. At $(L = 1)$, the bound grows linearly in (D) ; at $(L > 1)$, exponentially.

Empirical caveat.

The claim “trained LLM-scale maps typically have $\mathcal{L} > 1$ ” is an empirical conjecture, not a theorem. It should be tested for a given BDH model.

Classification.

Known / direct consequence.

H8. Finite-Task Reachability: A Conditional Construction, Otherwise Underdetermined

Assumptions.

- The task family is finite: $\{(x_i, y_i)\}$, $(i=1, \dots, N)$, with task inputs possibly grouped into tasks.
- The gate family has sufficient capacity to interpolate finite sets.
- For each task input $\{x\}$, the required output is reachable by the allowed gate class:

[

$y_i(x) \in \mathcal{R}_D(x) := \{\tilde{s}_D(x; \phi) : \phi \in \Phi\}.$

]

- If an input appears in more than one task, the target outputs agree on the overlap.

Mathematical object controlled.

The finite input–output relation of the gated composition.

Claim.

Under these assumptions, there exists a parameter vector ϕ realizing all finite task targets exactly.

Proof sketch.

The requirements form a finite set of constraints. Reachability gives a target gate output for each input; universal/interpolation capacity of the gate family fits those finite constraints. $\square$

Counterexamples showing underdetermination.

- If $F(s)=0$, then $\mathcal{R}_D(x)=\{0\}$ for every x , regardless of gating. Nontrivial outputs are impossible.
- If the same input belongs to two tasks with different target outputs, H3 makes simultaneous realization impossible.
- If $F(s)=s$ and the gate is multiplicative output gating with $(D=2)$, the reachable set is

$\{$

$\{g(x) \odot g(x) \odot x\}$.

$\setminus$

Many targets are reachable only if coordinatewise signs and roots are permitted; others are not.

Verdict.

Conditionally true; otherwise underdetermined. This is not a theorem about all nonlinear $\setminus(F)$.

Weakest additional assumptions.

Reachability of each target, disjoint task inputs, and gate capacity are sufficient. The reachability condition is the nontrivial one and must be checked for the concrete $\setminus(F)$.

Classification.

Plausible adaptation of finite interpolation and universal approximation; the reachability part is a **conjecture** unless proved for a specific $\setminus(F)$.

3. Summary of Verdicts

| Hypothesis | Verdict | Status |

| --- | --- | --- |

| H1: Global preservation forces identity | True | Exact theorem; direct consequence |

| H2: Asymmetric preservation constraint | True | Exact theorem; CL-relevant |

| H3: Overlapping inputs with different targets | Impossible | Exact theorem; identifiability |

| H4: Depth-agnostic gate-value limitation | Representational only | Direct consequence; behavioral claim false/underdetermined |

| H5: Projection rank condition | True | Exact linear algebra; not sufficient for isolation |

| H6: Gradient interference parameter budget | True | Exact under strong generality |

| H7: Error amplification bound | True | Exact; finite-depth vs uniform-depth distinction |

| H8: Finite-task reachability | Conditional | Plausible adaptation; reachability is conjecture |

4. Prior Art: Does It Address the Same Mathematical Problem?

| Prior art | What it actually does | Same problem? | Classification |

| --- | --- | --- | --- |

| Gradient projection / orthogonal gradient methods (GEM, A-GEM, GPM) | Projects gradients in parameter space so new-task updates do

not increase old-task losses | Partly: addresses training-time gradient interference; not forward depth-aliasing; usually modifies the main network | Known, partially relevant |

| Parameter isolation / masks (PackNet, Supermasks, Piggyback) | Allocates disjoint parameter subsets to tasks | No: creates per-task parameters/masks; constraint-violating in the stated setting | Known, related but different |

| Progressive networks / PathNet | Adds new task-specific columns/paths | No: per-task architectural growth | Known, different problem |

| Hypernetworks / fast weight programmers | Generate weights or modulations for a main network | Partly: can generate input-dependent modulations; typically per-layer outputs, which violates the “no per-depth parameters” constraint in spirit | Known, partially relevant |

| FiLM / feature-wise linear modulation | Input-dependent affine modulation of hidden features | Partly: same gating algebra, but usually per-layer parameters; not a frozen weight-tied path | Known, directly relevant |

| LSTM / GRU | Shared-parameter state-dependent gates in a recurrent path | Closest: same composition structure; but gates see the current state, not only the original input | Known, nearest prior art with a different information assumption |

| Mixture of experts / conditional computation | Input-dependent routing to expert parameter sets | No: per-layer routers and distinct experts | Known, structurally similar |

| Reservoir computing / echo state networks | Fixed random recurrent path with trained readout | Partly: frozen recurrent core plus trainable output; not input-dependent forward gating for task isolation | Known, relevant background |

| Weight-tied transformers (Universal Transformer, ALBERT) | Reuse a block across depth | Partly: same composition; no task-isolation gating | Known, background context |

| Nonlinear RNN observability / reachability theory | Characterizes what a recurrent map can reach | Yes for the reachability question, but not used in the draft | Known, should be used for H8 |

Conclusion. No surveyed prior art solves the exact problem: isolation of computations inside one frozen shared sequential composition using only an input-dependent gate, with no per-depth trainable parameters. The closest neighbors are LSTM/GRU, FiLM, and reservoir computing.

5. Novelty Ledger

| Claim | Classification |

| --- | --- |

| Formalization of two live interference channels (depth-aliasing and gradient interference) in a frozen weight-tied path | **Apparently novel as packaging**; ingredients known |

| H1: exact global preservation forces identity gate | **Direct consequence** |

| H2: asymmetric preservation constraint on old inputs | **Direct consequence**, but CL-relevant reframing |

| H3: overlapping-input task ambiguity | **Direct consequence** |

| H4: depth-agnostic gate-value representational limitation | **Direct consequence**; behavioral conclusion underdetermined/not generally true |

| H5: projection rank condition | **Direct consequence** |

| H6: parameter budget for exact zero gradient interference | **Direct consequence** |

| H7: Lipschitz error amplification | **Direct consequence** |

| H8: finite-task reachability via gate interpolation | **Plausible adaptation / conjecture** |

| Claim that state or depth information is necessary for robust gating | **Plausible adaptation**, consistent with LSTM/FiLM practice, not a general theorem |

| Empirical claim “LLM-scale $\setminus(F)$ is expansive” | **Empirical conjecture**, not theorem |

No experimental result is used as evidence for any theorem in this document.

6. Failure Condition and Main Obstructions

The research question asks whether input-path interference can be eliminated, bounded, or made controllable without per-depth trainable parameters and without retraining the frozen shared path.

The strictest reading is:

- depth-agnostic gate/projection,
- frozen $\setminus(F)$,

- exact preservation of the original computation on all inputs.

Under that strict reading:

- **Elimination is impossible** in the uninteresting sense of H1: the gate must be identity on all inputs, so it provides no isolation.
- **Bounding** is possible only under the dynamics condition of H7: uniform bounds require $\|L\| < 1$; otherwise errors can grow with depth.
- **Controllability** is not a well-posed theorem without a reachability analysis of the specific $\mathcal{F}$ and gate class.

However, the strict reading is **not** the relevant continual-learning formulation.

We therefore treat the **asymmetric continual-learning reading** as central:

- old tasks: exact preservation of $\mathcal{F}^D(x)$ on $\mathcal{S}_{\text{old}}$;
- new tasks: free to change;
- no task label in the input;
- no per-depth trainable parameters.

Under this reading, the results are:

- The identity obstruction restricts to $\mathcal{S}_{\text{old}}$ (H2).
- Overlapping task inputs with different demands are impossible (H3).

- Exact zero cross-task gradient interference imposes a rank/parameter budget if demanded for all losses (H6).
- Whether nontrivial old/new isolation is possible is **underdetermined** without a reachability analysis (H4, H8).

The main obstruction is therefore not a single theorem. It is a conjunction of:

- **Task identifiability:** input-only gates cannot separate same-input different-output tasks.
- **Preservation-identity tension:** global preservation kills the gate; asymmetric preservation restricts it to old-task zero-coordinate directions.
- **Gradient coupling:** shared gate parameters couple tasks unless heavy orthogonality/low-rank conditions are imposed.
- **Dynamics amplification:** if the gated composition is expanding, small deviations from ideal gate behavior are amplified exponentially in depth.

None of these is an algorithmic failure; they are representational and algebraic. But none of them, by itself, proves a general impossibility of asymmetric continual learning with a frozen shared path. The negative claim must therefore be scoped:

> **Exact elimination is impossible under the trivial global-preservation reading. Under the asymmetric continual-learning reading, the problem is open; this analysis provides necessary conditions and conditional constructions, not a general impossibility.**

7. What Would Make the Question Decidable?

7.1 Instantiate $\mathcal{F}$

The first step is to replace the abstract $\mathcal{F}$ with BDH's actual update from paper .tex. The gate placement then becomes concrete:

- If BDH uses a residual block $s_{t+1} = s_t + F(s_t)$, gating the whole block may gate the residual stream; the natural placement is $s_{t+1} = s_t + g(x) \odot F(s_t)$.
- If BDH has Hebbian-style multiplicative terms, a gate on those terms is not the same as a gate on the full block.
- BDH's native plasticity must be explicitly disabled for the frozen-path premise.

7.2 Resolve the $\mathcal{T}$ -conditioning interpretation

The deepest loophole is the meaning of “no per-depth/per-level trainable parameters”:

- A gate $g_\phi(x, t)$ where $\mathcal{T}$ is a **fixed, non-trainable encoding** adds $\mathcal{O}(1)$ trainable parameters and does not literally violate the constraint.
- A **learned depth embedding** adds $\mathcal{O}(D)$ trainable parameters and does violate the constraint.

Every theorem above assumes the former is excluded. If the team permits fixed depth indexing, the depth-agnostic obstruction disappears, and the question becomes a reachability/interpolation question (H8). The team must decide this before any experiment is interpreted.

7.3 Formalize controllability

The research question asks whether interference can be “made controllable.” That requires a definition. A workable definition is reachability of finite task targets by the gate class without modifying $\setminus$ (F). The control-theoretic machinery of nonlinear discrete-time reachability and observability should be applied to the specific $\setminus$ (F).

8. Direct Answer to the Research Question

> Can input-path interference arising from shared sequential composition be eliminated, bounded, or made controllable by an input-dependent gate/projection mechanism, without introducing per-depth/per-level trainable parameters and without retraining the frozen shared path?

Elimination.

- Under the trivial global-preservation reading: **impossible**, because the only gate that preserves everything is the identity gate.

- Under the meaningful asymmetric continual-learning reading: **not generally proved either possible or impossible**. The surviving necessary conditions are task-input disjointness (H3), identity restriction on old-task coordinates (H2), and gradient orthogonality/low-rank budgets for exact zero cross-task interference (H6).

Bounding.

- **Possible conditionally.** If the gated transition is non-expansive ($\|L\| \leq 1$), bounds are uniform in depth. If $\|L\| > 1$, deviations can grow exponentially in depth. A finite-depth bound always exists for finite $\|L\|$, but may be uselessly large.

Controllability.

- **Underdetermined without a reachability analysis.** If each task target is reachable by the gate class, finite task sets can be interpolated, subject to gate capacity and task-input disjointness. Whether BDH's frozen map admits such reachability is not known from the public material.

Overall.

The strict form of the proposed mechanism does not provide a general solution. The obstruction is not any single nonlinearity of $\mathcal{F}$; it is the combination of input-only task identification, shared-parameter gradient coupling, and preservation constraints. The productive next step is to instantiate the abstract $\mathcal{F}$ with BDH's concrete update,

decide the $\backslash(t\backslash)$ -conditioning question, and then either prove or disprove reachability for the specific gate class.

[MODERATOR Round 3]

Final Moderator Synthesis — Debate Closure on BDH Input-Path Isolation

Role: ox-alpha, moderator. This document consolidates three rounds of contributions (STRATEGIST, CRITIC, OPTIMIZER), adjudicates all disputes, records the final consensus state, and delivers the scoped answer to the research question together with the precise characterization of the obstruction.

Two closing factual verifications are warranted before sealing the record — one on the BDH block equations (the single most consequential unresolved grounding item), one on public novelty of the exact problem:

[SEARCH: "Dragon Hatchling" BDH-GPU equations LayerNorm ReLU Hebbian residual update rule outer product]

[SEARCH: BDH "baby dragon hatchling" continual learning catastrophic forgetting gating projection]

0. Evidence-Base Status (unchanged, recorded for the archive)

- The BDH paper (arXiv:2509.26507) and third-party implementations corroborate that BDH reuses **shared parameters across depth** ("recurrent depth") with sparse non-negative activations. The *literal* update equations from paper .tex were never retrieved in-debate; neither were the internal reports (2026-08-*, plan_v2, cl-h1-report). Consequently every result below is proved for the **abstraction** $s_{t+1} = \Phi(g(x), s_t)$ over a frozen map F , with a residual variant $s_{t+1} = s_t + g(x) \odot F(s_t)$ treated as the likely BDH-shaped instantiation. Instantiation remains a mandatory follow-up (§8).
- No experimental result from any report was admitted as evidence for any theorem. This discipline held through all three rounds and is confirmed in the audit (§9).

1. Trajectory of the Debate (process record)

The debate executed exactly the correction loop the brief demanded:

- **Round 2** produced a unified impossibility package — identity-forcing + diagonalization + rank budget — headlined as "exact elimination is impossible in general."
- **Round 3 CRITIC** landed five substantive defeaters: (i) H1 is true but near-tautological and drawn from a degenerate preservation requirement; (ii) the diagonalization argument is a statement about gate *values*, not reachable *behaviors* — constant-depth gates

emulate depth-varying effects through state feedback, and injectivity fails at zero activations; (iii) the projection capacity bound $\sum m_i \leq d$ committed a category error (image-subspace geometry $\neq$ parameter-space gradient geometry); (iv) the error-bound "iff $L < 1$ " misquantified (finite- D vs. uniform-in- D); (v) "controllability" was never defined. Plus: missing BDH grounding, missing prior art (FiLM, fast-weight programmers, reservoir computing), and the unresolved t -conditioning scope.

- **Round 3 responses:** OPTIMIZER conceded essentially all points and *restructured* — adding two genuinely new exact results (asymmetric-preservation constraint; overlapping-input identifiability), withdrawing the invalid corollary, fixing the quantifiers, and reframing the headline. STRATEGIST incorporated the asymmetric-preservation and t -conditioning flags into its own §7 while retaining a stronger headline.

Adjudication principle applied throughout: a claim enters the record only at the strength of its *weakest surviving proof*.

2. Final Adjudicated Verdicts

| # | Claim | Final status | Notes |

| --- | ----- | ----- | ----- |

| **V1** | Identity-forcing: exact trajectory preservation on input set S under depth-agnostic multiplicative gating forces $g_i(x)=1$ for every (x,i) with coordinate i active along x 's trajectory | **TRUE — exact theorem, direct consequence** | Near-tautological. Its real content is

locative: it identifies precisely where gate freedom lives (never-active coordinates; non-\$\$\$ inputs). |

| **V2** | Cross-input forward-pass contamination is structurally zero for stateless mechanisms on a frozen path | **ESTABLISHED** | Caveat added in synthesis: any *stateful* component inside the gate (running-statistics normalization, caches) revives this channel. Statelessness must be listed as an explicit design premise. |

| **V3** | Identifiability: a deterministic input-only mechanism cannot satisfy two tasks demanding different outputs at a shared input | **TRUE — exact theorem** | The sharpest structural limit of "input-dependent only," independent of depth, nonlinearity, capacity. Practical force in LM-CL depends on measure-theoretic overlap of task supports; do not overclaim either direction. |

| **V4** | Diagonalization: depth-agnostic gate confined to diagonal of modulation space | **Representational lemma TRUE; behavioral impossibility NOT established — downgraded** | Counterexample accepted ($F(s)=s$, $D=2$: constant gate matches depth-varying final outputs via state feedback); injectivity fails at zeros. Superseded by the open reachability question. |

| **V5** | Projection rank: $\text{rank} P(x) \geq m(x)$ necessary for exact preservation | **TRUE — exact linear algebra** | Capacity corollary $\sum_i m_i \leq d$ **WITHDRAWN** (invalid inference from forward-pass geometry to gradient geometry). Survives only under a special independent-subspace parameterization that violates the spirit of the constraints anyway. |

| **V6** | Gradient budget: zero first-order interference *for all smooth losses* $\Leftrightarrow$ mutually orthogonal task-Jacobian row spaces $\Rightarrow \sum_i \dim R_i \leq p$ | **TRUE — exact under stated generality; standard accounting** | Weak for *fixed specific* losses (inner product can vanish accidentally); non-binding in the realistic $p \gg d$ regime. Worst-case guarantee, not typical-case behavior. |

| **V7** | Amplification: $e_D \leq \delta \sum_{k < D} L_{\Phi^k}$; finite- D bound exists for every finite L_{Φ} ; **uniform-in-depth** bound iff $L_{\Phi} < 1$ | **TRUE — exact, quantifiers corrected** | "Trained LLM-scale

maps are expansive" reclassified as **empirical conjecture**, requiring measurement on the actual model. |

| **V8** | Reachability/interpolation: finite task sets realizable iff targets reachable by gate class + capacity + disjointness/agreement |

CONDITIONAL; crux is a conjecture | Reachability on a concrete frozen F is the decisive unproven assumption; counterexamples ($F \equiv 0$; sign/root obstructions) show it fails generically without hypotheses on F . |

3. Consensus Evaluation

Full consensus (all three roles, post-revision):

- The formal skeleton (shared composition, frozen path, gate/projection placements, elimination-vs-bounding distinction).
- V1–V3, V5–V8 as scoped above.
- The two-channel picture: the live interference channels are **forward depth-aliasing** (within a pass) and **cross-time gradient interference** (across passes); naive cross-input contamination is dead (V2).
- Prior-art triage, extended in Round 3: LSTM/GRU (nearest neighbor; state-conditioned), FiLM (same gating algebra, per-layer parameters), fast-weight programmers/hypernetworks (adjacent, per-layer outputs), reservoir computing (frozen core + trained readout), orthogonal-gradient methods (parameter-space, complementary — see §7-C1), progressive networks/PackNet/supermasks (constraint-violating), MoE (different problem), Universal Transformer/ALBERT (backdrop). **No surveyed work solves the exact problem:** isolation within one frozen weight-tied

composition via an input-only gate without per-depth trainable parameters.

Residual dissent, adjudicated:

- STRATEGIST's Round-3 headline ("elimination impossible in general") versus the OPTIMIZER/CRITIC scoped formulation.

Ruling: the scoped formulation prevails. The unscoped headline holds only under global preservation on *all* inputs — a reading under which the gate must be inert everywhere, hence no adaptation anywhere, hence not a continual-learning question at all. STRATEGIST's own §7.1 concedes the asymmetric setting is the honest one and open; that concession is entered into the record as binding. The debate's central epistemic event is precisely this downgrade: **the general impossibility theorem dissolved upon inspection; what survives is a set of exact necessary conditions plus an open reachability problem.**

4. Final Answer to the Research Question

> Can input-path interference arising from shared sequential composition be eliminated, bounded, or made controllable by an input-dependent gate/projection mechanism, without per-depth/per-level trainable parameters and without retraining the frozen shared path?

| Reading | Verdict | Epistemic status |

| --- | --- | --- |

| **Elimination**, global-preservation reading (mechanism preserves F 's behavior on *all* inputs) | Trivially impossible — the only admissible gate is the identity, so the mechanism does nothing | Proved; direct consequence (degenerate) |

| **Elimination**, asymmetric CL reading (exact preservation on S_{old} , free on S_{new}) | **Open**. Necessary conditions established: identity confinement on old-task active coordinates (V1-scoped); input-disjointness/target-agreement (V3); worst-case gradient budget (V6); dynamics amplification (V7). No general possibility or impossibility theorem | Partially characterized |

| **Bounding** | Finite-depth bounds exist for any finite L_{Φ} ; **uniform-in-depth** bounding iff gated dynamics contractive ($L_{\Phi} < 1$). Bounding is a property of the composed system, not achievable by gate design alone under expansion | Proved, conditional |

| **Controllability** | **Not yet a well-posed question**: requires formal definition (reachability of task targets by the gate class on the concrete frozen F) and BDH instantiation | Undefined pending instantiation |

The brief's failure condition is therefore met **only in the degenerate reading**. In the CL-relevant reading the correct outcome is: *not proven impossible, not proven possible; reduced to identifiable necessary conditions and one decisive open conjecture (reachability)*. Reporting anything stronger would violate the novelty discipline.

5. Obstruction Characterization (deliverable under the failure clause)

There is no single obstruction theorem in the meaningful setting. The failure surface is a **conjunction of four necessary conditions**, each independently capable of killing the program, none individually fatal:

- **Identifiability failure** — conflicting targets on overlapping inputs are unservable by any input-only mechanism (V3). In language-modeling continual learning, supports overlap extensively; escape requires context-sufficiency of the input or task conditioning.
- **Preservation-geometry confinement** — the gate is pinned to the identity on every (old input, active coordinate) pair (V1-scoped). Freedom survives only on never-active coordinates (substantial under sparse activations) and on new-task inputs.
- **Gradient coupling of shared ϕ** — new-task updates generically move g_ϕ off the identity on old inputs; worst-case immunity costs a rank budget (V6) that is either restrictive ($\leq d$) or vacuous ($\gg d$), and is unnecessary for specific loss pairs.
- **Depth amplification** — under expansive gated dynamics, per-depth modulation error grows geometrically in D (V7); bounding then demands architectural contractivity, which is an assumption on F , not a capability of the gate.

Plus three **scope conditionals** on which every result silently rests: (a) statelessness of the mechanism (else V2 dies); (b) the t -conditioning interpretation (below); (c) BDH's native Hebbian plasticity actually disabled (else the path is not frozen).

The t -conditioning fork (recorded as the pivotal scope decision): a shared-parameter gate $g_\phi(x, t)$ with t supplied by a *fixed, non-trainable* encoding adds $O(1)$ parameters and does not violate the constraint as literally worded — and it dissolves V1's depth-agnostic restrictions and V4's residue entirely, converting the problem into pure reachability (V8). A *learned* depth embedding adds $O(D)$ parameters and violates the constraint outright. Every impossibility-flavored

statement in this record applies only on the strict side of this fork. The team must rule before any experiment is interpreted.

6. Novelty Ledger (final, adjudicated)

| Claim | Classification |

| --- | --- |

| Problem formulation: isolation inside a frozen weight-tied composition via input-only gate, CL-framed, two-channel interference taxonomy |

Apparently novel (packaging only) |

| V1, V3, V5 (necessary-condition parts), V6, V7 | **Direct consequences** (textbook ingredients) |

| V4 representational lemma | Direct consequence; behavioral version **false as stated** |

| Necessity of state/depth/context information for effective gating | **Plausible adaptation** (consistent with LSTM/FiLM practice; proven only under injectivity hypotheses) |

| V8 reachability for concrete BDH $\$F\$$ | **Conjecture** |

| Expansiveness of trained BDH blocks | **Empirical conjecture** |

| Any observed effect in c_l-h₁-report | Inadmissible as theorem evidence; admissible only as a test of the conjectures above |

Contribution posture for write-up: *a new formalization and a clean map of elementary obstructions — not new mathematics.*

7. Constructive Directions Surfaced by Synthesis (all $\leq$ conjecture)

- **C1 — Preservation as a maintained invariant (rehabilitates dismissed prior art).** Once the protected object is recognized to be the *gate parameters* ϕ (not the frozen path θ), orthogonal-gradient machinery becomes directly applicable *compatibly with all constraints*: project new-task gradients onto the orthogonal complement of the old-task gradient subspace in ϕ -space, thereby holding $g_\phi|_{S_{\text{old}}} \approx \mathbf{1}$ to first order across sequential training. Classification: **plausible adaptation** of GEM/GPM to a new target object; no theorem claimed. Open statistical question: whether the invariant generalizes beyond the old-task sample.
- **C2 — Fixed-encoding t -conditioning** ($O(1)$ parameters): dissolves the depth-agnostic restrictions; legality contingent on the scope ruling (§5).
- **C3 — Delta-gate placement** for the residual BDH variant: gate $g(x) \odot \Delta(s_t)$ inside $s_{t+1} = s_t + \Delta$, never the residual stream itself.
- **C4 — Sparse-coordinate freedom:** exploit never-active coordinates of old-task trajectories (plausible under ReLU sparsity) as legal modification space even on preserved inputs.

8. Mandatory Follow-Ups Before Any Empirical Reconciliation

- **Instantiate F** from paper . tex: fix the literal update (residual form, Hebbian/multiplicative terms, normalization placement), choose gate placement (C3), and verify native plasticity is disabled. *(Closing searches issued above; see appended results — treat as corroboration only, not substitution for reading the TeX source.)*
- **Rule on the t -conditioning fork** (§5) in writing.
- **Define controllability formally** as reachability of finite task targets by the gate class on the instantiated F ; select the tool class (nonlinear discrete-time reachability).
- **Retrieve the internal reports**; map the implemented mechanism onto exactly one formal variant (depth-agnostic / t -conditioned / state-conditioned); reconcile cl-h1-report outcomes strictly as tests of the V7/V8 conjectures.
- **Measurement program**: empirical Lipschitz/expansiveness of the gated BDH block (tests V7's conjecture); empirical cross-task Jacobian row-space overlap in ϕ -space (tests whether V6's worst case binds in practice).

9. Discipline Audit

- No experimental observation was cited as evidence for any theorem at any point. ✓
- All overclaims identified by the CRITIC were either withdrawn on the record (capacity corollary; behavioral diagonalization; finite- D "iff") or explicitly rescoped (global $\rightarrow$ asymmetric preservation). ✓

- Novelty labels are consistent post-adjudication: packaging novel, mathematics derivative. ✓

Consensus summary: complete agreement on the formal framework, on the two live interference channels, on the exact necessary conditions (V1–V3, V5–V7 as scoped), on prior-art triage, and on the open status of the asymmetric continual-learning question. The single substantive disagreement — whether a general impossibility was established — was resolved against the stronger claim on the strength-of-weakest-proof principle. The investigation exits the debate not with an impossibility theorem but with something more useful: a precise map of where gate freedom lives, four independent ways the program can fail, one pivotal scope decision (t -conditioning), and one decisive open conjecture (reachability on the instantiated BDH map) whose resolution will determine whether the mechanism is possible, impossible, or merely unproven.

Web Research

- **The Dragon Hatchling: A New AI Architecture Bridging Transformers and the Brain | Deep Paper**

September 1, 2025 - Use LayerNorm and ReLU to keep activations positive and encourage sparsity. The BDH-GPU state-space update (informal) per layer and timestep can be summarized as: ... update neuron activations $\mathbf{x}$ with a ReLU-lowrank residual like $\mathbf{x} \mapsto \mathbf{x} + \text{ReLU}(\mathbf{L}\mathbf{y})$...

Source: <https://deep-paper.org/en/papers/2025-10/2509.26507/>

- **The Dragon Hatchling: The Missing Link between the Transformer and Models of the Brain**

September 30, 2025 - The observed emergent effects follow naturally from the design choices of the BDH and BDH-GPU architectures, and are largely attributable to the combination of: the choice of model dimensions with comparable model-to-state ratio, reliance on linear attention in high dimension, reliance on ReLU thresholds for ensuring that activation vectors are positive (trivially) and sparse (an effect empirically noted in (Haziza et al.

Source: <https://arxiv.org/html/2509.26507v1>

- **The Dragon Hatchling - ArXivIQ**

October 5, 2025 - The BDH-GPU architecture (Figure ... neuronal dimension n : ReLU-Lowrank Feed-Forward Network: The FFN uses the transformation $z \rightarrow (DEz)^+$, where E is an encoder ($n \rightarrow d$) and D is a decoder ($d \rightarrow n$)....

Source: <https://arxiviq.substack.com/p/the-dragon-hatchling>

- **Understanding Baby Dragon Hatchling (BDH): The Missing Link Between Transformers and the Brain | Colin McNamara**

October 18, 2025 - The dynamics of BDH are governed by what the authors call “equations of reasoning”—local update rules that bear a striking resemblance to Hebbian learning (“neurons that fire together, wire together”).

Source: <https://colinmcnamara.com/blog/understanding-baby-dragon-hatchling-bdh>

- **BDH Architecture Explained: Pathway’s Dragon Hatchling**

2 weeks ago - Chapter 2 derives BDH from the attention equations, reframing retrieval as synaptic memory in a high-dimensional sparse neuron space and showing how Hebbian updates and low-rank factorization lead to the practical BDH-GPU architecture.

Source: <https://pathway.com/research/bdh-explainer>

- **cMiraka/bdh-cl: Attempts extension of Baby Dragon Hatchling...**

Baby Dragon Hatchling Continual Learning (BDH-CL). Fork of: pathway.com/bdh. BDH-CL introduces practical, biologically inspired continual learning capabilities into the BDH architecture, enabling robust lifelong learning beyond the single-task limitations of the original.

Source: <https://github.com/cMiraka/bdh-cl>

- **Beyond Transformer LLM Models: How the BDH Architecture Solves...**

cMiraka/bdh-cl: Attempts extension of Baby Dragon Hatchling (BDH) with continual learning using pseudo-metaplasticity.

Source: <https://www.crvscience.com/post/beyond-transformer-llm-models-how-the-bdh-architecture-solves-the-context-window-crisis>

- **Dragon Hatchling: Linking Transformers & Brain Models**

BDH bridges Transformer LLMs with brain models via local graph dynamics and Hebbian learning, delivering interpretability and state-of-the-art performance.

Source: <https://www.emergentmind.com/papers/2509.26507>

• **Introducing BDH: A Brain-Inspired AI Model | Shivam... | LinkedIn**

Brain-Inspired Language Models Are Here Researchers have introduced BDH (Baby Dragon Hatchling)—a new large-language-model architecture inspired by biological networks and the human brain.

Source: https://www.linkedin.com/posts/imsane_airesearch-largelanguagemodels-neuroscienceinspired-activity-7387937972265893888-UzDm

10. We Watched a Brain Emerge..." The AI That Might Kill... - YouTube

Pathway Calls It Baby Dragon Hatchling 16:52 - Continual Learning and the Dragon Metaphor 17:20 - Learning Like a Brain: Neurons and Connections 21:27 - When a Brain Emerges Inside the Model 22:54 - Memory as Strengthened Connections 24:58...

Source: <https://www.youtube.com/watch?v=duw7RUif8hE>

[STRATEGIST Round 4]

Final Strategic Assessment: Input-Path Interference in BDH via Gating/Projection

Date: 2026-08-25 | **Status:** Complete (3-round debate, adjudicated)

Executive Summary

Under the strict constraints — depth-agnostic input-dependent gate/projection, frozen shared path, no per-depth trainable parameters, no retraining — **exact elimination of input-path interference is impossible only under the degenerate global-preservation reading** (the gate must be identity everywhere, doing nothing). Under the **continual-learning-relevant asymmetric reading** (exact preservation on old tasks, free on new tasks), the problem is **open**: no general impossibility *or* possibility theorem has been established. What has been established is a precise map of four independent necessary conditions, each capable of killing the program, plus one decisive open conjecture (reachability) whose resolution determines feasibility.

1. Formal Definitions

1.1 Shared Sequential Composition

Let $V = \mathbb{R}^d$ be the state space. A **depth- D shared sequential composition** is defined by a fixed map $F: V \rightarrow V$ and the recurrence:

$$s_0(x) = x, \quad s_{t+1}(x) = F(s_t(x)), \quad t = 0, \dots, D-1$$

The unperturbed trajectory is $\tau_F(x) = (s_0(x), s_1(x), \dots, s_D(x))$.

If BDH uses residual connections (likely, per the BDH-GPU architecture), the relevant recurrence is:

$$s_{t+1}(x) = s_t(x) + F(s_t(x))$$

Classification: Direct characterization of BDH's weight-tied recurrent depth, corroborated by BDH-MLX ("Same weights reused in all layers (recurrent depth)") and BDH-GPU descriptions ("ReLU-Lowrank residual like $x \mapsto \dots$ "). Instantiation with the literal BDH update from paper .tex remains mandatory.

1.2 Frozen Shared Path

F is **frozen** if $\partial F / \partial \theta = 0$ during the continual-learning phase. Only the gate/projection mechanism ϕ may be trained.

Classification: Known — standard continual learning constraint.

1.3 Input-Path Interference: Three Precise Notions

(A) Cross-input forward-pass contamination. With frozen θ and a stateless input-conditioned mechanism, each input's forward pass is computationally independent. This notion is **structurally zero** and not a live channel.

(B) Forward depth-aliasing. A depth-agnostic gate $g(x)$ applies the same modulation at every depth. If an oracle would apply $g^*(x,t)$ (varying with depth), the depth-aliasing error is:

$$\mathrm{DA}(g; x) = \sum_{t=0}^{D-1} \left| \Phi(g(x), s_t(x)) - \Phi(g^*(x,t), s_t(x)) \right|$$

(C) Training-time gradient interference (cross-task forgetting).

Because gate parameters ϕ are shared across all inputs, updating ϕ on task B may change g_ϕ on task- A inputs. First-order damage is proportional to $\langle \nabla_\phi \mathcal{L}_A, \nabla_\phi \mathcal{L}_B \rangle$.

The live channels are (B) and (C). Notion (A) is dead under the stated assumptions.

Classification: The depth-aliasing notion is **apparently novel** as a packaged definition; components are known from control theory. The gradient notion is **known** from continual learning.

1.4 Gating

An **input-dependent gate** $g: X \rightarrow \mathbb{R}^d$ is applied identically at every depth:

- Output gating: $s_{t+1} = g(x) \odot F(s_t)$
- Input gating: $s_{t+1} = F(g(x) \odot s_t)$
- Delta gating (for residual blocks): $s_{t+1} = s_t + g(x) \odot F(s_t)$

The gate has **no per-depth parameters**: the same $g(x)$ is used at every t .

Classification: Known mechanism (LSTM, MoE), but the depth-independence + frozen-path combination is novel.

1.5 Projection

An **input-dependent projection** $P: X \rightarrow \Pi(V)$ maps each input to an orthogonal projection, applied as $s_{t+1} = P(x) F(s_t)$ or $s_{t+1} = F(P(x) s_t)$.

Classification: Known (gradient projection methods), formalized for forward-pass application.

1.6 Computation-Isolability

A mechanism is **exactly computation-isolating** for a task family $\{(S_i, y_i)\}_{i=1}^K$ if: (1) each task output is produced exactly; (2) zero forward depth-aliasing error; (3) zero first-order cross-task gradient interference.

Classification: Apparently novel — this specific formulation combining shared composition, frozen parameters, and forward-pass isolation does not appear in surveyed literature.

1.7 Elimination vs. Bounding vs. Controllability

- **Elimination:** Exact equality of outputs and exact zero interference.
- **Bounding:** Deviation $\leq \epsilon$, controlled by Lipschitz constant, depth, gate error, parameter count.
- **Controllability:** Reachability of finite task targets by the gate class without modifying F .

Classification: Plausible adaptation of approximation theory and control-theoretic concepts.

2. Theorem-Shaped Hypotheses (Final Adjudicated Versions)

V1. Identity-Forcing Under Global Preservation

Assumptions: F defined on open $U \subseteq \mathbb{R}^d$; depth-independent output gating $s_{t+1} = g(x) \odot F(s_t)$; exact trajectory preservation on input set S : $\tilde{s}_t(x) = F^t(x)$ for all $x \in S$, all t .

Claim: For every $x \in S$ and every coordinate i : $g_i(x) = 1$ whenever $\exists t: F_i(s_t(x)) \neq 0$.

Proof: At depth t : $g(x) \odot F(s_t(x)) = F(s_t(x))$ implies $(g_i(x) - 1) F_i(s_t(x)) = 0$ componentwise. $\square$

Verdict: True as an exact theorem. Near-tautological under global preservation. Its real content is *locative*: it identifies precisely where gate freedom lives — never-active coordinates and non- S inputs.

Classification: Known / direct consequence.

V2. Asymmetric Preservation: The CL-Relevant Constraint

Assumptions: Old-task inputs S_{old} ; new-task inputs S_{new} ; output gating; exact preservation required only on S_{old} .

Claim: For $x \in S_{\text{old}}$: $g_i(x) = 1$ whenever $\exists t$: $F_i(s_t(x)) \neq 0$. On the zero-coordinate set $Z(x) = \{i: F_i(s_t(x)) = 0 \forall t\}$, the gate is unconstrained. On S_{new} , the gate is entirely unconstrained.

Proof: Same componentwise argument as V1, restricted to S_{old} . $\square$

Verdict: True as an exact theorem. If old-task trajectories have support in all coordinates, the gate is forced to identity on all of

S_{old} . If old and new inputs are disjoint and the gate network can interpolate, exact isolation is *not* ruled out by this obstruction alone.

Classification: Direct consequence; the asymmetric framing is the correct CL formalization.

V3. Identifiability: Overlapping Inputs Are Fatal

Assumptions: The mechanism is a deterministic function of the input: $\tilde{s}_D(x; \phi) = \mathcal{M}_\phi(x)$. Two tasks A, B share an input $x \in S_A \cap S_B$ with different targets: $y_A(x) \neq y_B(x)$.

Claim: No single ϕ can satisfy both tasks.

Proof: The map is a function: one input, one output. Two different required outputs are incompatible. $\square$

Verdict: True as an exact theorem. The sharpest structural limit of "input-dependent only," independent of depth, nonlinearity, and capacity.

Classification: Direct consequence; essentially an identifiability condition.

V4. Depth-Agnostic Gate-Value Limitation (Downgraded)

Assumptions: $\Phi(g,s)$ injective in g along the trajectory; ideal modulation $g^*(x,t)$ nonconstant in t .

Claim: No depth-independent gate $g(x)$ can reproduce the nonconstant gate-value sequence $g^*(x,t)$.

Proof: Equality at every depth gives $g(x) = g^*(x,t)$ for every t , contradicting nonconstancy. $\square$

Critical correction: This is a statement about **gate values**, not about **reachable input-output maps**. A constant-in-depth gate can produce depth-varying effective behavior because the state s_t changes.

Counterexample: $F(s) = s$, $D = 2$. Output gating: $s_2 = g(x) \odot g(x) \odot x$. Constant gate $g(x) = (\alpha, \beta)$ produces $(\alpha^2 x_1, \beta^2 x_2)$. Depth-varying sequence $g^0 = (c, 1)$, $g^1 = (1, c)$ produces (cx_1, cx_2) , also achievable by constant gate $(\sqrt{c}, \sqrt{c})$.

Verdict: Representational lemma TRUE; behavioral impossibility NOT established. The reachability question is open.

Classification: Direct consequence for the representational part; the broad behavioral impossibility is **false as stated**.

V5. Projection Rank Condition

Assumptions: Output projection $s_{t+1} = P(x) F(s_t)$; exact preservation of unperturbed trajectory required.

Claim: If $m(x) = \dim \operatorname{span}\{F(s_0(x)), \dots, F(s_{D-1}(x))\}$, then $\operatorname{rank} P(x) \geq m(x)$.

Proof: $P(x) F(s_t(x)) = F(s_t(x))$ forces $F(s_t(x)) \in \operatorname{Im} P(x)$. $\square$

Verdict: True as an exact linear-algebra theorem.

What this does NOT say: The capacity corollary $\sum_i m_i \leq d$ is **WITHDRAWN**. Orthogonality of image subspaces is a forward-pass geometric property; cross-task gradient interference is a property of the parameterization. These are different mathematical objects.

Classification: Direct consequence.

V6. Gradient Interference Parameter Budget

Assumptions: Gate has p trainable parameters ϕ ; task Jacobians $J_i \in \mathbb{R}^{d \times p}$; zero first-order interference required for *all smooth losses*.

Claim: Row spaces $\{R_i\}$ must be mutually orthogonal: $\sum_i \dim R_i \leq p$.

Proof: For arbitrary loss functions, vanishing inner products force row-space orthogonality. $\square$

Important caveats:

- For fixed specific losses, zero interference is weaker (can hold accidentally).
- If $p \gg d$ (typical for LLM-scale gates), the bound is non-binding.
- Worst-case guarantee, not typical-case behavior.

Verdict: True as an exact theorem under stated generality. Standard parameter-isolation accounting.

Classification: Direct consequence.

V7. Error Amplification Bound

Assumptions: Gated transition L -Lipschitz in state; per-depth gate error $\leq \delta$.

Claim:

$$\|s_D(x) - s_{D^*}(x)\| \leq \delta \sum_{k=0}^{D-1} L^k = \begin{cases} \delta \frac{1 - L^D}{1 - L} & L < 1 \\ \delta D & L = 1 \\ \delta \frac{L^D - 1}{L - 1} & L > 1 \end{cases}$$

Precise interpretation:

- For **fixed finite depth D** : bound exists for every finite L .
- **Uniform-in-depth** bound exists iff $L < 1$.

Empirical caveat: "Trained LLM-scale maps have $L > 1$ " is an **empirical conjecture**, not a theorem.

Verdict: True as an exact theorem.

Classification: Known / direct consequence.

V8. Finite-Task Reachability (Conditional)

Assumptions: Finite task set (x_i, y_i) ; gate family has sufficient capacity; each target reachable: $y_i(x) \in \mathcal{R}_D(x) := \{\tilde{s}_D(x; \phi) : \phi \in \Phi\}$; overlapping inputs have agreeing targets.

Claim: There exists ϕ realizing all finite task targets exactly.

Counterexamples showing underdetermination:

- $F(s) = 0$: $\mathcal{R}_D(x) = \{0\}$ for every x , regardless of gating.
- Same input, different targets: V3 kills it.
- $F(s) = s$, $D = 2$, multiplicative output gating: reachable set is $\{g(x) \odot g(x) \odot x\}$; sign/root obstructions limit reachability.

Verdict: Conditionally true; otherwise underdetermined. The reachability assumption is the nontrivial crux.

Classification: Plausible adaptation; reachability is a **conjecture** unless proved for specific F .

3. Summary of Verdicts

#	Claim	Final Status	
---	-----	-----	

- | V1 | Global preservation forces identity | TRUE — exact theorem; near-tautological |
- | V2 | Asymmetric preservation constraint | TRUE — exact theorem; CL-relevant |
- | V3 | Overlapping inputs with different targets | TRUE — exact theorem; identifiability |
- | V4 | Depth-agnostic gate-value limitation | Representational TRUE; behavioral FALSE/underdetermined |
- | V5 | Projection rank condition | TRUE — exact linear algebra; capacity corollary withdrawn |
- | V6 | Gradient interference parameter budget | TRUE — exact under stated generality |
- | V7 | Error amplification bound | TRUE — exact; finite- D vs uniform- D corrected |
- | V8 | Finite-task reachability | CONDITIONAL; reachability is conjecture |

4. Prior Art Analysis

| Prior Art | Same Mathematical Problem? | Classification |

| --- | --- | --- |

| **Orthogonal gradient methods** (GEM, A-GEM, GPM) | Partly — addresses gradient interference in parameter space; not forward depth-aliasing; usually modifies main network | Known, partially relevant |

| **Parameter isolation / masking** (PackNet, Supermasks) | No — per-task masks/parameters; constraint-violating | Known, related but different |

| **Progressive networks / PathNet** | No — per-task architectural growth | Known, different problem |

| **Hypernetworks / fast weight programmers** | Partly — can generate input-dependent modulations; typically per-layer outputs | Known, partially relevant |

| **FiLM** (feature-wise linear modulation) | Partly — same gating algebra, but usually per-layer parameters | Known, directly relevant |

| **LSTM / GRU** | **Closest** — shared-parameter gates at every time step; crucially state-dependent, not input-only | Known, nearest prior art |

| **Mixture of Experts** | No — per-layer routers, distinct experts | Known, structurally similar |

| **Reservoir computing / echo state networks** | Partly — frozen recurrent core + trained readout | Known, relevant background |

| **Universal Transformer / ALBERT** | Partly — weight-tied depth, no task isolation | Known, background context |

| **Nonlinear RNN reachability / observability** | Yes for the reachability question | Known, should be used for V8 |

No surveyed prior art solves the exact problem: isolation within one frozen weight-tied composition via an input-only gate without per-depth trainable parameters.

5. Obstruction Characterization

There is no single obstruction theorem in the CL-relevant setting. The failure surface is a **conjunction of four necessary conditions**, each independently capable of killing the program, none individually fatal:

- **Identifiability failure** (V3): Conflicting targets on overlapping inputs are unservable by any input-only mechanism. In language-modeling CL, supports overlap extensively. Escape requires context-sufficiency or task conditioning.
- **Preservation-geometry confinement** (V1/V2): The gate is pinned to identity on every (old input, active coordinate) pair. Freedom survives only on never-active coordinates (substantial under sparse activations) and on new-task inputs.
- **Gradient coupling of shared ϕ** (V6): New-task updates generically move g_ϕ off identity on old inputs. Worst-case immunity costs a rank budget that is either restrictive ($p \leq d$) or vacuous ($p \gg d$).
- **Depth amplification** (V7): Under expansive gated dynamics, per-depth modulation error grows geometrically in D . Bounding demands architectural contractivity — an assumption on F , not a capability of the gate.

Plus three **scope conditionals**: (a) statelessness of the mechanism; (b) the t -conditioning interpretation; (c) BDH's native Hebbian plasticity actually disabled.

6. The Pivotal Scope Decision: t -Conditioning

A shared-parameter gate $g_\phi(x,t)$ where t is a **fixed, non-trainable encoding** (e.g., sinusoidal) adds $O(1)$ parameters and **does not violate "no per-depth/per-level trainable parameters" as literally worded**. This dissolves V1's depth-agnostic restrictions and V4's residue entirely, converting the problem into pure reachability (V8).

A **learned depth embedding** adds $O(D)$ parameters and violates the constraint outright.

Every impossibility-flavored statement applies only on the strict side of this fork. **The team must rule before any experiment is interpreted.**

7. Constructive Directions (all $\leq$ conjecture)

- **C1 — Orthogonal gradient on ϕ -space:** Project new-task gradients onto the orthogonal complement of old-task gradient subspace in ϕ -space, holding $g_\phi|_{S_{\text{old}}}$ $\approx \mathbf{1}$ to first order. Classification: **plausible adaptation** of GEM/GPM.

- **C2 — Fixed-encoding t -conditioning** ($O(1)$ parameters): dissolves depth-agnostic restrictions; legality contingent on scope ruling.
- **C3 — Delta-gate placement** for residual BDH: gate $g(x) \odot \Delta(s_t)$ inside $s_{t+1} = s_t + \odot$, never the residual stream.
- **C4 — Sparse-coordinate freedom:** exploit never-active coordinates of old-task trajectories (plausible under ReLU sparsity) as legal modification space.

8. Novelty Ledger (Final, Adjudicated)

| Claim | Classification |

|-----|-----|

| Problem formulation: isolation inside frozen weight-tied composition via input-only gate, CL-framed, two-channel interference taxonomy |

Apparently novel (packaging only) |

| V1, V2, V3, V5, V6, V7 | **Direct consequences** (textbook ingredients) |

| V4 representational lemma | Direct consequence; behavioral version **false as stated** |

| Necessity of state/depth information for effective gating | **Plausible adaptation** (consistent with LSTM/FiLM practice) |

| V8 reachability for concrete BDH $\$F\$$ | **Conjecture** |

| Expansiveness of trained BDH blocks | **Empirical conjecture** |

Contribution posture: a new formalization and a clean map of elementary obstructions — not new mathematics.

9. Direct Answer to the Research Question

> Can input-path interference arising from shared sequential composition be eliminated, bounded, or made controllable by an input-dependent gate/projection mechanism, without introducing per-depth/per-level trainable parameters and without retraining the frozen shared path?

| Reading | Verdict | Epistemic Status |

|-----|-----|-----|

| **Elimination**, global-preservation | Trivially impossible — only admissible gate is identity | Proved; degenerate |

| **Elimination**, asymmetric CL | **Open** — necessary conditions established, no general impossibility or possibility theorem | Partially characterized |

| **Bounding** | Finite-depth bounds exist for any finite L ; uniform-in-depth iff $L < 1$ | Proved, conditional |

| **Controllability** | **Not yet well-posed** — requires formal reachability definition and BDH instantiation | Undefined pending instantiation |

The failure condition is met only in the degenerate reading. In the CL-relevant reading, the correct outcome is: *not proven impossible, not proven possible; reduced to identifiable necessary conditions and one decisive open conjecture (reachability).*

10. Mandatory Follow-Ups

- **Instantiate F** from paper . tex: fix the literal update (residual form, Hebbian/multiplicative terms, normalization), choose gate placement (C3), verify native plasticity is disabled.
- **Rule on the t -conditioning fork** in writing.
- **Define controllability formally** as reachability of finite task targets by the gate class on the instantiated F .
- **Retrieve internal reports**; map the implemented mechanism onto exactly one formal variant; reconcile cl - h1 - report outcomes strictly as tests of V7/V8 conjectures.
- **Measurement program**: empirical Lipschitz/expansiveness of the gated BDH block (tests V7's conjecture); empirical cross-task Jacobian row-space overlap in ϕ -space (tests whether V6's worst case binds).

Critical review of the BDH input-path isolation analysis

Overall assessment

The multi-round analysis is substantially better in its final form than its first draft. The main correction — that the headline impossibility theorem only holds in a degenerate “preserve everything on every input” reading, and that the continual-learning-relevant asymmetric setting remains open — is correct and important. The separation of three different meanings of “interference” is useful, and the final verdict table is epistemically honest.

However, the analysis still has several serious gaps. The largest is that it never instantiates the abstraction with the actual BDH equations from paper .tex, despite the paper reportedly being available in RAG. Consequently, many of the formal claims are placement-specific, and the most important placement for BDH — gating a residual delta rather than a full block output — is not treated in the proofs.

1. Strengths worth preserving

Interference taxonomy.

The tripartition is the most useful contribution:

- cross-input forward-pass contamination,
- forward depth-aliasing,
- training-time cross-task gradient interference.

For a frozen, stateless, input-conditional gate, cross-input forward contamination is indeed structurally zero. The caveat that stateful components such as running-statistics normalization or caches can revive it is well taken.

Rejection of the naive diagonalization impossibility.

The original claim that a depth-agnostic gate is confined to the diagonal of the modulation product space and therefore cannot express depth-varying behavior was correctly downgraded. A constant-in-depth gate can produce depth-dependent effective behavior because the state changes across depth. The explicit counterexample with $(F(s)=s)$ is convincing.

Withdrawal of the projection capacity corollary.

$(\text{rank } P(x) \geq m(x))$ is correct as a necessary condition for exact trajectory preservation under output projection. The later step to $(\sum_i m_i \leq d)$ was a category error: forward-pass subspace geometry does not automatically translate into gradient-parameter orthogonality. Withdrawing it was correct.

Correct scoping of the main claim.

The final statement that exact elimination is only proved impossible under the degenerate global-preservation reading, while the asymmetric CL reading remains open, is the right conclusion. The four-

part obstruction characterization is a useful map of failure modes, even though it is not a single impossibility theorem.

2. Major gaps and weaknesses

2.1 The analysis is not grounded in the actual BDH equations

The entire theory is built for

$\backslash[$

$$s_{t+1}=F(s_t)$$

$\backslash]$

with a few still-abstract variants. But the analysis did not retrieve and parse paper .tex. It repeatedly says the literal BDH update was not instantiated. This is not a minor omission; it changes the applicable theorem statements.

Public descriptions suggest BDH-GPU has a ReLU-lowrank residual structure, something like:

$\backslash[$

$$x \mapsto \operatorname{ReLU}\bigl(x + \text{low-rank FFN}(x)\bigr)$$

\]

with shared parameters across depth, LayerNorm, and possibly Hebbian-style multiplicative terms.

This matters for three reasons:

- **Gate placement.** If the real BDH block is residual, then the natural gate is not

\[

$$s_{t+1} = g(x) \odot F(s_t)$$

\]

but rather

\[

$$s_{t+1} = s_t + g(x) \odot F(s_t).$$

\]

Under this delta gate, exact preservation of the unperturbed trajectory does **not** force $\forall i(x) = 1$. It forces

\[

$$g_i(x) = 0$$

\]

on every coordinate where $\forall i(s_t(x)) \neq 0$.

The “identity-forcing” theorem therefore does not transfer directly. The gate is pinned to zero on active coordinates, and only zero-coordinate

directions are free. This is analogous but not identical, and it should be restated.

- **LayerNorm and running statistics.** If the frozen path contains normalization layers that depend on batch or sequence statistics, then the mechanism is not stateless. A gate depending only on the original input is no longer purely cross-input-independent in the forward pass. The statelessness premise needs explicit verification.
- **Native Hebbian or plastic components.** The analysis acknowledges that BDH's native plasticity must be disabled for the frozen-path premise. But if the genuine BDH path contains Hebbian-style outer-product updates, then "freezing" it is not just a hyperparameter choice; it changes the architecture. Until this is checked, theorems about a static $\mathcal{F}$ may not apply.

2.2 The internal reports were not reconciled

The prompt states that the plan and H1 report are available in RAG. The analysis did not retrieve them. As a result:

- The implemented mechanism cannot be mapped to one of the formal variants.
- The empirical outcomes in `2026-08-25_cl-h1-report.md` are neither explained nor refuted.
- It is unknown whether the implementation gates the full block, the delta, an intermediate feature, or something else.

This is a major evidence-base gap. The final answer explicitly recommends reconciling the reports, but the review should treat this as a defect of the current analysis, not an optional next step.

2.3 “Controllability” is still not adequately formalized

The final report defines controllability as reachability of finite task targets by the gate class. That is a reasonable start, but it is not a full formalization.

For an input-dependent gate $\backslash(g(x)\backslash)$, the control set is severely restricted. At each depth the same gate output is applied, so the state trajectory is

$\backslash[$

$$s_{\{t+1\}} = \backslash\Phi(g(x), s_t),$$

$\backslash]$

where $\backslash(g(x)\backslash)$ is fixed for the entire pass. This is not the usual discrete-time control problem with independent controls $\backslash(u_t\backslash)$ at each step. The reachable set for input $\backslash(x\backslash)$ is parametrized by a single vector $\backslash(g(x)\backslash)$, not by $\backslash(D\backslash)$ independent controls.

A proper definition should specify:

- the state space,
- the admissible gate value space,
- the transition rule $\backslash(\Phi\backslash)$,

- the class of target maps $\{y_i(x)\}$,
- and whether the same gate parameters $\{\phi\}$ must simultaneously realize all tasks.

Without this, claims that “controllability dissolves into reachability” are only a promissory note.

2.4 The $\{t\}$ -conditioning fork remains unresolved

The analysis correctly identifies this as pivotal:

- A fixed, non-learned depth encoding $\{t\}$ as an input to a shared-parameter gate adds $\mathcal{O}(1)$ trainable parameters and likely does not violate the constraint literally.
- A learned depth embedding adds $\mathcal{O}(D)$ trainable parameters and does violate the constraint.

Until the project rules on this, the strong statements remain conditional. But the current write-up does not commit to either side, and some earlier statements still treat the strict depth-agnostic reading as the default. For clarity, every theorem should be prefixed with:

> “Assuming the strict depth-agnostic reading...”

or

> “Assuming fixed non-learned $\{t\}$ -conditioning is allowed...”

2.5 The parameter-budget bound is generic and likely non-binding

The gradient-interference bound,

$$\sum_i \dim R_i \leq p,$$

is correct under the very strong condition of zero first-order interference for all smooth losses. But it is essentially the standard parameter-isolation accounting for any shared parameter mechanism. It is not specific to shared sequential composition, and it gives no practical bound until the actual Jacobian ranks are measured.

For a gate network attached to an LLM-scale BDH path, (p) may be much larger than (d) , in which case the bound is vacuous for practical purposes. Conversely, the fixed-loss inner product can vanish accidentally even when row spaces overlap. The analysis acknowledges this, but the final obstruction list should not overstate V_6 as a decisive barrier.

2.6 The overlapping-input impossibility may need more nuance

V_3 says a deterministic input-only mechanism cannot satisfy two tasks that assign different targets to the same input. This is mathematically true, but its practical force depends on the task formalization.

In language-modeling-style continual learning, inputs are token sequences. If two tasks have different target distributions for the same prefix, the task identity is not contained in the input, and any input-only mechanism fails. But if the input includes task-relevant context, the overlap may be measure-zero. The analysis should distinguish:

- exact input collision,
- distribution-level overlap,
- near-overlap in representation space.

Without that, “supports overlap extensively” is an empirical claim, not a theorem.

3. Specific technical corrections

3.1 Restate V1/V2 for delta gating

For a residual delta gate,

\[

$$s_{t+1} = s_t + g(x) \odot F(s_t),$$

\]

exact preservation of the unperturbed trajectory on (S_{old}) gives

[

$$g_i(x)F_i(s_t(x)) = 0.$$

]

Thus:

- if coordinate (i) is active at some depth (t) , then $(g_i(x)=0)$;
- if coordinate (i) is never active along the trajectory, $(g_i(x))$ is unconstrained.

This is the correct “zero-forcing” obstruction, not the “identity-forcing” obstruction stated for full output gating.

3.2 Input projection versus output projection

The projection rank theorem is proved for output projection,

[

$$s_{t+1} = P(x)F(s_t).$$

]

For input projection,

$\forall$

$$s_{t+1} = F(P(x)s_t),$$

$\forall$

exact preservation requires

$\forall$

$$F(P(x)s_t) = F(s_t).$$

$\forall$

Only if $\forall(F)$ is injective along the trajectory does this imply $\forall(P(x)s_t = s_t)$, hence $\forall(s_t \in \text{Im}\{P(x)\})$. The theorem only holds under that injectivity condition and must not be stated as if the two placements are interchangeable.

3.3 The error-amplification “iff” is now correct

The final version of V7 correctly distinguishes finite-depth bounds from uniform-in-depth bounds:

- for fixed finite $\forall(D)$, a finite $\forall(L)$ always yields a finite bound;
- uniform-in-depth bounding requires $\forall(L < 1)$.

This correction should be retained, and any phrase like “bounding is possible exactly when $\backslash(L<1\backslash)$ ” should be avoided in future drafts.

4. Assessment of the final answer

The final answer is appropriately scoped:

- **Global-preservation elimination:** trivially impossible.
- **Asymmetric CL elimination:** open.
- **Bounding:** conditional on contractivity or finite depth.
- **Controllability:** undefined pending formalization.

This is intellectually honest and avoids the original overclaim.

However, the direct answer to the research question is still weak in the following sense. The question asks whether interference can be eliminated, bounded, or made controllable *by an input-dependent gate/projection mechanism*. The analysis does not deliver a theorem about the actual BDH mechanism. It delivers necessary conditions for an abstract family of mechanisms. The most decisive remaining item — reachability of the frozen BDH map under the allowed gate class — is unproved.

So the correct bottom line is:

> The strict proposed mechanism is not yet shown possible or impossible. The analysis reduces the question to a set of necessary conditions and one open reachability conjecture, but those conditions are not instantiated for BDH.

5. Prior-art assessment

The triage is mostly sound. LSTM/GRU is the closest neighbor and differs by using state-dependent gates. FiLM and fast weight programmers are closer than the early rounds admitted. Reservoir computing and nonlinear RNN reachability theory are relevant for the controllability question.

One missing item in some versions was the distinction between **learned per-step controls** and **input-only open-loop modulation**. This is the central difference from most control-theoretic prior art. The final analysis should make this even more explicit.

6. Recommendations before any write-up

- **Retrieve and parse paper .tex.** Extract the actual BDH block update, including residual form, normalization, and any Hebbian terms.

- **Reinstantiate every theorem** for the correct gate placement. The likely relevant case is delta gating $\{s_{t+1}=s_t+g(x)\odot F(s_t)\}$, not full output gating.
- **Rule on $\{t\}$ -conditioning.** Write down whether fixed non-learned depth encoding is permitted.
- **Retrieve the internal reports.** Map plan_v2 and cl-h1-report to the formal variants.
- **Define controllability rigorously** as a reachability problem over the frozen BDH map and the exact gate class.
- **Measure** the Lipschitz constant of the gated BDH block and the cross-task Jacobian row-space overlap in $\{\phi\}$ -space. These are the only claimed barriers that can be tested empirically.
- **Do not claim a general impossibility** in the asymmetric CL setting. The current evidence supports only necessary conditions and an open conjecture.

Bottom line

The final debate record is a materially improved and cautiously scoped analysis. Its best contributions are the interference taxonomy, the correction of the diagonalization and projection overclaims, and the explicit acknowledgment that the meaningful continual-learning question is open. Its main defects are lack of BDH grounding, placement-specific theorems stated too generally, an unresolved $\{t\}$ -conditioning fork, and an only partially formalized controllability notion. The honest published conclusion should be: **strict global preservation makes nontrivial gating impossible; asymmetric continual learning with a frozen shared path remains open pending concrete reachability analysis of the actual BDH update.**

Input-Path Interference in a Frozen Shared Sequential Composition: A Refined Formal Analysis for BDH

Status: Final, revised after critical review

Date: 2026-08-25

0. Scope and Evidence-Base Notice

BDH (arXiv:2509.26507) is a scale-free, biologically inspired network of locally interacting neuron particles with recurrent-depth parameter sharing. The internal project files listed in the task were not retrievable during this analysis, and the literal BDH update equations from `paper.tex` were not available for instantiation. **This document therefore proves results for the following abstraction, which both the research question and the project reports assume:**

 $\setminus[$

$$s_0(x)=x, \text{ \textit{qqquad} } s_{t+1}(x)=F(s_t(x)), \text{ \textit{qqquad} } t=0, \text{ \textit{ldots} }, D-1,$$

\]

with a residual variant $(s_{t+1}(x)=s_t(x)+F(s_t(x)))$ treated where explicitly stated. All theorem-shaped claims are about this abstraction. They become theorems about BDH **only after** (F) is instantiated with BDH's concrete update and the gate placement is fixed. This is stated as a mandatory follow-up, not as an optional refinement.

No experimental result from the implementation reports is used as evidence for any theorem.

1. Formal Definitions

1.1 Shared Sequential Composition

Let $(V=\mathbb{R}^d)$ be the state space and $(F:V\rightarrow V)$ a fixed map. A **depth- (D) shared sequential composition** is

[

$$s_0(x)=x, \quad s_{t+1}(x)=F(s_t(x)).$$

]

The unperturbed trajectory is $(\tau_F(x)=(s_0(x), \dots, s_D(x)))$. If the architecture uses residual connections, the update is instead

$\backslash[$

$$s_{t+1}(x) = s_t(x) + F(s_t(x)).$$

$\backslash]$

1.2 Frozen Shared Path

$\backslash(F\backslash)$ is **frozen** if its parameters are not updated during the continual-learning phase. Only the gate/projection parameters $\backslash(\phi\backslash)$ are trainable:

$\backslash[$

$$\frac{\partial F}{\partial \theta} = 0.$$

$\backslash]$

This is the standard continual-learning constraint; no theorem below relies on retraining $\backslash(F\backslash)$.

1.3 Gating and Projection

Input-dependent gate: a map $\backslash(g:X\rightarrow\mathbb{R}^d\backslash)$ applied identically at every depth. Canonical placements:

- Output gating: $\backslash(\Phi(g,s) = g(x) \odot F(s)\backslash)$
- Input gating: $\backslash(\Phi(g,s) = F(g(x) \odot s)\backslash)$
- Delta gating (for residual blocks): $\backslash(\Phi(g,s) = s + g(x) \odot F(s)\backslash)$

The gate has **no per-depth trainable parameter vectors**: the same function (g_ϕ) is used at every depth. We separately consider the relaxed variant $(g_\phi(x,t))$, where the depth index is supplied as an input, using either a **fixed non-trainable encoding** (adds $(O(1))$ parameters) or a **learned depth embedding** (adds $(O(D))$ parameters and violates the stated constraint).

Input-dependent projection: a map $(P:X\rightarrow\mathcal{P}(V))$ applied as

$$\begin{aligned} &[\\ &s_{t+1}=P(x)F(s_t) \\ &\quad\text{or}\quad \\ &s_{t+1}=F(P(x)s_t). \\ &] \end{aligned}$$

1.4 Input-Path Interference: Three Distinct Notions

The word "interference" must be disambiguated.

(A) Cross-input forward-pass contamination.

If (F) is frozen and the gate is stateless and a function of (x) alone, then the trajectory of (x_1) is computationally independent of the presence of (x_2) :

$$\begin{aligned} &[\\ &\tilde{s}_t^{(M)}(x_1\mid x_2\text{ elsewhere})=\tilde{s}_t^{(M)}(x_1). \end{aligned}$$

\]

This notion is **structurally zero** in this setting. It becomes live again only if the mechanism contains stateful components (running statistics, caches, etc.).

(B) Forward depth-aliasing.

A depth-agnostic gate applies the same modulation at every depth. If an oracle with access to depth $\ell(t)$ would apply a reference modulation $\ell(g^*(x, t))$, the depth-aliasing error is

\[

$\mathrm{DA}(g; x) =$

$\sum_{t=0}^{D-1}$

$\left|$

$\Phi(g(x), s_t(x)) - \Phi(g^*(x, t), s_t(x))$

$\right|$.

\]

This is a representational mismatch within one forward pass.

(C) Training-time gradient interference.

The gate parameters $\ell(\phi)$ are shared across all inputs. Updating $\ell(\phi)$ on task $\ell(B)$ may change gate outputs on task- $\ell(A)$ inputs. First-order damage is proportional to

$\langle$

$\langle \nabla_{\phi} \mathcal{L}_A(\phi), \nabla_{\phi} \mathcal{L}_B(\phi) \rangle$

$\rangle$

The operationally relevant channels for continual learning are **(B)** and **(C)**. Channel **(A)** is dead under a frozen, stateless, input-only mechanism.

1.5 Computation-Isolability

A mechanism is **exactly computation-isolating** for a task family $\{(S_i, y_i)\}$ if:

- task outputs are produced exactly: $\tilde{s}_D(x; \phi) = y_i(x)$ for all $x \in S_i$;
- forward depth-aliasing error is zero on the specified reference trajectories;
- first-order cross-task gradient interference is zero:

$\langle \nabla_{\phi} \mathcal{L}_i, \nabla_{\phi} \mathcal{L}_j \rangle = 0$ for all $i \neq j$.

It **bounds interference** if all relevant errors are at most ϵ , with ϵ controlled by explicit quantities.

1.6 Elimination, Bounding, Controllability

- **Elimination:** exact task outputs, exact zero interference.

- **Bounding:** finite worst-case error, typically depending on Lipschitz constants, depth, gate error, capacity.
- **Controllability:** formalized as **reachability**:

> A target function $y_i: S_i \rightarrow \mathbb{V}$ is reachable by the gate class if there exists ϕ such that $s_D(x; \phi) = y_i(x)$ for all $x \in S_i$.

> The mechanism is **controllable** over a finite task sequence if every finite set of reachable tasks with compatible overlaps is simultaneously realizable by a single ϕ .

This is the natural control-theoretic reading: because the gate is input-only, the control $g(x)$ is fixed for the entire depth trajectory, rather than an independently chosen per-step control u_t . Any statement about controllability must be understood through this definition, not through unrestricted discrete-time controllability.

2. Theorem-Shaped Hypotheses

T1. Exact Preservation Forces the Gate to Be Inert on Active Coordinates

Assumptions:

- $\mathbb{V}$ is defined on an open set $U \subseteq \mathbb{R}^d$.
- The mechanism is a depth-agnostic output gate:

$$s_{t+1} = g(x) \odot F(s_t).$$

- Exact preservation of the unperturbed trajectory is required on an input set S :

$$\tilde{s}_t(x) = F^t(x) \text{ for all } (x \in S), \text{ all } t.$$

Claim. For every $(x \in S)$ and every coordinate (i) ,

$$\begin{aligned} &[(g_i(x) - 1)F_i(s_t(x)) = 0. \\ &] \end{aligned}$$

If coordinate (i) is active at any depth along the trajectory, then $(g_i(x) = 1)$. If (F) is componentwise nonzero along all trajectories in (S) , then $(g(x) = \mathbf{1})$ on (S) .

Proof. At each depth, exact preservation gives $(g(x) \odot F(s_t(x)) = F(s_t(x)))$, which is componentwise $((g_i(x) - 1)F_i(s_t(x)) = 0)$. ■

Delta-gate variant. For residual delta gating,

$$\begin{aligned} &[\\ s_{t+1} &= s_t + g(x) \odot F(s_t), \\ &] \end{aligned}$$

exact preservation forces

$\forall$

$$g_i(x), F_i(s_t(x)) = 0.$$

$\forall$

Thus the gate is **zero-forced** on active coordinates, not identity-forced. This is the placement likely relevant to BDH's residual structure.

Verdict. True as an exact theorem; near-tautological under global preservation. Its real content is *locative*: it identifies exactly where gate freedom survives.

Classification. Direct consequence.

T2. Asymmetric Preservation: The Continual-Learning-Relevant Setting

Assumptions:

- Old-task inputs $\mathcal{S}_{\text{old}}$, new-task inputs $\mathcal{S}_{\text{new}}$.
- Exact preservation is required only on $\mathcal{S}_{\text{old}}$.
- No preservation constraint is imposed on $\mathcal{S}_{\text{new}}$.

Claim. On (S_{old}) , the gate is pinned to identity (output gating) or zero (delta gating) on every coordinate active along the old trajectory. On never-active coordinates, and on all of (S_{new}) , the gate is unconstrained by preservation.

Verdict. True as an exact theorem. This is the correct formalization of the continual-learning constraint. The identity/zero obstruction does **not** by itself rule out nontrivial isolation when old and new task inputs are separated and the gate can interpolate.

Classification. Direct consequence; the asymmetric framing is the honest CL formalization.

T3. Identifiability: Same Input, Different Target Is Impossible

Assumptions:

- The mechanism is a deterministic function of the input: $\tilde{s}_D(x; \phi) = \mathcal{M}_\phi(x)$.
- Two tasks share an input $(x \in S_A \cap S_B)$ and demand different outputs: $(y_A(x) \neq y_B(x))$.

Claim. No single (ϕ) can satisfy both tasks.

Proof. A function returns one output for one input. Two different required outputs are incompatible. ■

Verdict. True as an exact theorem. It is the sharpest structural limit of an input-only mechanism, independent of depth, capacity, or nonlinearity.

Practical nuance. In language continual learning, the force of this theorem depends on whether task-relevant context is part of the input. If task identity is encoded in the context, exact input collisions may be rare; if not, they can be pervasive. The theorem is about exact collision, not distribution-level overlap.

Classification. Direct consequence; essentially an identifiability condition.

T4. Depth-Agnostic Gate Values: Representational Limit, Not Behavioral Impossibility

Assumptions:

- $\Phi(g, s)$ is injective in the gate argument g along the trajectory.
- An oracle demands a depth-varying gate-value sequence $(g^x(x, 0), \dots, g^x(x, D-1))$, nonconstant in t .

Claim. A depth-agnostic gate $\backslash(g(x)\backslash)$ cannot reproduce the exact gate-value sequence $\backslash(g^*(x,t)\backslash)$.

Proof. If it did, injectivity would force $\backslash(g(x)=g^*(x,t)\backslash)$ for every $\backslash(t)\backslash$, contradicting nonconstancy. ■

Important correction. This is a statement about **gate values**, not about **reachable input-output behavior**. Because the state changes across depth, a constant gate can produce depth-varying effective behavior.

Counterexample. Let $\backslash(F(s)=s)\backslash$, $\backslash(D=2)\backslash$. Output gating gives

$\backslash[$

$$s_2 = g(x) \odot g(x) \odot x.$$

$\backslash]$

A constant gate $\backslash(\alpha, \beta)\backslash$ produces $\backslash(\alpha^2 x_1, \beta^2 x_2)\backslash$. A depth-varying sequence $\backslash(g^*_0=(c, 1), g^*_1=(1, c)\backslash)$ produces $\backslash((cx_1, cx_2)\backslash)$, which the constant gate $\backslash(\sqrt{c}, \sqrt{c})\backslash$ reproduces. Depth-varying controls can therefore be emulated by depth-agnostic gates via state feedback in some cases.

Verdict. The representational lemma is true. The behavioral impossibility is **not established**; it was withdrawn in the final adjudication. The reachability question is open.

Classification. Direct consequence for the representational part; the broad behavioral claim is **false as stated**.

T5. Projection Mechanisms: Necessary Rank Condition

Assumptions:

- Output projection: $\{s_{t+1}\} = P(x)F(s_t)$.
- Exact preservation of the unperturbed trajectory is required for input $\{x\}$.

Claim. Let

$\{$

$m(x) = \dim \operatorname{span}\{F(s_0(x)), \dots, F(s_{D-1}(x))\}.$

$\}$

Then $\operatorname{rank} P(x) \geq m(x)$.

Proof. Exact preservation forces $\{P(x)F(s_t(x)) = F(s_t(x))\}$ for all $\{t\}$, so all $\{F(s_t(x))\}$ lie in $\operatorname{Im} P(x)$. ■

Input-projection caveat. For input projection $\{s_{t+1} = F(P(x)s_t)\}$, exact preservation means $\{F(P(x)s_t) = F(s_t)\}$. This implies $\{P(x)s_t = s_t\}$ only if $\{F\}$ is injective along the trajectory. The theorem as stated applies to output projection; the input-projection variant requires the injectivity assumption.

Withdrawn corollary. The earlier claim $\{\sum_i m_i \leq d\}$ for multiple tasks is **withdrawn**. Image-subspace orthogonality is a forward-pass geometric property; cross-task gradient interference is a property of the parameterization of $\{P(x)\}$. These are different objects, and the capacity bound does not follow without a special parameterization assumption.

Classification. Direct consequence (linear algebra); no isolation capacity bound follows.

T6. Gradient Interference and the Parameter Budget

Assumptions:

- The gate/projection has $\{p\}$ trainable parameters.
- For task $\{i\}$, the row space of the collection of output Jacobians is $\{R_i \subseteq \mathbb{R}^p\}$.
- Zero first-order interference is required for **all smooth losses**.

Claim. The row spaces must be mutually orthogonal, and

$\sum_i \dim R_i \leq p$.

Proof. For arbitrary losses, zero inner products across all linear functionals force row-space orthogonality; orthogonal subspaces of $\mathbb{R}^p$ have total dimension at most p . ■

Caveats.

- For fixed specific losses, zero interference can hold accidentally even when row spaces overlap. The orthogonality condition is a worst-case guarantee, not a typical-case description.
- If $p \gg d$, the bound is likely non-binding for practical LLM-scale gates.
- This is standard parameter-isolation accounting, not a result specific to shared sequential composition.

Classification. Direct consequence; not a decisive obstruction unless Jacobian ranks are measured.

T7. Error Amplification: Bounding Is Conditional on Dynamics

Assumptions:

- The gated transition is Lipschitz in state with constant L .
- The per-depth gate error against a reference is at most δ .

Claim.

$$\begin{aligned} & \mathbb{E} \\ & \mathbb{E} |s_D(x) - s_{D^*}(x)| \\ & \leq \\ & \delta \sum_{k=0}^{D-1} L^k \\ & = \\ & \begin{cases} \delta \frac{1-L^D}{1-L}, & L < 1, \\ \delta D, & L = 1, \\ \delta \frac{L^D-1}{L-1}, & L > 1. \end{cases} \\ & \end{aligned}$$

Precise interpretation.

- For **fixed finite depth** D , a finite bound exists for every finite L .
- A bound that is **uniform in depth** exists iff $L < 1$.

- The claim "trained LLM-scale maps have $\epsilon(L > 1)$ " is an **empirical conjecture**, not a theorem, and must be measured for the actual BDH block.

Classification. Direct consequence; quantifier-corrected in this revision.

T8. Finite-Task Reachability: Conditional Construction, Otherwise Underdetermined

Assumptions:

- The task family is finite: $\{(x_i, y_i)\}, \{i=1, \dots, N\}$.
- The gate family can interpolate finite sets.
- Each target is reachable: $\{y_i(x) \in \text{mathcal R}_D(x) := \{\tilde{s}_D(x; \phi) : \phi \in \Phi\}\}$.
- Overlapping task inputs have agreeing targets.

Claim. There exists $\phi \in \Phi$ realizing all finite task targets exactly.

Proof sketch. Reachability gives a finite set of constraints; interpolation capacity fits them. ■

Underdetermination examples.

- If $(F \equiv 0)$, the reachable set is $\{0\}$ for every (x) , regardless of gating.
- If the same input has different targets, T3 applies.
- If $(F(s)=s)$ and $(D=2)$ with multiplicative output gating, reachable outputs are of the form $(g(x) \odot g(x) \odot x)$, which excludes many targets.

Verdict. Conditionally true; otherwise underdetermined. The reachability assumption is the nontrivial crux and remains a **conjecture** for a concrete BDH map.

Classification. Plausible adaptation of universal interpolation; reachability is unproved.

Summary of Verdicts

#	Claim	Status
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T1	Exact preservation forces identity/zero on active coordinates	True; exact theorem
T2	Asymmetric preservation constrains only old-task active coordinates	True; CL-relevant
T3	Same input, different target impossible	True; identifiability

| T4 | Depth-agnostic gate-value representational limit |
Representational true; behavioral impossibility false/underdetermined |

| T5 | Projection rank necessary condition | True; capacity corollary
withdrawn |

| T6 | Gradient interference parameter budget | True under worst-case
generality; often non-binding |

| T7 | Error amplification bound | True; finite-depth vs uniform-depth
corrected |

| T8 | Finite-task reachability | Conditional; reachability is conjecture |

3. Prior Art: Does It Address the Same Problem?

| Prior art | What it does | Same mathematical problem? | Classification
|

| --- | --- | --- | --- |

| Gradient projection / orthogonal-gradient methods (GEM, A-GEM,
GPM) | Project gradients in parameter space during training | Partly:
cross-task gradient interference; not forward depth-aliasing on a frozen
path | Known, partially relevant |

| Parameter isolation / masks (PackNet, Supermasks, Piggyback) |
Allocate disjoint parameter subsets to tasks | No: creates per-task
parameters/masks, violating the input-only gate constraint | Known,
related but different |

| Progressive networks / PathNet | Add task-specific columns or paths |
No: per-task architectural growth | Known, different problem |

| Hypernetworks / fast weight programmers | Generate weights or modulations for a main network | Partly: can generate input-dependent modulations; typically per-layer outputs | Known, partially relevant |

| FiLM / feature-wise linear modulation | Input-dependent affine modulation of hidden features | Partly: same gating algebra, but usually per-layer parameters | Known, directly relevant |

| LSTM / GRU | Shared-parameter state-dependent gates in a recurrent path | Closest: same composition; but gates see the current state, not only the input | Known, nearest prior art |

| Mixture of experts / conditional computation | Input-dependent routing to expert parameter sets | No: per-layer routers and distinct experts | Known, structurally similar |

| Reservoir computing / echo state networks | Fixed random recurrent core plus trained readout | Partly: frozen core; not input-dependent forward gating for task isolation | Known, relevant background |

| Weight-tied transformers (Universal Transformer, ALBERT) | Reuse a block across depth | Partly: weight-tied composition; no task-isolation gating | Known, background context |

| Nonlinear RNN reachability / observability | Characterizes what a recurrent map can reach | Yes for the reachability question | Known, should be applied to T8 |

Conclusion. No surveyed prior art solves the exact problem: isolation of computations within a single frozen weight-tied composition using only an input-dependent gate, with no per-depth trainable parameters and no retraining of the shared path.

4. Novelty Classification

| Claim | Classification |

| --- | --- |

| Formalization of two live interference channels (depth-aliasing and cross-task gradient interference) in a frozen weight-tied path |

Apparently novel (packaging only) |

| T1–T3, T5–T7 | **Direct consequences** (textbook ingredients) |

| T4 representational lemma | Direct consequence; behavioral version false as stated |

| Necessity of state/depth/context information for effective gating |

Plausible adaptation (consistent with LSTM/FiLM practice) |

| T8 reachability for concrete BDH $\setminus (F)$ | **Conjecture** |

| Expansiveness of trained BDH blocks | **Empirical conjecture** |

| Any observed effect in c1-h1-report | Inadmissible as theorem evidence; admissible only as a test of the conjectures above |

Contribution posture for write-up: **a new formalization and a clean map of elementary obstructions, not new mathematics.**

5. Obstruction Characterization

The strict proposed mechanism is:

- input-dependent gate/projection,
- no per-depth/per-level trainable parameters,
- frozen shared path,
- no retraining of $\mathcal{F}$.

Under the **global-preservation reading** — the mechanism must preserve $\mathcal{F}$'s behavior on **all** inputs — the gate is forced to be inert everywhere, so exact elimination is **trivially impossible**.

Under the **asymmetric continual-learning reading** — exact preservation on old-task inputs, free behavior on new-task inputs — no general impossibility or possibility theorem has been established. What is established is a conjunction of four necessary conditions:

- **Identifiability failure (T3).** Conflicting targets on identical inputs cannot be served by any input-only mechanism. In language modeling, the force of this depends on how much task identity is contained in the input context.
- **Preservation-geometry confinement (T1/T2).** The gate is pinned to identity (or zero under delta gating) on every active old-task coordinate. Freedom survives only on never-active coordinates and on new-task inputs.
- **Gradient coupling of shared ϕ (T6).** New-task updates generically move g_ϕ away from the preserved behavior on old inputs. Worst-case immunity costs a rank budget that may be non-binding when $p \gg d$.
- **Depth amplification (T7).** If the gated dynamics are expansive, small per-depth gate errors grow geometrically in depth. Bounding then requires contractivity of the composed system — a property of $\mathcal{F}$, not a capability of the gate alone.

Three **scope conditionals** attach to every result:

- **Statelessness.** Any stateful component inside the gate revives cross-input forward contamination (channel A).
- **(t) -conditioning fork.** A gate $(g_\phi(x,t))$ with a fixed non-trainable depth encoding adds $(O(1))$ parameters and does not violate the constraint as literally worded. A learned depth embedding adds $(O(D))$ parameters and does violate it. The depth-agnostic obstruction dissolves on the permissive side of this fork, converting the problem into pure reachability (T8).
- **Frozen path.** BDH's native Hebbian-style plasticity must be explicitly disabled for the frozen-path premise to hold.

6. Direct Answer to the Research Question

> Can input-path interference arising from shared sequential composition be eliminated, bounded, or made controllable by an input-dependent gate/projection mechanism, without per-depth/per-level trainable parameters and without retraining the frozen shared path?

| Reading | Verdict | Epistemic status |

| --- | --- | --- |

| **Elimination**, global preservation on all inputs | Impossible — the only admissible gate is identity/zero on active coordinates | Proved; degenerate |

| **Elimination**, asymmetric CL (preserve (S_{old}) , free on (S_{new})) | **Open**. Necessary conditions established; no general impossibility or possibility theorem | Partially characterized |

| **Bounding** | Finite-depth bounds exist for any finite (L) ; uniform-in-depth bound iff $(L < 1)$ | Proved, conditional |

| **Controllability** | Not yet well-posed: requires formal reachability analysis on the concrete frozen BDH map with the exact gate class | Undefined pending instantiation |

The failure condition contemplated by the brief is met **only in the degenerate global-preservation reading**. In the continual-learning-relevant reading, the correct outcome is: *not proven impossible, not proven possible; reduced to identifiable necessary conditions and one decisive open conjecture — reachability*.

7. Mandatory Follow-Ups Before Empirical Reconciliation

- **Instantiate (F) from paper .tex**. Fix the literal BDH update, including residual form, LayerNorm placement, and any Hebbian-style multiplicative terms.
- **Reinstantiate every theorem for the correct gate placement**. For a residual block, the natural gate is $(s_{t+1} = s_t + g(x) \odot F(s_t))$, which changes the identity obstruction to a zero obstruction.
- **Rule on the (t) -conditioning fork**. State in writing whether fixed non-learned depth encoding is permitted.
- **Retrieve the internal reports**. Map 2026-08-25_bdh-continual-learning-plan_v2.md and 2026-08-25_cl-h1-

report.md onto exactly one formal variant: depth-agnostic, $\mathcal{T}$ -conditioned, or state-conditioned.

- **Define controllability formally** as reachability of finite task targets by the gate class on the instantiated BDH map.
- **Measure** (a) the Lipschitz constant of the gated BDH block, and (b) cross-task Jacobian row-space overlap in $\mathcal{H}$ -space. These are the only claimed barriers that can be tested empirically.

Bottom line. A strict input-only gate/projection on a frozen shared sequential composition does not provide a proven general solution. The obstruction is not a single theorem but a map of four necessary failure conditions and one open reachability problem. The honest conclusion for publication is: *global preservation makes nontrivial gating impossible; asymmetric continual learning with a frozen shared path remains open pending concrete reachability analysis of the actual BDH update.*

[MODERATOR Round 4]

[moderator] Round 4: LLM call failed (litellm.RateLimitError: RateLimitError: OpenrouterException - {"error":{"message":"Provider returned error","code":429,"metadata":{"raw":"stealth/ox-alpha is temporarily rate-limited upstream. Please retry shortly."},"provider_name":"Stealth","is_byok":false,"limit_source":"upstream_provider_sha shortly, add your own provider key (<https://openrouter.ai/settings/integrations>), or route to another provider with provider routing: <https://openrouter.ai/docs/features/provider-routing>}},"user_id":"user_3Ay1IY7xf9lrDwaN7Fq4EUjnQA9"})

[STRATEGIST Round 5]

Final Strategic Assessment: Input-Path Interference in BDH via Gating/Projection

Date: 2026-08-25 | **Status:** Complete (4-round debate, adjudicated)

Executive Summary

Under the strict constraints — depth-agnostic input-dependent gate/projection, frozen shared path, no per-depth trainable parameters, no retraining — **exact elimination of input-path interference is impossible only under the degenerate global-preservation reading** (the gate must be identity everywhere, doing nothing). Under the **continual-learning-relevant asymmetric reading** (exact preservation on old tasks, free on new tasks), the problem is **open**: no general impossibility *or* possibility theorem has been established. What has been established is a precise map of four independent necessary conditions, each capable of killing the program, plus one decisive open conjecture (reachability) whose resolution determines feasibility.

1. Formal Definitions

1.1 Shared Sequential Composition

Let $V = \mathbb{R}^d$ be the state space. A **depth- D shared sequential composition** is defined by a fixed map $F: V \rightarrow V$ and the recurrence:

$$s_0(x) = x, \quad s_{t+1}(x) = F(s_t(x)), \quad t = 0, \dots, D-1$$

The unperturbed trajectory is $\tau_F(x) = (s_0(x), s_1(x), \dots, s_D(x))$.

If BDH uses residual connections (likely, per the BDH-GPU architecture), the relevant recurrence is:

$$s_{t+1}(x) = s_t(x) + F(s_t(x))$$

Classification: Direct characterization of BDH's weight-tied recurrent depth, corroborated by BDH-MLX ("Same weights reused in all layers (recurrent depth)") and BDH-GPU descriptions ("ReLU-Lowrank residual like $x \mapsto \dots$ "). Instantiation with the literal BDH update from paper . remains mandatory.

1.2 Frozen Shared Path

F is **frozen** if $\partial F / \partial \theta = 0$ during the continual-learning phase. Only the gate/projection mechanism ϕ may be trained.

Classification: Known — standard continual learning constraint.

1.3 Input-Path Interference: Three Precise Notions

(A) Cross-input forward-pass contamination. With frozen θ and a stateless input-conditioned mechanism, each input's forward pass is computationally independent. This notion is **structurally zero** and not a live channel.

(B) Forward depth-aliasing. A depth-agnostic gate $g(x)$ applies the same modulation at every depth. If an oracle would apply $g^*(x, t)$ (varying with depth), the depth-aliasing error is:

$$\mathrm{DA}(g; x) = \sum_{t=0}^{D-1} \left| \Phi(g(x), s_t(x)) - \Phi(g^*(x, t), s_t(x)) \right|$$

(C) Training-time gradient interference (cross-task forgetting).

Because gate parameters ϕ are shared across all inputs, updating ϕ on task B may change g_ϕ on task- A inputs. First-order damage is proportional to $\angle \nabla_\phi \mathcal{L}_A, \nabla_\phi \mathcal{L}_B \angle$.

The live channels are (B) and (C). Notion (A) is dead under the stated assumptions.

Classification: The depth-aliasing notion is **apparently novel** as a packaged definition; components are known from control theory. The gradient notion is **known** from continual learning.

1.4 Gating

An **input-dependent gate** $g: X \rightarrow \mathbb{R}^d$ is applied identically at every depth:

- Output gating: $s_{t+1} = g(x) \odot F(s_t)$
- Input gating: $s_{t+1} = F(g(x) \odot s_t)$
- Delta gating (for residual blocks): $s_{t+1} = s_t + g(x) \odot F(s_t)$

The gate has **no per-depth parameters**: the same $g(x)$ is used at every t .

Classification: Known mechanism (LSTM, MoE), but the depth-independence + frozen-path combination is novel.

1.5 Projection

An **input-dependent projection** $P: X \rightarrow \Pi(V)$ maps each input to an orthogonal projection, applied as $s_{t+1} = P(x) F(s_t)$ or $s_{t+1} = F(P(x) s_t)$.

Classification: Known (gradient projection methods), formalized for forward-pass application.

1.6 Computation-Isolability

A mechanism is **exactly computation-isolating** for a task family $\{(S_i, y_i)\}_{i=1}^K$ if: (1) each task output is produced exactly; (2) zero forward depth-aliasing error; (3) zero first-order cross-task gradient interference.

Classification: Apparently novel — this specific formulation combining shared composition, frozen parameters, and forward-pass isolation does not appear in surveyed literature.

1.7 Elimination vs. Bounding vs. Controllability

- **Elimination:** Exact equality of outputs and exact zero interference.
- **Bounding:** Deviation $\leq \epsilon$, controlled by Lipschitz constant, depth, gate error, parameter count.
- **Controllability:** Reachability of finite task targets by the gate class without modifying F .

Classification: Plausible adaptation of approximation theory and control-theoretic concepts.

2. Theorem-Shaped Hypotheses (Final Adjudicated Versions)

V1. Identity-Forcing Under Global Preservation

Assumptions: F defined on open $U \subseteq \mathbb{R}^d$; depth-independent output gating $s_{t+1} = g(x) \odot F(s_t)$; exact trajectory preservation on input set S : $\tilde{s}_t(x) = F^t(x)$ for all $x \in S$, all t .

Claim: For every $x \in S$ and every coordinate i : $g_i(x) = 1$ whenever $\exists t: F_i(s_t(x)) \neq 0$.

Proof: At depth t : $g(x) \odot F(s_t(x)) = F(s_t(x))$ implies $(g_i(x) - 1) F_i(s_t(x)) = 0$ componentwise. $\square$

Delta-gate variant: For residual delta gating $s_{t+1} = s_t + g(x) \odot F(s_t)$, exact preservation forces $g_i(x) F_i(s_t(x)) = 0$. The gate is **zero-forced** on active coordinates, not identity-forced.

Verdict: True as an exact theorem. Near-tautological under global preservation. Its real content is *locative*: it identifies precisely where gate freedom lives — never-active coordinates and non- S inputs.

Classification: Known / direct consequence.

V2. Asymmetric Preservation: The CL-Relevant Constraint

Assumptions: Old-task inputs S_{old} ; new-task inputs S_{new} ; output gating; exact preservation required only on S_{old} .

Claim: For $x \in S_{\text{old}}$: $g_i(x) = 1$ whenever $\exists t: F_i(s_t(x)) \neq 0$. On the zero-coordinate set $Z(x) = \{i: F_i(s_t(x)) = 0 \forall t\}$, the gate is unconstrained. On S_{new} , the gate is entirely unconstrained.

Proof: Same componentwise argument as V1, restricted to S_{old} . $\square$

Verdict: True as an exact theorem. If old-task trajectories have support in all coordinates, the gate is forced to identity on all of S_{old} . If old and new inputs are disjoint and the gate network can interpolate, exact isolation is *not* ruled out by this obstruction alone.

Classification: Direct consequence; the asymmetric framing is the correct CL formalization.

V3. Identifiability: Overlapping Inputs Are Fatal

Assumptions: The mechanism is a deterministic function of the input: $\tilde{s}_D(x; \phi) = \mathcal{M}_\phi(x)$. Two tasks A, B share an input $x \in S_A \cap S_B$ with different targets: $y_A(x) \neq y_B(x)$.

Claim: No single ϕ can satisfy both tasks.

Proof: The map is a function: one input, one output. Two different required outputs are incompatible. $\square$

Verdict: True as an exact theorem. The sharpest structural limit of "input-dependent only," independent of depth, nonlinearity, and capacity.

Classification: Direct consequence; essentially an identifiability condition.

V4. Depth-Agnostic Gate-Value Limitation (Downgraded)

Assumptions: $\Phi(g,s)$ injective in g along the trajectory; ideal modulation $g^*(x,t)$ nonconstant in t .

Claim: No depth-independent gate $g(x)$ can reproduce the nonconstant gate-value sequence $g^*(x,t)$.

Proof: Equality at every depth gives $g(x) = g^*(x,t)$ for every t , contradicting nonconstancy. $\square$

Critical correction: This is a statement about **gate values**, not about **reachable input-output maps**. A constant-in-depth gate can produce depth-varying effective behavior because the state s_t changes.

Counterexample: $F(s) = s$, $D = 2$. Output gating: $s_2 = g(x) \odot g(x) \odot x$. Constant gate $g(x) = (\alpha, \beta)$ produces $(\alpha^2 x_1, \beta^2 x_2)$. Depth-varying sequence $g^0 = (c, 1)$, $g^1 = (1, c)$ produces (cx_1, cx_2) , also achievable by constant gate $(\sqrt{c}, \sqrt{c})$.

Verdict: Representational lemma TRUE; behavioral impossibility NOT established. The reachability question is open.

Classification: Direct consequence for the representational part; the broad behavioral impossibility is **false as stated**.

V5. Projection Rank Condition

Assumptions: Output projection $s_{t+1} = P(x) F(s_t)$; exact preservation of unperturbed trajectory required.

Claim: If $m(x) = \dim \operatorname{span}\{F(s_0(x)), \dots, F(s_{D-1}(x))\}$, then $\operatorname{rank} P(x) \geq m(x)$.

Proof: $P(x) F(s_t(x)) = F(s_t(x))$ forces $F(s_t(x)) \in \operatorname{Im} P(x)$. $\square$

Input-projection caveat: For input projection $s_{t+1} = F(P(x)s_t)$, exact preservation means $F(P(x)s_t) = F(s_t)$. This implies $P(x)s_t = s_t$ only if F is injective along the trajectory.

Withdrawn corollary: The capacity corollary $\sum_i m_i \leq d$ is **WITHDRAWN**. Orthogonality of image subspaces is a forward-pass geometric property; cross-task gradient interference is a property of the parameterization. These are different mathematical objects.

Verdict: True as an exact linear-algebra theorem.

Classification: Direct consequence.

V6. Gradient Interference Parameter Budget

Assumptions: Gate has p trainable parameters ϕ ; task Jacobians $J_i \in \mathbb{R}^{d \times p}$; zero first-order interference required for *all smooth losses*.

Claim: Row spaces $\{R_i\}$ must be mutually orthogonal: $\sum_i \dim R_i \leq p$.

Proof: For arbitrary loss functions, vanishing inner products force row-space orthogonality. $\square$

Important caveats:

- For fixed specific losses, zero interference is weaker (can hold accidentally).
- If $p \gg d$ (typical for LLM-scale gates), the bound is non-binding.
- Worst-case guarantee, not typical-case behavior.

Verdict: True as an exact theorem under stated generality. Standard parameter-isolation accounting.

Classification: Direct consequence.

V7. Error Amplification Bound

Assumptions: Gated transition L -Lipschitz in state; per-depth gate error $\leq \delta$.

Claim:

$$\|s_D(x) - s_{D^*}(x)\| \leq \delta \sum_{k=0}^{D-1} L^k = \begin{cases} \delta \frac{1 - L^D}{1 - L} & \text{if } L < 1 \\ \delta D & \text{if } L = 1 \\ \delta \frac{L^D - 1}{L - 1} & \text{if } L > 1 \end{cases}$$

Precise interpretation:

- For **fixed finite depth D** : bound exists for every finite L .
- **Uniform-in-depth** bound exists iff $L < 1$.

Empirical caveat: "Trained LLM-scale maps have $L > 1$ " is an **empirical conjecture**, not a theorem.

Verdict: True as an exact theorem.

Classification: Known / direct consequence.

V8. Finite-Task Reachability (Conditional)

Assumptions: Finite task set (x_i, y_i) ; gate family has sufficient capacity; each target reachable: $y_i(x) \in \mathcal{R}_D(x) := \{\tilde{s}_D(x; \phi) : \phi \in \Phi\}$; overlapping inputs have agreeing targets.

Claim: There exists ϕ realizing all finite task targets exactly.

Counterexamples showing underdetermination:

- $F(s) = 0$: $\mathcal{R}_D(x) = \{0\}$ for every x , regardless of gating.
- Same input, different targets: V3 kills it.

- $F(s) = s$, $D = 2$, multiplicative output gating: reachable set is $\{g(x) \odot g(x) \odot x\}$; sign/root obstructions limit reachability.

Verdict: Conditionally true; otherwise underdetermined. The reachability assumption is the nontrivial crux.

Classification: Plausible adaptation; reachability is a **conjecture** unless proved for specific F .

3. Summary of Verdicts

#	Claim	Final Status
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V1	Global preservation forces identity	TRUE — exact theorem; near-tautological
V2	Asymmetric preservation constraint	TRUE — exact theorem; CL-relevant
V3	Overlapping inputs with different targets	TRUE — exact theorem; identifiability
V4	Depth-agnostic gate-value limitation	Representational TRUE; behavioral FALSE/underdetermined
V5	Projection rank condition	TRUE — exact linear algebra; capacity corollary withdrawn

| V6 | Gradient interference parameter budget | TRUE — exact under stated generality |

| V7 | Error amplification bound | TRUE — exact; finite- D vs uniform- D corrected |

| V8 | Finite-task reachability | CONDITIONAL; reachability is conjecture |

4. Prior Art Analysis

| Prior Art | Same Mathematical Problem? | Classification |

| --- | --- | --- |

| **Orthogonal gradient methods** (GEM, A-GEM, GPM) | Partly — addresses gradient interference in parameter space; not forward depth-aliasing; usually modifies main network | Known, partially relevant |

| **Parameter isolation / masking** (PackNet, Supermasks) | No — per-task masks/parameters; constraint-violating | Known, related but different |

| **Progressive networks / PathNet** | No — per-task architectural growth | Known, different problem |

| **Hypernetworks / fast weight programmers** | Partly — can generate input-dependent modulations; typically per-layer outputs | Known, partially relevant |

| **FiLM** (feature-wise linear modulation) | Partly — same gating algebra, but usually per-layer parameters | Known, directly relevant |

| **LSTM / GRU** | **Closest** — shared-parameter gates at every time step; crucially state-dependent, not input-only | Known, nearest prior art |

| **Mixture of Experts** | No — per-layer routers, distinct experts | Known, structurally similar |

| **Reservoir computing / echo state networks** | Partly — frozen recurrent core + trained readout | Known, relevant background |

| **Universal Transformer / ALBERT** | Partly — weight-tied depth, no task isolation | Known, background context |

| **Nonlinear RNN reachability / observability** | Yes for the reachability question | Known, should be used for V8 |

No surveyed prior art solves the exact problem: isolation within one frozen weight-tied composition via an input-only gate without per-depth trainable parameters.

5. Obstruction Characterization

There is no single obstruction theorem in the CL-relevant setting. The failure surface is a **conjunction of four necessary conditions**, each independently capable of killing the program, none individually fatal:

- **Identifiability failure** (V3): Conflicting targets on overlapping inputs are unservable by any input-only mechanism. In language-modeling CL, supports overlap extensively. Escape requires context-sufficiency or task conditioning.

- **Preservation-geometry confinement** (V1/V2): The gate is pinned to identity (output gating) or zero (delta gating) on every (old input, active coordinate) pair. Freedom survives only on never-active coordinates (substantial under sparse activations) and on new-task inputs.
- **Gradient coupling of shared ϕ** (V6): New-task updates generically move g_ϕ off identity on old inputs. Worst-case immunity costs a rank budget that is either restrictive ($p \leq d$) or vacuous ($p \gg d$).
- **Depth amplification** (V7): Under expansive gated dynamics, per-depth modulation error grows geometrically in D . Bounding demands architectural contractivity — an assumption on F , not a capability of the gate.

Plus three **scope conditionals**: (a) statelessness of the mechanism; (b) the t -conditioning interpretation; (c) BDH's native Hebbian plasticity actually disabled.

6. The Pivotal Scope Decision: t -Conditioning

A shared-parameter gate $g_\phi(x, t)$ where t is a **fixed, non-trainable encoding** (e.g., sinusoidal) adds $O(1)$ parameters and **does not violate "no per-depth/per-level trainable parameters"** as

literally worded. This dissolves V1's depth-agnostic restrictions and V4's residue entirely, converting the problem into pure reachability (V8).

A **learned depth embedding** adds $O(D)$ parameters and violates the constraint outright.

Every impossibility-flavored statement applies only on the strict side of this fork. **The team must rule before any experiment is interpreted.**

7. Constructive Directions (all $\leq$ conjecture)

- **C1 — Orthogonal gradient on ϕ -space:** Project new-task gradients onto the orthogonal complement of old-task gradient subspace in ϕ -space, holding $g_\phi|_{S_{\text{old}}} \approx \mathbf{1}$ to first order. Classification: **plausible adaptation** of GEM/GPM.
- **C2 — Fixed-encoding t -conditioning** ($O(1)$ parameters): dissolves depth-agnostic restrictions; legality contingent on scope ruling.
- **C3 — Delta-gate placement** for residual BDH: gate $g(x) \odot \Delta(s_t)$ inside $s_{t+1} = s_t + \cdot$, never the residual stream.

- **C4 — Sparse-coordinate freedom:** exploit never-active coordinates of old-task trajectories (plausible under ReLU sparsity) as legal modification space.

8. Novelty Ledger (Final, Adjudicated)

| Claim | Classification |

|-----|-----|

| Problem formulation: isolation inside frozen weight-tied composition via input-only gate, CL-framed, two-channel interference taxonomy |

Apparently novel (packaging only) |

| V1, V2, V3, V5, V6, V7 | **Direct consequences** (textbook ingredients) |

| V4 representational lemma | Direct consequence; behavioral version **false as stated** |

| Necessity of state/depth information for effective gating | **Plausible adaptation** (consistent with LSTM/FiLM practice) |

| V8 reachability for concrete BDH $\$F\$$ | **Conjecture** |

| Expansiveness of trained BDH blocks | **Empirical conjecture** |

Contribution posture: a new formalization and a clean map of elementary obstructions — not new mathematics.

9. Direct Answer to the Research Question

> Can input-path interference arising from shared sequential composition be eliminated, bounded, or made controllable by an input-dependent gate/projection mechanism, without introducing per-depth/per-level trainable parameters and without retraining the frozen shared path?

| Reading | Verdict | Epistemic Status |

|-----|-----|-----|

| **Elimination**, global-preservation | Trivially impossible — only admissible gate is identity/zero on active coordinates | Proved; degenerate |

| **Elimination**, asymmetric CL | **Open** — necessary conditions established, no general impossibility or possibility theorem | Partially characterized |

| **Bounding** | Finite-depth bounds exist for any finite L ; uniform-in-depth iff $L < 1$ | Proved, conditional |

| **Controllability** | **Not yet well-posed** — requires formal reachability definition and BDH instantiation | Undefined pending instantiation |

The failure condition is met only in the degenerate reading. In the CL-relevant reading, the correct outcome is: *not proven impossible, not proven possible; reduced to identifiable necessary conditions and one decisive open conjecture (reachability).*

10. Mandatory Follow-Ups

- **Instantiate F** from paper . tex: fix the literal update (residual form, Hebbian/multiplicative terms, normalization), choose gate placement (C3), verify native plasticity is disabled.
- **Rule on the t -conditioning fork** in writing.
- **Define controllability formally** as reachability of finite task targets by the gate class on the instantiated F .
- **Retrieve internal reports**; map the implemented mechanism onto exactly one formal variant; reconcile cl - h1 - report outcomes strictly as tests of V7/V8 conjectures.
- **Measurement program**: empirical Lipschitz/expansiveness of the gated BDH block (tests V7's conjecture); empirical cross-task Jacobian row-space overlap in ϕ -space (tests whether V6's worst case binds).

[CRITIC Round 5]

Final Critical Review: BDH Input-Path Isolation Debate

Overall Assessment

The analysis improved materially across rounds. Its main correction is the right one: the strong claim “exact elimination is impossible in general” has been dissolved into a **scoped set of exact necessary conditions plus an open reachability conjecture**. The separation of three distinct meanings of “interference” is the most useful contribution, and the final verdict table is epistemically honest.

That said, the analysis is **not yet grounded in the actual BDH mechanism**. It operates on an abstraction $\{s_{t+1} = F(s_t)\}$ or a residual variant, but does not instantiate $\{F\}$ from paper .tex. Consequently, several theorem statements are placement-sensitive, and the most BDH-relevant placement — **delta gating of a residual block** — was only partially treated.

Strengths Worth Preserving

- **Interference taxonomy.**

The separation of

- cross-input forward-pass contamination,
- forward depth-aliasing,
- training-time cross-task gradient interference

is correct and necessary. Under a frozen, stateless, input-dependent mechanism, cross-input forward contamination is structurally zero. The live channels are depth-aliasing and cross-task parameter updating.

- **Rejection of the naive diagonalization impossibility.**

The claim that a depth-agnostic gate is confined to the diagonal of the modulation space was correctly downgraded. A constant-in-depth gate can still produce depth-varying effective behavior because the state changes. The counterexample with $\{F(s)=s\}$ is decisive.

- **Withdrawal of the projection capacity corollary.**

The rank condition

$\forall$

$\operatorname{rank} P(x) \geq m(x)$

$\forall$

is correct for output projection. The stronger cross-task capacity bound $\sum_i m_i \leq d$ was correctly withdrawn. It conflated forward-pass subspace geometry with gradient-parameter geometry.

- **Correct scoping of the main claim.**

Exact elimination is proved impossible only in the degenerate global-preservation reading. In the continual-learning-relevant asymmetric setting, the question is not resolved either way.

Remaining Weaknesses and Gaps

1. The actual BDH equations were never instantiated

This is the largest remaining defect. The analysis repeatedly acknowledges that `paper.tex` was not parsed. Public descriptions suggest a BDH-GPU structure such as

$\lceil$

$x \mapsto \operatorname{ReLU}(x + \text{low-rank FFN}(x)),$

$\rceil$

with shared parameters, normalization, and possibly Hebbian-style multiplicative terms.

This matters because:

- **Delta gating changes the preservation obstruction.**

For full output gating $(s_{t+1} = g(x) \odot F(s_t))$, preservation forces $(g_i(x) = 1)$ on active coordinates.

For residual delta gating $(s_{t+1} = s_t + g(x) \odot F(s_t))$, preservation forces

$\lceil$

$g_i(x)F_i(s_t(x)) = 0,$

$\rceil$

so the gate is pinned to **zero**, not identity, on active coordinates.

- **Input versus output projection.**

The projection rank theorem does not automatically extend from output projection to input projection without injectivity assumptions on (F) .

- **Statefulness.**

If the frozen path contains normalization with running statistics, the mechanism is not stateless, which can revive cross-input forward contamination.

- **Hebbian/plastic components.**

If BDH includes native plastic terms, “freezing” the path is an architectural change that must be verified.

2. Internal reports were not reconciled

The plan and H1 report were not retrieved. Without them:

- the implemented gate placement is unknown,
- the empirical outcomes of `cl-h1-report.md` are neither explained nor refuted,
- the implementation cannot be mapped to a formal variant.

This is a defect of the current analysis, not merely an optional future step.

3. “Controllability” is still not fully formalized

The final definition — reachability of finite task targets by the gate class — is a reasonable start. But the control structure is unusual: an input-dependent gate $g(x)$ is fixed for the entire depth trajectory, so per-input reachable sets are parameterized by a single vector, not by (D) independent controls.

A complete formalization should specify:

- the state space,
- the admissible gate range,
- the exact transition rule,
- the target map class,
- and the requirement that one parameter vector ϕ simultaneously realize all tasks.

Until then, “controllability reduces to reachability” is only a promissory note.

4. The t -conditioning fork remains unresolved

The pivotal scope decision is still open:

- a fixed, non-learnable depth encoding adds $O(1)$ trainable parameters and likely does not violate the constraint,
- a learned depth embedding adds $O(D)$ trainable parameters and violates it.

Every impossibility-flavored statement holds only on the strict depth-agnostic side. Future write-ups must label assumptions explicitly.

5. The gradient-interference bound is not specific to shared composition

The bound

$\sum_i \dim R_i \leq p$

is standard parameter-isolation accounting. It is correct only under the strong requirement of zero first-order interference for all smooth losses. For fixed losses, interference can vanish accidentally. More importantly, if $(p \gg d)$, the bound is often non-binding. It should not be presented as a decisive shared-sequential-composition obstruction until actual Jacobian ranks are measured.

6. The overlapping-input theorem needs practical nuance

Same-input-different-target impossibility is mathematically true. But its practical force depends on whether task identity is represented in the input context. Exact input collisions and distribution-level overlap should not be conflated.

Final Scoped Answer to the Research Question

> Can input-path interference arising from shared sequential composition be eliminated, bounded, or made controllable by an input-dependent gate/projection mechanism, without per-depth/per-level trainable parameters and without retraining the frozen shared path?

| Reading | Verdict | Status |

| --- | --- | --- |

| **Elimination**, global preservation on all inputs | Trivially impossible: the only admissible gate is inert on active coordinates | Proved; degenerate |

| **Elimination**, asymmetric CL: preserve (S_{old}) , free (S_{new}) | **Open**. Necessary conditions established; no general impossibility or possibility theorem | Partially characterized |

| **Bounding** | Finite-depth bounds exist for any finite Lipschitz (L) ; uniform-in-depth bounding requires $(L < 1)$ | Proved, conditional |

| **Controllability** | Not yet well-posed without reachability analysis of the concrete frozen BDH map | Undefined pending instantiation |

The failure condition in the brief is met only in the **degenerate global-preservation reading**. In the meaningful continual-learning setting, the correct conclusion is:

> Not proven impossible; not proven possible. The question reduces to identifiable necessary conditions and one decisive open conjecture — reachability of the frozen BDH map under the allowed gate class.

Novelty Ledger, Final Form

| Claim | Classification |

| --- | --- |

| Two-channel interference taxonomy in a frozen weight-tied path |
Apparently novel as packaging; ingredients known |

| Identity-forcing / zero-forcing preservation lemma | Direct
consequence |

| Asymmetric-preservation constraint | Direct consequence; CL-relevant
reframing |

| Overlapping-input identifiability impossibility | Direct consequence |

| Depth-agnostic gate-value representational limit | Direct
consequence; behavioral version false as stated |

| Projection rank necessary condition | Direct consequence |

| Gradient-interference parameter budget | Direct consequence |

| Lipschitz error-amplification bound | Direct consequence |

| Necessity of state/depth/context information for robust gating |
Plausible adaptation |

| Reachability of concrete BDH $\backslash(F)$ under gating | Conjecture |

| Expansiveness of trained BDH blocks | Empirical conjecture |

No experimental result from the implementation reports was used as
evidence for any theorem.

Required Next Steps Before Publication

- **Parse paper .tex** and instantiate the literal BDH update, including residual form, normalization, and Hebbian or multiplicative terms.
- **Re-prove every theorem for the actual gate placement**, especially residual delta gating.
- **Rule formally on $\backslash(t\backslash)$ -conditioning**: fixed non-learned depth encoding versus learned depth embedding.
- **Retrieve and map the internal reports** to exactly one formal variant.
- **Define controllability rigorously** as finite-target reachability over the concrete BDH map and exact gate class.
- **Measure:**
 - the Lipschitz constant of the gated BDH block;
 - cross-task Jacobian row-space overlap in $\backslash(\phi\backslash)$ -space.

The honest published conclusion remains:

> **Strict global preservation makes nontrivial gating impossible. Asymmetric continual learning with a frozen shared path remains open pending concrete reachability analysis of the actual BDH update.**

[OPTIMIZER Round 5]

Input-Path Interference in BDH via Input-Dependent Gating/Projection: A Scoped Formal Analysis

Date: 2026-08-25

Status: Final, after adversarial review

1. Scope, Abstraction, and Evidence Base

BDH (arXiv:2509.26507) is a scale-free, biologically inspired network of locally interacting neuron particles with **shared parameters across depth** (“recurrent depth”), corroborated by third-party implementations and public descriptions. The literal BDH update equations from paper .tex and the internal project reports were not retrievable within this analysis. Consequently, the formal results below are proved for the abstraction that the research question itself assumes:

$$\begin{aligned} & \backslash[\\ & s_0(x)=x,\quad s_{t+1}(x)=F(s_t(x)),\quad t=0,\dots,D-1, \\ & \backslash] \end{aligned}$$

with the residual variant

$$\begin{aligned} & \backslash[\\ & s_{t+1}(x)=s_t(x)+F(s_t(x)) \\ & \backslash] \end{aligned}$$

treated where explicitly stated. All theorem-shaped claims are about this abstraction; they become theorems about BDH only after $\mathcal{F}$ is replaced by the concrete BDH update and the gate placement is fixed. This instantiation is a mandatory follow-up, not an optional refinement.

No experimental observation from any implementation report is used as evidence for any theorem.

2. Formal Definitions

2.1 Shared Sequential Composition

Let $(V = \mathbb{R}^d)$ be the state space and $(F: V \rightarrow V)$ a fixed map. A **depth- D shared sequential composition** is

[

$$s_0(x) = x, \quad s_{t+1}(x) = F(s_t(x)).$$

]

The unperturbed trajectory is $(\tau_F(x) = (s_0(x), \dots, s_D(x)))$.

2.2 Frozen Shared Path

F is **frozen** if its parameters are not updated during the continual-learning phase:

$$\frac{\partial F}{\partial \theta} = 0.$$

Only the gate/projection parameters ϕ are trainable.

2.3 Gating

An **input-dependent gate** is a map $g: X \rightarrow \mathbb{R}^d$ applied identically at every depth. Canonical placements:

- **Output gating:** $\Phi(g, s) = g(x) \odot F(s)$
- **Input gating:** $\Phi(g, s) = F(g(x) \odot s)$
- **Delta gating** (for residual blocks): $\Phi(g, s) = s + g(x) \odot F(s)$

The gate has **no per-depth trainable parameter vectors**: the same function g_ϕ is used at every depth. A relaxed variant $g_\phi(x, t)$, where the depth index is supplied as an input, is considered separately:

- **Fixed non-trainable depth encoding:** adds $O(1)$ trainable parameters; does not violate the constraint as literally worded.
- **Learned depth embedding:** adds $O(D)$ trainable parameters; violates the constraint.

2.4 Projection

An **input-dependent projection** is a map $\pi: X \rightarrow \Pi(V)$ applied as

[

$$s_{t+1} = P(x)F(s_t)$$

or

$$s_{t+1} = F(P(x)s_t).$$

]

2.5 Input-Path Interference: Three Distinct Notions

The word “interference” is ambiguous. Three notions must be separated.

(A) Cross-input forward-pass contamination.

If F is frozen and the gate is stateless and a function of x alone, then the trajectory of x_1 is computationally independent of the presence of x_2 :

[

$$\tilde{s}_t^{(M)}(x_1 \mid x_2 \text{ processed elsewhere}) = \tilde{s}_t^{(M)}(x_1).$$

]

This notion is **structurally zero** in this setting. It becomes live only if the mechanism contains stateful components (running statistics, caches, etc.).

(B) Forward depth-aliasing.

A depth-agnostic gate applies the same modulation at every depth. If an oracle with access to depth t would apply a reference modulation $g^*(x, t)$, the depth-aliasing error is

$$\begin{aligned} \mathrm{DA}(g; x) = & \sum_{t=0}^{D-1} \\ & \left| \Phi(g(x), s_t(x)) - \Phi(g^*(x, t), s_t(x)) \right|. \end{aligned}$$

This is a representational mismatch within a single forward pass.

(C) Training-time gradient interference.

The gate parameters ϕ are shared across all inputs. Updating ϕ on task B may change gate outputs on task- A inputs. First-order damage is proportional to

$$\left| \right|$$

$\langle \nabla_{\phi} \mathcal{L}_A(\phi), \nabla_{\phi} \mathcal{L}_B(\phi) \rangle$.

]

The operationally relevant channels for continual learning are **(B)** and **(C)**. Channel **(A)** is dead under a frozen, stateless, input-only mechanism.

2.6 Computation-Isolability

A mechanism is **exactly computation-isolating** for a task family $\{(S_i, y_i)\}$ if:

- task outputs are produced exactly: $\tilde{s}_D(x; \phi) = y_i(x)$ for all $(x \in S_i)$;
- forward depth-aliasing error is zero on the specified reference trajectories;
- first-order cross-task gradient interference is zero:

$\langle \nabla_{\phi} \mathcal{L}_i, \nabla_{\phi} \mathcal{L}_j \rangle = 0$ for all $(i \neq j)$.

It **bounds interference** if all relevant errors are at most ϵ , with ϵ controlled by explicit quantities.

2.7 Elimination, Bounding, Controllability

- **Elimination** means exact task outputs and exact zero interference.

- **Bounding** means a finite worst-case deviation ϵ , controlled by Lipschitz constants, depth, gate error, and capacity.
- **Controllability** is formalized as **reachability**: a target function $y_i: S_i \rightarrow V$ is reachable by the gate class if there exists ϕ such that $\tilde{s}_D(x; \phi) = y_i(x)$ for all $x \in S_i$; the mechanism is controllable over a finite task sequence if every finite set of reachable tasks with compatible overlaps is simultaneously realizable by a single ϕ .

This is the natural control-theoretic reading: because the gate is input-only, the control $g(x)$ is fixed for the entire depth trajectory, rather than an independently chosen per-step control u_t .

3. Theorem-Shaped Hypotheses

T1. Exact Preservation Forces the Gate to Be Inert on Active Coordinates

Assumptions. F is defined on an open set $U \subseteq \mathbb{R}^d$. The mechanism is a depth-agnostic output gate:

$$s_{t+1} = g(x) \odot F(s_t).$$

Exact preservation of the unperturbed trajectory is required on an input set $\mathcal{S}$:

[

$$\tilde{s}_t(x) = F^t(x) \quad \text{for all } x \in \mathcal{S}, \text{ for all } t.$$

]

Claim. For every $(x \in \mathcal{S})$ and every coordinate (i) ,

[

$$(g_i(x) - 1) F_i(s_t(x)) = 0.$$

]

If coordinate (i) is active at any depth along the trajectory, then $(g_i(x) = 1)$. If (F) is componentwise nonzero along all trajectories in $\mathcal{S}$, then $(g(x) = \mathbf{1})$ on $\mathcal{S}$.

Delta-gate variant. For residual delta gating,

[

$$s_{t+1} = s_t + g(x) \odot F(s_t),$$

]

exact preservation forces

$\setminus$

$g_i(x), F_i(s_t(x))=0.$

$\setminus$

Thus the gate is **zero-forced** on active coordinates, not identity-forced. This is the placement likely relevant to BDH's residual structure.

Verdict. True as an exact theorem. Near-tautological under global preservation; its real content is *locative*: it identifies precisely where gate freedom survives.

Classification. Direct consequence.

T2. Asymmetric Preservation: The Continual-Learning-Relevant Setting

Assumptions. Old-task inputs $\setminus(S_{\text{old}}\setminus)$, new-task inputs $\setminus(S_{\text{new}}\setminus)$. Exact preservation is required only on $\setminus(S_{\text{old}}\setminus)$. No preservation constraint is imposed on $\setminus(S_{\text{new}}\setminus)$.

Claim. On $\setminus(S_{\text{old}}\setminus)$, the gate is pinned to identity (output gating) or zero (delta gating) on every coordinate active along the old trajectory. On never-active coordinates, and on all of $\setminus(S_{\text{new}}\setminus)$, the gate is unconstrained by preservation.

Verdict. True as an exact theorem. This is the correct formalization of the continual-learning constraint. The identity/zero obstruction does **not** by itself rule out nontrivial isolation when old and new task inputs are separated and the gate can interpolate.

Classification. Direct consequence; the asymmetric framing is the honest CL formalization.

T3. Identifiability: Same Input, Different Target Is Impossible

Assumptions. The mechanism is a deterministic function of the input:

$$\begin{aligned} & \backslash[\\ & \tilde{s}_D(x; \phi) = \text{mathcal{M}}_\phi(x). \\ & \backslash] \end{aligned}$$

Two tasks share an input $(x \in S_A \cap S_B)$ and demand different outputs: $(y_A(x) \neq y_B(x))$.

Claim. No single (ϕ) can satisfy both tasks.

Proof. A function returns one output for one input; two different required outputs are incompatible. ■

Verdict. True as an exact theorem. It is the sharpest structural limit of an input-only mechanism, independent of depth, capacity, or nonlinearity.

Practical nuance. In language continual learning, the force of this theorem depends on whether task identity is encoded in the input context. Exact input collisions and distribution-level overlap should not be conflated.

Classification. Direct consequence; essentially an identifiability condition.

T4. Depth-Agnostic Gate Values: Representational Limit, Not Behavioral Impossibility

Assumptions. $\Phi(g, s)$ is injective in the gate argument g along the trajectory. An oracle demands a depth-varying gate-value sequence $(g^*(x, 0), \dots, g^*(x, D-1))$, nonconstant in t .

Claim. A depth-agnostic gate $g(x)$ cannot reproduce the exact gate-value sequence $(g^*(x, t))$.

Proof. If it did, injectivity would force $\forall (g(x)=g^*(x,t))$ for every $\forall (t)$, contradicting nonconstancy. ■

Important correction. This is a statement about **gate values**, not about **reachable input-output behavior**. Because the state changes across depth, a constant gate can produce depth-varying effective behavior.

Counterexample. Let $\forall (F(s)=s)$, $\forall (D=2)$. Output gating gives

$\forall$

$$s_2 = g(x) \odot g(x) \odot x.$$

$\forall$

A constant gate $\forall (\alpha, \beta)$ produces $\forall (\alpha^2 x_1, \beta^2 x_2)$. A depth-varying sequence $\forall (g^0_0=(c, 1), g^1_1=(1, c))$ produces $\forall ((cx_1, cx_2))$, which the constant gate $\forall (\sqrt{c}, \sqrt{c})$ reproduces.

Verdict. The representational lemma is true. The behavioral impossibility is **not established**; it was withdrawn in final adjudication. The reachability question is open.

Classification. Direct consequence for the representational part; the broad behavioral claim is **false as stated**.

T5. Projection Mechanisms: Necessary Rank Condition

Assumptions. Output projection:

$\{$

$$s_{t+1} = P(x)F(s_t).$$

$\}$

Exact preservation of the unperturbed trajectory is required for input $\lambda(x)$.

Claim. Let

$\{$

$$m(x) = \dim \operatorname{span} \{F(s_0(x)), \dots, F(s_{D-1}(x))\}.$$

$\}$

Then

$\{$

$$\operatorname{rank} P(x) \geq m(x).$$

$\}$

Proof. Exact preservation forces $(P(x)F(s_t(x))=F(s_t(x)))$ for all (t) , so all $(F(s_t(x)))$ lie in $(\text{Im}\{P(x)\})$. ■

Input-projection caveat. For input projection $(s_{t+1}=F(P(x)s_t))$, exact preservation means $(F(P(x)s_t)=F(s_t))$. This implies $(P(x)s_t=s_t)$ only if (F) is injective along the trajectory. The theorem as stated applies to output projection; the input-projection variant requires the injectivity assumption.

Withdrawn corollary. The earlier claim $(\sum_i m_i \leq d)$ for multiple tasks is **withdrawn**. Image-subspace orthogonality is a forward-pass geometric property; cross-task gradient interference is a property of the parameterization of $(P(x))$. These are different mathematical objects, and the capacity bound does not follow without a special parameterization assumption.

Classification. Direct consequence (linear algebra); no isolation capacity bound follows.

T6. Gradient Interference and the Parameter Budget

Assumptions. The gate/projection mechanism has (p) trainable parameters. For task (i) , the row space of the collection of output Jacobians is $(R_i \subseteq \mathbb{R}^p)$. Zero first-order interference is required for **all smooth losses**.

Claim. The row spaces must be mutually orthogonal, and

$\{$

$\sum_i \dim R_i \leq p.$

$\}$

Proof. For arbitrary losses, zero inner products across all linear functionals force row-space orthogonality; orthogonal subspaces of $(\mathbb{R}^p)$ have total dimension at most p . ■

Caveats.

- For fixed specific losses, zero interference can hold accidentally even when row spaces overlap.
- If $p \gg d$, the bound is likely non-binding for practical LLM-scale gates.
- This is standard parameter-isolation accounting, not a result specific to shared sequential composition.

Classification. Direct consequence; not a decisive obstruction unless Jacobian ranks are measured.

T7. Error Amplification: Bounding Is Conditional on Dynamics

Assumptions. The gated transition is Lipschitz in state with constant L . The per-depth gate error against a reference is at most δ .

Claim.

$$\begin{aligned} & \mathbb{E} |s_D(x) - s_{D^*}(x)| \\ & \leq \delta \sum_{k=0}^{D-1} L^k \\ & = \begin{cases} \delta \frac{1-L^D}{1-L}, & L < 1, \\ \delta D, & L = 1, \\ \delta \frac{L^D - 1}{L - 1}, & L > 1. \end{cases} \end{aligned}$$

Precise interpretation.

- For **fixed finite depth** D , a finite bound exists for every finite L .
- A bound that is **uniform in depth** exists iff $L < 1$.
- The claim “trained LLM-scale maps have $L > 1$ ” is an **empirical conjecture**, not a theorem, and must be measured for the actual BDH block.

Classification. Direct consequence; quantifier-corrected.

T8. Finite-Task Reachability: Conditional Construction, Otherwise Underdetermined

Assumptions. The task family is finite: $\{(x_i, y_i)\}, (i=1, \dots, N)$. The gate family can interpolate finite sets. Each target is reachable:

$\{$

$y_i(x) \in \text{mathcal R}_D(x) := \{\tilde{s}_D(x; \phi) : \phi \in \Phi\}.$

$\}$

Overlapping task inputs have agreeing targets.

Claim. There exists $\phi \in \Phi$ realizing all finite task targets exactly.

Proof sketch. Reachability gives a finite set of constraints; interpolation capacity fits them. ■

Underdetermination examples.

- If $(F \equiv 0)$, the reachable set is $\{0\}$ for every (x) , regardless of gating.

- If the same input has different targets, T3 applies.
- If $(F(s)=s)$ and $(D=2)$ with multiplicative output gating, reachable outputs are of the form $(g(x)\odot g(x)\odot x)$, which excludes many targets.

Verdict. Conditionally true; otherwise underdetermined. The reachability assumption is the nontrivial crux and remains a **conjecture** for a concrete BDH map.

Classification. Plausible adaptation of universal interpolation; reachability is unproved.

4. Summary of Verdicts

#	Claim	Status	
---	-----	-----	
T1	Exact preservation forces identity/zero on active coordinates	True; exact theorem	
T2	Asymmetric preservation constrains only old-task active coordinates	True; CL-relevant	
T3	Same input, different target impossible	True; identifiability	
T4	Depth-agnostic gate-value representational limit	Representational true; behavioral impossibility false/underdetermined	

| T5 | Projection rank necessary condition | True; capacity corollary withdrawn |

| T6 | Gradient interference parameter budget | True under worst-case generality; often non-binding |

| T7 | Error amplification bound | True; finite-depth vs uniform-depth corrected |

| T8 | Finite-task reachability | Conditional; reachability is conjecture |

5. Prior Art: Does It Address the Same Problem?

| Prior art | What it does | Same mathematical problem? | Classification |

| --- | --- | --- | --- |

| Gradient projection / orthogonal-gradient methods (GEM, A-GEM, GPM) | Project gradients in parameter space during training | Partly: cross-task gradient interference; not forward depth-aliasing on a frozen path | Known, partially relevant |

| Parameter isolation / masks (PackNet, Supermasks, Piggyback) | Allocate disjoint parameter subsets to tasks | No: creates per-task parameters/masks, violating the input-only gate constraint | Known, related but different |

| Progressive networks / PathNet | Add task-specific columns or paths | No: per-task architectural growth | Known, different problem |

| Hypernetworks / fast weight programmers | Generate weights or modulations for a main network | Partly: can generate input-dependent modulations; typically per-layer outputs | Known, partially relevant |

| FiLM / feature-wise linear modulation | Input-dependent affine modulation of hidden features | Partly: same gating algebra, but usually per-layer parameters | Known, directly relevant |

| LSTM / GRU | Shared-parameter state-dependent gates in a recurrent path | Closest: same composition; but gates see the current state, not only the input | Known, nearest prior art |

| Mixture of experts / conditional computation | Input-dependent routing to expert parameter sets | No: per-layer routers and distinct experts | Known, structurally similar |

| Reservoir computing / echo state networks | Fixed random recurrent core plus trained readout | Partly: frozen core; not input-dependent forward gating for task isolation | Known, relevant background |

| Weight-tied transformers (Universal Transformer, ALBERT) | Reuse a block across depth | Partly: weight-tied composition; no task-isolation gating | Known, background context |

| Nonlinear RNN reachability / observability | Characterizes what a recurrent map can reach | Yes for the reachability question | Known, should be applied to T8 |

Conclusion. No surveyed prior art solves the exact problem: isolation of computations within a single frozen weight-tied composition using only an input-dependent gate, with no per-depth trainable parameters and no retraining of the shared path.

6. Novelty Classification

| Claim | Classification |

| --- | --- |

| Formalization of two live interference channels (depth-aliasing and cross-task gradient interference) in a frozen weight-tied path |

Apparently novel (packaging only) |

| T1–T3, T5–T7 | **Direct consequences** (textbook ingredients) |

| T4 representational lemma | Direct consequence; behavioral version false as stated |

| Necessity of state/depth/context information for effective gating |

Plausible adaptation (consistent with LSTM/FiLM practice) |

| T8 reachability for concrete BDH $\setminus (F)$ | **Conjecture** |

| Expansiveness of trained BDH blocks | **Empirical conjecture** |

| Any observed effect in c1-h1-report | Inadmissible as theorem evidence; admissible only as a test of the conjectures above |

Contribution posture: **a new formalization and a clean map of elementary obstructions, not new mathematics.**

7. Obstruction Characterization

The strict proposed mechanism is:

- input-dependent gate/projection,
- no per-depth/per-level trainable parameters,

- frozen shared path,
- no retraining of $\mathcal{F}$.

Under the **global-preservation reading** — the mechanism must preserve $\mathcal{F}$'s behavior on **all** inputs — the gate is forced to be inert everywhere, so exact elimination is **trivially impossible**.

Under the **asymmetric continual-learning reading** — exact preservation on old-task inputs, free behavior on new-task inputs — no general impossibility or possibility theorem has been established. What is established is a conjunction of four necessary conditions:

- **Identifiability failure (T3)**. Conflicting targets on identical inputs cannot be served by any input-only mechanism. In language modeling, the force of this depends on how much task identity is contained in the input context.
- **Preservation-geometry confinement (T1/T2)**. The gate is pinned to identity (or zero under delta gating) on every active old-task coordinate. Freedom survives only on never-active coordinates and on new-task inputs.
- **Gradient coupling of shared ϕ (T6)**. New-task updates generically move g_ϕ away from the preserved behavior on old inputs. Worst-case immunity costs a rank budget that may be non-binding when $p \gg d$.
- **Depth amplification (T7)**. If the gated dynamics are expansive, small per-depth gate errors grow geometrically in depth. Bounding then requires contractivity of the composed system — a property of $\mathcal{F}$, not a capability of the gate alone.

Three **scope conditionals** attach to every result:

- **Statelessness.** Any stateful component inside the gate revives cross-input forward contamination.
- **ϕ -conditioning fork.** A gate $g_\phi(x,t)$ with a fixed non-trainable depth encoding adds $O(1)$ parameters and does not violate the constraint as literally worded. A learned depth embedding adds $O(D)$ parameters and does violate it. The depth-agnostic obstruction dissolves on the permissive side of this fork, converting the problem into pure reachability (T8).
- **Frozen path.** BDH's native Hebbian-style plasticity must be explicitly disabled for the frozen-path premise to hold.

8. Direct Answer to the Research Question

> Can input-path interference arising from shared sequential composition be eliminated, bounded, or made controllable by an input-dependent gate/projection mechanism, without per-depth/per-level trainable parameters and without retraining the frozen shared path?

| Reading | Verdict | Epistemic status |

| --- | --- | --- |

| **Elimination**, global preservation on all inputs | Impossible — the only admissible gate is identity/zero on active coordinates | Proved; degenerate |

| **Elimination**, asymmetric CL (preserve S_{old}), free on S_{new} | **Open**. Necessary conditions established; no general impossibility or possibility theorem | Partially characterized |

| **Bounding** | Finite-depth bounds exist for any finite $\mathcal{L}$; uniform-in-depth bound iff $|\mathcal{L}| < \aleph_1$ | Proved, conditional |

| **Controllability** | Not yet well-posed: requires formal reachability analysis on the concrete frozen BDH map with the exact gate class | Undefined pending instantiation |

The failure condition contemplated by the brief is met **only in the degenerate global-preservation reading**. In the continual-learning-relevant reading, the correct outcome is: *not proven impossible, not proven possible; reduced to identifiable necessary conditions and one decisive open conjecture — reachability*.

9. Mandatory Follow-Ups Before Empirical Reconciliation

- **Instantiate $\mathcal{F}$ from paper .tex.** Fix the literal BDH update, including residual form, LayerNorm placement, and any Hebbian-style multiplicative terms.
- **Reinstantiate every theorem for the correct gate placement.** For a residual block, the natural gate is $s_{t+1} = s_t + g(x) \odot F(s_t)$, which changes the identity obstruction to a zero obstruction.
- **Rule on the $\mathcal{L}$ -conditioning fork.** State in writing whether fixed non-learned depth encoding is permitted.
- **Retrieve the internal reports.** Map 2026-08-25_bdh-continual-learning-plan_v2.md and 2026-08-25_cl-h1-report.md onto exactly one formal variant: depth-agnostic, $\mathcal{L}$ -conditioned, or state-conditioned.
- **Define controllability formally** as reachability of finite task targets by the gate class on the instantiated BDH map.

- **Measure** (a) the Lipschitz constant of the gated BDH block, and (b) cross-task Jacobian row-space overlap in ϕ -space. These are the only claimed barriers that can be tested empirically.

Bottom line. A strict input-only gate/projection on a frozen shared sequential composition does not provide a proven general solution. The obstruction is not a single theorem but a map of four necessary failure conditions and one open reachability problem. The honest conclusion for publication is: *global preservation makes nontrivial gating impossible; asymmetric continual learning with a frozen shared path remains open pending concrete reachability analysis of the actual BDH update.*

[MODERATOR Round 5]

Final Moderator Synthesis — Debate Closure: Input-Path Isolation in BDH via Gating/ Projection

Role: ox-alpha, moderator. This document consolidates five rounds of adversarial deliberation (STRATEGIST, CRITIC, OPTIMIZER), records the final consensus state, adjudicates all surviving disputes, and delivers the scoped answer to the research question together with the precise characterization of the obstruction.

0. Process Record and Evidence-Base Status

The debate executed a full correction loop:

- **Rounds 1–2** built a unified formal skeleton and initially headlined a strong claim: *"exact elimination of input-path interference is impossible in general"* via three obstructions (identity-forcing, diagonalization, rank budget).
- **Round 3–4 CRITIC reviews** landed decisive defeaters: (i) the headline rested on a *degenerate* global-preservation reading; (ii) the diagonalization argument conflated gate **values** with reachable **behaviors** (counterexample: $F(s)=s$, $D=2$); (iii) the projection capacity corollary $\sum_i m_i \leq d$ committed a category error (forward-pass image-subspace geometry $\neq$ parameter-space gradient geometry); (iv) the amplification "iff $L < 1$ " misquantified finite- D vs. uniform-in- D bounds; (v) controllability was undefined; (vi) the BDH-relevant **delta-gate placement** converts identity-forcing into **zero-forcing**, changing the obstruction's form.
- **Round 5:** all three roles converged on the same scoped verdict set (STRATEGIST V1–V8, OPTIMIZER T1–T8 — substantively identical), with the CRITIC accepting the scoping while flagging residual gaps.

Adjudication principle throughout: a claim enters the record only at the strength of its weakest surviving proof.

Evidence-base limitations (recorded for honesty): the internal project files (2026-08-* reports, paper-parity-plan, bdh-continual-learning-plan_v2.md, cl-h1-report.md) and the literal equations from paper.tex were **never retrieved in-debate** despite repeated attempts. All results below are therefore proved for the abstraction $s_{t+1} = \Phi(g(x), s_t)$ over a frozen map F (with the residual/delta variant treated explicitly). Instantiation against the actual BDH update remains a mandatory follow-up, not an optional refinement. No

experimental observation from any report was admitted as theorem evidence at any point.

1. Consensus Evaluation

Full consensus (all three roles, post-revision)

| Item | Status |

| --- | --- |

| Formal skeleton: shared sequential composition, frozen path, gate/projection placements (output / input / delta), elimination-vs-bounding distinction | Agreed verbatim |

| **Interference taxonomy (three channels):** (A) cross-input forward-pass contamination — **structurally zero** for stateless mechanisms on a frozen path; (B) forward **depth-aliasing** within a pass — live; (C) training-time **cross-task gradient interference** through shared ϕ — live | Agreed; channel (A)'s revival conditions (stateful components: running statistics, caches) noted |

| Verdicts T1–T3, T5–T8 (see §3) | Agreed |

| Prior-art triage: LSTM/GRU nearest neighbor (state-conditioned gates); FiLM same gating algebra but per-layer parameters; fast-weight programmers/hypernetworks adjacent; reservoir computing background; GEM/A-GEM/GPM parameter-space complementary; PackNet/supermasks/progressive nets constraint-violating; MoE different problem; Universal Transformer/ALBERT backdrop. **No surveyed work solves the exact problem.** | Agreed |

Adjudicated disputes (all resolved)

- **"Impossibility in general" vs. scoped openness** → the **scoped formulation prevails**. The unscoped headline holds only under global preservation on *all* inputs — a reading under which the gate must be inert everywhere and hence cannot adapt anywhere, dissolving the continual-learning question itself. STRATEGIST's own §7 concession is entered as binding.
- **Diagonalization obstruction** → downgraded from behavioral impossibility to a true-but-representational lemma about gate values; superseded by the open reachability question (T8).
- **Projection capacity corollary** → withdrawn; survives only under a special independent-subspace parameterization that violates the spirit of the constraints anyway.
- **Delta-gate placement** → incorporated into T1/T2: under $s_{t+1} = s_t + g(x) \odot F(s_t)$, preservation forces $g_i(x) = 0$ (not 1) on active old-task coordinates. This is the placement likely correct for BDH's residual structure.

Residual dissent (non-blocking)

The CRITIC maintains two emphasis-level caveats, both accepted into the record: (a) the practical force of T3 depends on measure-theoretic overlap of task supports — exact collision, distributional overlap, and representation-space near-overlap must not be conflated; (b) T6 is worst-case accounting that is likely non-binding when $p \gg d$ and must not be presented as a decisive barrier until Jacobian ranks are measured.

2. Consolidated Definitions (Final)

- **Shared sequential composition:** $s_0 = x$, $s_{t+1} = F(s_t)$, D steps; residual variant $s_{t+1} = s_t + F(s_t)$.
- **Frozen shared path:** $\partial F / \partial \theta = 0$ during adaptation; only ϕ (gate parameters) trainable.
- **Gate:** input-dependent $g_\phi(x)$ applied identically at every depth; placements: output $g \odot F(s)$, input $F(g \odot s)$, $\Delta s + g \odot F(s)$. No per-depth parameter vectors.
- **Projection:** input-dependent orthogonal $P(x)$; output $P(x)F(s)$ or input $F(P(x)s)$.
- **Computation-isolability (exact):** exact task outputs + zero depth-aliasing error + zero first-order cross-task gradient interference.
- **Elimination vs. bounding:** exact equality/zero interference vs. finite worst-case deviation controlled by explicit quantities (L_ϕ , D , Δ , p).
- **Controllability (partial formalization):** reachability of finite task targets by the gate class on the frozen map — with the crucial caveat that an input-only control $g(x)$ is fixed for the entire pass (open-loop modulation), not an independent per-step control u_t . A fuller formalization specifying state space, admissible gate range, transition rule, target class, and joint-realizability requirements remains outstanding.

3. Final Adjudicated Verdict Table

| # | Claim | Status | Notes |

| --- | --- | --- | --- |

| **T1** | Exact preservation forces the gate **inert** on active coordinates: $g_i = \mathbf{1}$ (output gating) or $g_i = 0$ (**delta gating**) wherever $F_i(s_t(x)) \neq 0$ along the preserved trajectory | **TRUE — exact theorem; direct consequence** | Near-tautological globally; real content is *locative*: identifies precisely where gate freedom lives (never-active coordinates; non- S inputs) |

| **T2** | Under **asymmetric preservation** (exact on S_{old} , free on S_{new}), the confinement restricts to old-task active coordinates | **TRUE — exact theorem; the honest CL formalization** | Does not by itself preclude nontrivial isolation when supports are separable and the gate can interpolate |

| **T3** | Deterministic input-only mechanism cannot serve two tasks demanding different outputs at a shared input | **TRUE — exact theorem (identifiability)** | Sharpest limit of "input-dependent only"; practical force depends on context-sufficiency of inputs |

| **T4** | Depth-agnostic gate confined to diagonal of modulation space | Representational lemma **true**; behavioral impossibility **not established** (downgraded after counterexample) | Superseded by open reachability question |

| **T5** | Output projection preserving trajectory requires $\text{rank } P(x) \geq m(x)$ | **TRUE — exact linear algebra** | Capacity corollary withdrawn; input-projection variant requires injectivity of F along the trajectory |

| **T6** | Zero first-order interference *for all smooth losses* $\Leftrightarrow$ mutually orthogonal task-Jacobian row spaces $\Rightarrow \sum_i \dim R_i \leq p$ | **TRUE — exact under stated generality** | Worst-case guarantee; likely non-binding for $p \gg d$; standard accounting, not composition-specific |

| **T7** | $e_D \leq \delta \sum_{k < D} L_{\Phi^k}$; finite- D bound exists for every finite L_{Φ} ; **uniform-in-depth** bound iff $L_{\Phi} < 1$ | **TRUE —**

exact, quantifiers corrected | "Trained BDH blocks are expansive" reclassified as empirical conjecture requiring measurement |

| **T8** | Finite task sets realizable iff targets reachable by gate class + interpolation capacity + overlap agreement | **CONDITIONAL; crux is a conjecture** | Counterexamples ($F \equiv 0$; sign/root obstructions) show reachability fails generically without hypotheses on F |

4. Final Answer to the Research Question

> *Can input-path interference arising from shared sequential composition be eliminated, bounded, or made controllable by an input-dependent gate/projection mechanism, without per-depth/per-level trainable parameters and without retraining the frozen shared path?*

| Reading | Verdict | Epistemic status |

| --- | --- | --- |

| **Elimination**, global-preservation (mechanism preserves F 's behavior on *all* inputs) | Trivially impossible — the only admissible gate is inert everywhere | Proved; degenerate |

| **Elimination**, asymmetric CL (exact preservation on S_{old} , free on S_{new}) | **OPEN**. Necessary conditions established (T1–T3, T5–T7); no general possibility *or* impossibility theorem | Partially characterized |

| **Bounding** | Finite-depth bounds exist for any finite L_Φ ; **uniform-in-depth** bounding iff gated dynamics contractive. Bounding is

a property of the composed system, not achievable by gate design alone under expansion | Proved, conditional |

| **Controllability** | **Not yet well-posed**: requires formal reachability analysis on the concrete frozen BDH map with the exact gate class | Undefined pending instantiation |

The brief's failure condition is met **only in the degenerate reading**. In the CL-relevant reading the correct outcome is: *not proven impossible, not proven possible; reduced to identifiable necessary conditions plus one decisive open conjecture (reachability)*. Reporting anything stronger would violate the novelty discipline.

5. Obstruction Characterization (Deliverable Under the Failure Clause)

There is no single obstruction theorem in the meaningful setting. The failure surface is a **conjunction of four necessary conditions**, each independently capable of killing the program, none individually fatal:

- **Identifiability failure (T3)** — conflicting targets on overlapping inputs are unservable by any input-only mechanism. In language-modeling CL, escape requires context-sufficiency of the input or task conditioning.
- **Preservation-geometry confinement (T1/T2)** — the gate is pinned to identity (output gating) or zero (delta gating) on every (old input, active coordinate) pair. Freedom survives on never-active coordinates — potentially substantial under BDH's ReLU sparsity — and on new-task inputs.

- **Gradient coupling of shared ϕ (T6)** — new-task updates generically move g_ϕ off the preserved behavior on old inputs; worst-case immunity costs a rank budget that is restrictive ($\leq d$) or vacuous ($\leq \text{rank } d$).
- **Depth amplification (T7)** — under expansive gated dynamics, per-depth modulation error grows geometrically in D ; bounding then demands architectural contractivity, an assumption on F , not a capability of the gate.

Plus **three scope conditionals** silently underlying every result: (a) **statelessness** of the mechanism (any running-statistics component revives channel A); (b) the **ϕ -conditioning fork** — a shared-parameter gate $g_\phi(x,t)$ fed by a *fixed, non-trainable* depth encoding adds $O(1)$ parameters and does not violate the constraint as literally worded, and it dissolves the depth-agnostic restrictions entirely, converting the problem into pure reachability (T8); a *learned* depth embedding adds $O(D)$ parameters and violates the constraint outright. Every impossibility-flavored statement applies only on the strict side of this fork; (c) BDH's native Hebbian-style plasticity must be explicitly disabled, else the path is not frozen.

6. Novelty Ledger (Final)

| Claim | Classification |

| --- | --- |

| Problem formulation: isolation inside a frozen weight-tied composition via input-only gate, CL-framed, three-channel interference taxonomy |
Apparently novel (packaging only) |

| T1–T3, T5–T7 | **Direct consequences** (textbook ingredients) |

| T4 representational lemma | Direct consequence; behavioral version false as stated |

| Necessity of state/depth/context information for effective gating |

Plausible adaptation (consistent with LSTM/FiLM practice; proven only under injectivity assumptions) |

| Reachability of the concrete BDH map under the gate class (T8 crux) |

Conjecture |

| Expansiveness of trained BDH blocks | **Empirical conjecture** |

| Any observed effect in c_l-h₁-report | Inadmissible as theorem evidence; admissible only as a test of the above conjectures |

Write-up posture: claim a new formalization and a clean map of elementary obstructions — not new mathematics.

7. Constructive Directions Surfaced (all $\leq$ conjecture)

- **C1 — Orthogonal-gradient protection of ϕ :** once the protected object is recognized as the *gate parameters*, GEM/GPM-style projection applies compatibly with all constraints: project new-task gradients onto the orthogonal complement of the old-task gradient subspace in ϕ -space, holding $g_\phi|_{S_{\text{old}}}$ $\approx$ inert to first order. Plausible adaptation; open statistical question about generalization beyond the old-task sample.

- **C2 — Fixed-encoding $\text{t}\text{\$}$ -conditioning** ($O(1)$ parameters): dissolves depth-agnostic restrictions; legality contingent on the scope ruling.
- **C3 — Delta-gate placement:** gate $g(x) \odot \Delta(s_t)$ inside $s_{t+1} = s_t + \cdot$; never the residual stream itself.
- **C4 — Sparse-coordinate freedom:** exploit never-active coordinates of old-task trajectories (plausible under ReLU sparsity) as legal modification space even on preserved inputs.

8. Mandatory Follow-Ups Before Any Empirical Reconciliation

- **Instantiate $\text{F}\text{\$}$ from paper . tex:** fix the literal update (residual form, LayerNorm placement, Hebbian/multiplicative terms), choose gate placement (C3), verify plasticity disabling and statelessness.
- **Re-prove every theorem for the actual placement** — especially the zero-forcing variant under delta gating.
- **Rule in writing on the $\text{t}\text{\$}$ -conditioning fork** (fixed non-learned encoding permitted?).
- **Retrieve and map the internal reports** (plan_v2, cl-h1-report) onto exactly one formal variant: depth-agnostic / $\text{t}\text{\$}$ -conditioned / state-conditioned.
- **Define controllability rigorously** as finite-target reachability over the instantiated map and exact gate class, drawing on nonlinear discrete-time reachability theory.
- **Measurement program:** empirical Lipschitz/expansiveness of the gated BDH block (tests T7's conjecture); empirical cross-task Jacobian row-space overlap in ϕ -space (tests whether T6's worst case binds).

9. Discipline Audit

- No experimental observation cited as evidence for any theorem at any point. ✓
- All overclaims identified were withdrawn on the record (capacity corollary; behavioral diagonalization; finite- D "iff"; global-preservation headline) or rescoped. ✓
- Novelty labels consistent post-adjudication: packaging novel, mathematics derivative. ✓

10. Closing Statement

Consensus is complete on the formal framework, the three-channel interference taxonomy, the exact necessary conditions T1–T3/T5–T7 as scoped, the conditional bounding result, the prior-art triage, and the open status of the asymmetric continual-learning question. The single substantive disagreement — whether a general impossibility had been established — was resolved **against** the stronger claim on the strength-of-weakest-proof principle.

The investigation exits the debate not with an impossibility theorem but with something more useful: a precise map of where gate freedom lives (never-active sparse coordinates, new-task inputs), four independent

ways the program can fail, one pivotal scope decision (\$t\$-conditioning), and one decisive open conjecture — **reachability of the instantiated BDH map under the allowed gate class** — whose resolution will determine whether the proposed mechanism is possible, impossible, or merely unproven.

Audit-Trail

Zeitstempel	Ereignis	Knoten	Aktor	L
2026-08-25T17:43:52.802045+00:00	node_completed	node-strategist	strategist	m p
2026-08-25T17:44:15.437125+00:00	node_completed	node-critic	critic	d v (L
2026-08-25T17:44:44.860621+00:00	node_completed	node-optimizer	optimizer	d v (L
2026-08-25T17:44:45.878329+00:00	node_failed	node-moderator	moderator	o (C
2026-08-25T17:49:14.096188+00:00	node_completed	node-strategist	strategist	m p
2026-08-25T17:49:38.216061+00:00	node_completed	node-critic	critic	d v (L
2026-08-25T17:53:48.267247+00:00	node_completed	node-optimizer	optimizer	d v (L
2026-08-25T17:57:15.960663+00:00	node_completed	node-moderator	moderator	o (C

Zeitstempel	Ereignis	Knoten	Aktor	L
2026-08-25T17:58:43.530593+00:00	node_completed	node-strategist	strategist	m p
2026-08-25T18:01:53.464601+00:00	node_completed	node-critic	critic	d v (L
2026-08-25T18:04:51.060386+00:00	node_completed	node-optimizer	optimizer	d v (L
2026-08-25T18:09:21.254543+00:00	node_completed	node-moderator	moderator	o (C
2026-08-25T18:10:46.592547+00:00	node_completed	node-strategist	strategist	m p
2026-08-25T18:12:37.994648+00:00	node_completed	node-critic	critic	d v (L
2026-08-25T18:13:41.556736+00:00	node_completed	node-optimizer	optimizer	d v (L
2026-08-25T18:13:43.779694+00:00	node_failed	node-moderator	moderator	o (C
2026-08-25T18:14:49.819694+00:00	node_completed	node-strategist	strategist	m p
2026-08-25T18:15:39.195477+00:00	node_completed	node-critic	critic	d v (L
2026-08-25T18:16:14.994091+00:00	node_completed	node-optimizer	optimizer	d v (L
2026-08-25T18:18:31.354724+00:00	node_completed		moderator	

Zeitstempel	Ereignis	Knoten	Aktor	L
		node-moderator		o (0
2026-08-25T18:18:31.365369+00:00	workflow_completed	None	system	
2026-08-25T18:18:32.136803+00:00	workflow_completed	None	system	